Finite semidefinite representation of the simplest quantum correlation set

Complete proofs

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A Bell contact balances eight angular monomials. Their elementary square identities determine the angular positivity. The remaining argument keeps every state direction and rescales its coefficients until a positive real Gram matrix controls the full polynomial.

Chapters 2–6 prove the physical reduction and the algebra used in that construction. Appendix A treats bounded source coefficients. Appendices B and C organize the singular contacts by their source scales, with the local identities and complementary measurement charts proved together.

Chapter 7 combines these certificates and proves the universal sum-of-squares statement that gives finite semidefinite representability. Every singular case remains part of this volume.

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Chapter 1

Finite semidefinite representation

Consider two separated quantum systems, with two binary measurements on each. Their behaviour is specified by the eight expectations

$$q = (a_0, a_1, b_0, b_1, E_{00}, E_{01}, E_{10}, E_{11}), \qquad p(\alpha, \beta \mid x, y) = \frac{1 + \alpha a_x + \beta b_y + \alpha\beta E_{xy}}{4}.$$

Let $\mathcal{Q}_{222}$ contain all such behaviours obtained from finite-dimensional quantum states and binary positive-operator-valued measurements. Local dimensions are unrestricted.

Theorem 1.1. *There are finite integers N, m and real symmetric matrices $A_0, A_1, \dots, A_8, B_1, \dots, B_m$ of order N such that*

$$\mathcal{Q}_{222} = \left\{ q \in \mathbb{R}^8 : \ \exists y \in \mathbb{R}^m, \quad A_0 + \sum_{\nu=1}^{8} q_\nu A_\nu + \sum_{j=1}^{m} y_j B_j \succeq 0 \right\}.$$

Thus every Bell functional in this scenario can be optimized over one fixed finite semidefinite representation. We prove existence by certifying all supporting inequalities. The certificate must remain valid when the inequality and the quantum realization are perturbed independently. This independence is expressed by a tensor product.

Theorem 1.2 (Separated sums of squares)**.** *Let K_c, K_t be labelled real closed extensions of $\mathbb{R}$. If $f = f_0 + \sum_{\nu=1}^{8} f_\nu q_\nu$ is nonnegative on $(\mathcal{Q}_{222})_{K_c}$ and $q \in (\mathcal{Q}_{222})_{K_t}$, then*

$$[f, q] := f_0 \otimes 1 + \sum_{\nu=1}^{8} f_\nu \otimes q_\nu \in \Sigma(K_c \otimes_{\mathbb{R}} K_t)^2.$$

Here ΣT^2 denotes finite sums of squares in T. The two fields may be labelled copies of the same extension.

The proof proceeds from the physical model to the supporting inequalities, and then to their singular contacts.

Chapter 2 first reduces the quantum set to $\operatorname{conv}\Phi(S^1 \times S^1 \times S^3)$, where Φ is the polynomial expectation map for two real qubits. At most nine such points are needed in a mixture. This gives a compact convex semialgebraic set and a finite formula for its extension to any real closed field.

Chapter 4 then reduces a valid inequality to nonnegative combinations of events and canonical supporting inequalities. Two closed families have explicit small matrix representations. Every other extreme inequality has a stationary zero and, after relabelling, one of two event supports:

$$\begin{aligned} \mathsf{C}_1 &: \quad \lambda = n = 0, \quad d, m, C, A, \alpha > 0, \quad D, \beta \geq 0, \quad D + \beta > 0, \\ \mathsf{C}_2 &: \quad \lambda = \beta = 0, \quad d, m, C, A, D, \alpha > 0, \quad n \geq 0. \end{aligned}$$

Chapter 5 derives the signs and rational coordinates of these contacts from the original stationary equations.

It remains to certify the corresponding quadratic form in the state amplitudes. Chapter 3 provides the common algebraic step: a matrix with bounded tensor coefficients and positive-definite real residue gives an exact sum of squares. The local calculations express the full quadratic form in roots for which this criterion applies. Chapter 7 gives an exhaustive source partition; the subsequent proofs establish its identities at every boundary. Vanishing event weights and commuting measurement limits remain part of the argument.

Finally, finite mixtures preserve the certificates. If $f = \sum_j a_j f_j$, $q = \sum_i w_i q_i$, $a_j, w_i \geq 0$ and $\sum_i w_i = 1$, and if $[f_j, q_i] = \sum_k h_{ijk}^2$, then

$$[f, q] = \sum_{i,j,k} \left[(\sqrt{a_j} \otimes \sqrt{w_i}) h_{ijk} \right]^2. \tag{M2}$$

The normalization of the testing weights preserves the constant term of f. Theorem 1.2 follows. Applying Scheiderer's tensor-evaluation criterion [1, Corollary 3.19] to two labelled copies of each real closed field gives Theorem 1.1.

Throughout, source coefficients and contact coordinates belong to K_c; testing measurements and states belong to K_t. The symbols $\mathrm{I}, \mathrm{U}, \mathrm{O}$ distinguish positive infinitesimals, positive bounded elements with bounded inverse, and positive unbounded elements. Formula tags are local to each section. Every sum and algebraic operation in a certificate is finite.

Chapter 2

The finite real-qubit model and exact normalization

2.1 An exact finite-dimensional physical model

Let $Q \subset \mathbb{R}^8$ be the set of correlations obtained from arbitrary finite-dimensional complex bipartite tensor-product states and two binary measurements per party. The coordinates are

$$(a_0, a_1, b_0, b_1, E_{00}, E_{01}, E_{10}, E_{11}). \tag{RFA1}$$

A binary measurement is represented by an effect $0 \preceq E \preceq I$, with observable $2E - I$. Put $Z = \mathrm{diag}(1, -1)$ and $X = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. For $(c_A, s_A), (c_B, s_B) \in S^1$ and $\psi \in S^3 \subset \mathbb{R}^4$, let Φ be the eight expectations of

$$A_0 = B_0 = Z, \qquad A_1 = c_A Z + s_A X, \qquad B_1 = c_B Z + s_B X \tag{RFA2}$$

in the state ψ. Every coordinate of Φ is polynomial in these eight real parameters.

Theorem 2.1 (Real-qubit representation)**.** *With $\mathcal{Z} = S^1 \times S^1 \times S^3$,*

$$Q = \mathrm{conv}\, \Phi(\mathcal{Z}) = \left\{ \sum_{i=1}^{9} w_i \Phi(z_i) : z_i \in \mathcal{Z},\ w_i \geq 0,\ \sum_i w_i = 1 \right\}. \tag{RFA3}$$

In particular Q is compact, convex and semialgebraic over the real field.

Proof. First, both effects of a party can be dilated with the same embedding. For an effect E on $\mathcal{H}$ define, on $\mathcal{H} \oplus \mathcal{H}$,

$$\widehat{P}_E = \begin{pmatrix} E & \sqrt{E(I-E)} \\ \sqrt{E(I-E)} & I - E \end{pmatrix}, \qquad Vh = (h, 0). \tag{RFA4}$$

The square root commutes with E by its spectral construction. Direct block multiplication gives $\widehat{P}_E^2 = \widehat{P}_E$ and $V^* \widehat{P}_E V = E$. Thus for both measurement settings the same V compresses the two projections to the original effects. Applying this construction to each party and replacing ρ by $(V_A \otimes V_B)\rho(V_A \otimes V_B)^*$ preserves every marginal and joint expectation. We may therefore assume the two observables of each party are Hermitian involutions.

Here is the needed two-dimensional block decomposition. If U, V are Hermitian involutions, $C = UV$ is unitary and $UCU = C^*$. For an eigenvector e of C with nonreal eigenvalue λ, Ue has eigenvalue $\overline{\lambda}$. The vectors are orthogonal and their span is invariant under U and $V = UC$. Its orthogonal complement is invariant too. Repeating on orthogonal eigenvectors gives two-dimensional blocks. On the real eigenvalue spaces $\lambda = \pm 1$ one has $V = \pm U$; diagonalizing U gives one-dimensional blocks. This proves the decomposition in finite dimension without an assumption on multiplicities.

On a two-dimensional block U has eigenvalues $1, -1$. In its eigenbasis, the other nontrivial involution has the form $\left(\begin{smallmatrix} c & \overline{z} \\ z & -c \end{smallmatrix}\right)$, where c is real and $c^2 + |z|^2 = 1$. A diagonal change of phase leaves $U = Z$ and makes z real. Thus the pair has the form $Z, cZ + sX$. A one-dimensional block with outcomes $a_0, a_1 \in \{\pm 1\}$ embeds into this form: use the a_0 eigenvector of Z and take $c = a_0 a_1, s = 0$. The unused basis vector does not affect the experiment.

Apply both local decompositions. All measurement operators are block diagonal, so their expectations depend only on the diagonal blocks of the state. The traces of these positive blocks are nonnegative weights summing to one. Every nonzero block, after normalization and the harmless one-dimensional embedding, is a two-qubit state with real measurement matrices. Decompose that state into finitely many complex pure states. For a real symmetric matrix M and a vector $r + is$,

$$(r + is)^* M (r + is) = r^T M r + s^T M s, \qquad \|r + is\|^2 = \|r\|^2 + \|s\|^2. \tag{RFA5}$$

The cross terms cancel by symmetry. The nonzero real and imaginary parts, normalized separately, are real pure states with their squared norms as weights. This proves $Q \subseteq \mathrm{conv}\, \Phi(\mathcal{Z})$.

Conversely each atom is a physical two-qubit experiment. A finite convex combination is realized by a finite direct sum on each party and a state with the corresponding classical diagonal weights in matching summands. Therefore the reverse inclusion holds.

Any convex combination of more than nine points in $\mathbb{R}^8$ can be shortened. The augmented vectors $(1, q_i) \in \mathbb{R}^9$ are linearly dependent, so there are α_i, not all zero, with $\sum_i \alpha_i = 0$ and $\sum_i \alpha_i q_i = 0$. Change their common sign

if needed so that some $\alpha_i > 0$, and replace the positive weights w_i by $w_i - \tau\alpha_i$, where $\tau = \min_{\alpha_i>0} w_i/\alpha_i$. They remain nonnegative, keep the same sum and point, and at least one vanishes. Iterate. Zero weights allow a fixed list of nine atoms. The right side of (RFA3) is the continuous image of the compact product of a simplex and nine copies of $\mathcal{Z}$, hence is compact. It is semialgebraic by the projection theorem for real closed fields applied to its displayed finite polynomial formula. Convexity has already been proved. □

For any real closed extension $K \supset \mathbb{R}$, define Q_K by (RFA3) with all variables in K. Define $P = \{f \in \mathbb{R}[q]_{\le 1} : f \ge 0 \text{ on } Q\}$. The defining formula $\forall q\,(q \in Q \Rightarrow f(q) \ge 0)$, together with quantifier elimination and elementary transfer, gives exactly

$$P_K = \{f \in K[q]_{\le 1} : f \ge 0 \text{ on } Q_K\}. \tag{RFA6}$$

Compactness was used only over $\mathbb{R}$. No compactness assertion for the order topology of a non-Archimedean field is needed.

2.2 Separated fields and the full homogeneous parent

Fix arbitrary real closed extensions $K_c, K_t \supset \mathbb{R}$ and put $T = K_c \otimes_{\mathbb{R}} K_t$. The factors are labelled; they may be two copies of the same extension. There is no assumption of algebraic disjointness and no chosen ordering on T. For $f = f_0 + \sum_{\nu=1}^{8} f_\nu q_\nu$ define

$$\mathscr{S}(f,q) = f_0 \otimes 1 + \sum_{\nu=1}^{8} f_\nu \otimes q_\nu. \tag{RFA7}$$

The event and CHSH convention is

$$S = E_{00} - E_{01} + E_{10} + E_{11}, \quad H = 1 - S/2, \quad f = dH + 4\sum_{\xi} \xi\, p_\xi, \quad \xi \in \{\lambda, n, m, C, A, D, \alpha, \beta\}. \tag{RFA8}$$

The homogeneous parent is explicit.

Choose projective angle pairs (a_0, a_1) and (b_0, b_1) in K_t^2, each nonzero, and set $\Delta_A = a_0^2 + a_1^2 > 0$, $\Delta_B = b_0^2 + b_1^2 > 0$. The corresponding circle coordinates are

$$c_A = (a_0^2 - a_1^2)/\Delta_A, \quad s_A = 2a_0a_1/\Delta_A, \tag{RFA9}$$

and similarly for Bob. These cover every circle point over K_t: if $c \ne -1$ use $(1 + c, s)$, and if $c = -1$ use $(0, 1)$. No affine half-angle chart is therefore needed at infinity. For $\Psi = (W, X, Y, Z)$ put

$$\begin{aligned} L_m &= a_0b_1W - a_0b_0X + a_1b_1Y - a_1b_0Z, \\ L_D &= a_1b_0W + a_1b_1X - a_0b_0Y - a_0b_1Z. \end{aligned}$$

Products of a coefficient-field element with a testing-field expression in the next identity mean their separated tensor product. The complete homogeneous parent is

$$\begin{aligned} \mathcal{F} = {} & 4\lambda\Delta_A(b_0W + b_1X)^2 + 4n\Delta_A\Delta_BX^2 + 4mL_m^2 \\ & + 4C\Delta_B(a_1W - a_0Y)^2 + 4A\Delta_A\Delta_BY^2 \\ & + (4D + 2d)L_D^2 + (4\alpha + 2d)\Delta_A(b_1Y - b_0Z)^2 \\ & + (4\beta + 2d)\Delta_B(a_0X + a_1Z)^2 - 2d\Delta_A\Delta_BZ^2. \end{aligned} \tag{RFA10}$$

For a nonzero state let $N = \|\Psi\|^2 > 0$ and let z be the atom with the circle coordinates (RFA9) and unit state $\Psi/\sqrt{N}$. Then

$$\mathcal{F} = (1 \otimes \Delta_A\Delta_BN)\,\mathscr{S}(f, \Phi(z)). \tag{RFA11}$$

To check the normalization, first take $a_0 = b_0 = 1$, $u = a_1$ and $v = b_1$. The plus/minus measurement eigenvectors are $(1, u)/\sqrt{1 + u^2}$ and $(u, -1)/\sqrt{1 + u^2}$. Their product amplitudes give, in the order of (RFA8),

$$\begin{gathered} \frac{(W + vX)^2}{1 + v^2}, \quad X^2, \quad \frac{(vW - X + uvY - uZ)^2}{(1 + u^2)(1 + v^2)}, \quad \frac{(uW - Y)^2}{1 + u^2}, \quad Y^2, \\ \frac{(uW + uvX - Y - vZ)^2}{(1 + u^2)(1 + v^2)}, \quad \frac{(vY - Z)^2}{1 + v^2}, \quad \frac{(X + uZ)^2}{1 + u^2}. \end{gathered} \tag{RFA12}$$

For a normalized state these are exactly the eight selected event probabilities. For a general state they are multiplied by N. Substitution of the matrices (RFA2) into H gives its numerator

$$2\{(uW + uvX - Y - vZ)^2 + (1 + u^2)(vY - Z)^2 + (1 + v^2)(X + uZ)^2 - (1 + u^2)(1 + v^2)Z^2\}.$$

This proves (RFA11) in the affine chart. Homogenization gives (RFA10); it is a polynomial identity and therefore also gives the zero and infinite projective coordinates without dividing by a_0 or b_0. The zero state has zero parent separately.

If $\mathcal{F}$ or $(s \otimes 1)\mathcal{F}$ has an exact finite SOS certificate, with $s \in K_c^{>0}$, the original slack is restored by multiplying each root by

$$s^{-1/2} \otimes (\Delta_A \Delta_B N)^{-1/2}. \tag{RFA13}$$

All these inverses are taken within one field before forming a tensor. The identity does not permit cancellation of an arbitrary mixed tensor. Additional positive source gauges are handled in the same way. A state chart or a change of frame must first prove its own denominator nonzero.

Literal commuting tests. These require no limiting local certificate. If one party's two involutions commute, on a simultaneous eigenspace the CHSH operator $S = A_0(B_0 - B_1) + A_1(B_0 + B_1)$ is either $\pm 2B_0$ or $\pm 2B_1$. Thus $H = I - S/2$ is positive semidefinite in the testing field. For $d \geq 0$, its contribution is a pure-source square times a pure-testing square. Every nonnegative event packet is also separated SOS. This covers the actual commuting angle axes and infinities for every nonnegative canonical coefficient tuple. It does not cover merely infinitesimal nonzero angles, which are covered by the canonical certificate theorem.

Chapter 3

Exact sums of squares in the separated tensor ring

3.1 Two labelled fields and their residues

Let K_c, K_t be real closed extensions of $\mathbb{R}$ and put

$$T = K_c \otimes_{\mathbb{R}} K_t, \qquad B_i = \{x \in K_i : |x| < M \text{ for some } M \in \mathbb{R}_{>0}\}, \quad i \in \{c, t\}.$$

Write $\mathfrak{m}_i$ for the elements whose absolute value is smaller than every positive real number. The order on K_i is its unique field order. No order, embedding into a field, or algebraic-disjointness assumption is imposed on T.

Every $x \in B_i$ has a unique real standard part $\bar{x}$ with $x - \bar{x} \in \mathfrak{m}_i$. Indeed, the real cut $\{r \in \mathbb{R} : r \leq x\}$ is nonempty and bounded; its supremum has the claimed property. Two such real numbers differ by a real infinitesimal and hence agree. Infinitesimals form an ideal of B_i, so $x \mapsto \bar{x}$ is an $\mathbb{R}$-algebra homomorphism with kernel $\mathfrak{m}_i$. In particular $B_i = \mathbb{R} \oplus \mathfrak{m}_i$ as real vector spaces.

Since tensoring real vector spaces preserves injections, $B := B_c \otimes_{\mathbb{R}} B_t$ embeds into T. Its residue map is

$$\rho : B \longrightarrow \mathbb{R}, \qquad \rho\Big(\sum_i a_i \otimes b_i\Big) = \sum_i \bar{a}_i \bar{b}_i. \tag{TF1}$$

The direct-sum decompositions above show that

$$\ker \rho = \mathfrak{m}_c \otimes B_t + B_c \otimes \mathfrak{m}_t. \tag{TF2}$$

Thus every tensor of zero residue has a finite expression in which each elementary term has at least one infinitesimal factor. None of these observations requires the two field extensions to be disjoint.

3.2 A strict real reserve gives an exact tensor sum of squares

Lemma TF1 (positive residue). If $a \in B$ and $\rho(a) > 0$, then a is a finite sum of squares in T. In fact its square roots may all be chosen to be elementary tensors.

Proof. By (TF2), write

$$a = \varepsilon + \sum_{j=1}^{N} x_j \otimes y_j, \qquad \varepsilon = \rho(a) > 0, \tag{TF3}$$

with bounded pure factors and an infinitesimal in at least one factor of each term. If $N = 0$, the assertion is immediate. Otherwise choose positive real numbers ε_j summing to ε. For a term with x_j infinitesimal, choose a positive real $M_j > |y_j|$. Direct multiplication gives

$$\varepsilon_j + x_j \otimes y_j = \frac{1}{2}\left[(\varepsilon_j/M_j + x_j) \otimes (M_j + y_j) + (\varepsilon_j/M_j - x_j) \otimes (M_j - y_j)\right]. \tag{TF4}$$

All four pure factors are strictly positive in their respective fields. Each elementary summand therefore is a square: for example $\frac{1}{2}u \otimes v = (\sqrt{u/2} \otimes \sqrt{v})^2$. If instead y_j is infinitesimal, interchange the roles of the two factors in this calculation. Sum (TF4) over j. Every operation is a field operation in one labelled factor; no mixed radical or mixed inverse occurs. □

Lemma TF2 (strict bounded Gram). Let $Q \in \mathrm{Sym}_r(B)$ and suppose its entrywise real residue $\rho(Q)$ is positive definite. Then

$$g^{\mathsf{T}} Q g \in \Sigma T^2 \qquad \text{for every } g \in T^r. \tag{TF5}$$

The entries of g need not be bounded, independent, or polynomial.

Proof. Choose a real invertible matrix C with $C^{\mathsf{T}} \rho(Q) C = I$. Replacing Q by $C^{\mathsf{T}} Q C$ and g by $C^{-1} g$ preserves the quadratic expression, so assume $\rho(Q) = I$. Choose a real $\delta > 0$ such that $(r-1)\delta < 1$. With e_i the real

coordinate vectors,

$$\begin{aligned} Q = &\sum_i \big(Q_{ii} - (r-1)\delta\big) e_i e_i^{\mathsf{T}} \\ &+ \sum_{i<j} \frac{\delta + Q_{ij}}{2} (e_i + e_j)(e_i + e_j)^{\mathsf{T}} \\ &+ \sum_{i<j} \frac{\delta - Q_{ij}}{2} (e_i - e_j)(e_i - e_j)^{\mathsf{T}}. \end{aligned} \tag{TF6}$$

The off-diagonal entries agree because the last two contributions have difference Q_{ij}; their diagonal contributions sum to δ for every incident pair. The residues of all scalar coefficients in (TF6) are positive. Lemma TF1 expresses each coefficient as a finite sum of squares in T. Evaluation on g then expresses (TF5) as a finite sum of squares of those scalar roots times g_i or $g_i \pm g_j$. This is a polynomial identity in the commutative ring T, so no size condition on g is involved. □

3.3 Legal normalization and finite assembly

Lemma TF3 (pure-factor cancellation). For $s \in K_c^{>0}$, $t \in K_t^{>0}$, and $F \in T$,

$$F \in \Sigma T^2 \quad \Longleftrightarrow \quad (s \otimes t)F \in \Sigma T^2. \tag{TF8}$$

Indeed multiply each root by $\sqrt{s} \otimes \sqrt{t}$ in one direction and by $s^{-1/2} \otimes t^{-1/2}$ in the other. These are inverse units of T. Nonvanishing must be proved before this lemma is used. A homogeneous coordinate that is literally zero cannot be cancelled, and the lemma does not authorize cancellation of a general mixed tensor.

Lemma TF4 (positive matrix pairing). If $X \in \mathrm{Sym}_r(K_c)$ and $Y \in \mathrm{Sym}_r(K_t)$ are positive semidefinite in their own fields, then

$$\sum_{i,j} X_{ij} \otimes Y_{ij} = \sum_{a,b} \left(\sum_i U_{ia} \otimes V_{ib} \right)^2, \qquad X = UU^{\mathsf{T}}, \quad Y = VV^{\mathsf{T}}. \tag{TF9}$$

To obtain the factorizations over a real closed field, pivot on a positive diagonal entry and complete its square. The remaining Schur complement is positive semidefinite. If every diagonal entry is zero, every off-diagonal entry is zero by its two-dimensional quadratic restriction. Induction, and positive square roots of the pivots, finish the factorization even for singular matrices. Expansion of the right side proves (TF9).

Lemma TF5 (conic and convex closure). For $f = f_0 + \sum_{\nu=1}^n f_\nu q_\nu$ set $[f, q] = f_0 \otimes 1 + \sum_\nu f_\nu \otimes q_\nu$. Suppose in the source field $f = \sum_j a_j f_j$ with $a_j \geq 0$, and in the testing field $q = \sum_i w_i q_i$ with $w_i \geq 0$ and $\sum_i w_i = 1$. If $[f_j, q_i] = \sum_k h_{ijk}^2$ in T, then

$$[f, q] = \sum_{i,j,k} \big[(\sqrt{a_j} \otimes \sqrt{w_i}) h_{ijk} \big]^2. \tag{TF10}$$

This follows from linearity in f and affineness in q; the equation $\sum_i w_i = 1$ preserves the constant coefficient. All sums are finite. Zero weights give zero roots and are never inverted.

3.4 From separated squares to a finite representation

Let $S \subset \mathbb{R}^n$ be closed, convex, and semialgebraic, and let P_S be its cone of nonnegative affine functions. Fix a finite real formula defining S. Over any real closed extension K, its transfer S_K is the set defined by that same formula. The formula

$$\forall q \; (q \in S \implies f(q) \geq 0) \tag{TF11}$$

shows, by quantifier elimination and elementary transfer, that $(P_S)_K$ consists exactly of the K-affine functions nonnegative on S_K. This is a statement about formulas; it makes no claim that S_K is compact in the order topology of K.

Scheiderer's Corollary 3.19 [1] says that S has a finite semidefinite lift if and only if, for every real closed extension $K/\mathbb{R}$, every $f \in (P_S)_K$, and every $q \in S_K$, the point-first tensor evaluation is a finite sum of squares in $K \otimes_{\mathbb{R}} K$. The coefficient-first convention used here is equivalent by the factor-flip automorphism, which preserves squares. Therefore a proof of

$$[f, q] \in \Sigma (K \otimes_{\mathbb{R}} K)^2 \quad \text{for every } K, \; f \in (P_S)_K, \; q \in S_K \tag{TF12}$$

is sufficient. Two labelled copies of each one arbitrary K suffice; a theorem for all unequal pairs K_c, K_t is stronger than necessary. The labels must still remain separate in every identity. In particular, algebraic disjointness of the copies must not be imposed.

The number and tensor ranks of the roots in (TF12) may depend on K, f, q. Corollary 3.19 converts this universal assertion into one finite semidefinite representation by a finite-dimensional algebraic construction and compactness. This is the implication used here.

Chapter 4

The source cone: two closed faces and two contact families

4.1 The exact physical polynomial

Use the affine functional $H = 1 - (E_{00} - E_{01} + E_{10} + E_{11})/2$. Order the eight losing events as

$$(\lambda, n, m, C, A, D, \alpha, \beta) = \big((++|01), (+-|00), (+-|11), (-+|10), (-+|00), (-+|11), (--|01), (+-|10)\big).$$

Every event vanishes at the chosen PR point. Each is one at its own deterministic CHSH-facet neighbour and zero at the other seven. Consequently every affine functional has the unique expansion

$$f = dH + 4\sum_i a_i p_i, \qquad d = -f(\mathrm{PR}), \qquad a_i = f(v_i)/4. \tag{SG1}$$

For a quantum-valid functional negative at that PR point, $d > 0$ and $a_i \geq 0$. If a quantum-valid functional is nonnegative at every PR point, it is nonnegative on the entire nonsignalling polytope and is a nonnegative combination of its probability inequalities. The latter sources have immediate separated square certificates. Here is a complete short vertex argument. The nonsignalling affine space has dimension eight. If no local marginal is deterministic, each two-by-two outcome table has positive row and column sums, hence at most two zero entries. A vertex requires at least eight active probability inequalities. Thus every table has exactly two zeros, necessarily in opposite corners. Write its perfect correlation or anticorrelation as $a_x = s_{xy} b_y$, with $s_{xy} = \pm 1$. If $\prod_{x,y} s_{xy} = -1$, these equations force all four marginals to vanish and give a PR vertex. If the product is $+1$, one marginal remains free and the table is a mixture of two deterministic points, so an interior marginal value is not a vertex. If instead one marginal is deterministic, say A_0, a local model is obtained by sampling A_1 and then sampling B_0, B_1 independently conditional on A_1, using the two given joint tables. Nonsignalling makes their A_1 marginals agree; the tables involving fixed A_0 are reproduced automatically. Such a vertex is therefore deterministic. This proves that the only vertices are the sixteen deterministic points and the eight PR points. Linear separation (equivalently finite-dimensional Farkas duality) now gives the asserted event representation.

Put $D_u = 1 + u^2$, $D_v = 1 + v^2$, and write the real state as $\psi = (W, X, Y, Z)$. The entire cleared parent is

$$\begin{aligned} F = {} & 4\lambda D_u (W + vX)^2 + 4n D_u D_v X^2 + 4m(vW - X + uvY - uZ)^2 \\ & + 4C D_v (uW - Y)^2 + 4A D_u D_v Y^2 + (4D + 2d)(uW + uvX - Y - vZ)^2 \\ & + (4\alpha + 2d) D_u (vY - Z)^2 + (4\beta + 2d) D_v (X + uZ)^2 - 2d D_u D_v Z^2. \end{aligned} \tag{SG2}$$

It is $D_u D_v \|\psi\|^2 f(q)$ for a nonzero unnormalised state. All these factors are pure testing factors. Write $P(u, v)$ for its symmetric state matrix. If $d > 0$ and $uv \neq 0$, its $Z = 0$ principal block is positive definite, since the signed part alone is

$$2d\{(uW + uvX - Y)^2 + D_u v^2 Y^2 + D_v X^2\}. \tag{SG3}$$

Thus P has at least three positive eigenvalues, its kernel at a zero is one-dimensional with $Z \neq 0$, and

$$P \succeq 0 \quad \Longleftrightarrow \quad \det P \geq 0. \tag{SG4}$$

Here is the entire determinant datum needed for the geometric reduction. Set

$$\begin{aligned} t_1 &= d + 2(\lambda + n + D), & t_2 &= d + 2(\alpha + m + A), \\ t_3 &= d + 2(\lambda + C + D), & t_4 &= d + 2(\alpha + \beta + m), \\ t_5 &= d + 2(\lambda + \beta + n), & t_6 &= d + 2(\alpha + C + A), \\ t_7 &= d + 2(\beta + n + m), & t_8 &= d + 2(C + A + D). \end{aligned}$$

All $t_i > 0$. Directly expanding the four-dimensional determinant of SG2, or applying Cauchy–Binet to its eight positive and one negative rank-one packets, gives

$$\det P = 16 D_u^2 D_v^2 u^2 v^2 L(u^2, v^2), \qquad L(U, V) = h + \sum_{(i,j) \neq (1,1)} q_{ij} U^{i-1} V^{j-1}, \tag{SG5}$$

where

$$\begin{aligned} q_{00} &= 4\lambda\alpha t_7 t_8, & q_{01} &= 4(\lambda D t_7 t_6 + \alpha m t_5 t_8), \\ q_{02} &= 4m D t_5 t_6, & q_{10} &= 4(\alpha A t_7 t_3 + \lambda n t_8 t_4), \\ q_{12} &= 4(\beta m t_1 t_6 + C D t_5 t_2), & q_{20} &= 4n A t_3 t_4, \\ q_{21} &= 4(\beta n t_3 t_2 + C A t_1 t_4), & q_{22} &= 4\beta C t_1 t_2. \end{aligned} \tag{SG6}$$

The central coefficient is fixed without a long expansion by $h = \det P(1,1)/256 - \sum_{(i,j)\neq(1,1)} q_{ij}$. The eight nonnegative outer coefficients determine the possible contact geometries.

4.2 The twelve elementary angular identities

The determinant of the state matrix has eight nonconstant Laurent monomials. Their coefficients are nonnegative. A stationary zero therefore has a particularly simple algebraic decomposition, which also explains the cubic and quartic identities used below.

Let $E = \{-1, 0, 1\}^2 \setminus \{(0, 0)\}$ and write

$$L(U, V) = h + \sum_{e \in E} q_e U^{e_1} V^{e_2}, \qquad q_e \geq 0.$$

Suppose $U_c, V_c > 0$ satisfy $L = L_U = L_V = 0$. Put $w_e = q_e U_c^{e_1} V_c^{e_2}$. The value and derivative equations are exactly

$$h = -\sum_e w_e, \qquad \sum_e w_e e = 0.$$

Thus the positive coefficients balance around the origin in the exponent plane.

Lemma 4.1 (Balanced Laurent polynomial)**.** *Let K_c, K_t be labelled real closed extensions of $\mathbb{R}$. If the coefficients and contact above belong to K_c, then for every $U, V > 0$ in K_t the separated evaluation of L is a finite sum of squares in $K_c \otimes_{\mathbb{R}} K_t$.*

Proof. First decompose the coefficient vector w using its balance equation. If $w \neq 0$, the origin lies in the convex hull of its active exponents. A minimal subset containing the origin has two or three elements. For the eight exponents in E, the possibilities, up to rotation and reflection of the square, are

exponents	positive integer weights	number
$\{e, -e\}$	$(1, 1)$	4
$\{(1, 1), (-1, 0), (0, -1)\}$	$(1, 1, 1)$	4
$\{(1, 1), (-1, 1), (0, -1)\}$	$(1, 1, 2)$	4.

Indeed a two-element set must be opposite. A triangle containing the origin contains either one corner and its two opposite axial points, or two adjacent corners and their opposite axial point. These are the last two rows. For its integer weight vector b, subtract λb, where $\lambda = \min_{b_e > 0} w_e / b_e$. The remainder is nonnegative and balanced, and has fewer active entries. Repeating at most eight times gives $w = \sum_j \lambda_j b^{(j)}$ with $\lambda_j > 0$ in K_c.

Set $X = U_c^{-1} \otimes U$ and $Y = V_c^{-1} \otimes V$. Their rational powers mean separate positive roots in the two fields. For each row of the table the monomials $p_i = X^{e_{i1}} Y^{e_{i2}}$ satisfy $\prod_i p_i^{b_i} = 1$. The required elementary identities are

$$\begin{aligned} p + p^{-1} - 2 &= (\sqrt{p} - p^{-1/2})^2, \\ a^3 + b^3 + c^3 - 3abc &= \tfrac{1}{2}(a + b + c)\big[(a - b)^2 + (b - c)^2 + (c - a)^2\big], \\ a^4 + b^4 + c^4 + d^4 - 4abcd &= (a^2 - b^2)^2 + (c^2 - d^2)^2 + 2(ab - cd)^2. \end{aligned}$$

For weights $(1, 1, 1)$ take $a = p_1^{1/3}$, $b = p_2^{1/3}$, $c = p_3^{1/3}$; then $abc = 1$. Each of a, b, c is an elementary tensor square, so the middle line is a sum of tensor squares. For weights $(1, 1, 2)$ take $a = p_1^{1/4}$, $b = p_2^{1/4}$ and $c = d = p_3^{1/4}$; then $abcd = 1$. Finally multiply the squares by $\sqrt{\lambda_j} \otimes 1$ and sum. Equation (4.2) makes the result exactly L. □

In a single real closed field this proves that a stationary zero of L is a global minimum, using only the three displayed identities. For the Bell state matrix, its positive three-dimensional principal block then turns determinant nonnegativity into matrix positivity. The same calculation proves a separated sum-of-squares identity for the determinant at independent tests. To obtain a certificate for the whole state polynomial we must also control the state-normal terms; the following matrix lemmas perform that step.

4.3 Stationary zeros and physicality

Contact lemma. Suppose $d > 0$, all event weights are nonnegative, and there exist positive u_c, v_c and a nonzero real state ψ_c such that

$$P(u_c, v_c)\psi_c = 0, \qquad \psi_c^T P_u(u_c, v_c)\psi_c = 0, \qquad \psi_c^T P_v(u_c, v_c)\psi_c = 0. \tag{SG7}$$

Then the source is physical at every angle and every state. This statement holds over every real closed field.

Proof. SG3 gives three positive complementary state eigenvalues. The adjugate at contact is a positive multiple of $\psi_c \psi_c^T$, so SG7 and SG5 give $L = L_U = L_V = 0$ there. The balanced Laurent identity just proved makes L nonnegative at every positive pair in the same field. Equation SG4 therefore gives physicality for $uv \neq 0$. At a

commuting or projective endpoint, simultaneous diagonalization of the commuting local pair reduces the CHSH operator to ± 2 times a remote reflection, so $H \succeq 0$. The event terms are nonnegative there as well. Homogenizing SG2 includes all endpoints. This proves the claim over every real closed field. □

This lemma is also the justification for deleting a positive event whose amplitude vanishes at a contact. Its square and all first derivatives vanish; deleting it preserves SG7. It is not legitimate to delete an event whose amplitude is nonzero.

4.4 The geometry of a stationary zero

Coercivity at every physical finite contact. Every physical source with $d > 0$ and a finite noncommuting zero has coercive Newton support. Indeed, put $a_e = q_e U_c^{e_1} V_c^{e_2} > 0$ for the active exponent vectors. The two contact derivatives give

$$\sum_e a_e e = 0. \tag{SG17}$$

If zero is not in the interior of their convex hull, a separating linear functional nonnegative on every active vector must vanish on all of them, by SG17. Thus the active support is empty or lies on one of the four lines through the origin in the square grid. A nonempty one-sided subset cannot satisfy SG17. The horizontal opposite pair $q_{01}, q_{21} > 0$ has four possible product pairings in SG6. These respectively force λn, CD, βm or αA positive, activating an off-line coefficient q_{10} or q_{12}. The vertical pair similarly forces αm, CA, βn or λD and activates q_{01} or q_{21}.

For the diagonal pair $q_{00}, q_{22} > 0$, all off-line coefficients can vanish only on the alternating matching $\lambda, \alpha, \beta, C > 0$, $n = m = A = D = 0$. For the other diagonal they can vanish only on the opposite matching $n, m, A, D > 0$, $\lambda = C = \alpha = \beta = 0$. The latter is not physical: SG2 at $u = v = 1$ and $\psi = (1, -\varepsilon, \varepsilon, 1)$ gives

$$F = 16\varepsilon\{(d + n + m + A + D)\varepsilon - d\} < 0 \quad \text{for} \quad \varepsilon = \frac{d}{2(d + n + m + A + D)}.$$

The first matching is its physical one-step cycle rotation. If every outer coefficient is zero, SG6 forbids every adjacent or opposite positive pair in the cycle $(\lambda, D, C, A, \alpha, m, \beta, n)$. Any maximal remaining set is a triple with cyclic gaps $(2, 3, 3)$; smaller sets extend to such a triple. The eight triples are rotations of $\{\lambda, m, A\}$. At $u = v = 1$, $\psi = (1, -1, 0, 2)$ annihilates precisely the needed three event rows and has signed-core value -4, giving $F = -4d < 0$. This excludes empty support too.

For completeness, the physical cycle symmetries used here are the CHSH-preserving maps on event coordinates (x, y, a, b)

$$\begin{aligned} g_A &: (x, y, a, b) \mapsto (1 - x, y, a, (-1)^y b), \\ g_B &: (x, y, a, b) \mapsto (x, 1 - y, (-1)^{1-x} a, b), \\ g_T &: (x, y, a, b) \mapsto (y, x, (-1)^y b, (-1)^x a). \end{aligned}$$

In the displayed cycle, indexed from zero, they act as $i \mapsto 5 - i, 3 - i, 2 - i$ modulo eight. Their compositions give a one-step rotation and a reflection. The assertions follow by inserting the eight event definitions in SG1.

Strict single-field contact Hessian. The logarithmic angular Hessian is now

$$\mathcal{H}_L = \sum_e a_e e e^T \succ 0, \tag{SG18}$$

since the active vectors span the plane. Write the state matrix as $P = \left(\begin{smallmatrix} H_3 & b \\ b^T & c \end{smallmatrix}\right)$ in the coordinates $(W, X, Y; Z)$ and set $Z = 1$. The positive definite state block SG3 gives, within the single source field,

$$F = (z + H_3^{-1} b)^T H_3 (z + H_3^{-1} b) + s(u, v), \qquad s = \det P / \det H_3, \quad z = (W, X, Y)^T. \tag{SG19}$$

At contact $s = 0$ and its gradient vanishes. Its angular Hessian is the positive scalar $16 D_u^2 D_v^2 u^2 v^2 / \det H_3$ times the congruence of SG18 by $\mathrm{diag}(2/u, 2/v)$. Consequently the full Hessian of F in (W, X, Y, u, v) has positive state block $2H_3$ and a positive angular Schur complement; it is positive definite. All formulas for derivatives and positive definiteness are algebraic. The finite real assertion therefore transfers to every real closed field; logarithms are only notation for $U\partial_U, V\partial_V$. The inverse in SG19 belongs to the source field.

Positive contact state orientation. Let F_H denote SG2 with $d = 1$ and all eight event weights zero. Its state matrix satisfies

$$\det P_H = -16 D_u^2 D_v^2 u^2 v^2 \neq 0. \tag{SG20}$$

This follows by a four-dimensional determinant expansion of the three signed-core squares and the negative Z^2 term in SG2. If all positive-weight event rows vanished at a physical contact, their state matrices would annihilate the contact state. The kernel equation would then give $P_H\psi_c = 0$, contradicting SG20. Hence at least one event contributes positively. Since $F(\psi_c) = 0$ and $d > 0$, the core value $F_H(\psi_c)$ is strictly negative.

For positive u, v, the X, Y block of F_H is $2\left(\begin{smallmatrix} h_0 & -uv \\ -uv & h_0 \end{smallmatrix}\right)$, where $h_0 = 1 + v^2 + u^2v^2$. Its Schur form is exactly SG15. It is nonnegative when $WZ \le 0$. Thus the negative core value forces

$$W_cZ_c > 0, \qquad Z_c = 1 \Longrightarrow W_c > 0. \tag{SG21}$$

The forbidden-deletion implication. If a positive event row vanished at a contact, deleting that coefficient would preserve the four kernel equations and both angular derivatives. SG7 would make the resulting nonnegative source physical, with the same finite contact. SG17–SG18 therefore make its support coercive. We obtain the exact rule

a positive contact event cannot vanish if deleting it makes the Newton support noncoercive. (SG22)

In particular every deletion introducing an exclusive omitted pair is forbidden by SG9. This rule will fix the signs of the contact coordinates in Chapter 5.

4.5 Reduction to two contact families

The active exponent vectors in SG5 lie on the boundary of $\{-1, 0, 1\}^2$. The exponential sum is coercive exactly when zero is in the interior of their convex hull. One direction follows from uniform positive support in every unit direction; the other follows from a separating direction along which every active exponential is nonincreasing.

For this eight-point configuration a separating direction can be chosen from $(\pm1, 0), (0, \pm1), (\pm1, \pm1)$. For example the axis condition in SG6 gives $\lambda\alpha = \lambda D = \alpha m = mD = 0$, whose minimal omitted event pairs are $\{\lambda, m\}$ and $\{\alpha, D\}$. For a diagonal the additional minimal pair is $\{\lambda, \alpha\}$; the remaining minimal triples already contain an axis pair. Rotating the event cycle

$$(\lambda, D, C, A, \alpha, m, \beta, n) \tag{SG8}$$

gives exactly the twelve pairs of exclusive events. Therefore

coercive exponent support $\iff$ every two omitted losing events are compatible. (SG9)

Compatibility means agreement on each shared local measurement. Pairwise compatible events extend to a single deterministic assignment. Such an assignment loses one or three CHSH inputs; there are consequently at most three compatible omissions. In SG8 compatible pairs have cyclic distance one or two, and a compatible triple is three consecutive vertices.

Extreme-source lemma. Normalise the canonical source by $d + \sum a_i = 1$ and impose physicality and nonnegativity. This is a compact convex subset of an eight-dimensional simplex. Every extreme point with $d > 0$ and coercive support has a finite stationary zero and exactly five or six positive event weights.

Proof. Coercivity gives a finite minimum $\mu \ge 0$ of L. If $\mu > 0$, take sufficiently small opposite nonradial perturbations inside the exact active coefficient face, preserving the normalisation. Every positive q_e changes by relative error at most ε, every zero q_e remains zero, and h changes by absolute error at most ε. Thus, uniformly at all positive angles,

$$L' \ge (1 - \varepsilon)L + \varepsilon h - \varepsilon \ge (1 - \varepsilon)\mu + \varepsilon h - \varepsilon > 0.$$

The perturbations remain physical by SG4 and the direct endpoint argument, contradicting extremality. The perturbation exists: SG9 permits at most three omissions, hence at least six active coordinates including d. Thus $\mu = 0$ and there is a contact.

Fix its angles and state. SG7 consists of six homogeneous linear equations in the source coefficients. If at least seven events were positive, there would be at least eight active coordinates; their solution space has dimension at least two and contains the source. Intersecting its directions with coefficient sum zero leaves a nonzero direction. Small opposite perturbations preserve all signs and SG7, and are physical by the contact lemma. This again contradicts extremality. Hence at most six events are positive. The lower bound five follows from SG9. □

The CHSH-preserving event relabellings generate the dihedral group of SG8. Thus the coercive extreme sources, including every allowed third omission, reduce to the two patterns

$$\begin{aligned} &\mathsf{C}_1: \quad \lambda = n = 0, \quad d, m, C, A, \alpha > 0, \quad D, \beta \ge 0, \quad D + \beta > 0, \\ &\mathsf{C}_2: \quad \lambda = \beta = 0, \quad d, m, C, A, D, \alpha > 0, \quad n \ge 0. \end{aligned} \tag{SG10}$$

The C1 third omission is $D = 0$ or $\beta = 0$; the C2 third omission is $n = 0$, already a C1 orbit. This overlap is harmless. It is not legitimate to replace positive infinitesimal coefficients by zeros.

Every point of the normalised source body is a convex combination of at most nine extreme points. The fixed nine-term assertion, the midpoint definition of extremality, SG7, and physicality all have first-order polynomial formulations. Their real proof therefore transfers to a real closed coefficient field, without using topological compactness or arbitrary infinite decompositions in that field. A nonnegative decomposition preserves each literal zero coefficient term by term.

4.6 Noncoercive sources have explicit lifts

Every noncoercive source omits an exclusive pair. Up to the same finite symmetries there are only two possibilities: $\lambda = m = 0$ and $\lambda = \alpha = 0$.

We first record the elementary quadratic matrix lemma. A real symmetric matrix polynomial $K(t) = K_0 + tK_1 + t^2K_2$ is PSD for every real t exactly when there is a skew-symmetric S such that

$$\begin{pmatrix} K_0 & K_1/2 + S \\ K_1/2 - S & K_2 \end{pmatrix} \succeq 0. \tag{SG11}$$

For completeness, the cone of Gram-representable coefficient triples is closed because the trace of a representing PSD Gram is $\operatorname{tr}(K_0 + K_2)$. A nonnegative dual functional has block Hankel matrix $\left(\begin{smallmatrix} U & V \\ V & W \end{smallmatrix}\right) \succeq 0$, with U, V, W symmetric. Diagonalising $U^{+/2}VU^{+/2}$ on $\operatorname{ran} U$ writes its moment part as $\sum_j (z_j, t_j z_j)(z_j, t_j z_j)^T$; the remainder is $\operatorname{diag}(0, W - VU^{+}V) \succeq 0$. Its value on K is consequently $\sum_j z_j^T K(t_j) z_j + \sum_i w_i^T K_2 w_i \geq 0$. Separation proves SG11, including singular U. The fixed-size statement transfers to real closed fields.

For $\lambda = m = 0$, use SG11 with

$$\begin{aligned} K_0 &= \operatorname{diag}(4C + 4D + 2d, 4A, 4\alpha), \\ K_2 &= \operatorname{diag}(4C, 4A + 4\alpha + 2d, 4D), \\ K_1 &= \begin{pmatrix} 0 & 0 & -(4D + 2d) \\ 0 & 0 & -(4\alpha + 2d) \\ -(4D + 2d) & -(4\alpha + 2d) & 0 \end{pmatrix}. \end{aligned} \tag{SG12}$$

Necessity follows by setting $W = a/u$ and letting $u \to 0$ in the homogeneous state form. The surviving form, after $R = a - Y$, is $2t_5 D_v X^2 + (R, Y, Z)^T K(v)(R, Y, Z)$. For sufficiency the determinant identity gives

$$\det K(v) = \frac{8D_v}{t_5}(q_{10} + hv^2 + q_{12}v^4),$$

whereas $L = h + q_{10}/V + q_{12}V + U(q_{20}/V + q_{21} + q_{22}V)$. PSD of K makes the first three terms nonnegative; all remaining terms are nonnegative. SG4 and the endpoint argument finish the equivalence. When $d = 0$ all forms are event-square sums. The Gram is strictly feasible at $C = A = 2$, $D = \alpha = d = n = \beta = 1$, $S = 0$: after permutation its two Schur margins are $125/56 > 0$.

For $\lambda = \alpha = 0$, put $p = 4C + 4D + 2d$, $q = 4m + 4\beta + 2d$, $r = 4D + 2d$. The exact lift is

$$B_1 = \begin{pmatrix} 4m & -4m & \xi \\ -4m & q + 4n & q \\ \xi & q & q \end{pmatrix} \succeq 0, \quad B_2 = \begin{pmatrix} p + 4A & 4D & -p \\ 4D & 4D & -(4m + r) - \xi \\ -p & -(4m + r) - \xi & p \end{pmatrix} \succeq 0. \tag{SG13}$$

Indeed set $u = \varepsilon s$, $v = \varepsilon$, $(W, Z) = (w, z)/\varepsilon$, $X = x$, $Y = y$. The surviving form is

$$\begin{aligned} G_s = {} & 4nx^2 + 4m(w - x - sz)^2 + 4C(sw - y)^2 + 4Ay^2 \\ & + r(sw - y - z)^2 + (4\beta + 2d)(x + sz)^2 - 4dyz - 2dz^2. \end{aligned}$$

Its determinant is $16(q_{02} + hs^2 + q_{20}s^4)$. The same nonnegative-remainder argument proves equivalence with physicality. Applying SG11 to its state matrix, the zero diagonal entries on sx, sy force their Gram rows to vanish. Coefficient matching leaves only ξ, and the remaining Gram splits into the two blocks SG13. Equivalently $G_s = (w, x, sz)B_1(w, x, sz)^T + (y, z, sw)B_2(y, z, sw)^T$; the ξ terms cancel because $w(sz) = z(sw)$. At $C = A = n = \beta = 2$, $m = D = d = 1$, $\xi = -5$, the successive principal minors are $(4, 72, 234)$ and $(22, 72, 234)$. Both lifts retain all nonnegative coefficient constraints.

Why these source lifts give separated certificates. Represent either source cone as $a \geq 0$ and $\exists z\, M(a, z) \succeq 0$, absorbing the signs into diagonal blocks. The displayed points make this entire homogeneous lift strictly feasible.

The image of its dual PSD cone is closed: pairing with a fixed strict feasible matrix bounds the trace of every PSD preimage of a convergent dual sequence. Separation therefore proves attained dual representation for every functional nonnegative on the source cone. This is a finite-dimensional argument, and its fixed matrix-size quantified conclusion transfers.

Given a source in K_c, choose its lift $A_c \succeq 0$ there. Given a physical test in K_t, its evaluation functional belongs to the dual source cone in K_t; choose its attained dual matrix $B_t \succeq 0$ there. Their separated trace pairing equals the slack because all auxiliary-coordinate coefficients cancel. Factoring $A_c = RR^T$ and $B_t = SS^T$ in their respective fields gives

$$\langle A_c, B_t\rangle_{\otimes} = \sum_{i,j}\left(\sum_k R_{ki}\otimes S_{kj}\right)^2. \tag{SG14}$$

No mixed inverse or limiting certificate is involved. Literal zero coefficients and singular matrices are included.

4.7 The remaining quadratic form

Let K_c, K_t be separately labelled real closed extensions of $\mathbb{R}$ and $T = K_c \otimes_{\mathbb{R}} K_t$. The source partition in Chapter 7 reduces the following assertion to explicit local calculations.

Canonical assertion $\mathcal{C}$. Take $d = 1$ and source coefficients in either pattern SG10. Assume only that there exist $u_c, v_c > 0$, $W_c > 0$, and $X_c, Y_c \in K_c$ for which SG7 holds with $\psi_c = (W_c, X_c, Y_c, 1)$. For every independent $u, v, W > 0$ and $X, Y \in K_t$, the full expression SG2 with $Z = 1$ belongs to ΣT^2.

Physicality follows from SG7. Every source produced by the extreme reduction has this form after the angle signs and state normalization are fixed in its own field. The independent testing state remains arbitrary in $\mathcal{C}$.

Assembly lemma. Once $\mathcal{C}$ is proved, every transferred quantum-valid affine functional and every independent transferred quantum behaviour have separated slack in ΣT^2. Therefore the universal tensor-SOS finite-lift criterion applies to the full eight-coordinate quantum set.

Proof. The source reduction gives at most nine canonical source atoms, each of which is an event atom, a closed-face atom, or SG10 with an actual contact. Event atoms and closed faces were handled above. Restore d by multiplying each root by $\sqrt{d}\otimes 1$.

For a canonical contact source, finite angle signs are harmless. The polynomial is unchanged by

$$(u; W, X, Y, Z) \mapsto (-u; W, X, -Y, -Z), \qquad (v; W, X, Y, Z) \mapsto (-v; W, -X, Y, -Z).$$

Literal zero/infinite angles have nonnegative pure-testing H and nonnegative event squares, as already proved. For $u, v > 0$, the signed core is nonnegative whenever $WZ \le 0$. To check this, put $h_0 = 1 + v^2 + u^2v^2$; its X, Y block is $2\left(\begin{smallmatrix} h_0 & -uv \\ -uv & h_0\end{smallmatrix}\right) \succ 0$, and the remaining Schur form on W, Z is

$$\frac{2D_uD_v}{h_0^2 - u^2v^2}\{u^2v^2(W^2 + Z^2) - 2uvh_0WZ\} \ge 0. \tag{SG15}$$

This is a pure testing calculation. Its square root in K_t, together with $\sqrt{d}$ in K_c, certifies the signed core; every event remains a square. It includes $Z = 0$. Otherwise $Z \ne 0$ and $W/Z > 0$; divide the state by Z inside K_t and apply $\mathcal{C}$. Restore Z^2 afterwards. Real and imaginary state parts add because the measurements are real. All homogeneous angle normalisations are pure testing squares and have nonzero denominators on their stated charts.

Finally every testing behaviour is a mixture of at most nine real two-qubit behaviours. If source and testing weights are a_j, w_i and atom certificates are $\sum_k h_{ijk}^2$, the entire slack is

$$\sum_{i,j,k}[(\sqrt{a_j}\otimes\sqrt{w_i})h_{ijk}]^2.$$

The affine constant is preserved because $\sum_i w_i = 1$. Zero weights are never inverted. This proves the universal tensor assertion; applying the criterion in Chapter 3 is the final qualitative existence step. □

Chapter 5

Coordinates of the contact families

The source support, physicality, curvature and deletion facts used below were proved in Chapter 4, including (SG17)–(SG22).

5.1 Signs and stationary coordinates

The stationary equations give rational formulas for the event weights. To use those formulas at a singular limit, we first determine the signs of every denominator. The contact geometry of Chapter 4 provides the required strict inequalities.

Statement and the real maximal-contact graph

Use the raw full parent F, with $d > 0$. Every source below is physical and has an actual finite noncommuting zero. The two allowed coefficient domains are

$$\begin{aligned} &\mathrm{C1}: \quad \lambda = n = 0, \quad m, C, A, \alpha > 0, \quad D, \beta \geq 0, \quad D + \beta > 0, \\ &\mathrm{C2}: \quad \lambda = \beta = 0, \quad m, C, A, D, \alpha > 0, \quad n \geq 0. \end{aligned} \tag{EN1}$$

Make the actual angle gauges positive and put $Z_c = 1$, as permitted by SG3–SG4. Then $W_c > 0$ by SG15 and SG21. We prove the strict signs

$$\begin{aligned} X_c < 0, \quad Y_c > 0, \quad u_c W_c - Y_c > 0, \quad 1 - v_c Y_c > 0, \\ R_{m,c} := v_c W_c - X_c + u_c v_c Y_c - u_c > 0, \\ R_{D,c} := u_c W_c + u_c v_c X_c - Y_c - v_c < 0, \\ X_c + u_c > 0, \qquad W_c + v_c X_c > 0. \end{aligned} \tag{EN2}$$

These inequalities hold over every real closed coefficient field. In particular they are strict ordered-field statements, with no assertion that a gap has positive real residue.

Here is the common connected-graph argument, proved over $\mathbb{R}$ before first-order transfer. In either exact two-omission family, initially require all six retained events positive, normalize $C = 2$, and let their other five weights vary over the positive orthant. Put

$$d_*(w) = \max\{d \geq 0 : (d, w) \text{ is physical}\}. \tag{EN3}$$

The admissible set is a closed interval by convexity. It has positive length: at $d = 0$ the central Laurent coefficient is positive, its nonnegative Cauchy–Binet expansion containing $16mCAD$; the outer coefficients remain nonnegative for $d \geq 0$. Thus sufficiently small positive d is physical by SG3–SG6 and the homogeneous endpoint limits. A fixed strictly CHSH-violating real test bounds d above.

For these positive six-event supports the active outer exponents are constant and have zero in the interior of their convex hull. The logarithmic Laurent sum is strictly convex and coercive. At d_* its unique minimum is zero. Indeed if its minimum were $\mu > 0$, a small increase of d preserves every active outer coefficient within a relative error ε and the central coefficient within absolute error ε. As in the extreme-source argument, this preserves a positive uniform lower bound, contradicting maximality. There is therefore one positive squared-angle contact, and SG3–SG4 gives one state with $Z = 1$.

At that contact the core value is strictly negative (SG20–SG21). The Jacobian of the six equations $\nabla_z F = 0, F = 0$, in $z = (W, X, Y, u, v)$ and d, is

$$\begin{pmatrix} \nabla_z^2 F & \partial_d \nabla_z F \\ 0 & \partial_d F \end{pmatrix}, \qquad \nabla_z^2 F \succ 0, \quad \partial_d F < 0. \tag{EN4}$$

The real analytic implicit-function theorem gives a local analytic contact branch. SG7 makes it physical and the negative core value makes its d maximal. Uniqueness makes these branches agree. Thus the normalized contact is analytic on the entire connected positive orthant. Every physical finite zero with those event weights lies on this graph, because its negative core value prevents increasing d at that same test.

C1: strict signs including the two literal boundary faces

Write

$$(W, X, Y, Z) = (S, u(\theta - 1), uY_0, 1). \tag{EN5}$$

We first determine the sign of θ from an adjugate entry. With $B = \operatorname{adj} P$,

$$\begin{aligned} uB_{44} + B_{24} &= 16uv^2 D_u^2 D_v \mathcal{F}, \\ \mathcal{F} &= (d+2D)\{(d+2\alpha)(mD_v + Cu^2v^2) + 2AD_v(m+Cu^2)\} \\ &\quad + 2mC\{d(u^2v^2+D_v) + 2\alpha + 2Dv^2D_u\} > 0. \end{aligned} \tag{EN6}$$

Since P has one-dimensional kernel and three positive eigenvalues, $B = \gamma\psi\psi^T$ with $\gamma > 0$. Equation EN6 gives $\theta > 0$.

Neither $Y_0 = 0$ nor $S - Y_0 = 0$ is possible. These walls would delete the positive A-event or C-event, respectively, while preserving the kernel and both angle jets. The resulting omission set contains an exclusive pair, so has noncoercive support. SG7 and SG22 rule this out. The remaining cross-multiplied contact identity is

$$m\,[S + uv(\theta-1)]\,[v(S+u^2Y_0) - u\theta] = 2u^3(S-Y_0)(1-\theta)D_v. \tag{EN7}$$

For precision about exceptional source minors, use the four raw kernel equations and the two full angular derivatives. Their selected six-column system, for $(d, m, A, D, \alpha, \beta)$ with $C = 2$, has determinant

$$-8192\theta u^2 v D_u^2 D_v^2\,\mathcal{D}, \qquad \mathcal{D} = Y_0 D_u (uY_0 + v)(1 - uvY_0)[S + uv(\theta-1)]R_m(-R_D). \tag{EN8}$$

EN7 follows first from its rational solution wherever this selected minor is nonzero: the Cramer determinant for the m-column, multiplied by $[S + uv(\theta-1)]R_m$, equals this determinant times $2u^3(S-Y_0)(1-\theta)D_v$. This is an exact polynomial determinant identity. The equivalent native formula is $m = 2ae^2c_2/(Mc_3)$, with the definitions in EN13 below: $c_2/c_3 = (1-\theta)D_v/E$, $M = R_m/(Sv)$, $E = [S + uv(\theta-1)]/S$, $a = u(S-Y_0)/v$, $e = u/S$. These substitutions give EN7 exactly. The native contact equations are polynomial identities, so this argument does not assume the signs being proved.

At the exact positive seed

$$(u, v; S, Y_0, \theta) = (1, 1; 1, \tfrac{1}{4}, \tfrac{1}{2}), \qquad (d, m, C, A, D, \alpha, \beta) = (\tfrac{27}{2}, 4, 2, 9, \tfrac{5}{4}, \tfrac{13}{4}, \tfrac{9}{4}), \tag{EN9}$$

all six stationary equations hold and the selected minor is nonzero. Consequently that analytic minor is not identically zero on the connected contact graph. Its nonzero set is dense. Continuity extends EN7 to every contact on the graph. At $\theta = 1$, EN7 and $S, m > 0$ would imply $R_m = 0$, permitting deletion of m and creating an exclusive omitted pair, again impossible. The three continuous nonzero functions $Y_0, S - Y_0, \theta - 1$ therefore have constant signs, fixed by EN9:

$$S > 0, \qquad 0 < Y_0 < S, \qquad 0 < \theta < 1. \tag{EN10}$$

The full-parent C1 reflection swaps $(\lambda, n), (m, C), (A, \alpha), (D, \beta)$, sends u to $1/u$, and sends the state to

$$(vS - u(\theta-1),\ -S - uv(\theta-1),\ 1 - uvY_0,\ uY_0 + v).$$

Its state matrix G has $G^TG = D_vI$; the full core and every event obey $u^2F_{\sigma_1 c}(1/u, v; G\psi) = D_vF_c(u, v; \psi)$. Its last coordinate is positive. Applying EN10 to the transformed normalized state proves successively $1 - uvY_0 > 0$, $S + uv(\theta-1) > 0$, $R_D < 0$, $R_m > 0$. Together with EN10 this is precisely EN2.

For a literal third omission $D = 0$ or $\beta = 0$, start at the given real contact and open that coefficient to $\varepsilon > 0$. EN4 is still invertible by the strict contact Hessian (SG18–SG19). The local analytic branch stays at positive d, has nonnegative event weights, and is physical by SG7. The already proved inequalities hold on it. At the limit, EN6 and SG15 and SG21 stay strict; the still positive A, C, m and the same deletion arguments make the other weak inequalities strict. EN7 passes to the limit, and so does the reflection. Thus EN2 holds on both boundary faces. This is an implicit-function limit at a fixed finite real contact, not an assertion that an optimizer cannot escape to infinity. First-order transfer now gives the entire implication over every real closed field.

C2: all signs and a positive omitted-row cofactor

Use the same positive six-event graph. Five retained-amplitude walls are excluded directly by SG22:

$$Y = 0, \quad uW - Y = 0, \quad R_m = 0, \quad R_D = 0, \quad vY - 1 = 0. \tag{EN11}$$

Deleting A, C, m, D, α, respectively, introduces an exclusive omitted pair in addition to $\lambda = \beta = 0$. The wall $X = 0$ deletes n; the resulting source is on the C1 three-consecutive-omission face $\lambda = n = \beta = 0$. It is physical by SG7, so EN10 gives $X < 0$, a contradiction. Thus all six retained event rows have fixed signs on the connected C2 graph. The exact seed

$$(n, m, C, A, D, \alpha) = (\tfrac{1}{8}, \tfrac{3}{8}, \tfrac{1}{6}, \tfrac{5}{8}, \tfrac{3}{8}, \tfrac{1}{6}), \quad d = 1, \qquad (u, v; W, X, Y, Z) = (1, 1; 1, -\tfrac{1}{4}, \tfrac{1}{4}, 1)$$

has every required positive weight and every stationary equation. It fixes their signs to the first six signs in EN2.

Here is a direct proof of the omitted-row sign, including $n = 0$. Let P_0 be this raw C2 matrix with n set to zero. Changing n adds only $4nD_uD_v$ to its XX entry. The cofactor B_{24} does not contain that column, whereas B_{44} changes by $4nD_uD_v \det(P_0|_{W,Y})$. Consequently

$$\begin{aligned} uB_{44} + B_{24} = 16uv^2D_u^2D_v\,\mathcal{F} \\ + 4unD_uD_v \det(P_0|_{W,Y}) > 0. \end{aligned} \tag{EN12}$$

Here $\mathcal{F}$ is EN6 with $\beta = 0$. The determinant in the second line is positive: the $Z = 0$ state block already has the strictly positive signed-core form SG3–SG4. Thus $X + u > 0$. The C2 full-parent reflection

$$(u, v; W, X, Y, Z) \mapsto (1/v, 1/u; Z, X, Y, W), \quad (m, D), (C, \alpha), (\lambda, \beta)$$

has the exact homogeneous identity $u^2v^2F_{\sigma_2 c}(1/v, 1/u; Z, X, Y, W) = F_c(u, v; W, X, Y, Z)$. The transformed contact has positive last coordinate W. Applying the just-proved $X' + u'Z' > 0$ gives $W + vX > 0$. This proves EN2 for positive n. For literal $n = 0$, EN1 is already a C1 triple face and its signs apply directly; EN12 remains strict as well.

Exact native-coordinate transport and uniqueness

For C1, reflect if needed so $0 < k = u_c \le 1$, and define

$$\begin{gathered} S = W_c, \quad r = Y_c/(kS), \quad \theta = 1 + X_c/k, \quad h = 1 - \theta, \\ \eta = v_c/(kS), \quad \zeta = Sv_c/k, \quad f = \sqrt{\eta}, \quad g = \sqrt{\zeta}, \quad e = k/S = kf/g, \\ a = (1 - r)/\eta, \quad t = \theta/\zeta, \quad x = k^2, \\ R = 1 + xr, \quad T = 1 + xh, \quad M = R - t, \quad J = T - a, \\ E = 1 - x\eta h, \qquad Q = 1 - x\zeta r. \end{gathered} \tag{EN13}$$

By EN2, $0 < r, \theta < 1$, all η, ζ, a, t are positive, and

$$R_m = k\zeta M > 0, \quad R_D = -v_cJ < 0, \quad W_c + v_cX_c = SE > 0, \quad 1 - v_cY_c = Q > 0.$$

Thus $M, J, E, Q > 0$, and $0 < a, t < 2$. The selected determinant EN8 is now strictly nonzero. The native seven coquotients and source coefficients SD1 in Section B.2.1, with their exact stationarity identities SD9–SD11, satisfy exactly the six stationary equations at this same contact. Uniqueness therefore identifies them with the original normalized source, including $D = 0$ or $\beta = 0$. No asymptotic hypothesis is used for these polynomial identities.

For C2, normalize $d = 1$, and set

$$\begin{gathered} c = u_c, \quad a = v_c/u_c, \quad b = aW_c, \quad r = -X_c/u_c, \quad t = (u_cW_c - Y_c)/v_c, \quad z = c^2, \\ h = b - a^2t = aY_c/u_c, \quad g = (1 + z)b - (1 - r + a^2zt) = R_m/u_c, \quad j = 1 + zr - t = -R_D/v_c. \end{gathered} \tag{EN14}$$

The complete strict native domain follows:

$$a, b, c, t > 0, \quad 0 < r < 1, \quad h, g, j > 0, \quad 1 - zh = 1 - v_cY_c > 0, \quad b - a^2zr = a(W_c + v_cX_c) > 0. \tag{EN15}$$

The actual six-column stationary determinant is, in the unweighted event convention of the retained determinant proof,

$$-64ac^6rgt\,hj(1 - zh)(1 + z)^2(1 + a^2z)^2\Delta,$$

$$\Delta = 1 + (1 + z)b + z(1 + z)b^2 + 2z(1 + z)br + z^2r^2 + 2zr + a^2z\{(1 + z)b + 1 + zr(2 - r)\} \ge 1. \tag{EN16}$$

That body provides its explicit reduction to a three-by-three determinant. The six rational coefficients in Section C.1.2 solve these six equations at the contact EN14. EN16 makes the solution unique, so they are exactly the original source coefficients. This remains valid at $n = 0$; one does not delete its column from the square system.

All gauge actions, positivity statements and source equations used above are finite polynomial or rational ordered-field statements with denominators explicitly nonzero. The real connected-graph proof therefore transfers as one fixed first-order implication. Together, EN1–EN16 discharge the native sign and positive-gap interface for both families in SG10. They do not prove an all-tests tensor assertion; that assertion is supplied by the source partitions and their proofs in Appendices B and C.

Chapter 6

Algebraic tools for singular contacts

A singular contact has vanishing quadratic terms. The following identities express its higher-order terms using roots with a positive residue Gram. Chapter 3 then turns that Gram into an exact sum of squares.

6.1 Finite algebra and contact completion

Finite penalty lemma. Let R be a real symmetric quadratic form and L a real linear map. If R is positive definite on $\ker L$, then $R + M\|Lz\|^2$ is positive definite for some real $M > 0$.

Proof. Choose a linear complement of $\ker L$. The restriction of L to that complement has a positive least singular value. In these coordinates the kernel block of R is positive definite, and adding ML^TL makes the other Schur complement positive for sufficiently large M. □

For an opening argument one must additionally exhibit a bounded remainder after subtracting the divergent squares, and a bounded right inverse for their independent hard rows. If a family has a uniform positive lower bound before subtraction, its limit on the limiting kernel is positive: lift a kernel vector to exact kernels with that right inverse and take the real limit. The finite penalty lemma then supplies finite caps. The applications in Appendices B and C give the required bounded minimizers and independent rows.

Small-cost helper lemma. Suppose the main roots have a strict real residue Gram and an exact identity has the form $A(z, H) - H^2 = \sum R_i^2$, where every coefficient of the quadratic form A on the main roots z and H is infinitesimal. Then any prescribed finite positive diagonal budget cH^2 can be introduced by the exact replacement

$$F = (F - cA + cH^2) + c\sum R_i^2.$$

It changes the main residue Gram by zero and retains the displayed positive squares. Finitely many helper roots can be treated in a stated order, after which a real Schur complement handles their bounded cross coefficients. When the identities are triangular, introduce the auxiliary roots in that order. In the double-infinitesimal C1 case this means obtaining the radial pivot before the angular roots.

6.2 Finite polynomial division at a contact

Write $d = \delta$ in this section; this local parameter is distinct from the global Bell coefficient d_{B}. Put

$$R_0 = \mathbb{Q}[d, k, b, a, t], \qquad R = R_0[u, w, v], \qquad J = (u - k, w - dk, v - dk^2b^2).$$

For $F \in R$, write $F_c = F(k, dk, dk^2b^2)$. Define

$$A = wv, \quad B = dv, \quad C = db^2uw, \quad D_0 = d^2k^3b^2, \qquad M = (A, B, C),$$
$$x = B - C, \qquad y = A - kC, \qquad z = uB - kC, \qquad h = kC - D_0.$$

Equivalently, write $g_0 = x$, $h_{13} = y$, $h_{23} = z$, and $h_{34} = h$. Then $D_0 = a_4^2$, whereas the source monomial in g_4 is kD_0. Adjoin the separated radicals used in the four-monomial identity,

$$a_1 = \sqrt{wv}, \quad a_2 = \sqrt{duv}, \quad a_3 = b\sqrt{dkuw}, \quad a_4 = dk^{3/2}b, \qquad q = a_1a_2 - a_3a_4.$$

Only $q^2 \in R$ is needed in the scalar lift. The following three identities are the entire conductor calculation:

$$C(u - k) = z - ux, \tag{6.1}$$
$$C(w - dk) = dy - wx, \tag{6.2}$$
$$C(v - dk^2b^2) = b^2(wz + dky - kwx). \tag{6.3}$$

Thus $CJ \subset (x, y, z)$ without inverting any parameter. For

$$r_* = y + z - h - 2kx$$

we also have the two exact identities

$$r_* = (v - dk^2b^2)\big((w - dk) + d(u - k)\big) - dkb^2(u - k)(w - dk), \tag{6.4}$$
$$kCr_* = q^2 - yz. \tag{6.5}$$

The first says $r_* \in J^2$. The second supplies the otherwise missing conductor factor through one of the original circuit squares.

Explicit monic division. Every $S \in R$ has a polynomial decomposition

$$S = S_c + (u-k)T_u + (w-dk)T_w + (v-dk^2b^2)T_v. \tag{CC1}$$

One obtains it by substituting $u = k$, then $w = dk$, then $v = dk^2b^2$, and using $X^n - Y^n = (X-Y)\sum_{j=0}^{n-1} X^{n-1-j}Y^j$ at each step. There are no parameter denominators. If $S_c = 0$, equations (6.1)–(6.3) give

$$CS = x(-uT_u - wT_w - b^2kwT_v) + y(dT_w + dkb^2T_v) + z(T_u + b^2wT_v). \tag{CC2}$$

Lemma 6.1 (First-order and restricted second-order contact lifts)**.** *In the above ring, the following assertions hold, with polynomial coefficients in R:*

$$M \cap J \subset (x,y,z), \qquad \big(M^2 + (D_0^2)\big) \cap J^2 \subset (x,y,z,h,q)^2. \tag{CC3}$$

Here (D_0^2) is the principal ideal generated by D_0^2; the second assertion does not replace it by all mixed products involving D_0.

Proof. For the first assertion, write $F = R_A A + R_B B + R_C C$. Since $A = y + kC$ and $B = x + C$,

$$F = R_A y + R_B x + CS, \qquad S = kR_A + R_B + R_C.$$

The contact value is $F_c = C_c S_c$, where $C_c = d^2k^2b^2$ is a nonzero polynomial. Since R_0 is a domain, $F_c = 0$ implies $S_c = 0$. Apply (CC2). This cancellation proves a zero identity; it introduces no coefficient $1/C_c$.

For the second assertion choose any displayed representation $F = \sum_{i\le j} f_{ij} m_i m_j + \rho D_0^2$, with $(m_1, m_2, m_3) = (A, B, C)$. Substitute

$$A = y + kC, \qquad B = x + C, \qquad D_0 = kC - h.$$

This gives

$$F = Q_2 + C(xH_x + yH_y + zH_z + hH_h) + C^2 S, \qquad H_h = -2k\rho, \tag{CC4}$$

where $Q_2 \in (x,y,h)^2$ and initially $H_z = 0$. The contact value again implies $S_c = 0$. Use (CC2) on CS and absorb C^2S into the four coefficients in (CC4). This leaves $H_h = -2k\rho$ unchanged and gives the same formula without C^2S.

The first derivatives of $F \in J^2$ vanish at contact. The first three root gradients have determinant

$$\det\left(\frac{\partial(x,y,z)}{\partial(u,w,v)}\right)_c = -d^4k^4b^4 \neq 0. \tag{CC5}$$

Equation (6.4) gives a gradient relation with coefficient vector $(-2k, 1, 1, -1)$. Its coefficient at h is -1. Set $\lambda = 2k\rho_c$. Subtract $\lambda(-2k, 1, 1, -1)$ from (H_x, H_y, H_z, H_h). The last resulting contact value is zero, and the first three satisfy the homogeneous linear system with determinant (CC5). Multiplication by its adjugate and cancellation in the domain R_0 show that those three contact values are zero too. No inverse of the determinant is used in a coefficient: each resulting polynomial J_i lies in J and is divided only by the monic procedure (CC1). Consequently

$$F = Q_2 + 2k\rho_c C r_* + C\sum_{i=1}^{4} r_i J_i = Q_2 + 2\rho_c(q^2 - yz) + \sum_{i=1}^{4} r_i (CJ_i), \tag{CC6}$$

where $(r_1, r_2, r_3, r_4) = (x, y, z, h)$. Each CJ_i is explicitly a combination of x, y, z by (CC2), which proves the assertion in the unlocalized polynomial ring. □

The explicit scalar and both state-normal coefficients in Section B.3.1 satisfy these ideal conditions. There the same monic divisions also preserve a positive coefficient weight. This supplies bounded corrections with zero residue, which can be added to the strict Gram matrix.

6.3 A real two-block allocation

A rank-one allocation lemma. Let P be a positive definite real symmetric matrix, let ℓ, b be real columns, and let $H, c_0 > 0$. Set

$$d = \ell^T P^{-1} \ell, \qquad e = \ell^T P^{-1} b, \qquad h = c_2 - b^T P^{-1} b.$$

Suppose

$$p(T) = c_0 + (h + 4Hdc_0)T + 4H(dh + e^2)T^2 > 0 \quad (T \ge 0), \qquad dh + e^2 > 0. \tag{UA1}$$

There exists a real λ such that

$$|\lambda| < \sqrt{4Hc_0}, \qquad h + 2e\lambda - d\lambda^2 > 0. \tag{UA2}$$

Indeed put $s = \sqrt{4Hc_0}$. If $d > 0$ and $|e| < ds$, take $\lambda = e/d$. If $|e| \geq ds$, the value at the appropriate endpoint is $h + 2|e|s - ds^2$. Were this nonpositive, writing $x = |e| - ds \geq 0$ would give $h + ds^2 \leq -2sx$ and $dh + e^2 \leq x^2$; the quadratic in (UA1) would then have a nonpositive value at a positive argument. Thus a nearby interior λ works. This includes equality $|e| = ds$. The scalar interval argument also holds in a limiting case $d = 0$; then $e \neq 0$ follows from $dh + e^2 > 0$. (For a finite positive P itself, $d = 0$ would force $\ell = e = 0$, so this case occurs only when using the continuous scalar extension at a boundary.)

Consequently the two real matrices

$$\begin{pmatrix} P & b - \lambda\ell \\ (b - \lambda\ell)^T & c_2 \end{pmatrix}, \qquad \begin{pmatrix} 4H & \lambda \\ \lambda & c_0 \end{pmatrix} \tag{UA3}$$

are positive definite. This is an elementary real lemma. Its inverse and square root are never taken in the tensor ring.

6.4 Noncommuting testing residues

Use the two labelled fields and the bounded tensor ring of Chapter 3. The strict-Gram lemma applies to arbitrary root columns, including unbounded rows.

6.4.1 Parent, physicality and contact curvature

For real homogeneous state coordinates $\psi = (W, X, Y, Z)$ and half angles u, v, set

$$D_u = 1 + u^2, \quad D_v = 1 + v^2, \quad M = vW - X + uvY - uZ, \quad J = uW + uvX - Y - vZ.$$

All nine source coefficients $d, \lambda, n, m, c, A, D, \alpha, \beta$ are nonnegative. The entire parent is

$$\begin{aligned} F = {} & 4\lambda D_u(W + vX)^2 + 4nD_uD_vX^2 + 4mM^2 + 4cD_v(uW - Y)^2 + 4AD_uD_vY^2 \\ & + (4D + 2d)J^2 + (4\alpha + 2d)D_u(vY - Z)^2 + (4\beta + 2d)D_v(X + uZ)^2 - 2dD_uD_vZ^2. \end{aligned} \tag{NC1}$$

Write C for the coefficient of d in this expression. A source is *physical* if its state matrix is positive semidefinite at every measurement pair over its own field K_c. The claim below concerns evaluation of (NC1) at an independent test over K_t.

Theorem 6.2. *For every physical source in (NC1), its tensor evaluation is a finite sum of squares at every independent test for which both measurement pairs are noncommuting in residue, and every homogeneous real or complex state. No relation between source scales or between source and testing scales is assumed.*

Local frame changes and a projective state chart are pure-testing operations. At a noncommuting residue one may take u, v to be bounded units, and normalize one state coordinate to a bounded unit. The original slack is restored by positive pure-testing norm factors. For complex states, apply the real result separately to real and imaginary parts, since the state matrix is real symmetric. We prove the real case.

If $d = 0$, the parent is already a sum of event squares. For the remaining $d > 0$ case, use the eight outer coefficients q_{ij} and the positive quantities t_i in (SG5)–(SG6), put $q_{11} = h$, and denote the event coefficient by c here. Equivalently

$$\det P = 16D_u^2D_v^2 \sum_{i,j=0}^{2} q_{ij}u^{2i}v^{2j}. \tag{NC2}$$

Thus every coefficient in the constrained determinant calculations below is specified by that displayed finite dictionary.

For $U = u^2, V = v^2 > 0$, let

$$L(U, V) = q_{11} + \sum_{(i,j)\neq(1,1)} q_{ij}U^{i-1}V^{j-1}. \tag{NC4}$$

Every nonconstant coefficient is nonnegative. Over the real residue field, a stationary zero of such a Laurent polynomial satisfies

$$\sum_{e\in E} w_e e = 0, \qquad \nabla^2_{\log U, \log V} L = \sum_{e\in E} w_e e e^T, \quad w_e = q_e U^{e_1}V^{e_2} > 0. \tag{NC5}$$

Consequently its angular Hessian is positive definite exactly when the active exponent vectors span $\mathbb{R}^2$. Logarithms are used only for this real calculation; the displayed formulas are algebraic.

When $d > 0$ and $uv \neq 0$, the core restricted to $Z = 0$ is

$$C|_{Z=0} = 2\{(uW + uvX - Y)^2 + D_u v^2 Y^2 + D_v X^2\}, \tag{NC6}$$

which is positive definite in (W, X, Y). Thus a physical state matrix at a nonzero kernel has exactly three positive transverse directions. The same observation supplies two or one positive transverse directions after imposing one or two independent state-linear event constraints, respectively.

6.4.2 Exact local lifting with unequal hard weights

We record the local argument that will turn real curvature into an exact tensor certificate. Suppose, in a real coordinate neighborhood, the complete physical parent has the form

$$F(x, y) = \sum_{i=1}^{k} a_i b_i(y) x_i^2 + \sigma G(x, y), \qquad \varepsilon_i = \sigma / a_i > 0 \text{ infinitesimal}, \tag{NC7}$$

where a_i, σ are pure-source, all coefficients of b_i, G are bounded, $\bar{x} = 0$, $\bar{y} = y_0$, and $\bar{b}_i(y_0) > 0$. The case $k = 0$ is allowed. Rational expressions are also allowed when their denominators are products of pure-source units and real testing polynomials nonzero at the residue. Multiply first by their squares; the resulting polynomial has the same physical sign and the same strictness at a zero, and the factors are canceled afterward.

Let $g(y) = \overline{G}(0, y)$. Physicality on the exact hard kernel implies $g \geq 0$ in its real neighborhood. If $g(y_0) > 0$, the roots $(x_i/\sqrt{\varepsilon_i}, 1)$ give an exact bounded Gram with residue $\operatorname{diag}(\bar{b}_i(y_0), g(y_0)) \succ 0$: every term of $G - G(0, y)$ containing x_i acquires a factor $\sqrt{\varepsilon_i}$ in this frame.

If $g(y_0) = 0$ and $\nabla^2 g(y_0) \succ 0$, physicality also gives $\nabla g(y_0) = 0$. Put $x_i = \varepsilon_i s_i$ in the stationary equations for F/σ and divide the x_i equations by their nonzero factors. The exact equations are

$$2b_i(y)s_i + G_{x_i}(\varepsilon s, y) = 0, \qquad \sum_i \varepsilon_i s_i^2 \nabla b_i(y) + G_y(\varepsilon s, y) = 0. \tag{NC8}$$

At residue they have a block upper triangular Jacobian with diagonal blocks $2\operatorname{diag}(\bar{b}_i(y_0))$ and $\nabla^2 g(y_0)$. The real implicit-function theorem gives a unique local solution in fixed semialgebraic neighborhoods. Existence, uniqueness and continuity bounds in those neighborhoods are fixed first-order statements and transfer to K_c. They give an exact source-field stationary point (x_c, y_c) with $x_{ci} = \varepsilon_i s_{ci}$ and bounded s_c, of the prescribed residue. No center depends on a testing infinitesimal.

Finite polynomial Taylor expansion at this center, using the rows

$$h = \big((x_i - x_{ci})/\sqrt{\varepsilon_i}\big)_i \oplus (y - y_c),$$

gives exactly

$$F/\sigma = a + h^T Q h, \qquad a = F(x_c, y_c)/\sigma \geq 0, \qquad \bar{Q} = \operatorname{diag}(\bar{b}_i(y_0), \tfrac{1}{2}\nabla^2 g(y_0)) \succ 0. \tag{NC9}$$

Indeed two hard deviations are allocated to their two scaled rows; one hard deviation and a tangent deviation leave $\sqrt{\varepsilon_i}$ in the coefficient; a hard packet's pure tangent term leaves ε_i, because x_{ci}/ε_i is bounded. The G terms have the same or smaller factors. All higher Taylor terms leave an infinitesimal deviation in the coefficient. The exact linear terms vanish at the center. This proves the full-parent certificate, including all state normals and higher terms. In particular, $k = 0$ proves the ordinary regular-contact case.

6.4.3 Geometry of the event constraints

The losing events form one bipartite cycle of local outcome rays, in edge order

$$(\lambda, D, c, A, \alpha, m, \beta, n). \tag{NC10}$$

For completeness its consecutive vertices may be taken as $A_0^+, B_1^+, A_1^-, B_0^+, A_0^-, B_1^-, A_1^+, B_0^-$. The four rays at either party are distinct at noncommuting residue angles. CHSH-preserving relabelings act on (x, y, a, b) as

$$\begin{aligned} g_A &: (x, y, a, b) \mapsto (1 - x, y, a, (-1)^y b), \\ g_B &: (x, y, a, b) \mapsto (x, 1 - y, (-1)^{1-x} a, b), \\ g_T &: (x, y, a, b) \mapsto (y, x, (-1)^y b, (-1)^x a). \end{aligned}$$

They act on the edge indices of (NC10) by $i \mapsto 5 - i, 3 - i, 2 - i$ modulo eight. Hence they generate a cycle rotation and reflection. They preserve the parent interpretation; chart changes contribute only positive pure-testing homogeneous factors.

Product residue states need no further argument. For a product state the CHSH expression satisfies

$$S = a_0(b_0 - b_1) + a_1(b_0 + b_1) \le |b_0 - b_1| + |b_0 + b_1| = 2\max(|b_0|, |b_1|) \le 2.$$

The inequality is strict when both measurement pairs are noncommuting. If a Bob modulus is one, the other is strictly less than one, and equality would require Alice to be a common eigenstate of her two noncommuting reflections. Thus C has positive testing residue and its actual pure-testing value is positive. The term dC and all event packets are then sums of squares using separate roots.

For an entangled real residue state, write its amplitudes as an invertible two-by-two matrix Ψ. A fixed Alice ray a has exactly one Bob projective ray satisfying $a^T \Psi b = 0$, and conversely. The vanishing events therefore share no endpoint in (NC10): they form a matching and there are at most four of them. Any three matching event vectors are state-linearly independent. To see this, the span of two rank-one matrices with independent left and right factors contains no other rank-one projective point; three dependent rank-one matrices must therefore share a left or a right factor, contrary to the matching condition. For $k \le 3$, the event amplitudes are consequently exact local projective-state coordinates. Their coordinate inverse has only pure-testing unit denominators and their packets have the diagonal form (NC7).

There are two perfect matchings. For $\{n, m, A, D\}$ their equations are $X = Y = 0$, $vW = uZ$, $uW = vZ$. Thus $u^2 = v^2$; on either sign branch $W \neq 0$ and, after a sign gauge, $W = Z = 1$, $v = u = t \neq 0$. With $t = u$ and $\delta = v - u$, the independent event-normal differentials are

$$dX, \quad dY, \quad dR_m = d\delta - dX + t^2 dY - t\, dZ, \quad dR_D = t^2 dX - dY - d\delta - t\, dZ.$$

Their rank is four; the remaining tangent is t. The core is zero on the curve and has differential

$$dC = 2t\{(1 + t^2)dX - (1 + t^2)dY - dR_m + dR_D\}. \tag{NC11}$$

Every coefficient here is nonzero. The other perfect matching is $\{\lambda, c, \alpha, \beta\}$; with $W = 1$ its equations give $X = -1/v$, $Y = u$, $Z = uv$, $u^2v^2 = 1$. It is the symmetry image of the first and has the same rank and differential properties.

6.4.4 Curvature when the signed coefficient is visible

All statements in this subsection are over $\mathbb{R}$ with $d > 0$. We prove that every physical zero of the full parent, or of its restriction to at most three hard matching events, has strictly positive Hessian in its remaining angle and projective-state coordinates. This supplies precisely the soft Hessian in (NC8).

No hard event. At a zero the transverse state block is positive by (NC6). If the active exponents in (NC4) have rank one, stationarity (NC5) requires an opposite pair on one of the four lines through the Newton square. A horizontal opposite pair combines λD or αm with βn or cA; in the four cases it also activates $\lambda n, cD, \beta m, \alpha A$, respectively, off that line. The vertical case is identical: combining αA or λn with βm or cD activates $\alpha m, cA, \beta n, \lambda D$. The two diagonal pairs force exactly the supports $\{n, m, A, D\}$ and $\{\lambda, c, \alpha, \beta\}$. Any further event activates an exponent off the diagonal.

Neither diagonal support is physical. At $u = v = 1$ the core matrix is

$$\begin{pmatrix} 2 & 2 & -2 & -2 \\ 2 & 6 & -2 & 2 \\ -2 & -2 & 6 & -2 \\ -2 & 2 & -2 & 2 \end{pmatrix}. \tag{NC12}$$

The corresponding vectors $(1, 0, 0, 1)$ and $(1, -1, 1, 1)$ are annihilated by all selected events, have zero core value, and have nonzero core image. A positive semidefinite matrix cannot have zero quadratic value and nonzero image at the same vector.

If no outer exponent is active, the event support contains neither neighboring nor opposite pairs in (NC10); these are exactly the twelve activating pairs in (SG6). Such a set extends to a triple with cyclic gaps $(2, 3, 3)$. Indeed an independent four-set in the cycle would be alternating and contain an opposite pair, and the only possible independent triple gap patterns are $(2, 3, 3)$ and $(2, 2, 4)$, of which the latter contains an opposite pair. All the allowed triples are cyclic images of $\{\lambda, m, c\}$. The state

$$\psi_H = (uv, -u, u^2v, 1 + v^2 + u^2v^2)$$

annihilates this triple and has $C(\psi_H) = -2u^2v^2 D_u D_v < 0$. This contradicts physicality. Thus the active exponents span the plane.

One hard event: explicit coefficients. By symmetry take the hard event to be λ. Its kernel is $W = -vX$. The constrained matrix in (X, Y, Z) is positive definite on $Z = 0$, so it has two positive state pivots at a zero. Extracting the coefficient of λ from (NC2) gives its determinant as $4D_u D_v^2 UV L_\lambda(U, V)$, whose outer coefficients are

$$\begin{aligned} a_{00} &= 4\alpha t_7 t_8, & a_{01} &= 4(Dt_7 t_6 + 2\alpha m t_8), \\ a_{02} &= 8mDt_6, & a_{10} &= 4(2\alpha A t_7 + n t_8 t_4), \\ a_{12} &= 8(\beta m t_6 + cDt_2), & a_{20} &= 8nAt_4, \\ a_{21} &= 8(\beta n t_2 + cAt_4), & a_{22} &= 8\beta c t_2. \end{aligned} \tag{NC13}$$

The factor follows directly by using first the normal coordinate $W + vX$, whose packet is $4\lambda D_u(W + vX)^2$, and then the three kernel coordinates. All t_i in (NC13) are strictly positive.

The four rank-one possibilities are exhausted as follows. On the horizontal line, off-line vanishing forces $\alpha = n = 0$; its negative end requires $D > 0$, forcing $m = c = 0$, so its positive end is absent. On the vertical line, off-line vanishing gives $\alpha = D = 0$; its negative end requires $n > 0$, forcing $\beta = 0$, so its positive end is absent. The slope-minus-one pair requires $m, D, n, A > 0$, which activates a_{01} off the line. The slope-plus-one pair forces $\alpha, \beta, c > 0$ and $m = D = n = A = 0$: it completes the perfect matching containing the hard λ event.

This last possibility is not nonnegative on the hard kernel. On the matching curve every event and the core vanish, but (NC11) has a nonzero derivative after only one event-normal direction is fixed. All event squares have zero derivative there, so the constrained parent changes sign. If no exponent is active, (NC13) gives $\alpha = D = n = 0$ and $\beta m = cA = \beta c = 0$. Its support, including the hard event, is contained in one of

$$\{\lambda, A, \beta\}, \quad \{\lambda, m, c\}, \quad \{\lambda, m, A\}.$$

These are cyclic images of the negative-core triple above, so each has a kernel witness contradicting constrained physicality. The active exponents of L_λ therefore span the plane.

Two hard events: exhaustive positivity table. Matching pairs have cyclic distances two, three or four, represented by $\{\lambda, c\}$, $\{\lambda, m\}$ and $\{\lambda, \alpha\}$. Extract the coefficient of the two hard weights from (SG6). An adapted state basis shows that it is a positive angle-dependent factor times the determinant of the constrained two-by-two state form. After the monomial UV is removed, the exact positivity conditions of its eight outer coefficients are the following. Juxtaposition means both entries are positive; "or" means at least one condition holds.

e	$\{\lambda, m\}$	$\{\lambda, c\}$	$\{\lambda, \alpha\}$
$(-1,-1)$	α	α	always
$(-1,0)$	D or α	D or αm	D or m
$(-1,1)$	D	Dm	Dm
$(0,-1)$	n or $A\alpha$	n	A or n
$(0,1)$	β or Dc	D or βm	Dc or βm
$(1,-1)$	An	0	An
$(1,0)$	Ac or βn	A	Ac or βn
$(1,1)$	βc	β	βc

(NC14)

This table is obtained by differentiating (SG6), not by a sign approximation. For example the first two entries in the $\{\lambda, m\}$ column, apart from the positive factor four, are $2\alpha(2A + 2D + 2c + d)$ and $2(2AD + 2A\alpha + 4D\alpha + 2Dc + Dd + 2\alpha c + \alpha d)$.

For $\{\lambda, m\}$, either diagonal pair activates the off-line exponent $(-1, 0)$. A horizontal pair combines D or α with Ac or βn and activates an off-line exponent in all four cases. A vertical pair combines n or $A\alpha$ with β or Dc and does the same. Rank zero forces $\alpha = D = n = \beta = 0$ and $Ac = 0$, so only A or only c may remain.

For $\{\lambda, c\}$ the slope-minus-one pair is impossible because one end is identically zero. A horizontal pair requires $A > 0$ and $D > 0$ or $\alpha m > 0$, activating $(0, 1)$ or $(-1, -1)$ off the line. A vertical pair requires $n > 0$ and either $D > 0$ or $\beta m > 0$, activating $(-1, 0)$ or $(1, 1)$ off the line. The slope-plus-one pair forces $\alpha, \beta > 0$ and $D = n = A = m = 0$. Rank zero leaves only m.

For $\{\lambda, \alpha\}$ the exponent $(-1, -1)$ is always active. A rank-one stationary support must be the slope-plus-one pair, forcing $c, \beta > 0$ and $D = m = A = n = 0$. These last two rank-one possibilities complete $\{\lambda, c, \alpha, \beta\}$. Imposing only two hard events still leaves a nonzero core derivative in (NC11), so the constrained parent changes sign on the matching curve. The rank-zero possibilities are contained in $\{\lambda, m, A\}$ or $\{\lambda, m, c\}$. At $u = v = 1$, their kernel vectors $(1, -1, 0, 2)$ and $(1, -1, 1, 3)$ have core values $-4, -8$. Thus every obstruction is excluded by constrained physicality. The active exponents again span the plane.

Three hard events: two complete Laurent restrictions. A three-edge matching has cyclic gaps $(2, 2, 4)$ or $(2, 3, 3)$; rotations and reflections are transitive on each pattern. The two orbits have representatives and kernel states

$$M_N = \{\lambda, m, A\}, \quad \psi_N = (uv, -u, 0, 1 + v^2); \qquad M_E = \{n, m, A\}, \quad \psi_E = (u, 0, 0, v).$$

Direct substitution into the entire parent gives

$$\frac{F(\psi_N)}{2UV(1+V)} = \frac{2n(1+U)}{V} + 2cU + \frac{2D(1+V)}{U} + \frac{2\alpha(1+U)(1+V)}{UV} + 2\beta V - d, \tag{NC15}$$

$$\frac{F(\psi_E)}{2UV} = d(U/V-1) + \frac{2\lambda(1+U)}{V} + \frac{2cU(1+V)}{V} + \frac{2D(U-V)^2}{UV} + \frac{2\alpha(1+U)}{U} + 2\beta(1+V). \tag{NC16}$$

For (NC15), any positive n, D or α supplies two independent exponents. If they vanish, $2cU + 2\beta V - d$ cannot have a stationary zero: stationarity forces $c = \beta = 0$ and the value is $-d < 0$. For (NC16), $d > 0$ supplies $(1, -1)$. Rank less than two forces $\lambda = c = \alpha = \beta = 0$. The remaining function of $x = U/V$ has infimum

$$2\sqrt{2D(d+2D)} - d - 4D < 0, \qquad (d+4D)^2 - 8D(d+2D) = d^2 > 0.$$

The formula also gives a negative infimum when $D = 0$. This contradicts physicality. Both constrained Laurent Hessians are therefore strict at every zero.

From angular determinants to the full soft Hessian. For zero, one or two hard events, (NC6) supplies a positive state block on the remaining projective-state coordinates. In a fixed real kernel chart write the constrained state form as $a + 2b^T y + y^T C_0 y$, with $C_0 \succ 0$ at contact. Its state Schur complement is $a - b^T C_0^{-1} b$, and its determinant equals this scalar times $\det C_0$. At a stationary zero, differentiating a positive prefactor does not change the Hessian sign. Equations (NC5), (NC12)–(NC14) therefore give a positive angular Schur complement and hence the full positive soft Hessian. For three hard events only the two angles remain and (NC15)–(NC16) apply directly. All these matrix inverses are ordinary real residue calculations. The exact certificate itself is (NC7)–(NC9), with no tensor inverse.

6.4.5 Four matching weights, with no comparability assumption

Suppose the four positive weights of a perfect matching dominate d and every complementary event. By symmetry use $\{n, m, A, D\}$ and arrange that $D = \mu = \min(n, m, A, D)$. Set

$$\eta_1 = \mu/n, \quad \eta_2 = \mu/A, \quad \eta_3 = \mu/m, \qquad 0 < \eta_i \le 1.$$

Their residues may vanish independently. Work near the matching curve in $W = 1$ and introduce the exact state coordinates

$$R = v - X + uvY - uZ, \qquad Z = \frac{v - X + uvY - R}{u}, \qquad \delta = v - u,$$

using only the testing unit u^{-1}. The fourth event is exactly $f + L$, where

$$f = \frac{u^2 - v^2}{u} = -2\delta - \delta^2/u, \qquad L = v(u + u^{-1})X - (1+v^2)Y + (v/u)R. \tag{NC17}$$

The first three hard packets are $4nD_uD_vX^2 + 4AD_uD_vY^2 + 4mR^2$. On $X = Y = R = 0$ the core is exactly $2uf$, so

$$4\mu f^2 + 2duf = 4\mu\left(f + \frac{du}{4\mu}\right)^2 - \frac{d^2u^2}{4\mu}. \tag{NC18}$$

Let $\tau = \max(\lambda, c, \alpha, \beta)$. Testing (NC18) at $u = 1$, $v = \sqrt{1 + d/(4\mu)}$ proves

$$\tau > 0, \qquad \rho = \frac{d^2}{\mu\tau} \text{ bounded.} \tag{NC19}$$

Indeed all complementary amplitudes are bounded at this source-field test; if their scale were infinitesimal relative to d^2/μ, the complete parent would be negative. The case $d = 0$ was already an event-square sum. Put $\varepsilon = d/\mu$, which is infinitesimal.

If $\bar\rho = 0$, use the five rows

$$(\sqrt{n/\tau}X, \sqrt{A/\tau}Y, \sqrt{m/\tau}R, \sqrt{\mu/\tau}\delta, 1).$$

Their four-normal residue is, with $B_0 = 1 + u_0^2$,

$$4B_0^2z_1^2 + 4B_0^2z_2^2 + 4z_3^2 + 4(-2z_4 + B_0\sqrt{\bar\eta_1}z_1 - B_0\sqrt{\bar\eta_2}z_2 + \sqrt{\bar\eta_3}z_3)^2, \tag{NC20}$$

which is positive definite. The constant residue is $4(1+u_0^2)(P + Qu_0^2) > 0$, where $P = \overline{(\lambda+\alpha)/\tau}$, $Q = \overline{(c+\beta)/\tau}$ and $P + Q > 0$. The signed linear coefficients are $\sqrt{\rho\eta_i}$ and $\sqrt{\rho}$, so vanish. All other normalized coefficients are bounded by $\varepsilon, \eta_i, \tau/\mu$ and bounded testing units. This is an exact strict bounded Gram of the whole parent.

Now suppose $\bar\rho > 0$. In the source-field stationary equations put

$$X = \varepsilon\eta_1 s_1, \quad Y = \varepsilon\eta_2 s_2, \quad R = \varepsilon\eta_3 s_3, \quad \delta = \varepsilon r, \quad u = t. \tag{NC21}$$

Divide the three state equations and the δ equation by d, and the tangent equation by τ. All coefficients are bounded after clearing powers of the unit t: the factors η_i cancel the large weights μ/η_i, and $\tau = \mu\varepsilon^2/\rho$. Put, in residue,

$$B_t = 1 + t^2, \quad \ell = (B_t, -B_t, 1), \quad b = (B_t^2, B_t^2, 1), \quad k = -2r + \sum_i \bar\eta_i \ell_i s_i.$$

Direct differentiation of (NC1), (NC17) gives the four normal equations

$$8b_i s_i + 8k\ell_i + (4tB_t, -4tB_t, 0)_i = 0, \qquad -16k - 4t = 0. \tag{NC22}$$

Eliminating the last row from the first three gives diagonal pivots $8b_i$, with the last pivot 32. The determinant is $32 \cdot 8^3 B_t^4 > 0$, even if some $\bar\eta_i = 0$. Their solution is

$$s_1^* = -\frac{t}{4B_t}, \quad s_2^* = \frac{t}{4B_t}, \quad s_3^* = \frac{t}{4}, \quad r^* = \frac{t}{8}(1 - \bar\eta_1 - \bar\eta_2 + \bar\eta_3). \tag{NC23}$$

The complete normalized residue objective is

$$\begin{aligned}\Psi = \bar\rho\{4\sum_i \bar\eta_i b_i s_i^2 + 4k^2 + 4tB_t(\bar\eta_1 s_1 - \bar\eta_2 s_2) - 4tr\} \\ + 4(1 + t^2)(P + Qt^2).\end{aligned}$$

Completing its four normal squares by (NC23) leaves exactly

$$g(t) = 4(1 + t^2)(P + Qt^2) - \frac{\bar\rho t^2}{4}(1 + \bar\eta_1 + \bar\eta_2 + \bar\eta_3). \tag{NC24}$$

This completion is polynomial in the $\bar\eta_i$ and remains valid when one vanishes; its scaled equation in (NC22) still has the displayed positive pivot. Substituting the corresponding normal values into the physical source and taking residue proves $g(t) \geq 0$ for every real $t \neq 0$.

If $g(t_0) > 0$, shifting by (NC23) gives the positive normal form (NC20) and a positive constant, hence a strict bounded Gram. At a zero, divide (NC24) by $4t^2$:

$$P(1 + t^{-2}) + Q(1 + t^2) - \frac{\bar\rho}{16}(1 + \bar\eta_1 + \bar\eta_2 + \bar\eta_3).$$

Stationarity gives $P/t_0^2 = Qt_0^2$. Since $P + Q > 0$, both are positive; the logarithmic second derivative is positive, and thus $g''(t_0) > 0$. Append the tangent equation to (NC22). Its Schur complement is $g''(t_0)$, so its real Jacobian is invertible. The semialgebraic implicit-function argument gives an exact source stationary center of the form (NC21), with bounded s_c, r_c, t_c. Its normalized value is nonnegative by physicality.

The exact full Taylor rows are

$$\big(\sqrt{n/\tau}(X - X_c), \sqrt{A/\tau}(Y - Y_c), \sqrt{m/\tau}(R - R_c), \sqrt{\mu/\tau}(\delta - \delta_c), t - t_c\big). \tag{NC25}$$

Here is the required coefficient control, including all normal terms. For $a_i = \mu/\eta_i$ the two center factors in a hard packet are

$$\frac{a_i x_{ci}^2}{\tau} = \rho\eta_i s_{ci}^2, \qquad \frac{a_i x_{ci}}{\sqrt{a_i\tau}} = \sqrt{\rho\eta_i}\, s_{ci}. \tag{NC26}$$

The fourth packet has the analogous factors ρr_c^2 and $\sqrt{\rho} r_c$. The core belongs to the ideal (X, Y, R, δ) because it vanishes identically on the matching curve. Its center-linear factors after division by τ are therefore $dx_{ci}/\tau = \rho\eta_i s_{ci}$ and $d\delta_c/\tau = \rho r_c$; its normalized normal factors are $d/\sqrt{a_i\tau} = \sqrt{\rho\eta_i}$ and $d/\sqrt{\mu\tau} = \sqrt{\rho}$. Its quadratic normal factors are bounded by $d/\mu = \varepsilon$. Complementary coefficients are at most τ and give smaller factors. Each packet is at most quadratic in the state coordinates, and all remaining higher Taylor factors are actual angle or state deviations with zero residue. The only testing denominators are powers of the unit t. Thus exact packetwise Taylor allocation is a bounded Gram on (NC25). Its normal residue is (NC20) and its tangent Schur complement is $g''(t_0)/2 > 0$, as follows by differentiating the completed squares (NC23)–(NC24). This assertion also holds when $\bar\eta_i = 0$: the normal form (NC20), its cross entries and the completed Schur expression are continuous polynomial expressions in $\sqrt{\bar\eta_i}$, with a strict normal block. The strict-Gram lemma, the nonnegative center value and pure-factor restoration prove the four-event case without any restriction on its internal coefficient ratios.

6.4.6 A terminating induction on event constraints

Normalize the nonzero source by $d + \lambda + n + m + c + A + D + \alpha + \beta = 1$. The factor is positive and pure-source. Literal $d = 0$ is already an event-square sum; positive slack residue is covered by the positive residue lemma. If $\bar{d} > 0$ and the residue slack is zero, the unconstrained curvature argument and (NC9) with $k = 0$ apply.

It remains that $d > 0$, $\bar{d} = 0$ and the residue slack is zero. Product residue states were settled above. At an entangled residue, let S be all events with positive coefficient residue. This is a nonempty matching. If it has four edges, every complementary coefficient and d is infinitesimal relative to its smallest weight, and (NC17)–(NC26) apply. Otherwise use its at most three independent event amplitudes as the exact hard state coordinates. Set

$$\sigma = \max\big(\{d\} \cup \{c_e : e \notin S\}\big) > 0.$$

Every hard weight a_i satisfies σ/a_i infinitesimal. The parent has the diagonal form (NC7), and its constrained next-scale residue $g = \overline{G}(0, y)$ is nonnegative by physicality.

If $g(y_0) > 0$, (NC9)'s positive-value argument finishes. If $g(y_0) = 0$ and $\overline{d/\sigma} > 0$, the one-, two- or three-hard curvature calculation above gives a strict full soft Hessian; (NC8)–(NC9) finish without assuming comparability of the hard weights.

The only remaining case has $g(y_0) = 0$ and $\overline{d/\sigma} = 0$. Then g is a nonnegative event-square sum, and at least one remaining event coefficient has positive normalized residue. All such event amplitudes vanish at the same entangled residue state. Add them all to S. The enlarged set is still a matching; every coefficient left outside it is infinitesimal relative to each of its weights. If it has at most three edges, repeat. If it has four, apply the four-event proof above, which allows arbitrary source scales. Each repetition adds an edge, and a matching has at most four edges. There are therefore at most three enlargements. Every step retains the entire original parent. This proves Theorem 6.2.

Chapter 7

Assembly of the local certificates

The preceding chapters reduce every valid inequality to events, the closed faces, and the two contact families C1 and C2. Events and closed faces already have separated certificates. We now prove the same assertion for the contact families.

Theorem 7.1 (Canonical certificate)**.** *For every source in either family SG10 with an actual finite noncommuting stationary zero, the complete homogeneous state polynomial is a finite sum of squares in $K_c \otimes_{\mathbb{R}} K_t$ at every independent projective testing pair and every real or complex state.*

Here and below, a source coefficient times a testing expression denotes its separated tensor product. With $D_u = 1 + u^2$, $D_v = 1 + v^2$, the full polynomial to be certified is

$$\begin{aligned} F = {} & 4\lambda D_u (W + vX)^2 + 4n D_u D_v X^2 + 4m(vW - X + uvY - uZ)^2 \\ & + 4C D_v (uW - Y)^2 + 4A D_u D_v Y^2 + (4D + 2d)(uW + uvX - Y - vZ)^2 \\ & + (4\alpha + 2d) D_u (vY - Z)^2 + (4\beta + 2d) D_v (X + uZ)^2 - 2d D_u D_v Z^2. \end{aligned} \tag{LC2}$$

Its homogeneous form defines the literal projective endpoints.

Proof. Normalize $d = 1$. Every original coefficient can be restored by multiplying the roots by $\sqrt{d} \otimes 1$. The source parameter

$$\delta_{\mathrm{tot}} = \frac{d}{d + \lambda + n + m + C + A + D + \alpha + \beta} \tag{LC3}$$

separates two possibilities. If it is a positive unit, all coefficients after normalization are bounded. The proof in Chapter A covers this domain. Otherwise δ_{tot} is a positive infinitesimal, and the contact coordinates from Chapter 5 give the source partitions of Chapters B and C.

For reference, the C1 coordinates satisfy

$$\begin{gathered} 0 < k \le 1, \quad \eta, \zeta > 0, \quad x = k^2, \\ r = 1 - a\eta \in (0,1), \quad \theta = t\zeta \in (0,1), \quad h = 1 - \theta, \\ 0 < a, t < 2, \quad E = 1 - x\eta h > 0, \quad Q = 1 - x\zeta r > 0, \\ M = 1 + xr - t > 0, \quad J = 1 + xh - a > 0, \quad e = k\sqrt{\eta/\zeta} > 0. \end{gathered} \tag{LC4}$$

The C1 partition first separates bounded and infinitesimal k, then the relative source scales η, ζ, e. When ζ is noninfinitesimal, the product $L = x\zeta$ gives the remaining three alternatives. Chapter B proves their source inequalities and constructs the corresponding full quadratic forms. The literal faces $D = 0$ and $\beta = 0$ are included wherever the stationary equations permit them.

For C2, the reflection $(u, v; W, X, Y, Z) \mapsto (1/v, 1/u; Z, X, Y, W)$ exchanges (m, D) and (C, α). Chapter C uses it to reduce the source angle types. The case of two weak complementary events has a separate radial calculation. With one weak event, the directed source partition distinguishes bounded and unbounded ratios; the bounded ratios share one full matrix completion. The remaining unbounded cases and every $n = 0$ boundary are proved there. All these partitions concern coordinates in K_c alone.

Each local identity certifies the full state quadratic form, including its two first-normal terms and its state Hessian. Positive source excesses are retained as whole squares. The remaining coefficient matrix is bounded and has positive-definite real residue, so the lemma of Chapter 3 gives a finite tensor sum of squares. The explicit complementary charts cover all independent testing angles and state ratios.

The sign changes $(u, Y, Z) \mapsto (-u, -Y, -Z)$ and $(v, X, Z) \mapsto (-v, -X, -Z)$ preserve LC2. Its homogeneous identities include zero and infinite half-angles. State normalizations use a nonzero testing coordinate and are restored by its square. Finally, real and imaginary state parts add, since the measurements are real. This proves the theorem. □

To finish the global argument, write an arbitrary valid source as a nonnegative combination of at most nine extreme sources, and an arbitrary testing behaviour as a mixture of at most nine real-qubit behaviours. Events, closed faces and the theorem above certify every atom pair. Equation M2 combines their squares and preserves the affine constant. This proves the separated-slack theorem, and hence the finite semidefinite representation.

Chapter A

Bounded source coefficients

A.1 Two quartic contacts

When one measurement pair becomes commuting, a quadratic angular coefficient can vanish. The next coefficient is then quartic. We treat the two such contacts together: the source determines a critical centre, and a finite polynomial expansion about that centre gives the required Gram matrix. All coefficients in this section are bounded, d is a positive unit, and $\lambda = 0$. The source is physical. Write (n, m, C, A, D) for the coefficients denoted $(e_1, e_2, e_3, e_4, e_5)$ in the event order of SG1.

A finite commuting edge. Suppose the testing residues are $u = 0$, $v = v_0 \neq 0$ and $[W : X : Y : Z] = [1 : 0 : 0 : 0]$, and m is infinitesimal. On the residue source the determinant numerator is

$$Q(U, V) = UE(V) + U^2 B(V), \quad E(V) = q_{10} + hV + q_{12}V^2, \quad B(V) = q_{20} + q_{21}V + q_{22}V^2.$$

Assume $q_{10}, q_{12}, B(v_0^2) > 0$ and $E(v_0^2) = 0$. The unit event amplitude $vW - X + uvY - uZ$ can be set to one, so $W = (1 + X - uvY + uZ)/v$. The opened event contributes the constant $4m$. The state block at $u = 0$ is

$$M_0(v) = \begin{pmatrix} 2(d + 2n + 2\beta)(1 + v^2) & 0 & 0 \\ 0 & 2d(1 + v^2) + 4(C + A)(1 + v^2) + 4D + 4\alpha v^2 & 4v(D - \alpha) \\ 0 & 4v(D - \alpha) & 4\alpha + 4Dv^2 \end{pmatrix}.$$

Assume $M_0(v_0) \succ 0$. In the applications below its lower minor is positive because $\alpha d + 2\alpha C + 2\alpha A + 2\alpha D(1 + v_0^2) + Dv_0^2(d + 2C + 2A) > 0$.

The substitution $X = u^2 x$, $Y = ua$, $Z = ub$, $U = u^2$ defines the polynomial K by the exact identity

$$F(u^2 x, ua, ub, u, v) = 4m + u^2 K(x, a, b, v, U). \tag{FE1}$$

Its coefficients are Laurent polynomials in v. Substitution in SG2 shows that $K_x = UE_x$ and

$$(E_x)_x\big|_{U=0} = 4(d + 2n + 2\beta)(1 + v^2) > 0. \tag{FE2}$$

The residue of K at $U = 0$ is independent of x. Minimizing its positive (a, b) block leaves

$$r(v) = \frac{16(1 + v^2)^2 E(v^2)}{v^2 \det M_0(v)}.$$

Since $E \geq 0$ and has a zero at $v_0^2 > 0$, $E(V) = q_{12}(V - v_0^2)^2$. Thus the Hessian in (a, b, v) is positive definite at the residue zero. Equation FE2 then determines x. The equations $\mathcal{E} = (E_x, K_a, K_b, K_v) = 0$ have an invertible Jacobian in $q = (x, a, b, v)$. They define a source-field branch $q_*(c, U)$. Its minimized profile $k(c, U) = K(q_*(c, U), U)$ satisfies

$$k(c_0, 0) = 0, \qquad \gamma = k_U(c_0, 0) = \frac{16(1 + v_0^2)^2 B(v_0^2)}{v_0^2 \det M_0(v_0)} > 0. \tag{FE3}$$

The last equality follows by taking the Schur complement of the three-state block in the event-normalized parent. Its determinant is $\det P/v^2$; the coefficients of U and U^2 give $r(v)$ and FE3 respectively.

A finite-ratio corner. Now let α be infinitesimal. At the residue source suppose

$$Q(U, V) = q_{20}U^2 + hUV + q_{02}V^2 + q_{21}U^2 V + q_{12}UV^2 + q_{22}U^2 V^2,$$

where $q_{20}, q_{02} > 0$. Choose a nonzero real t_0 such that

$$q_{20}t_0^4 + ht_0^2 + q_{02} = 0, \qquad q_{21}t_0^4 + q_{12}t_0^2 > 0.$$

Normalize the unit amplitude $vY - Z$ to -1, giving $Z = 1 + vY$. The substitution $X = vx$, $Y = vy$, $u = vt$, $W = w$, $V = v^2$ defines a polynomial K by

$$F = 4\alpha + v^2 K(x, y, t, w, V). \tag{FR1}$$

Its constant coefficient in V is

$$\begin{aligned}K_0 &= 4nx^2 + 4m(w-x-t)^2 + 4C(tw-y)^2 + 4Ay^2\\ &\quad + 4D(tw-y-1)^2 + 4\beta(x+t)^2\\ &\quad + 2d\{(tw-y)^2 + (x+t)^2 - 2tw\} + 4\alpha t^2.\end{aligned}$$

Assume the residue quadratic form at t_0 has kernel $(w_0, x_0, y_0, 1)$ and a positive (w, x, y) principal block $M_J(t_0)$. Its minimized scalar is $16(q_{20}t^4 + ht^2 + q_{02})/\det M_J(t)$. Physicality makes the numerator $q_{20}(t^2 - t_0^2)^2$. The full Hessian in $q = (x, y, t, w)$ is therefore positive definite. The source equations $\nabla_q K = 0$ define $q_*(c, V)$, and

$$\gamma = \partial_V K(q_0, 0) = \frac{16(q_{21}t_0^4 + q_{12}t_0^2)}{\det M_J(t_0)} > 0. \tag{FR2}$$

This is again the next coefficient of the full determinant after the positive state block is minimized. The testing chart has u, v infinitesimal and $[W : X : Y : Z]$ of residue $[w_0 : 0 : 0 : 1]$. The actual testing ratio u/v is unrestricted.

The common choice of centre. In either case write $S = U$ or V, and put $\kappa = k(c, 0)$. If $\kappa \geq 0$, choose $S_c = 0$ and $q_c = q_*(c, 0)$. If $\kappa < 0$, solve

$$\mathcal{E}(q_c, S_c) = 0, \qquad K(q_c, S_c) + S_c K_S(q_c, S_c) = 0, \tag{FC1}$$

where $\mathcal{E} = \nabla_q K$ in the corner case. The Jacobian has the invertible normal block above and last pivot $2\gamma > 0$. The semialgebraic implicit-function theorem and real closed transfer give a unique nearby source solution. The function $k(c, S) + Sk_S(c, S)$ increases from the negative value κ, so $S_c > 0$. Realizing the corresponding source measurement and state shows that

$$\mu = c_* + S_c K(q_c, S_c) \geq 0, \qquad c_* = 4m \text{ or } 4\alpha. \tag{FC2}$$

Every use of the implicit-function theorem here is within the source field. The tensor identity uses only the resulting source coordinates and finite polynomial divisions.

Full state expansion. At the edge set

$$N_s = (X - u^2 x_c, Y - ua_c, Z - ub_c), \qquad N_v = u(v - v_c).$$

Quadratic state dependence gives exactly

$$F = 4m + UK(x_c, a_c, b_c, v, U) + UE_x N_x + uK_a N_y + uK_b N_z + N_s^T M N_s.$$

The centre equations make each derivative vanish at (v_c, U_c). Difference division in v and U therefore gives only the cross terms

$$N_x U(U - U_c), \quad N_y u(U - U_c), \quad N_z u(U - U_c), \quad N_x u N_v, \quad N_y N_v, \quad N_z N_v,$$

and N_v^2 and $N_v u(U - U_c)$ from the scalar profile. All coefficients are bounded. For negative powers of the unit v, use $(v^{-j} - v_c^{-j})/(v - v_c) = -\sum_{i=0}^{j-1} v^{-i-1} v_c^{-(j-i)}$.

At the corner set

$$N = (X - vx_c, Y - vy_c, u - vt_c, v(W - w_c)), \qquad z = W - w_c.$$

The linear part is $v\nabla_q K(q_c, V) \cdot N$. The rest belongs to $(N_1, N_2, N_3, vz)^2$: in SG2, a term containing one z and only one of the first three normals carries v, and a term containing z^2 and no such normal carries v^2. Terms with two of the first three normals already belong to this ideal, with bounded z left in their coefficients. This gives the whole quadratic remainder without dividing by v.

In both cases the derivative coefficients are divisible by $S - S_c$. Their cross coefficients with that last generator are infinitesimal. The fixed-centre scalar profile has a double zero at S_c when $\kappa < 0$. Finite division thus yields

$$\begin{aligned}&\kappa \geq 0: \quad F = c_* + \kappa S + G^T H G, \quad G = (N, S),\\ &\kappa < 0: \quad F = \mu + G^T H G, \quad G = (N, S - S_c).\end{aligned} \tag{FC3}$$

At the edge $N = (N_s, N_v)$ and the normal residue block is the positive X entry together with half the positive (a, b, v) Hessian. At the corner it is half the positive four-variable Hessian. The last diagonal is $\gamma > 0$, and its cross entries vanish in residue. The strict bounded-Gram lemma applies. The remaining scalars in FC3 are nonnegative in the source field; S is a testing square. Restoring the event normalization completes both certificates, including arbitrary relative infinitesimal rates.

For the edge with $E(v_0^2) > 0$, the four-normal quadratic form in (X, Y, Z, u) is already strict, with $4m$ retained separately. This supplies the other finite-edge alternative.

A.2 Positive total scale: the finite-pitchfork case

The finite-edge calculation applies whenever d, C, A, D, α are positive units. We establish its hypotheses and then treat the remaining commuting and projective limits.

Theorem A.1. *Let the original normalized C1 or C2 source be bounded, nonnegative and physical over a real closed coefficient extension of $\mathbb{R}$. Assume d, C, A, D, α are positive bounded units. Retain its exact canonical zero coefficients, in particular $\lambda = 0$, and every other nonnegative original coefficient n, m, β. Then its entire parent is tensor-SOS at every independent projective testing pair and every real or complex testing state.*

Noncommuting testing residues are covered by Theorem 6.2.

The finite edge

The only possible zero at $u = 0$, $v = v_0 \neq 0$ is the W state ray when m is infinitesimal. We verify the conditions of Section A.1 on that domain.

At $\lambda = m = 0$, the exact determinant polynomial is

$$Q(U, V) = UE(V) + U^2 B(V), \quad E(V) = q_{10} + q_{11}V + q_{12}V^2, \quad B(V) = q_{20} + q_{21}V + q_{22}V^2.$$

Use the positive quantities $t_1, \ldots, t_8$ in SG6.

All eight have positive bounded-unit residues. At the residue $m = 0$, the required coefficients are

$$q_{10} = 4\alpha A t_7 t_3 > 0, \quad q_{12} = 4CDt_5 t_2 > 0, \quad q_{21} = 4(\beta n t_3 t_2 + CAt_1 t_4) > 0,$$

and $q_{20}, q_{22} \geq 0$. Therefore $B(v_0^2) > 0$ for every $v_0 \neq 0$. Physicality implies $E(V) \geq 0$ for all real $V > 0$: divide $Q(U, V) \geq 0$ by $U > 0$ and let the real variable U tend to zero. Consequently a zero $E(v_0^2) = 0$ is a double zero, $E(V) = q_{12}(V - v_0^2)^2$, and otherwise $E(v_0^2) > 0$.

The state complement $M_0(v_0)$ in Section A.1 is positive definite: its lower determinant has the positive factor $\alpha d + 2\alpha C + 2\alpha A + 2\alpha D(1 + v_0^2) + Dv_0^2(d + 2C + 2A)$. Thus that section proves the full state certificate at this edge, including both possible signs of the optimized source value.

The other measurement limits

Noncommuting residue tests use A6. It remains that at least one angle has residue zero or infinity. For finite $u_0 \neq 0$ and $v_0 = 0$, the real residue parent is a sum of nonnegative event terms and the core $2d[(uW - Y)^2 + (X + uZ)^2]$. The unit A, α, C terms force $Y = Z = W = 0$, and the core then forces $X = 0$. Hence the residue state matrix is positive definite.

For $u_0 = 0$ and finite $v_0 \neq 0$, the unit C, A, D, α terms and $d > 0$ leave only the W ray if m is weak, while unit m removes that ray too. In the weak case, positive residue slack is immediate; at the W ray, the previous section applies for either $E(v_0^2) > 0$ or $E(v_0^2) = 0$.

For $u = \infty$ and finite nonzero v, set $p = 1/u$ and homogenize as $F_p = p^2 F(1/p, v)$. Its real residue has core

$$2d[(W + vX)^2 + (vY - Z)^2]$$

and positive $4C(1 + v^2)W^2 + 4A(1 + v^2)Y^2$ squares. These force every state coordinate to vanish at a zero, so the matrix is positive definite. For $v = \infty$ and finite nonzero u, the homogenized residue has core $2d(1 + u^2)(X^2 + Y^2)$ together with $4C(uW - Y)^2 + 4D(uX - Z)^2$. Again it is positive definite. These are exact residue identities and use no density argument.

It remains to cover the four double corners. The following displayed quadratic forms are complete, with coefficient letters interpreted in real residue. All coefficients that are merely nonnegative remain so. At a unit state coordinate, normalize it in the testing field.

At $(u, v) = (0, 0)$ the only possible zero state is the W ray. Put $W = 1$ and retain the entire m square. The remainder lies exactly in $(u, X, Y, Z)^2$, and its residue quadratic is

$$2(d + 2n + 2\beta)X^2 + (4C + 4D + 2d)(u - Y)^2 + 4AY^2 + 4\alpha Z^2.$$

It is strict. The testing v is bounded infinitesimal in its coefficients and is never divided out.

At $(u, v) = (\infty, 0)$ the only possible zero state is the X ray when n is weak; if n is a unit there is no zero. Put $X = 1$, retain the entire n square in F_p, and use roots (p, v, W, Y, Z). The whole remainder belongs to their square ideal and its residue is

$$4CW^2 + 4AY^2 + (4D + 2d)(W + v)^2 + 4\alpha Z^2 + [4(m + \beta) + 2d](p + Z)^2 \succ 0.$$

At $(u,v)=(0,\infty)$ the only possible zero state is the W ray when m is weak. Put $q=1/v$, $W=1$, and retain the entire m square in $F_q=q^2F(u,1/q)$. The whole remainder is in $(u,X,Y,Z)^2$ and its residue quadratic is

$$(4n+4\beta+2d)X^2+4C(u-Y)^2+[4(A+\alpha)+2d]Y^2+4DZ^2\succ 0.$$

The testing q remains a bounded infinitesimal coefficient.

At $(u,v)=(\infty,\infty)$ the only possible zero state is the Z ray when β is weak. Put $Z=1$ and retain the entire β square in $F_{pq}=p^2q^2F(1/p,1/q)$. The complete remainder lies in $(p,q,W,X,Y)^2$ and its residue quadratic is

$$4CW^2+4AY^2+4D(X-p)^2+(2d+4n)X^2+[2d+4(m+\alpha)](Y-q)^2\succ 0.$$

For each corner, allocate two normal factors from every monomial to the Gram roots; every remaining factor is bounded. Terms above quadratic order have infinitesimal coefficients. The displayed full quadratic residues are strict, so the separated strict-Gram lemma applies. Restore each retained square. Homogeneous angle factors and state normalizations are pure testing squares. The identities include literal zero and infinity and every original nonnegative opening rate. For nonzero state residues away from the listed rays, positivity gives the scalar strict-residue exit. The zero state is immediate and complex states split into real and imaginary evaluations. This proves the theorem.

A.3 Bounded C2 sources with infinitesimal alpha

Theorem A.2. *Let the original bounded normalized C2 source be physical, with $\lambda=\beta=0$, all event coefficients nonnegative, d,n,m,C,A,D positive bounded units, and α weak, meaning $\bar{\alpha}=0$. Then the complete original parent is tensor-SOS for every independent projective testing pair and every real or complex testing state.*

The finite-ratio calculation of Section A.1 treats the remaining corner. We first identify the zero states to which it applies.

Only the zero-zero corner needs a new argument

A6 covers noncommuting testing residues. On finite $u=0$, $v\neq 0$, the unit m removes the W ray, and the (X,Y,Z) complement is strict because d,D,C,A are units and $v\neq 0$. On finite $v=0$, $u\neq 0$, the units n,A force $X=Y=0$ at a zero; the C square forces $W=0$, and the signed core $(X+uZ)^2$ forces $Z=0$.

The exact single-infinity core identities in Section A.2 require no unit α: at $u=\infty$, $v\neq 0$ use $2d[(W+vX)^2+(vY-Z)^2]$ and the C,A squares; at $v=\infty$, $u\neq 0$ use $2d(1+u^2)(X^2+Y^2)$ and the C,D squares. Both state matrices are strictly positive.

At $(\infty,0)$ the unit n removes the only possible X zero ray, and at $(0,\infty)$ the unit m removes the W ray. At (∞,∞), put $p=1/u,q=1/v$, normalize the possible Z zero ray to $Z=1$, and use (p,q,W,X,Y) as roots. The exact full-parent square-ideal allocation given in Section A.2 has residue

$$4CW^2+4AY^2+4D(X-p)^2+(2d+4n)X^2+[2d+4(m+\alpha)](Y-q)^2\succ 0.$$

This form remains strict for weak α. Thus every projective test except u,v infinitesimal is already covered, including literal zero/infinite endpoints and all signs.

The exact two state charts at zero-zero

Write F for the original parent, with $\lambda=\beta=0$, and retain the whole nonnegative α square $4\alpha(1+u^2)(vY-Z)^2$ whenever using an ordinary strict jet. At $u=v=0$ the residue is strictly positive on the (X,Y) state subspace. Hence X- or Y-unit states have positive residue slack. For all other states, normalize a maximal coordinate. Then either $W=1$ with Z infinitesimal, or $Z=1$ with W bounded.

In the first chart the entire remaining parent belongs to $(X,Y,u,v)^2$. Its quadratic residue is

$$(4n+2d)X^2+4m(v-X)^2+[4(C+D)+2d](u-Y)^2+4AY^2\succ 0. \tag{1}$$

Consequently the finite normal allocation and restoration of the complete α square give tensor-SOS. No testing ratio is used.

In the second chart, set $w=\bar{W}\in\mathbb{R}$. The full residue quadratic jet on (X,Y,u,v) is

$$\begin{aligned}Q_w={}&4nX^2+4m(vw-X-u)^2+4C(uw-Y)^2+4AY^2\\&+4D(uw-Y-v)^2+2d[(uw-Y)^2+(X+u)^2-2uvw].\end{aligned} \tag{2}$$

All coefficient letters in (1)–(2) now denote their real residues. Physicality implies $Q_w\succeq 0$: evaluate the physical residue parent at $(X,Y,u,v)=s(x,y,a,b)$ and let the real s tend to zero. The (X,Y) principal block is strictly

positive, already from the nX^2 and AY^2 terms. If Q_w is strict, the same ordinary four-normal allocation applies, with $W - w$ infinitesimal in its coefficients, and the whole α square restored.

Suppose instead that Q_w is singular. It has no nonzero kernel vector with $v = 0$: the nonnegative sum at $v = 0$ includes $4nX^2 + (4m + 2d)(X + u)^2 + 4AY^2$, so $X = u = Y = 0$. It also has no nonzero kernel vector with $u = 0$: the sum at $u = 0$ includes $4nX^2 + 4AY^2 + 4D(Y + v)^2$, so $X = Y = v = 0$. Its real angular Schur complement therefore has positive diagonal entries. Being singular and positive semidefinite, it has rank one and a kernel with finite nonzero ratio $t_0 = u/v$. Normalize a real kernel vector by $v = 1$ and write it as $(x_0, y_0, t_0, 1)$. This constructs a real coefficient-side jet center; it imposes no condition on the actual independent testing value of u/v.

Application of the finite-ratio contact calculation

The homogeneous state jet at a real ratio t is

$$J_t(W, x, y, Z) = 4nx^2 + 4m(W - x - tZ)^2 + 4C(tW - y)^2 + 4Ay^2 \\ + 4D(tW - y - Z)^2 + 2d[(tW - y)^2 + (x + tZ)^2 - 2tWZ].$$

Its (W, x, y) complement at $Z = 0$ is strict: it contains $4nx^2 + 4m(W - x)^2 + 4Ay^2$, with the remaining terms nonnegative. At $t = t_0$, its kernel is $(w, x_0, y_0, 1)$ by (2).

In the original determinant notation, at $\lambda = \beta = \alpha = 0$,

$$q_{20} = 4nAt_3t_4 > 0, \qquad q_{02} = 4mDt_5t_6 > 0, \qquad q_{21} = 4CAt_1t_4 > 0, \qquad q_{12} = 4CDt_5t_2 > 0.$$

Here the t_i are the original positive sums of bounded coefficients printed in the finite-pitchfork fit; all are units since d is a unit. The full jet determinant is

$$\det J_t = 16(q_{20}t^4 + q_{11}t^2 + q_{02}).$$

Thus the singular jet gives precisely the finite nonzero root required by Section A.1, and its next coefficient satisfies $q_{21}t_0^4 + q_{12}t_0^2 > 0$. The state complement, both endpoint coefficients, and the positive next coefficient are all verified.

That calculation normalizes the pure-testing event $vY - Z$ to -1, setting $Z = 1 + vY$. It keeps the complete opened α square, including its $4\alpha u^2$ term, in the exact profile

$$F = 4\alpha + v^2K(x, y, t, w, V), \qquad (X, Y, u, W, V) = (vx, vy, vt, w, v^2)$$

used only to construct coefficient-field critical centers. The positive state complement and the double finite root give a strict full Hessian in (x, y, t, w), with positive next coefficient

$$\frac{16(q_{21}t_0^4 + q_{12}t_0^2)}{\det M_J(t_0)}.$$

The two signs of the optimized constant are handled by coefficient-only implicit centers exactly as in the finite-edge proof; physicality makes the remaining scalar nonnegative.

Its actual tensor roots are the un-divided expressions

$$X - vx_c, \quad Y - vy_c, \quad u - vt_c, \quad v(W - w_c), \quad v^2 - V_c.$$

All four full first-normal coefficients are $v\nabla K$ on this frame. Monic difference division makes their coupling to $v^2 - V_c$ infinitesimal. All 55 full-tail monomials have an exact allocation to the first four roots; factors of $W - w_c$ that remain in coefficients are infinitesimal. There is no substitution dividing $v(W - w_c)$ by v. The full residue Gram is the positive four-variable Hessian and the positive last scalar pivot. Hence the theorem applies to every testing point with the stated residue state, regardless of whether its actual tested u/v is bounded or approaches t_0. This includes literal $v = 0$ and arbitrary independent opening rates.

The two zero-corner state charts and the strict/singular jet split are exhaustive. Together with the complementary projective cases, this proves the theorem. The zero state is direct, and complex states split into real and imaginary quadratic evaluations.

A.4 The complete positive-scale C1 reciprocal case

The source domain. Let the canonical C1 source be physical, with an actual finite noncommuting zero contact, and normalize $d = 1$. Suppose every source coefficient is bounded, $\lambda = n = 0$, C, A, α are positive bounded units, $m > 0$ and $D \geq 0$ are infinitesimal, and $\beta \geq 0$ is arbitrary bounded. Retain the original canonical source gaps and intersections. Then the entire parent has a finite sum-of-squares representation in $K_c \otimes_{\mathbb{R}} K_t$ for every independent

projective test and every real or complex state. The assertion includes literal $D = 0$ and literal $\beta = 0$ when permitted by the actual contact; they cannot both vanish.

We use $O(1)$ and $o(1)$ for bounded and infinitesimal elements of one of the two fields, and $a \asymp b$ for positive source quantities whose ratio is a bounded unit. A bounded tensor Gram with a strictly positive real residue is a sum of squares by the separated strict-Gram lemma. Its rows need not be bounded. No order on the tensor product, mixed inverse, or cancellation of a determinant multiplier is used below.

Source balances at the reciprocal corner

Both C1 and C2 have a reciprocal corner when m, D are infinitesimal and C, A, α are positive units. All coefficients are bounded, $d = 1$, $\lambda = 0$, and $n\beta = 0$. At a stationary source contact put $U = u_c^2$, $P = v_c^{-2}$. The determinant formula SG5–SG6 becomes

$$L(U, P^{-1}) = h + \frac{ma_1}{U} + \frac{mDa_2}{UP} + a_3 P + \frac{\beta m a_4 + Da_5}{P} + a_6 U + n a_8 UP + \frac{\beta a_7 U}{P},$$

where the positive units are explicitly

$$\begin{array}{llll} a_1 = 4\alpha t_5 t_8, & a_2 = 4t_5 t_6, & a_3 = 4\alpha A t_7 t_3, & a_4 = 4t_1 t_6, \\ a_5 = 4C t_5 t_2, & a_6 = 4CA t_1 t_4, & a_7 = 4C t_1 t_2, & a_8 = 4A t_3 t_4. \end{array}$$

Here h is bounded. At a zero, every positive Laurent term is bounded by $-h$, so U, P are bounded. Differentiating at the stationary contact gives the exact balances

$$\begin{aligned} U^2(a_6 + n a_8 P + \beta a_7/P) &= m(a_1 + Da_2/P), \\ P^2(a_3 + n a_8 U) &= mDa_2/U + \beta m a_4 + Da_5 + \beta a_7 U. \end{aligned} \tag{A.1}$$

These equations force both U and P to be infinitesimal. Indeed, if U were a unit and P a unit, the first balance would have a positive-unit left side and an infinitesimal right side. If U were a unit and P infinitesimal, the second balance would give $mDa_2/P \le UP(a_3 + n a_8 U)$, again making the first right side infinitesimal. Thus U is infinitesimal. If P were a unit, the first balance would then give

$$\frac{mDa_2}{U} = UP(a_6 + n a_8 P) + \beta a_7 U - m a_1 P/U \le UP(a_6 + n a_8 P) + \beta a_7 U,$$

which is infinitesimal. Every term on the second right side would be infinitesimal, contradicting its positive-unit left side.

The nonnegative terms in the second balance now give $D/P^2 = O(1)$ and $\beta U/P^2 = O(1)$. The first consequently gives

$$\frac{m}{U^2} \asymp 1 + \frac{\beta}{P}, \qquad \frac{m}{U},\ \frac{D}{P},\ \frac{\beta U}{P} \in \mathrm{Inf}\,. \tag{A.2}$$

In C2, $\beta = 0$, so m/U^2 is a positive unit and

$$\frac{D}{P^2} = \frac{a_3 + n a_8 U}{a_5 + m a_2/U}$$

is a positive unit as well. Thus $m \asymp U^2$, $D \asymp P^2$. For C1 the preceding bounded estimates also allow literal $D = 0$ or $\beta = 0$ whenever the source contact permits it.

Source entry before any state-ratio division

Write the actual contact angles as $u_c = k > 0$, $b_c = 1/v_c = p > 0$, and put $U = k^2$, $P = p^2$. Equation (A.1), with $n = 0$, gives U, P positive infinitesimals and

$$D/P^2 = O(1), \qquad \beta U/P^2 = O(1), \qquad m/U^2 \asymp 1 + \beta/P. \tag{CR1}$$

The same balances give

$$m/U = o(1), \qquad D/P = o(1), \qquad \beta U/P = o(1). \tag{CR2}$$

At a physical noncommuting contact the signed core is negative, and the exact core Schur identity SE9 gives $W_c Z_c > 0$. Normalize $W_c = 1$ and write $Z_c = q > 0$. Define

$$Q = q^2, \quad L = PQ, \qquad r = \frac{k}{pq}, \quad y = \frac{Y_c}{pq}, \quad x = \frac{X_c}{pq^2}. \tag{CR3}$$

We now prove these divisions are bounded wherever they are used. In the reciprocal quarter-parent CR9 below the hard X, Y matrix at the source has residue $\mathrm{diag}(\bar\beta + 1/2, H_0)$, where $H = C + A + \alpha + 1/2$ is a positive unit. Eliminate this two-by-two block inside the source field. Its Schur block on W, Z satisfies

$$\frac{S_{WW}}{U} \sim a := C - C^2/H > 0, \qquad -\frac{S_{WZ}}{kp} \sim b_* := 1/2 + C\alpha/H > 0. \tag{CR4}$$

Here and below a displayed positive expression in an asymptotic formula is evaluated in residue. To check CR4 without a shape assumption, the original hard couplings are

$$\begin{aligned} P_{XW} &= p\{-m + (D + 1/2)U\}, & P_{YW} &= k\{m - C(1 + P) - (D + 1/2)P\},\\ P_{XZ} &= k\{\beta - D + P(m + \beta + 1/2)\}, & P_{YZ} &= p\{D - \alpha - U(m + \alpha + 1/2)\},\\ P_{WZ} &= -(m + D + 1/2)kp. \end{aligned}$$

CR2 shows that the X corrections in CR4 vanish in residue and the Y corrections are respectively C^2/H and $C\alpha/H$. The kernel equation gives $q = S_{WW}/(-S_{WZ})$, so r is a positive bounded unit.

The scaled X, Y kernel equations in the unknowns x, y have bounded entries and a diagonal positive residue. For example their X equation is exactly

$$\begin{aligned} 0 = {} & \{(\beta + 1/2)(1 + P) + Pm + (D + 1/2)r^2PQ\}x\\ & + r\{\beta - D + P[\beta + 1/2 - m(y - 1) + (D + 1/2)(r - y)]\} - m/Q. \end{aligned} \tag{CR5}$$

The quantity $m/Q = (m/U)r^2P$ is bounded before this equation is used. The y coefficient in CR5 is infinitesimal; the x coefficient in the scaled Y equation is $O(PQ)$ and its y pivot is $H + o(1)$. Thus x, y are bounded. Taking the Y and Z kernel equations in residue, using CR1–CR2, yields

$$Hy \sim Cr + \alpha, \qquad C(r - y) \sim 1/2, \qquad \alpha(1 - y) \sim r/2. \tag{CR6}$$

Consequently $y, r - y, 1 - y$ are positive bounded units. Equivalently their real residues satisfy

$$C_0 = \frac{1}{2(r_0 - y_0)}, \quad A_0 = \frac{1 + r_0 - y_0}{2y_0}, \quad \alpha_0 = \frac{r_0}{2(1 - y_0)}. \tag{CR7}$$

For the last equation in CR6, divide the original Z kernel equation by Pq. Its extra X contribution is $O(x\beta Q + xPQ)$; CR1 and the already proved unit r give $\beta Q = O(P)$, while $PQ = U/r^2$ is infinitesimal. All the omitted terms therefore vanish in residue. This proves CR6 in the unbounded-q case as well.

In CR5, $D = o(P)$ and $m/Q = o(P)$. Its numerator for $-x$ is the sum of $r\beta$ and a positive-unit multiple of P, up to $o(P)$. Hence the following bounds hold on the whole source domain:

$$\begin{aligned} L = PQ = o(1), \qquad \beta Q/P = O(1), \qquad -x \asymp P + \beta, \qquad r + x \text{ is a positive unit},\\ m \asymp PQ^2(P + \beta), \qquad m/(PQ) = O(PQ + \beta Q) = o(1), \qquad xQ = o(1). \end{aligned} \tag{CR8}$$

The statement about $r + x$ follows directly from CR5: $x/r = -\beta/(\beta + 1/2) + o(1)$.

The entire scalar and all normal coefficients

Use $b = 1/v$, $D_u = 1 + u^2$, $D_b = 1 + b^2$. The full reciprocal quarter-parent is

$$\begin{aligned} F = {} & m(W - bX + uY - ubZ)^2 + CD_b(uW - Y)^2 + AD_uD_bY^2\\ & + (D + 1/2)(ubW + uX - bY - Z)^2 + (\alpha + 1/2)D_u(Y - bZ)^2\\ & + (\beta + 1/2)D_b(X + uZ)^2 - D_uD_bZ^2/2. \end{aligned} \tag{CR9}$$

All further local identities are expansions of this full expression. In the chart $W = 1$, $Z = z$ put

$$u_g = rbz, \qquad X_g = xbz^2, \qquad Y_g = ybz, \qquad V = b^2, \quad T = z^2. \tag{CR10}$$

Its entire scalar restriction is

$$\Phi = m + DT + hVT + \mathcal{C}V^2T + EVT^2 + \mathcal{F}V^2T^2 + GV^3T^2 + JV^2T^3, \tag{CR11}$$

where

$$\begin{aligned} \mathcal{C} &= Ay^2 + (C + D + 1/2)(r - y)^2,\\ E &= \beta(r + x)^2 + x^2/2 - 2Drx, \qquad G = Ar^2y^2, \qquad J = (D + 1/2)r^2x^2,\\ \mathcal{F} &= Ar^2y^2 + (\alpha + 1/2)r^2(y - 1)^2 + \beta(r + x)^2 + m[r(y - 1) - x]^2\\ &\qquad + (2D + 1)rx(r - y) + rx + x^2/2,\\ h &= Ay^2 + C(r - y)^2 + \alpha(y - 1)^2 + y^2/2 - r - 2D(r - y) + 2m[r(y - 1) - x]. \end{aligned} \tag{CR12}$$

In particular $\mathcal{C}, G$ are positive units, $E \asymp \beta + P^2 > 0$, $J \geq 0$, and all these coefficients are bounded. If β is infinitesimal, $\mathcal{F}$ is a positive unit, with residue $r_0^2(1 + r_0 - y_0)/2$. For unit β, $\mathcal{F}$ can have either sign; it is not discarded or assumed positive.

Actual contact and its two angular/state jets give $\Phi(P,Q) = \Phi_V(P,Q) = \Phi_T(P,Q) = 0$. Therefore

$$\begin{aligned} D &= P^2\mathcal{C} - PQE + P^3QG - P^2Q^2J, \\ m &= PQ^2E + P^2Q^2\mathcal{F} + P^3Q^2G + 2P^2Q^3J, \\ h &= -2P\mathcal{C} - QE - 2PQ\mathcal{F} - 3P^2QG - 2PQ^2J. \end{aligned} \tag{CR13}$$

Set

$$\delta = D/P^2 \geq 0, \qquad \theta = EQ/P > 0. \quad \delta, \theta = O(1), \qquad \delta + \theta = \mathcal{C} + GL - JQ^2 = \mathcal{C} + o(1), \tag{CR14}$$

where $L = PQ$. The bounds follow either from CR1 and CR8 or from CR13; $JQ^2 = O((xQ)^2) = o(1)$. At least one of δ, θ is a unit. Define

$$R = z(V - P), \quad K = VT - PQ, \quad M = V^2T + (P/Q)VT^2 + P^2Q - 3PVT.$$

The exact complete balanced scalar is

$$\begin{aligned} \Phi = \delta R^2 + \theta M + \mathcal{F}K^2 + G\{PK^2 + VzRK\} \\ + J\{(QR - zK)^2 + 2QK^2\}. \end{aligned} \tag{CR15}$$

The entire J line is retained as positive squares.

To retain the complete first-normal terms, Let L_u, L_X, L_Y be half the first derivatives of CR9 at CR10. Then

$$L_u = bzA_u(V,T), \quad L_Y = bzA_y(V,T), \quad L_X = b\{-m + TB_X(V,T)\}, \tag{CR16}$$

where

$$\begin{aligned} A_u &= a_{00} + a_{10}V + a_{01}T + a_{11}VT + a_{21}V^2T + a_{12}VT^2, \\ A_y &= e_{00} + e_{10}V + e_{11}VT + e_{21}V^2T, \\ B_X &= c_0 + c_1V + c_2VT, \\ c_0 &= -Dr + \beta(r + x) + x/2, \\ c_1 &= (D + 1/2)r(r - y) + (\beta + 1/2)(r + x) + m[x - r(y - 1)], \\ c_2 &= (D + 1/2)r^2x. \end{aligned} \tag{CR17}$$

Every displayed coefficient is bounded. The only bounds beyond this needed below are

$$a_{01} = \beta(r + x) - Dx = O(E), \quad a_{12} = (D + 1/2)rx^2, \quad a_{12}Q = O(1), \qquad c_2 = O(P + E). \tag{CR18}$$

They follow from CR8 and CR12. Every other coefficient is the corresponding monomial coefficient of the explicit polynomial CR9.

State and angle stationarity give $A_u(P,Q) = A_y(P,Q) = 0$ and $B_X(P,Q) = m/Q$. Put $E_T = bz(T - Q)$. Finite monic division therefore gives exactly

$$\begin{aligned} L_i = A_{i,V}(V,T)bR + B_i(T)E_T,\ B_u(T) &= a_{01} + P\{a_{11} + Pa_{21} + a_{12}(T + Q)\}, \\ B_y(T) = P(e_{11} + Pe_{21}), \\ L_X = b\{[m/(PQ) + c_2T]K + z[c_1 - m/(PQ)]R\}. \end{aligned} \tag{CR19}$$

The first line is for $i = u, y$ and $A_{i,V} = [A_i(V,T) - A_i(P,T)]/(V - P)$, an exact polynomial. In particular, using $P(T - Q) = K - zR$, each of L_u, L_Y, L_X is a bounded linear combination of

$$bR, \qquad bzK, \qquad b[m/(PQ) + c_2T]K, \qquad EE_T. \tag{CR20}$$

Every occurrence of K outside the last product has an infinitesimal coefficient. CR19 remains literal at $b = 0$ or $z = 0$.

The cubic conductors and the bounded source-ratio chart

All radicals in this subsection are products of radicals in the separate fields. Set

$$\mathsf{A} = (V^2T)^{1/3}, \quad \mathsf{B} = ((P/Q)VT^2)^{1/3}, \quad \mathsf{C} = (P^2Q)^{1/3}, \qquad \rho = (P/Q)^{1/3}.$$

Because $\mathsf{ABC} = PVT$,

$$M = \tfrac{1}{2}(\mathsf{A} + \mathsf{B} + \mathsf{C})[(\mathsf{A} - \mathsf{B})^2 + (\mathsf{A} - \mathsf{C})^2 + (\mathsf{B} - \mathsf{C})^2]. \tag{CR21}$$

Let $\mathcal{Z}$ be its nine roots multiplied by $\sqrt{\theta}$, so that $\theta M = \frac{1}{2}\sum_{j=1}^{9} \mathcal{Z}_j^2$. For nonnegative testing b, z, the following exact identities hold:

$$\begin{aligned} bR &= b\sqrt{\mathsf{A}}(\mathsf{A}-\mathsf{B}) + \rho b^{1/3} z^{2/3}\sqrt{\mathsf{A}}(\mathsf{A}-\mathsf{C}), \\ EE_T &= \sqrt{E}\, z\,[\sqrt{\theta}\sqrt{\mathsf{B}}(\mathsf{B}-\mathsf{A})] \\ &\quad + \theta^{1/3} E^{1/6} b^{2/3} z^{1/3}[\sqrt{\theta}\sqrt{\mathsf{B}}(\mathsf{B}-\mathsf{C})]. \end{aligned} \tag{CR22}$$

Negative signs are handled by the exact testing gauges; absolute values give the same monomial identities if desired. If θ is a unit, the first line has infinitesimal coefficients on $\mathcal{Z}$: $\rho = (E/\theta)^{1/3}$ is bounded and b is infinitesimal. If δ is a unit, bR instead has the infinitesimal coefficient b on the strict row R. The second line has infinitesimal coefficients whenever E is infinitesimal or z is infinitesimal. Its displayed coefficients use no comparison of a source rate with a testing rate.

Suppose first that actual q is bounded and β is infinitesimal. Use K and $\mathcal{Z}$, with R added when δ is a unit. If only θ is a unit, retain δR^2 as a separate positive square. In CR15 the K pivot $\mathcal{F}$ is a positive unit. The only signed extra term is $GVzRK = Gbz(bR)K$, an infinitesimal Gram correction by CR22. The term GPK^2 is nonnegative. Thus CR15 has a strict residue Gram on the stated roots. CR19–CR22 give infinitesimal lifts of all three complete first normals to these same roots.

If β is a unit, CR8 forces Q infinitesimal and E a unit. The only chart still requiring CR10 will be z infinitesimal. The second line of CR22 still has infinitesimal coefficients. When δ is a unit, use the exact identity

$$2QM + TR^2 - K^2 = (K - zR)^2 + 2QR^2. \tag{CR23}$$

Choose a positive real a with $\overline{\mathcal{F}} + a > 1$. Subtracting a times the corresponding zero identity changes the first three terms of CR15 into

$$(\delta - aT + 2aQ)R^2 + (1 - 2aP/E)\theta M + (\mathcal{F} + a)K^2 + a(K - zR)^2. \tag{CR24}$$

The first three pivots have strictly positive residues; both costs T and P/E are infinitesimal. This handles the signed $\mathcal{F}$ exactly.

When instead θ is a unit, ρ is also a unit and

$$K = \rho^{-1}\{\sqrt{\mathsf{A}}\,[\sqrt{\mathsf{A}}(\mathsf{B}-\mathsf{C})] + \sqrt{\mathsf{C}}\,[\sqrt{\mathsf{C}}(\mathsf{A}-\mathsf{C})]\}. \tag{CR25}$$

Both coefficients on $\mathcal{Z}$ are infinitesimal. Thus the entire signed $\mathcal{F}K^2$ is an infinitesimal correction on the strict circuit Gram. The G cross term and every first normal are controlled by CR22 and CR25; δR^2 and the J line remain positive squares. No inverse of δ is used.

An unbounded actual state ratio in the same testing chart

Now let actual q be unbounded. CR8 and CR12 imply

$$\beta/P = O(1/Q) = o(1), \quad -x \asymp P, \quad m \asymp L^2, \quad E/P = o(1), \quad JQ^2 = O(L^2) = o(1). \tag{CR26}$$

Here β is infinitesimal and $\mathcal{F}$ is a positive unit. In the testing chart $W = 1$, z bounded, the exact relation

$$\theta M = \theta R^2 + (E/P)(K^2 - zRK) \tag{CR27}$$

makes CR15 a strict two-row Gram on R, K, plus its whole positive J line. Its residue is $\operatorname{diag}(\overline{\delta+\theta}, \overline{\mathcal{F}})$. Every correction coefficient is infinitesimal: E/P, P and Vz are infinitesimal. In CR19, B_u/P and B_y/P are bounded, because $a_{01}/P = O(\beta/P + Dx/P)$ and $a_{12}Q$ are bounded. Using $PE_T = bzK - bz^2R$ shows that all three full normals are infinitesimal linear combinations of R, K.

The complementary testing state chart

Use $Z = 1$, $W = w$ infinitesimal, and write $a = u/r$. The graph is

$$b = aw, \qquad X = xa, \qquad Y = yaw. \tag{CR28}$$

The entire scalar is

$$\Phi' = D + mw^2 + Ea^2 + ha^2w^2 + \mathcal{C}a^4w^4 + \mathcal{F}a^4w^2 + Ga^6w^4 + Ja^4. \tag{CR29}$$

First assume actual q bounded and β infinitesimal. If δ is a unit, use the three strict rows $P, a^2w, \sqrt{E}\,a$. The sole signed cross in CR29 is $ha^2w^2 = (h/P)w\,P(a^2w)$, and h/P is bounded by CR13. Every other term is positive or has the positive-unit $\mathcal{F}$ pivot. If θ is a unit, $P/E = Q/\theta$ is bounded; use instead $\sqrt{E}\,a, a^2w$ and retain D, mw^2 whole. The same signed term is the infinitesimal correction $(h/E)w^2(\sqrt{E}\,a)^2$.

Here are the complete complementary first normals, which also prove the bounds for unbounded actual q. Define

$$R' = a^2w^2 - P, \qquad K' = w(a^2 - L). \tag{CR30}$$

Half the b, Y derivatives of the original full CR9 at CR28 have the respective forms

$$L'_i = aw(d_{i0} + d_{i1}a^2 + d_{i2}a^2w^2 + d_{i3}a^4w^2), \qquad i = b, Y, \tag{CR31}$$

with bounded coefficients. At the actual contact $a^2 = L, w^2 = 1/Q$, they satisfy $d_{i0} + Ld_{i1} + Pd_{i2} + PLd_{i3} = 0$. Consequently the entire, exact divisions are

$$L'_i = (d_{i1} + Pd_{i3})aK' + (d_{i2} + a^2d_{i3})awR'. \tag{CR32}$$

The full X derivative is

$$L'_X = a(c_0 + c_2a^2 - mw^2 + c_1a^2w^2), \qquad c_0 + c_2L - m/Q + c_1P = 0. \tag{CR33}$$

For bounded Q, c_0/P and m/P are bounded, and $c_2 \in PB_c + EB_c$ by CR18. The products aK' and awR' are infinitesimal linear combinations of P, a^2w. Expanding CR33 gives infinitesimal coefficients on $P, a^2w, \sqrt{E}\,a$. If θ is a unit, use $P/E = O(1)$: $c_0/\sqrt{E} = o(1)$, $c_2 = O(E)$ and $m/E = O(1)$, giving the same assertion on the two rows $\sqrt{E}\,a, a^2w$. These are pure-source bounded quotients; no testing w is inverted.

For unbounded actual q, CR29 has the exact alternative form

$$\begin{aligned} \Phi' &= \delta R'^2 + \theta M' + \mathcal{F}K'^2 + G\{PK'^2 + a^2wR'K'\} \\ &\quad + J\{a^4(Qw^2 - 1)^2 + 2QK'^2\}, \\ M' &= a^4w^4 + (P/Q)a^2 + P^2Qw^2 - 3Pa^2w^2. \end{aligned} \tag{CR34}$$

Define nonnegative elementary radicals

$$\mathsf{A}' = |aw|^{4/3}, \quad \mathsf{B}' = (P/Q)^{1/3}|a|^{2/3}, \quad \mathsf{C}' = (P^2Q)^{1/3}|w|^{2/3}.$$

They give the same nine-square cubic identity for M'. The exact conductor needed for the complete normals is

$$|a|R' = |a|\sqrt{\mathsf{A}'}(\mathsf{A}' - \mathsf{C}') + \sqrt{L}\sqrt{\mathsf{B}'}(\mathsf{A}' - \mathsf{B}'). \tag{CR35}$$

Its coefficients are infinitesimal. If δ is a unit, use R', K' and retain the circuit as positive squares. If θ is a unit, use its scaled nine roots and K', retaining $\delta R'^2$ whole. In either case CR34 is strict: its sole signed cross is $Gaw(aR')K'$, controlled by CR35, and its last line is retained as squares.

CR26 implies $c_2/P = O(1)$ and $c_0/P = m/(PQ) - Qc_2 - c_1 = O(1)$. The full exceptional normal has the exact division

$$\begin{aligned} L'_X = &-[c_0/P + (c_2/P)a^2]\,aR' \\ &+[(c_2/P)(a^2 + L) + c_0/P + c_1]\,awK'. \end{aligned} \tag{CR36}$$

Together with CR32, this uses only bounded multiples of aR', awR', aK'. Each has infinitesimal coefficients on the strict soft rows by CR35. This also proves the result at literal $w = 0$.

All higher normals and the remaining unit-beta states

The preceding scalar and first-normal formulas certify the entire parent. Here are the required hard blocks and the justification for every higher term. Normalize the testing state so its coordinates are bounded with at least one unit. At the reciprocal angle corner $u, b = o(1)$ the state residue hard block is $(\bar{\beta} + 1/2)X^2 + H_0Y^2$. If X or Y is a unit, the entire parent has positive real residue and the strict-element lemma applies immediately. Otherwise X, Y are infinitesimal and one of W, Z is a unit.

The exact sign gauges make $u, b \geq 0$. If $WZ \leq 0$, the signed core is a nonnegative pure-testing scalar by the full core Schur identity SE9 and its literal homogeneous endpoints, so its source multiple and all event squares are separated squares. Otherwise choose either $W = 1$, $z = Z$ bounded, or $Z = 1$, $w = W$ infinitesimal. Both have X, Y infinitesimal.

In the $W = 1$ chart, set $N_u = u - rbz$, $N_X = X - xbz^2$, $N_Y = Y - ybz$. They are infinitesimal. The complete expansion of CR9 is its scalar CR11, twice the three first normals CR16, and a polynomial quadratic Gram in N_u, N_X, N_Y. Terms of normal degree at least three have infinitesimal Gram entries. For weak β, or for unit β with z infinitesimal, the hard residue is

$$C_0(N_u - N_Y)^2 + (A_0 + \alpha_0 + 1/2)N_Y^2 + (\beta_0 + 1/2)N_X^2 \succ 0. \tag{CR37}$$

All first-normal lifts established above have infinitesimal coefficients. The full hard-and-soft residue is therefore positive definite. In the $Z = 1$ chart use $N_b = b - aw$, $N_X = X - xa$, $N_Y = Y - yaw$. For weak β its hard residue is

$$N_X^2/2 + H_0 N_Y^2 - 2\alpha_0 N_b N_Y + \alpha_0 N_b^2 \succ 0, \tag{CR38}$$

since $H_0 - \alpha_0 = C_0 + A_0 + 1/2 > 0$. The same finite expansion handles every higher normal term. Thus none of CR15, CR27 or CR34 is being used merely as a scalar restriction.

It remains only unit β and a testing state with Z a unit relative to W. Normalize $Z = 1$ and $W = \omega$ bounded. Retain the entire m, D squares. The remainder belongs exactly to the square of the ideal (u, b, X, Y). Its quadratic real residue is

$$\begin{aligned}(\beta_0 + 1/2)\left(X + \frac{\beta_0}{\beta_0 + 1/2}u\right)^2 + \frac{\beta_0}{1 + 2\beta_0}u^2 \\ + H_0\left(Y - \frac{C_0\omega u + \alpha_0 b}{H_0}\right)^2 + a_0(\omega u - r_0 b)^2, \qquad a_0 = C_0 - C_0^2/H_0 > 0.\end{aligned} \tag{CR39}$$

CR7 is precisely the critical relation needed for this identity. The expression is positive definite in (u, b, X, Y) for every real ω, including zero. Its complete bounded polynomial matrix is consequently strict; every higher degree is an infinitesimal correction. This completes all local states, including the unit-beta case where $\mathcal{F}$ was signed.

Every other test and restoration

The exact outer identities RC18–RC25 from the newly rederived C2 reciprocal theorem apply here as polynomial identities, with $n = 0$ and the entire extra β square retained. Their strictness uses only C, A, α positive units and the critical residue relations CR7; no C2 source membership is used. Explicitly:

- For bounded original u, v, the four rows (uW, Y, X, Z) give RC18. After CR7 its real determinant is $(r_0 + u^2v^2)D_u^2 D_v^2(1 + r_0 - y_0)^2/[16y_0(r_0 - y_0)(1 - y_0)] > 0$.
- For Alice reciprocal and Bob bounded, use the four rows $(W, Y, vX, sX + Z)$ and the full core RC20. Its real residue RC21 is strict using $C, A > 0$.
- For Bob reciprocal and Alice a bounded unit, use $(uW - Y, Y, X, bZ, \sqrt{D}\,Z)$ and the entire D square as in RC22–RC23. CR7 makes the three-by-three angular determinant strict. Literal $D = 0$ causes its last row to be zero, which is allowed in a strict coefficient Gram.
- For both angles reciprocal, RC24–RC25 give the five rows $(W, X, Y, bZ, \sqrt{D}\,sZ)$ and a strictly positive real residue.

The only remaining angle chart is exactly $u, b = o(1)$, treated above. These are whole-parent identities with every omitted event square retained, and include all literal axes and infinities. The exact source and testing sign gauges, positive pure-field homogeneous normalization factors and real/imaginary state splitting restore the stated projective tensor-SOS theorem.

A.5 Exhaustion of the bounded source domain

Exact statement. Let K_c, K_t be labelled real closed extensions of $\mathbb{R}$. The original canonical coefficient vector is physical over K_c, has nonnegative coefficients, and has an actual finite noncommuting zero contact. In C1 the exact omissions are $\lambda = n = 0$, with $m, C, A, \alpha > 0$, $D, \beta \geq 0$, $D + \beta > 0$; in C2 they are $\lambda = \beta = 0$, with $m, C, A, D, \alpha > 0$, $n \geq 0$. Use the actual native sign chart of this contact, with its explicitly positive source gaps, as supplied by canonical entry. Assume that normalization by $d > 0$ makes all coefficients bounded. Equivalently the normalized total-scale coordinate $d/(d + \lambda + n + m + C + A + D + \alpha + \beta)$ is a positive unit. Then the entire original homogeneous slack is a sum of squares in $K_c \otimes_{\mathbb{R}} K_t$ for every independent projective testing pair and every real or complex testing state. Chapter 5 supplies these contact coordinates for every source in the two families.

Division by d is a positive pure-source normalization. Hence take $d = 1$ below. Every nonnegative bounded source quantity is either infinitesimal (including literal zero where permitted) or a positive unit. These two alternatives give the source partition below.

We use the following two cases established above:

	positive-unit coefficients	proof
S	d, C, A, D, α	Section A.2
CU	d, n, m, C, A, D (α infinitesimal)	Section A.3.

Every other nonnegative coefficient may have arbitrary residue in the stated canonical support. Both proofs certify the full state polynomial at all independent tests. Their noncommuting case is Theorem 6.2; their finite-edge and finite-ratio cases were proved at the start of this chapter.

Symmetries and the source partition

Both transformations below preserve the core and nonnegative event coefficients, actual physicality, finite nonzero contact, and pure total-scale normalization. They act separately in source and testing fields, so transport a full tensor certificate.

For C1 let

$$\sigma_1 = (\lambda\ n)(m\ C)(A\ \alpha)(D\ \beta), \qquad G_v(W,X,Y,Z) = (vW - X, -W - vX, Z - vY, Y + vZ).$$

The entire raw parent has the polynomial identity

$$u^2 F_{\sigma_1 c}(1/u, v; G_v\psi) = (1+v^2)F_c(u,v;\psi), \qquad G_v^{\mathsf{T}} G_v = (1+v^2)I. \tag{AU12}$$

Homogenization interchanges the two Alice angle coordinates; the Bob-homogeneous G has norm factor $v_0^2+v_1^2 > 0$. Thus literal endpoints are transported without dividing zero.

For C2 let

$$\sigma_2 = (\lambda\ \beta)(m\ D)(C\ \alpha), \qquad J(W,X,Y,Z) = (Z,X,Y,W).$$

The full identity is

$$u^2 v^2 F_{\sigma_2 c}(1/v, 1/u; J\psi) = F_c(u,v;\psi). \tag{AU13}$$

It is the physical party interchange with both first observables negated, followed by the local real X basis changes. The homogeneous identity includes both projective endpoints. In particular the source squared half-angles transform as $(U,V) \mapsto (1/V, 1/U)$.

The two actual forbidden weak faces are

$$\begin{aligned} h|_{\lambda=\beta=A=0} &= -(1+2n)(2D+2C+1)(2\alpha+1+2m), \\ h|_{\lambda=n=m=C=0} &= -(2D+1)(2\beta+1)(2A+2\alpha+1). \end{aligned} \tag{AU14}$$

In C2 with A weak, the only positive Laurent terms are $q_{01}/U + q_{02}V/U + q_{12}V$, which tend to zero along $U = t^2, V = 1/t,\ t \to \infty$. In C1 with both m, C weak the only positive term is q_{10}/V, which tends to zero as $V \to \infty$. The negative constants contradict physicality. Thus C2 has A unit, and C1 has at least one of m, C unit.

C2. If α is weak, WA1–WA6 force C, A, D to be units. There are precisely three disjoint alternatives:

n, m units	CU
n unit, m weak	WA1−WA27
n weak, m arbitrary	WN1−WN44.

(AU15)

If C is weak, apply σ_2 and the same table. Every unit requirement in the transformed table is a consequence of the transformed physical source, including its required $D' = m$ bound; it is not assumed for an excluded original case.

It remains that C, A, α are units. The actual dictionary then has

$$q_{01} = ma_1,\ q_{02} = mDa_2,\ q_{10} = a_3,\ q_{12} = Da_4,\ q_{20} = na_5,\ q_{21} = a_6,$$

where every a_i is a positive bounded unit. The contact equation bounds $U, V^{-1}, m/U, DV$. The two angular equations give exactly

$$\begin{aligned} DV^2(a_4 + a_2 m/U) &= a_3 + na_5 U, \\ U^2(a_6 + na_5/V) &= m(a_1 + a_2 DV). \end{aligned}$$

All parentheses are positive bounded units, so $DV^2 \asymp 1$, $U^2 \asymp m$. Consequently the exhaustive partition is

D unit, m arbitrary	S
D weak, m unit	σ_2(S)
D, m weak	RC1−RC25.

(AU16)

RC is the direct full-parent C2 reciprocal theorem; it permits arbitrary $n \geq 0$. No unit-n condition has entered this part of the partition.

C1. If α is weak, WN applies directly, with all its common source-entry implications proved in WA1–WA10. If A is weak, apply σ_1, making $\alpha' = A$ weak, and use WN on the transformed physical source. Otherwise A, α are units. By AU14, orient with σ_1, if necessary, so that C is a unit. The exact dictionary becomes

$$\begin{gathered} q_{01} = ma_1,\ q_{02} = mDa_2,\ q_{10} = a_3, \quad q_{12} = \beta m a_4 + Da_5, \quad q_{21} = a_6, \quad q_{22} = \beta a_7, \\ U^2(a_6 + \beta a_7 V) = m(a_1 + a_2 DV), \\ a_3 = V^2\{mDa_2/U + \beta m a_4 + Da_5 + \beta a_7 U\}. \end{gathered} \tag{AU17}$$

The contact equation again bounds $U, V^{-1}, m/U, DV$. If m is weak, U is infinitesimal. If m is a unit, U is a positive unit. Indeed, the contact equation also bounds βUV; if U were infinitesimal, then $U^2(a_6 + \beta a_7 V)$ would be infinitesimal, contradicting the positive unit ma_1 on the right of AU17. For unit m, the last braces are comparable with $D + \beta$. For weak m, bounded V forces m/U weak in the preceding equation, and then the last equation forces D unit; conversely unit D forces bounded V. Thus the following source partition is exhaustive:

D unit, m arbitrary	S
D weak, m unit, β unit	σ_1(S)
m unit, D, β weak	PP1−PP19
m, D weak, β arbitrary	CR1−CR39.

(AU18)

In the second line all of m, C, A, α are units before reflection, so the reflected strong theorem has every required unit. PP is the C1 projective-pitchfork theorem, with actual contact U unit and V unbounded. CR is the C1 reciprocal all-tests theorem, including unit β and literal allowed $D = 0$ or $\beta = 0$. The last line treats infinitesimal m, D. No boundary is promoted by a density argument.

AU15–AU18 cover every permitted bounded positive-d canonical source. Each leaf proves the entire original parent at every independent projective test, with the exact same source coefficients. The full-parent symmetries, positive pure-source d scaling, positive pure-testing homogeneous factors and complex-state split restore the assertion in the opening paragraph. This completes A5 at its canonical-entry interface.

The source estimates in each listed case use precisely its physical contact equations and the displayed coefficient bounds. The finite partition therefore proves the all-tests assertion for every bounded canonical source.

A.6 A three-event face and its stationary zeros

The three-event face has an explicit matrix representation. Its dual gives a separated sum of squares. We then use the stationary equations to show that this face contains no source with a finite noncommuting contact.

The selected full parent

Write $D_u = 1 + u^2$, $D_v = 1 + v^2$, and retain the coefficient order $(d, \lambda, n, m, C, A, D, \alpha, \beta)$. The original complete parent is

$$\begin{aligned} F = {} & 4\lambda D_u(W + vX)^2 + 4nD_uD_vX^2 + 4m(vW - X + uvY - uZ)^2 \\ & + 4CD_v(uW - Y)^2 + 4AD_uD_vY^2 + (4D + 2d)(uW + uvX - Y - vZ)^2 \\ & + (4\alpha + 2d)D_u(vY - Z)^2 + (4\beta + 2d)D_v(X + uZ)^2 - 2dD_uD_vZ^2. \end{aligned} \tag{CAP1}$$

All nine coefficients are nonnegative. Let F_* be the selected parent with $\lambda = n = m = \alpha = \beta = 0$, so that only the consecutive event triple (D, C, A) remains. For $d, C, A, D \geq 0$ define

$$\Delta = 8ADC - 4ADd - d^2\{d + 2(C + A + D)\}. \tag{CAP2}$$

Proposition A.3 (Exact physical wall and strict interior-angle positivity)**.** *If $d > 0$, the selected parent F_* is positive semidefinite at every projective measurement pair if and only if $\Delta \geq 0$. Whenever this holds, its state matrix is positive definite at every finite pair $u, v \neq 0$. In particular, the equality wall $\Delta = 0$ has no finite noncommuting contact.*

Proof. Let $P_*(u, v)$ be the symmetric state matrix. Direct expansion of the full polynomial (CAP1), without a state or angle truncation, gives

$$\det P_* = 16D_u^2D_v^2u^2v^2\{d\Delta + 4CDd(d + 2A)v^2 + 4CAd(d + 2D)u^2\}. \tag{CAP3}$$

Its restriction to $Z = 0$ is the sum of the nonnegative event squares and

$$2d\{(uW + uvX - Y)^2 + D_uv^2Y^2 + D_vX^2\}. \tag{CAP4}$$

For $uv \neq 0$ these three squared linear forms are independent. Hence the upper 3×3 state block is positive definite. A Schur complement taken in this one ordered field proves that $P_* \succeq 0$ if and only if its determinant is nonnegative, and that a positive determinant gives positive definiteness. This is a statement about ordinary field-valued source or testing matrices; it takes no inverse in a tensor product.

Suppose $\Delta < 0$. Put $r = 4CDd(d + 2A)$ and $s = 4CAd(d + 2D)$, both nonnegative. Taking

$$u^2 = v^2 = \frac{-d\Delta}{2(r + s + 1)} > 0$$

makes the expression in braces in (CAP3) negative. This is an explicit ordered-field witness to nonphysicality, so no limiting argument is needed. Conversely, $\Delta \geq 0$ makes all three terms in braces nonnegative. It also forces $C, A, D > 0$: if one vanished, (CAP2) would be strictly negative because $d > 0$. Thus $r, s > 0$ and the determinant is strictly positive whenever $uv \neq 0$.

At an actual commuting angle, including a projective infinity, one party's two reflections commute. Simultaneous diagonalization makes the CHSH operator on each local eigenspace either ± 2 times one remote reflection. The signed core $I - S/2$ is then positive semidefinite. Every event square is positive semidefinite too. This covers all literal endpoints directly. The argument holds over every real closed field, because it uses real diagonalization of symmetric matrices and explicit ordered-field identities. If $d = 0$, every parent in (CAP1) is its event-square sum separately. □

A small exact lift of the selected source cone

Set

$$p = 4C + 4D + 2d, \qquad r = 4D + 2d, \qquad B(d, C, A, D) = \begin{pmatrix} p + 4A & 4D & -p \\ 4D & 4D & -r \\ -p & -r & p \end{pmatrix}. \tag{CAP5}$$

For nonnegative coefficients the selected source is physical exactly when $B \succeq 0$. Here is a direct verification of this matrix description. When $d > 0$, use the positive last diagonal entry p. The Schur complement of that entry is

$$\begin{pmatrix} 4A & -2d \\ -2d & 4b/p \end{pmatrix}, \qquad b = 4CD - 2Dd - d^2, \qquad \det B = 8\Delta. \tag{CAP6}$$

Moreover

$$\Delta = 2Ab - d^2(d + 2C + 2D). \tag{CAP7}$$

If $\Delta \geq 0$, then $A, b > 0$, and (CAP6) is positive semidefinite. If $B \succeq 0$, its determinant is nonnegative, hence $\Delta \geq 0$. For $d = 0$ one has the exact quadratic identity

$$(y, z, x)B(0, C, A, D)(y, z, x)^T = 4C(y - x)^2 + 4Ay^2 + 4D(y + z - x)^2. \tag{CAP8}$$

Thus the entire selected physical cone, including all zero faces, is

$$\mathcal{C} = \{(d, C, A, D) \in \mathbb{R}^4_{\geq 0} : B(d, C, A, D) \succeq 0\}. \tag{CAP9}$$

It has a strict feasible point $(1, 2, 2, 2)$: the leading principal minors of B there are $26, 144, 280$.

Lemma A.4 (Separated certificate for the selected wall)**.** *For arbitrary labelled real closed fields $K_c, K_t \supset \mathbb{R}$, every source $c \in \mathcal{C}_{K_c}$, and every projective angle pair and real state in K_t, the complete selected parent is a finite sum of squares in $K_c \otimes_{\mathbb{R}} K_t$. The full parent (CAP1) has the same property if its selected coefficients satisfy (CAP9).*

Proof. We first establish the exact dual representation used in this lemma. For a real linear functional ℓ nonnegative on $\mathcal{C}$, there are $\mu \in \mathbb{R}^4_{\geq 0}$ and a real symmetric $Q \succeq 0$ such that, for all $c = (d, C, A, D)$,

$$\ell(c) = \mu \cdot c + \operatorname{tr}(B(c)Q). \tag{CAP10}$$

Indeed let $\mathcal{D}$ be the cone of functionals on the right. It is closed: if a sequence of these functionals converges, evaluation at $c_0 = (1, 2, 2, 2)$ bounds both $\sum \mu_i$ and $\operatorname{tr} Q$, because all coordinates of c_0 and the smallest eigenvalue of $B(c_0)$ are strictly positive. Positive semidefinite matrices with bounded trace are bounded, so there is a convergent subsequence of the representing pairs. The dual cone $\mathcal{D}^*$ is exactly (CAP9). Finite-dimensional separation of closed convex cones gives $\mathcal{D} = \mathcal{C}^*$ and proves (CAP10). This statement has a fixed finite first-order formulation using the principal-minor inequalities for a 3×3 matrix. Consequently it holds over every real closed field by transfer.

Fix an arbitrary testing tuple in K_t. Its homogeneous selected parent is a linear functional ℓ_t on the four source coefficients. The physical equivalence just proved says that $\ell_t \geq 0$ on $\mathcal{C}_{K_t}$. Apply (CAP10) in K_t to obtain $\mu_t \geq 0$, $Q_t \succeq 0$. Evaluating the resulting linear coefficient identity in separated fields gives

$$F_*(c,t) = \sum_{i=1}^{4} c_i \otimes \mu_{t,i} + \sum_{j,k=1}^{3} B(c)_{jk} \otimes (Q_t)_{jk}. \tag{CAP11}$$

The first terms are squares by separate square roots. Factor $B(c) = UU^T$ within K_c and $Q_t = VV^T$ within K_t. The last term in (CAP11) is exactly

$$\sum_{a,b} \left(\sum_j U_{ja} \otimes V_{jb} \right)^2 .$$

This proves the tensor certificate with no mixed denominator. The five omitted event squares in (CAP1) are separate positive-source factors times testing squares and can be restored exactly. Every literal projective endpoint is already included in the homogeneous testing tuple. Splitting a complex state into its real and imaginary parts gives the same assertion for complex states. □

Every actual finite contact lies strictly below all eight walls

The fixed event cycle is

$$(a_0, \ldots, a_7) = (\lambda, D, C, A, \alpha, m, \beta, n).$$

For indices modulo eight put

$$\Delta_i = 8a_i a_{i+1} a_{i+2} - 4d a_i a_{i+2} - d^2\{d + 2(a_i + a_{i+1} + a_{i+2})\}. \tag{CAP12}$$

The cyclic equivalence can be verified without an unspecified symmetry import. In the CHSH convention $S = A_0B_0 - A_0B_1 + A_1B_0 + A_1B_1$, the Bell relabeling

$$(A_0', A_1', B_0', B_1') = (B_1, -B_0, -A_0, -A_1) \tag{CAP13}$$

preserves S and maps each event in the displayed cycle to the next. For example the transformed $\lambda = (+,+ \mid 0,1)$ is the original $D = (-,+ \mid 1,1)$. The complete eight-event permutation and the CHSH identity are checked exactly. Relabeling preserves whether each local pair is noncommuting, so the strict assertion of the first proposition applies to every cyclic triple.

Theorem A.5 (Automatic contact exclusion)**.** *Let $d > 0$, let every event coefficient be nonnegative, and suppose the original parent (CAP1) has a zero on a nonzero source-field state at an actual finite noncommuting source pair. Then*

$$\Delta_i < 0 \qquad (i = 0, \ldots, 7). \tag{CAP14}$$

Proof. If some $\Delta_i \geq 0$, the selected parent for that triple is positive definite at the specified pair. The full parent is this selected parent plus five nonnegative event squares. It is therefore positive on every nonzero state at that pair, contradicting the assumed zero. This uses only ordinary ordered-field positivity at the source contact. It does not evaluate a source contact in an independent testing field. □

More generally, suppose a proposed cap keeps the same signed coefficient d and removes nonnegative amounts e_j from event coefficients. At an actual zero t_c one has the exact identity

$$F_{\mathrm{cap}}(t_c) = -4 \sum_j e_j \, \omega_j(t_c) R_j(t_c)^2, \tag{CAP15}$$

where ω_j is the positive angle factor multiplying the event row in (CAP1). Thus a globally physical cap can remove a strictly positive amount only from an event whose row vanishes at that contact. If one removed positive square has nonzero row, (CAP15) is already a strictly negative physicality witness. In the special three-event cap, (CAP14) rules out even the case in which all removed rows vanish. The same argument permits smaller retained event coefficients: add their nonnegative excesses back as well. It does not apply to a modification of d, whose difference is a signed CHSH term rather than a nonnegative event square.

Exact consequences for source normalization and routing

For $d > 0$ put

$$\delta_{\mathrm{tot}} = \frac{d}{d + \sum_{i=0}^{7} a_i}, \qquad b_i = a_i/d. \tag{CAP16}$$

In a real closed extension, "bounded" means bounded in absolute value by a real number, and "infinitesimal" means smaller in absolute value than every positive real number. Since all b_i are nonnegative and the sum is finite,

$$\begin{aligned} &\delta_{\mathrm{tot}} \text{ has positive real residue} &&\iff \text{every } b_i \text{ is bounded,} \\ &\delta_{\mathrm{tot}} \text{ is infinitesimal} &&\iff \text{at least one } b_i \text{ is unbounded.} \end{aligned} \tag{CAP17}$$

These alternatives are disjoint and exhaustive because $0 < \delta_{\mathrm{tot}} \le 1$. They are invariant under a positive source rescaling and under Bell relabeling. Normalizing $d = 1$ is legal in the coefficient field, and multiplying each tensor root by the pure-source factor $\sqrt{d}$ restores the original scale.

There cannot be three consecutive unbounded b_i at an actual finite contact. Indeed division of (CAP12) by its positive triple product gives

$$\frac{\Delta_i}{a_i a_{i+1} a_{i+2}} = 8 - \frac{4d}{a_{i+1}} - \frac{d^3}{a_i a_{i+1} a_{i+2}} - 2d^2 \left\{ \frac{1}{a_i a_{i+1}} + \frac{1}{a_i a_{i+2}} + \frac{1}{a_{i+1} a_{i+2}} \right\}. \tag{CAP18}$$

If the three ratios a_j/d were unbounded, its residue would be 8, contradicting (CAP14). Equivalently, every consecutive triple contains an event bounded relative to d.

If a later chart uses $C = 2$, the rescaling factor is $2/C$, and its new signed weight is $2d/C$; (CAP16) remains unchanged. The implication d/C infinitesimal $\Rightarrow \delta_{\mathrm{tot}}$ infinitesimal is valid, but the reverse implication does not follow from normalization alone. A different event may be the unbounded one. An entry map that claims more must prove its extra source inequalities explicitly.

For a more precise cap screen with $d > 0$, (CAP7) yields

$$b > 0 \quad \Longrightarrow \quad \Delta < 0 \iff A < \frac{d^2(d + 2C + 2D)}{2b}. \tag{CAP19}$$

If $b \le 0$, the strict inequality $\Delta < 0$ is automatic. Every inverse in (CAP19) is within the source field, and the polynomial condition (CAP14) remains the primary contact condition.

A.7 Weak-alpha source entry and the unit-n case

The source domain. All coefficients are bounded in the source field, $d = 1$, and the original canonical source is physical with an actual finite noncommuting zero contact. The C2 convention is $\lambda = \beta = 0$, and the C1 convention is $\lambda = n = 0$. The first subsection applies to either family with α infinitesimal. The all-tests theorem proved below has the additional C2 hypotheses that n is a positive unit and m is infinitesimal. It includes all independent testing states, signs and literal projective endpoints. The weak-n and C1 cases are separate.

Source entry, proved before any shape quotient is used

Physicality forces C, A, D to be positive units. Here are exact residue-face tests proving this assertion. In C2 the negative central coefficient at $A = 0$, and its vanishing positive Laurent terms along $U = t^2, V = 1/t$, are the displayed forbidden-face proof in the actual positive-d source partition. If also $C = 0$ in residue, the source has only the four alternating events n, m, A, D. At $u = v = 1$ the full parent at state $(1, -\varepsilon, \varepsilon, 1)$ is exactly

$$16\varepsilon\{(1 + n + m + A + D)\varepsilon - 1\} < 0 \tag{WA1}$$

for a sufficiently small positive real ε. Thus C cannot be weak. The remaining central coefficient identities are

$$\begin{aligned} h|_{\alpha=D=0} &= (2A + 2C + 1)\{8\beta mn - 2\beta - 4mn - 2m - 2n - 1\}, \\ h|_{\alpha=C=n=0} &= -(2A + 1)(2D + 1)(2\beta + 2m + 1), \\ h|_{\alpha=A=n=0} &= -(2C + 2D + 1)(2\beta + 2m + 1). \end{aligned} \tag{WA2}$$

In the first line $n\beta = 0$, so it is strictly negative. For C2 with $\alpha = D = 0$, the other Laurent terms are $q_{20}U/V + q_{21}U$, killed by $U = t^2, V = t$. For C1 with $\alpha = D = 0$, they are $q_{12}V + q_{21}U + q_{22}UV$, killed by $U = V = t$. For C1 with $\alpha = C = 0$, they are $q_{02}V/U + q_{12}V$, killed by $U = 1/t, V = t$. For C1 with $\alpha = A = 0$, they are $q_{02}V/U + q_{12}V + q_{22}UV$, killed by $U = t, V = t^2$. Here t tends to zero through positive reals. These contradictions prove all the asserted unit bounds without a source-contact assumption.

Write $U = u_c^2, V = v_c^2 > 0$. The actual determinant dictionary now has

$$q_{01} = \alpha m a_1, \quad q_{02} = m a_2, \quad q_{10} = \alpha a_3, \quad q_{12} = a_4, \quad q_{20} = n a_5, \quad q_{21} = a_6, \quad q_{22} = \beta a_7, \tag{WA3}$$

where the a_i are positive bounded units. The two angular equations and their sum are

$$\begin{aligned} U^2(na_5 + a_6V + \beta a_7V^2) &= mV(\alpha a_1 + a_2V),\\ V^2(ma_2 + a_4U + \beta a_7U^2) &= \alpha a_3U + na_5U^2,\\ a_6U + a_4V + 2\beta a_7UV &= \alpha(ma_1/U + a_3/V). \end{aligned} \tag{WA4}$$

Every t_i in the dictionary is a positive unit because it includes $d = 1$. At contact all noncentral Laurent terms are nonnegative, and their sum is bounded. In particular $U, V, mV/U, nU/V$ are bounded. Multiplying the last equation by V shows $\alpha/V = O(U + V)$. The first two equations then exclude either U or V having nonzero residue: if U is a unit the first forces V a unit, and conversely the second forces U a unit; the last equation then contradicts weak α. Both are infinitesimal. Consequently

$$\alpha/V = o(1), \qquad m \asymp (U/V)^2(n + V), \qquad \alpha \asymp V(U + V). \tag{WA5}$$

For the last comparison, the denominator $a_3 + ma_1V/U$ is a positive bounded unit. Actual finite contact forces $m, \alpha > 0$ in WA4. If n is a unit and m weak,

$$U/V \asymp \sqrt{m} = o(1). \tag{WA6}$$

If n is weak, then, whether U/V is bounded or unbounded, WA5 gives $mV/U, nU/V = o(1)$. These same two conclusions hold under WA6. They are the common source bounds used in the shape argument.

Orient the actual signs positively, use $W_c = 1$, and put

$$p = v_c, \quad q = Z_c, \quad P = p^2, \quad Q = q^2, \qquad u_c = rpq, \quad X_c = xpq^2, \quad Y_c = ypq. \tag{WA7}$$

The exact core sign identity SE9 ensures $W_cZ_c > 0$, so this normalization is legitimate. To prove boundedness before using these quotients, eliminate the X, Y block inside the source field. Set

$$H_X = n + m + \beta + \tfrac{1}{2}, \quad K = C + D + \tfrac{1}{2}, \quad H = K + A.$$

The block has diagonal positive residue H_X, H and an infinitesimal off-diagonal entry. The source Schur entries satisfy

$$S_{WW}/U \sim a = AK/H > 0, \qquad -S_{WZ}/(u_cv_c) \sim b = \tfrac{1}{2} + AD/H > 0. \tag{WA8}$$

Indeed the only additional leading contributions are multiples of $mV/U, nU/V, mn$, all infinitesimal on the domains just stated. The kernel equation gives $r \sim b/a$, a positive unit. The scaled X, Y equations are then a bounded system with positive triangular residue; its apparent m/Q term is bounded since $m/Q = r^2mV/U$. Thus x, y are bounded before taking residues. Those equations and the two remaining kernel equations give

$$Ay \sim \tfrac{1}{2}, \quad K(r - y) \sim D + \tfrac{1}{2}, \quad D(1 + y - r) \sim r/2, \qquad x \sim -\frac{m + \beta + 1/2}{H_X}\, r. \tag{WA9}$$

For the last kernel division the possible extra α/V, nU/V and mV/U terms vanish by WA5. In particular $y, r - y$ and $1 + y - r$ are positive units, and the critical residue dictionary is

$$A_0 = \frac{1}{2y_0}, \qquad C_0 = \frac{1 + y_0}{2(r_0 - y_0)}, \qquad D_0 = \frac{r_0}{2(1 + y_0 - r_0)}. \tag{WA10}$$

In the unit-n, weak-m domain, $\beta = 0$, and hence

$$P, Q = o(1), \qquad m \asymp Q^2, \qquad \alpha \asymp P^2, \qquad x_0 = -r_0/(1 + 2n_0). \tag{WA11}$$

No independent testing coordinate occurred in this entry proof.

The exact scalar and every first normal

For reference the full finite quarter-parent is

$$\begin{aligned} F = {} & nD_uD_vX^2 + m(vW - X + uvY - uZ)^2 + CD_v(uW - Y)^2 + AD_uD_vY^2\\ & + (D + \tfrac{1}{2})(uW + uvX - Y - vZ)^2 + (\alpha + \tfrac{1}{2})D_u(vY - Z)^2\\ & + (\beta + \tfrac{1}{2})D_v(X + uZ)^2 - \tfrac{1}{2}D_uD_vZ^2. \end{aligned} \tag{WA12}$$

Here $D_u = 1 + u^2, D_v = 1 + v^2$. We retain β symbolically to make the exact common graph explicit, but the current case has $\beta = 0$. In the testing chart $W = 1, Z = z$, use

$$u_g = rvz, \qquad X_g = xvz^2, \qquad Y_g = yvz, \qquad V_t = v^2, \quad T = z^2. \tag{WA13}$$

In the rest of this subsection write $V = V_t$ only. The full graph restriction is

$$\begin{aligned}
\Phi &= mV + \alpha T + hVT + \mathcal{C}V^2T + EVT^2 + \mathcal{F}V^2T^2 + GV^3T^2 + JV^2T^3 + \mathcal{K}V^3T^3,\\
\mathcal{C} &= Ay^2 + C(r-y)^2 + (\alpha + \tfrac{1}{2})y^2 + 2mry,\\
E &= \alpha r^2 + (m + \beta + \tfrac{1}{2})(r+x)^2 + nx^2,\\
G &= r^2y^2(A + \alpha + m + \tfrac{1}{2}), \qquad J = nr^2x^2, \qquad \mathcal{K} = (D + n + \tfrac{1}{2})r^2x^2,\\
\mathcal{F} &= Ar^2y^2 + (2D+1)rx(r-y-1) - (2\alpha+1)r^2y\\
&\quad - 2mry(r+x) + \beta(r+x)^2 + nx^2 + rx + x^2/2.
\end{aligned} \tag{WA14}$$

The coefficient h is defined by this entire expansion; it is bounded. Every displayed coefficient is bounded. In the current unit-n domain $\mathcal{C}, E$ are positive units, since

$$\mathcal{C}_0 = r_0(1+y_0)/2, \qquad E_0 = n_0r_0^2/(1+2n_0). \tag{WA15}$$

The coefficient $\mathcal{F}$ need not be positive.

The actual contact gives $\Phi(P,Q) = \Phi_V(P,Q) = \Phi_T(P,Q) = 0$. Solving these three linear equations in m, α, h is exact:

$$\begin{aligned}
m &= EQ^2 + \mathcal{F}PQ^2 + GP^2Q^2 + 2JPQ^3 + 2\mathcal{K}P^2Q^3,\\
\alpha &= \mathcal{C}P^2 + \mathcal{F}P^2Q + 2GP^3Q + JP^2Q^2 + 2\mathcal{K}P^3Q^2,\\
h &= -2\mathcal{C}P - 2EQ - 3\mathcal{F}PQ - 4GP^2Q - 4JPQ^2 - 5\mathcal{K}P^2Q^2.
\end{aligned} \tag{WA16}$$

Put

$$R = z(V-P), \quad S = v(T-Q), \quad \Psi = QR^2 + PS^2 + vzRS.$$

With the exact balanced monomials

$$\begin{aligned}
B_{32} &= 2PQR^2 + (vzR + PS)^2,\\
B_{23} &= 2PQS^2 + (QR + vzS)^2,\\
B_{33} &= (2PQ^2 + VT^2)R^2 + (2P^2Q + PVT)S^2 + (3PQ + PT)vzRS,
\end{aligned} \tag{WA17}$$

the whole scalar, not a leading truncation, is

$$\Phi = \mathcal{C}R^2 + ES^2 + \mathcal{F}\Psi + GB_{32} + JB_{23} + \mathcal{K}B_{33}. \tag{WA18}$$

For v, z infinitesimal, all entries after the first two positive unit pivots are infinitesimal, by $P, Q = o(1)$. This is a strict two-row Gram even when $\mathcal{F}$ is signed.

Let L_u, L_X, L_Y be half the full first derivatives of WA12 at WA13. The complete polynomials have the forms

$$\begin{aligned}
L_u &= vzA_u(V,T), \qquad L_Y = vzA_y(V,T),\\
A_u &= a_{00} + a_{10}V + a_{01}T + a_{11}VT + a_{12}VT^2 + a_{21}V^2T + a_{22}V^2T^2,\\
A_y &= e_{00} + e_{10}V + e_{11}VT + e_{21}V^2T,\\
L_X &= v[-m + TB_X(V,T)], \qquad B_X = c_0 + c_1V + c_2VT + c_3V^2T.
\end{aligned} \tag{WA19}$$

All coefficients are bounded, and the expansion fixes every one unambiguously. In particular

$$c_2 = nr^2x, \qquad c_3 = (D + n + \tfrac{1}{2})r^2x.$$

Their actual source jets are $A_u(P,Q) = A_y(P,Q) = 0$ and $B_X(P,Q) = m/Q$. Finite monic difference division gives

$$\begin{aligned}
L_i &= vA_{i,V}(V,T)R + zB_i(T)S, \qquad i = u, Y,\\
L_X &= vzB_{X,V}(V,T)R + \{T(Pc_2 + P^2c_3) + m/Q\}S.
\end{aligned} \tag{WA20}$$

Here $A_{i,V} = [A_i(V,T) - A_i(P,T)]/(V-P)$, and $B_{X,V}$ is defined identically. The other divided polynomials are

$$\begin{aligned}
B_u(T) &= a_{01} + Pa_{11} + Pa_{12}(T+Q) + P^2a_{21} + P^2a_{22}(T+Q),\\
B_Y(T) &= Pe_{11} + P^2e_{21}.
\end{aligned}$$

These are literal polynomial identities with bounded coefficients, including $v = 0$ or $z = 0$. Since $m/Q = O(Q) = o(1)$, every full first normal is an infinitesimal linear combination of R, S.

Set $N_u = u - rvz, N_X = X - xvz^2, N_Y = Y - yvz$. In the chart under consideration these are infinitesimal. The full higher-normal remainder is a polynomial in the three normals with no terms of normal degree below two; all terms of degree at least three give infinitesimal Gram entries. Its quadratic residue is

$$K_0(N_u - N_Y)^2 + A_0N_Y^2 + (n_0 + \tfrac{1}{2})N_X^2 \succ 0. \tag{WA21}$$

WA18 and WA20 therefore give a strict bounded Gram for the entire parent on the three hard and two soft rows. The strict-Gram lemma proves its tensor-SOS assertion, with no mixed inverse.

The complementary local states

Normalize a maximal testing state coordinate. At $u, v = o(1)$ the state residue has positive pivots on X, Y; if one is a unit, the full parent has positive residue immediately. Otherwise one of W, Z is a unit. The preceding proof covers $W = 1, Z = o(1)$. Every remaining state can be normalized to $Z = 1, W = \omega$ bounded. Retain the entire positive m, α squares. The remainder lies exactly in the square of (u, v, X, Y). Its quadratic real residue is

$$\begin{aligned}&(n_0 + \tfrac{1}{2})\left(X + \frac{u}{1+2n_0}\right)^2 + \frac{n_0}{1+2n_0}u^2 \\ &+ H_0\left(Y - \frac{K_0\omega u - D_0 v}{H_0}\right)^2 + a_0(\omega u - r_0 v)^2, \qquad a_0 = A_0K_0/H_0 > 0.\end{aligned} \tag{WA22}$$

The critical identities WA10 prove the equality. This quadratic form is positive definite for every real ω, including zero. All higher normal terms are infinitesimal corrections on these four rows, while $\omega - \bar{\omega}$ is retained in coefficients. Thus every state at the double-infinitesimal angle corner is covered.

All other testing angles, including the literal endpoints

Again retain the entire m, α squares and call the remainder F_0. Its source residue is the critical C, A, D, n source. For bounded u, v, F_0 is exactly a bounded quadratic form in the four rows (uW, X, Y, Z). Its real $Z = 0$ block is positive: the nX^2 and AY^2 squares and $C(uW - Y)^2$ suffice. The determinant is

$$\det \mathrm{Mat}_{(uW,X,Y,Z)}(\bar{F}_0) = D_u^2 D_v^2\{q_{20}u^2 + q_{12}v^4 + q_{21}u^2v^2\}/16 > 0 \tag{WA23}$$

whenever at least one of the two real angle residues is nonzero. Here $q_{20}, q_{12}, q_{21} > 0$ are the actual residue determinant coefficients, and $h = 0$ follows from WA10. This proves strictness of the full four-row Gram. Literal $u = 0$ is allowed.

For Bob reciprocal $b = 1/v$ infinitesimal and Alice bounded, use the same rows (uW, X, Y, Z) in $b^2F_0(u, 1/b)$. At $b = 0$ its whole real expression is

$$C_0(uW - Y)^2 + (A_0 + \tfrac{1}{2})D_uY^2 + D_0(uX - Z)^2 + (n_0 + \tfrac{1}{2})D_uX^2 \succ 0. \tag{WA24}$$

It is strict at every real u, including zero. For Alice reciprocal $s = 1/u$ infinitesimal and Bob bounded, the original state rows suffice. The expression at $s = 0$ is

$$n_0D_vX^2 + C_0D_vW^2 + A_0D_vY^2 + (D_0 + \tfrac{1}{2})(W + vX)^2 + \tfrac{1}{2}(vY - Z)^2 \succ 0. \tag{WA25}$$

This includes $v = 0$, using the positive n_0 pivot.

For both reciprocal angles $s, b = o(1)$, use the five rows (W, X, Y, sZ, bZ). The whole reciprocal polynomial belongs to their square ideal, and its real coefficient Gram is

$$C_0W^2 + A_0Y^2 + D_0(X - sZ)^2 + (n_0 + \tfrac{1}{2})X^2 + \tfrac{1}{2}(Y - bZ)^2 \succ 0. \tag{WA26}$$

Every remaining coefficient is infinitesimal. These exact finite square-ideal allocations include $s = 0$, $b = 0$ and arbitrary states; no projective endpoint is obtained by a limiting argument.

The four angle charts and the two local state charts are exhaustive. Testing sign gauges, positive pure-field homogeneous normalization factors and real/imaginary state splitting restore the original parent:

$$F_c(u, v; \psi) \ \in \ \Sigma(K_c \otimes_{\mathbb{R}} K_t)^2 \quad \text{for every independent projective test and state.} \tag{WA27}$$

This closes exactly the C2 unit-n, weak-m, α case. The common source-entry subsection does not by itself close the weak-n or C1 weak-α branches.

A.8 Weak-alpha weak-n and C1: the complete parent

The source domain. The source is an actual physical canonical finite noncommuting contact, all coefficients are bounded in the source field, and $d = 1$. Either it is C2 with $\lambda = \beta = 0$ and n, α infinitesimal, or it is C1 with $\lambda = n = 0$, α infinitesimal and arbitrary bounded $\beta \geq 0$. All strict source gaps retain their original meaning. We prove the tensor-SOS statement for the entire parent at every independent projective testing angle and every state. The common source-entry proof WA1–WA10 is used on precisely its weak-n domain. In particular $C, A, D, r, y, r - y, 1 + y - r$ are positive units, x is bounded, and $x_0 = -r_0$. Boundedness, infinitesimality and residue refer to the separated bounded tensor ring and its real residue map. The strict-Gram lemma is applied only after a finite explicit row allocation; no mixed tensor is inverted.

Use the full quarter-parent WA12 and its exact source coordinates

$$p = v_c, \quad q = Z_c > 0, \quad P = p^2, \quad Q = q^2, \qquad W_c = 1, \quad u_c = rpq, \quad X_c = xpq^2, \quad Y_c = ypq. \tag{WN1}$$

Thus P, PQ are positive infinitesimals. Put

$$H_0 = m + \beta + \tfrac{1}{2}, \quad H_X = H_0 + n, \quad K = C + D + \tfrac{1}{2}, \quad H_Y = K + A. \tag{WN2}$$

These are positive units. The exact scalar coefficients $\mathcal{C}, E, \mathcal{F}, G, J, \mathcal{K}$, the contact balances, and the complete first normal polynomials are those displayed in WA14, WA16 and WA19; those identities retain β and n. Their relevant residues are

$$\mathcal{C}_0 = \frac{r_0(1 + y_0 + 4m_0y_0)}{2} > 0, \qquad \mathcal{F}_0 = \frac{r_0^2(1 + y_0)}{2} > 0, \qquad G, J, \mathcal{K} \geq 0. \tag{WN3}$$

Both $\mathcal{C}, \mathcal{F}$ are consequently positive units. The exact source X-jet gives, with the WA19 coefficients,

$$\begin{gathered} c_0 = H_0(r + x) + nx, \quad c_2 = nr^2x, \quad c_3 = (D + n + \tfrac{1}{2})r^2x, \\ c_0 = m/Q - Pc_1 - PQc_2 - P^2Qc_3, \\ E = \alpha r^2 + c_0^2/H_X + nH_0r^2/H_X. \end{gathered} \tag{WN4}$$

The last equality is an exact completion inside the source field.

Two finite positive circuit identities

For $V = v^2, T = z^2$ put

$$R = z(V - P), \quad S = v(T - Q), \quad \Psi = V^2T^2 + PQ^2V + P^2QT - 3PQVT.$$

Then $\Psi = QR^2 + PS^2 + vzRS$. It is an explicit six-square cubic circuit: if $L_1^3 = V^2T^2, L_2^3 = PQ^2V, L_3^3 = P^2QT$, then $L_1L_2L_3 = PQVT$, and

$$\Psi = \sum_{i=1}^{3} L_i\big\{(L_1 - L_2)^2 - (L_1 - L_2)(L_1 - L_3) + (L_1 - L_3)^2\big\}. \tag{WN5}$$

The real 2-by-2 Gram in each brace is strictly positive. All radicals in this formula are elementary separated monomials: source radicals and testing radicals are taken in their own real closed fields and then tensored.

The balanced degree-$(3, 3)$ term has the polynomial identity

$$a^{10} + 2b^{10} + 2c^{10} - 5a^2b^4c^4 = 2(b^5 - c^5)^2 + (a^2 - bc)^2(a^6 + 2a^4bc + 3a^2b^2c^2 + 4b^3c^3). \tag{WN6}$$

For nonnegative elementary monomials a, b, c, the bracket is the sum of squares of the four rows

$$a^3, \qquad \sqrt{2}a^2\sqrt{bc}, \qquad \sqrt{3}abc, \qquad 2(bc)^{3/2}.$$

This is a five-square separated certificate, including vanishing variables. Applied with

$$a^{10} = V^3T^3, \quad b^{10} = P^2Q^3V, \quad c^{10} = P^3Q^2T,$$

it proves positivity of the entire B_{33} in WA17.

A further exact identity supplies any fixed positive real pivot on the row PS:

$$2P\Psi + VTR^2 - (PS)^2 = (PS + vzR)^2 + 2PQR^2. \tag{WN7}$$

Indeed, for every fixed $b > 0$,

$$\begin{aligned} \mathcal{C}R^2 + \mathcal{F}\Psi = (\mathcal{C} - bVT + 2bPQ)R^2 + (\mathcal{F} - 2bP)\Psi \\ + b(PS)^2 + b(PS + vzR)^2. \end{aligned} \tag{WN8}$$

On the charts where this is used, VT, P, PQ are infinitesimal, so the original two unit residues are unchanged.

Bounded source ratio Q

From WA5, WN4 and bounded Q,

$$m \asymp Q^2(n+P), \quad \alpha \asymp P^2, \quad r+x = O(n+P), \quad E \asymp n+P^2. \tag{WN9}$$

In particular m, n, E, α are infinitesimal. None of these estimates contains an independent testing coordinate.

First take $W = 1, Z = z$ bounded and u, v infinitesimal. Use the entire graph $u = rvz, X = xvz^2, Y = yvz$. The exact scalar is

$$\Phi = \mathcal{C}R^2 + ES^2 + \mathcal{F}\Psi + GB_{32} + JB_{23} + \mathcal{K}B_{33}, \tag{WN10}$$

where the last three terms retain their positive full certificates. The strict soft rows are R, the six cubic rows in WN5, $\sqrt{E}\,S$, and PS; WN8 supplies any needed real pivot on the last row.

The exact monic normal divisions WA20 apply unchanged. Their R-coefficients contain v or vz, hence are infinitesimal. Their S-coefficients are $O(E+P)$: in particular

$$a_{01} = \alpha r + H_0(r+x) = O(E+P), \qquad a_{12} = nrx^2, \qquad m/Q = O(E+P). \tag{WN11}$$

Every ES contribution has infinitesimal coefficient $\sqrt{E}$ on $\sqrt{E}S$. Coefficients on PS may be bounded units. There are only finitely many of them, so choose b in WN8 larger than their finite real Schur cost. This uses a real constant, not an inverse of a mixed tensor.

The three hard normals $N_u = u - rvz, N_X = X - xvz^2, N_Y = Y - yvz$ are infinitesimal. Their complete quadratic residue is

$$K_0(N_u - N_Y)^2 + A_0 N_Y^2 + (\beta_0 + \tfrac{1}{2})(N_X + \bar{z}N_u)^2 \succ 0. \tag{WN12}$$

All higher-normal terms have degree at least three and bounded coefficients, hence allocate as infinitesimal Gram entries on these hard rows. WN10–WN12 prove the full parent in this chart.

For the complementary local states take $Z = 1, W = w$ infinitesimal, set $a = u/r$, and use

$$v = aw, \quad X = xa, \quad Y = yaw, \qquad U = a^2, \quad T = w^2. \tag{WN13}$$

The entire scalar is

$$\Phi' = \alpha + mUT^2 + EU + hUT + \mathcal{C}U^2T^2 + \mathcal{F}U^2T + GU^3T^2 + JU^2 + \mathcal{K}U^3T. \tag{WN14}$$

Here $a = (1/r) \otimes u$ is elementary; its roots remain separated. The strict rows are $P, a^2w, \sqrt{E}\,a$. The term hUT is harmless: the contact balances give $h = O(P+E)$, and writing $h = Ph_P + Eh_E$ with bounded source coefficients gives the infinitesimal Gram terms $h_P w\,P(a^2w) + h_E w^2(\sqrt{E}a)^2$. All other terms of WN14 are positive.

For completeness, half the entire v, Y, X derivatives have the polynomial forms

$$\begin{aligned} L'_v &= aw(d_0 + mT + d_1U + d_2UT + d_3U^2 + d_4U^2T), \\ L'_Y &= aw(e_0 + e_1U + e_2UT + e_3U^2T), \\ L'_X &= a(c_0 - mT + c_2U + c_1UT + c_3U^2T). \end{aligned} \tag{WN15}$$

Every coefficient is bounded. Source jets at $U = PQ, T = 1/Q$ give

$$\begin{aligned} d_0 &= -m/Q - d_1PQ - d_2P - d_3P^2Q^2 - d_4P^2Q = O(n+P), \\ e_0 &= -e_1PQ - e_2P - e_3P^2Q = O(P). \end{aligned}$$

Also $c_0, m = O(n+P)$ and $c_2 = O(n)$. Splitting $n + P = O(E+P)$ in these displayed complete polynomials allocates all first normals with infinitesimal coefficients to $P, a^2w, \sqrt{E}a$. For example awP, awE, and a^3w have coefficients aw, $w\sqrt{E}$, and a on those three rows; the other monomials have still more small testing factors. The complete hard residue is

$$(\beta_0 + \tfrac{1}{2})N_X^2 + D_0N_v^2 + 2D_0N_vN_Y + H_{Y,0}N_Y^2 \succ 0, \tag{WN16}$$

where $N_v = v - aw, N_Y = Y - yaw, N_X = X - xa$. The higher-normal argument is again exact polynomial allocation. This proves all local states when Q is bounded.

Unbounded source ratio: the W-chart

Suppose Q is unbounded. The source estimates and positive contact balances give

$$PQ = o(1), \quad m \asymp Q^2(n+P), \quad nQ^2, PQ^2 = O(1), \quad \alpha \asymp P^2Q. \tag{WN17}$$

WN4 now gives

$$c_0 \sim m/Q, \quad r+x \sim m/(H_XQ), \quad E \asymp n + (m/Q)^2, \quad h/m = o(1). \tag{WN18}$$

To justify the first comparison before its use, divide the three errors Pc_1, PQc_2, P^2Qc_3 by m/Q. Their absolute values are $O(1/Q), O(P), O(P)$, respectively; $n/(m/Q) = O(1/Q)$. The positive term $\alpha/(m/Q)^2 = O(1/Q)$ follows from $m \ge cPQ^2$. The last comparison follows term by term from WA16 using these same inequalities.

Take $W = 1, Z = z$ bounded, u, v infinitesimal. If m is a unit, retain the entire n, α squares. The remainder lies in the square of (u, v, X, Y). Its real quadratic form is

$$\begin{aligned} &H_0\{X + zu - (m/H_0)v\}^2 + \frac{m(\beta + 1/2)}{H_0}v^2 \\ &\quad + H_Y\{Y - (Ku - Dzv)/H_Y\}^2 + a_*(u - rzv)^2, \qquad a_* = AK/H_Y > 0, \end{aligned} \tag{WN19}$$

with coefficients interpreted in residue. WA10 verifies this identity. It is strict for every bounded z, so the complete higher terms give a strict Gram.

If m is weak, use the graph WA13. Its entire scalar has strict rows $\sqrt{m}\,v, v^2z, \sqrt{\alpha}\,z$, since $mV + hVT = mV[1 + (h/m)T]$, while $\mathcal{C}, \mathcal{F}, E, G, J, \mathcal{K}$ are nonnegative. The complete source jet equations yield

$$a_{00} = O(m), \quad e_{00} = O(PQ), \quad a_{01}, c_0 = O(m/Q). \tag{WN20}$$

Thus $a_{00}/\sqrt{m}$, $e_{00}/\sqrt{m}$, $a_{01}/\sqrt{m}$, $c_0/\sqrt{m}$ are infinitesimal. For e_{00}, use $PQ/\sqrt{m} = O(\sqrt{P})$. Every other monomial of the complete normals WA19 has a factor v^3z or v^3z^2, so it has an infinitesimal coefficient on v^2z. The hard residue is WN12; all higher terms are allocated exactly as there. This proves the whole W-chart.

Unbounded source ratio: the complementary scalar

Use WN13 with a, w infinitesimal. Introduce only source quantities

$$\begin{gathered} \epsilon = Q^{-1} = o(1), \quad \Lambda = PQ = o(1), \quad \tau = \Lambda/\epsilon = PQ^2 = O(1), \\ \nu = n/\epsilon^2 = O(1), \qquad N = E/\epsilon^2 = O(1). \end{gathered} \tag{WN21}$$

The quantities $\epsilon, \Lambda, \tau, N$ are positive; $\nu \ge 0$. Let

$$A_1 = a(T - \epsilon), \quad K_1 = UT - \Lambda\epsilon, \quad \mathcal{M} = U^2T + \tau UT^2 + \Lambda^2\epsilon - 3\Lambda UT. \tag{WN22}$$

The whole scalar WN14, using exactly the three contact balances, is

$$\begin{aligned} \Phi' &= \mathcal{C}K_1^2 + NA_1^2 + \mathcal{F}\mathcal{M} \\ &\quad + G\{2\Lambda K_1^2 + UT^2(U - \Lambda)^2\} \\ &\quad + J\{(U - \Lambda)^2 + 2\Lambda A_1^2/\epsilon^2\} + \mathcal{K}\mathcal{B}_{33}, \\ \mathcal{B}_{33} &= U^3T + 2(\Lambda^2/\epsilon)UT^2 + 2\Lambda^3\epsilon - 5\Lambda^2UT. \end{aligned} \tag{WN23}$$

The apparent $1/\epsilon^2$ in the positive J square is a source-field quantity, and $J/\epsilon^2 = \nu r^2x^2$ is bounded. All positive terms are retained. WN6 applies to $\mathcal{B}_{33}$ with $a^{10} = U^3T, b^{10} = (\Lambda^2/\epsilon)UT^2, c^{10} = \Lambda^3\epsilon$. Here $\Lambda^2/\epsilon = \tau\Lambda$ is bounded.

Write $a = t^3, w = s^3, \epsilon = h^6, \tau = c^6$, taking positive elementary roots. Then t, s, h are infinitesimal and c is bounded; it is allowed to be a unit. Put

$$\begin{gathered} L_1 = t^4s^2, \quad L_2 = c^2t^2s^4, \quad L_3 = h^6c^4, \\ f_1 = t^2s, \quad f_2 = cts^2, \quad f_3 = h^3c^2, \\ R_{2i-1} = f_i(L_1 - L_2), \qquad R_{2i} = f_i(L_1 - L_3) \quad (i = 1, 2, 3). \end{gathered} \tag{WN24}$$

Then $\mathcal{M} = \sum_{i=1}^{3}(R_{2i-1}^2 - R_{2i-1}R_{2i} + R_{2i}^2)$, a strict six-row constant Gram. The following are exact identities:

$$\begin{aligned} aw(U - \Lambda) &= t^3R_1 + c^2ts^2R_2, \\ a^2w(T - \epsilon) &= -h^6R_1 + s^6R_2 - h^3s^3R_5, \\ \tau aw(T - \epsilon) &= c^2ts^2(R_2 - R_1) - c^3s^3R_3, \\ \tau\sqrt{\epsilon}\,A_1 &= t^3R_6 - t(t^2 + c^2s^2)R_5. \end{aligned} \tag{WN25}$$

Every coefficient on the right is infinitesimal, even when c is a unit. These are finite linear conductors, not limiting approximations. The further quadratic conductor is

$$\boxed{\Lambda(U - \Lambda)K_1 = \Lambda\mathcal{M} - (\tau\sqrt{\epsilon}\,A_1)^2.} \tag{WN26}$$

Its right side is an infinitesimal Gram on the six cubic rows. It remains so after multiplication by any bounded signed coefficient.

If τ is a unit, then K_1 itself has an infinitesimal linear representation on those rows:

$$K_1 = \tau^{-1/3}\{\sqrt{L_1}(R_2 - R_1) + \sqrt{L_3}R_6\}. \tag{WN27}$$

This follows by expansion as $\tau^{-1/3}(L_1L_2 - L_3^2) = UT - \Lambda\epsilon$. Consequently any fixed real pivot on K_1 can be supplied at infinitesimal cost to the strict $\mathcal{M}$ Gram in this case.

Every complementary first normal

Let D_U, E_U denote the monic U-divided differences of the bracket polynomials in the first two lines of WN15. The source jets at $U=\Lambda, T=\epsilon$ give the exact identities

$$\begin{aligned} L'_v &= aw\{D_U(U,T)(U-\Lambda)+(m+\Lambda d_2+\Lambda^2 d_4)(T-\epsilon)\},\\ L'_Y &= aw\{E_U(U,T)(U-\Lambda)+(\Lambda e_2+\Lambda^2 e_3)(T-\epsilon)\},\\ L'_X &= -mA_1+a(c_2+c_3\Lambda\epsilon)(U-\Lambda)+a(c_1+c_3U)K_1. \end{aligned} \tag{WN28}$$

There are no omitted monomials. Define

$$d_* = (c_2+c_3\Lambda\epsilon)/\epsilon = \nu\epsilon r^2 x+\Lambda c_3.$$

Since $\epsilon(U-\Lambda)=K_1-U(T-\epsilon)$, the last line is

$$L'_X = -mA_1+P_X, \qquad P_X = a(c_1+c_3U+d_*)K_1-d_*UA_1. \tag{WN29}$$

The exact source equations also give

$$\begin{aligned} m &= N+\tau(\mathcal{F}+G\Lambda\epsilon+2\mathcal{K}\Lambda)+2\nu r^2x^2\Lambda,\\ c_0/\epsilon &= m-\Lambda B, \qquad B=c_1+\Lambda c_3+\nu\epsilon r^2x,\\ N &= \alpha r^2/\epsilon^2+(m-\Lambda B)^2/H_X+\nu H_0r^2/H_X. \end{aligned} \tag{WN30}$$

In particular

$$N,\nu,\tau=O(m), \quad \Lambda=O(m\epsilon), \quad N=m^2/H_X+\nu H_0r^2/H_X+O(m\Lambda). \tag{WN31}$$

For the error term use $\alpha/\epsilon^2=O(\tau\Lambda)$, which is the exact positive alpha balance WN23/WA16. All these comparisons are inside the source field. We split by the bounded source ratio ν/m; $m>0$.

Substantial ν/m. If ν/m has positive residue, keep $S_0=\sqrt{m}A_1$. The N_X, S_0 Gram, including the entire $-mA_1$ first normal, has residue from

$$\begin{pmatrix} H_X & -\sqrt{m}\\ -\sqrt{m} & N/m \end{pmatrix}. \tag{WN32}$$

Its determinant is $(\nu/m)H_0r^2+O(\Lambda)$, a positive unit. The v,Y first normals in WN28 use the first conductor WN25; their remaining coefficients on S_0 are $O(\sqrt{m}\,w)$, hence infinitesimal. In WN29 the K_1-coefficient contains a, and $d_*U/\sqrt{m}=O(\sqrt{m}\,\epsilon U)$ is infinitesimal. Thus all other first normals have infinitesimal coefficients on K_1, the six cubic rows, and S_0. The v,Y hard block in WN16 is strict. All higher terms have at least three infinitesimal hard normals and bounded coefficients. This proves the full parent on this source subdomain.

Thin ν/m, including literal $n=0$. Assume $\nu/m=o(1)$. Equations WN30–WN31 imply τ/m is a positive unit: after division by m,

$$1=N/m+(\tau/m)\mathcal{F}+o(1), \qquad N/m=m/H_X+o(1),$$

and $1-m/H_X=(n+\beta+1/2)/H_X$ has positive residue. Write

$$N-m^2/H_X=\nu\xi+\eta, \qquad \xi=H_0r^2/H_X>0, \qquad \eta=O(m\Lambda).$$

Complete only the source coefficient H_X:

$$\begin{aligned} H_XN_X^2-2mN_XA_1+NA_1^2 &= H_X\widetilde{N}_X^2+\nu\xi A_1^2+\eta A_1^2,\\ \widetilde{N}_X &= N_X-(m/H_X)A_1. \end{aligned} \tag{WN33}$$

Retain the strict row $\sqrt{\nu}A_1$ when $\nu>0$; if $\nu=0$, all terms allocated to it vanish literally. The signed error is an infinitesimal cubic Gram, since

$$m\Lambda A_1^2=(m/\tau)(\tau\sqrt{\epsilon}\,A_1)^2. \tag{WN34}$$

The first two normals in WN28 use WN25 and bounded m/τ. For P_X, its K_1-coefficient contains a, while its two other contributions have infinitesimal allocations

$$\nu\epsilon UA_1=(\sqrt{\nu}\,\epsilon U)(\sqrt{\nu}A_1), \qquad \Lambda UA_1=(\sqrt{\epsilon}\,U)(\tau\sqrt{\epsilon}\,A_1). \tag{WN35}$$

Therefore $2P_X\widetilde{N}_X$ has infinitesimal Gram entries.

The shift in WN33 also creates the entire scalar term $2(m/H_X)A_1P_X$, which must not be dropped. Its main factor is controlled by the exact identity

$$mU(T-\epsilon)K_1 = mK_1^2 - (m/\tau)\Lambda(U-\Lambda)K_1. \tag{WN36}$$

The second term is infinitesimal by WN26. If m is infinitesimal, the first is an infinitesimal K_1-Gram. If m is a unit, then τ is a unit and WN27 makes it an infinitesimal cubic Gram. In that case a positive real K_1-pivot may be reinserted by WN27 at infinitesimal cubic cost. This allocation remains valid with each bounded factor $c_1 + c_3U + d_*$. The remaining scalar factors are multiples of $m\nu\epsilon UA_1^2$ and $m\Lambda UA_1^2$; they have infinitesimal Gram entries on $\sqrt{\nu}A_1$ and the fourth conductor in WN25, respectively. This accounts for the whole scalar effect of the completion.

The entire higher-normal restoration in the thin case

The full parent is quadratic in X. Its coefficient beyond the source constant H_X is exactly the nonnegative polynomial

$$D_X = (n+\beta+\tfrac{1}{2})v^2 + nu^2 + (n+D+\tfrac{1}{2})u^2v^2. \tag{WN37}$$

Retain the entire square $D_XN_X^2$, evaluated at the actual independent u, v, as separated squares. Each monomial coefficient is a nonnegative source scalar times a testing square. It remains a positive whole square after WN33. The remaining polynomial has no quadratic N_X term except $H_XN_X^2$. At the complementary graph its half mixed derivatives have the exact forms

$$[N_XN_v] = w[-m + a^2B_v], \qquad [N_XN_Y] = a^2wB_Y, \tag{WN38}$$

where B_v, B_Y are bounded polynomials fixed by the full parent expansion. After the shift, the new first-normal terms are bounded multiples of $\tau aw(T-\epsilon)$, using $m/\tau = O(1)$; WN25 gives infinitesimal coefficients. Every other remaining monomial linear in N_X has degree at least two in the other hard normals. Replacing N_X by $\widetilde{N}_X + (m/H_X)A_1$ gives an infinitesimal coefficient on a product of at least two hard normals, since $A_1 = o(1)$. Terms involving $\widetilde{N}_X$ have total hard-normal degree at least three and are handled in the same way. Terms independent of N_X are unchanged. The full real hard block is

$$H_{X,0}\widetilde{N}_X^2 + D_0N_v^2 + 2D_0N_vN_Y + H_{Y,0}N_Y^2 \succ 0. \tag{WN39}$$

Equations WN23–WN39 therefore give a finite strict Gram for the whole parent, including all first normals, every higher term, all β-openings and literal $n = 0$. This completes the double-infinitesimal testing corner for both source ratios and all local states.

Every other angle, all states and literal endpoints

Retain the entire m, n, α, β squares and call the remainder F_0. Only its residue is the critical C, A, D source in WA10; no global physicality of the subtracted finite source is asserted. The following are complete polynomial row allocations, so their strict residue Grams apply to the original nearby coefficients.

For bounded u, v with v a unit use (uW, X, Y, Z). The $Z = 0$ principal block is positive definite and

$$\det \mathrm{Mat}_{(uW,X,Y,Z)}\, \bar{F}_0 = D_u^2D_v^2(q_{12}v^4 + q_{21}u^2v^2)/16 > 0. \tag{WN40}$$

Both q_{12}, q_{21} are positive units. If u is a unit and v infinitesimal, use instead $(uW - Y, Y, X + uZ, vZ)$. The entire parent is polynomial in these rows with bounded coefficients; the only algebraic coefficient divisions are by the pure testing unit u. Its determinant at $v = 0$ is $D_u^2q_{21}u^2/16 > 0$, and the first three-row principal block is strict. This covers all bounded angle pairs outside the double-infinitesimal corner.

For Bob reciprocal $b = 1/v = o(1)$, Alice bounded, the rows (uW, X, Y, Z) give the real form

$$C_0(uW - Y)^2 + (A_0 + \tfrac{1}{2})D_uY^2 + D_0(uX - Z)^2 + \tfrac{1}{2}D_uX^2 \succ 0. \tag{WN41}$$

For Alice reciprocal $s = 1/u = o(1)$ and Bob a bounded unit, the original state rows give

$$C_0D_vW^2 + A_0D_vY^2 + (D_0 + \tfrac{1}{2})(W + vX)^2 + \tfrac{1}{2}(vY - Z)^2 \succ 0.$$

If s, v are both infinitesimal, use (W, Y, vX, Γ), where $\Gamma = sX + Z$. The entire reciprocal parent is polynomial in these four un-divided rows, and its real form is

$$C_0W^2 + A_0Y^2 + (D_0 + \tfrac{1}{2})(W + vX)^2 + \tfrac{1}{2}\Gamma^2 \succ 0. \tag{WN42}$$

The coefficient-extraction substitution $X = R/v, Z = \Gamma - sR/v$ cancels all denominators; the asserted identity itself contains no v^{-1}, and includes literal $v = 0$.

For both reciprocal angles $s, b = o(1)$, the full polynomial belongs to the square ideal of (W, X, Y, sZ, bZ), with real form

$$C_0W^2 + A_0Y^2 + D_0(X - sZ)^2 + \tfrac{1}{2}X^2 + \tfrac{1}{2}(Y - bZ)^2 \succ 0. \tag{WN43}$$

These finite identities include $s = 0$ and $b = 0$ exactly.

It remains to check that the local state charts were exhaustive. Normalize a maximal state coordinate in the testing field. At $u, v = o(1)$, a unit X or Y gives positive full residue immediately. Otherwise one of W, Z is a unit. The exact sign gauges give $u, v \geq 0$. If $WZ \leq 0$, the full-core sign identity SE9 supplies separated nonnegative terms; all event squares are positive. If $WZ > 0$, choose $W = 1, Z$ bounded, or $Z = 1, W = o(1)$. These are exactly the states covered above; vanishing angles or states are included by the same polynomial identities (or by the nonpositive-sign case). Positive pure-field homogeneous normalization and real/imaginary state splitting now restore the original full parent:

$$F_c(\text{any independent projective test and state}) \ \in \ \Sigma(K_c \otimes_{\mathbb{R}} K_t)^2. \tag{WN44}$$

This proves precisely the weak-n C2 and all-β C1 weak-α source domains in the source domain stated above. It supplies the missing local leaves usually called WB, WU and CW. It does not, by itself, assert the global source partition or the finite-lift conclusion.

A.9 A source-exact projective pitchfork for C1

Theorem and precise source domain. Take a bounded physical canonical C1 source with $d = 1$, $\lambda = n = 0$, m, C, A, α positive bounded units and $D, \beta \geq 0$ infinitesimal. Assume the original actual finite noncommuting stationary zero contact, and retain every original source gap and intersection. Then the entire parent has a finite sum-of-squares certificate in $K_c \otimes_{\mathbb{R}} K_t$ for every independent projective test and every real or complex state. This includes literal $D = 0$ or $\beta = 0$ whenever allowed by the source domain. The two cannot vanish together at an actual contact.

Opening-safe source entry

Write the positive source contact as $(u_c, w_c) = (c, 1/v_c)$, and $V = w_c^2$. The retained positive-d balance proof gives c a positive bounded unit and V a positive infinitesimal. Its second stationarity equation, in the notation of that proof, is

$$a_3 = V^{-2}\{mDa_2/c^2 + \beta m a_4 + Da_5 + \beta a_7 c^2\}, \tag{PP1}$$

where all a_i are positive bounded units. Therefore

$$D = V^2\delta, \quad \beta = V^2\eta, \qquad \delta, \eta \geq 0 \text{ bounded}, \qquad \delta + \eta \text{ a positive unit}. \tag{PP2}$$

No inverse of either opening is used.

Normalize the source state by $Z_c = 1$. Set

$$p = W_c/w_c, \qquad y = Y_c/w_c, \qquad x = X_c/V. \tag{PP3}$$

All three are bounded source elements. To prove this without a shape hypothesis, restrict the reciprocal full parent PP6 to (W, Y, X) at $w = 0$, and then take residues. Its positive hard form is

$$H_u(W, Y, X) = m(W + uY)^2 + C(uW - Y)^2 + (A + \alpha + 1/2)(1 + u^2)Y^2 + (1 + u^2)X^2/2. \tag{PP4}$$

It is positive definite for every bounded real u. Scale the two W, Y kernel equations at the source by w_c, and the X equation by V. In the unknowns (p, y, x) the resulting bounded matrix has a block triangular residue: its p, y block is the positive W, Y block of PP4, and its x pivot is $(1 + c^2)/2$. The terms $(\beta - D)/V = V(\eta - \delta)$ are bounded. All right hand sides are bounded, proving PP3 by an ordinary source-field matrix inverse with unit determinant.

Let $q_{01}, q_{10}, q_{21}, h$ be the original Laurent coefficients. Their first three residues are positive units because m, C, A, α are units. The actual zero and angular stationarity equations, using PP2, give

$$\bar{q}_{01} = \bar{q}_{21}\bar{c}^{\,4}, \qquad \bar{h} = -2\bar{q}_{21}\bar{c}^{\,2}. \tag{PP5}$$

Every other nonconstant Laurent contribution apart from $q_{10}w^2$ vanishes in this residue calculation. These are equations wholly in the coefficient field followed by its real residue map.

A complete local normal and divided-difference argument

It remains locally to treat testing w infinitesimal or zero and u with the same positive real residue as c. The full reciprocal quarter-parent is

$$\begin{aligned} F = {} & m(W - wX + uY - uwZ)^2 + CD_w(uW - Y)^2 + AD_uD_wY^2 \\ & + (D + 1/2)(uwW + uX - wY - Z)^2 + (\alpha + 1/2)D_u(Y - wZ)^2 \\ & + (\beta + 1/2)D_w(X + uZ)^2 - D_uD_wZ^2/2, \qquad D_u = 1 + u^2, \quad D_w = 1 + w^2. \end{aligned} \tag{PP6}$$

Use three unrestricted normals and two scalar conductors

$$N_W = W - pwZ, \quad N_Y = Y - ywZ, \quad N_X = X - xw^2Z, \qquad Q = w(u - c)Z, \quad L = (w^2 - V)Z. \tag{PP7}$$

The centers agree with the actual source contact; no equation is imposed on an independent state. Write the full normal expansion

$$F = N^T H(u, w)N + 2Z\sum_i N_i\ell_i(u, w) + E(u, w^2)Z^2, \quad N = (N_W, N_Y, N_X)^T. \tag{PP8}$$

It is an exact expansion of every term of PP6. The matrix H has bounded entries and residue PP4 at $u = \bar{c}$. Parity and direct inspection of PP6 give bounded polynomials a_W, a_Y, a_X such that

$$\ell_W = wa_W(u, b), \qquad \ell_Y = wa_Y(u, b), \qquad \ell_X = u(\beta - D) + ba_X(u, b), \quad b = w^2. \tag{PP9}$$

Actual state stationarity gives $a_W(c, V) = a_Y(c, V) = 0$ and $\ell_X(c, w_c) = 0$. Finite monic division thus gives bounded polynomials A_i, B_i with

$$a_i(u, b) = (u - c)A_i(u, b) + (b - V)B_i(b), \qquad i = W, Y.$$

Consequently the entire first two normal couplings are $Z\ell_i = A_iQ + wB_iL$.

For the third normal put $r(u) = u(\beta - D)/V + a_X(u, V)$. This is a bounded source polynomial by PP2, and $r(c) = 0$. Hence $r = (u - c)R(u)$ for a bounded polynomial R. Division in b gives another bounded polynomial B_X with

$$Z\ell_X = \{B_X(u, b) - (u - c)R(u)\}L + wR(u)Q. \tag{PP10}$$

Here the identity $V(u - c)Z = wQ - (u - c)L$ is exact. Moreover $B_X(c, 0)$ has zero residue: the source equation $u(\beta - D) + Va_X(u, V) = 0$ at $u = c$, divided by V, gives $a_X(c, V)$ infinitesimal, and $B_X(c, 0)$ has the same residue. Thus the third normal has no residue coupling to L or Q. The first two have no residue coupling to L.

For completeness, the scalar coefficient obeys

$$E(u, 0) = D + \beta u^2, \qquad E(c, V) = E_u(c, V) = E_b(c, V) = 0. \tag{PP11}$$

The first assertion follows by setting $w = 0$ in the entire PP6 after the graph substitution. The other three follow from the actual full source contact and its two angle derivatives; neither normal term is discarded in making this inference. Define $d(u) = \delta + \eta u^2$ and the bounded polynomial

$$J(u, b) = \{E(u, b) - d(u)(b - V)^2\}/b. \tag{PP12}$$

This notation means exact monic division: PP11–PP2 make the constant term zero. No testing b is inverted. At (c, V) the value and both first derivatives of J vanish. A second finite monic Taylor division gives bounded polynomials J_{20}, J_{11}, J_{02} with

$$J(u, b) = (u - c)^2J_{20}(u, b) + (u - c)(b - V)J_{11}(b) + (b - V)^2J_{02}(b). \tag{PP13}$$

Substitution yields the complete exact scalar identity

$$E(u, w^2)Z^2 = J_{20}(u, w^2)Q^2 + wJ_{11}(w^2)QL + \{d(u) + w^2J_{02}(w^2)\}L^2. \tag{PP14}$$

All divisions preceded tensor evaluation, with monic divisors and bounded source coefficients. These formulas hold literally at $w = 0$.

Equations PP8–PP14 represent the entire parent by a bounded Gram on the five rows N_W, N_Y, N_X, Q, L. Its L pivot in residue is $\bar{\delta} + \bar{\eta}\bar{c}^2 > 0$, and its L crosses have zero residue. It remains to check the four-row hard and angular residue; the following exact determinant computation provides that check without taking a mixed inverse.

Strictness of the angular residue

At $D = \beta = 0$, replace Z by T/w in the whole reciprocal parent and take its polynomial value at $w = 0$. This gives a real four-state quadratic form $B_u(W, Y, X, T)$, with hard block PP4. The original determinant identity gives

$$\det B_u = \frac{(1+u^2)^2}{16}(q_{01} + hu^2 + q_{21}u^4). \tag{PP15}$$

Using PP5, its real Schur coefficient at T is

$$g(u) = \frac{(1+u^2)^2 q_{21}}{16 \det H_u}(u^2 - c^2)^2. \tag{PP16}$$

Here all entries and inverses are real residues; this calculation proves positivity of a real matrix and does not factor a matrix in the tensor ring. The contact centers PP3 give the unique real kernel of B_c, so the four-row residue in PP8–PP14 is precisely the quadratic Taylor coefficient in hard normals and the change $u - c$. Its angular Schur pivot is

$$g''(c)/2 = \frac{(1+c^2)^2 q_{21} c^2}{4 \det H_c} > 0. \tag{PP17}$$

The hard block is PP4 and is positive definite. Thus this full four-row residue is positive definite, as is the independent L pivot. The separated strict bounded-Gram lemma proves that PP6 is tensor SOS in the complete local chart. The rows may be unbounded or linearly dependent at a literal endpoint; only the coefficient Gram needs a strict real residue.

Explicit testing complement and projective endpoints

If u, v are bounded original testing coordinates, use the ordinary state matrix except near the double zero corner. At finite nonzero real residues, the residue Laurent polynomial is

$$q_{21}(u^2 - c^2)^2/u^2 + q_{10}/v^2 > 0, \tag{PP18}$$

so its state matrix is positive definite by the hard $Z = 0$ block. At $u = 0$, $v \neq 0$ the real quarter-parent is

$$m(vW - X)^2 + (C + A + 1/2)(1 + v^2)Y^2 + \alpha(vY - Z)^2 + (1 + v^2)X^2/2 \succ 0.$$

At $v = 0$, $u \neq 0$ it is $m(X + uZ)^2 + (C + 1/2)(uW - Y)^2 + A(1 + u^2)Y^2 + \alpha(1 + u^2)Z^2 + (X + uZ)^2/2$, which is likewise strict. These are residue identities; the bounded full matrices therefore give tensor SOS on their respective charts.

When both u, v are infinitesimal or zero, use the five rows (uW, vW, X, Y, Z) in the entire original finite parent. The exact polynomial substitution is bounded and its residue is

$$m(vW - X)^2 + (C + 1/2)(uW - Y)^2 + AY^2 + \alpha Z^2 + X^2/2 \succ 0 \tag{PP19}$$

as a coefficient Gram on those five rows. This treats that corner with no state normalization.

For reciprocal Alice and bounded Bob, the complete core identity RC20 applies unchanged with all m, D, α, β event squares retained whole. Its residue RC21 is strict using only $C, A > 0$. For two reciprocal testing angles, retain those positive squares and use RC24 for the four rows W, X, Y, wZ. The residue $CW^2 + AY^2 + X^2/2 + (Y - wZ)^2/2$ is strictly positive. These identities allow the reciprocal coordinate to be literally zero.

For Bob reciprocal and bounded Alice, retain the whole D, β squares, put $T = wZ$, and use the full four-row expression

$$m(W - wX + uY - uT)^2 + CD_w(uW - Y)^2 + AD_uD_wY^2 + \alpha D_u(Y - T)^2 + K/2,$$

where K is the full core RC22 with $\rho_* = uW - Y$. Its residue is exactly B_u. By PP15–PP16 it is strictly positive whenever u^2 has residue different from $\bar{c}^2$, including u infinitesimal. The remaining matched residues are PP7–PP17 after the exact Alice sign gauge if necessary.

Both signs of either angle are allowed by the exact simultaneous state sign gauges of the original parent. Every normalization is a positive square in a single source or testing field. Complex states split into real and imaginary parts. The finite and reciprocal homogeneous identities include all literal axes and projective endpoints. This completes the all-tests theorem on the stated positive-d C1 projective source domain.

A.10 A direct certificate for the positive-d C2 reciprocal corner

The source domain. Normalize the original C2 source by $d = 1$ and $\lambda = \beta = 0$. Assume that all source coefficients are bounded, that C, A, α are positive bounded units, that m, D are positive infinitesimals, and that

$n \geq 0$ is bounded. Retain the original physicality, finite noncommuting stationary zero contact and positive native gaps. This section proves the complete parent is tensor SOS at every independent projective test and every state. The contact is used only inside the source field. No mixed determinant is cancelled.

Equation (A.1), with $\beta = 0$, gives, for the reciprocal Bob contact coordinate $w_c = 1/v_c$,

$$U = u_c^2 \in \mathrm{Inf}_{>0}, \qquad V = w_c^2 \in \mathrm{Inf}_{>0}, \qquad m \asymp U^2, \qquad D \asymp V^2. \tag{RC1}$$

The positive-unit ratios follow from the final formula in Section A.4.

The actual source has a bounded shape chart

Choose the native positive contact chart with $Z_c = 1$, and set

$$S = (u_c/w_c)W_c, \qquad y = Y_c/w_c, \qquad R = -X_c/(u_c V). \tag{RC2}$$

Then $S, y, S-y, 1-y, R$ are positive bounded units. Here is an entry proof that does not assume those facts as extra source hypotheses. In the reciprocal parent RC6 below, take the ordinary source-field Schur complement of the X, Y block. That block has residue $\mathrm{diag}(n+1/2, H)$, where $H = C+A+\alpha+1/2$. Its W, W Schur entry divided by U has residue $a = C - C^2/H > 0$, while its W, Z entry divided by $u_c w_c$ has residue $-b = -(1/2 + C\alpha/H) < 0$. The balance $m/U \in \mathrm{Inf}$ and $D/V \in \mathrm{Inf}$ justifies these divisions in the source field. The zero-state equation therefore gives $S \sim b/a$, a positive bounded unit.

The two hard state equations, scaled in the source field by $u_c V$ and w_c, are a bounded linear system in R, y. Its residue is triangular with positive diagonal $n + 1/2, H$. Consequently both quantities are bounded. The second equation gives $Hy = CS + \alpha$ in residue, so y is a positive unit. The W equation divided by $u_c w_c$ gives $C(S-y) = 1/2$ in residue. The Z equation divided by V gives $\alpha(1-y) = S/2$ in residue: its apparently constant U terms cancel exactly before the division. Finally the first hard equation gives $(n+1/2)R = (1+S-y)/2$ in residue. These relations prove all five claimed positive-unit bounds. The exact native gaps already give $0 < y < 1$ and $S - y > 0$ before taking residues.

The unchanged native source formulas are converted by the following dictionary. Their letters are used only in this display:

$$\begin{gathered} z = U, \quad a^2 = 1/(UV), \quad b = S/U, \quad t = (S-y)V, \quad r = VR, \quad h = y/U, \\ g = S/U + y - 1 + VR, \qquad j = 1 + UVR - V(S-y). \end{gathered} \tag{RC3}$$

Substitution into the six exact native coefficients has denominators with positive-unit leading parts after the displayed powers of U, V are removed. Thus $\mu = m/U^2$ and $\delta = D/V^2$ are bounded units, and the source residues are

$$\begin{aligned} C_0 &= \frac{1}{2(S-y)}, & A_0 &= \frac{1+S-y}{2y}, & \alpha_0 &= \frac{S}{2(1-y)}, \\ n_0 &= \frac{1+S-y-R}{2R}, & \mu_0 &= \frac{1+S-y}{2S^2}, & \delta_0 &= \frac{S(1+S-y)}{2}. \end{aligned} \tag{RC4}$$

In residue formulas such as RC4 the shape letters mean their real residues. In particular

$$C_0\{\alpha_0(A_0 - 1/2) - 1/4\} = (A_0 + \alpha_0 + 1/2)/4. \tag{RC5}$$

These are source statements only. They do not identify any source infinitesimal with an independent testing infinitesimal.

The complete local parent and both unrestricted normals

For u, w infinitesimal or zero in the independent testing field, put $D_u = 1 + u^2$, $D_w = 1 + w^2$. The full quarter-parent after the pure testing reciprocal Bob normalization is

$$\begin{aligned} F = {} & m(W - wX + uY - uwZ)^2 + nD_uD_wX^2 + CD_w(uW - Y)^2 + AD_uD_wY^2 \\ & + (D+1/2)(uwW + uX - wY - Z)^2 + (\alpha+1/2)D_u(Y - wZ)^2 \\ & + \tfrac{1}{2}D_w(X + uZ)^2 - \tfrac{1}{2}D_uD_wZ^2. \end{aligned} \tag{RC6}$$

All four state coordinates are arbitrary. Introduce tensor rows

$$\begin{gathered} N_x = X + uRw^2Z, \qquad N_y = Y - ywZ, \\ \rho = uW - SwZ, \qquad k = UW - SuwZ, \qquad \ell = (w^2 - V)Z. \end{gathered} \tag{RC7}$$

The testing-dependent w^2 in N_x is essential; it agrees with the source contact and removes the unbounded coefficient produced by a contact-fixed center. The exact conductor identities are

$$U\rho - uk = Sw(u^2 - U)Z, \qquad u\rho - k = (u^2 - U)W. \tag{RC8}$$

Expanding the entire RC6 gives

$$F = F_0 + H_x N_x^2 + H_y N_y^2 + 2H_{xy} N_x N_y + 2N_x L_x + 2N_y L_y, \tag{RC9}$$

where F_0 is only the constant coefficient of this unrestricted normal expansion. The hard coefficients are exactly

$$\begin{aligned} H_x &= mw^2 + nD_u D_w + (D + 1/2)u^2 + D_w/2, \\ H_y &= mu^2 + CD_w + AD_u D_w + (D + 1/2)w^2 + (\alpha + 1/2)D_u, \\ H_{xy} &= -(m + D + 1/2)uw. \end{aligned} \tag{RC10}$$

Set $J_x = n(1 + w^2) + D + 1/2$ and $K_x = n(1 + U) + 1/2 + m$. The complete first normal is

$$L_x = (D + 1/2)uw\rho - (m/U)wk + u\{D/V - Rw^2 K_x\}\ell - (R/S)uwJ_x(U\rho - uk). \tag{RC11}$$

Every coefficient on ρ, k, ℓ is bounded infinitesimal. For the other normal define

$$\begin{aligned} a_{10} &= Ay + (m + \alpha + 1/2)(y - 1), \\ a_{01} &= -(C + D + 1/2)S + (C + A + D + 1/2)y, \\ a_{11} &= R(m + D + 1/2) + Ay. \end{aligned}$$

Then the exact expression is

$$\begin{aligned} L_y = \{&m - C - (C + D + 1/2)w^2 + (U/S)(a_{10} + a_{11}w^2)\}\rho \\ &- (u/S)(a_{10} + a_{11}w^2)k + w(a_{01} + a_{11}U)\ell. \end{aligned} \tag{RC12}$$

Its only nonzero residue coupling is $-C_0\rho$. RC11–RC12 follow by full expansion and the two actual source state equations; they account for every term linear in the two normals in RC6.

A scalar certificate by finite monic divisions

Put

$$c = C + (C + D + 1/2)w^2, \quad b_2 = (D + 1/2)R, \quad b_1 = (C + D + 1/2)(y - S) - mR, \quad \gamma = b_1 + b_2 U.$$

Define

$$F_{\text{base}} = (c - 2Ub_2w^2/S)\rho^2 + \mu k^2 + (2ub_2w^2/S)\rho k - 2w\gamma\rho\ell. \tag{RC13}$$

The source W state equation removes the remaining WZ coefficient. Direct expansion then gives $F_0 - F_{\text{base}} = E(u^2, w^2)Z^2$, with $E(a, b)$ a bounded-coefficient polynomial of degree at most two in a. Its exact properties are

$$\begin{aligned} E(a, 0) = D, \qquad E(U, V) = E_a(U, V) = E_b(U, V) = 0, \\ E(a, b) = E_0(b) + (a - U)E_1(b) + (a - U)^2 E_2(b), \\ E_2(b) = R^2 b^2\{n(1 + b) + D + 1/2\}. \end{aligned} \tag{RC14}$$

The three vanishing jets follow from the actual full contact: every row in RC7 vanishes there, the complete scalar and both angle derivatives vanish, and RC13 has zero first derivatives there. They are also checked directly from the native source coefficients. Since $E_1(0) = E_1(V) = 0$ and E_0 has a double zero at V, finite monic polynomial division gives bounded polynomials J_0, J_1 such that

$$E_0(b) = (b - V)^2 J_0(b), \qquad E_1(b) = b(b - V)J_1(b). \tag{RC15}$$

These divisions take place in the source polynomial algebra before tensor evaluation. They introduce no inverse of V or of a mixed quantity. In particular $V^2 J_0(0) = D$, and the other coefficients of J_0 remain bounded; hence $J_0(w^2)$ has residue δ_0.

RC8 now gives the exact scalar identity

$$\begin{aligned} F_0 = F_{\text{base}} &+ \frac{R^2 w^2\{n(1 + w^2) + D + 1/2\}}{S^2}(U\rho - uk)^2 \\ &+ \frac{w}{S}J_1(w^2)(U\rho - uk)\ell + J_0(w^2)\ell^2. \end{aligned} \tag{RC16}$$

This polynomial formula also holds at $u = 0$ or $w = 0$; neither is inverted. Combining RC9–RC16 represents the entire parent by a bounded Gram on $(N_x, N_y, \rho, k, \ell)$ with real residue

$$(n_0 + 1/2)N_x^2 + H_0 N_y^2 - 2C_0 N_y\rho + C_0\rho^2 + \mu_0 k^2 + \delta_0\ell^2, \qquad H_0 = C_0 + A_0 + \alpha_0 + 1/2. \tag{RC17}$$

This is positive definite: the N_y, ρ determinant is $C_0(A_0 + \alpha_0 + 1/2) > 0$, and the other three pivots are strictly positive. The separated strict bounded-Gram lemma therefore gives a finite tensor SOS of the complete RC6. It requires bounded matrix entries, not bounded rows or states.

All other testing charts

The remaining measurement charts follow from three matrix identities. In each one we retain the indicated event squares and prove positivity of the remaining bounded Gram matrix.

Both original testing coordinates bounded. Put $p = uW$. Retain the whole m, n, D squares and use the remaining Gram on (p, Y, X, Z):

$$\begin{aligned} CD_v(p-Y)^2 + AD_uD_vY^2 + \tfrac{1}{2}(p+uvX-Y-vZ)^2 + (\alpha+1/2)D_u(vY-Z)^2 \\ + \tfrac{1}{2}D_v(X+uZ)^2 - \tfrac{1}{2}D_uD_vZ^2. \end{aligned} \tag{RC18}$$

The $Z = 0$ hard block is positive definite by its displayed C, A, X squares. After RC4 the determinant of the real residue is

$$\frac{(S+u^2v^2)(1+u^2)^2(1+v^2)^2(1+S-y)^2}{16y(S-y)(1-y)} > 0. \tag{RC19}$$

Here also u, v mean real testing residues. Thus RC18 is a strict bounded Gram, even when u or v has zero residue.

Alice reciprocal, Bob bounded. Let $s = 1/u$ be infinitesimal or zero, set $D_s = 1 + s^2$, and use rows $(W, Y, V_x, R_x) = (W, Y, vX, sX + Z)$. Retain the entire m, n, D, α squares. The remaining expression is $CD_v(W - sY)^2 + AD_sD_vY^2 + K/2$, where exactly

$$\begin{aligned} K = \{W - sY + D_sV_x - svR_x\}^2 + D_vR_x^2 \\ + D_s\{v^2Y^2 - 2Y(vR_x - sV_x) - (vR_x - sV_x)^2\}. \end{aligned} \tag{RC20}$$

Its residue is

$$C_0D_vW^2 + A_0D_vY^2 + \tfrac{1}{2}(W+V_x)^2 + \tfrac{1}{2}(vY-R_x)^2 \succ 0. \tag{RC21}$$

This single formula includes all bounded v, including literal zero and arbitrary competing small s, v. There is no s/v coefficient.

Bob reciprocal, Alice a bounded unit. Let w be infinitesimal or zero, and u a bounded unit. Use $\rho_* = uW - Y$, $T = wZ$, $S_D = \sqrt{D}\,Z$, and the five rows (ρ_*, Y, X, T, S_D). Retain m, n whole. The D square is exactly $\{\sqrt{D}(uX + w\rho_*) - S_D\}^2$. The other terms are

$$CD_w\rho_*^2 + AD_uD_wY^2 + \alpha D_u(Y-T)^2 + K/2,$$

where the complete core is

$$K = (D_u + w^2)X^2 + 2uwXT + w^2\rho_*^2 + 2uw\rho_*X - 2\rho_*T - T^2 + D_u(Y-T)^2. \tag{RC22}$$

The X, S_D residue pivots are $D_u/2, 1$. On (ρ_*, Y, T) the residue matrix is

$$\begin{pmatrix} C_0 & 0 & -1/2 \\ 0 & GD_u & -(\alpha_0+1/2)D_u \\ -1/2 & -(\alpha_0+1/2)D_u & \alpha_0D_u + u^2/2 \end{pmatrix}, \qquad G = A_0 + \alpha_0 + 1/2. \tag{RC23}$$

Its first two leading minors are C_0, C_0GD_u. By RC5 its determinant is $GD_uu^2(1 + 2C_0)/4 > 0$. This proves strictness. The alternative u infinitesimal or zero is precisely RC7–RC17.

Both testing coordinates reciprocal. Let s, w be infinitesimal or zero. Set $T = wZ$ and $S_D = \sqrt{D}\,sZ$. Use the five rows W, X, Y, T, S_D, and retain the entire m, n, α squares. The D square is $\{\sqrt{D}(wW + X - swY) - S_D\}^2$. The other terms are $CD_w(W - sY)^2 + AD_sD_wY^2 + K/2$, where exactly

$$\begin{aligned} K = (wW + X - swY)^2 + D_s(Y-T)^2 + s^2D_wX^2 \\ - 2sWT + 2s^2YT + 2swXT - s^2T^2. \end{aligned} \tag{RC24}$$

The full five-row residue is

$$C_0W^2 + A_0Y^2 + X^2/2 + (Y-T)^2/2 + S_D^2 \succ 0. \tag{RC25}$$

This gives the final strict bounded Gram.

Every ordered-field testing coordinate is bounded or has infinitesimal reciprocal. RC18–RC25 and the local proof therefore cover both projective lines. The identities are polynomial in the chosen homogeneous coordinates, so literal infinity is included by $s = 0$ or $w = 0$. Both signs are allowed throughout; no state ratio or state normalization is used. Positive pure testing square normalizations restore the original homogeneous parent, and the original source scale is restored by a pure source square root. Real and imaginary parts of a complex state are handled separately.

Chapter B

The C1 coefficient domain

Let K_c, K_t be separately labelled real closed extensions of $\mathbb{R}$ and let $T = K_c \otimes_{\mathbb{R}} K_t$. Source coefficients belong to K_c; independent testing coordinates belong to K_t. A coefficient is bounded if it is a finite sum of products of bounded elements of the two fields. Its residue is obtained by taking the real residues in both factors. All sums of squares below are finite sums in T.

We prove the complete C1 assertion under the native source hypotheses of the contact-entry theorem: $d, m, C, A, \alpha > 0$, $\lambda_{\mathrm{ev}} = n = 0$, $D, \beta \geq 0$, $D + \beta > 0$, the original strict native gaps, and positive infinitesimal total scale. Normalize $d = 1$ by a pure-source square. The full target is the original four-state quadratic parent in Chapter 7, with its homogeneous value used at projective testing endpoints. No testing angle is identified with the source contact.

B.1 Source coordinates and polynomial operations

The contact equations determine the source coefficients. Their positive denominators and the lower bound for $E = 1 - x\eta h$ determine which rescalings remain bounded. We establish these facts before choosing the roots of the certificate.

B.1.1 Finite polynomial operations used in the proof

All coefficient arrays in this chapter have the following precise meaning. If Π is a displayed sum of retained positive squares, s is a vector of the remaining state coordinates and $s = s_0 + A\xi$ is the displayed change of state coordinates, put

$$\mathcal{Q}(\xi) = (F - \Pi)(s_0 + A\xi), \qquad q_0 = \mathcal{Q}(0), \quad q_i = [\xi_i]\mathcal{Q}, \quad H_{ij} = \frac{1}{2}\partial_i\partial_j\mathcal{Q}.$$

Then exactly $\mathcal{Q} = q_0 + \sum_i q_i\xi_i + \xi^T H\xi$. Thus the scalar term, both linear state couplings and the entire state quadratic form are computed from the same polynomial. Multiplying by the displayed positive pure-source and pure-testing common denominators makes these coefficient extractions polynomial. All later divisions are divisions by the specified monic polynomials.

For a polynomial f define its finite difference quotient by

$$\Delta_{z,a} f = \frac{f(z) - f(a)}{z - a}.$$

It is the polynomial determined by $(z^j - a^j)/(z - a) = \sum_{i=0}^{j-1} z^{j-1-i}a^i$. More generally, a quotient/remainder by $z^r - a$ is determined uniquely by repeatedly replacing z^r by a and collecting the difference. Thus every quotient defined below is an explicit finite polynomial operation, even where a source opening tends to zero. Weighted degree is checked on the original polynomial before division; division by a weighted-homogeneous monic polynomial preserves the filtration with its stated degree shift. The support and weight bounds specified below ensure that the resulting Gram coefficients remain bounded after rescaling.

Lemma B.1 (Auxiliary square allocation)**.** *Let r be roots with a bounded symmetric Gram whose real residue G_0 is positive definite. Let $h = (h_1, \ldots, h_s)$ be additional roots and suppose there are exact certificates*

$$Q_j(r, h) - h_j^2 \in \Sigma T^2,$$

where every Gram coefficient of Q_j is bounded with zero residue. Suppose the remaining terms of the polynomial are a bounded quadratic form in (r, h) whose r, r residue block is G_0. Then the complete polynomial is a tensor sum of squares.

Proof. For a sufficiently large fixed real $M > 0$, add $M\sum_j(h_j^2 - Q_j)$ to the remaining Gram. Its real residue is

$$\begin{pmatrix} G_0 & B_0 \\ B_0^T & C_0 + MI \end{pmatrix} \succ 0.$$

Existence of such an M follows by a Schur complement of the finite real matrix G_0. The strict positive-residue Gram lemma applies. Restore $M\sum_j(Q_j - h_j^2)$ using the displayed exact certificates. Only a finite real matrix was inverted in this proof. □

We also use the common product-relation allocation, the restricted monic contact lift CC1–CC6, and the finite real two-block allocation UA1–UA3. Their hypotheses are checked in the source cases below. The arbitrary-scale noncommuting theorem is used only for the original physical source, after its strong-event source conditions have been proved. The generic LII and RR polynomial identities are used only with the explicit C1 substitutions given below.

Whenever a proof below retains a nonnegative coefficient multiplying a whole square, an unbounded positive excess stays outside its bounded Gram. A strict leading block is obtained by clipping whole positive squares at fixed real lower bounds, never by clipping individual matrix entries. The real constrained-limit argument gives finite real caps where a source opening makes a retained coefficient unbounded. Every coefficient correction and both complete first state normals are included before this argument is applied.

B.1.2 The native source partition

Use the native coordinates $x = k^2$, $0 < k \le 1$, $\eta, \zeta > 0$, $r = 1 - a\eta$, $\theta = t\zeta$, $h = 1 - \theta$, $E = 1 - x\eta h > 0$, $Q = 1 - x\zeta r > 0$, $M = 1 + xr - t > 0$ and $J = 1 + xh - a > 0$. All satisfy $0 < r, \theta < 1$ and $0 < a, t < 2$. Write I for positive infinitesimal, U for positive bounded with nonzero residue and O for positive unbounded. Put $e = k\sqrt{\eta/\zeta}$. The exhaustive source cases are

k	ζ	other conditions	proof below
U	any	η, ζ : I	Section B.3.2
U	any	η, ζ not both I	Sections B.3.4, B.1.3, B.3.6
I	U or O	$\eta > 0$	Section B.1.2
I	I	η : I	Section B.4.7
I	I	η : O	Section B.4.8
I	I	η : U, e : I	Sections B.4.18, B.4.19
I	I	η, e, r : U	Section B.4.13
I	I	η, e : U, r : I	Section B.4.15
I	I	η, r : U, e : O	Section B.4.17
I	I	η : U, e : O, r : I	Section B.4.15

The source identities below justify every listed implication, including the ordinary UI/UH alternatives and the bound needed to split C1.1. The last two centered-A rows share the same bounded-reciprocal construction. The source field alone determines this partition.

The $L = x\zeta$ cases

For k infinitesimal and ζ noninfinitesimal, put $L = x\zeta$. When L is unbounded, $Q > 0$ forces r infinitesimal and $1 - r = a\eta$, $a < 2$ exclude infinitesimal η. Unit η is treated in Section B.5.1; unbounded η is covered for every testing pair by Theorem B.24. It combines the two-small-angle certificate with one full-state completion on the inverse-Alice chart and the adjacent-event bound for the remaining Bob angles. For unit L, infinitesimal and unit η are covered in Sections B.5.2 and B.5.3. For unbounded η, Theorem B.25 covers every testing angle. Its proof controls the state matrix away from the double infinitesimal corner and uses Theorem B.23 at that corner.

When L is infinitesimal, unbounded η gives

$$S = \sigma + L, \quad \sigma = 1/\eta, \quad \kappa = x/S, \quad \nu = \sigma/S, \quad \ell = L/S, \quad \nu + \ell = 1, \quad \zeta = \ell/\kappa.$$

The identity $\kappa Q_s + \ell M + \nu E = 1$, proved below, makes κ bounded and gives the five source cases

κ	ν	ℓ	proof
U	U	U	Sections B.7.5, B.7.6
U	I	U	Section B.7.7
I	U	U	Section B.7.8
I	U	I	Section B.7.9
I	I	U	Sections B.7.11, B.7.12

In the last row $q_\nu = \nu/\ell$ and $\tau = h/q_\nu$ are defined only there. BN proves every bounded τ and UN every unbounded τ. No separate restriction on β/q_ν^2 is made.

If L is infinitesimal and η bounded, unbounded ζ with unit η is Section B.6.9. Small η with unit or unbounded ζ uses Sections B.6.10, B.6.11 and B.6.13. The remaining η, ζ are units. Put

$$j = \eta(1 + xh) - 1 + r = \eta J > 0, \qquad j_0 = \eta + r - 1, \qquad j - j_0 = \eta x h \in \mathrm{Inf}_c\,.$$

Unit j is Section B.6.4; small j with unit r is Section B.6.5; small j, r is Section B.6.6. Thus every positive source opening and each permitted literal $D = 0$ or $\beta = 0$ has a case in the proof.

B.1.3 The native coefficient formulas and their uniform gap

For the physical C1 source, $\lambda = n = 0$, $d > 0$, $m, C, A, \alpha > 0$, $D, \beta \geq 0$ with at most one zero, and the source has its actual finite noncommuting stationary contact. Use the common global sign entry and normalize $C = 2$. The exact full-parent Alice reflection permits $0 < k \leq 1$. Put

$$x = k^2, \quad r = 1 - a\eta, \quad \theta = t\zeta, \quad h = 1 - \theta, \quad y = v_c^2 = x\eta\zeta,$$

$$K = 1 + x,\ R = 1 + xr,\ T = 1 + xh,\ G = h + \zeta, \quad M = R - t,\ J = T - a,\ E = 1 - x\eta h,\ Q = 1 - x\zeta r. \tag{NC1}$$

The common exact signs give $0 < r, \theta < 1$, $E, Q, M, J > 0$ and $0 < a, t < 2$. There is no size assumption on η, ζ, e or a positive lower bound on any opening.

Three short positive source identities. The original coquotient polynomials obey

$$\begin{aligned} P_4 &= TE + x(1-r)G, \\ P_\alpha &= Rh(1+y) + \zeta ME, \\ P_d &= p_0 + \eta p_1, \qquad p_0 = 1 + 2xh + rx^2G + x\zeta, \quad p_1 = x\{K\zeta + xh\theta\}. \end{aligned} \tag{NC2}$$

These are obtained by substituting $a\eta = 1 - r$ and $t\zeta = \theta$ in the unchanged seven native coquotients. In particular $P_d \geq 1$. The original nonnegative D, β are precisely the inequalities

$$B_5 := K(1+y)(r+\eta) - (\eta T - 1 + r)P_d \geq 0, \qquad B_\beta := Kx(r+\eta)G - \theta P_d \geq 0. \tag{NC3}$$

The exact relative coefficients are

$$\begin{aligned} \frac{C}{d} &= \frac{K(r+\eta)E}{2(1-r)P_d}, & \frac{A}{d} &= \frac{\eta P_4}{2rP_d}, \\ \frac{m}{d} &= \frac{Kx(r+\eta)h(1+y)}{2M\zeta P_d}, & \frac{\alpha}{d} &= \frac{x^2\zeta(r+\eta)P_\alpha}{2QP_d}, \\ \frac{D}{d} &= \frac{B_5}{2J\eta P_d}, & \frac{\beta}{d} &= \frac{B_\beta}{2\theta P_d}. \end{aligned} \tag{NC4}$$

The source inequalities make every denominator positive; their sizes will enter the bounds below.

Lemma B.2 (A uniform native gap)**.** *On every physical native C1 contact oriented by* $0 < k \leq 1$*,*

$$E \geq \frac{1}{18}. \tag{NC5}$$

This includes the literal boundaries $D = 0$ *or* $\beta = 0$*.*

Proof. Define

$$\mathcal{A} = \theta T + K\zeta > 0, \qquad \mathcal{C} = RG(1 + rxh),$$

$$\begin{aligned} \mathcal{B} = {} & h^2rx^2 + hrx(2-h) + h^2x + hrx^2\zeta + hx\zeta + hx + h \\ & + rx\zeta + x\zeta + 2\zeta + 1. \end{aligned}$$

Direct multiplication of (NC2)–(NC3), using $x\eta h = 1 - E$, gives the polynomial identity

$$hB_5 = -\mathcal{A}E^2 + \mathcal{B}E - \mathcal{C}. \tag{NC6}$$

Every summand in $\mathcal{B}$ is nonnegative. Since $0 < x, r, h \leq 1$ and $h(2-h) \leq 1$, its six terms without ζ sum to at most 6 and the coefficients of ζ sum to at most 6. Hence $\mathcal{B} \leq 6(1+\zeta)$. The native inequality $\theta = t\zeta < 2\zeta$ gives $1 < h + 2\zeta$, so $1 + \zeta < 3G$. Also $\mathcal{C} \geq G > 0$. From $B_5 \geq 0$ and (NC6),

$$G \leq \mathcal{C} \leq \mathcal{B}E \leq 18GE.$$

Cancel the positive pure source G. This proves (NC5) without inverting an opening or assuming any of η, ζ bounded. □

An elementary division-free elimination. Keep p_0, p_1 from (NC2), and set

$$a_2 = Kx\zeta - Tp_1, \qquad a_1 = K(1 + x\zeta r) - Tp_0 + (1 - r)p_1, \qquad a_0 = Kr + (1 - r)p_0,$$
$$b_1 = KxG - \theta p_1, \qquad b_0 = KxrG - \theta p_0.$$

Then $B_5 = a_2\eta^2 + a_1\eta + a_0$ and $B_\beta = b_1\eta + b_0$. Expanding these two expressions, or just eliminating their quadratic and linear terms successively, gives

$$\begin{aligned} b_1^2 B_5 - a_2 B_\beta^2 - (b_1 a_1 - 2a_2 b_0) B_\beta \\ &= b_1^2 a_0 - b_1 a_1 b_0 + a_2 b_0^2 \\ &= x^2 h^2 K^2 R G^2 \{1 + 2xh + rx^2 h^2 + (1 - r)x\zeta\}. \end{aligned} \tag{NC7}$$

The last equality is a short polynomial identity in x, r, h, ζ; all five of its input coefficients are given above. No inverse of b_1, no resultant premise, and no localization is used. For reference, $b_1 = xh\{K(1+\zeta) - x\theta^2\} > 0$ as well.

Theorem B.3 (Actual unit/unit C1 contacts enter the five-event exit)**.** *If both actual C1 half-angles k, v_c are positive bounded units, the original source has five consecutive events bounded below relative to d. Consequently its entire original parent has the inherited all-tests tensor-SOS certificate. All upper event scales and the retained zero-coefficient boundaries are allowed.*

Proof. After the exact Alice orientation, x, y are bounded positive units. Thus $\eta\zeta = y/x$ is a bounded positive unit. If η were unbounded, ζ would be infinitesimal. Then $\theta = t\zeta$ would be infinitesimal, contradicting $x\eta(1 - \theta) < 1$. If ζ were unbounded, η would be infinitesimal and $r = 1 - a\eta$ would have residue one, contradicting $x\zeta r < 1$. Therefore η, ζ are both bounded positive units.

All polynomials in (NC2)–(NC3) now have bounded coefficients and bounded arguments. If h were infinitesimal, the exact specialization

$$B_\beta\big|_{\theta=1} = -1 - x\zeta(1 - r) \le -1$$

would contradict $B_\beta \ge 0$. Thus h is a positive bounded unit. The uniform cap gives that E is a unit. Equations (NC2) make P_4, P_α, P_d positive bounded units: use $P_4 \ge E$, $P_\alpha \ge Rh(1 + y)$, and $P_d \ge 1$.

The first four equations of (NC4) now show that C, A, m, α are all strong relative to d. Their denominators are positive bounded quantities; they may be infinitesimal. Suppose both D/d and β/d were infinitesimal, including a possible literal zero. Their last two bounded denominator products in (NC4) would make B_5, B_β infinitesimal. Every coefficient on the left of (NC7) is bounded, so that side would be infinitesimal. Its right side is a positive unit, since x, h, K, R, G are units and the last bracket is at least one. This is a contradiction.

Thus D or β is strong. The original cycle $(\lambda, D, C, A, \alpha, m, \beta, n)$ has respectively the five consecutive strong events (D, C, A, α, m) or (C, A, α, m, β). Apply the common five-consecutive-event theorem, whose stated scope allows arbitrary upper weights and every independent testing chart. No independent test is identified with an actual source angle in this source-entry argument. □

B.1.4 Consequences of the native source inequalities

This section records source implications of (NC1)–(NC7), not further all-tests certificates. All quantities are in the coefficient field. Write $f \asymp g$ for two strictly positive quantities whose quotient and reciprocal quotient are bounded. Set

$$L = x\zeta, \qquad S = 1 + L + y, \qquad \rho = r + \eta.$$

The native signs and (NC5) imply the uniform estimates

$$S \le P_d \le 6S, \qquad \frac{1 + L}{54} \le P_4 \le 3(1 + L), \qquad P_\alpha \asymp h(1 + y) + \zeta M. \tag{NI1}$$

Indeed $rx^2\zeta < x$ follows from $Q > 0$, and $\eta x^2 h\theta = x\theta(1 - E) \le 1$, giving the P_d bounds from (NC2). For P_4, if $L \le 2$ use $E \ge 1/18$; if $L \ge 2$, then $r < 1/L \le 1/2$ and $P_4 \ge L/2$. Its upper bound follows immediately from (NC2). The last comparison uses $1 \le R \le 2$ and $1/18 \le E < 1$. Consequently the unchanged original source satisfies

$$C/d \asymp \frac{\rho}{(1 - r)S}, \qquad A/d \asymp \frac{\eta(1 + L)}{rS},$$
$$m/d \asymp \frac{x\rho h(1 + y)}{M\zeta S}, \qquad \alpha/d \asymp \frac{x^2\zeta\rho\{h(1 + y) + \zeta M\}}{QS}. \tag{NI2}$$

No opening has been bounded away from zero in these statements. In particular, E and P_4 do not supply additional hidden small source denominators on C1.1, C1.4 or C1.5.

C1.1: two unavoidable large weights and its actual angle. Here x, ζ are infinitesimal and η is unbounded. The common original-D cap gives $q = x\eta$ bounded with residue at most $1/2$. Hence $y = q\zeta$ is infinitesimal: the actual angle type is necessarily (I, I), independently of e. The exact formulas give

$$C/d \asymp \frac{\eta}{1-r}, \qquad A/d \asymp \frac{\eta}{r}, \qquad m/d \asymp \frac{e^2}{M}, \qquad \alpha/d \asymp xq\zeta, \qquad \beta/d + \frac{1}{2} \asymp \frac{e^2}{t}. \tag{NI3}$$

Thus C, A are unbounded relative to d, and α is weak. The common bounded polynomial formula for B_5 gives $D/d = O(1/\eta)$, so D is weak too, including its literal zero. In (NI3), $\beta \geq 0$ forces $t = O(e^2)$. Therefore when e is infinitesimal, t is infinitesimal, M is a unit and m is weak. When e is a unit, m is strong; when e is unbounded, both m, β are unbounded. These are restrictions on the original C1.1 row, not new certificates. Its positive infinitesimal δ_{tot} condition is automatic because $C/d, A/d$ are unbounded.

C1.4: the natural scale is $L = x\zeta$. Now x is infinitesimal and ζ is noninfinitesimal. The following implications follow uniformly through all retained coefficient boundaries.

1. If L is unbounded, $Q > 0$ forces r infinitesimal, and $1 - r = a\eta < 2\eta$ makes η noninfinitesimal. Hence $y = L\eta$ is unbounded: the actual type is (I, H). Here ζ is unbounded, $t = \theta/\zeta$ is infinitesimal, and M is a unit. Since $hy/\zeta = x\eta h < 1$, (NC2) gives $P_\alpha \asymp \zeta$. Equations (NI1)–(NI2) then yield

$$C/d \asymp L^{-1}, \qquad A/d \asymp r^{-1}, \qquad \alpha/d \asymp L/Q, \qquad m/d \asymp (x\eta h)/\zeta \leq O(\zeta^{-1}). \tag{NI4}$$

   In particular C, m are weak, A, α unbounded, and $(m/d)/(C/d) = O(x)$ is infinitesimal. The original total-scale intersection is automatic.

2. If L is a bounded unit, ζ is unbounded, M is a unit, $P_\alpha \asymp \zeta$, and $S \asymp 1 + \eta$. The identity $1 = r + a\eta$ gives $\rho/(1+\eta)$ a positive bounded unit. Thus

$$C/d \asymp (1-r)^{-1}, \qquad \alpha/d \asymp Q^{-1}, \qquad A/d \asymp \frac{\eta}{r(1+\eta)}, \qquad m/d = O(x). \tag{NI5}$$

   So C, α are strong and m weak. Moreover A is weak exactly when η is infinitesimal; that case has r a unit and D/d unbounded, by $B_5 \sim 1$ in (NC3). The actual Bob angle is infinitesimal, a unit, or unbounded exactly as η is, because $y = L\eta$. No restriction on D, β is added in the other cases.

3. If L is infinitesimal and η is bounded, then y is infinitesimal. The $B_\beta \geq 0$ equation bounds $\theta \leq Kx\rho G/P_d = O(x+L)$, so h, M are units. Now (NI2) gives C strong, m, α weak, and A weak exactly when η is infinitesimal. In that last case D/d is unbounded, as in the preceding item. The actual type is (I, I).

4. If L is infinitesimal and η is unbounded, then

$$C/d \asymp \frac{\eta}{(1-r)(1+y)}, \qquad A/d \asymp \frac{\eta}{r(1+y)}. \tag{NI6}$$

   Both are unbounded since $\eta/(1+y) = (\eta^{-1} + L)^{-1}$. Also Q is a unit and (NI2) bounds

$$\alpha/d \leq O\left(x^2\zeta\eta h + \frac{x^2\zeta^2\eta}{1+y}\right) = O(L),$$

   so α is weak. Finally $J \sim 1$ and $P_d/(1+y) \sim 1$ directly in (NC2); (NC4) gives $D/d + 1/2 \sim 1/2$, hence D weak. This extends the C, A-large/D, α-weak source pattern of C1.1 into C1.4, without claiming that its existing conductors apply. Here y may be infinitesimal, a unit, or unbounded. The total-scale intersection is again automatic.

These four alternatives exhaust C1.4, but they are necessary source reductions only. In particular no entire C1.4 leaf is closed.

C1.5 after the actual unit/unit certificate. Here k is a unit. The new theorem removes actual (U, U); the common both-η, ζ-infinitesimal theorem is also subtracted. Two useful remaining source types can be specified more tightly.

If v_c is infinitesimal, $\eta\zeta$ is infinitesimal. Neither η nor ζ can be unbounded: ζ infinitesimal forces θ infinitesimal and excludes unbounded η by $E > 0$; η infinitesimal gives $r \sim 1$ and excludes unbounded ζ by $Q > 0$. Thus exactly one of η, ζ is a unit in this residual subcase. The exact stabilizer (AS5) exchanges these two alternatives. It keeps k a bounded unit, although a second orientation need not keep $k \leq 1$. Choose the representative η a unit and ζ infinitesimal. Then

$$m/d, \beta/d \text{ are unbounded}, \qquad C/d, A/d \text{ are strong}, \qquad \alpha/d \text{ is weak}. \tag{NI7}$$

Indeed $\theta = t\zeta$ is infinitesimal, E is a unit, $P_d, P_4, P_\alpha, r+\eta$ are units, and (NC4) gives $m/d \asymp 1/(M\zeta)$ and $\beta/d + 1/2 \asymp 1/\theta$. The reflected representative also has a unit E: its exact value is $E' = h(1+y)/G$ in the original coordinates, with h a unit by the same $\theta = 1$ exclusion used in the ordinary-angle proof. There is no conclusion about the size of D/d in (NI7).

If v_c is unbounded, both η, ζ are noninfinitesimal and at least one is unbounded. To see the first assertion, an infinitesimal ζ would force $h \sim 1$ and η bounded by $E > 0$; an infinitesimal η would force $r \sim 1$ and ζ bounded by $Q > 0$. Two rate regimes are then useful.

- If η is unbounded and ζ a unit, then h is infinitesimal. The exact alternative numerator identity

$$B_\beta = (x\eta h)\{K(1+\zeta) - x\theta^2\} + xrGT - \theta(1 + 2xh + x\zeta) \tag{NI8}$$

  shows that $x\eta h$ is a unit: otherwise its residue at $\theta = 1$ would be $-1 - x\zeta(1-r) < 0$. Thus P_α is a unit and $P_d \asymp \eta$. Equations (NC4) give

$$C/d \asymp (1-r)^{-1}, \quad A/d \asymp r^{-1}, \quad m/d \asymp M^{-1}, \quad \alpha/d \asymp Q^{-1}, \quad D/d, \beta/d \text{ weak}. \tag{NI9}$$

  For the last assertion use $J \sim 1$, $\theta \sim 1$ and $P_d/(Ky) \sim 1$ in the original $D/d + 1/2, \beta/d + 1/2$ formulas. The total-scale condition δ_{tot} infinitesimal is therefore equivalent on this typed branch to at least one of $r, 1-r, M, Q$ being infinitesimal. If all four are units, the inherited positive-total-scale theorem already applies.

- If ζ is unbounded, r is infinitesimal and M a unit. The noninfinitesimal η obeys $\eta h \asymp 1$. For if θ is infinitesimal this follows from $E > 0$ and $h \sim 1$; otherwise divide (NI8) by ζ, use $xrG/\zeta = O(r)$, and obtain that $x\eta h$ has a positive-unit lower bound. Therefore (NC2) and (NC4) give

$$C/d \asymp \zeta^{-1}, \quad m/d \asymp \zeta^{-1}, \quad A/d \asymp r^{-1}, \quad \alpha/d \asymp \zeta/Q. \tag{NI10}$$

  Thus C, m are weak and A, α unbounded. No size restriction on D, β is inferred. This branch already has infinitesimal δ_{tot}.

All implications retain the original physicality and exact source openings.

B.1.5 The bound on $x\eta$

Throughout, d is the original positive signed coefficient. The event coefficient at the fourth cycle position is denoted A_{ev}, and $D_{\mathrm{ev}} = e_5$. These must not be confused with actual state coordinates. The source has the canonical omissions $\lambda = n = 0$ and is normalized by $c = e_3 = 2$. We use the canonical actual-contact sign entry, including its covered single-zero D_{ev} or β boundaries. Explicitly, the retained $m, c, A_{\mathrm{ev}}, \alpha$ are positive, and D_{ev}, β are positive or exactly one of them is zero. The noncoercive further-zero faces are already separate exits. The sign and nonsingularity input is the native-entry theorem EN in Chapter 5, proved for every finite noncommuting physical contact in this exact coefficient domain.

Exact transport from the actual contact. Write its positive-angle contact as

$$u_c = k, \quad v_c = \ell, \qquad \psi_c = (S, k(\theta-1), kSr, 1), \quad S > 0, \quad 0 < r, \theta < 1.$$

The exact C1 reflection lets us orient $0 < k \le 1$. Define source-only parameters

$$\eta = \frac{\ell}{kS}, \quad \zeta = \frac{S\ell}{k}, \quad f = \sqrt{\eta}, \quad g = \sqrt{\zeta}, \quad e = \frac{k}{S} = \frac{kf}{g}, \quad a = \frac{1-r}{\eta}, \quad t = \frac{\theta}{\zeta}. \tag{CE1}$$

Thus the classifier $\mathfrak{T} = k/(S\ell)$ is exactly ζ^{-1}, and its old state-ratio variable B is exactly $\eta^{-1/2}$. Put $\kappa = k^2$, $K = 1 + \kappa$. The original amplitude signs imply

$$R = 1 + \kappa r, \quad T = 1 + \kappa(1-\theta), \quad M = R - t > 0, \quad J = T - a > 0,$$

$$E = 1 - \kappa\eta(1-\theta) > 0, \qquad Q_c = 1 - \kappa\zeta r > 0. \tag{CE2}$$

Indeed the positive second amplitude is $k\zeta M$, the negative fifth amplitude is $-\ell J$, the state gap is SE, and the remaining state gap is Q_c. In particular $0 < a, t < 2$, without any shape-unit hypothesis.

The seven native coquotients used by the uniform theorem are valid under (CE1), not just asymptotically. Their constructed contact is

$$(W, X, Y, Z) = (\Delta k, -k(1 - t\zeta), \Delta k^2(1 - a\eta), 1), \qquad \Delta = e^{-1} = S/k,$$

which is exactly the original contact. The native contact identities prove all six kernel and angular equations. To make the nonsingularity premise explicit, put $Y_0 = Sr$, $h_0 = S + k\ell(\theta - 1)$, $R_2 = \ell(S + k^2 Y_0) - k\theta$, and $R_5 = k(S - Y_0) + k^2\ell(\theta - 1) - \ell$. The common selected six-column contact system, on $(d, m, A_{\mathrm{ev}}, D_{\mathrm{ev}}, \alpha, \beta)$ with $c = 2$, has determinant

$$-8192\theta k^2 \ell (1 + k^2)^2 (1 + \ell^2)^2 \mathcal{D},$$

$$\mathcal{D} = Y_0(1 + k^2)(kY_0 + \ell)(1 - k\ell Y_0) h_0 R_2(-R_5) > 0.$$

Each factor is strictly positive by the stated source sign theorem. Thus the normalized coefficients equal the unique original source coefficients in that common system. This establishes full original-coefficient transport, including the exact zero openings; it does not use an archived completion label.

A bounded polynomial consequence of the original fifth event. Assume now

$$k \text{ infinitesimal}, \qquad \zeta = \mathfrak{T}^{-1} \text{ infinitesimal}. \tag{CE3}$$

Set $q = \kappa\eta$. Since $t < 2$ and $E = 1 - q(1 - t\zeta) > 0$, q is bounded and $\bar{q} \le 1$. The original fifth-event formula is

$$D_{\mathrm{ev}} = \frac{a}{Jc_3}(2c_5 - J\eta c_d).$$

Using $L_c = r$, $H_c = r + \eta$ and $V_c = q\zeta$, its sign is precisely

$$\mathcal{B}_5 := K(1 + q\zeta)(r + \eta) - J\eta P_d \ge 0. \tag{CE4}$$

Although η need not be bounded, this whole expression has the following bounded polynomial form. Define

$$\begin{aligned} H &= 2 + \kappa r + \zeta[K - (\kappa + 2)t] + q\zeta(-\zeta t^2 + t + 1) + \kappa\zeta(1 - r)(t - 1), \\ P_d &= 1 + q\zeta + \kappa H. \end{aligned}$$

Then exact cancellation using $a\eta = 1 - r$ and $q = \kappa\eta$ gives

$$\mathcal{B}_5 = Kr(1 + q\zeta) + (1 - r)P_d + q\{\zeta t(1 + q\zeta) - (K - \kappa\zeta t)H\}. \tag{CE5}$$

Every variable on the right is bounded. Hence (CE4) gives

$$1 - 2\bar{q} \ge 0, \qquad 0 \le \bar{q} \le \tfrac{1}{2}. \tag{CE6}$$

This is obtained before dividing by $1 - q$ or by a coefficient opening. In particular E is a positive unit with residue at least $1/2$.

The fourth coquotient polynomial likewise has the bounded expression

$$\begin{aligned} P_4 = {} & K(1 - q) + \kappa(1 - r) - \kappa\zeta t \\ & + q\zeta(-\kappa\zeta t^2 + 2\kappa t + t) + \kappa\zeta(1 - r)(1 - t). \end{aligned} \tag{CE7}$$

Thus $\bar{P}_d = 1$ and $\bar{P}_4 = 1 - \bar{q} \ge 1/2$. These exact low-degree identities are the physicality input needed for the entry, rather than an estimate on a truncated coefficient.

B.2 Contact identities and complementary measurement charts

The scalar contact polynomial supplies the first square roots. Polynomial division then expresses state couplings through these roots. The complementary measurement charts follow directly from the positive event terms and will be used throughout the remaining proof.

B.2.1 Polynomial source coordinates and exact contact balances

The letter d in the contact–conductor proof denotes the new parameter δ, not the Bell coefficient. Here the latter is denoted d_{B}, and the original event coefficient e_5 is denoted D_{e}. The testing factor is always $D = 1 + u^2$. The conductor monomial $C = \delta b^2 uw$ is distinct from the clearing multiplier $\mathcal{C} = K^6 Q^4 T_n$.

Scope of the polynomial identities. The seven coquotient formulas and the contact identities (SD9)–(SD11) are exact for arbitrary positive native parameters. The size assumptions below are needed only for the displayed bounded-Gram application.

Domain and seven native coquotients. The parameters $k, f, g > 0$ belong to the source field; put $e = kf/g$ and $\Delta = e^{-1}$. The formulas are exact before a size condition is imposed. The shapes a, t are positive bounded elements;

$$R = 1 + k^2(1 - af^2), \qquad T = 1 + k^2 - k^2 g^2 t, \qquad M = R - t > 0, \quad J = T - a > 0.$$

All original event coefficients are assumed nonnegative. In particular, the formulas below retain the requirements $D_{\mathrm{e}}, \beta \geq 0$. No relation compares a testing rate with a coefficient rate. Put

$$\kappa = k^2, \quad \eta = f^2, \quad \zeta = g^2, \quad V_c = \kappa\eta\zeta, \quad L_c = 1 - a\eta, \quad H_c = 1 + (1 - a)\eta, \quad Q_c = 1 - \kappa\zeta + aV_c.$$

Define

$$\begin{aligned}
P_4 &= 1 + \kappa + \eta\kappa(a - \kappa - 1) - \zeta\kappa t + V_c[-\zeta\kappa t^2 + 2\kappa t + a(1 - t) + t],\\
P_\alpha &= 1 + \kappa - a\eta\kappa + \zeta[(1 + \kappa) - (2 + \kappa)t] + V_c[-\zeta t^2 + a(t - 1) + t],\\
P_d &= (1 + \kappa)^2 - a\eta\kappa^2 + \zeta\kappa[(\kappa + 1) - (\kappa + 2)t]\\
&\quad + V_c[-\zeta\kappa t^2 + \kappa a(t - 1) + \kappa t + \kappa + 1].
\end{aligned}$$

The complete seven coquotients are

$$\begin{aligned}
c_2 &= 2(1 + \kappa)(1 - t\zeta)L_c(1 + V_c)H_cQ_c,\\
c_b &= 2(1 + \kappa)L_c[1 + (1 - t)\zeta]H_cQ_c,\\
c_3 &= 2(1 + \kappa)L_cH_cQ_c(1 - \kappa\eta + tV_c),\\
c_5 &= 2(1 + \kappa)L_cH_cQ_c(1 + V_c),\\
c_4 &= 2Q_cP_4, \qquad c_\alpha = 2L_cH_cP_\alpha, \qquad c_d = 4L_cQ_cP_d.
\end{aligned}$$

At $\eta = \zeta = 0$, the first six equal $2(1 + k^2)$ and $c_d = 4(1 + k^2)^2$. Here c_b is the original c_β, and the subscript on c_d is a name rather than the conductor parameter.

Original source and full parent. The original event coefficients and multiplier are

$$\begin{aligned}
&\lambda = n = 0, \qquad c = 2, \qquad d_{\mathrm{B}} = \frac{2a\eta c_d}{c_3}, \qquad m = \frac{2ae^2c_2}{Mc_3}, \qquad A = \frac{2a\eta^2c_4}{c_3},\\
&\alpha = \frac{2a\Delta^2k^6\eta^2c_\alpha}{c_3}, \qquad D_{\mathrm{e}} + \frac{d_{\mathrm{B}}}{2} = \frac{2ac_5}{Jc_3}, \qquad \beta + \frac{d_{\mathrm{B}}}{2} = \frac{2ae^2c_b}{tc_3}, \qquad \Lambda = \frac{RTc_3}{2a}.
\end{aligned} \tag{SD1}$$

Let F denote the original complete parent, including its signed core. For a testing state (W, X, Y, Z) define

$$\begin{aligned}
D = 1 + u^2, \quad D_v = 1 + v^2, \quad B = X + uZ, \quad Q_A = W + uY,\\
r = uW - Y, \quad r_2 = vQ_A - B, \quad r_5 = r - v(Z - uX).
\end{aligned}$$

With $H_2 = 4Tc_2$, $H_b = 4Tc_b$, $H_3 = 4Rc_3$, $H_5 = 4Rc_5$, the exact entire parent is

$$\begin{aligned}
\Phi := \Delta^2\Lambda F = {} & H_2\frac{R}{M}r_2^2 + H_b\frac{R}{t}D_vB^2 + \Delta^2H_3\frac{T}{a}D_vr^2 + \Delta^2H_5\frac{T}{J}r_5^2\\
&+ 4RT\Delta^2\eta^2c_4DD_vY^2 + 4RT\Delta^4k^6\eta^2c_\alpha D(vY - Z)^2\\
&+ 2RT\Delta^2\eta c_dD\{(vY - Z)^2 - D_vZ^2\}.
\end{aligned} \tag{SD2}$$

Thus every original source event is accounted for.

Retained terms and state coordinates. In the remaining chart $Z = 1$ with u, v, W, X, Y infinitesimal, put

$$p = t/R, \quad h = a/T, \quad q_A = v(W + uY), \quad q_B = v(1 - uX), \quad N = B - pq_A, \quad M_B = r - hq_B.$$

Retain the four nonnegative terms

$$\Pi_1 = H_2\frac{t}{M}N^2, \quad \Pi_2 = H_b\frac{M}{t}D_vN^2, \quad \Pi_3 = \Delta^2H_3\frac{J}{a}D_vM_B^2, \quad \Pi_4 = \Delta^2H_5\frac{a}{J}M_B^2. \tag{SD3}$$

Their subtraction is the exact identity

$$\begin{aligned}
G = \Phi - \sum_j \Pi_j = {} & H_2\{N^2 - 2q_AN + (1 - p)q_A^2\} + H_bD_v\{N^2 + 2q_AN + pq_A^2\}\\
&+ \Delta^2H_3D_v\{M_B^2 + 2q_BM_B + hq_B^2\} + \Delta^2H_5\{M_B^2 - 2q_BM_B + (1 - h)q_B^2\} + \mathcal{W},\\
\mathcal{W} = {} & 4RT\Delta^2\eta^2c_4DD_vY^2 + 4RT\Delta^4k^6\eta^2c_\alpha D(vY - 1)^2\\
&+ 2RT\Delta^2\eta c_dD\{(vY - 1)^2 - D_v\}.
\end{aligned} \tag{SD4}$$

The testing coordinate $w = (W + uY)/D$ has the full inverse

$$\begin{aligned} X &= N + pDvw - u, \qquad q_B = v(D - uN - puDvw),\\ W &= w + \frac{u(M_B + hq_B)}{D}, \qquad Y = uw - \frac{M_B + hq_B}{D}. \end{aligned} \tag{SD5}$$

Its only testing denominator is the positive unit D. On the negative-core branch, sign gauges give $u, v > 0$, and $w > 0$: after eliminating the two positive core squares the remaining scalar is

$$\frac{uvDD_v}{\mathcal{D}}\{Duvw^2 - 2DD_vw + uv\}, \qquad \mathcal{D} = u^2v^4 + u^2v^2 + u^2 + v^4 + 2v^2 + 1 > 0,$$

which is nonnegative if $w \le 0$. The eliminated matrix is $\left(\begin{smallmatrix} 1+v^2+u^2v^2 & uv \\ uv & (D+v^2)/D \end{smallmatrix}\right) \succ 0$. The nonnegative-core branch and literal axes already give direct SOS. Thus the radicals in the conductor proof separate between the two fields.

Short native proof of the exact contact identities. Write $V = 1 + V_c$ and $K = 1 + \kappa$ in this paragraph. The factored coquotients above give immediately

$$c_b - c_2 = \zeta c_3, \qquad c_3 - c_5 = -\kappa\eta c_b, \qquad c_2 = (1 - t\zeta)c_5. \tag{SD9}$$

Hence $Vc_b - c_2 = \zeta c_5$ and $Vc_3 - c_5 = -\kappa\eta c_2$. The remaining three balances are the following short polynomial identities; substituting the displayed P_4, P_α, P_d proves them by multiplication and collecting terms:

$$\begin{aligned} \mathfrak{Y} &:= 2\kappa c_2 + 2VL_cc_4 - 2\kappa^2\zeta Q_cc_\alpha - Q_cc_d = 0,\\ \mathfrak{S} &:= \kappa L_cc_2 + c_5 + VL_c^2c_4 + \kappa Q_c^2c_\alpha + \tfrac{1}{2}(\kappa\zeta L_c^2 - 2L_c - \eta)c_d = 0,\\ \mathfrak{V} &:= \kappa Rc_2 + t\kappa V_cc_b + aV_cc_3 + Tc_5 + KV_cL_c^2c_4 - \kappa^2\zeta KQ_cL_cc_\alpha - \tfrac{K}{2}(Q_cL_c + \eta)c_d = 0. \end{aligned} \tag{SD10}$$

For example, after the common factor $4L_cQ_c$ is removed, the first balance is just

$$\kappa K(1 - t\zeta)VH_c + VP_4 - \kappa^2\zeta H_cP_\alpha - Q_cP_d = 0.$$

The two other balances are equally direct identities in the same three short polynomials; no expanded scalar or ideal calculation enters.

The exact contact, in the original coordinates of (SD2), is

$$u = k, \quad v = \Delta k^2\eta, \qquad (W, X, Y, Z) = (\Delta k, -k(1 - t\zeta), \Delta k^2L_c, 1). \tag{SD11}$$

At this point

$$r_2 = Mk\zeta, \quad B = tk\zeta, \quad r = av, \quad r_5 = -Jv, \quad vY - 1 = -Q_c, \quad D = K, \quad D_v = V.$$

Set $\Psi = \Phi/(4RT)$ and let $E_s = \frac{1}{2}\partial_s\Psi$, with all other original testing coordinates held fixed. Direct differentiation of the seven terms in (SD2), followed by (SD9), gives at (SD11)

$$\begin{aligned} \Psi &= K\zeta^2\mathfrak{S},\\ E_W &= kv\{\zeta c_2 + \Delta^2(Vc_3 - c_5)\} = 0,\\ E_X &= k\zeta\{Vc_b - c_2 - \zeta c_5\} = 0,\\ E_Y &= \tfrac{1}{2}K\Delta^3\kappa\eta^2\mathfrak{Y},\\ E_u &= k\zeta^2\{\mathfrak{S} - c_2 + (1 - t\zeta)c_5\},\\ E_v &= v\Delta^2\mathfrak{V}. \end{aligned}$$

They all vanish by (SD9)–(SD10). Homogeneity in (W, X, Y, Z) also gives $E_Z = 0$. This proves contact and full first-order stationarity directly in the native source.

B.2.2 A finite four-monomial circuit

Let

$$a_1 = \sqrt{wv}, \quad a_2 = \sqrt{\delta uv}, \quad a_3 = b\sqrt{\delta kuw}, \quad a_4 = \delta k^{3/2}b.$$

The radicals are separated coefficient/testing radicals. Put

$$h_{13} = a_1^2 - a_3^2, \quad h_{23} = a_2^2 - a_3^2, \quad h_{34} = a_3^2 - a_4^2, \qquad q_1 = a_1a_2 - a_3a_4, \quad q_2 = a_1a_3 - a_2a_4.$$

Averaging the three-square arithmetic–geometric-mean identities for pairings 12|34 and 13|24 gives exactly

$$\sum_{i=1}^{4} a_i^4 - 4a_1a_2a_3a_4 = h_{13}^2 + h_{23}^2 + h_{34}^2 - h_{13}h_{23} + h_{23}h_{34} + q_1^2 + q_2^2. \tag{B.1}$$

The contrast block is

$$G = \begin{pmatrix} 1 & -1/2 & 0 \\ -1/2 & 1 & 1/2 \\ 0 & 1/2 & 1 \end{pmatrix} \succ 0; \qquad \det G_{[1]} = 1, \quad \det G_{[2]} = 3/4, \quad \det G = 1/2.$$

Unlike a single pairing, the two pairings control all three independent contrasts of the four squared monomials.

Write $K = 1 + k^2$, $D = 1 + u^2$ and

$$g_0 = \delta(v - b^2uw), \qquad g_4 = uwv - \delta^2k^4b^2.$$

The leading scalar used in the complete source is

$$P = D(w^2 + \delta^2)v^2 + \delta^2Kb^4u^2w^2 + \delta^4Kk^6b^4 - 2\delta^2K^2b^2uwv = g_0^2 + g_4^2 + \sum_i a_i^4 - 4\prod_i a_i.$$

There is a second exact simplification:

$$g_4 = uh_{13} + kh_{23} - kug_0 + kh_{34}. \tag{B.2}$$

If k, u are infinitesimal, the four conductor coefficients are infinitesimal. Otherwise retain g_4^2 as a separate square. The two-pairing contrast matrix displayed above is strict in either case.

B.2.3 The restricted three-monomial contact lift

Work in the same tensor ring $T = K_c \otimes_{\mathbb{R}} K_t$ and bounded subring B as throughout this package. Every inverse and radical below is of a pure-field or positive separated element. Let $k > 0$ be a pure-source infinitesimal and $s > 0$ a positive pure-source bounded unit. Let $u > 0$ be a pure-testing infinitesimal and $t > 0$ a positive pure-testing bounded element (possibly infinitesimal). Define

$$A_1 = u^2t, \qquad A_2 = \frac{kut^2}{s}, \qquad A_3 = k^2s, \qquad G = kut, \qquad d_i = A_i - G.$$

Then $A_1A_2A_3 = G^3$. Two exact conductor identities are

$$G(u - k) = kd_1, \qquad G(t - s) = sd_2. \tag{B.3}$$

They permit a unit testing ratio t without comparing a mixed tensor.

Put $Q = G\sum_i d_i$ and $L = \sum_i A_i^2 - 3G^2$. For $a_i = A_i^{1/3}$, the elementary cubic identity gives

$$Q = \frac{1}{2}\sum_{1\le i<j\le 3}\sum_{\ell=1}^{3}\{\sqrt{Ga_\ell}(a_i - a_j)\}^2, \tag{B.4}$$
$$L = \sum_i d_i^2 + 2Q.$$

Thus L has a positive constant diagonal Gram on a finite list of three difference roots and nine cubic-identity roots. No assertion that every nonnegative tensor is SOS is used.

Exact finite lift. Let $\lambda \in B_c$ have positive real residue, and consider

$$F = \sum_{i=1}^{3}(\lambda + \epsilon_i)A_i^2 - (3\lambda + \epsilon_g)G^2, \qquad h = \sum_i \epsilon_i - \epsilon_g.$$

Write $\Delta_u = u - k$, $\Delta_t = t - s$. Assume explicit finite identities

$$\epsilon_i - \epsilon_{i0} = K_{iu}\Delta_u + K_{it}\Delta_t, \qquad \epsilon_i, \epsilon_{i0} \in B_{\mathrm{Inf}}, \tag{B.5}$$
$$h = b_u\Delta_u + b_t\Delta_t + \sum_{j,\ell\in\{u,t\}} H_{j\ell}\Delta_j\Delta_\ell, \qquad b_u, b_t \in K_c, \tag{B.6}$$
$$2(\epsilon_{10} - \epsilon_{30}) + kb_u = 0, \qquad 2(\epsilon_{20} - \epsilon_{30}) + sb_t = 0. \tag{B.7}$$

Require $K_{iu}, H_{uu}, H_{ut}, H_{tu} \in B$ and $K_{it}, H_{tt} \in B_{\mathrm{Inf}}$. The distinction is essential: the second conductor has a unit coefficient, so boundedness alone of the second-coordinate arrays does not suffice.

Then F is tensor SOS. Indeed, expanding $A_i = G + d_i$ and applying (B.3) and (B.7) yields the exact identity

$$\begin{aligned} F - \lambda L = {} & \sum_i \epsilon_i d_i^2 + 2\epsilon_{30} Q + 2\sum_i (kK_{iu} d_1 d_i + sK_{it} d_2 d_i) \\ & + k^2 H_{uu} d_1^2 + ks(H_{ut} + H_{tu}) d_1 d_2 + s^2 H_{tt} d_2^2. \end{aligned} \tag{B.8}$$

In detail, the two bare-G first-order terms are cancelled by $G^2 b_u \Delta_u = k b_u G d_1$ and $G^2 b_t \Delta_t = s b_t G d_2$. All remaining coefficients in (B.8) are bounded infinitesimals. The strict diagonal Gram for λL in (B.4) therefore remains strictly positive in real residue. The bounded-Gram lemma supplies a finite tensor SOS. This argument uses no power series or mixed inverse.

Normal extension. Suppose a coupling to a normalized hard state normal has the exact form $\ell = \sum_i c_i A_i$ with $c_i \in B_{\mathrm{Inf}}$, $\sum_i c_{i0} = 0$, and

$$\sum_i (c_i - c_{i0}) = J_u \Delta_u + J_t \Delta_t, \qquad J_u \in B, \quad J_t \in B_{\mathrm{Inf}}.$$

Then

$$\ell = \sum_i c_i d_i + k J_u d_1 + s J_t d_2$$

has infinitesimal coefficients on the same root list. It can be joined to a hard normal block with strictly positive real residue. Extra already-controlled scalar roots may be appended, provided their coupling coefficients are likewise infinitesimal. The source-specific arrays and hard-block bounds remain separate proof obligations.

Small-L coordinates. For $x = k^2$, $s = \sqrt{\eta\zeta}$ and $t = v/u$, this gives

$$(A_1, A_2, A_3, G) = \left(uv, \sqrt{\frac{x}{\eta\zeta}} \frac{v^2}{u}, \sqrt{\eta\zeta}\, x, \sqrt{x}\, v \right).$$

The lemma applies only when the stated pure-field size assumptions and all finite scalar and normal identities have been proved. A leading three-term AM–GM form alone does not close a source domain.

B.2.4 The complementary projective testing charts

Normalize the original parent by its positive pure-source coefficient d. Write $D_z = 1 + z^2$, and write the signed core as $2K$, where

$$K = (uW + uvX - Y - vZ)^2 + D_u(vY - Z)^2 + D_v(X + uZ)^2 - D_u D_v Z^2. \tag{CE1}$$

The event terms used below are

$$4m(vW - X + uvY - uZ)^2, \quad 4CD_v(uW - Y)^2, \quad 4AD_u D_v Y^2, \quad 4\alpha D_u(vY - Z)^2.$$

Every omitted event and every nonnegative excess is retained as its exact original square. Source and testing fields remain independent. “Strong” means bounded below by a positive real number; “unbounded” means larger than every real number. Neither assertion constrains upper scales.

Theorem B.4 (Two adjacent unbounded events, with complete raw frames)**.** *If C, A are unbounded, the full original parent is tensor SOS at every projective test for which u is not infinitesimal, and every state. No physicality assumption beyond nonnegative event weights is needed for this particular assertion. By the exact cyclic relabeling below, if A, α are unbounded, the conclusion holds whenever v is not projectively unbounded.*

Proof. Select $C = A = \mu$, with a positive real μ chosen after the real testing residues; unboundedness permits every such choice. The following are exact full quadratic forms, before any restriction to a state ray or contact. They therefore retain every first state normal and hard-normal coupling.

For u a bounded nonzero unit and v bounded, use the four raw roots

$$(W, Y, B, V), \qquad B = X + uZ, \quad V = vZ.$$

The entire core equals

$$K = (uW - Y + uvB)^2 + D_v B^2 + D_u v^2 Y^2 + D_u u^2 V^2 - 2D_u uV(W + vB). \tag{CE2}$$

The two selected event forms remain $4\mu D_v(uW-Y)^2$ and $4\mu D_uD_vY^2$. In their common real kernel $W=Y=0$, the core has matrix on (B,V)

$$\begin{pmatrix} 1+v^2D_u & -D_uuv \\ -D_uuv & D_uu^2 \end{pmatrix}, \qquad \det = D_uu^2>0. \tag{CE3}$$

Here and in residue matrices, bounded testing letters mean their real residues. Thus a sufficiently large finite μ gives a strictly positive residue on the entire four-root space, even when $v=0$ literally.

For $p=1/u$ infinitesimal and v bounded, homogenize by p^2 and take

$$(W,Y,B,R), \qquad B=pX+Z, \quad R=vX.$$

The exact full core is

$$K_p = B^2-2pvBW-2vBY+D_pR^2+2D_pRW+(W-pY)^2+D_pv^2Y^2. \tag{CE4}$$

The selected terms are $4\mu D_v(W-pY)^2$ and $4\mu D_pD_vY^2$. At $p=0$ their sum with $2K_p$ is

$$4\mu D_v(W^2+Y^2)+2(W+R)^2+2(B-vY)^2, \tag{CE5}$$

strict for every $\mu>0$. The formula allows literal $p=0$ and $v=0$; the raw roots need not form independent state coordinates.

For projectively unbounded v, put $w=1/v$, homogenize by w^2, and use (W,X,Y,H) with $H=wZ$. The exact full core is

$$\begin{aligned} K_w = {} & u^2H^2-2u^2HY-2uHW+2uwHX+u^2w^2W^2 \\ & +2u^2wWX-2uw^2WY+(D_u+w^2)(X^2+Y^2)-2uwXY. \end{aligned} \tag{CE6}$$

The selected terms are $4\mu D_w(uW-Y)^2$ and $4\mu D_uD_wY^2$. If u is a bounded nonzero unit, at $w=0$ their common kernel has $W=Y=0$ and $K_w=D_uX^2+u^2H^2$ strictly positive. A sufficiently large real μ again works. If also $p=1/u$ is infinitesimal, homogenize (CE6) by p^2. The exact full core is

$$\begin{aligned} K_{pw} = {} & H^2-2pHW+2pwHX-2HY+w^2W^2+2wWX-2pw^2WY \\ & +(D_p+p^2w^2)(X^2+Y^2)-2pwXY. \end{aligned} \tag{CE7}$$

At $p=w=0$ the full selected residue is

$$4\mu(W^2+Y^2)+2X^2+2(H-Y)^2>0. \tag{CE8}$$

All these exact Gram coefficients are bounded. Strictness in (CE3) and its counterpart following (CE6) implies a finite cap by the elementary real Schur-complement argument: the event Gram is positive definite on the complement of its kernel, and the core is positive definite on the kernel. No source/testing rate comparison is involved. Apply the bounded positive-residue Gram lemma, and restore every retained original square.

For the second assertion, compose the common Alice reflection with the swap/reflection. The resulting one-step inverse cycle rotation is

$$(u',v')=(1/v,u), \quad (W',X',Y',Z')=(Y+vZ,-W-vX,Z-vY,vW-X), \tag{CE9}$$

and sends event index i to $i-1$ in $(\lambda,D,C,A,\alpha,m,\beta,n)$. In particular original A,α become new C,A. Its full-parent identity is

$$v^2F(\mathrm{rot}^{-1}c;1/v,u;\psi')=D_vF(c;u,v;\psi). \tag{CE10}$$

The state map has determinant of absolute value D_v^2, and its homogeneous version at $v=\infty$ is invertible as well. Thus $u'=1/v$ not infinitesimal is exactly v not projectively unbounded. Only pure testing factors are restored. Both signs, all literal endpoints, and complex states (by real/imaginary splitting) are included. □

Theorem B.5 (Four consecutive strong events)**.** *For an original physical parent with C,A,α,m strong, the full parent is tensor SOS at every test except possibly u a bounded nonzero unit and v projectively unbounded. All original upper coefficient scales and all nonnegative unselected coefficients are allowed.*

Proof. Cap these four events at any positive real μ smaller than all four weights, and retain every excess. Noncommuting-residue tests use the common physical-source theorem, applied to the original full parent. For commuting-residue tests use the capped parent.

First suppose v is bounded. On the three open commuting axes, the core and the four caps have zero common kernel: at $u=0$, the core forces $X=Y=0$, then m and α force $W=Z=0$ when v is a unit; at $u=\infty$,

C, A force $W = Y = 0$, then the core forces $X = Z = 0$; at $v = 0$, A, α force $Y = Z = 0$, then the core forces $W = X = 0$ when u is a unit. These are strictly positive bounded state Grams.

At $(u, v) = (0, 0)$ the residue is

$$(2 + 4\mu)X^2 + (2 + 8\mu)Y^2 + 4\mu Z^2.$$

Only the W ray remains. Normalize $W = 1$ and take the five infinitesimal roots (u, v, X, Y, Z). The full polynomial has no constant or linear term; its quadratic part is

$$2(u - Y)^2 + 2X^2 + 4\mu\{(v - X)^2 + (u - Y)^2 + Y^2 + Z^2\}, \tag{CE11}$$

which is strictly positive. At $(u, v) = (\infty, 0)$ only the X ray remains. Normalize $X = 1$ and use (p, v, W, Y, Z). Its quadratic part is

$$2(W + v)^2 + 2(p + Z)^2 + 4\mu\{(p + Z)^2 + W^2 + Y^2 + Z^2\}, \tag{CE12}$$

also strictly positive. All higher monomials in each chart allocate to products of two displayed roots with infinitesimal bounded coefficients. This is an exact finite Gram, not a truncation certificate.

For $w = 1/v$ infinitesimal use the full raw frame of (CE6). All four selected terms, with every normal present, are exactly

$$4\mu\{(W - wX + uY - uH)^2 + D_w(uW - Y)^2 + D_u D_w Y^2 + D_u(Y - H)^2\}. \tag{CE13}$$

At $u = w = 0$ their sum with $2K_w$ has residue

$$4\mu W^2 + 2X^2 + (8\mu + 2)Y^2 + 4\mu(Y - H)^2, \tag{CE14}$$

strict on all four roots. If $p = 1/u$ is infinitesimal, homogenize again; the exact terms become

$$4\mu\{(pW - pwX + Y - H)^2 + D_w(W - pY)^2 + D_p D_w Y^2 + D_p(Y - H)^2\}. \tag{CE15}$$

Together with (CE7), at $p = w = 0$ the residue is

$$4\mu W^2 + 2X^2 + 4\mu Y^2 + (8\mu + 2)(Y - H)^2, \tag{CE16}$$

again strict. These frames require no state normalization, and no division by either small testing coordinate. Literal $w = 0$ is included. The only chart not covered is ordinary u and infinitesimal w. Restore the original terms and all pure factors as above. □

Lemma B.6 (Matched weak outer events at the large-Bob corner)**.** *Suppose $q > 0$ is a pure-source infinitesimal and*

$$C \geq cq^2, \quad m \geq m_* q^2, \quad A \geq a_*/q^2, \quad \alpha \geq b_*/q^2 \tag{CE17}$$

for positive real c, m_*, a_*, b_*. At projectively unbounded v, the complete parent is tensor SOS whenever u is infinitesimal or projectively unbounded. In the latter chart the lower bound on m is unnecessary.

Proof. Keep $w = 1/v$ and $H = wZ$ as in (CE6)–(CE7). If u is infinitesimal, use the five raw roots

$$E = qW, \quad X, \quad F = Y/q, \quad G = H/q, \quad V = uwW.$$

The exact whole core (CE6) becomes

$$\begin{aligned} K_w = {} & V^2 + (D_u + w^2)X^2 + q^2\{u^2(G - F)^2 + D_w F^2\} \\ & - 2uGE + 2quwX(G - F) + 2uVX - 2uw^2 EF. \end{aligned} \tag{CE18}$$

Cap only m, A, α at the indicated weights. Their complete terms are respectively

$$4m_*(E - qwX + uq^2(F - G))^2, \quad 4a_* D_u D_w F^2, \quad 4b_* D_u(F - G)^2.$$

The selected residue is

$$2V^2 + 2X^2 + 4m_* E^2 + 4a_* F^2 + 4b_*(F - G)^2 > 0. \tag{CE19}$$

Every coefficient in (CE18) is bounded independently of all relative source/testing rates.

If instead $p = 1/u$ is infinitesimal, use

$$E = qW, \quad F = Y/q, \quad G = H/q, \quad R = X + wW, \quad S = pX.$$

The exact whole core (CE7) is

$$K_{pw} = R^2 + D_w S^2 + q^2(G - F)^2 + p^2 D_w q^2 F^2 - 2pE(G + w^2 F) + 2qwS(G - F). \tag{CE20}$$

Cap only C, A, α. Their complete terms are

$$4cD_w(E - pq^2 F)^2, \quad 4a_* D_p D_w F^2, \quad 4b_* D_p(F - G)^2.$$

Together with $2K_{pw}$ their residue is

$$2R^2 + 2S^2 + 4cE^2 + 4a_* F^2 + 4b_*(F - G)^2 > 0. \tag{CE21}$$

Only the pure-source inverse $1/q$ occurs in a root. These exact raw frames include every state normal and every literal endpoint. The strict-residue lemma and restoration of excess terms prove the claim. □

Exact application to the original C1 source rows. The source premises here are precisely the already proved original formulas (NC1)–(NC7), (NI3)–(NI10); no independent source family is substituted. These common theorems give the following new quantified test reductions, always intersected with prior certificates:

1. C1.1, every e type: (NI3) gives C, A unbounded; only u infinitesimal remains.
2. C1.4, L infinitesimal and η unbounded: (NI6) again gives C, A unbounded; only u infinitesimal remains.
3. C1.4, L unbounded: (NI4) and (CE20) leave only u bounded and v unbounded.
4. C1.5 actual UH, η unbounded and ζ unit: (NI9) gives four strong events; only u unit and v unbounded remains.
5. C1.5 actual UH, ζ unbounded: (NI10) and (CE17) leave the same sole strip, u unit and v unbounded.

For the last line choose $q = \zeta^{-1/2}$. Formula (NI10) gives $C, m \asymp q^2$, $\alpha \asymp 1/(q^2 Q)$ and $A \asymp 1/r$; the original $Q = 1 - x\zeta r > 0$, with unit x, bounds Aq^2 below. Thus (CE17) holds with positive real constants. The adjacent-event theorem covers bounded v and (CE18)–(CE21) remove both end types of u at unbounded v. Both actual UH source patterns now have the same sole untreated testing strip, ordinary u and unbounded v.

In the L-unbounded C1.4 line take $q = L^{-1/2}$. Then (NI4) gives $C \asymp q^2$, $\alpha \asymp 1/(q^2 Q)$ and $Aq^2 \asymp 1/(Lr) > 1$. The p-chart (CE20), which needs no lower bound on m, removes the projectively unbounded-u tests there.

All bounds are relative to original d. The four-source-event result does not require the additional NI9 total-scale condition that at least one opening be infinitesimal. Every literal permitted $D = 0$ or $\beta = 0$ is retained throughout, since neither coefficient was capped. No claim of nonemptiness or full-row completion is made. In particular, the L unit and the L infinitesimal/η bounded parts of C1.4 are unchanged, and these remote results do not supply the missing critical contact conductors.

B.2.5 Two exact complementary-current identities

Let $\mathcal{H} = 2[(uW + uvX - Y - vZ)^2 + D_u(vY - Z)^2 + D_v(X + uZ)^2 - D_u D_v Z^2]$ be the original testing core.

Lemma B.7 (Two strong currents)**.** *Suppose the original event coefficients are nonnegative, $d > 0$, and $d/c, d/D$ are infinitesimal, where $c = e_3$, $D = e_5$. Then every testing chart with $1/v$ bounded is tensor-SOS, for every other angle and every state.*

Proof. Put $g = cD/(c + D)$, $h = D/(c + D)$, $R = uW - Y$, $C = v(uX - Z)$, $B = X + uZ$, and $N = R + hC$. Let $D_u = 1 + u^2$, $D_v = 1 + v^2$ and set

$$K_0 = 2\sqrt{g}\, C, \qquad B_0 = \sqrt{2d}\sqrt{D_v/D_u}\, B, \quad c_u = D_u^{-1/2}, \quad t_u = uD_u^{-1/2}, \quad \eta_v = \sqrt{D_v}/v.$$

The complete selected parent is exactly

$$\begin{aligned} 4cD_v R^2 + 4D(R + C)^2 + d\mathcal{H} = {} & 4(c + D)N^2 + 4cv^2 R^2 + 2d(R + C)^2 \\ & + 2dD_u(vY - Z)^2 + (K_0, B_0)Q(K_0, B_0)^T, \end{aligned}$$

where

$$Q = \begin{pmatrix} 1 - \dfrac{d}{2g} c_u^2 \eta_v^2 & \sqrt{\dfrac{d}{2g}}\, t_u \eta_v \\ \sqrt{\dfrac{d}{2g}}\, t_u \eta_v & 1 \end{pmatrix}.$$

To check the identity, use $Z = (uB - C/v)/D_u$ and complete the two current squares. Since $d/g = d/c + d/D$ is infinitesimal and c_u, t_u, η_v are bounded, $\bar{Q} = I_2$. Restore every unselected positive event. All roots separate between fields. Clearing the original pure-testing homogeneous angle factors gives the same identity at literal infinity: $\eta_v^2 = 1 + (1/v)^2$, and c_u, t_u are the normalized homogeneous Alice coordinates. Thus no infinite-angle endpoint is excluded. □

Lemma B.8 (The critical current coefficient)**.** *Under the preceding strength assumptions, suppose*

$$\gamma = \frac{Ag}{d^2} \quad \textit{is bounded}, \qquad 4\bar{\gamma} > \bar{c}_u^2,$$

where $A = e_4$ is the original event coefficient. Then every independent test with v infinitesimal satisfying this residue inequality is tensor-SOS, for every state, including the homogeneous Alice-infinite endpoint.

Proof. Keep $R, C, B, N, g, h, c_u, t_u$ from the preceding proof, and put $\epsilon = d/g$, $q = \sqrt{\epsilon}$ and

$$K_1 = 2\sqrt{g}\,C + \sqrt{d\epsilon}\,Y, \quad B_1 = \sqrt{d}(B - uvY), \quad V_0 = \sqrt{d}\,vY, \quad Y_0 = \sqrt{d\epsilon}\sqrt{D_u}\,Y.$$

Define

$$L_0 = \frac{qc_u}{2}(K_1 - c_uY_0) - t_uvB_1 + c_uvV_0.$$

Direct multiplication gives $L_0 = \sqrt{dD_u}\,v(vY - Z)$ and the exact full selected-parent identity

$$\begin{aligned} &4cD_vR^2 + 4D(R+C)^2 + 4AD_uD_vY^2 + d\mathcal{H} \\ &\quad = 4(c+D)N^2 + 4cv^2R^2 + 2d(R+C)^2 + \mathcal{G}, \\ \mathcal{G} &= K_1^2 + 2D_v(B_1^2 + V_0^2) + (4\gamma D_v - c_u^2)Y_0^2 \\ &\quad + 2qv(K_1 - c_uY_0)V_0 - 2L_0^2. \end{aligned}$$

The four-root Gram of $\mathcal{G}$ has residue $\operatorname{diag}(1,2,2,4\bar{\gamma} - \bar{c}_u^2)$. Its last entry is positive by the stated residue inequality. Every correction is polynomial in bounded q, v, c_u, t_u, γ and has zero residue. The strict-residue lemma and restoration of the other events prove the claim. The displayed bounded homogeneous factors also cover literal Alice infinity. □

B.3 Bounded reciprocal contacts and a nonvanishing Alice scale

For infinitesimal $\eta = f^2, \zeta = g^2$, one contact calculation covers bounded k and bounded $\Delta = 1/e$, including infinitesimal k. For unit k, a source reflection covers every positive ratio g/f. The remaining ordinary-Alice sources use the state completions derived below.

B.3.1 One contact calculation for bounded k and bounded reciprocal e

Use the original C1 source in the normalization $C = 2$, with the seven native coquotients $c_2, c_b, c_3, c_5, c_4, c_\alpha, c_d$ and exact source/parent SD1–SD2. Suppose $k > 0$ is bounded, $f, g > 0$ are infinitesimal, and $\Delta = g/(kf)$ is bounded. Assume the original positive shapes a, t and positive gaps $M = R - t$, $J = T - a$. They make a, t bounded because $R, T \le 1 + k^2$. No bound $a, t < 2$ is used; in particular the statement allows a bounded unit $k > 1$. Then the complete parent is tensor SOS on the testing chart with u bounded, v infinitesimal, Z a unit and all state coordinates bounded. The proof allows bounded unit u, W and both unit and infinitesimal k, Δ.

Normalize $Z = 1$ in the testing field. Retain exactly the four positive squares SD3 and use the full change of state coordinates SD5, with $w = (W + uY)/(1 + u^2)$. Put

$$K = 1 + k^2, \quad \delta = \Delta R/K, \quad b^2 = \eta K/R, \quad Q = K + ak^2b^2, \quad T_n = K^2 - t\delta^2k^4Qb^2.$$

Here δ, k, a, t, u, w are bounded, b and v are infinitesimal, and

$$\eta = \frac{Kb^2}{Q}, \quad \Delta = \frac{\delta Q}{K}, \quad \zeta = \frac{\delta^2k^2Qb^2}{K}, \quad R = \frac{K^2}{Q}, \quad T = \frac{T_n}{K}. \tag{SD6}$$

Every denominator K, Q, T_n, R, T is a positive pure-source unit; its residue is respectively K, K, K^2, K, K. The testing denominator $D = 1 + u^2$ is a positive bounded testing unit. The positive source multiplier $\Delta^2\Lambda$ need not be a unit; it is restored using pure-source square roots after the construction.

Define the polynomial pullback $C_i = K^2Q^3c_i$ by this substitution. It is polynomial because a native monomial $k^jf^{2r}g^{2s}a^\alpha t^\tau$ becomes

$$\delta^{2s}k^{j+2s}b^{2(r+s)}a^\alpha t^\tau K^{2+r-s}Q^{3-r+s}, \tag{SD7}$$

and the seven native polynomials have $-2 \le r - s \le 3$. Thus no shape inverse or negative exponent is introduced. Let $\mathcal{C} = K^6Q^4T_n$ and define

$$\begin{gathered} A = wv, \quad B = \delta v, \quad C_0 = \delta b^2uw, \quad D_0 = \delta^2k^3b^2, \quad D_v = 1 + v^2, \\ V_n = K^2 - tQuvw, \quad Y_n = KT_nuw - avV_n, \quad B_n = vY_n - KT_n, \quad A_c = C_5T_n + aK(C_3D_v - C_5). \end{gathered}$$

Put $G = \Delta^2\Lambda F - \sum_i \Pi_i$ exactly as in SD4. The complete scalar of $\mathcal{C}G$ at the two state normals $N = \delta M_B = 0$ is the following five-term polynomial:

$$\begin{aligned} F_0 &= 4KQT_n^2[C_2(K^2 - tQ) + C_bD_vtQ]D^2A^2 + 4Q^2A_cD^2V_n^2B^2 \\ &\quad + 4K^3C_4DD_v(KT_nC_0 - ab^2BV_n)^2 + 4KQ^2C_\alpha DB_n^2D_0^2 \\ &\quad + 2K^2QC_dD\{b^2B^2(Y_n^2 - K^2T_n^2 + 2aKT_nV_n) - 2K^2T_n^2BC_0\}. \end{aligned}$$

This follows by the state substitution in SD4. In particular it lies in $(A,B,C_0)^2+(D_0^2)$ before any contact division. The complete two first-normal coefficients are

$$\begin{aligned}
S_B &= -8aK^2b^4C_4D_vV_n+8\delta^2Q^2k^6b^4C_\alpha B_n+4KQb^2C_dB_n,\\
S_C &= 8K^3T_nb^2C_4D_v,\\
L_1 &= A[8K^3QT_n^2(C_bD_v-C_2)D]\\
&\quad +B[-8\delta K^2Q^2uDvV_nA_c+\delta K^3auvS_B]+C_0[\delta K^3auvS_C],\\
L_2 &= B[8K^2Q^2T_nDV_n(C_3D_v-C_5)-K^2T_nS_B]-C_0K^2T_nS_C.
\end{aligned}\tag{SD8}$$

They include every first-order state term, and belong to (A,B,C_0) before contact division.

A single positive-weight estimate. Give (δ,k,b,a,t,u,w,v) weights $(0,0,1,0,0,0,0,2)$. The zero-weight quantities are bounded; only b,v require positive weights. Each of A,B,C_0,D_0 has weight two. Directly from the seven source polynomials,

$$C_i=2K^6+O_{\mathrm{wt}}(2)\ (i=2,b,3,5,4,\alpha),\quad C_d=4K^7+O_{\mathrm{wt}}(2),\quad Q=K+O_{\mathrm{wt}}(2),$$

$$T_n=K^2+O_{\mathrm{wt}}(2),\quad V_n=K^2+O_{\mathrm{wt}}(2),\quad B_n=-K^3+O_{\mathrm{wt}}(2),\quad A_c=2K^8+O_{\mathrm{wt}}(2).$$

The five scalar lines therefore have leading terms respectively

$$8K^{14}D^2A^2,\quad 8K^{14}D^2B^2,\quad 8K^{15}DC_0^2,\quad 8K^{15}DD_0^2,\quad -16K^{16}DBC_0.$$

These are identities modulo the coefficient ideal (b^2,v) for arbitrary bounded k,δ,u,w, not identities only at contact. The mixed term in the third line has a factor b^2; so does the separate B^2 term in the fifth line. Consequently, with

$$P=D(A^2+B^2)+K(C_0^2+D_0^2)-2K^2BC_0,\qquad E_0=F_0-8K^2D\mathcal{C}P,$$

one has $\mathrm{wt}(E_0)\ge 6$. For the complete normals use the exact differences

$$K(C_b-C_2)=\delta^2k^2Qb^2C_3,\qquad Q(C_3-C_5)=-k^2Kb^2C_b.$$

Since K,Q have positive unit constant terms in b, these give weight at least two for both differences; also S_B,S_C have weight at least two. The displayed full normal expressions imply $\mathrm{wt}(L_1),\mathrm{wt}(L_2)\ge 4$.

The exact native contact is $(u,w,v,N,M_B)=(k,\delta k,\delta k^2b^2,0,0)$. The six native balances SD9–SD11 and the contact-zero retained squares prove $E_0\in\mathfrak{J}^2$ and $L_i\in\mathfrak{J}$, where $\mathfrak{J}=(u-k,w-\delta k,v-\delta k^2b^2)$. Apply the restricted contact lift CC1–CC6. For completeness, its coefficient filtration is particularly simple here. The three contact substitutions preserve weight. Monic division by $u-k$ and $w-\delta k$ loses no weight. A division by $v-\delta k^2b^2$ loses two, but its quotient is always multiplied by b^2 in the CC2 formula for C_0S. Thus that formula preserves coefficient weight. The normal monomial quotients have weight at least $4-2=2$; the scalar monomial quotients have weight at least $6-4=2$. The successive CC reductions, including the gradient subtraction $2k\rho_c(-2k,1,1,-1)$, preserve this bound. Therefore every final scalar and normal correction coefficient has weight at least two. It is a bounded tensor coefficient of zero residue. No determinant inverse is a lifting coefficient.

The complete state-normal block. Before multiplication by $\mathcal{C}$, let

$$C_B=H_3D_vh_s+H_5(1-h_s),\quad h_s=a/T,$$

$$B_Y=4RT\Delta^2\eta^2c_4DD_v+4RT\Delta^4k^6\eta^2c_\alpha Dv^2+2RT\Delta^2\eta c_dDv^2.$$

The full matrix on $(N,\delta M_B)$ is defined equivalently by coefficient extraction of SD4 after SD5. Its entries are

$$\begin{aligned}
A_{11}&=H_2+H_bD_v+\Delta^2C_Bu^2v^2+B_Yh_s^2u^2v^2/D^2,\\
A_{12}&=-(\Delta^2/\delta)(H_3D_v-H_5)uv-(B_Y/\delta D^2)h_suv,\\
A_{22}&=(\Delta^2/\delta^2)(H_3D_v+H_5)+B_Y/(\delta^2D^2).
\end{aligned}$$

All negative powers cancel: $\Delta/\delta=Q/K$ is a source unit, and B_Y/δ^2 is an explicit bounded infinitesimal because each summand contains η^2 or ηv^2 after that cancellation. Also $H_i\sim 8K^2$, C_B is bounded and $H_3D_v-H_5=4R(c_3-c_5)+H_3v^2$ is infinitesimal. Hence the block after multiplication by $\mathcal{C}$ has exact residue $16\bar{K}^{14}I_2$, for arbitrary bounded u,w.

Finally use the six roots

$$x=B-C_0,\quad y=A-kC_0,\quad z=uB-kC_0,\quad h=kC_0-D_0,\quad q_1=a_1a_2-a_3a_4,\quad q_2=a_1a_3-a_2a_4,$$

where $a_1 = \sqrt{wv}$, $a_2 = \sqrt{\delta uv}$, $a_3 = b\sqrt{\delta kuw}$ and $a_4 = \delta k^{3/2} b$. Writing $g_4 = uwv - \delta^2 k^4 b^2$, the exact identities are

$$\begin{aligned} P &= x^2 + g_4^2 + \sum_i a_i^4 - 4a_1 a_2 a_3 a_4 \\ &= x^2 + g_4^2 + y^2 + z^2 + h^2 - yz + zh + q_1^2 + q_2^2. \end{aligned}$$

The (y, z, h) matrix has successive principal minors $1, 3/4, 1/2$; the other three coefficients are one. Thus the six-root Gram is strictly positive and constant. Retain g_4^2 separately; its coefficients need not be infinitesimal when k, u are units. The full scalar and both normals are thus infinitesimal corrections of this strict six-root Gram and the two strict state-normal roots. The positive-residue Gram lemma applies. Restore g_4^2 and every retained square, and undo the positive pure-source multiplier. Literal axes and a nonnegative testing core have their direct original event certificates. On the negative-core chart the exact SD core calculation gives $u, v, w > 0$, so all radicals above are separated nonnegative radicals. This proves the stated local result for every real state, and separately for both parts of a complex state.

B.3.2 Every test in the bounded-reciprocal and ordinary-Alice sources

The preceding local calculation gives the complete projective result when k is positive bounded, $\eta = f^2, \zeta = g^2$ are infinitesimal and $\Delta = g/(kf)$ is bounded. It also gives the entire source with unit k and arbitrary positive g/f, after the source reflection below. We give the complementary testing and state cases explicitly.

Complementary testing angles. Let $g_c = CD/(C + D)$. The source coefficients SD1 give, with $D_1 = 2c_5/c_3 - J\eta c_d/c_3 \sim 2$,

$$\frac{g_c}{a} = \frac{2D_1}{2J + aD_1}, \qquad 2J + aD_1 \sim 2K,$$

$$d/C,\ d/D \in \mathrm{Inf}_c, \qquad \gamma = \frac{Ag_c}{d^2} \sim \frac{1}{4K^3} > 0.$$

Lemma B.7 therefore applies when $1/v$ is bounded, including Bob infinity, for every Alice angle. When v is infinitesimal and Alice is unbounded, Lemma B.8 has residue diagonal

$$\mathrm{diag}(1, 2, 2, 4\bar{\gamma} - \bar{c}_u^2), \qquad c_u = (1 + u^2)^{-1/2}.$$

Here $\bar{c}_u = 0$ and $\bar{\gamma} > 0$, so the matrix is strict. Its homogeneous identity includes Alice infinity. Only bounded u and infinitesimal v remain. No upper bound on an event coefficient was used; the complementary identities retain all original positive excess squares.

The complementary state chart. Normalize a nonzero state by a coordinate of maximal absolute value. Then all state coordinates are bounded. The testing signed core is a pure-testing element. If it is nonnegative, it is a testing square and every event term is already a separated square.

Suppose the core is negative. At $v = 0$ it is $2\{(X + uZ)^2 + (uW - Y)^2\}$. Thus both displayed currents have zero residue. If Z is a unit, use Section B.3.1; its coefficient bounds allow bounded u, w and every bounded W, X, Y after $Z = 1$ normalization. If Z is infinitesimal, then X is infinitesimal and $Y - uW$ infinitesimal. Some state coordinate is a unit, so W is a unit and $Q_A = W + uY$ is a unit with residue $(1 + \bar{u}^2)\bar{W}$.

On this chart the exact five-root current identity uses

$$z = (r_2, B, \Delta r, \Delta r_5, \Delta\eta Y).$$

Retain the four positive source-gap excesses, the whole alpha term and the positive $v^2 Y^2$ part of the core. The remaining diagonal is

$$H_2 z_1^2 + H_b D_v z_2^2 + H_3 D_v z_3^2 + H_5 z_4^2 + 4RTc_4 D_u D_v z_5^2,$$

and its complete negative correction is

$$-4RTc_d D_u \frac{\Delta Z}{Q_A} z_5(z_1 + z_2) - 2RTc_d D_u \frac{\Delta^2 \eta Z^2}{Q_A^2}(z_1 + z_2)^2.$$

The five positive coefficients are units and both correction coefficients are infinitesimal because Δ is bounded and Z infinitesimal. This proves the state chart. The literal testing axes have nonnegative core directly. The local and complementary state arguments are therefore exhaustive.

Ordinary Alice with arbitrary state scale. Suppose k is a positive bounded unit and f, g are infinitesimal. Only Δ unbounded remains. The exact complete-parent reflection is

$$u' = 1/u, \quad v' = v, \quad (W', X', Y', Z') = (vW - X, -W - vX, Z - vY, Y + vZ), \tag{AS3}$$

$$u^2 F(\mathrm{ref}(c); 1/u, v; \psi') = (1 + v^2) F(c; u, v; \psi), \tag{AS4}$$

with event permutation $(\lambda_{\mathrm{ev}}\ n)(m\ C)(A\ \alpha)(D\ \beta)$. Put $H_f = 1 + (1-a)\eta$, $H_g = 1 + (1-t)\zeta$ and $V = 1 + k^2\eta\zeta$. At the actual source contact its native parameters are

$$\begin{gathered} k' = 1/k, \quad \eta' = k^2 \zeta H_f / H_g, \quad \zeta' = k^2 \eta H_g / H_f, \\ a' = M/(k^2 H_f), \quad t' = J/(k^2 H_g), \quad M' = aV/(k^2 H_g), \quad J' = tV/(k^2 H_f). \end{gathered} \tag{AS5}$$

These identities follow from $H_g - Q_c = \zeta M$ and $H_f - E = \eta J$ and preserve all original signs and gaps. Here H_f, H_g are positive units, k' is a unit, η', ζ' are infinitesimal, and

$$\Delta' = \frac{\sqrt{\zeta'/\eta'}}{k'} = \frac{H_g}{\Delta H_f}$$

is infinitesimal. Normalize the reflected coefficients by a positive pure-source factor to $C' = 2$ and apply the bounded-reciprocal result. The homogeneous reflection restores the original projective endpoints as well. This proves every ordinary-Alice source with small f, g, without a separate comparable-ratio or shape-unit argument.

B.3.3 The positive Laurent form at the ordinary/small source

This is the positive Laurent-model lemma used in the complete UI proof. Write $x = k^2$, $r = 1 - a\eta$, $K = 1 + x$, $R = 1 + xr$ and

$$E_0 = 1 - x\eta, \quad J = K - a, \quad P_4 = KE_0 + x(1 - r), \quad P_d = 1 + 2x + rx^2.$$

At a real interior source point the leading coefficients of the parent divided by a are

$$\begin{aligned} h &= \frac{2x\eta}{RE_0}, & C &= \frac{2}{a}, & A &= \frac{2\eta^2 P_4}{Kr(r+\eta)E_0}, \\ d &= \frac{4\eta P_d}{K(r+\eta)E_0}, & D &= \frac{2}{JE_0} - \frac{d}{2}, & \alpha &= \frac{2x^2\eta R}{KE_0}. \end{aligned} \tag{CL1}$$

All are positive except $D \geq 0$, including its literal zero. Here α is the coefficient after extracting the weak g^2 scale. For a test (u, V) put $D_u = 1 + u^2$ and define

$$\begin{aligned} f = {} & 4hV^2(W + uY)^2 + 4C(uW - Y)^2 + 4AD_uY^2 \\ & + (4D + 2d)(uW - Y - VD_uZ)^2 + 4\alpha D_uZ^2 \\ & - 4dD_uVYZ - 2dD_uV^2Z^2. \end{aligned} \tag{CL2}$$

Let M_f be its symmetric three-state matrix and set $U = u^2$, $T = V^2$. Direct collection gives

$$\det M_f = 64 D_u^2 UT\, \mathcal{L}(U, T), \tag{CL3}$$

where

$$\begin{aligned} \mathcal{L} = {} & AC(D + d/2)U + \frac{h\alpha(A + C + D + d/2)}{U} + \frac{A\alpha(C + D + d/2)}{T} \\ & + hD(A + C + d/2)\frac{T}{U} + hC(D + d/2)UT \\ & + h(AD + 2CD + Ad/2 + Cd/2 + Dd/2 + d^2/4)T + L_0, \\ L_0 = {} & ACD - ADd/2 + h\alpha(C + D + d/2) - d^2(A + C + D)/4 - d^3/8. \end{aligned} \tag{CL4}$$

Every nonconstant coefficient is nonnegative. The coefficients of U, U^{-1}, T, T^{-1} are strictly positive, also when $D = 0$. Thus in coordinates $(\log U, \log T)$ the Hessian is strictly positive and the function is coercive. Substitution of (CL1) gives

$$\mathcal{L}(x, x\eta) = \partial_U \mathcal{L}(x, x\eta) = \partial_T \mathcal{L}(x, x\eta) = 0. \tag{CL5}$$

Strict convexity proves that $\mathcal{L}$ is positive away from this single contact. The (W, Y) principal block of (CL2) is strictly positive for $U, T > 0$, because its C and A terms already have trivial common kernel. Hence f is positive

semidefinite everywhere and positive definite off the contact locus. Literal endpoints follow by continuity. This proof uses no numerical determinant table.

There is a useful corrected formula at the large-D boundary. Impose its hard equation $uW - Y - VD_uZ = 0$, eliminating W. The resulting two-state determinant is $16D_u^2T\mathcal{L}_D$, where

$$\begin{aligned}\mathcal{L}_D &= hT\{(C + A + d/2)/U + (2C + A + d/2) + CU\}\\ &\quad + h\alpha(1 + 1/U) + AC(U + 1) + A\alpha/T - Ad/2 - d^2/4.\end{aligned} \tag{CL6}$$

In particular the coefficients of T/U and UT are $h(C + A + d/2)$ and hC, respectively. Interchanging these coefficients is incorrect. All three identities follow by polynomial collection in the displayed model.

B.3.4 The ordinary/small source: the complete state quadratic

Use the unchanged native C1 source in the representative of (NI7): k, η are positive bounded units and $\zeta = g^2$ is positive infinitesimal. All original openings and $D = 0$ are retained. The exact stabilizer recovers the other representative; no condition $k \le 1$ is imposed after that change. This section proves the missing ordinary/small testing strip, and joins the remote theorem (CR1)–(CR13).

Exact bounded remainder. Normalize by the positive pure-source a. Write $x = k^2$, $r = 1 - a\eta$, $K = 1 + x$, $R = 1 + xr$, $M = R - t$, and

$$J_s = K - xtg^2 - a, \quad E = 1 - x\eta + tx\eta g^2, \quad Q_c = 1 - xg^2r, \quad y = x\eta g^2.$$

Here $a, t, r, 1-r, M, J_s$ are positive bounded openings; $E, Q_c, R, K, r+\eta$ are positive units. The exact polynomials P_4, P_d, P_α are (NC2). The coefficients after division by a are

$$\begin{aligned}&c = 2/a, && A_0 = \frac{2\eta^2P_4}{Kr(r+\eta)E}, && d_0 = \frac{4\eta P_d}{K(r+\eta)E},\\ &D_0 = \frac{2(1+y)}{J_sE} - d_0/2, && \alpha_0 = \frac{2x^2\eta P_\alpha}{KQ_cE}, && \alpha/a = g^2\alpha_0.\end{aligned} \tag{CU1}$$

In particular c, A_0 are strong; d_0, α_0 are units; and ca, A_0r, D_0J_s are bounded with continuous rational expressions through every opening. Put

$$h_2 = \frac{2x\eta(1 - tg^2)(1 + y)}{E}, \qquad h_b = \frac{2x\eta[1 + (1 - t)g^2]}{E}, \qquad h_b - h_2 = 2x\eta g^2.$$

Both h_2, h_b are positive units. The exact coefficients are $m/a = h_2/(g^2M)$ and $\beta/a + d_0/2 = h_b/(g^2t)$.

For the test put $D_u = 1 + u^2, D_v = 1 + v^2$ and

$$Q = W + uY, \quad B = X + uZ, \quad p = t/R, \quad N = (B - pvQ)/g, \quad L = vQ/g.$$

The inverse $W = Q - uY$, $X = gN + pvQ - uZ$ is polynomial. Retain the two positive squares

$$\frac{4h_2t}{MR}N^2, \qquad \frac{4h_bM}{tR}D_vN^2. \tag{CU2}$$

Their remainder is exactly (CL7). Choose bounded caps $c_* \le c$, $A_* \le A_0$, $D_* \le D_0$, with $c_*, A_* > 0$, $D_* \ge 0$; bounded original weights can be kept unchanged. Retain the three positive centered excesses

$$\begin{aligned}&4(c - c_*)D_v(r_t - avZ)^2, \quad 4(A_0 - A_*)D_uD_v(Y - rvZ/\eta)^2,\\ &4(D_0 - D_*)(J_t + J_svZ)^2,\end{aligned} \tag{CU3}$$

where

$$r_t = uQ - D_uY, \quad J_t = r_t - D_uvZ + ugvN + upv^2Q.$$

All five retained rows vanish at the actual contact

$$(u, v) = (k, k\sqrt{\eta}g), \quad (W, X, Y, Z) = (g/\sqrt{\eta}, k(tg^2 - 1), krg/\sqrt{\eta}, 1). \tag{CU4}$$

Set $c_a = 2 - c_*a$, $A_r = A_0r - A_*r$, $D_J = D_0J_s - D_*J_s$. After removing (CU2)–(CU3), the entire parent divided by a is

$$\begin{aligned}G = \frac{4}{R}&\{(h_2 + h_bD_v)N^2 + 2(h_bD_v - h_2)LN + [h_2(1 - p) + h_bpD_v]L^2\}\\ &+ 4c_*D_vr_t^2 + 8c_aD_vvZr_t - 4c_aaD_vv^2Z^2\\ &+ 4A_*D_uD_vY^2 + \frac{8A_r}{\eta}D_uD_vvZY - \frac{4A_rr}{\eta^2}D_uD_vv^2Z^2\\ &+ 4D_*J_t^2 - 8D_JvZJ_t - 4D_JJ_sv^2Z^2\\ &+ 4\alpha_0D_u(gvY - gZ)^2 + 2d_0\{J_t^2 + D_u(v^2Y^2 - 2vYZ - v^2Z^2)\}.\end{aligned} \tag{CU5}$$

Thus G is an exact bounded Gram on (N, Q, Y, vZ, gZ, L) whenever u is a bounded unit and v infinitesimal. Every source denominator in its coefficients is a unit. Formula (CU5), rather than a scalar restriction, supplies all normal terms below.

Local contact lifting with all first normals. Put $\kappa = k\sqrt{\eta}$, $\varepsilon = \kappa g$, $s = R/(k\eta)$, $\delta = u - k$, and define

$$E_Q = Q - svZ, \quad E_Y = Y - rvZ/\eta, \quad R_h = (vQ - s\varepsilon^2 Z)/\varepsilon, \quad S = v\delta Z, \quad T = \varepsilon\delta Z.$$

Let $c_L = 4[h_2(1-p) + h_b p D_v]/R$, a positive bounded unit. The sole potentially unbounded L row can be replaced exactly:

$$L = \kappa R_h + kRgZ, \qquad L^2 - \kappa^2 R_h^2 = 2kRvQZ - k^2R^2g^2Z^2. \tag{CU6}$$

Consequently (CU5) equals $c_L\kappa^2 R_h^2$ plus $8\kappa(h_bD_v - h_2)NR_h/R$ and a polynomial G_0. Explicitly, in (CU5) replace its first line by

$$\frac{4}{R}(h_2 + h_bD_v)N^2 + 8k(h_bD_v - h_2)gNZ + c_L(2kRvQZ - k^2R^2g^2Z^2). \tag{CU7}$$

All coefficients of G_0 are bounded polynomials in u, v, g with pure-source unit denominators. The coefficient of NR_h is infinitesimal, since $h_bD_v - h_2 = 2x\eta g^2 + h_bv^2$.

Here are exact prescriptions for every normal coefficient, avoiding any implicit omission. Substitute $Q = E_Q + svZ$ and $Y = E_Y + rvZ/\eta$ in G_0 and collect the quadratic polynomial in (N, E_Q, E_Y). The constant coefficient is $\phi(u,\xi)Z^2$, where $\xi = v^2$ and $\phi \in (g^2, \xi)$. The linear coefficients of E_Q, E_Y are $vL_Q(u,\xi)Z, vL_Y(u,\xi)Z$; that of N is $gL_N(u,\xi)Z$, where $L_N \in (g^2, \xi)$. These parity and ideal assertions follow term by term in (CU5),(CU7). The remaining quadratic normal block is the full coefficient collection of the same expression. For explicit pen-and-paper formulas, set

$$b = r/\eta, \quad F_0 = us - D_ub, \quad G_0^{\mathrm{sc}} = F_0 - D_u + up\xi s, \quad H_0 = (8D_* + 4d_0)G_0^{\mathrm{sc}} - 8D_J.$$

Then, writing $D_v = 1 + \xi$ in these polynomials,

$$\begin{aligned}
L_Q &= 2c_LkR + 8D_vu(c_*F_0 + c_a) + u(1 + p\xi)H_0,\\
L_Y &= -8D_vD_u(c_*F_0 + c_a) + 8D_uD_v(A_*b + A_r/\eta) - D_uH_0\\
&\quad + 8\alpha_0D_ug^2(b\xi - 1) + 4d_0D_u(b\xi - 1),\\
L_N &= 8k(h_bD_v - h_2) + u\xi H_0.
\end{aligned}$$

The complete scalar coefficient is

$$\begin{aligned}
\phi = {} & c_L(2kRs\xi - k^2R^2g^2) + 4c_*D_vF_0^2\xi + 8c_aD_vF_0\xi - 4c_aaD_v\xi\\
& + 4A_*D_uD_vb^2\xi + 8(A_r/\eta)D_uD_vb\xi - 4A_rrD_uD_v\xi/\eta^2\\
& + 4D_*(G_0^{\mathrm{sc}})^2\xi - 8D_JG_0^{\mathrm{sc}}\xi - 4D_JJ_s\xi\\
& + 4\alpha_0D_ug^2(b\xi - 1)^2 + 2d_0\{(G_0^{\mathrm{sc}})^2\xi + D_u(b^2\xi^2 - 2b\xi - \xi)\}.
\end{aligned}$$

These formulas display the two asserted ideals directly. They also give a short explicit input for the monic divisions, without an unprinted state-normal computation.

The source is an exact stationary zero and every removed square vanishes there. Therefore

$$\phi = \phi_u = \phi_\xi = 0, \qquad L_Q = L_Y = L_N = 0 \quad \text{at } (u,\xi) = (k, \varepsilon^2).$$

Successive monic polynomial division gives, with bounded coefficients,

$$\phi = \delta^2(\varepsilon^2B_0 + \xi B_1) + \delta(\xi - \varepsilon^2)B_2 + (\xi - \varepsilon^2)^2B_3, \tag{CU8}$$

$$L_i = \delta L_{i0} + (\xi - \varepsilon^2)L_{i1} (i = Q, Y), \quad L_N = \delta(g^2L_{N0} + \xi L_{N1}) + (\xi - \varepsilon^2)L_{N2}.$$

For clarity, each quotient is obtained by subtracting the value at $u = k$ and dividing by the monic $u - k$, then by the monic $\xi - \varepsilon^2$. The second u quotient of ϕ stays in (g^2, ξ); its constant and first u terms have the stated double and simple zeros in ξ. The same argument gives the L_N ideal. No small source number is inverted in these divisions.

Put $R_s = (v^2 - \varepsilon^2)Z/\varepsilon$. The exact root identities

$$\varepsilon R_s = (\varepsilon R_h - vE_Q)/s, \quad R_sT = (R_hT - SE_Q)/s, \quad S^2 - T^2 = \delta(R_hT - SE_Q)/s \tag{CU9}$$

allocate (CU8) and every first normal on the six roots (N, E_Q, E_Y, R_h, S, T). For example $v\delta Z = S$, $\delta(\xi - \varepsilon^2)Z^2 = (R_hT - SE_Q)/s$, and $(\xi - \varepsilon^2)^2Z^2 = (\varepsilon R_h - vE_Q)^2/s^2$. The N normal has only infinitesimal couplings. All quadratic normal coefficients are already bounded. This is the required exact full lift.

Strict matched Hessian and source boundary caps. First take finite interior source residues. The leading form is (CL2). Put $V = k\sqrt{\eta}V_1$, $W = W_1/\sqrt{\eta}$, $Y = kY_1/\sqrt{\eta}$, $Z = 1$. At the contact $(W_1, Y_1, V_1, u) = (1, r, 1, k)$, the Schur complement of the two state coordinates in half the Hessian, in coordinates (V_1, u), is

$$\begin{aligned} G_{11} &= \frac{32\eta x^2 P_4}{KE_0}, & G_{12} &= \frac{16\eta k^3(R^2 - 2KE_0)}{KE_0 R}, \\ G_{22} &= \frac{32\eta x(P_4 + xrP_d)}{R^2 P_4}, & \det G &= \frac{256\eta^2 x^3(4K^2E_0 - R^2)}{K^2 E_0^2}. \end{aligned} \tag{CU10}$$

Here $E_0 = 1 - x\eta$, $P_4 = KE_0 + x(1 - r)$, $P_d = 1 + 2x + rx^2$. There are no opening denominators. The leading $D \geq 0$ condition is

$$K(2 + xr)E_0 - R^2 \geq 0,$$

so $4K^2E_0 - R^2 \geq K(4K - 2 - xr)E_0 > 0$. The state block is uniformly bounded below since c, A_0 are strong. Thus the matched Hessian is strict, uniformly through every permitted source boundary after the appropriate hard directions are retained.

We give the boundary justification explicitly. Divergences can occur only at $a = 0$, $r = 0$, or $J_s = 0$, for c, A_0, D_0, respectively. Since $a + J_s \sim K$, c and D_0 cannot both diverge. If $a = 0$, then $r = 1$, so c is the only divergent weight. The only possible divergent pair is A_0, D_0. Their hard-state rows are E_Y and $kE_Q - KE_Y$, with nonzero determinant k. All angular parts of these centered rows are bounded. Solving these at most two real linear equations therefore supplies a bounded competitor for the minimizing state vector. The bounded finite part and (CU10) imply that the state minimizer is bounded as well, by completing its positive state square. The full matched Hessian hence has a uniform positive lower bound on the Euclidean norm in a nearby finite-source family. On a vector in the limiting hard kernel, adjust only the two state coordinates to annihilate the nearby exact hard rows. This adjustment tends to zero, and the divergent penalties vanish exactly. The bounded finite part is consequently strictly positive on that kernel. The elementary real finite-penalty lemma supplies finite caps.

Every allowed residue is approachable by such finite sources. For fixed $x > 0$, the real leading inequalities are

$$0 \leq r \leq 1, \qquad \eta_{\min}(r) = \frac{1 - r}{K} \leq \eta \leq \eta_{\max}(r) = \frac{1 - R^2/[K(2 + xr)]}{x},$$

with positive interval width $R/[xK(2 + xr)]$. Perturb r into $(0, 1)$ and its interval coordinate into $(0, 1)$. This makes a, r, J and the D numerator strictly positive while converging to the given residue. Taking $t = R/2$ and then sufficiently small real $g > 0$ restores all original source signs, including $\beta > 0$. This construction, rather than substitution into a singular leading denominator, justifies the boundary limit.

In the six-root residue Gram, N is an isolated positive pivot. The (E_Q, E_Y) block is strict, R_h has positive pivot $c_L\kappa^2$, S couples only to E_Q, E_Y, and T only to R_h. Setting $S = T$ is the matched Hessian under an invertible real change of coordinates: R_h/g has relative-angle derivative $2s\kappa \neq 0$, and $T/g = \kappa\delta$ supplies the ordinary-angle derivative. After the hard Schur complements, the sum of the S, T diagonals is therefore strictly positive. Add a real multiple of the last zero in (CU9) to redistribute this sum into two positive diagonals. The extra coefficients contain δ and vanish in the residue. The bounded strict-residue Gram lemma proves the full parent at $u - k$ and v infinitesimal, including every source opening.

Ordinary angles away from contact. Let now $\bar{u}^2 \neq \bar{k}^2$, with u a unit and v infinitesimal. Use the bounded frame (N, r_t, Y, vZ, gZ, L) of (CU5), and the exact zero

$$L(gZ) - Q(vZ) = 0, \qquad Q = (r_t + D_u Y)/u. \tag{CU11}$$

For finite leading source weights, N is an isolated positive pivot. The remaining residue is

$$q^T P q + 2Ub^T q + c_2 U^2 + 4hL^2 + c_0 T^2, \quad q = (r_t, Y)^T,\ U = vZ,\ T = gZ,$$

where the source coefficients are (CL1) and

$$\begin{aligned} s_D &= 4D + 2d, \quad s_C = 4C + s_D, \quad P = \operatorname{diag}(s_C, 4AD_u), \\ \ell &= (1/u, D_u/u)^T, \quad b = (-s_D D_u, -2dD_u)^T, \\ c_0 &= 4\alpha D_u, \qquad c_2 = s_D D_u^2 - 2dD_u. \end{aligned} \tag{CU12}$$

Apply the real rank-one allocation lemma (UA1)–(UA3), using (CU11). We verify its hypotheses uniformly through all source boundaries. Writing $d_a = \ell^T P^{-1}\ell$, $e_a = \ell^T P^{-1} b$, $h_a = c_2 - b^T P^{-1} b$, direct inversion yields

$$\begin{aligned} d_a &= \frac{aJE_0}{8u^2(JE_0+a)} + \frac{D_u}{4Au^2}, \\ e_a &= -\frac{aD_u}{u(JE_0+a)} - \frac{dD_u}{2Au}, \\ h_a &= \frac{8D_u^2}{JE_0+a} - 2dD_u - \frac{d^2 D_u}{A}, \\ P^{-1}b &= (-aD_u/(JE_0+a), -d/(2A))^T. \end{aligned} \tag{CU13}$$

All are bounded continuous, since JE_0+a is a positive unit and $1/A$ has a continuous bounded formula proportional to r. The same holds for $P^{-1}\ell$.

Normalize the Laurent polynomial (CL4) by As_C. Its U and T^{-1} coefficients are, respectively, $1/[2(JE_0+a)]$ and $\alpha/4$, positive units. Its T coefficient is

$$h\{(D+d/2)/s_C + 2CD/(As_C) + d/(8A)\},$$

also a positive unit: when $1/A = 0$, $r = 0$ forces a a unit, so $(D+d/2)/s_C = a/[4(JE_0+a)]$ is a unit; otherwise the last term is positive. The U^{-1} coefficient is $h\alpha(1/s_C + 1/(4A))$. If it vanishes, $r = 0$ and $J = 0$, and the T/U coefficient has the positive term $hD/s_C = h/4$. All remaining normalized nonconstant coefficients are bounded and nonnegative. The constant is bounded by the exact contact equation. Thus strict convexity and coercivity in logarithmic coordinates persist at every boundary. The normalized Laurent function retains its unique stationary zero $(U,T) = (x, x\eta)$.

For fixed $u^2 \neq x$, the rank-one determinant polynomial is

$$p(T) = c_0 + (h_a + 4hd_a c_0)T + 4h(d_a h_a + e_a^2)T^2 = \frac{16D_u}{As_C} T\mathcal{L}(u^2, T) > 0.$$

Its constant and leading coefficients are positive units by the preceding coefficient bounds. The strict allocation interval therefore survives at every source boundary. The two real Gram blocks have a uniform positive lower bound; (CU13) controls their state minimizers even if P is unbounded. The centered divergent rows have hard parts r_t, Y, r_t; the only simultaneous pair is Y, r_t, whose determinant is one. Exact small state corrections annihilate the large rows in the nearby finite family. The same finite-penalty argument as above supplies caps in (CU3). The exact bounded remainder (CU5), after (CU11), has strict residue. This proves every ordinary/small test away from contact.

Exhaustive certificate. The two arguments cover the whole ordinary/small testing strip; the exact gauges cover negative contact signs. The remote theorem (CR1)–(CR13), including CR5a, covers its complement. All transformations are homogeneous in the state; no state coordinate was divided out in the tensor proof. Real and imaginary state parts are treated separately. Hence the entire actual UI subdomain of C1.5, and its exact stabilizer images, has an all-tests tensor-SOS proof. The actual UH subdomain of C1.5 remains a separate obligation.

B.3.5 The ordinary/small source: complementary testing charts

The source has actual ordinary Alice angle and infinitesimal Bob angle. Apply the exact native stabilizer (AS5) as in (NI7) and take η a positive bounded unit, $\zeta = g^2$ infinitesimal and k a bounded unit. The reflected orientation need not satisfy $k \leq 1$. Its E remains a positive unit by the exact source entry in (NI7). Normalize by the positive pure-source d. Equations (NC2)–(NC4) give

$$C/d,\ A/d,\quad g^2 m/d,\quad g^2\beta/d,\quad \alpha/(dg^2)\ \text{ strong},\qquad D/d \geq 0. \tag{CR1}$$

For example $P_d, P_\alpha, Q, r+\eta$ are units, whereas the positive openings $M, t, r, 1-r$ are bounded and may be infinitesimal. The $-1/2$ in β/d disappears after multiplication by g^2. In this particular source subdomain $\beta = 0$ is impossible; $D = 0$ is allowed throughout. Choose positive real constants μ, ν, c, a, b below the five quantities in (CR1), respectively in the order $g^2 m/d, g^2\beta/d, C/d, A/d, \alpha/(dg^2)$. Whenever needed we cap these coefficients at

$$m = \mu/g^2,\quad \beta = \nu/g^2,\quad C = c,\quad A = a,\quad \alpha = bg^2. \tag{CR2}$$

Every unused term, every positive excess, and the complete D term is retained as a square. The signed core, divided by two, is

$$K = J^2 + D_u(vY - Z)^2 + D_v(X + uZ)^2 - D_u D_v Z^2,\quad J = uW + uvX - Y - vZ. \tag{CR3}$$

Only the core and the explicitly selected caps need a bounded Gram. This observation keeps all upper coefficient scales out of the proof.

Theorem B.9 (C1 UI remote tests)**.** *For every source satisfying (CR1), the complete original C1 parent is tensor SOS at every independent testing pair except possibly u a bounded nonzero unit and v infinitesimal. The conclusion includes literal projective endpoints, both signs, all states, and $D = 0$.*

Proof. All constants in residue forms below are real residues. Roots may be unbounded, but all coefficients in their exact Gram are bounded. First put $Q = W + uY$, $B = X + uZ$; then

$$M = vQ - B, \quad uW - Y = uQ - D_uY, \quad J = uQ - D_uY + uvB - vD_uZ. \tag{CR4}$$

These identities account for every state normal.

If u is a bounded unit and v projectively infinite, put $w = 1/v$ and homogenize by w^2. Use roots $(E_Q, E_B, Y, Z) = (Q/g, B/g, Y, Z)$ and cap only m, β. The exact homogenized core is

$$[w(ugE_Q - D_uY) + ugE_B - D_uZ]^2 + D_u(Y - wZ)^2 + D_w g^2 E_B^2 - D_u D_w Z^2.$$

Its residue is $D_u(Y^2 + u^2Z^2)$. Including the two caps gives

$$4\mu E_Q^2 + 4\nu E_B^2 + 2D_u(Y^2 + u^2Z^2) > 0. \tag{CR5}$$

If u is infinitesimal and v projectively infinite, the preceding four-root frame loses the Z pivot. Keep $w = 1/v$ and instead use (E_Q, E_B, Y, U, T), where $U = uZ$ and $T = gwZ$. Put $A_0 = w(ugE_Q - D_uY) + ugE_B$. The exact homogenized core is

$$K_w = A_0^2 + D_uY^2 + D_w g^2 E_B^2 + D_uU^2 - 2D_u g(wE_Q + E_B)U + 2wD_u uYU.$$

All its coefficients are bounded, with residue $Y^2 + U^2$. Cap m, β, α; their amplitudes after normalization are $E_Q - wE_B$, E_B, and $gY - T$, with the positive factors D_w, D_u in the last two terms. The full selected residue is

$$4\mu E_Q^2 + 4\nu E_B^2 + 2Y^2 + 2U^2 + 4bT^2 > 0. \tag{CR5a}$$

This includes literal $u = 0$ or $w = 0$; no division by either was used.

If u is infinitesimal and v a bounded unit, use $E_Q = Q/g, E_B = B/g, Y, U = uZ, Z_g = gZ$, and cap m, β, α. Substitute $W = gE_Q - uY$, $X = gE_B - uZ$ in (CR3). All terms involving Z collect into uZ; the resulting exact core is a polynomial in $(u, v, g; E_Q, E_B, Y, U)$. Its residue is $D_vY^2 + v^2U^2$. The full selected residue is

$$4\mu(vE_Q - E_B)^2 + 4\nu D_v E_B^2 + 2D_vY^2 + 2v^2U^2 + 4bZ_g^2 > 0. \tag{CR6}$$

The α amplitude is $gvY - Z_g$, so no inverse of u was used in constructing this Gram.

If both u, v are infinitesimal, set

$$U = uQ, \quad V = vQ/g, \quad E_B = B/g, \quad Z_g = gZ, \quad S = uvZ,$$

and include Y as the sixth root. The exact core is

$$\begin{aligned} K = {} & (U - D_uY + uvgE_B)^2 + D_u v^2 Y^2 + D_v g^2 E_B^2 + D_u S^2 \\ & + 2D_u uYS - 2D_u uVZ_g - 2D_u vgE_B S. \end{aligned} \tag{CR7}$$

In particular the apparently problematic term UvZ is exactly uVZ_g, with no mixed inverse. Cap all five weights in (CR2). The entire selected residue is

$$4\mu(V - E_B)^2 + 4\nu E_B^2 + (4c + 2)(U - Y)^2 + 4aY^2 + 4bZ_g^2 + 2S^2 > 0. \tag{CR8}$$

It remains to cover u projectively infinite. Put $p = 1/u$, homogenize by p^2, and use $D_p = 1 + p^2$, $Q = pW + Y$, $B = pX + Z$. Then

$$M = vQ - B, \quad r = W - pY, \quad J = D_p(W + vX) - p(Q + vB). \tag{CR9}$$

For v a bounded unit, use $(Q/g, B/g, W, X)$ and cap m, β, C. The core after $Y = gE_Q - pW$, $Z = gE_B - pX$ is a bounded polynomial with residue $(W + vX)^2$. The selected residue is

$$4\mu(vE_Q - E_B)^2 + 4\nu D_v E_B^2 + 4cD_v W^2 + 2(W + vX)^2 > 0. \tag{CR10}$$

For infinitesimal v, use (W, R, Y, V, E_B) with $R = vX$, $V = vQ/g$, $E_B = B/g$. Cap m, β, C, A, retaining α entirely. The exact core is

$$\begin{aligned} K_p = {} & (W + D_pR - pY - pvgE_B)^2 + D_v g^2 E_B^2 \\ & + D_p\{v^2Y^2 - 2vgYE_B + 2pYR - (vgE_B - pR)^2\}. \end{aligned} \tag{CR11}$$

Every coefficient is bounded. The selected residue is

$$4\mu(V-E_B)^2+4\nu E_B^2+4cW^2+4aY^2+2(W+R)^2>0. \tag{CR12}$$

Finally, when both testing angles are infinite, also put $w=1/v$ and homogenize by p^2w^2. Use $(Q/g,B/g,W,X)$ and cap m,β,C. Substitution $Y=gE_Q-pW$, $Z=gE_B-pX$ in the homogenized core is polynomial and has residue X^2. The selected residue is

$$4\mu E_Q^2+4\nu E_B^2+4cW^2+2X^2>0. \tag{CR13}$$

Each displayed positive form is strict on the complete root space. The bounded positive-residue Gram lemma therefore supplies tensor SOS. Restoring the retained terms proves the original parent. The bounded unit/unit testing chart is already covered by the common noncommuting-residue theorem. This leaves exactly the ordinary/small testing strip stated in the theorem. The exact gauges $(u,W,X)\mapsto(-u,-W,-X)$ and $(v,W,Y)\mapsto(-v,-W,-Y)$ preserve every event square and the core. All homogeneous formulas allow $p=0$ or $w=0$ literally. Real and imaginary state components are treated separately. □

The ordinary/small testing strip is completed in Section B.3.4. Together with the remote charts proved here, it covers the entire actual UI source. The actual UH source is treated by the separate complete conductor in Section B.3.6.

B.3.6 The ordinary/unbounded source

All coefficients below are divided by the original positive pure-source d. We treat precisely (NI9) and (NI10), with no alteration of the native source (NC1)–(NC7). The revision-6 common-event theorem already covers all tests except positive unit u and $w=1/v$ infinitesimal. Put $x=k^2$, $K=1+x$, $\sigma=1/\eta$, $z=1/\zeta=q^2$, $b=r/z$, $H=h/\sigma$, and retain $\sigma>0$ (it need not be a unit). On (NI9), σ is infinitesimal and z is a unit; on (NI10), z is infinitesimal and σ is bounded. Throughout, k,H are positive units, b,z,σ are bounded, $Q=1-xb>0$, and

$$r=zb,\quad R=1+xzb,\quad M=R-z+\sigma zH,\quad J=1+\sigma(xH-1+zb),\quad \theta=1-\sigma H,\quad E_s=1-xH.$$

The exact source polynomials, all with bounded arguments, are

$$\begin{aligned}P&=xK+\sigma x+\sigma z+\sigma zbx^2+x^2\sigma zH(1-\sigma H)+2x\sigma^2zH+\sigma^2z^2bx^2H,\\ V&=z(1+x\sigma H)E_s+x(1-zb)(1+\sigma zH),\\ B_\alpha&=R\sigma zH+RxH+ME_s.\end{aligned}$$

Writing $m=m_{\mathrm{old}}/z$, $c=C/z$, $a=Az$, $\ell=\alpha z$,

$$c=\frac{K(1+\sigma zb)E_s}{2(1-zb)P},\quad m=\frac{Kx(1+\sigma zb)H(\sigma z+x)}{2MP},\quad a=\frac{V}{2bP},\quad \ell=\frac{x^2(1+\sigma zb)B_\alpha}{2QP}. \tag{UH1}$$

Define the polynomial quotients

$$N_D=\frac{K(\sigma z+x)(1+\sigma zb)-JP}{\sigma},\qquad N_\beta=\frac{Kx(1+\sigma zb)(1+\sigma zH)-\theta P}{\sigma}.$$

Then $D=\sigma N_D/(2JP)$ and $\beta=\sigma N_\beta/(2\theta P)$, exactly.

The bounded full parent. Let $D_u=1+u^2$, $D_w=1+w^2$. Its six event amplitudes are

$$M_w=W-wX+uY-uwZ,\ R_w=uW-Y,\ Y,\ J_w=uwW+uX-wY-Z,\ Y-wZ,\ B_w=X+uZ.$$

Thus the exact parent is

$$F=4mzM_w^2+4czD_wR_w^2+4(a/z)D_uD_wY^2+4DJ_w^2+4(\ell/z)D_u(Y-wZ)^2+4\beta D_wB_w^2+2K_w, \tag{UH2}$$

where, without any restriction to a state ray,

$$\begin{aligned}K_w=&\,u^2w^2Z^2-2u^2wYZ-2uwWZ+2uw^2XZ+u^2w^2W^2+2u^2wWX-2uw^2WY\\&+(D_u+w^2)(X^2+Y^2)-2uwXY.\end{aligned}$$

For a potentially unbounded weight choose a finite cap, keeping bounded weights unchanged. Subtract and retain the following nonnegative centered excesses:

$$\begin{aligned}&4(m-m_*)[qM_w-kM(wZ/q)]^2,\\&4(c-c_*)D_w[qR_w-x(1-r)(wZ/q)]^2,\\&4(a-a_*)D_uD_w[(Y-xbwZ)/q]^2,\quad 4(\ell-\ell_*)D_u[(Y-xbwZ)/q]^2.\end{aligned} \tag{UH3}$$

The first line uses qM_w and the second qR_w; thus these are exactly excesses of the scaled weights in (UH1). Their remainders are bounded because $mM, c(1-r), ab, \ell Q$ are bounded. When σ is a unit, additionally retain

$$4(D-D_*)(J_w+JZ)^2, \qquad 4(\beta-\beta_*)D_w(X+uhZ)^2. \tag{UH4}$$

Here DJ and $\beta\theta$ are bounded. When σ is infinitesimal, no (UH4) cap is needed: $D, \beta = O(\sigma^2)$, as proved below. Denote the complete remainder by G. Every retained row vanishes at the actual contact

$$u = k, \quad w = \frac{q\sqrt{\sigma}}{k}, \qquad (W, X, Y, Z) = (\frac{\sqrt{\sigma}}{q}, -kh, kbq\sqrt{\sigma}, 1). \tag{UH5}$$

No comparison between independent source and test rates occurs.

Exact homogeneous conductor, including every first normal. Put $h = \sigma H$, $\delta = u - k$, and use the six roots

$$\begin{gathered} N_x = X + khZ, \quad N_y = (Y - kbq^2W)/q, \\ U = wW - \sigma Z/k, \quad N = (wZ - q^2W/k)/q, \\ S = q\delta W, \quad T = \sigma\delta Z. \end{gathered} \tag{UH6}$$

Substitute $X = N_x - khZ$, $Y = qN_y + kbq^2W$ in the entire bounded remainder G. Its complete quadratic normal block is the coefficient collection in (N_x, N_y). Its scalar and first-normal coefficients have exactly the following supports:

$$\begin{aligned} G_{00} &= \frac{A}{q^2}w^2Z^2 + Bw^2W^2 + C\sigma^2Z^2 + D_1wWZ + Eq^2W^2, \\ G_{10} &= A_xwW + B_xZ + C_xw^2Z, \\ G_{01} &= q(A_y + B_yw^2)W + C_ywZ/q. \end{aligned} \tag{UH7}$$

All eleven displayed coefficient families and the quadratic normal coefficients are bounded polynomials in u, with bounded source coefficients; B_x is divisible by σ. In the unit-σ case divisibility means a bounded quotient, not a new polynomial identity. Formulas below specify every coefficient before caps, and (UH3)–(UH4) specify their exact changes. No state normal is omitted.

Set $s = \sigma/k$, $a_0 = q^2/k$ and

$$L = 2A/k + 2B\sigma/k + D_1, \quad J_0 = C - B/k^2, \quad K_0 = A/k^2 + 2B\sigma/k^2 + D_1/k + E.$$

The scalar is identically

$$AN^2 + BU^2 + qLWN + \sigma^2J_0Z^2 + q^2K_0W^2. \tag{UH8}$$

Exact source contact and all stationary jets give

$$\begin{gathered} L(k) = J_0(k) = K_0(k) = 0, \qquad K_0'(k) = -\sigma J_0'(k), \\ B_x(k) + s[A_x(k) + a_0C_x(k)] = 0, \qquad A_y(k) + C_y(k)/k + sa_0B_y(k) = 0. \end{gathered} \tag{UH9}$$

These hold separately for every centered excess, hence for every cap. Write the monic divisions

$$\begin{gathered} L = \delta L_1, \quad J_0 = \delta j + \delta^2 J_2, \quad K_0 = -\sigma\delta j + \delta^2 K_2, \\ B_x + s(A_x + a_0C_x) = \sigma\delta H_x, \qquad A_y + C_y/k + sa_0B_y = \delta H_y. \end{gathered}$$

All quotients are bounded. The scalar and full first normals become

$$\begin{aligned} G_{00} &= AN^2 + BU^2 + (L_1 + k\sigma j)SN - kjTU + J_2T^2 + K_2S^2, \\ G_{10} &= (A_x + a_0C_x)U + C_xqwN + H_xT, \\ G_{01} &= (C_y + sq^2B_y)N + qB_ywU + H_yS. \end{aligned} \tag{UH10}$$

Together with the unchanged full quadratic normal block, this is an exact bounded six-root Gram for G, not a scalar restriction. The only root relation needed for strict completion is

$$T^2 = \sigma S^2 + k\sigma\delta SN - k\delta TU. \tag{UH11}$$

Its verification is one multiplication using (UH6). Adding any bounded multiple of its zero changes no parent. No mixed inverse, root, order, or source/test identification occurs in (UH6)–(UH11).

For the cap argument it is useful to print every centered row directly in this same frame. In the order $m, c, (a, \ell), D, \beta$, they are

$$\begin{aligned} &uzN_y - (kR + z\delta)N - qwN_x - (zQ/k)S, \\ &S - zN_y - x(1-r)N, \qquad N_y - xbN, \\ &(u - kbz)U + uN_x - qwN_y + (E_s/k)T, \qquad N_x + HT. \end{aligned} \tag{UH20}$$

These are exact identities before taking any residue.

Explicit coefficient input. For (UH2) before caps put $j_u = u - kbz$, $p_u = 1 + ukbz$, $t_u = kh - u$. The full scalar coefficients in (UH7) are

$$\begin{aligned}
A &= 4mz^2t_u^2 + 4\ell D_u + (4\beta+2)zt_u^2,\\
B &= (4cz + 4D + 2)j_u^2 + 4aD_uxb^2z,\\
C &= [4D(1+ukh)^2 + 4\beta(u-kh)^2 + 2D_uxh^2]/\sigma^2,\\
D_1 &= 8mzp_ut_u - 8Dj_u(1+ukh) - 8\ell D_ukb - 4u^2k(h+bz) - 4u + 4uxhbz,\\
E &= 4mp_u^2 + 4cj_u^2 + 4(a+\ell)D_uxb^2 + 2D_uxb^2z.
\end{aligned}$$

The first normals are

$$\begin{aligned}
A_x &= -8mzp_u + (8D+4)uj_u,\\
B_x &= -8Du(1+ukh) + 8\beta(u-kh) - 4D_ukh,\\
C_x &= -8mzt_u + (8\beta+4)(u-kh),\\
A_y &= 8mzup_u - 8czj_u + 8(a+\ell)D_ukb + 4D_ukbz,\\
B_y &= -8czj_u + 8aD_ukb - (8D+4)j_u,\\
C_y &= 8mz^2ut_u - 8\ell D_u + 8Dz(1+ukh) + 4uzt_u.
\end{aligned}$$

The coefficients of N_x^2, N_y^2, N_xN_y are respectively

$$4mzw^2 + 4Du^2 + 4\beta D_w + 2(D_u + w^2),$$

$$4mz^2u^2 + 4cz^2D_w + 4aD_uD_w + 4\ell D_u + 4Dzw^2 + 2z(D_u+w^2), \qquad -8mzuqw - 8Duqw - 4uqw. \tag{UH12}$$

Here products such as Du mean $D\,u$, whereas $D_u = 1 + u^2$.

Infinitesimal σ: source collapse and strict residue. The exact polynomial $P_m = (N_D + xN_\beta)/\sigma$ is bounded. At $\sigma = 0$, $N_D = -xN_\beta$ and

$$N_\beta = x(K+z)H + bxz - x - z.$$

Nonnegativity of D, β forces

$$H = H_0 + O(\sigma), \quad H_0 = \frac{x+zQ}{x(K+z)}, \qquad E_0 = 1 - xH_0 = \frac{R}{K+z}. \tag{UH13}$$

Indeed $JD + x\theta\beta = \sigma^2 P_m/(2P)$ and J, θ are units. Thus $D, \beta = O(\sigma^2)$, including either literal zero. Moreover

$$(D + x\beta)/\sigma^2 \longrightarrow KH_0E_0/2 > 0.$$

The other limiting coefficients are

$$c_0 = \frac{R}{2x(1-zb)(K+z)}, \quad m_0 = \frac{xH_0}{2(1-zQ)}, \quad a_0^{\mathrm{wt}} = \frac{H_0}{2b}, \quad \ell_0 = \frac{xR}{2Q(K+z)}. \tag{UH14}$$

At a matched angle $u = k$, the residue Gram (UH10) splits into (N_y, N, S) and (N_x, U, T), because their only off-block coefficients contain w. Before caps, the (N_y, N) block is positive definite; its angular Schur complement is

$$\Gamma = 8RH_0/K > 0. \tag{UH15}$$

The (N_x, U) block is precisely

$$2K(N_x + xH_0U)^2 + 2KxH_0E_0U^2. \tag{UH16}$$

The remaining T couplings are bounded. As σ, δ have zero residue, (UH11) permits an arbitrarily large positive T^2 pivot without changing any other residue coefficient. This proves a strict full six-root residue Gram. For $u^2 \neq x$, use the full raw frame $(E_t, F_t, G_t, R_t, X, T_0) = (qW, Y/q, wZ/q, wW, X, \sigma Z)$. Its leading first block is

$$\begin{aligned}
&4m_0[E_t + uz(F_t - G_t)]^2 + 4c_0(uE_t - zF_t)^2 + 4a_0^{\mathrm{wt}}D_uF_t^2 + 4\ell_0D_u(F_t - G_t)^2\\
&\quad + 2z\{u^2(G_t - F_t)^2 + F_t^2\} - 4uE_tG_t.
\end{aligned}$$

Put $v_0 = u^2$ and $M_0 = 1 - zQ$. In $(F_t, F_t - G_t)$ the hard block is diagonal, with weights

$$B_h = \frac{2(x+zQ)[v_0(1-r)+R]}{bx(1-r)(K+z)}, \qquad L_h = \frac{2R[v_0(x+zQ)+xM_0]}{Q(K+z)M_0}.$$

The exact E_t Schur complement is

$$\frac{2(v_0-x)^2(1+v_0)RH_0}{[v_0(1-r)+R][v_0(x+zQ)+xM_0]}. \tag{UH17}$$

It has a positive-unit limit through every opening when $v_0 \neq x$. The hard minimizers are

$$\frac{F_t}{E_t} = \frac{ubK}{v_0(1-r)+R}, \qquad \frac{F_t-G_t}{E_t} = \frac{-uKQ}{v_0(x+zQ)+xM_0},$$

also bounded through every opening. The other raw blocks are $2\{u^2R_t^2+2u^2R_tX+D_uX^2\}$ and $4(D+u^2\beta)T_0^2/\sigma^2$, both strictly positive. This gives the full bounded away-angle lift, without a determinant-only claim or any omitted first state normal.

Unit σ: angular Schur sign. Here (NI10) gives $z=0$ at residue. Put $h=\sigma H$, $J=1+\sigma(xH-1)$, $\theta=1-\sigma H$. The source signs imply

$$0<\sigma<K, \qquad \frac{1}{K+\sigma} \le H \le \frac{x+\sigma}{x(K+\sigma)}. \tag{UH18}$$

These inequalities are stated on the interior; its closure includes $\sigma=K$ when $J=\theta=0$. The limits are

$$m=\frac{KxH}{2(K+\sigma)}, \quad c=\frac{K(1-xH)}{2x(K+\sigma)}, \quad a=\frac{1}{2b(K+\sigma)}, \quad \ell=\frac{x}{2Q(K+\sigma)},$$

$$D=\frac{K}{2J(K+\sigma)}-\tfrac{1}{2}, \quad \beta=\frac{K}{2\theta(K+\sigma)}-\tfrac{1}{2}.$$

The two hard blocks in (UH10) are positive definite. Their angular Schur coefficients γ_S,γ_T have no ST term and satisfy

$$\gamma_S+\sigma\gamma_T=\frac{-2p(H)}{\sigma(K+\sigma)}>0, \qquad p(H)=Kx(K+4\sigma)H^2-2K(x+2\sigma)H+x. \tag{UH19}$$

Convexity of p and (UH18) prove strict negativity: its endpoint values are respectively

$$\frac{-\sigma(3\sigma x+4\sigma+4x+4)}{(K+\sigma)^2}, \quad \frac{-\sigma(4\sigma x+3\sigma+4x^2+4x)}{x(K+\sigma)^2}.$$

Use (UH11) to distribute this positive total between S^2,T^2; for example choose its multiple so both resulting Schur coefficients are positive. This yields the required strict six-root Gram.

For completeness, angles $u^2 \neq x$ are strictly positive as well. Eliminating the two hard state variables from the leading form, multiplying by the positive source factor $(K+\sigma)/2$, and writing $U_0=u^2$, $P_0=(w/q)(Z/q)$, $T_0=(Z/q)^2$ gives a positive Laurent function

$$A+BP_0^2+CP_0^2/T_0+D_0T_0-2EP_0,$$

where

$$A=KxH+K(1-xH)U_0/x,\ B=x(1+U_0),\ C=\frac{KU_0}{J+\theta U_0},$$

$$D_0=(1+U_0)\left[\frac{K(1+U_0)}{J+\theta U_0}-(K+\sigma)\right], \quad E=\frac{K\sqrt{U_0}(1+U_0)}{J+\theta U_0}.$$

The strict criterion is $E<\sqrt{AB}+\sqrt{CD_0}$. Squaring only to analyze these positive real residue quantities gives

$$(E^2-AB-CD_0)^2-4ABCD_0=\frac{K^2(1+U_0)^2}{(J+\theta U_0)^2}(U_0-x)^2Q_2(U_0).$$

The quadratic Q_2 has positive constant $H^2x^2J^2$, positive leading coefficient $\theta^2(1-xH)^2$, and $Q_2(x)=xp(H)<0$. Explicitly, with $E_s=1-xH$,

$$Q_2(U_0)=E_s^2\theta^2U_0^2+2E_s\{x\theta^2+E_s\theta J+(K+\sigma)\theta-2K\}U_0+x^2H^2J^2.$$

The concave quadratic $U_0(K+\sigma)-[x^2H+(1-xH)U_0](J+\theta U_0)$ is negative at zero and infinity and positive at x; at each of its two zeros the preceding discriminant is negative. Consequently its positive interval lies inside $Q_2<0$. This proves the criterion for all $U_0 \neq x$. Sign charts with $P_0 \le 0$ are immediate. These square roots are an analysis of real residues only and are not tensor roots or mixed localizations.

This inequality is directly a bounded Gram allocation, not a comparison of mixed ratios. Before choosing projective coordinates, put $E_t = qW$, $G_t = wZ/q$, $R_t = wW$. The exact relation $E_tG_t = R_tZ$ permits distributing the cross term between the two quadratic blocks (E_t, G_t) and (R_t, Z). The strict inequality above is exactly the condition that both 2×2 blocks can be made positive definite by a finite real allocation. It therefore handles every independent rate of q, w. At $J = 0$ or $\theta = 0$, the root-interval argument permits one endpoint at zero or infinity. At the double face $J = \theta = 0$, necessarily $\sigma = K$, $H = 1/K$. As $J + \theta U_0 \to 0$,

$$E - \sqrt{CD_0} \longrightarrow K\sqrt{U_0}, \qquad AB - K^2U_0 = (U_0 - x)^2.$$

Thus the constrained allocation remains strictly positive away from $U_0 = x$, including this double face without dividing by a vanishing residue denominator.

Opening faces, tensor conclusion, and exact scope. The finite-cap argument is applied to the full Gram above. For σ infinitesimal the possible diverging rows have openings $1 - zb$, $1 - zQ$, b, Q; at most two can diverge together. The pairs (c, m), (m, a), (c, ℓ) have independent hard rows; other pairs are excluded by $0 < zb, xb < 1$ and $1 - zQ > 0$. In the (N_y, N) columns of (UH20), their determinants have absolute values kzK, k, x, respectively; each is a unit on its compatible face. The hard block is uniformly coercive, since it contains positive-unit multiples of N_y^2 and $(N_y - N)^2$. Its minimizing hard coordinates for fixed angular S are explicitly

$$\frac{N_y}{S} = \frac{b(2bxz - x + 1)}{K}, \qquad \frac{N}{S} = \frac{4bxz - x - 2z + 1}{xK}.$$

They are bounded through every opening. Thus the limit of the uncapped Schur complement is the Schur complement restricted to the diverging centered rows' kernel. The quotients (UH15)–(UH16) stay positive units there. The elementary finite-dimensional penalty lemma now applies: a bounded symmetric form strictly positive on the common kernel becomes strictly positive after sufficiently large finite positive multiples of these rows' squares are added. This proves the finite-cap claim, with the entire angular right-hand sides in (UH20) retained. Away from $u^2 = x$, (UH17) and its printed bounded hard minimizers give the same constrained strictness directly.

For unit σ, only b, Q, J, θ can be openings. The b, Q rows coincide in the hard direction $N_y - xbN$ and cannot both diverge. The J, θ rows on (N_x, U) are independent even if both vanish; their determinant is the unit u. Formula (UH19) remains a positive unit through these faces. More explicitly, the two hard blocks dominate positive-unit multiples of $N_y^2 + (N_y - N)^2$ and $N_x^2 + (N_x + U)^2$. Their angular minimizers are

$$\frac{N_y}{S} = \frac{b(1 - HxK)}{K}, \quad \frac{N}{S} = \frac{1 - HxK}{xK},$$

$$\frac{N_x}{T} = \frac{\sigma H^2 xK - HxK - \sigma H(2x + 1) + x}{\sigma K}, \quad \frac{U}{T} = \frac{HxK(K + \sigma) - xK - \sigma}{x\sigma K}.$$

Their D and β row defects divided by T are respectively $k(HK - 1)J/(\sigma K)$ and $-x\theta(HK - 1)/(\sigma K)$. Hence every relevant constrained limit exists with bounded minimizers; the same penalty argument applies. On the angular relation $T^2 = \sigma S^2$, its strictly positive Schur value is (UH19), and after taking finite caps the bounded zero (UH11) distributes that value into two strict angular pivots. The away-angle allocation and its single/double-face limits above give the same conclusion there. All removed excesses remain exact positive source/test squares. No positive lower bound on an original opening has been imposed.

A strict bounded residue Gram lifts by the common tensor Gram lemma. Every root in (UH6) is a finite sum of separated source and testing factors. After restoring the retained squares this certifies the complete original parent, including $D = 0$ or $\beta = 0$. Together with revision-6 remote tests, this closes both (NI9) and (NI10). Those are the entire actual UH portion of C1.5. The already proved UU and UI portions and exact source stabilizers therefore close the entire C1.5 row, subject to the exact checks and mathematical review recorded alongside this section.

B.4 Vanishing contact scales and the state normal directions

When k vanishes, some contact directions become singular. The following polynomial identities retain their exact factors before any rescaling. They control the scalar term and both state normal directions, including the complementary state charts.

B.4.1 The endpoint polynomial source

Seven short source polynomials. The following definitions make the endpoint input reproducible without a stored expansion. Put

$$K = k^2, \quad \eta = e^2b^2, \quad \zeta = k^2b^2, \quad V_c = K\eta\zeta.$$

$$L_c = 1 - a\eta, \qquad H_c = 1 + (1-a)\eta, \qquad Q_c = 1 - K\zeta + aV_c.$$

Define

$$\begin{aligned}
P_4 &= 1 + K + \eta K(a - K - 1) - \zeta K t + V_c[-\zeta K t^2 + 2Kt + a(1-t) + t],\\
P_\alpha &= 1 + K - a\eta K + \zeta[(1+K) - (2+K)t] + V_c[-\zeta t^2 + a(t-1) + t],\\
P_d &= (1+K)^2 - a\eta K^2 + \zeta K[(K+1) - (K+2)t]\\
&\quad + V_c[-\zeta K t^2 + Ka(t-1) + Kt + K + 1].
\end{aligned}$$

The seven coquotients are

$$\begin{aligned}
c_2 &= 2(1+K)(1-t\zeta)L_c(1+V_c)H_cQ_c,\\
c_\beta &= 2(1+K)L_c[1 + (1-t)\zeta]H_cQ_c,\\
c_3 &= 2(1+K)L_cH_cQ_c(1 - K\eta + tV_c),\\
c_5 &= 2(1+K)L_cH_cQ_c(1+V_c),\\
c_4 &= 2Q_cP_4, \qquad c_\alpha = 2L_cH_cP_\alpha, \qquad c_d = 4L_cQ_cP_d.
\end{aligned}$$

Their residues are respectively $2, 2, 2, 2, 2, 2, 4$, so all are positive coefficient units on this domain. With $M = R_c - t$ and $J = T_0 - a$, the unchanged source is

$$\lambda = n = 0, \quad c = 2, \qquad d = \frac{2a\eta c_d}{c_3}, \quad m = \frac{2ae^2c_2}{Mc_3}, \quad A = \frac{2a\eta^2c_4}{c_3}, \quad \alpha = \frac{2aK^2\eta\zeta c_\alpha}{c_3},$$
$$D + \frac{d}{2} = \frac{2ac_5}{Jc_3}, \qquad \beta + \frac{d}{2} = \frac{2ae^2c_\beta}{tc_3}, \qquad \Lambda = \frac{R_cT_0c_3}{2a} > 0.$$

The theorem retains the hypothesis $D, \beta \geq 0$. Write

$$r_2 = v - X + uvY - uZ, \quad r_\beta = X + uZ, \quad r_3 = u - Y, \quad r_5 = u + uvX - Y - vZ, \quad r_\alpha = vY - Z,$$

and put $q_0 = v_g$, $z_0 = \eta u$. Then G is exactly the sum of

$$\begin{aligned}
G_2 &= 4T_0e^2c_2[(r_2 + tq_0)^2 - tR_cq_0^2],\\
G_\beta &= 4T_0e^2c_\beta(1+v^2)[(r_\beta + Mq_0)^2 - R_cMq_0^2],\\
G_3 &= 4R_cc_3(1+v^2)[(r_3 + Jz_0)^2 - T_0Jz_0^2],\\
G_5 &= 4R_cc_5[(r_5 - az_0)^2 - aT_0z_0^2],\\
G_4 &= 4R_cT_0\eta^2c_4(1+u^2)(1+v^2)Y^2,\\
G_\alpha &= 4R_cT_0K^2\eta\zeta c_\alpha(1+u^2)r_\alpha^2,\\
G_d &= 2R_cT_0\eta c_d(1+u^2)[r_\alpha^2 - (1+v^2)Z^2].
\end{aligned}$$

These seven factored expressions agree separately with the defining source. Their proof requires multiplication of short polynomials only.

B.4.2 Finite monic divisions at the endpoint

Deriving the scalar and first-normal inputs by monic division. Here is a finite construction of every coefficient in the preceding table. It uses the seven factored terms, differentiation, and division by monic polynomials. Thus neither a stored expansion nor a rational-function ideal calculation is an input.

Let $\mathcal{R} = \mathbb{Q}[a, t, b, e, k]$ and use an indeterminate U for u^2 in this paragraph only. Put $K = k^2$. Substitute $(X, Y, v) = (X_g, Y_g, v_g)$ in the peeled polynomial E. Directly from the seven terms, the resulting expression has the form

$$\frac{E_g}{e^2b^4} = e^2\mathcal{A}(U) + ekZ\mathcal{B}(U) + k^2Z^2\mathcal{C}(U), \qquad \mathcal{A}, \mathcal{B}, \mathcal{C} \in \mathcal{R}[U].$$

The displayed division means coefficientwise polynomial extraction: the constant, linear and quadratic coefficients in Z have the factors e^2, ek and k^2, respectively. No evaluated tensor is divided. Substituting the graph into the factored terms gives the five short contact and axis identities

$$(\mathcal{A} + \mathcal{B} + \mathcal{C})(K) = 0, \qquad (\mathcal{B} + 2\mathcal{C})(K) = 0, \qquad (\mathcal{A} + \mathcal{B} + \mathcal{C})'(K) = 0,$$
$$\mathcal{A}(0) = \mathcal{B}(0) = 0.$$

Consequently the following monic divisions have zero remainder:

$$\mathcal{N} = \frac{\mathcal{A}+\mathcal{B}+\mathcal{C}}{(U-K)^2}, \qquad \mathcal{D} = \frac{\mathcal{B}+2\mathcal{C}}{U-K},$$
$$\mathcal{R}_0 = \frac{\mathcal{C}+K\mathcal{D}+K^2\mathcal{N}}{U}, \qquad \mathcal{J} = \frac{-\mathcal{D}-2K\mathcal{N}}{U}.$$

For the last two divisions, evaluate the numerators at $U=0$ and use the first two division identities together with $\mathcal{A}(0)=\mathcal{B}(0)=0$. This is a polynomial-ring argument; the symbols K and U are not inverted after evaluation. Set

$$A_{00} = \mathcal{R}_0(u^2), \qquad A_{01} = u\mathcal{J}(u^2), \qquad A_{11} = \mathcal{N}(u^2).$$

Comparing the three coefficients of $1, Z, Z^2$ proves exactly

$$\frac{E_g}{e^2b^4} = A_{00}\,[u(e-kZ)]^2 + A_{01}\,u(e-kZ)(eu^2-k^3Z) + A_{11}(eu^2-k^3Z)^2.$$

Multiplying by e^2b^4 gives the scalar identity used above.

For each $\xi \in \{X, Y, v\}$, let R_ξ be the graph restriction of $\partial_\xi E$, divided coefficientwise by e^2b^2, eb^2, e^2b^2, respectively. The first two are linear in Z. The third has one quadratic coefficient, supplied solely by the signed term:

$$[Z^2]R_v = -4R_cT_0c_d\,ekb^2u(1+u^2).$$

Define

$$C_X = C_Y = 0, \qquad C_v = 4R_cT_0c_d\,eb^2(1+u^2).$$

Then $\widetilde{R}_\xi = R_\xi - C_\xi Z\,u(e-kZ)$ is linear in Z and has the polynomial parity form

$$\widetilde{R}_\xi = eu\,L_{\xi,0}(U) + kuZ\,L_{\xi,1}(U), \qquad L_{\xi,i} \in \mathcal{R}[U].$$

In particular, subtracting the quadratic term also leaves its corrected linear coefficient divisible by ku; this follows upon collecting the signed term with the two current terms. The contact identity is $\big(L_{\xi,0}+L_{\xi,1}\big)(K)=0$. Form the monic quotient

$$J_\xi = \frac{L_{\xi,0}+L_{\xi,1}}{U-K}.$$

The promised full first-normal coefficients are now defined by the short formulas

$$P_\xi = (L_{\xi,0} - UJ_\xi)(u^2) + C_\xi Z, \qquad Q_\xi = uJ_\xi(u^2).$$

Indeed direct multiplication gives $R_\xi = P_\xi u(e-kZ) + Q_\xi(eu^2-k^3Z)$. This proves the full first-normal identities, including the quadratic-in-Z term; no first-normal term has been omitted.

For completeness, the weighted splits have an equally explicit finite rule. In a polynomial expanded in (e,k,b,u), place a monomial with k-exponent at least r in the first bucket and divide it by k^r; place every remaining monomial in the second bucket and divide it by e^sb^s. Use $(r,s)=(4,2)$ for A_{00} and $(r,s)=(2,1)$ for

$$A_{01}, \quad P_X, \quad P_Y, \quad P_v - 16kP_0.$$

Substitution of the seven short terms into the monic formulas shows that every monomial left for the second bucket has the required e^sb^s factor. Equivalently, the only polynomial remainders to check are

$$A_{00} \equiv 0 \pmod{(k^4, e^2b^2)}, \qquad A_{01}, P_X, P_Y, P_v - 16kP_0 \equiv 0 \pmod{(k^2, eb)}.$$

These are ordinary finite polynomial truncations: discard the monomials divisible by the displayed generators. The two bucket quotients have respectively residues $(-8,-8)$ for A_{00} and $(0,0)$ for each of the other four entries; A_{11} has residue 8, and all Q_ξ have residue zero. To obtain these numbers, after the divisions set $e=k=u=Z=0$, keeping a,t,b free. In particular the new term C_vZ belongs to $eb\,\mathcal{R}[u,Z]$ and its second-bucket quotient contains the infinitesimal testing factor Z. No assumption $\bar{b}=0$ is used.

This is a reproducible coefficient construction: the source consists of seven short terms, the scalar construction of four monic quotients, and each first-normal construction of one monic quotient. The finite checks accompany these formulas as a convenience; the definitions and the multiplication identities, rather than expanded coefficient arrays, are the proof objects.

B.4.3 The endpoint product relation

The new boundary mechanism is algebraic. A troublesome root is retained, and a second relation controls its product with an additional root. This supplies two positive pivots without comparing an infinitesimal in one field with an infinitesimal in the other.

Lemma B.10 (Product-relation completion)**.** *Let z be a root vector over T, and $h, r \in T$. Suppose*

$$F = (z,h,r)^T Q(z,h,r), \qquad Q \in \mathrm{Sym}(B), \qquad \bar{Q} = \mathrm{diag}(Q_0, -a, 0),$$

where Q_0 is a real positive-definite matrix and $a \in \mathbb{R}_{\geq 0}$. Suppose $p \in B$ has nonzero residue, A is a row with entries in $\mathfrak{n}$, and

$$ph = Az + r, \qquad phr = (z,h,r)^T E(z,h,r), \quad E \in \mathrm{Sym}(\mathfrak{n}).$$

Then $F \in \Sigma T^2$.

Proof. For a sufficiently large positive real M, add the exact zero

$$M\left[p^2h^2 + r^2 - (Az)^2 - 2phr\right] = 0.$$

The residue Gram becomes $\mathrm{diag}(Q_0, M\bar{p}^2 - a, M) \succ 0$. The subtracted terms have zero residue by hypothesis. Lemma 3.2 applies. In particular p is never inverted. □

Exact contact scope and full-polynomial input

This application uses the canonical C1 contact model, with $\lambda = n = 0$. Its source parameters are positive e, k, b, a, t satisfying

$$e, k \in \mathfrak{m}_c, \quad b \in B_c, \qquad R_c = 1 + k^2(1 - ae^2b^2), \quad T_0 = 1 + k^2 - tk^4b^2,$$
$$0 < a < T_0, \qquad 0 < t < R_c.$$

The signed weight satisfies $d > 0$, and all eight event coefficients are nonnegative original physical C1 contact coefficients. This is a specified contact chart, not every coefficient vector with the same omissions. The independent testing chart is $W = 1$ with u, v, X, Y, Z infinitesimal. In this section there is no restriction on k^3/e or k/e.

The complete seven-term contact polynomial is denoted G. Its short factored definition and unchanged original coefficient vector are given below.

Define

$$v_g = ekb^2u, \quad X_g = -uZ + tv_g, \quad Y_g = (1 - ae^2b^2)u,$$
$$N_1 = X - X_g, \quad N_2 = Y_g - Y, \quad \delta v = v - v_g, \quad D_t = Z - uX,$$
$$U_0 = 1 + u^2, \quad c_0 = 1 - tk^2b^2, \quad P_0 = T_0 + u^2 = U_0 + k^2c_0,$$
$$S = u + uvX - Y - vZ + (T_0 - a)e^2b^2u,$$
$$s_0 = eb^2u(e - kZ), \qquad q = eb^2(eu^2 - k^3Z).$$

Polynomial source. The source is defined (Section B.4.1).

Direct substitution gives the exact nonlinear relation

$$L = P_0s_0 - S + N_2 - D_t\delta v + uv_gN_1 - uc_0q = 0. \tag{E1}$$

Peel the two complete current squares:

$$E = G - H_3(1 + v^2)N_2^2 - H_5S^2, \qquad H_3 = 4R_cc_3, \quad H_5 = 4R_cc_5, \quad \bar{H}_3 = \bar{H}_5 = 8.$$

Each peeled term divided by e^2 is polynomial in bounded variables. The full graph scalar has the identity

$$E_g = A_{00}s_0^2 + A_{01}s_0q + A_{11}q^2,$$

and the entire first normals, with $g_0 = s_0/(eb^2)$ and $g_2 = q/(eb^2)$ denoting their displayed polynomial expressions, satisfy

$$E_X = e^2b^2(P_Xg_0 + Q_Xg_2), \quad E_Y = eb^2(P_Yg_0 + Q_Yg_2), \quad E_v = e^2b^2(P_vg_0 + Q_vg_2).$$

The exact polynomial splits and residues needed are

Quantity	Polynomial split	Quotient residues
A_{00}	$k^4a_0 + e^2b^2a_1$	$-8, -8$
A_{01}	$k^2c_1 + ebc_2$	$0, 0$
P_X	$k^2p_X + ebq_X$	$0, 0$
P_Y	$k^2p_Y + ebq_Y$	$0, 0$
$P_v - 16kP_0$	$k^2p_v + ebq_v$	$0, 0$

Also $\bar{A}_{11} = 8$ and $\bar{Q}_X = \bar{Q}_Y = \bar{Q}_v = 0$. These equalities remain valid with b symbolic; boundedness, rather than infinitesimality, suffices for b.

Coefficient construction. The finite coefficient construction is given (Section B.4.2).

Exact finite Taylor expansion in $(N_1, -N_2, \delta v)$ accounts for every higher-normal term. Because E/e^2 is polynomial, every term of normal degree at least two allocates to $x = eN_1$, $y = e\delta v$, N_2 with bounded coefficients. Terms of degree at least three have an extra infinitesimal normal. Terms involving N_2 retain an e factor. The only additional nonzero residue is

$$B_0 = \begin{pmatrix} 16 & -8 \\ -8 & 8 \end{pmatrix} \quad \text{on } (x, y).$$

It is visible without expansion of the full Hessian: at $u = Y = Z = 0$, the quadratic residue of E/e^2 is $8(X - v)^2 + 8X^2$, with b still free.

The new identity and its nine-root proof

If the signed testing core is nonnegative, its pure testing square root and the original event terms already give SOS. Otherwise the sign gauges $u, v \geq 0$ preserve every term, and the core Schur complement implies $u, v > 0$ and $D_t > 0$. To check the latter, after $Z = D_t + uX$ the Schur complement on (W, D_t) is

$$\frac{2(1+v^2)}{N_0} \begin{pmatrix} u^2v^2(1+u^2) & -uv(1+u^2)(1+v^2) \\ -uv(1+u^2)(1+v^2) & u^2v^2 \end{pmatrix},$$

where $N_0 = u^2v^4 + u^2v^2 + u^2 + v^4 + 2v^2 + 1 > 0$. The eliminated block is positive definite. For $W = 1, D_t \leq 0$, the remaining form is nonnegative, proving the claim.

Consequently $w = \sqrt{ek}\sqrt{D_t}\,\delta v$ is a legal separated root. Set

$$h = k^2 s_0 - uq = e^2b^2u(k^2 - u^2), \qquad z = ebs_0, \qquad r = k^2 D_t \delta v,$$
$$A_0 = k^2(S - N_2) - k^3b^2u^2x - uU_0q.$$

Multiplication of (E1) by k^2 gives $P_0h = A_0 + r$. The additional exact identity is

$$\begin{aligned} P_0hr = {} & ku^2(P_0 - k^2)y(S - N_2) - ku(P_0 + c_0k^2u^2)yq \\ & + u^2(P_0 - k^2)w^2 - k^4b^2u^2T_0xy. \end{aligned} \tag{E2}$$

Every displayed coefficient has zero residue and is a polynomial in separately bounded factors. To derive (E2), substitute the two soft roots to obtain

$$u(P_0 - k^2)s_0 - (1 + c_0u^2)q = ekb^2(k^2 - u^2)(D_t + uN_1).$$

Then use $D_t = U_0Z - uN_1 - tuv_g$, $T_0 = 1 + k^2c_0$, and (E1) to replace $S - N_2$ in (E2). The terms in $D_t(\delta v)^2$ cancel; the remaining terms give P_0hr. This is a whole-polynomial calculation.

Let J_0 denote the right-hand side of (E2). Thus

$$Z_0 = P_0^2h^2 + r^2 - A_0^2 - 2J_0 = 0.$$

Add to the entire G the exact zero expression

$$-16kyL + 16(ebL)^2 + 16Z_0.$$

The first term cancels the exceptional $16kP_0$ first normal and supplies $16w^2$. Use $k^2s_0 = h + uq$ in every other scalar and first-normal term. The scalar becomes

$$a_0(h + uq)^2 + a_1z^2 + [c_1(h + uq) + c_2z]q + A_{11}q^2.$$

Its only negative residue diagonals are -8 on h, z. The safe relation

$$ebL = P_0z - ebS + ebN_2 - bD_ty + ekb^3u^2x - ebuc_0q$$

has only infinitesimal coefficients apart from P_0; hence $16(ebL)^2$ changes the z diagonal to 8. Every coefficient of A_0 is infinitesimal, as is every coefficient in (E2). Thus $16Z_0$ changes the h diagonal to 8 and supplies a new r diagonal 16, with no other residue change.

The full Gram on

$$\boxed{(x, y, N_2, S, w, q, h, z, r)}$$

therefore has residue

$$\boxed{B_0 \oplus \operatorname{diag}(8, 8, 16, 8, 8, 8, 16) \succ 0.}$$

Its first two LDL pivots are $16, 4$; the other seven pivots are given above. The strict-residue lemma proves SOS of the complete seven-term polynomial for every coefficient ratio k^3/e.

Restoration and coverage

The complete original-parent restoration is

$$\Lambda F = G + 4T_0e^2c_2\frac{t}{M}(r_2 - Mq_0)^2 + 4T_0e^2c_\beta\frac{M}{t}(1+v^2)(r_\beta - tq_0)^2 + 4R_cc_3\frac{J}{a}(1+v^2)(r_3 - az_0)^2 + 4R_cc_5\frac{a}{J}(r_5 + Jz_0)^2.$$

Every additional term is a nonnegative coefficient factor times a pure-testing factor times a square. This identity is four instances of

$$S^2 + 2(A+B)qS + (A+B)Aq^2 + \frac{B}{A}S^2 = \frac{A+B}{A}(S+Aq)^2.$$

Use respectively $(A,B,q) = (M,t,q_0), (t,M,q_0), (a,J,z_0), (J,a,-z_0)$ and $S = r_2 - Mq_0, r_\beta - tq_0, r_3 - az_0, r_5 + Jz_0$. Comparing the seven resulting coefficients with the displayed source proves the identity. Since Λ is a positive pure coefficient factor, the pure-factor restoration in RFA13 (p. 2) recovers the original parent. Normalizing the original state to $W = 1$ and undoing that normalization use only testing factors.

The result applies to every positive bounded b, with no condition on k^3/e. The unbounded-b case is proved separately in Section B.4.5; the full source and state partition is given in Section B.4.6.

B.4.4 The uniform small-e source on the bounded Z chart

Domain and conclusion. Use the original physical uniform C1 source with $d = 1$, $\lambda = n = 0$, $k, f, g, e = kf/g$ positive infinitesimals, and all original nonnegative event weights. Retain

$$r = 1 - af^2, \quad R = 1 + k^2r, \quad T = 1 + k^2 - tk^2g^2, \quad M = R - t > 0, \quad J = T - a > 0, \qquad a, t > 0. \tag{EZ1}$$

In particular a, t are bounded, r, R, T are positive units, and M, J may approach zero at arbitrary rates. For every independent testing u, v, X, Y infinitesimal and $Z = 1$, W arbitrary bounded of either sign, the entire original parent is a separated tensor SOS. This includes both bounded and unbounded $b = g/k$, and every permitted $\beta = 0$ face. It does not claim the complementary $W = 1, Z$-infinitesimal chart.

The exact source and a common set of bounded coefficients

Use the seven short native coquotients $c_2, c_\beta, c_3, c_5, c_4, c_\alpha, c_d$ in the retained Section B.4.1, now with $\eta = f^2, \zeta = g^2$. Their residues are $2, 2, 2, 2, 2, 2, 4$. In the present $d = 1$ normalization the exact weights are

$$C = \frac{c_3}{af^2c_d}, \quad A = \frac{f^2c_4}{c_d}, \quad m = \frac{k^2c_2}{g^2Mc_d}, \quad \alpha = \frac{k^4g^2c_\alpha}{c_d}, \qquad D + \tfrac{1}{2} = \frac{c_5}{Jf^2c_d}, \qquad \beta + \tfrac{1}{2} = \frac{k^2c_\beta}{g^2tc_d}. \tag{EZ2}$$

Every use below is of these actual weights. In particular $\beta \geq 0$ is retained, not replaced by a formal source condition.

The following two pure-source coordinate choices are exhaustive:

	s	ω	P	G	$\kappa = P/G$
b bounded	e	b^2	k^2	k^2	1
b unbounded	f	1	k^2	g^2	$(k/g)^2$

(EZ3)

In both rows $f^2 = s^2\omega$, $g^2 = \omega G$, $s, \sqrt{P}, \sqrt{G}$ are infinitesimal, and ω, κ are positive bounded. In the second row κ is infinitesimal. Here $\beta \geq 0$ gives exactly $t \leq 2\kappa c_\beta/c_d$, so $t = \kappa\tau$ with $\tau > 0$ bounded; consequently M is a unit in this row.

Define bounded source coefficients

$$c_* = \frac{c_3}{Tc_d}, \quad d_* = \frac{c_5}{Tc_d}, \quad a_* = \frac{c_4}{c_d}, \quad \gamma_* = \frac{c_\alpha}{c_d}, \qquad B_* = \omega\beta t = \frac{\kappa c_\beta}{c_d} - \frac{\omega t}{2}. \tag{EZ4}$$

The first four have residue $1/2$; $B_* \geq 0$ may vanish. Use the further table

	μ	χ	ψ
b bounded	$c_2/(Rc_d)$	t	tM
b unbounded	$\kappa c_2/(Mc_d)$	0	0.

(EZ5)

All entries are bounded. In the first row μ has residue $1/2$; in the second μ is infinitesimal.

Exact removal of the source poles

The original quarter-parent is

$$\begin{aligned} F = {} & mr_2^2 + CD_v r_3^2 + AD_u D_v Y^2 + (D+\tfrac{1}{2})r_5^2 + (\alpha+\tfrac{1}{2})D_u(vY-1)^2 \\ & + (\beta+\tfrac{1}{2})D_v(X+u)^2 - \tfrac{1}{2}D_u D_v, \end{aligned} \tag{EZ6}$$

where $r_2 = vW - X + uvY - u$, $r_3 = uW - Y$, $r_5 = uW + uvX - Y - v$, $D_u = 1+u^2$, $D_v = 1+v^2$. Put

$$z = sW, \qquad q_0 = f^2 uW^2 = \omega u z^2, \qquad z_0 = f^2 uW = s\omega uz.$$

Retain the following complete nonnegative terms of the original polynomial:

$$\frac{CJ}{T}D_v(r_3 - az_0)^2, \qquad \frac{(D+1/2)a}{T}(r_5 + Jz_0)^2, \qquad \beta D_v(X + u - tq_0)^2. \tag{EZ7}$$

For bounded b, also retain

$$\frac{mt}{R}(r_2 - Mq_0)^2. \tag{EZ8}$$

Each vanishes at the actual source contact. They are literal positive squares, even when $\beta = 0$ or a source opening is small. The first two removals use $a + J = T$ and the exact identity

$$\frac{a}{T}[(L+Jq)^2 - TJq^2] + \frac{J}{T}(L - aq)^2 = L^2.$$

The last removal uses $r_2^2 - (t/R)(r_2 - Mq_0)^2 = (M/R)(r_2^2 + 2tq_0 r_2 - tMq_0^2)$. There is no mixed denominator.

Use all three unrestricted normal coordinates

$$N_x = X + u - tq_0, \qquad N_y = (Y - ruW)/s, \qquad N_v = (v - f^2 uW)/s. \tag{EZ9}$$

Only N_x is necessarily bounded. The exact bounds needed later are $N_x, sN_y, sN_v \in \mathrm{I}$; they follow from the stated testing chart and bounded W. No boundedness of N_y, N_v is assumed. Define

$$X_* = -u + tq_0 + N_x, \qquad y_* = ruz + s^2 N_y = sY, \qquad v_* = \omega uz + N_v = v/s,$$

$$L_2 = zv_* + uv_* y_* - X_* - u, \quad L_3 = T\omega uz - N_y, \quad L_5 = -N_y + v_*(uX_* - 1).$$

After multiplying the original quarter-parent by the positive pure-source ω and removing precisely the scaled terms EZ7–EZ8, the *entire* remaining polynomial is

$$\begin{aligned} \mathcal{F} = {} & \mu(L_2^2 + 2\chi q_0 L_2 - \psi q_0^2) + c_* D_v(L_3^2 - TJ\omega^2 u^2 z^2) \\ & + \omega^2 a_* D_u D_v y_*^2 + d_*(L_5^2 - aT\omega^2 u^2 z^2) \\ & + (\omega^2 P^2 G\gamma_* + \omega/2)D_u(v_* y_* - 1)^2 + \frac{\omega}{2}D_v(X_* + u)^2 \\ & + B_* D_v\{2q_0(X_* + u) - tq_0^2\} - \frac{\omega}{2}D_u D_v, \qquad D_v = 1 + s^2 v_*^2. \end{aligned} \tag{EZ10}$$

This is a short full-parent definition of every scalar and normal coefficient used below. In particular no normal was set to zero in obtaining it. All its coefficient arrays are bounded through a, J, M, t; the unbounded original weights occur only in the retained complete squares.

The entire scalar is a cubic circuit with finite corrections

Set $U = u^2, V = z^2$. The polynomial obtained from EZ10 at $N_x = N_y = N_v = 0$ is exactly

$$\mathcal{F}_0 = \omega^2 \Phi(U, V), \qquad \Phi = \gamma_* P^2 G(1+U) + UV\, H(U, V), \tag{EZ11}$$

where H has bounded finite polynomial coefficient arrays. The apparent divisions by ω^2 and UV in this definition are coefficientwise polynomial extractions: substitution in EZ10 shows those factors before any evaluation. No testing coordinate is inverted.

At the actual source contact $u^2 = P, z^2 = G$ all three normal coordinates and all retained terms vanish. The original contact value and stationary equations give

$$\Phi(P, G) = \Phi_U(P, G) = \Phi_V(P, G) = 0. \tag{EZ12}$$

The exact native coquotients in EZ2 independently verify these identities. Substituting EZ11, and cancelling nonzero pure-source factors in the source field, gives

$$H(P, G) = -\gamma_* P(1+P), \quad H_U(P, G) = \gamma_*, \quad H_V(P, G) = \gamma_* \kappa (1+P).$$

Write $\Delta = U - P$, $\Gamma = V - G$. Finite monic division, first in $U - P$ and then in $V - G$, therefore gives the complete scalar identity

$$\begin{aligned}\Phi &= \gamma_*\{\mathcal{M} + P\kappa U\Gamma^2\} \\ &\quad + UV\{\Delta^2 H_{20} + \Delta\Gamma H_{11} + \Gamma^2 H_{02}\}, \\ \mathcal{M} &= U^2V + \kappa UV^2 + P^2G - 3PUV.\end{aligned} \tag{EZ13}$$

All quotients are finite bounded coefficient polynomials; no inverse of P, G is introduced by their monic construction. The extra term $\gamma_* P\kappa U\Gamma^2$ is retained as a nonnegative square.

In the inverse-b row one further coefficient factor is needed:

$$H_{02} = \kappa\widetilde{H}_{02}, \qquad \widetilde{H}_{02} \text{ has bounded arrays.} \tag{EZ14}$$

Here is a finite source-division certificate, rather than an estimate involving independent testing rates. Write $P = \kappa G$, $t = \kappa\tau$ before substituting the native coquotients into EZ10. Every denominator is among the pure-source units M, T, c_d and their factors. At formal $\kappa = 0$,

$$P = t = \mu = B_* = 0, \quad c_* = d_* = \frac{1 + (1-a)s^2}{2}, \quad a_* = \frac{1}{2(1 - as^2)}.$$

The formula gives $H(0, V) = 0$ identically in V. After clearing unit denominators, ordinary polynomial division by κ therefore has zero remainder. Apply the two monic $V - G$ divisions to this quotient. This proves EZ14 with bounded arrays. In the bounded-b row $\kappa = 1$, so it needs no such factor.

Six explicit strict roots and their conductors

The following identities apply to both rows of EZ3. Put

$$\mathsf{h} = G^{1/6}, \quad \mathsf{c} = \kappa^{1/6}, \quad \mathsf{r} = \sqrt[3]{u}, \quad \mathsf{s} = \sqrt[3]{z},$$

$$(L_1, L_2, L_3) = (\mathsf{r}^4\mathsf{s}^2,\ \mathsf{c}^2\mathsf{r}^2\mathsf{s}^4,\ \mathsf{h}^6\mathsf{c}^4), \qquad (f_1, f_2, f_3) = (\mathsf{r}^2\mathsf{s},\ \mathsf{crs}^2,\ \mathsf{h}^3\mathsf{c}^2).$$

Define $R_{2i-1} = f_i(L_1 - L_2)$ and $R_{2i} = f_i(L_1 - L_3)$ for $i = 1, 2, 3$. Then, exactly,

$$\mathcal{M} = \sum_{i=1}^{3}(R_{2i-1}^2 - R_{2i-1}R_{2i} + R_{2i}^2). \tag{EZ15}$$

Each pair has the fixed strictly positive Gram $\left(\begin{smallmatrix} 1 & -1/2 \\ -1/2 & 1 \end{smallmatrix}\right)$. Real cube roots of signed testing quantities are permitted: $z = sW$ has the separated cube root $s^{1/3}\sqrt[3]{W}$. The $L_i = f_i^2$ identity is valid for either sign of u, z. No mixed radical occurs.

For explicit finite allocation define

$$J_1 = \mathsf{r}^3R_1 + \mathsf{c}^2\mathsf{rs}^2R_2, \qquad J_2 = -\mathsf{h}^6R_1 + \mathsf{s}^6R_2 - \mathsf{h}^3\mathsf{s}^3R_5.$$

Direct multiplication gives

$$\begin{aligned} uz\Delta &= J_1, & u^2z\Gamma &= J_2, \\ \kappa uz\Gamma &= \mathsf{c}^2\mathsf{rs}^2(R_2 - R_1) - \mathsf{c}^3\mathsf{s}^3R_3, \\ UV\Delta^2 &= J_1^2, \\ UV\Delta\Gamma &= \mathsf{s}^6R_2^2 - \mathsf{h}^6R_1^2 - \mathsf{h}^3\mathsf{s}^3R_1(R_5 + R_6), \\ \kappa UV\Gamma^2 &= \mathsf{s}^6(2R_1^2 - 3R_1R_2 + R_2^2 + R_3^2) + \mathsf{h}^3\mathsf{s}^3R_1(R_5 - R_6). \end{aligned} \tag{EZ16}$$

All coefficients on the right are infinitesimal: $\mathsf{r}, \mathsf{s}, \mathsf{h}$ are infinitesimal and c is bounded. It is harmless that $\mathsf{c} = 1$ in the first row. Equations EZ13–EZ16 therefore put the entire scalar, apart from its retained positive square, on the six rows ωR_i with a bounded Gram whose real residue is $\frac{1}{2}$ times the strict matrix in EZ15.

Every complete first normal and all higher terms

Define L_x, L_y, L_v as half the three derivatives of the whole polynomial EZ10 at zero normals. Expanding its short factored form gives exactly

$$L_x = \omega uz^2T_x(U, V), \qquad L_y = \omega uzT_y(U, V), \qquad L_v = \omega uzT_v(U, V), \tag{EZ17}$$

where each T_i is a polynomial with bounded coefficient arrays. This supplies a finite construction of every coefficient: differentiate EZ10, set the three normals to zero, and extract the displayed monomial. No derivative is dropped. The exact actual-source contact equations give $T_i(P, G) = 0$. Consequently the monic divisions are

$$T_i = (U - P)A_i(U, V) + (V - G)B_i(V). \tag{EZ18}$$

For inverse b, each $B_i = \kappa\widetilde{B}_i$ with bounded arrays. Indeed the same source substitution used in EZ14 gives $T_i(0, V) = 0$ identically at $\kappa = 0$. Divide the cleared source polynomial by κ and then by the monic $V - G$. This is an exact coefficient construction; it uses neither $1/u$, $1/z$, nor an evaluated mixed quotient. For bounded b, $\kappa = 1$ again suffices.

Combining EZ16–EZ18 proves that both complete state normals and the complete testing-angle normal have infinitesimal coefficients on the strict six-row list ωR_i. The additional factor z in L_x is bounded and infinitesimal. This argument uses the full first-normal polynomial, not just its value or linear approximation at contact.

The full normal expansion of EZ10 has a bounded quadratic Gram whose real residue, in order (N_x, N_y, N_v), is

$$\begin{pmatrix} \bar{\mu} + \bar{\omega}/2 & 0 & 0 \\ 0 & 1 & 1/2 \\ 0 & 1/2 & 1/2 \end{pmatrix} \succ 0. \tag{EZ19}$$

For bounded b, $\bar{\mu} = 1/2$; for inverse b, $\omega = 1$ and $\bar{\mu} = 0$. Both give a strictly positive first pivot. The other block has pivots $1, 1/4$.

Here is also a finite justification for all higher terms, since the scaled normals N_y, N_v may be unbounded. Expanding EZ10 in the three normals gives degree at most four. A monomial of normal degree at least three containing N_x has total N_y, N_v degree at most two. Assign those factors, and any needed extra N_x factor, to two Gram rows; the remaining factor contains $N_x \in \mathrm{I}$. For a higher monomial without N_x, each of the N_y, N_v exponents is at most two, and its coefficient has a factor s^2. If its degree is three, the unassigned coefficient is a bounded multiple of $s(sN_y)$ or $s(sN_v)$. If its degree is four, it is a bounded multiple of $(sN_y)(sN_v)$ (or of one of their squares). All are infinitesimal by EZ9. These support and divisibility statements follow directly from the factored quadratics in EZ10 and are checked on the entire polynomial. Thus no unbounded scaled normal is hidden in a Gram coefficient.

The scalar, all first normals, and the complete hard block now give a finite bounded Gram with strictly positive real residue. The separated strict-Gram lemma supplies tensor SOS. Restore the whole retained terms EZ7–EZ8 and the positive square in EZ13, and divide by the pure-source positive ω. This proves the stated original-parent theorem on both source rows, without a comparison of source and testing rates.

State and boundary scope. The signs of W and of the testing angles were not restricted in EZ10–EZ19. The polynomial and cube-root identities include literal zero testing coordinates. If a bounded normalized $Z = 1$ state has X or Y a unit, the original pure-testing core has positive residue at the double-small angle corner; its positive source multiple is already a separated square, and all original event terms are nonnegative squares. Thus this proof supplies the entire Z-unit state chart. Complex states split into their real and imaginary quadratic evaluations. Section B.4.6 combines this certificate with the two strong-current outer testing identities and the complementary W-unit chart to cover all tests.

B.4.5 The uniform small-e source at unbounded b

Statement and certificate ring. Let K_c, K_t be independent real closed extensions of $\mathbb{R}$, with bounded rings B_c, B_t and infinitesimal ideals $\mathfrak{m}_c, \mathfrak{m}_t$. The certificate ring is $\mathcal{T} = K_c \otimes_{\mathbb{R}} K_t$, with bounded coefficient ring $\mathcal{B} = B_c \otimes_{\mathbb{R}} B_t$ and zero-residue ideal $\mathfrak{n} = \mathfrak{m}_c \otimes B_t + B_c \otimes \mathfrak{m}_t$. We use the finite strict-residue Gram lemma in this ring. All radicals and inverses below are pure source, pure testing, or products of separately specified such factors.

Use the actual uniform native C1 stationary source, with $\lambda = n = 0$, $d > 0$, and the original nonnegative event coefficients. Normalize $d = 1$ by a positive pure-source scale. Suppose

$$k, f, g > 0 \text{ are infinitesimal}, \qquad k/g \in \mathfrak{m}_c, \qquad r = 1 - af^2, \quad h = 1 - tg^2,$$

$$R = 1 + k^2 r, \quad T_0 = 1 + k^2 h, \quad 0 < a < T_0, \quad 0 < t < R, \quad \beta \geq 0. \tag{IW1}$$

All original strict gaps remain imposed. Then the complete original parent is tensor-SOS whenever the independent positive testing angles u, v are infinitesimal, for every real or complex testing state. The proof below works in the rotated chart $w = 1, z > 0$ with z arbitrary; in particular it includes the previously missing original W-unit, Z-infinitesimal chart. It uses no bound on g/f and no comparison between any testing and source infinitesimals. The exact sign, zero-axis and state reductions are given at the end.

The actual C1 source and bounded source arrays

For clarity, the six weights used here come from the following unchanged native formulas. Put $x_n = k^2$, $\eta = f^2$, $\zeta = g^2$, $y_n = x_n\eta\zeta$, $K = 1 + x_n$, $G = h + \zeta$, and

$$M = R - t, \quad J = T_0 - a, \quad E_n = 1 - x_n\eta h, \quad Q_n = 1 - x_n\zeta r, \quad \theta = t\zeta.$$

Define

$$\begin{aligned} P_d &= 1 + 2x_n h + r x_n^2 G + x_n\zeta + \eta x_n(K\zeta + x_n h\theta),\\ P_4 &= T_0 E_n + x_n(1-r)G, \qquad P_\alpha = Rh(1+y_n) + \zeta M E_n. \end{aligned}$$

The actual relative coefficients, now with $d = 1$, are

$$\begin{aligned} m &= \frac{Kx_n(r+\eta)h(1+y_n)}{2M\zeta P_d}, & C &= \frac{K(r+\eta)E_n}{2(1-r)P_d},\\ A &= \frac{\eta P_4}{2rP_d}, & D + \tfrac12 &= \frac{K(1+y_n)(r+\eta)}{2J\eta P_d},\\ \beta + \tfrac12 &= \frac{Kx_n(r+\eta)G}{2\theta P_d}, & \alpha &= \frac{x_n^2\zeta(r+\eta)P_\alpha}{2Q_n P_d}. \end{aligned} \tag{IW2}$$

No auxiliary C2 stationary source is used. Since a, t are bounded and k, f, g are infinitesimal, r, h, P_d, R, T_0 have residue one. The inequality $\beta + \frac12 \ge \frac12$ in (IW2) gives $t = O(k^2/g^2)$. Thus t and $p = t/R$ are infinitesimal, and $M = R(1-p)$ is a positive unit. The two independent openings a and $J = T_0 - a$ are not capped.

Introduce the exact compact coordinates

$$j = \frac{T_0}{g^2R}, \quad B = \frac{k^2g^2R}{T_0}, \quad T = \frac{f^2T_0}{R}, \quad p = \frac{t}{R}, \quad H = \frac{a}{T_0}, \qquad x = jB, \quad y = TB, \quad q_z = jT. \tag{IW3}$$

Here j is unbounded, $0 < H < 1$, and

$$T, x, B, p, \sigma := 1/j, \delta := j^2B \in \mathfrak{m}_c, \qquad \delta \sim k^2/g^2, \qquad p = O(\delta). \tag{IW4}$$

The quantity q_z can have any positive size. The exact inverse is

$$\begin{aligned} R &= \frac{1+x}{1+HTx}, \quad T_0 = \frac{1+x}{1+Bp}, \quad r = \frac{1-HT}{1+HTx}, \quad h = \frac{j-p}{j(1+Bp)},\\ f^2 &= \frac{T(1+Bp)}{1+HTx}, \quad g^2 = \frac{1+HTx}{j(1+Bp)}, \quad a = HT_0, \quad t = pR. \end{aligned} \tag{IW5}$$

In particular both strict gaps are retained exactly.

Here is a short rational dictionary sufficient to check every source estimate. Write $D_+ = D + \frac12$, $b_+ = \beta + \frac12$, $\mu = m(1-p)$, and set

$$\begin{aligned} P &= 1 + B(j+1-p) + BT(1+Hj)\\ &\quad + B^2T(Hj + jp - p^2 + p) + B^2HT^2j(1+Bp),\\ L &= 1 + T(1 - H + Bp), \qquad E = 1 + B(HT + p - 1),\\ F_\alpha &= j + 1 - 2p + BHTj(1-p) + BTp(j-p). \end{aligned}$$

Substitution of (IW5) in (IW2) gives exactly

$$\begin{aligned} \mu &= \frac{Bj(j-p)(1+y)(1+Bp)L}{2(1+x)P},\\ C &= \frac{(1+Bp)L[1 + BTp - BTj(1-H)]}{2HT(1+x)P},\\ A &= \frac{T(1+Bp)[1 + BT(2Hj - Hp + H - j + p) + B^2H^2T^2j]}{2(1-HT)P},\\ D_+ &= \frac{(1+y)(1+Bp)(1+HTx)L}{2T(1-H)(1+x)P},\\ b_+ &= \frac{Bj(1+Bp)L(j+1-p+HTx)}{2p(1+x)P},\\ \alpha &= \frac{B^2LF_\alpha}{2EP}. \end{aligned} \tag{IW6}$$

The denominators $P, E, 1+x, 1-HT$ are positive bounded units. After $j = 1/\sigma$, $B = \delta\sigma^2$, each of the following normalized arrays is a rational function with polynomial numerator and denominator in the bounded variables σ, δ, T, p, H; its denominator has a nonzero real constant at $\sigma = \delta = T = p = 0$, independently of H:

$$\frac{2\mu}{\delta}, \quad \frac{2b_+p}{\delta}, \quad 2TCH, \quad 2TD_+(1-H), \quad \frac{2A}{T}, \quad \frac{2\alpha}{B^2j} \quad \text{all have residue 1.} \tag{IW7}$$

This finite polynomial criterion proves boundedness and the stated residues at every relative rate and at both H openings. It also shows $\beta p \le b_+p = O(\delta)$, including $\beta = 0$. A useful exact cancellation is

$$K_d := CH - D_+(1-H) = -\frac{b_+p}{j}. \tag{IW8}$$

The entire parent and both state normals

Put $D_u = 1+u^2$, $D_v = 1+v^2$. The quarter-parent is

$$\begin{aligned}\mathcal{P} &= m(vW - X + uvY - uZ)^2 + CD_v(uW - Y)^2 + AD_uD_vY^2 \\ &\quad + D_+(uW + uvX - Y - vZ)^2 + (\alpha + \tfrac12)D_u(vY - Z)^2 \\ &\quad + b_+D_v(X + uZ)^2 - \tfrac12 D_uD_vZ^2.\end{aligned} \tag{IW9}$$

On the rotated chart $w = (W + uY)/D_u = 1$, $z = (Z - uX)/D_u > 0$, use the two unrestricted normals

$$N = \frac{X + uZ - pD_uv}{D_u}, \qquad M_0 = \frac{uW - Y - HD_uvz}{D_u}.$$

Their exact inverse is

$$X = N + pv - uz, \quad Z = z + uN + puv, \quad W = 1 + u(M_0 + Hvz), \quad Y = u - M_0 - Hvz. \tag{IW10}$$

Only the pure-testing unit D_u is inverted. Retain the positive term $mpD_u^2N^2$, and put $G = (\mathcal{P} - mpD_u^2N^2)/D_u$. Direct substitution, before any state specialization, gives

$$\begin{aligned}G &= \mu D_u[N^2 - 2vN + (1-p)v^2] + CD_uD_v(M_0 + Hvz)^2 + AD_v(u - M_0 - Hvz)^2 \\ &\quad + D_+D_u(M_0 - (1-H)vz)^2 \\ &\quad + (\alpha + \tfrac12)[u(1-p)v - (1 + Hv^2)z - uN - vM_0]^2 \\ &\quad + b_+D_uD_v(N + pv)^2 - \tfrac12 D_v(z + uN + puv)^2.\end{aligned} \tag{IW11}$$

In particular G is exactly quadratic in the two normals:

$$G = F + 2L_NN + 2L_MM_0 + H_NN^2 + H_MM_0^2 + 2(\alpha + \tfrac12)uvNM_0, \tag{IW12}$$

where $F = G|_{N=M_0=0}$ and

$$\begin{aligned}h_0 &= \mu + b_+ \ge \tfrac12, \\ H_N &= h_0D_u + b_+v^2 + \beta u^2v^2 + \alpha u^2, \\ H_M &= D_u(CD_v + D_+) + AD_v + (\alpha + \tfrac12)v^2.\end{aligned} \tag{IW13}$$

The potentially unbounded β is fully retained in (IW11)–(IW13). There is no omitted higher-normal polynomial.

An exact separated scalar circuit

Define the rows

$$U = u/\sqrt{x}, \quad V = v/\sqrt{y}, \quad Z_* = z/\sqrt{q_z}, \quad Q_* = U - VZ_*, \quad B_* = V - UZ_*, \quad R_* = Z_* - UV,$$
$$\Phi = U^2 + V^2 + Z_*^2 + U^2V^2Z_*^2 - 4UVZ_*.$$

None of U, V, Z_* is assumed bounded. On the positive chart,

$$\Phi = (U-V)^2 + Z_*^2(1-UV)^2 + 2UV(1-Z_*)^2 \tag{IW14}$$

is a legal separated SOS; for example $\sqrt{UV} = (xy)^{-1/4}\sqrt{uv}$.

For an explicit finite definition of the actual masses, put $K_0 = CH^2 + D_+(1-H)^2$ and

$$\begin{aligned}&M_a = Ax, && M_b = [\mu(1-p) + b_+p^2]y, \\ &M_c = \alpha q_z, && M_{bc} = [K_0 + AH^2 + 2H\alpha + H - \tfrac12]yq_z, \\ &M_{b^2} = b_+p^2y^2, && M_{b^2c} = H^2(A + C + \alpha + \tfrac12)y^2q_z, \\ &M_{ab^2c} = CH^2xy^2q_z, && M_{ab^2} = \beta p^2xy^2.\end{aligned}$$

Set

$$f_c = M_b, \quad h_c = M_{ab^2}, \quad k_c = (M_a - M_b + M_{ab^2c} - M_{bc} + h_c)/2,$$
$$r_c = M_c - M_b - k_c, \quad q_c = M_a - f_c, \quad g_c = M_{ab^2c}, \quad d_c = M_{b^2c} - k_c, \quad e_c = b_+p^2y^2 - g_c.$$

The complete scalar from (IW11) satisfies the exact identity

$$\begin{aligned} F = {} & q_cQ_*^2 + r_cR_*^2 + d_cV^2Q_*^2 + e_cV^2B_*^2 \\ & + f_c\Phi + g_cV^2\Phi + k_cZ_*^2(V^2-1)^2 + h_cV^2R_*^2. \end{aligned} \tag{IW15}$$

The last term is the extra nonnegative square supplied by the actual β event.

The same finite bounded-array criterion as (IW7), with the displayed mass definitions, gives

$$\frac{2f_c}{Tx^2} \sim 1, \quad \frac{2q_c}{Tx} \sim 1, \quad \frac{2r_c}{TjB^2} \sim 1, \quad \frac{k_c}{T^2B^2j} \in \mathfrak{m}_c, \quad \frac{g_c}{f_c} \in \mathfrak{m}_c. \tag{IW16}$$

Here ~ 1 means residue one. In particular the three principal masses f_c, q_c, r_c are positive, and $|k_c|/r_c$, $|k_c|/(q_cy)$ are infinitesimal. All normalized mass denominators remain units after $j = 1/\sigma$, $B = \delta\sigma^2$; the companion finite-array checks test their residues independently of H.

For the signed terms use the exact identities

$$\begin{aligned} (1+U^2)\Phi &= R_*^2 + B_*^2 + U^2Q_*^2 + U^2(1-UVZ_*)^2, \\ (1+V^2)\Phi &= Q_*^2 + R_*^2 + V^2B_*^2 + V^2(1-UVZ_*)^2, \\ Z_*(V^2-1) &= -VQ_* - R_*. \end{aligned} \tag{IW17}$$

The e_c, g_c combination in (IW15) is exactly

$$b_+p^2y^2V^2B_*^2 - g_c\Phi + g_c[Q_*^2 + R_*^2 + V^2(1-UVZ_*)^2].$$

A k_c of either sign costs at most $3|k_c|V^2Q_*^2 + 2|k_c|R_*^2$, including its occurrence in d_c. The corresponding relative costs are $3|k_c|v^2/(q_cy)$ and $2|k_c|/r_c$, both in $\mathfrak{n}$. Every inequality here abbreviates a finite SOS identity, using $2A^2 + 2B^2 - (A+B)^2 = (A-B)^2$ and nonnegative pure-source factors. Consequently the scalar supplies strict fixed real fractions of

$$q_cQ_*^2, \qquad r_cR_*^2, \qquad f_c\Phi. \tag{IW18}$$

All other nonnegative terms remain available as whole squares.

Finite budgets for the complete normal rows

Call a row free if any fixed positive real multiple of its square can be subtracted at infinitesimal cost from (IW18), by a finite separated SOS identity. This definition allows a bounded normal coupling to be absorbed by a sufficiently large fixed real helper pivot. Put

$$q = vz - Tu, \quad b = v - uz/j, \quad R_z = z - uv/B, \quad R_u = v(u^2-x), \quad R_v = u(v^2-y),$$
$$J_u = u(u^2-x), \qquad J_v = v(v^2-y), \qquad \widehat{q} = q/\sqrt{T}.$$

Since $Q_* = -q/(T\sqrt{x})$, the first mass in (IW18) has a positive unit coefficient on $\widehat{q}^2$.

First, $R_u = \sqrt{y}\,x(-B_* - UR_*)$. Applying the first identity of (IW17) to B_*^2, and the actual r_c pivot to $U^2R_*^2$, shows that R_u is free. The three relative costs are bounded by constant multiples of

$$B, \qquad u^2/j, \qquad u^2, \tag{IW19}$$

all infinitesimal. Second, $R_v = \sqrt{x}\,y(-Q_* - VR_*)$, so $\sqrt{T}R_v$ is free; its two costs are $O(y^2)$ and $O(Tv^2)$.

The same-angle rows have exact decompositions

$$\begin{aligned} J_v &= \sqrt{y}\,v(v + \sqrt{y/x}\,u)(V-U) + (y/x)R_u, \\ J_u &= \sqrt{x}\,u(u + \sqrt{x/y}\,v)(U-V) + (x/y)R_v. \end{aligned} \tag{IW20}$$

If $\rho = O(\delta)$ is a nonnegative source factor, then ρJ_v is free. Indeed $\rho\sqrt{y/f_c} = O(\sqrt{\delta})$, $\sqrt{y/x} = \sqrt{T/j}$ is bounded, and $\rho y/x$ is infinitesimal. Thus every coefficient after division by the strict $\sqrt{f_c}(V-U)$ pivot is infinitesimal. Also $\sqrt{T}\,TxJ_u$ is free: the two coefficients of its first summand relative to $\sqrt{f_c}(U-V)$ are bounded multiples of

$$T\sqrt{x}\,u^2, \qquad \sqrt{T\delta}\,uv,$$

and its second summand is exactly $\delta(\sqrt{T}R_v)$, because $Tx^2/y = j^2B = \delta$. This last use of $\delta \in \mathfrak{m}_c$ is the additional source fact specific to the inverse-b sector.

Finally,

$$j\alpha b = -(\alpha/B)R_u - \alpha uR_z. \tag{IW21}$$

Here $\alpha/B = O(x)$, and the square of αuR_z costs $O(x^2u^2)$ relative to $r_cR_*^2$. Hence the entire left side of (IW21) is free. These identities involve no bound on z.

The unrestricted first normal. Define the bounded source quantities

$$K_z = H(\alpha + \tfrac{1}{2}) - \tfrac{1}{2}, \qquad K_R = -\mu - \alpha(1-p) - \tfrac{1}{2} + b_+ p + TK_z,$$
$$\rho = b_+ p + \beta p x \geq 0.$$

Direct differentiation of the entire (IW11), followed by the actual source contact identity, gives

$$L_N = -j\alpha b + (K_R + \beta p v^2) R_u + \rho J_v + K_z uvq. \tag{IW22}$$

For completeness, before imposing the source, the remainder is vS_N, with

$$S_N = j\alpha - \mu + b_+ p + K_R x + b_+ p y + \beta p x y = 0$$

identically under (IW6). Thus no constant or first-normal term is silently dropped. Equations (IW7) imply $K_R \sim -\frac{1}{2}$, $\rho = O(\delta)$ and $\beta p = O(\delta)$. The first three terms in (IW22) are free by (IW19)–(IW21); the last has coefficient $K_z\sqrt{T}\,uv \in \mathfrak{n}$ on $\widehat{q}$. Dividing by $\sqrt{h_0}$ preserves all these bounds because $h_0 \geq \frac{1}{2}$.

The unrestricted second normal. Use the exact K_d from (IW8), and put

$$C_q = K_d D_u + AH + \alpha + \tfrac{1}{2} + v^2[CHD_u + AH + (\alpha + \tfrac{1}{2})H],$$
$$c_2 = T[CH + AH + (\alpha + \tfrac{1}{2})H] - A - (\alpha + \tfrac{1}{2})(1-p).$$

The complete second normal has the exact division

$$L_M = C_q q + (c_2 + TCHu^2) R_v + T(K_d + CHy) J_u. \tag{IW23}$$

Before substituting the actual source its only remainder is uS_M, where

$$c_0 = T[K_d + AH + \alpha + \tfrac{1}{2}] - A,$$
$$S_M = c_0 + TK_d x + y c_2 + TCHxy = 0.$$

The last equality is an exact identity from (IW6). Furthermore c_2 is bounded (in fact infinitesimal), TCH is a unit, and $(K_d + CHy)/x$ is bounded with residue $-\frac{1}{2}$. It follows from the two free rows in (IW20) that

$$\sqrt{TH(1-H)}[(c_2 + TCHu^2) R_v + T(K_d + CHy) J_u]$$

is free with bounded coefficients. For the other term,

$$\sqrt{TH(1-H)}\, C_q q = T\sqrt{H(1-H)}\, C_q \widehat{q}.$$

Its coefficient is in $\mathfrak{n}$: multiplying C_q by T clears the only CH pole, and every term then contains either T or v^2. This statement is made on the full expression in (IW23), before any relative source/testing rate is considered.

Strict full Gram and state restoration

Normalize the hard rows by pure-source factors,

$$\widehat{N} = \sqrt{h_0}\, N, \qquad M_0 = \sqrt{TH(1-H)}\, \widehat{M}.$$

Their full hard matrix is bounded: $b_+/h_0 \leq 1$, $\beta/h_0 \leq 1$, $\alpha/h_0 = O(\alpha)$, and the two source poles of H_M are exactly cleared by $TH(1-H)$. Equations (IW7), (IW13) give its real residue

$$\operatorname{diag}(1, \tfrac{1}{2}). \tag{IW24}$$

Its off-diagonal coefficient has the factor $uv\sqrt{TH(1-H)}/\sqrt{h_0}$ and is infinitesimal. The $\widehat{q}$ couplings in (IW22)–(IW23) are infinitesimal. All remaining complete-normal couplings are bounded coefficients of free rows. Buy sufficiently large fixed real pivots on the finitely many free rows, use the elementary real Young square completion, and apply the strict-residue Gram lemma to the remaining finite matrix. The strict scalar pivots (IW18) lose only infinitesimal amounts. This proves $G \in \Sigma\mathcal{T}^2$. No state normal has been set to zero in this conclusion. Restoring $mpD_u^2 N^2$, the pure-testing factor D_u, and the positive source scale d proves the assertion for (IW9).

The proof has used $u, v > 0$ and $w = 1, z > 0$, with arbitrary z. The exact sign/state lemma SJ1–SJ10 applies to the same original core: sign gauges make u, v nonnegative while preserving every term; a negative core at positive angles forces the rotated w, z to have the same nonzero sign. Dividing all state coordinates by the nonzero pure-testing w gives precisely this chart. When the core is nonnegative it is itself a pure-testing square, and every event term is already positive. Zero angles have the direct core SOS exits. The zero state is immediate, and a complex state splits into its real and imaginary quadratic evaluations. Thus (IW1) is proved for all states at the double-infinitesimal angles. The existing two-current outer-angle identities give the separate all-tests join if needed; they are not required as a premise of the local conductor proved here.

B.4.6 The uniform small-e testing and state partition

Theorem and unchanged source scope. Use the original physical canonical C1 stationary source with $\lambda = n = 0$, $d > 0$, every original nonnegative event coefficient, and every original strict source gap. Suppose

$$k, f, g, e = kf/g > 0 \text{ are infinitesimal}, \qquad r = 1 - af^2, \quad R = 1 + k^2 r, \quad T_0 = 1 + k^2 - tk^2 g^2,$$

$$a, t > 0, \qquad J = T_0 - a > 0, \quad M = R - t > 0. \tag{UJ1}$$

Then the entire original parent is a finite tensor sum of squares for every independent projective testing pair and every real or complex testing state. There is no restriction on $b = g/k$, g/f, or any relative infinitesimal rate. In particular the original $\beta = 0$ face is retained.

This is a mathematical join of the actual bounded-b endpoint product-repair proof, the whole Z-chart proof EZ1–EZ19, the inverse-b local proof IW1–IW24, and the two full-parent outer-current identities below. No unavailable EZ or IW coefficient witness is used as a premise.

Exact original-source bounds for the two outer identities

Normalize $d = 1$ by a positive pure-source scale. The retained Section B.4.1 prints seven original coquotients. Their respective residues on (UJ1) are

$$(c_2, c_\beta, c_3, c_5, c_4, c_\alpha, c_d) \sim (2, 2, 2, 2, 2, 2, 4).$$

Their arguments are bounded, so they are positive source units. The actual weights satisfy

$$C = \frac{c_3}{af^2 c_d}, \qquad D = \frac{B_c}{2Jf^2 c_d}, \qquad A = \frac{f^2 c_4}{c_d}, \qquad B_c = 2c_5 - Jf^2 c_d \sim 4.$$

Consequently

$$C^{-1}, D^{-1} \in \mathfrak{m}_c, \qquad g_B = \frac{CD}{C + D} > 0, \qquad \gamma = A g_B = \frac{c_4 c_3 B_c}{c_d^2 (aB_c + 2Jc_3)} \sim \frac{1}{4}. \tag{UJ2}$$

This is uniform in both openings a, J: the last denominator has residue $16 \cdot 4(\bar{a} + \bar{J}) = 64$, since $a + J = T_0 \sim 1$. No positive lower bound on either opening is assumed. These bounds use the actual original C, D, A, not selected substitute weights. Any other positive event is restored whole.

Complementary testing angles. The actual source estimates UJ2 verify the hypotheses of Lemmas B.7 and B.8. For v infinitesimal and $1/u$ bounded, $\bar{\gamma} = 1/4$ and $\bar{c}_u^2 < 1$, so the latter real residue condition is strict. Their whole homogeneous identities include both projective endpoints. Only the two infinitesimal testing angles remain.

Every remaining source and state chart

At u, v infinitesimal normalize a nonzero testing state by a coordinate of largest absolute value. This is a pure-testing operation; all four coordinates are bounded and at least one is a unit. If X or Y is a unit, the original pure-testing core has positive real residue $2(X^2 + Y^2)$. It is a positive testing square, and every original event remains a positive separated square. We can therefore assume X, Y infinitesimal.

If Z is a unit, normalize $Z = 1$. The remaining W is arbitrary bounded and may have either sign. EZ1–EZ19 applies in each of its two actual-source coordinate rows:

source	exact coordinates	actual proof
$b = g/k$ bounded	$f = eb, \quad g = kb$	EZ1−EZ19, bounded b
$b = g/k$ unbounded	$(k, f, g), \quad k/g \in \mathfrak{m}_c$	EZ1−EZ19, inverse b.

(UJ5)

The extra inverse-b source inequality used by EZ is derived from the actual $\beta \geq 0$: $tb^2 \leq 2c_\beta/c_d$. Thus it adds no source hypothesis to (UJ1). Its full proof retains both state normals and the testing-angle normal, including unbounded scaled normal coordinates through their exact sN_y, sN_v coefficients.

If Z is infinitesimal, W must be a unit. Normalize $W = 1$; then u, v, X, Y, Z are all infinitesimal. For bounded b, use the actual retained Section B.4.3 proof. Its source variables are exactly (e, k, b, a, t) with $f = eb$, $g = kb$; its gaps $R_c = 1 + k^2(1 - ae^2 b^2)$ and $T_0 = 1 + k^2 - tk^4 b^2$ coincide with (UJ1). It allows every bounded positive b, including infinitesimal b, both strict-gap openings, and all original coefficient faces. Its whole seven-term polynomial, complete first normals, higher-normal support and nine-root product identity are printed with Section B.4.1 and Section B.4.2. Its final original-parent restoration is an exact sum of complete positive terms. There is no remaining condition on k^3/e or k/e in that proof.

For unbounded b, use IW1–IW24. Its exact source hypotheses $k, f, g, k/g \in \mathfrak{m}_c$, physical $\beta \geq 0$, and the original gaps follow directly from (UJ1) and this ratio class. It proves the complete original parent with both unrestricted state normals for every positive rotated state ratio, so it includes the present W-unit chart. The retained β term, including its zero face, and both source openings are part of that proof. The state and sign reduction is the actual SJ1–SJ10 identity, not a chart label.

These cases exhaust every source ratio and every state. The zero state is immediate. A complex state splits into its real and imaginary evaluations because the parent matrix is real. Undo all pure-testing normalizations and the positive source scale d. This proves the theorem on (UJ1) at all independent projective tests.

B.4.7 Every positive e in the uniform source

Use the original canonical C1 source and the exact native transport EN13, with every original coefficient sign and strict gap retained. Suppose

$$k, f, g > 0 \text{ are infinitesimal}, \qquad e = kf/g > 0. \tag{UE1}$$

There is no additional restriction on the size of e, either source ratio $g/k, g/f$, or either separately labelled testing field. Then the entire original parent is tensor SOS at every independent projective testing pair and every real or complex testing state.

Here are the actual proof inputs and the complete join. EN13 gives $0 < a, t < 2$, $r = 1 - af^2 \sim 1$, and the original strict $M = R - t > 0, J = T - a > 0$. Split the positive pure-source e into

$$e \text{ infinitesimal} \quad \text{or} \quad \Delta = 1/e \text{ bounded}. \tag{UE2}$$

In the first case UJ1–UJ5 is the full-parent theorem. It includes both source ratios g/k bounded and unbounded, both state charts, both complete state normals and all angular/projective joins; the actual bounded-ratio product-repair, EZ and IW proof bodies are its inputs.

In the second case use the actual retained Section B.3.2, Theorem *Uniform C1, bounded reciprocal e*. Its hypothesis is exactly UE1 with Δ bounded, and its proof is complete in that body. For clarity, the scope extension used there is substantive. After its pure-source substitution

$$\delta = \Delta R/(1 + k^2), \quad b^2 = f^2(1 + k^2)/R,$$

both δ and the pure-testing rotated state coordinate w are allowed to be bounded units. In its polynomial ring the weights

$$(\delta, k, b, a, t, u, w, v) = (0, 2, 6, 0, 0, 2, 0, 16) \tag{UE3}$$

make the full scalar error and both complete first normals have weights at least $42, 32, 26$. The displayed unlocalized monic conductor, restricted to $(A_m, B_m, C_m)^2 + (D_0^2)$, gives a scalar Gram whose every correction coefficient has weight at least 2; the first-normal coefficient bounds are $12, 6$. The exact hard block on its two normalized state normals has residue $16I_2$, including every quadratic term. Its explicit eleven-soft-root identity supplies a strict positive diagonal. Thus the entire parent, with all four retained terms restored, has a bounded strict-residue Gram. This is a finite polynomial division proof; neither δ^{-1} nor a mixed element is a lifting coefficient.

The same body proves the complementary $W = 1, Z$ infinitesimal chart with five exact current roots. Its two full correction coefficients contain respectively ΔZ and $\Delta^2 f^2 Z^2$, so vanish in residue even when Δ is a unit. The original source signs and both strict gap openings remain intact.

For the outer angle joins, no conclusion from the old infinitesimal-e assertion is imported. Use the two exact whole-parent identities in Lemmas B.7 and B.8. Their source hypotheses hold on all of UE1. Indeed the seven native coquotients have residues $2, 2, 2, 2, 2, 2, 4$, regardless of e. Writing $B_c = 2c_5 - Jf^2 c_d \sim 4$, the original weights after normalizing $d = 1$ satisfy

$$C^{-1}, D^{-1} \text{ infinitesimal}, \qquad A\frac{CD}{C + D} = \frac{c_4 c_3 B_c}{c_d^2(aB_c + 2Jc_3)} \sim \frac{1}{4}. \tag{UE4}$$

Here $a + J = T \sim 1$, so the denominator has residue 64 without an individual lower bound on either opening. Those identities therefore cover every testing chart outside u, v infinitesimal, including literal projective infinities. Literal axes, the zero state, signs and complex states are covered by their exact full-parent normalization identities.

The two alternatives UE2 exhaust every positive source e. All scalings are within one labelled field, and every subtracted term is restored. This proves UE1 at all tests, with literal $\beta = 0$ and every allowed original coefficient boundary. The two real closed extensions are arbitrarily chosen labelled factors; no algebraic-disjointness assumption is imposed.

B.4.8 The source with infinitesimal k, ζ and unbounded η

This section concerns the original C1.1 source, not an enlarged family. All coefficients are divided by $d > 0$. Put

$$b = \eta^{-1/2}, \quad g = \sqrt{\zeta}, \quad q = k^2\eta, \qquad x = b^2 q, \quad a = b^2(1-r). \tag{SC1}$$

Here b, g are positive infinitesimals, $q > 0$ is bounded, and $\overline{q} \le 1/2$. The new bounded two-normal reduction below is asserted for q a positive unit. The exact polynomial identities remain valid more generally, but their indicated boundedness argument would not control $g/\sqrt{q}$ when q is infinitesimal. No all-tests closure is inferred from the reduction.

Keep $K, R, T, M, J, E, Q, P_4, P_d, P_\alpha$ from (NC1)–(NC2), and set

$$\begin{aligned} c_b &= b^2 C/d = \frac{K(1+b^2r)E}{2(1-r)P_d}, & a_b &= b^2 A/d = \frac{P_4}{2rP_d}, \\ D_b &= D/(db^2), & \alpha_b &= \alpha/(db^2g^2), \\ h_2 &= \frac{Kq(1+b^2r)(1-tg^2)(1+qg^2)}{2P_d}, & h_b &= \frac{Kq(1+b^2r)[1+(1-t)g^2]}{2P_d}. \end{aligned}$$

Thus $m/d = h_2/(g^2M)$ and $\beta/d + 1/2 = h_b/(g^2 t)$. The exact relation

$$h_b - h_2 = \frac{Kq(1+b^2r)Eg^2}{2P_d} \tag{SC2}$$

removes the apparent opening dependence from the first normal of the paired m, β currents. The quantities D_b, α_b, h_2, h_b are bounded; α_b, h_2, h_b are units in the stated q-unit domain. Their residues are respectively $(1-2q)/2, q^2/2, q/2, q/2$, with bounded source letters on the right understood at residue.

Exact full-parent reduction. For bounded affine tests define

$$Q_b = (W + uY)/b, \quad Y_b = Y/b^2, \quad p = t/R, \quad N = \frac{(X+uZ)/b - pvQ_b}{g},$$

$$H = uQ_b/b, \quad L = vQ_b/g, \quad V = uvZ/b, \quad B_b = gN + pvQ_b, \quad D_u = 1+u^2, \quad D_v = 1+v^2.$$

All roots use only pure-source inverses. Retain the two nonnegative original terms

$$\frac{4h_2 t}{MR}N^2, \qquad \frac{4h_b M}{tR}D_v N^2.$$

The complete original parent divided by db^2, minus these terms, is exactly

$$\begin{aligned} G = {} & \frac{4}{R}\{(h_2 + h_b D_v)N^2 + 2(h_b D_v - h_2)LN + [h_2(1-p) + h_b p D_v]L^2\} \\ & + 4D_v\{c_b(H - D_u Y_b)^2 + a_b D_u Y_b^2\} + 4D_b J_b^2 + 4\alpha_b g^2 D_u(vb^2 Y_b - Z)^2 + 2K_b, \\ J_b = {} & b^2(H - D_u Y_b) + ubvB_b - D_u vZ, \\ K_b = {} & b^2(H - D_u Y_b)^2 + 2buv(H - D_u Y_b)B_b + u^2v^2B_b^2 \\ & + D_u v^2 b^2 Y_b^2 + D_u V^2 - 2D_u V(Q_b + vB_b) + 2D_u u^2 vZY_b. \end{aligned} \tag{SC3}$$

These are entire quadratic polynomials in the state; no restriction of a state normal has been made. In particular, (SC3) is a direct input for collecting both complete first normals and their full hard quadratic, rather than an identity solely on a contact graph.

A harmonic normal removes both r openings. Put

$$s = c_b + a_b, \quad \rho = c_b/s, \quad h = c_b a_b/s, \quad \rho_0 = Kr/R, \qquad N_y = \sqrt{s}\,(D_u Y_b - \rho_0 H).$$

Then the complete C, A line in (SC3), divided by $4D_v$, equals

$$\left[N_y + \sqrt{s}(\rho_0 - \rho)H\right]^2 + hH^2 - \frac{a_b u^2}{D_u}\left(\frac{N_y}{\sqrt{s}} + \rho_0 H\right)^2. \tag{SC4}$$

The inverse $1/D_u$ is pure testing, and every coefficient in (SC4) is bounded uniformly through either $r \to 0$ or $r \to 1$. Indeed s is bounded below, h is a positive unit, and a_b/s, $a_b\rho_0/\sqrt{s}$ and $a_b\rho_0^2$ are bounded. There is no mixed harmonic denominator.

Here is the exact source cancellation that also centers the normal without an opening loss. Define

$$S_0 = Kr(1+b^2r)E + (1-r)P_4,$$
$$T_0 = q^2(1-tg^2)^2 + qrg^2 + 2qtg^2 - qg^2 - 2q + r.$$

Then

$$\rho = \frac{Kr(1+b^2r)E}{S_0}, \qquad Kr - \rho R = -\frac{b^2r(1-r)KT_0}{S_0}, \qquad h = \frac{K(1+b^2r)EP_4}{2P_dS_0}. \tag{SC5}$$

Both S_0 and the other denominators are positive units. Also $s \asymp [r(1-r)]^{-1}$, so $\sqrt{s}(\rho_0 - \rho)$ is infinitesimal without a lower bound on either opening. At the actual contact,

$$(u, v) = (b\sqrt{q}, \sqrt{q}g), \quad Q_b = RgZ, \quad Y_b = r\sqrt{q}gZ,$$

and both N, N_y vanish exactly. Thus (SC3)–(SC5) remove every unbounded source weight while retaining the actual contact and every state normal.

The leading angular circuit. For u, v infinitesimal, the leading scalar after the harmonic state normal has been removed is twice

$$(1-q)H^2 + qL^2 + V^2 + (1-2q)(vZ)^2 + q^2(gZ)^2 - 2VQ_b.$$

Here the leading relation $H(vZ) = VQ_b$ is exact for the displayed roots. It splits as

$$(1-2q)(H - vZ)^2 + qH^2 + qL^2 + V^2 + q^2(gZ)^2 - 4qVQ_b. \tag{SC6}$$

The final four positive terms form one AM–GM circuit, since

$$(\sqrt{q}H)(\sqrt{q}L)V(qgZ) = (qVQ_b)^2.$$

On the positive negative-core chart its radical factors separate between the source and testing fields. This proves positivity of the leading scalar, including $q = 1/2$, where its separate binomial drops out. The leading hard normal coefficients are $4q$ on N and 4 on N_y. This organizing form alone is not yet a certificate for all the higher terms in (SC3).

A useful exact conductor can be stated without comparing any source and testing rate. Normalize $Z = 1$ on its pure-testing state chart, put $c = \sqrt{q}$, $w = (W + uY)/R$, and now use the contact-exact roots

$$H_0 = uw/b^2,\ L_0 = vw/(bg),\ V_0 = uv/b,\ D_0 = c^2g, \quad A_0 = cH_0,\ B_0 = cL_0,\ C_0 = V_0.$$

Their contact is $(u, v, w) = (bc, cg, bg)$. Set

$$E_0 = H_0 - v, \quad S_v = A_0 - D_0 - cE_0, \quad S_u = C_0 - D_0 - S_v, \quad S_w = B_0 - D_0 - S_v.$$

Direct multiplication gives

$$\begin{aligned} H_0(u - bc) &= (u - bc)E_0 + bS_u, \\ H_0(v - cg) &= (v - cg)E_0 + (v/c)S_v, \\ H_0(w - bg) &= (w - bg)E_0 + (bg/c)S_w. \end{aligned} \tag{SC7}$$

Only the positive-unit inverse $1/c$ occurs. These identities lift contact-vanishing coefficients times H_0 to the displayed root module with bounded coefficients on the small pure-testing chart. They do not alone prove a full scalar square lift: a first-jet syzygy among the redundant angular roots still has to be allocated, particularly when the coefficient of E_0^2 vanishes at $q = 1/2$.

Exact small-cost angular pivots. The four-term circuit has the exact symmetric decomposition

$$\sum_{i=1}^{4} A_i^2 - 4\sqrt{A_1A_2A_3A_4} = \frac{1}{3}\sum_{i<j}(A_i - A_j)^2 + \frac{2}{3}\sum_{\{ij\}|\{kl\}}(\sqrt{A_iA_j} - \sqrt{A_kA_l})^2,$$

where the second sum is over the three pair partitions. Use $(A_1, A_2, A_3, A_4) = (A_0, B_0, C_0, D_0)$ above and assume the positive chart $u, v, w > 0$. Then the following identities are exact:

$$\begin{aligned} c\sqrt{uw}\,E_0 &= \sqrt{uw}(A_0 - D_0) + b\sqrt{cg}(\sqrt{A_0D_0} - \sqrt{B_0C_0}), \\ \sqrt{vw}(u - bc) &= \frac{b^{3/2}}{\sqrt{c}}(\sqrt{A_0C_0} - \sqrt{B_0D_0}), \\ \sqrt{uw}(v - cg) &= b\sqrt{g/c}(\sqrt{B_0C_0} - \sqrt{A_0D_0}), \\ \sqrt{uv}(w - bg) &= \frac{b^{3/2}\sqrt{g}}{c}(\sqrt{A_0B_0} - \sqrt{C_0D_0}). \end{aligned} \tag{SC9}$$

The coefficient factors on the right are infinitesimal; all radicals separate between the source and testing fields. Consequently any fixed positive real multiples of the squares of the four left-hand roots can be inserted as additional positive Gram pivots at an infinitesimal cost in the displayed strict circuit Gram. To see this without a limiting assertion, write the four identities as $P = TZ$, with Z the displayed circuit roots and every entry of T in the bounded maximal ideal. Then $Z^T G Z = \tau P^T P + Z^T (G - \tau T^T T) Z$, and the second matrix retains the strictly positive real residue of G for every fixed real $\tau > 0$. This is a genuine full-rates Gram mechanism. It still requires showing that the entire first-normal and scalar remainders of (SC3) allocate on these weighted roots; the identities do not make that last membership automatic.

A quantified remote improvement when e is unbounded. The source identity $e^2 = q/g^2$ and (NI3) imply that m, β are both unbounded when e is unbounded. Three inverse-cycle rotations send this adjacent pair to C, A and transform tests as $(u, v) \mapsto (v, 1/u)$. Therefore the adjacent-event theorem covers all tests with v not infinitesimal. Combined with the previous C, A certificate, only u, v infinitesimal remain in this e type. This includes every q-unit source considered above.

There is also a complete state-chart certificate there. Normalize the pure-testing state by a coordinate of maximal absolute value. If W is a unit, cap C, A, m, β at a sufficiently large positive real μ, retain all excesses, and leave every other event as its original nonnegative square. At $u = v = X = Y = 0$, the quadratic part in (u, v, X, Y) is

$$2(uW - Y)^2 + 2(X + uZ)^2 - 4uvWZ + 4\mu\{(uW - Y)^2 + Y^2 + (vW - X - uZ)^2 + (X + uZ)^2\}. \tag{SC8}$$

The event part is strictly positive for every real $W \neq 0$ and bounded real Z: its zero equations give successively $Y = u = X = v = 0$. A sufficiently large finite μ therefore absorbs the entire signed quadratic core. The exact remainder has degree at least three in these four small roots, with bounded coefficients in W, Z, so it allocates with zero residue. This proves the entire original parent on this chart. If the core has negative sign and W is infinitesimal, then Z is a unit, since otherwise the core residue forces $X = Y = 0$, contrary to maximal-coordinate normalization. Thus the remaining state chart has $Z = 1$, W infinitesimal, and both testing half-angles infinitesimal. This is a full quantified narrowing, not closure of C1.1.

Completion and exact outstanding obligation. The following section supplies a finite scalar square lift and both complete first normals for the q-unit C1.1 subdomain, including $D = 0$. It uses a contact-centered D term followed by the geometric mean conductor; an unweighted E_0^2 perturbation by itself is not controlled by the circuit uniformly at arbitrary relative rates. For q infinitesimal, the uniform normal and protected contact lift in Sections B.4.11–B.4.12 resolve the additional $g/\sqrt{q}$ issue and all remote tests. No assumption of bounded e or $1/e$ is imposed.

B.4.9 Positive-unit $q = x\eta$

Theorem B.11. *On the original C1.1 source, suppose $q = k^2\eta$ is a positive bounded unit. The entire original parent is tensor SOS at every independent projective test and every real or complex state. All positive opening rates and the permitted literal boundary $D = 0$ are included.*

The proof uses the exact full quadratic (SC3), not its leading scalar. The q-infinitesimal part of C1.1 is not included in this theorem.

One more centered term. Use the variables of (SC1)–(SC5), and put $s_D = J/R$. Retain the additional nonnegative square

$$4D_b(J_b + s_D H)^2. \tag{SC10}$$

It vanishes at actual contact, where $H = RvZ$ and $J_b = -JvZ$. Its removal replaces $4D_b J_b^2$ in (SC3) exactly by

$$-8D_b s_D H J_b - 4D_b s_D^2 H^2.$$

In particular it removes the entire separate $(vZ)^2$ term. This step allows $D_b = 0$ literally and does not divide by D_b.

Set $Z = 1$ on the remaining state chart. Put $c = \sqrt{q}$ and $w = (W + uY)/R$, and use the four separated positive monomials

$$A = \frac{cuw}{b^2}, \quad B = \frac{cvw}{bg}, \quad C = \frac{uv}{b}, \quad D = c^2 g, \quad \Gamma = (ABCD)^{1/4}. \tag{SC11}$$

Their exact contact is $(u, v, w) = (k, cg, bg)$, with $k = bc$, and $A = B = C = D$ there. Notice that $H = RA/c$, $L = RB/c$, $V = C$, $g = D/q$, and $Hv = R\Gamma^2/q$.

Here are explicit formulas for the entire scalar at $N = N_y = 0$, after (SC10). Write $a_t = 1 - \rho_0 + pv^2$. Define

$$\begin{aligned} P_H &= 4D_v\{c_b(1-\rho_0)^2 + a_b\rho_0^2/D_u\} + 2b^2\{a_t^2 + v^2\rho_0^2/D_u\} \\ &\quad + 4\alpha_b g^2 b^4 v^2 \rho_0^2/D_u - 8D_b s_D b^2 a_t - 4D_b s_D^2, \\ P_L &= 4[h_2(1-p) + h_b p D_v]/R, \\ P_{Hv} &= -4[D_u(1+pv^2) - u^2\rho_0] - 8\alpha_b g^2 b^2 \rho_0 + 8D_b s_D D_u. \end{aligned}$$

The exact scalar is

$$f = f_A A^2 + f_B B^2 + f_C C^2 + f_D D^2 - f_\Gamma \Gamma^2, \tag{SC12}$$

where

$$f_A = R^2 P_H/q, \quad f_B = R^2 P_L/q, \quad f_C = 2D_u, \quad f_D = 4\alpha_b D_u/q^2, \quad f_\Gamma = -RP_{Hv}/q.$$

These are rational functions of u, v over bounded source coefficients, with only positive-unit denominators. For example,

$$c_b(1-\rho_0)^2 = \frac{K(1+b^2 r)E(1-r)}{2P_d R^2}, \qquad a_b\rho_0^2 = \frac{P_4 K^2 r}{2P_d R^2}.$$

Thus neither r opening survives in their denominators. For u, v, b, g infinitesimal their residues are

$$\overline{f}_A = \overline{f}_B = \overline{f}_C = \overline{f}_D = 2, \qquad \overline{f}_\Gamma = 8. \tag{SC13}$$

This uses the exact source residues in (SC2), including $\overline{D}_b = (1 - 2\overline{q})/2$. It does not assume a positive lower bound on D_b.

Here is an explicit finite denominator certificate used below. Every function just displayed is $P(u, v)/(1 + u^2)$ for a polynomial P with bounded pure-source coefficient array. For a monomial use

$$x^m - x_0^m = (x - x_0) \sum_{j=0}^{m-1} x^{m-1-j} x_0^j.$$

Applying this identity to each monomial supplies finite first and second difference arrays with bounded source coefficients. For the denominator use only the exact identity

$$\frac{1}{1+u^2} - \frac{1}{1+k^2} = -\frac{(u-k)(u+k)}{(1+u^2)(1+k^2)}.$$

Thus the first and second finite Taylor arrays about (k, cg) are bounded and use only pure-testing $1 + u^2$ and pure-source $1 + k^2$ denominators, both units. The same construction after $u = U^2, v = V^2$ gives the half-variable arrays below. This is a finite polynomial identity construction, not an analytic series.

Every first normal and the complete hard block. Let $\delta_\rho = \rho_0 - \rho$. The full first-normal coefficients of the remaining quadratic, with no normal suppressed, are

$$\begin{aligned} \mathcal{L}_N &= \frac{8}{R}(h_b D_v - h_2)L + 4buvg(a_t - 2D_b s_D)H - 4D_u v g V, \\ \mathcal{L}_y &= 8D_v\sqrt{s}\,\delta_\rho H \\ &\quad + \frac{H}{\sqrt{s}}\left[-\frac{8D_v a_b u^2 \rho_0}{D_u} + 8D_b s_D b^2 + \frac{8\alpha_b g^2 b^4 v^2 \rho_0}{D_u} - 4b^2 a_t + \frac{4b^2 v^2 \rho_0}{D_u}\right] \\ &\quad + \frac{4bu}{\sqrt{s}} V - \frac{8\alpha_b g b^2 v}{\sqrt{s}}(gZ). \end{aligned} \tag{SC14}$$

Each is a bounded infinitesimal-coefficient linear combination of A, B, C, D. At actual contact it is zero exactly, since the source is a stationary zero and every removed term vanishes there. The coefficients of N^2, N_y^2, NN_y are respectively

$$\begin{aligned} H_{NN} &= 4(h_2 + h_b D_v)/R + 2g^2 u^2 v^2, \\ H_{yy} &= 4D_v\left(1 - \frac{a_b u^2}{D_u s}\right) + \frac{4\alpha_b g^2 b^4 v^2}{D_u s} + \frac{2b^2}{s}\left(1 + \frac{v^2}{D_u}\right), \\ H_{Ny} &= -4buvg/\sqrt{s}. \end{aligned} \tag{SC15}$$

Thus the hard Gram has residue $\mathrm{diag}(4q, 4)$. In (SC15), H_{Ny} denotes the polynomial coefficient of NN_y, not the off-diagonal matrix entry.

A finite geometric-mean square lift. Here is a detailed lift for the particular raw module (SC12). It is not a claim about arbitrary contact-vanishing polynomials. Define $\alpha_i = A_i - \Gamma$, with $(A_1, A_2, A_3, A_4) = (A, B, C, D)$, and put

$$T_4 = \sum_i A_i^2 - 4\Gamma^2.$$

A fixed positive Gram on the following finite roots represents T_4:

$$\alpha_i, \qquad Q_{ij|kl} = \sqrt{A_iA_j} - \sqrt{A_kA_l}, \qquad \sqrt{\Gamma}(\sqrt{A_i} - \sqrt{A_j}), \qquad \sqrt{\Gamma}((A_iA_j)^{1/4} - (A_kA_l)^{1/4}). \tag{SC16}$$

Use all four α_i, all three pair partitions, and all six unordered differences in the third group. To verify strictness, average positive multiples of the symmetric circuit decomposition in (SC9) and

$$T_4 = \sum_i \alpha_i^2 + 2\Gamma\left(\sum_i A_i - 4\Gamma\right).$$

over the three decompositions

$$\sum_i A_i - 4\Gamma = (\sqrt{A_i} - \sqrt{A_j})^2 + (\sqrt{A_k} - \sqrt{A_l})^2 + 2((A_iA_j)^{1/4} - (A_kA_l)^{1/4})^2.$$

The additional differences $A_i - A_j$ in the first decomposition are $\alpha_i - \alpha_j$ and give nonnegative corrections to the already strict α block.

Write $\epsilon_i = f_i - 2$, $\epsilon_\Gamma = f_\Gamma - 8$, and $\sigma = \sum_i \epsilon_i - \epsilon_\Gamma$. Then

$$f - 2T_4 = \sum_i \epsilon_i \alpha_i^2 + 2\Gamma \sum_i \epsilon_i \alpha_i + \Gamma^2 \sigma. \tag{SC17}$$

All ϵ have zero residue. The exact contact value implies $\sigma(k, cg) = 0$. Clear the pure-testing denominator D_u if needed; monic division gives $\sigma = (u - k)\sigma_u + (v - cg)\sigma_v$ with bounded coefficients. For the monomials (SC11), the exact conductors are

$$\begin{aligned} \Gamma(u - k) &= \sqrt{uk}\, Q_{13|24}, \\ \Gamma(v - cg) &= \sqrt{vcg}\, Q_{23|14}, \\ \Gamma(w - bg) &= \sqrt{wbg}\, Q_{12|34}. \end{aligned} \tag{SC18}$$

Thus the last term in (SC17) becomes Γ times a linear combination of the Q roots with infinitesimal bounded coefficients. The exact identity

$$Q_{ij|kl} - \tfrac{1}{2}(A_i + A_j - A_k - A_l) = -\tfrac{1}{2}(\sqrt{A_i} - \sqrt{A_j})^2 + \tfrac{1}{2}(\sqrt{A_k} - \sqrt{A_l})^2 \tag{SC19}$$

changes these terms into infinitesimal quadratic corrections on (SC16) and $\Gamma \sum_i \mu_i \alpha_i$, where each μ_i is infinitesimal and is rational in $\sqrt{u}, \sqrt{v}$ with bounded coefficient arrays and positive-unit denominators.

The exact three contact jets imply that the four source values $\mu_i(k, cg)$ coincide. Indeed the differential vectors of α_i/Γ at contact are the four rows

$$(1/2, -1/2, 1/2), \quad (-1/2, 1/2, 1/2), \quad (1/2, 1/2, -1/2), \quad (-1/2, -1/2, -1/2)$$

in logarithmic coordinates $(u/k, v/(cg), w/(bg))$. Their sole linear relation is the sum of all four rows. Cancelling the nonzero source factors in this equality establishes coincidence; no inverse of a small number enters a Gram coefficient. Call the common infinitesimal value μ_*. The common term is

$$\mu_* \Gamma \sum_i \alpha_i = \mu_* \Gamma \left(\sum_i A_i - 4\Gamma\right),$$

an infinitesimal correction on the explicitly displayed variance roots. For each remaining $\mu_i - \mu_*$, monic division in $\sqrt{u}, \sqrt{v}$ has bounded coquotients. Apply the half-variable conductors

$$\begin{aligned} \Gamma(\sqrt{u} - \sqrt{k}) &= (uk)^{1/4}\sqrt{\Gamma}((AC)^{1/4} - (BD)^{1/4}), \\ \Gamma(\sqrt{v} - \sqrt{cg}) &= (vcg)^{1/4}\sqrt{\Gamma}((BC)^{1/4} - (AD)^{1/4}). \end{aligned} \tag{SC20}$$

Their coefficient factors are infinitesimal. Multiplying by the remaining α_i proves that the entire scalar correction in (SC17), not merely its low-order jet, is a bounded zero-residue quadratic form on (SC16). This uses only two finite monic divisions. The boundedness of their coefficient arrays follows from the explicit rational functions (SC12), whose test denominators are powers of D_u, and from the displayed square-root substitutions; bounded function values alone are not being used as a substitute for this fact.

The same argument for a first normal is shorter. If $\mathcal{L} = \sum_i a_i A_i$, write $\mathcal{L} = \sum_i a_i \alpha_i + \Gamma \sum_i a_i$. Its exact contact zero gives $\sum_i a_i(k, cg) = 0$; one monic division and (SC18) allocate it on (SC16), with infinitesimal coefficients. Apply this to both complete expressions (SC14). Together with (SC15), this yields a bounded strictly positive-residue Gram for the entire original remainder. Restore (SC10), the paired terms, and all positive pure-source and pure-testing normalizations.

Exhaustion and legal radicals. For q a unit, $e^2 = q/g^2$ is unbounded. The exact adjacent-event rotations and the W-unit state argument (SC8) therefore reduce every uncovered negative-core test to u, v, W infinitesimal and Z a unit. Literal axes are direct original-core exits. Use exact sign gauges and normalize $Z = 1$. The common pure-testing Schur identity forces $W + uY > 0$ on the remaining negative-core chart: with $w_t = (W + uY)/D_u$, the eliminated scalar is

$$\frac{uvD_uD_v}{\mathcal{D}}(D_uuvw_t^2 - 2D_uD_vw_t + uv), \qquad \mathcal{D} = u^2v^4 + u^2v^2 + u^2 + v^4 + 2v^2 + 1 > 0,$$

which is nonnegative when $w_t \le 0$; its eliminated two-square matrix is strictly positive. Hence $w = (W+uY)/R > 0$. Every monomial in (SC11), and every product under a radical in (SC16)–(SC20), factors into a positive pure-source quantity and a positive pure-testing quantity. The occurrence of R in w is a positive pure-source unit, not a mixed denominator. All test coefficients remain bounded because u, v, w are infinitesimal. If the general toric contact lemma is used, its third coordinate is literally the pure-testing $W + uY$, with pure-source contact *Rbg*; this direction is inactive in every coefficient function. Complex state evaluations split into real and imaginary parts. This supplies every independent test and proves the theorem.

B.4.10 A uniform normal at infinitesimal q

This intermediate lemma by itself is not an certificate of the full q-infinitesimal C1.1 source. The following two completion sections supply the scalar allocation and remote joins. Keep (SC1)–(SC3), but do not discard the whole second paired term. Its addition to the N^2 coefficient in (SC3) gives exactly

$$\frac{4}{R}(h_2 + h_bD_v) + \frac{4h_bM}{tR}D_v = H_0 + \frac{4h_b}{t}v^2, \qquad H_0 = \frac{4h_2}{R} + \frac{4h_b}{t}.$$

Retain the last summand as the positive square $4h_b(vN)^2/t$, and use the pure-source normalization

$$\widehat{N} = \sqrt{H_0}\,N.$$

The exact original inequality $\beta \ge 0$ gives $h_b/t \ge g^2/2$. The explicit formula for h_2 gives h_2/q a positive bounded unit, uniformly through both openings. Consequently $H_0 \ge c_0(q + g^2)$ for some positive real c_0. This argument also permits $\beta = 0$ literally. It does not require $e = \sqrt{q}/g$ or its reciprocal to be bounded.

The coefficient of $\widehat{N}^2$ from this source part is exactly one. Its remaining contribution $2g^2u^2v^2/H_0$ is bounded and infinitesimal on the small-angle testing chart. Moreover

$$\frac{g}{\sqrt{H_0}}, \quad \frac{\sqrt{q}}{\sqrt{H_0}}$$

are bounded. Since $h_b - h_2 = O(qg^2)$, the complete first normal (SC14), after this normalization, is an infinitesimal-coefficient linear combination of B, H, V for u, v infinitesimal. The first coefficient is bounded because

$$\frac{h_bD_v - h_2}{\sqrt{qH_0}} = O(g^2 + v^2),$$

and the other two coefficients are bounded by constant multiples of buv and v. The hard cross coefficient with N_y is likewise infinitesimal.

The issue left by this lemma is scalar and angular: H is unweighted in this statement, whereas the four-term circuit controls $\sqrt{q}H$. One must retain a strict D-based contact normal and allocate every unweighted H coefficient on that normal and the circuit. The naive full extraction (SC10) fails in this branch: its raw P_H satisfies the exact formal specialization

$$P_H\big|_{q=0,\,v=0} = -\frac{2ru^2}{1+u^2},$$

so P_H/q need not be bounded. Thus the corrected hard-normal normalization removes the former $g/\sqrt{q}$ obstruction. The later protected-square lift establishes the missing scalar allocation.

B.4.11 Infinitesimal q at the two small testing angles

Theorem B.12. *On the original C1.1 source with $q = k^2\eta$ infinitesimal, the entire original parent is tensor SOS whenever both independent testing half-angles u, v are infinitesimal. All original opening rates and $\beta = 0$ are retained. The boundary $D = 0$ is empty here because D_b has residue $1/2$.*

This statement includes every state; it does not yet assert the ordinary or unbounded v testing charts.

Use the preceding source notation and put $c = \sqrt{q}$, $e = c/g$, $h = H/R$, and $E_* = h - v$. Work first with $Z = 1$ and $u, v, h > 0$. Define

$$A = ch, \quad B = cL/R, \quad C = uv/b, \quad D = qg, \quad \Gamma = (ABCD)^{1/4}, \quad d_i = A_i - \Gamma.$$

Then

$$\Gamma^2 = qhv, \qquad BC = ev^2h, \qquad uh = uE_* + bC. \tag{QI1}$$

Here $h = u(W + uY)/(Rb^2)$ is separated into a positive pure-source factor and a positive pure-testing factor. Every radical below is therefore legal; no mixed inverse or order comparison is used. The exact original inequality $\beta \geq 0$ gives $t = O(e^2)$, while t is bounded. Hence t/e and $\sqrt{t/e}$ are bounded.

Exact scalar, with the D term restored. Write $U = u^2, V = v^2$, $D_u = 1 + U, D_v = 1 + V$, and $a_t = 1 - \rho_0 + (t/R)V$. At both state normals zero the entire original scalar is

$$ah^2 + b_*hv + c_*v^2 + P_LL^2 + 2D_uC^2 + 4\alpha_bg^2D_u, \tag{QI2}$$

where

$$\begin{aligned} a &= 4D_vR^2\{c_b(1-\rho_0)^2 + a_b\rho_0^2/D_u\} + 2b^2R^2\{a_t^2 + V\rho_0^2/D_u\} \\ &\quad + 4\alpha_bg^2b^4VR^2\rho_0^2/D_u + 4D_bb^4R^2a_t^2, \\ b_* &= -4R\{D_u(1 + tV/R) - U\rho_0\} - 8\alpha_bg^2b^2R\rho_0 - 8D_bb^2Ra_tD_u, \\ c_* &= 4D_bD_u^2. \end{aligned}$$

All these rational functions have bounded source coefficient arrays; only D_u and positive pure-source units occur in their denominators. The r openings cancel exactly as in (SC12). In particular

$$D_b = \frac{N_D}{2JP_d}, \qquad N_D = \frac{K(1 + qg^2)(1 + b^2r) - JP_d}{b^2}$$

has a polynomial numerator: the numerator before division vanishes identically at $b^2 = 0$, where $P_d = 1 + qg^2$. This is a finite monic division, so it introduces no unbounded source coefficient array. Also α_b/q^2, h_2/q, and h_b/q have explicit bounded arrays from their source formulas. These observations justify the q coquotients below uniformly through both source openings.

Use formal specialization at $q = 0$ only to construct finite polynomial coquotients; it is not a new physical source. Put $z = b^2$, $J_0 = 1 - z(1 - r)$, and $S = 1 - r + tV$. The three specialized coefficients are

$$\begin{aligned} a_0 &= 2D_v\{(1 + zr)(1 - r) + r/D_u\} + 2z\{S^2 + Vr^2/D_u\} + 2z^2S^2/J_0, \\ b_0 &= -4\{D_u(1 + tV) - Ur\} - 4zSD_u/J_0, \\ c_0 &= 2D_u^2/J_0. \end{aligned}$$

Define $\widetilde{b}_0 = b_0 + 4tD_uV$, and the bounded finite coquotients

$$X = a_0 + \widetilde{b}_0 + c_0 = UX_U + VX_V, \qquad Y = -\widetilde{b}_0 - 2c_0 = UY_U + VY_V. \tag{QI3}$$

For definiteness, first subtract the value at $U = 0$, divide by U, and then divide that value by V. Their two V coquotients are

$$X_V = 2 + 2zr(1 - 2t) + \frac{2z(1 + zr)t^2V}{J_0}, \qquad Y_V = \frac{4zt}{J_0}. \tag{QI4}$$

Thus $X_V = 2 + \varepsilon_V$, $\varepsilon_V \in B_{\mathrm{Inf}}$; X_U, Y_U have bounded arrays. Likewise write $a = a_0 + qa_1$, $b_* = b_0 + qb_1$, $c_* = c_0 + qc_1$ by finite numerator division. All a_1, b_1, c_1 have bounded arrays. Direct source specialization gives

$$\overline{a_1 - c_1} = 2, \qquad \overline{b_1 + 2c_1} = -8, \qquad \overline{R^2P_L/q} = 2, \qquad \overline{4\alpha_bD_u/q^2} = 2. \tag{QI5}$$

The following is the exact scalar identity to be allocated:

$$\begin{aligned} \text{(QI2)} &= (c_0 + qc_1)E_*^2 + 2D_uC^2 + X_U(uE_* + bC)^2 + Y_Uu(uE_* + bC)E_* \\ &\quad + X_V(vh)^2 + Y_Vv(vh)E_* - 4(tv/e)D_uBC \\ &\quad + (a_1 - c_1)A^2 + (b_1 + 2c_1)\Gamma^2 + (R^2P_L/q)B^2 + (4\alpha_bD_u/q^2)D^2. \end{aligned} \tag{QI6}$$

It uses only (QI1) and $qv^2 = 2\Gamma^2 - A^2 + qE_*^2$. In particular the apparently dangerous negative t term is already an infinitesimal-coefficient BC term, since v is infinitesimal.

Two extra contact squares and their finite first jets. Set

$$F_* = vh - gD, \quad R_F = \frac{\sqrt{g\Gamma}}{c}(\Gamma - D), \quad V_0 = cv, \quad R_E = \sqrt{\Gamma}(\sqrt{A} - \sqrt{V_0}).$$

All four roots vanish at actual contact. Exact multiplication gives

$$\begin{aligned} (vh)^2 &= F_*^2 + 2R_F^2 + 2gF_*d_4 + g^2D^2 - 4g^2\Gamma d_4, \\ c\Gamma E_* &= 2\Gamma d_1 - R_E^2, \\ \Gamma F_* &= -2g\Gamma d_4 + R_F^2/g. \end{aligned} \tag{QI7}$$

The root R_E receives a strict real square pivot from E_*^2:

$$qE_*^2 = (A - V_0)^2 = 4R_E^2 + (\sqrt{A} - \sqrt{V_0})^4. \tag{QI8}$$

Reserve this fraction q of the strict E_* pivot; the remaining coefficient still has positive real residue. The leading $2(vh)^2$ in (QI6), by (QI7), gives strict F_* and R_F pivots, with only infinitesimal quadratic corrections and the printed first-jet term. The remaining leading angular form is $2(\sum A_i^2 - 4\Gamma^2)$, with the strict finite toric square list from (SC16).

For precision, the only additional bare-Γ normal couplings left by (QI6), apart from ordinary toric first jets, have the form $\Gamma E_* a_* + \Gamma F_* b'_*$ with

$$a_* = bu(2X_U + Y_U) + Y_V vg, \qquad b'_* = 2\varepsilon_V g.$$

At contact $u = bc, v = cg$ these satisfy

$$a_{*0}/c = b^2(2X_{U0} + Y_{U0}) + Y_{V0}g^2 \in B_{\mathrm{Inf}}, \qquad b'_{*0}/g = 2\varepsilon_{V0} \in B_{\mathrm{Inf}}.$$

Their differences from contact have finite bounded monic Taylor arrays. The conductors (SC18) allocate those differences against E_*, F_* with infinitesimal coefficients. Apply (QI7) to their constant terms. What remains is a bounded infinitesimal toric quadratic plus $\Gamma \sum \mu_i d_i + \Gamma^2\sigma$. The exact scalar value and three original first jets imply its contact value and first-jet balances. The same two finite divisions as in (SC17)–(SC20), or the Hadamard contact-lift identity, allocate it on the strict toric root list. No infinite series or unsupported ideal-membership assertion is used: (QI3), the q coquotients, and the two displayed contact coefficient ratios are the finite certificates needed for the additional roots. Every coefficient function in these Taylor steps depends only on u, v. Thus only the first two conductors in (SC18)–(SC20) are used; the third state-coordinate jet supplies the inactive Hadamard balance. No bound on $(W + uY)/R$ after Z normalization is assumed.

Both complete state normals. Use $\widehat{N} = \sqrt{H_0}N$, $H_0 = 4h_2/R + 4h_b/t$, as in the preceding uniform-normal lemma. Add back the centered D term from (SC10), including its normals. Its scalar root is exactly

$$J_* = (T + b^2tV)h - D_u v = E_* + b^2c(1 - tg^2)A - buC + b^2tvF_* + b^2tvgD. \tag{QI9}$$

Its full root is $J_* + ubvgN - b^2N_y/\sqrt{s}$. All additional hard coefficients are infinitesimal after the $\widehat{N}$ normalization.

For the first N normal in (SC14), its H term is a bounded infinitesimal multiple of $uh = uE_* + bC$; its other terms are infinitesimal multiples of B, C. For the harmonic normal, multiply its H coefficient by $\sqrt{s}$. After the exact opening cancellations, its value at $q = U = V = 0$ is zero. Finite numerator coquotients therefore give

$$\sqrt{s}\,[H\text{-coefficient in } \mathcal{L}_y] = qK_q + UK_U + VK_V \tag{QI10}$$

with bounded arrays. Use

$$qH = RcA, \quad u^2H = Ru(uE_* + bC), \quad v^2H = Rv(F_* + gD).$$

The two remaining terms of $\mathcal{L}_y$ have infinitesimal coefficients on C, D since α_b/q^2 is bounded. Thus both complete first normals, including (QI9), are infinitesimal combinations of E_*, F_*, A, B, C, D. Their exact contact zeros make the coefficient sum of A, B, C, D zero at contact. One finite monic division and (SC18) allocate these on the same strict square list. The hard block has residue $\mathrm{diag}(1, 4)$ and all its other entries have zero residue. This proves a bounded positive residue Gram for the entire original parent, after restoring all retained positive terms.

States and signs. The restriction $Z = 0$ is positive by the original pure-testing state restriction. For $Z \neq 0$, divide the state by the pure-testing Z; its size is immaterial, since every coefficient array above depends only on u, v. On the negative-core chart, exact sign gauges give $u, v > 0$ and the common pure-testing Schur identity forces $W + uY > 0$. Hence $h > 0$, as required. The other core-sign chart and literal testing axes are direct original exits. Complex states split into real and imaginary parts. This completes the asserted small-u, small-v theorem.

B.4.12 Infinitesimal q at the other Bob testing angles

Theorem B.13. *The original C1.1 source with $q = k^2\eta$ infinitesimal is tensor SOS at every independent test. Together with the q-unit theorem this closes the entire original C1.1 source.*

The common C, A-large certificate already handles u not infinitesimal. The preceding local theorem handles u, v infinitesimal. Here are the two remaining v charts. Every positive opening rate and the literal permitted coefficient boundary is kept.

Ordinary v: the full normal residue is strict. Let v be a positive bounded unit, use $Z = 1$ on the negative-core chart, and retain the roots of (QI1). Normalize N by H_0 as before, but now restore the positive term $4h_b(vN)^2/t$. Write

$$\tau = \operatorname{res} t, \qquad \rho = \operatorname{res}(t/e^2), \qquad a^2 = \frac{\tau}{2(1+\tau)}, \quad \gamma^2 = \frac{\rho}{2(1+\tau)}.$$

The original $\beta \geq 0$ equation implies $0 \leq \rho \leq 1$; $0 \leq \tau \leq 1$ follows from $M = R - t > 0$ and $R \sim 1$. The entire residue on $(\widehat{N}, B, C)$ has diagonal entries

$$1 + \frac{v^2}{1+\tau}, \qquad 2(1+\tau v^2), \qquad 2,$$

and polynomial cross coefficients

$$4av^2, \qquad -4\gamma v, \qquad -4\sqrt{\tau\rho}\, v.$$

Its exact square decomposition is

$$\begin{aligned} &2(C - \gamma v\widehat{N} - \sqrt{\tau\rho}\, vB)^2 + \widehat{N}^2 + 2B^2 \\ &\quad + (1-\rho)v^2 \left(\frac{\widehat{N}}{\sqrt{1+\tau}} + \sqrt{2\tau}\, B \right)^2 . \end{aligned} \tag{QR1}$$

Thus it is strictly positive definite, also at $\rho = 1$, the possible $\beta = 0$ endpoint, and at $\tau = 0$ or 1. The other residue blocks are exactly

$$2E_*^2 + 2(vh)^2 + 4D_v N_y^2.$$

There are no nonzero leading cross entries with these blocks.

For completeness, the boundedness check uses the entire raw scalar (QI2) and both full normal formulas (SC14), with (QI9) restored. Terms containing uh use $uh = uE_* + bC$; all their coefficients are infinitesimal. The N–B and N–C coefficients with nonzero residue are precisely those retained in (QR1). The harmonic normal has infinitesimal coefficient on h for bounded v: after the opening cancellation its coefficient lies in (b^2, q, u^2) with bounded arrays on this chart. Its other coefficients are infinitesimal as well. All remaining source-constant terms and weak monomials enter the same strict Gram through the exact bounded pure-testing identities

$$h = (vh)/v, \qquad 1 = ((vh) - vE_*)/v^2.$$

For example $A = ch$ and $D = qg$ are infinitesimal-coefficient combinations of these two roots. Consequently the actual full Gram has the positive residue just displayed. Restore all retained terms. The same sign, $Z = 0$, and complex-state joins as in the local theorem apply; no restriction on the size of the normalized state was used.

Unbounded v: a five-root endpoint frame. Put $w = 1/v$, so w is infinitesimal, and multiply the original parent by w^2. The exact source formulas give positive real constants μ, d_0 with

$$C/d \geq \mu/b^2, \qquad A/d \geq \mu/b^2, \qquad D/d \geq d_0 b^2.$$

For instance sufficiently small fixed constants below $1/4$ work, since $b^2C/d, b^2A/d$ are uniformly bounded below and $D/(db^2) \sim 1/2$. Retain all excesses and the entire m, α, β terms. No assumption on e is needed. Use the five roots

$$P = uW/b, \quad Q = Y/b, \quad X, \quad J_b' = bZ, \quad V_b = uwZ.$$

The homogenized signed core is exactly

$$\begin{aligned} K = {} & V_b^2 - 2buV_bQ - 2wPJ_b' + 2wV_bX + b^2w^2P^2 + 2buwPX - 2b^2w^2PQ \\ & + (D_u + w^2)(X^2 + b^2Q^2) - 2buwXQ. \end{aligned} \tag{QR2}$$

The three selected event terms are

$$4\mu D_w(P-Q)^2 + 4\mu D_u D_w Q^2, \qquad 4d_0(b^2wP + buX - b^2wQ - J_b')^2. \tag{QR3}$$

Combining these with $2K$, the bounded Gram has strictly positive residue

$$4\mu\{(P-Q)^2 + Q^2\} + 4d_0(J_b')^2 + 2X^2 + 2V_b^2.$$

All other displayed coefficients have zero residue. This is a full state-quadratic certificate, not a restriction to a contact graph. It includes the homogeneous endpoint $w = 0$ directly and uses no inverse involving independent source and testing variables.

The infinitesimal, unit, and unbounded v charts exhaust all tests remaining after the common C, A certificate. Hence the entire q-infinitesimal C1.1 source, and consequently C1.1 itself, is closed.

B.4.13 Unit η, e, r

Theorem B.14. *Use the unchanged physical native C1 source, $c = 2$, $\lambda = n = 0$, with all original event signs and positive openings a, t, M, J. Suppose k, ζ are positive infinitesimals, and*

$$\eta, \quad r = 1 - a\eta, \quad e = k\sqrt{\eta/\zeta}$$

are positive bounded units. Then the complete original parent is tensor SOS at every independent projective test and every real or complex state. The openings may be arbitrarily small, and $\beta = 0$ is included.

The double-infinity corner. With $p = 1/u$, $q = 1/v$, the exact homogeneous parent is

$$\begin{aligned} F_{\infty\infty} = {} & 4m(pW - pqX + Y - qZ)^2 + 4c(1+q^2)(W - pY)^2 + 4A(1+p^2)(1+q^2)Y^2 \\ & + (4D + 2d)(qW + X - pqY - pZ)^2 + (4\alpha + 2d)(1+p^2)(Y - qZ)^2 \\ & + (4\beta + 2d)(1+q^2)(pX + Z)^2 - 2d(1+p^2)(1+q^2)Z^2. \end{aligned}$$

At $Z = 1$, after retaining α, β, its complete quadratic part is

$$4cW^2 + 2dX^2 + 4D(X-p)^2 + 4AY^2 + (4m + 2d)(Y-q)^2. \tag{BE1}$$

In particular the (Y, q) determinant is $4A(4m + 2d) > 0$. This is an exact full-parent expansion.

An even shorter full-parent certificate retains every event except c, A. The remaining polynomial belongs to $(W, X, Y, qZ)^2$, with bounded coefficients at $p, q = 0$. Its residue Gram is the quadratic form

$$4cW^2 + 2dX^2 + 4AY^2 + 2d(Y - qZ)^2. \tag{BE2}$$

It is strictly positive when $c/d, A/d$ are bounded below by positive units. All cross terms of the complete polynomial are allocated on these four roots, so (BE2) handles arbitrary states and both literal projective endpoints, without a normal restriction.

Uniform lower bounds and outer tests. Write $H = r + \eta$. The short coquotients have residues $2rH$ for $c_2, c_b, c_3, c_5, c_\alpha$, 2 for c_4, and $4r$ for c_d. The source identities imply

$$\frac{m}{d} = \frac{e^2 c_2}{M\eta c_d}, \quad \frac{c}{d} = \frac{c_3}{a\eta c_d}, \quad \frac{A}{d} = \frac{\eta c_4}{c_d}, \quad \frac{D}{d} = \frac{2c_5 - J\eta c_d}{2J\eta c_d}. \tag{BE3}$$

Every ratio is bounded below by a positive real number. For the last, $J + a = T \sim 1$ and

$$\overline{2c_5 - J\eta c_d} = 4\bar{r}(\bar{H} - \bar{J}\bar{\eta}) = 4\bar{r} > 0.$$

No reciprocal of an opening is required to be bounded. Choose positive real lower bounds for the four ratios, cap these four coefficients at the corresponding multiples of d, and retain their nonnegative excess event squares. All selected-parent Gram coefficients divided by d are then bounded.

The physical-source theorem covers every noncommuting residue test. For u infinitesimal and v a bounded unit, the selected residue form is

$$4m(vW - X)^2 + 2d(1+v^2)(X^2 + Y^2) + 4(c + A)(1+v^2)Y^2 + 4D(Y + vZ)^2 \succ 0.$$

For $u = \infty$ in residue and v a bounded unit, retain the signed core at that commuting residue and use the four positive squares cW^2, AY^2, $m(vY - Z)^2$, $D(W + vX)^2$; their common kernel is zero. For $v = \infty$ in residue and u

a bounded unit, use the core $2d(1+u^2)(X^2+Y^2)$ and the $c(uW-Y)^2, D(uX-Z)^2$ terms. The $(0,\infty)$ corner has the strictly positive state residue

$$4mW^2+2dX^2+\{4(c+A)+2d\}Y^2+4DZ^2.$$

The (∞,∞) corner is (BE2). At $(\infty,0)$ retain all events except c, A and use the exact four-root allocation on

$$(pX+Z,\ W,\ Y,\ vX), \qquad p=1/u.$$

Its residue is

$$2d(pX+Z)^2+2d(W+vX)^2+4cW^2+4AY^2 \succ 0.$$

The full selected polynomial lies in the square of this displayed root ideal; the pure-testing change $Z=(pX+Z)-pX$ makes this immediate. The other coefficients have zero residue. These homogeneous identities include literal infinities. Literal zero axes have a nonnegative pure-testing core and are immediate original-event SOS. Only bounded u and infinitesimal v remain.

State entry and the W endpoint. Normalize the state by a maximal coordinate. Set $D_u=1+u^2$ and $w=(W+uY)/D_u$. On the negative pure-core branch, if w is infinitesimal then Z is a unit: otherwise the residue core $2\{(X+uZ)^2+(uW-Y)^2\}$ forces every normalized state coordinate to have zero residue. Normalize $Z=1$; the exact pure-testing Schur identity in (SD5) gives $w>0$ after the standard angle gauges.

If w is a unit and u is a unit, a zero of that core residue has $Y=uw\neq 0$. The positive lower bound for A/d gives a strict state residue after clipping the positive events. Thus only u, v infinitesimal and $W\sim w$ a unit need an additional state check. When Z is infinitesimal, retain all events except clipped m, A and use the roots (X, Y, uW, vW). Since Z/W is a pure-testing infinitesimal, the complete selected polynomial has residue

$$2d\{X^2+(uW-Y)^2\}+4m(vW-X)^2+4AY^2 \succ 0. \tag{BE4}$$

Every remaining coefficient contains u, v or Z/W; there is no constant or first-normal term outside this frame. This includes literal $Z=0$. If Z is a unit, normalize $Z=1$ and use the next raw scalar frame with w a unit.

The exact remainder and its complete normals. Use exactly (SD3)–(SD8), their four positive terms, and the positive clearing factor $\mathcal{C}$. Put

$$\delta=\Delta R/K, \quad E=b^2=\eta K/R, \quad A_0=wv,\ B_0=\delta v,\ C_0=\delta Euw,\ D_0=\delta^2k^3E, \quad x=B_0-C_0.$$

Here δ, E are positive bounded units, and a, t are bounded. The factored five-term scalar is in $(A_0,B_0,C_0)^2+(D_0^2)$, and the two complete first normals are in (A_0,B_0,C_0). These are the full formulas, without restricting normals or replacing native coefficients by leading surrogates.

At $k=v=0$ and at the relevant small-w or small-u endpoint, put $r_0=1-aE$ and $H_0=r_0+E$. Direct collection of the short formulas gives the raw scalar residue

$$8r_0H_0D_u^2A_0^2+8D_u\{x^2+u^2r_0H_0B_0^2\}+8r_0H_0D_uD_0^2. \tag{BE5}$$

The hard block on $(N,\delta M_B)$ is

$$\operatorname{diag}(16r_0H_0,\ 16r_0H_0+8E^2/D_u),$$

and the complete first normals have residues $0, 16Ex$. The coupled $(\delta M_B, x)$ determinant is $128D_ur_0H_0>0$.

If u is a unit and w infinitesimal, (BE5) is a strict Gram on $(A_0,x,B_0,D_0,N,\delta M_B)$. If u is infinitesimal and w a unit, use the exact relations

$$B_0=(\delta/w)A_0, \qquad C_0=(\delta/w)A_0-x.$$

The factor δ/w is a product of a pure-coefficient unit and a pure-testing unit; it is not a mixed inverse. The resulting residue is the strict form

$$8r_0H_0A_0^2+8x^2+8r_0H_0D_0^2+16r_0H_0N^2+(16r_0H_0+8E^2)(\delta M_B)^2+16E(\delta M_B)x. \tag{BE6}$$

This supplies precisely the unit-w chart for which the older Δ-small retained-current argument is not applicable.

For transparency, the only nontrivial raw coefficient collection in (BE5) is

$$r_0H_0D_u+a^2E^2+r_0E(2a-1)=1+u^2r_0H_0, \qquad aE+r_0=1.$$

The cross coefficient is $-16D_u(aE+r_0)=-16D_u$. In the full normal formula S_B has residue $-16E$ and S_C has residue $16E$. These prove the claimed full normal residues directly.

The sole contact chart. For u, w, v infinitesimal and $Z = 1$, use the unchanged monic contact–conductor identities and the two-pairing circuit. Use weights

$$(\delta, k, b, a, t, u, w, v): \quad (0, 1, 0, 0, 0, 1, 1, 2).$$

The exact source collection is

$$(E_0)_{\le 6} = 8\rho T_4 - 8(\rho + E)k^2x^2, \quad (L_1)_{\le 3} = 0, \quad (L_2)_{\le 3} = 16Ex, \qquad \rho = r_0H_0 - 1.$$

The source/contact substitutions preserve this grading. After the displayed terms are removed, the scalar and normal lift coefficients have positive weights, hence zero residues. The term k^2x^2 also has a zero-residue coefficient. Thus the soft leading form is $8x^2 + 8r_0H_0T_4$, with the same strict two-pairing circuit frame. The hard block and its only first-normal coupling are those of (BE6), and their determinant is $128r_0H_0 > 0$. Retain the old positive g_4^2 square. The full eight-root remainder Gram is strict. This invokes only the polynomial monic conductor already proved; no opening or contact difference is inverted.

Finally restore the four complete positive terms, the retained positive square, and all pure-coefficient and pure-testing factors. The cases above exhaust both projective angle lines and every projective state. Complex states split into their real and imaginary evaluations. Neither this theorem nor the correction of (BE1) asserts coverage when r is infinitesimal, η unbounded, or e infinitesimal; those are separate source domains.

B.4.14 The centered-A source with unit e

Theorem B.15 (The centered-A unit-e slice)**.** *Use the physical native C1 source with its unchanged original events, $c = 2$, $\lambda = n = 0$, all positive openings a, t, M, J, and all original nonnegative signs, including literal $\beta = 0$. Suppose $k, \zeta, r = 1 - a\eta$ are positive infinitesimals, while η and $e = k\sqrt{\eta/\zeta}$ are positive bounded units. Then the full parent is tensor SOS at every independent projective test and every real or complex state. No bound on k^2/r is imposed.*

The theorem is restricted to unit e. Its proof uses δ^{-1} as a bounded pure coefficient number; it does not assert the cases of infinitesimal or unbounded e.

Centering the sole large event. The exact native contact has $Y_c = ru_cW_c$. Consequently

$$L_A = Y - ruW$$

vanishes at contact with no asymptotic replacement. The native formulas give a a positive unit, A unbounded and Ar a positive unit of residue 2. Put $A_* = 1/a$, a positive bounded pure coefficient unit. The complete term

$$4(A - A_*)D_uD_vL_A^2 \tag{CA1}$$

is positive and has zero value and full first derivative at the exact native contact. Retain it as a square.

It is essential to center this term before taking residues. The remaining A channel is exactly

$$4A_*D_uD_vL_A^2 + 4(Ar)D_uD_v\{2uWY - ru^2W^2\}. \tag{CA2}$$

Only Ar and r occur in its coefficients. In particular this is a bounded expression for every positive rate of r relative to k.

A source chart with no hidden k^2/r quotient. Keep the exact contact variables $\delta, k, E = b^2$ from (SD6), and put

$$K = 1 + k^2, \quad \ell = K - aE = Qr > 0, \quad Q = K^2 - \ell k^2, \quad a = (K - \ell)/E.$$

Here δ, E, a are positive bounded units and k, ℓ are positive infinitesimals. The exact source substitution is

$$\eta = KE/Q, \quad r = \ell/Q, \quad \Delta = \delta Q/K, \quad \zeta = \delta^2k^2QE/K, \quad R = K^2/Q, \quad T = K - k^2t\zeta. \tag{CA3}$$

All denominators in (CA3) are positive pure coefficient units. For $i = 2, b, 3, 5, \alpha, d$ let $\widetilde{c}_i = c_i/\ell$, defined by cancelling the explicit $L_c = \ell/Q$ factor in the short native formulas. This cancellation gives rational expressions with unit denominators; it does not leave an inverse of ℓ. For example, writing $H_c = r + \eta$ and $V_c = k^2\eta\zeta$,

$$\widetilde{c}_3 = \frac{2KH_cQ_c(1 - k^2\eta + tV_c)}{Q}, \qquad \widetilde{c}_d = \frac{4Q_cP_d}{Q}.$$

Their residues are $2E$ for $i = 2, b, 3, 5, \alpha$, and 4 for $i = d$; c_4 has residue 2.

Define the following bounded pure coefficient quantities:

$$\begin{aligned} B_2 &= \frac{8a\widetilde{c}_2}{R\widetilde{c}_3\Delta^2}, & B_b &= \frac{8a\widetilde{c}_b}{R\widetilde{c}_3\Delta^2},\\ B_3 &= 8a/T, & B_5 &= \frac{8a\widetilde{c}_5}{T\widetilde{c}_3},\\ L &= Ar = \frac{2a\eta^2 c_4}{Q\widetilde{c}_3}, & d &= \frac{2a\eta\widetilde{c}_d}{\widetilde{c}_3},\\ \alpha &= \frac{2a\Delta^2 k^6\eta^2\widetilde{c}_\alpha}{\widetilde{c}_3}. \end{aligned} \tag{CA4}$$

These formulas explicitly account for the cancellation needed below.

The exact full remainder, before any collection. Retain the four original (SD3) terms, divided by the positive source factor $\Delta^2\Lambda$, and retain (CA1). Use the unchanged pure testing coordinate $w = (W + uY)/D_u$ and the hard normals N, M_B. For $Z = 1$ put $p = t/R$, $h = a/T$ and

$$q_A = vD_u w, \quad q_B = v(D_u - uN - puD_u vw),$$

$$W = w + u(M_B + hq_B)/D_u, \qquad Y = uw - (M_B + hq_B)/D_u.$$

The complete remainder is exactly

$$\begin{aligned} \mathscr{G} = {} & B_2\{N^2 - 2q_A N + (1-p)q_A^2\} + B_b D_v\{N^2 + 2q_A N + pq_A^2\}\\ & + B_3 D_v\{M_B^2 + 2q_B M_B + hq_B^2\} + B_5\{M_B^2 - 2q_B M_B + (1-h)q_B^2\}\\ & + 4A_* D_u D_v(Y - ruW)^2 + 4LD_u D_v(2uWY - ru^2W^2)\\ & + 4\alpha D_u(vY - 1)^2 + 2dD_u\{(vY-1)^2 - D_v\}. \end{aligned} \tag{CA5}$$

There are no normal terms above degree two. Every coefficient of (CA5) is bounded. The only testing denominator is a power of the positive unit D_u; every source denominator in (CA3)–(CA5) is a positive unit. The possibly unbounded opening ratios remain exclusively in the four retained positive terms.

The exact low-weight collection. The outer/state lemma in Section B.4.16 reduces the independent tests to $Z = 1$ and u, w, v infinitesimal, with $w > 0$ on the negative pure-core branch. Use the unchanged circuit monomials

$$A_0 = wv, \quad B_0 = \delta v, \quad C_0 = \delta Euw, \quad D_0 = \delta^2 k^3 E,$$

$$x = B_0 - C_0, \quad y = A_0 - kC_0, \quad z = uB_0 - kC_0, \quad h_0 = kC_0 - D_0.$$

Let q_1, q_2 be the same two separated radical pairings as in the contact–conductor proof, and let

$$T_4 = y^2 + z^2 + h_0^2 - 2yz + 2q_1^2.$$

Give $(\delta, k, E, \ell, t, u, w, v)$ weights $(0, 1, 0, 1, 0, 1, 1, 2)$. Then x has weight two and y, z, h_0, q_1, q_2 have weight three. Denote the scalar of (CA5) by S, and its two complete first normals by L_N, L_M. Direct multiplication of (CA3)–(CA5) through the indicated weights gives

$$\begin{aligned} S_{\le 6} &= \frac{4P}{E^4\delta^2}x^2 - \frac{16(k+u)}{E^2\delta^2}xz + \frac{8}{E\delta^2}T_4,\\ (L_N)_{\le 3} &= 0,\\ (L_M)_{\le 3} &= \frac{8\{E^2 + 2E - 2\ell(E+1)\}}{E^2\delta}\,x, \end{aligned} \tag{CA6}$$

where the short coefficient polynomial is

$$\begin{aligned} P = {} & E^2(E+4) - E(E^2+6E+4)\ell + 2(E+1)(E+2)\ell^2\\ & + E^2(8-3E)k^2 + 4E^2ku + E^2(E+4)u^2. \end{aligned}$$

The full hard-normal block, from the quadratic part of (CA5), has residue

$$\operatorname{diag}\left(\frac{16}{E\delta^2}, \frac{4(E^2+4)}{E}\right). \tag{CA7}$$

Here and in the residue Grams, E, δ denote their positive residues. Formula (CA6), rather than a large expansion, is the finite coefficient identity used by the proof. Its 37 raw scalar monomials collect to the displayed three terms.

Why all higher terms have harmless coefficients. The scalar of (CA5), after clearing the stated units, belongs to $(A_0, B_0, C_0)^2 + (D_0^2)$. This follows directly from $q_A = wvD_u$, $q_B|_{N=M_B=0} \in (v)$, and Y, W in (CA5): the centered scalar $Y - ruW$ lies in (v, uw); its square is in the same raw monomial square ideal since δ, E are units. The sole constant state term comes from α, which has the factor k^6, and is a bounded unit multiple of D_0^2. Both full first normals belong to (A_0, B_0, C_0) by the same displayed expression.

All retained terms vanish to first order at the exact contact

$$(u, w, v, N, M_B) = (k, \delta k, \delta k^2 E, 0, 0).$$

Thus the scalar belongs to the square of its contact ideal and the first normals belong to that ideal. The unchanged monic contact–conductor identities therefore apply, over the coefficient ring in which only the displayed positive units have been inverted. They never divide by k, ℓ, M, J, t, or a contact difference; the displayed $1/a$ is the inverse of a known positive unit. The contact substitutions preserve the new weights. After (CA6) is subtracted, scalar weights are at least seven and normal weights at least four. Their lifted coefficients consequently have positive weights, hence zero residues. Clearing the finitely many unit denominators before this polynomial lift and restoring them afterward changes no residue sign and introduces no mixed inverse.

Strictness, full normals, and restoration. The leading scalar is

$$\frac{4(E+4)}{E^2\delta^2}x^2 + \frac{8}{E\delta^2}T_4.$$

The standard two-pairing circuit supplies a strict Gram on (y, z, h_0, q_1, q_2). The only nonzero first-normal residue is $8(E+2)x/(E\delta)$, the coefficient of the full normal M_B. Accordingly the coupled (M_B, x) Gram is

$$\begin{pmatrix} 4(E^2+4)/E & 4(E+2)/(E\delta) \\ 4(E+2)/(E\delta) & 4(E+4)/(E^2\delta^2) \end{pmatrix}, \qquad \det = \frac{256}{E^3\delta^2} > 0. \tag{CA8}$$

Together with the independent N pivot in (CA7), this gives a bounded strict eight-root Gram for the entire remainder (CA5). Restore (CA1), all four original positive terms, and all separated normalization factors. The outer/state lemma exhausts the remaining tests; real and imaginary state evaluations give the complex case.

This proves precisely the stated unit-e, unit-η, infinitesimal-r slice. Combined with the bounded-unit-r theorem, it covers every positive r when η, e are bounded units and k, ζ infinitesimal. It makes no assertion about unbounded η or either nonunit alternative for e.

B.4.15 The centered-A source with bounded reciprocal e

Theorem B.16. *Use the unchanged physical native C1 source with $k, \zeta, r = 1 - a\eta$ positive infinitesimals and η a positive bounded unit. Suppose $e = k\sqrt{\eta/\zeta}$ is bounded below by a positive real number; it may be unbounded. Then the entire original parent is tensor SOS at every independent projective test and every real or complex state. Every positive opening rate, literal $\beta = 0$, and every rate of k^2/r is included.*

Use all exact definitions and the centered positive term (CA1) in Section B.4.14. The sole change is that $\delta = \Delta R/K$ is now any positive bounded coefficient, so it must not be inverted in a bounded Gram. The source chart (CA3), $\ell = Qr = K - aE$, and the exact cancelled coquotients remain unchanged. Their only denominators are positive units, independent of the rate of ℓ relative to k.

Scale the entire remainder and its full normal. Let $\mathscr{G}$ be the exact full remainder (CA5). Put

$$\widehat{G} = \delta^2\mathscr{G}, \qquad \widehat{M} = \delta M_B, \quad \widehat{Y} = \delta Y, \quad \widehat{W} = \delta W.$$

The original testing variables have not been rescaled against a coefficient rate; these displayed quantities are explicit separated tensor roots. Their complete formulas are

$$q_A = vD_u w, \quad q_B = v(D_u - uN - puD_u vw),$$

$$\widehat{W} = \delta w + u(\widehat{M} + \delta h q_B)/D_u, \quad \widehat{Y} = \delta uw - (\widehat{M} + \delta h q_B)/D_u.$$

Set

$$\widehat{B}_2 = \frac{8a\widetilde{c}_2 K^2}{R\widetilde{c}_3 Q^2}, \qquad \widehat{B}_b = \frac{8a\widetilde{c}_b K^2}{R\widetilde{c}_3 Q^2}.$$

The entire exact scaled remainder is

$$\begin{aligned}\widehat{G} &= \widehat{B}_2\{N^2 - 2q_AN + (1-p)q_A^2\} + \widehat{B}_bD_v\{N^2 + 2q_AN + pq_A^2\}\\ &\quad + B_3D_v\{\widehat{M}^2 + 2\delta q_B\widehat{M} + \delta^2 hq_B^2\} + B_5\{\widehat{M}^2 - 2\delta q_B\widehat{M} + \delta^2(1-h)q_B^2\}\\ &\quad + 4A_*D_uD_v(\widehat{Y} - ru\widehat{W})^2 + 4LD_uD_v(2u\widehat{W}\widehat{Y} - ru^2\widehat{W}^2)\\ &\quad + 4\alpha D_u(v\widehat{Y} - \delta)^2 + 2dD_u\{(v\widehat{Y} - \delta)^2 - \delta^2 D_v\}. \end{aligned} \tag{CS1}$$

All coefficients are bounded; every occurrence of δ has nonnegative exponent. There is no inverse of δ, ℓ, k, M, J or t. The inverse $1/a$ in A_* is a positive unit. Every first-normal and hard-normal term is retained in (CS1).

Uniform contact grading. Keep the exact roots $A_0, B_0, C_0, D_0, x, y, z, h_0, q_1, q_2$ and polynomial P in (CA6). Use the same weights $\mathrm{wt}(\delta, k, E, \ell, t, u, w, v) = (0, 1, 0, 1, 0, 1, 1, 2)$. Weight zero means bounded here, not bounded away from zero. The exact finite coefficient collection from (CS1) is

$$\begin{aligned}(\widehat{G}|_{N=\widehat{M}=0})_{\le 6} &= \frac{4P}{E^4}x^2 - \frac{16(k+u)}{E^2}xz + \frac{8}{E}T_4,\\ (\widehat{L}_N)_{\le 3} &= 0,\\ (\widehat{L}_M)_{\le 3} &= \frac{8\{E^2 + 2E - 2\ell(E+1)\}}{E^2}x. \end{aligned} \tag{CS2}$$

The full hard block on $(N, \widehat{M})$ has residue

$$\mathrm{diag}(16/E, 4(E^2+4)/E). \tag{CS3}$$

These are precisely the old identities after the indicated scaling, but their validity for an infinitesimal δ requires the nonnegative-exponent assertion in (CS1), not cancellation of a purported unit δ.

The scalar is in $(A_0, B_0, C_0)^2 + (D_0^2)$ over the ring with E and the displayed positive units inverted. Both complete first normals are in (A_0, B_0, C_0). Direct monomial allocation in (CS1) proves these memberships with nonnegative δ exponents. At zero normals, $q_A = A_0D_u$, $\delta q_B = B_0D_u(1 - puvw)$, and both $\widehat{Y}, u\widehat{W}$ belong to (B_0, C_0) with bounded-unit coefficients. The sole constant from the α term satisfies $\alpha\delta^2/D_0^2 = 2a\widetilde{c}_\alpha/\widetilde{c}_3$ exactly. Thus these are memberships of the entire rational expression, not merely its finite collection. The unchanged monic contact conductor lifts them without a δ inverse. Its contact substitutions preserve the weights, even if δ is infinitesimal. After (CS2), all remaining scalar weights are at least seven and all remaining normal weights at least four, hence the lifted correction coefficients have zero residue.

The leading soft form is

$$\frac{4(E+4)}{E^2}x^2 + \frac{8}{E}T_4.$$

Use the strict two-pairing circuit on T_4. Its only hard coupling is the full first normal $8(E+2)x/E$ on $\widehat{M}$, and

$$\det\begin{pmatrix} 4(E^2+4)/E & 4(E+2)/E \\ 4(E+2)/E & 4(E+4)/E^2 \end{pmatrix} = 256/E^3 > 0. \tag{CS4}$$

Together with $16/E$ on N, this gives the strict eight-root Gram.

Exhaustion and restoration. The independent outer/state lemma Section B.4.16 uses the unit-e hypothesis only to infer a positive lower bound on m/d in (CAO2). When e is unbounded, m/d is instead unbounded, so that same lower bound and every clipping argument hold. Its $c/d, A/d, D/d$ bounds, exact centered root identity, ordinary-u certificate, unit-w certificate, and literal projective endpoint arguments do not contain e or δ. Thus it reduces every remaining independent test to exactly the contact chart settled above. Restore δ^2 times every retained original term, including (CA1), and undo the positive pure coefficient factor δ^2. All factors and radicals remain separated between the two fields. Complex states split into real and imaginary evaluations.

This closes the formerly missing infinitesimal-r, unbounded-e slice, and incorporates the previous unit-e theorem. Together with the common unit-r results, it covers all positive native r when η is a unit and e is bounded below. It does not yet claim the infinitesimal-e slice.

B.4.16 Complementary centered-A testing and state charts

Lemma B.17. *Use the unchanged physical native C1 source, normalized by $c = 2$ and $\lambda = n = 0$, with all original event signs and positive openings a, t, M, J. Suppose $k, \zeta, r = 1 - a\eta$ are positive infinitesimals and $\eta, e = k\sqrt{\eta/\zeta}$ are positive bounded units. Then the outer-test and state-entry arguments reduce the all-tests problem to precisely*

$$u, v, w > 0 \text{ infinitesimal}, \qquad Z = 1, \qquad w = (W + uY)/(1 + u^2). \tag{CAO1}$$

All other independent projective tests and states admit a full-parent tensor SOS. No comparison of k^2 with r is made, and arbitrarily small positive openings and literal $\beta = 0$ are included.

Source ratios with the common factor cancelled. Here r always denotes the native source parameter, not the event root $uW - Y$. Put

$$K = 1 + k^2, \quad H_c = r + \eta, \quad V_c = k^2\eta\zeta, \quad S_3 = 1 - k^2\eta + tV_c,$$

The two required short native polynomials are

$$\begin{aligned} P_4 &= 1 + k^2 + \eta k^2(a - k^2 - 1) - \zeta k^2 t \\ &\quad + V_c[-\zeta k^2 t^2 + 2k^2 t + a(1 - t) + t], \\ P_d &= (1 + k^2)^2 - a\eta k^4 + \zeta k^2[(k^2 + 1) - (k^2 + 2)t] \\ &\quad + V_c[-\zeta k^2 t^2 + k^2 a(t - 1) + k^2 t + k^2 + 1]. \end{aligned}$$

They are positive bounded units with residue one. The exact source identities, after cancelling the common factor r in the coquotients, are

$$\begin{aligned} \frac{d}{} &= \frac{4a\eta P_d}{KH_cS_3}, & Ar &= \frac{2a\eta^2 P_4}{KH_cS_3}, \\ \frac{m}{d} &= \frac{e^2K(1 - t\zeta)(1 + V_c)H_c}{2M\eta P_d}, & \frac{c}{d} &= \frac{KH_cS_3}{2a\eta P_d}, \\ \frac{A}{d} &= \frac{\eta P_4}{2rP_d}, & \frac{D}{d} &= \frac{KH_c(1 + V_c) - J\eta P_d}{2J\eta P_d}. \end{aligned} \tag{CAO2}$$

The positive openings obey $M + t = 1 + k^2 r$ and $J + a = 1 + k^2 - k^2\zeta t$; hence a, t, M, J are bounded. Since $a\eta = 1 - r$, one has

$$\bar{\eta} \geq 1, \qquad \bar{d} = 4/\bar{\eta}, \qquad Ar \sim 2, \qquad A \text{ positive unbounded.} \tag{CAO3}$$

The numerator of D/d in (CAO2) has residue $\bar{H}_c - \bar{J}\bar{\eta} = 1$. Therefore $m/d, c/d, D/d, A/d$ all have positive real lower bounds. This conclusion uses no inverse bound on an opening. Moreover,

$$\begin{cases} \bar{D} = \dfrac{2}{\bar{\eta}(\bar{\eta} - 1)}, & \bar{\eta} > 1, \\ D \text{ positive unbounded}, & \bar{\eta} = 1. \end{cases} \tag{CAO4}$$

The previously proved outer routing still applies. Clip m, c, A, D at positive real multiples of d, retaining all their nonnegative excess original event squares. The resulting coefficients divided by d are bounded. The outer-test argument in Section B.4.13, through (BE2)–(BE4), uses only these lower bounds and the independent physical-source certificate at noncommuting residue tests. Its exact homogeneous identities and four-root ideal memberships are unchanged. It therefore includes all projective infinity charts, the $(0, \infty)$, $(\infty, 0)$ and (∞, ∞) corners, and the unit-W, infinitesimal-Z state chart. The original-event SOS on literal zero axes is unchanged as well. This invokes none of the bounded-r scalar or normal conclusions (BE5)–(BE6). Only bounded u and infinitesimal v remain.

An exact centered root identity. Write $D_u = 1 + u^2$, $D_v = 1 + v^2$ and introduce

$$h = A^{-1/2}, \quad L = Ar, \qquad H = \sqrt{A}(Y - ruW), \quad B = X + uZ, \quad U = uW, \quad V = vZ. \tag{CAO5}$$

Thus $Y = rU + hH$, while h, r are infinitesimal and $L \sim 2$. All coefficient radicals in (CAO5) lie in the coefficient field; B, U, V lie in the testing field. The root H is their finite tensor combination, with no mixed inverse or mixed radical.

For clarity, the complete original finite parent used here is

$$\begin{aligned} F &= 4m(vW + uvY - X - uZ)^2 + 8D_v(uW - Y)^2 + 4AD_uD_vY^2 \\ &\quad + (4D + 2d)(uW + uvX - Y - vZ)^2 + (4\alpha + 2d)D_u(vY - Z)^2 \\ &\quad + (4\beta + 2d)D_v(X + uZ)^2 - 2dD_uD_vZ^2. \end{aligned}$$

Choose $0 < D_* \leq D$ bounded, and let $m_* = 0$ or choose a positive real $m_* \leq m$, as specified below. Retain the four nonnegative original event contributions with weights $m - m_*, D - D_*, \alpha, \beta$. For $C = 4D_* + 2d$, the exact

remaining parent is

$$\begin{aligned} F_* &= 8D_v[(1-r)U - hH]^2 + 4D_uD_vH^2 + 4LrD_uD_vU^2 + 8LhD_uD_vUH \\ &\quad + C[(1-r)U - hH + uvB - D_uV]^2 + 2dD_vB^2 \\ &\quad + 2dD_u\{v^2(rU+hH)^2 - 2(rU+hH)V - V^2\} + 4m_*R_m^2, \\ R_m &= vW + uv(rU+hH) - B. \end{aligned} \tag{CAO6}$$

In particular, the centered A contribution is the exact identity

$$4AD_uD_vY^2 = 4D_uD_vH^2 + 4LrD_uD_vU^2 + 8LhD_uD_vUH.$$

This retains its complete hard root. The last two coefficients have zero residue. Formula (CAO6) follows directly from the original parent and includes the complete signed core; it is not an eliminated scalar or a restriction to state normals. Every displayed coefficient is bounded. The term involving Z in that core has already cancelled to the displayed expression in $V = vZ$.

Every state when u is a bounded unit. Set $m_* = 0$, so the entire original m event is retained. Formula (CAO6) is an exact Gram allocation on (H, B, U, V). At infinitesimal v its residue is

$$4D_uH^2 + 2dB^2 + 8U^2 + C(U - D_uV)^2 - 2dD_uV^2. \tag{CAO7}$$

Residues are understood in this and the following determinant calculations. The U, V determinant is

$$D_u\{8CD_u - 2d(8+C)\}.$$

If $\bar{\eta} > 1$, take $D_* = D$, which is bounded by (CAO4). Then $C = 8/(\bar{\eta} - 1)$ in residue and this determinant is

$$\frac{64u^2(1+u^2)}{\bar{\eta} - 1} > 0.$$

If $\bar{\eta} = 1$, choose a sufficiently large positive real D_*; this is permitted because D is unbounded. The determinant is then

$$32(1+u^2)(D_*u^2 + 2u^2 - 2) > 0.$$

The first diagonal is $8 + C > 0$, so the complete four-root residue Gram is positive definite in both cases. The cap is a real constant chosen from the nonzero testing residue of u; no division of tensor elements is introduced. The bounded full Gram is therefore tensor SOS, and restoring the retained events proves the claim for every state in this chart.

The remaining unit-w state chart. Suppose u, v are infinitesimal and w is a unit. Then W is a unit on the negative-core branch. When Z is infinitesimal the inherited (BE4) argument already applies. Otherwise normalize the state so that Z is a unit and put $q = W/Z$, a pure-testing unit. Choose a positive real lower bound $m_* \le m$. Now

$$R_m = qV + uv(rU + hH) - B,$$

so (CAO6) is again an exact bounded Gram on (H, B, U, V), with residue

$$4H^2 + 2dB^2 + 4m_*(qV - B)^2 + 8U^2 + C(U - V)^2 - 2dV^2. \tag{CAO8}$$

The B pivot is positive. Its Schur complement adds

$$\mu V^2, \qquad \mu = \frac{4dm_*q^2}{d + 2m_*} > 0$$

to the U, V block. This is just a residue determinant computation; no mixed inverse is used to define a root or a full-parent coefficient. If $\bar{\eta} > 1$, take $D_* = D$ as before. The determinant becomes

$$\frac{8\bar{\eta}\mu}{\bar{\eta} - 1} > 0.$$

If $\bar{\eta} = 1$, choose a sufficiently large positive real cap D_* so that $\mu(C + 8) - 64 > 0$. The unboundedness of D permits this choice. Thus the complete residue Gram in (CAO8) is strict, including all hard-soft terms, and restoring the retained events gives the full-parent tensor SOS.

Completion of state entry. Normalize the state by a maximal coordinate and first remove the nonnegative pure-testing-core branch, which has its direct original SOS. On the negative-core branch with v infinitesimal, the residue core is

$$2\{(X+uZ)^2+(uW-Y)^2\}.$$

Consequently these two roots have zero residue. If w and Z were both infinitesimal, the exact inverse relations

$$W=w+\frac{u(uW-Y)}{D_u},\quad Y=uw-\frac{uW-Y}{D_u},\quad X=(X+uZ)-uZ$$

would make every normalized state coordinate infinitesimal. Hence small w forces Z to be a unit. The bounded-unit u chart was settled by (CAO7), and all unit-w states by (CAO8) and (BE4). Normalize the only remaining states by $Z=1$. The same exact pure-testing Schur identity (SD5) and testing sign gauges give $u,v,w>0$. This is exactly (CAO1). Literal axes and literal state zeros have already been included. Complex states split into their real and imaginary evaluations because the parent is real.

B.4.17 Unit η, r and unbounded e

Theorem B.18 (Unit η and unbounded e)**.** *Use the unchanged physical native C1 source, its original event signs, and the positive shapes and gaps a,t,M,J. Suppose k,ζ are positive infinitesimals, η is a positive bounded unit, $r=1-a\eta$ is a positive bounded unit, and $e=k\sqrt{\eta/\zeta}$ is unbounded. Then the full original parent is tensor SOS at every independent projective test and every real or complex state. No positive lower bound on a,t,M,J is imposed.*

Write $H=r+\eta$ and $\Delta=1/e$. The seven native coquotients have residues $2rH$ for c_2,c_b,c_3,c_5,c_α, 2 for c_4, and $4r$ for c_d. All are positive units. In particular the four retained terms of the contact–conductor proof remain positive and the remainder has bounded coefficients. We reuse their exact definitions, not asymptotic substitutes for the original source.

Outer tests. The exact formulas give

$$\frac{m}{d}=\frac{e^2c_2}{M\eta c_d},\qquad \frac{\beta}{d}=\frac{e^2c_b}{t\eta c_d}-\frac{1}{2}.$$

Both ratios are unbounded. Apply the exact full-parent Alice reflection (AS3)–(AS4), which exchanges $m\leftrightarrow c$ and $\beta\leftrightarrow D$ while leaving v unchanged. The already proved two-strong-current identity then covers every test with $1/v$ bounded, for every Alice angle and every state. Its homogeneous form includes both literal Alice endpoints, so no inverse-angle point is lost.

For v infinitesimal and Alice infinity, use the critical-current identity with $g_c=cD/(c+D)$, $\epsilon=d/g_c$ and $\gamma=Ag_c/d^2$. Its actual sufficient conditions on this chart are bounded ϵ and positive-unit γ: in its exact four-root identity the correction L_0 has factors c_u or v, where $c_u=(1+u^2)^{-1/2}$ is infinitesimal. Thus $\sqrt{\epsilon}$ need only be bounded. The residue Gram is $\operatorname{diag}(1,2,2,4\bar{\gamma})$. Use the exact relation $J+a=T$ with $T-1$ infinitesimal, and the cancelled original coefficient formula

$$\frac{D}{a}=\frac{2c_5/c_3-J\eta c_d/c_3}{J}.$$

Its numerator has residue $2/\bar{H}$. The unchanged native formulas give

$$\frac{d}{a}\sim\frac{4\eta}{H},\quad \frac{A}{a}\sim\frac{2\eta^2}{rH},\quad \frac{D}{a}\sim\frac{2}{JH},\qquad \epsilon\sim\frac{2\eta(JH+a)}{H},\quad \gamma\sim\frac{H}{4r(JH+a)}.$$

The identity $\overline{JH+a}=1+\bar{\eta}(1-\bar{a})^2$ shows that the two required conditions hold even when a or J is infinitesimal. The calculation uses the cancelled harmonic coefficient g_c/a, so no lower bound on an opening is inserted. Homogeneous angle normalization includes literal Alice infinity. It remains to treat bounded u and infinitesimal v.

State entry. Normalize the state by a coordinate of maximal absolute value and put $D_u=1+u^2$ and $w=(W+uY)/D_u$. If w is a unit, the exact five-root current identity (UW2)–(UW3) applies with $Q=W+uY=D_uw$ a unit. Its first four diagonal residues are $8rH$ and its fifth is $8D_u$; both negative corrections have factors Δ or $\Delta^2\eta$. Thus this chart is strict.

If w is infinitesimal, the nonnegative pure-testing core is a direct exit. On the negative-core branch Z is a unit. Indeed its residue at $v=0$ is $2\{(X+uZ)^2+(uW-Y)^2\}$; if Z,w and both currents had zero residue, every normalized state coordinate would have zero residue. Normalize $Z=1$. The exact core Schur identity gives $w>0$ after the usual positive-angle sign gauges. All transformations here belong solely to the testing field.

The ordinary-u chart needs no contact division. Use the exact source change

$$\delta = \Delta R/K, \quad b^2 = \eta K/R, \quad K = 1 + k^2, \quad Q = K + ak^2b^2, \quad T_n = K^2 - t\delta^2 k^4 Q b^2.$$

All source denominators are positive units, δ, k are infinitesimal and $b^2 \sim \eta$. Let

$$A_0 = wv, \quad B_0 = \delta v, \quad C_0 = \delta b^2 uw, \quad D_0 = \delta^2 k^3 b^2, \qquad x = B_0 - C_0.$$

The five-term exact scalar belongs to $(A_0, B_0, C_0)^2 + (D_0^2)$ and both complete first normals belong to (A_0, B_0, C_0). Collecting these displayed factors before taking residues is legitimate over the bounded coefficient ring. At $\delta = k = w = v = 0$, keeping bounded u free, the scalar Gram is

$$8rHD_u^2 A_0^2 + 8D_u\{x^2 + u^2 rHB_0^2\} + 8rHD_u D_0^2. \tag{UE1}$$

The hard block on $(N, \delta M_B)$ is $\mathrm{diag}(16rH, 16rH + 8\eta^2/D_u)$, and the complete first normals have residues 0 and $16\eta x$. The latter is the coefficient of δM_B in the full polynomial, so the symmetric Gram off-diagonal entry between δM_B and x is 8η. The coupled $(\delta M_B, x)$ block has determinant

$$(16rH + 8\eta^2/D_u)\, 8D_u - (8\eta)^2 = 128 D_u rH > 0. \tag{UE2}$$

If u is a unit, the remaining B_0 pivot in (UE1) is strict, as are the A_0, D_0, N pivots. Every omitted coefficient has zero residue. This proves the chart by a bounded strict Gram on these six roots, without any mixed inverse. Equation (UE1) follows directly from $a\eta + r = 1$: its (B_0, C_0) part before collection is $8D_u\{(1 + u^2 rH)B_0^2 - 2B_0C_0 + C_0^2\}$.

The infinitesimal-u chart uses the old conductor. Now δ, k, u, w, v are infinitesimal, while b, a, t are bounded. Use the unchanged roots

$$y = A_0 - kC_0, \quad z = uB_0 - kC_0, \quad h = kC_0 - D_0, \quad q_1 = a_1a_2 - a_3a_4,$$

and the second pairing $q_2 = a_1a_3 - a_2a_4$ of the four separated radicals. Set

$$T_4 = \sum_{i=1}^{4} a_i^4 - 4\prod_{i=1}^{4} a_i = y^2 + z^2 + h^2 - 2yz + 2q_1^2, \qquad \rho = (1 - ab^2)(1 + (1 - a)b^2) - 1.$$

Give $(\delta, k, b, a, t, u, w, v)$ weights $(1, 1, 0, 0, 0, 1, 2, 3)$. The root x has weight four and the other soft roots have weight five. Substitution at the exact contact preserves this grading. Writing E_0 for the complete scalar correction relative to the old leading polynomial, the five-term factorization and the two complete normal formulas give

$$\begin{aligned} E_0 &= 8\rho T_4 - 8(\rho + b^2)k^2x^2 + O_{\mathrm{wt}}(12), \\ L_1 &= O_{\mathrm{wt}}(6), \qquad L_2 = 16b^2x + O_{\mathrm{wt}}(6). \end{aligned} \tag{UE3}$$

These identities require only multiplication of the short source polynomials. In particular the residue correction is the single factor ρ multiplying the same circuit, rather than a new polynomial positivity problem.

The unlocalized monic contact–conductor lift applies unchanged. After the terms explicitly displayed in (UE3) are removed, scalar coefficient weights are at least two and normal coefficient weights are at least one. Therefore they all have zero residue. There is no division by k, b, δ, a, t or a contact difference. The leading strict soft Gram is now

$$8x^2 + 8rHT_4.$$

The two-pairing circuit identity supplies a strict Gram on (y, z, h, q_1, q_2), and rH is a positive unit. The complete hard block is $\mathrm{diag}(16rH, 16rH + 8\eta^2)$; its only nonzero first-normal residue is $16\eta x$. Its coupling to the $8x^2$ pivot has determinant $128rH > 0$. Retain the old positive g_4^2 square. The resulting eight-root remainder Gram is strict.

Finally restore all four positive terms, the positive square just retained, and the positive pure-coefficient multiplier. The state and angle cases above exhaust the independent tests. Complex states split into their real and imaginary quadratic evaluations. The result closes the entire stated source domain, rather than only an optimizing or matching testing chart.

B.4.18 Unit η and infinitesimal e: local polynomial

Proposition B.19 (Local part of the small-e theorem)**.** *Use the unchanged physical native C1 source, with k, ζ, e positive infinitesimals and η a positive bounded unit. Every native positive $r = 1 - a\eta$, and every positive opening rate, is allowed. At every independent test with u, v, w infinitesimal, $Z = 1$, and $w > 0$, the entire original parent is tensor SOS. Literal $\beta = 0$ is included.*

This proposition is the contact part only. The independent outer/state join is supplied separately; no assertion here identifies a testing rate with a source rate.

The source entry forced by the original sign. Keep K, Q, E, ℓ, δ from (CA3), with $E = b^2$ and $\ell = Qr = K - aE$. Here E is a positive unit, $0 < \ell < K$, and no lower bound on a or ℓ is assumed. Put

$$\varepsilon = 1/\delta, \qquad j = \delta k, \qquad t = \varepsilon^2 \tau.$$

Both ε, j are infinitesimal: $e = \varepsilon K/Q$ and $\zeta = j^2 QE/K$. The original condition $\beta \geq 0$ gives

$$\frac{t}{e^2} \leq \frac{2c_b}{\eta c_d},$$

whose right side is bounded after cancelling the common native r factor. Thus τ is positive bounded. The exact chart is

$$k = \varepsilon j, \quad K = 1 + \varepsilon^2 j^2, \quad Q = K^2 - \ell \varepsilon^2 j^2, \quad a = (K - \ell)/E,$$

$$\begin{aligned} &\eta = KE/Q, \quad r = \ell/Q, \quad H = r + \eta, \quad e^2 = \varepsilon^2 K^2/Q^2, \\ &\zeta = j^2 QE/K, \quad R = K^2/Q, \quad T = K - k^2 t\zeta. \end{aligned} \tag{SE1}$$

All denominators displayed here are positive units. The cancelled coquotients $\widetilde{c}_i = c_i/\ell$ are the same short exact formulas as in (CA3); the cancellation introduces no inverse of ℓ. Their residues are $2(E + \ell)$ for $i = 2, b, 3, 5, \alpha$, and 4 for $i = d$; c_4 has residue 2. These formulas hold uniformly as ℓ tends to zero or to one.

One bounded centered remainder, including both normals. Set

$$A_* = \frac{ArH}{1 + rH}, \qquad A - A_* = \frac{A}{1 + rH} > 0. \tag{SE2}$$

Retain the entire square $4(A - A_*)D_u D_v (Y - ruW)^2$. It and its full first derivative vanish at exact native contact. Divide the remaining parent by the positive pure coefficient a; every factor a cancels before coefficients are used. Write

$$\begin{aligned} &b_2 = 8\widetilde{c}_2 e^2/(R\widetilde{c}_3), && b_b = 8\widetilde{c}_b e^2/(R\widetilde{c}_3), \\ &b_3 = 8/T, && b_5 = 8\widetilde{c}_5/(T\widetilde{c}_3), \\ &L = Ar/a = 2\eta^2 c_4/(Q\widetilde{c}_3), && d_0 = d/a = 2\eta\widetilde{c}_d/\widetilde{c}_3, \\ &A_0 = A_*/a = LH/(1 + rH), && \alpha_0 = \alpha/a = 2\varepsilon^4 j^6 Q^2 \eta^2 \widetilde{c}_\alpha/(K^2 \widetilde{c}_3). \end{aligned} \tag{SE3}$$

Every quantity in (SE3) is bounded with unit denominators, and b_2, b_b contain ε^2. Use the original normals N, M_B, put $p = t/R$, $h = a/T$, and write

$$\begin{aligned} &q_A = vD_u w, \qquad q_B = v(D_u - uN - puD_u vw), \\ &W = w + u(M_B + hq_B)/D_u, \qquad Y = uw - (M_B + hq_B)/D_u. \end{aligned}$$

The entire bounded remainder is

$$\begin{aligned} G_s = {} & b_2\{N^2 - 2q_A N + (1 - p)q_A^2\} + b_b D_v\{N^2 + 2q_A N + pq_A^2\} \\ & + b_3 D_v\{M_B^2 + 2q_B M_B + hq_B^2\} + b_5\{M_B^2 - 2q_B M_B + (1 - h)q_B^2\} \\ & + 4A_0 D_u D_v (Y - ruW)^2 + 4LD_u D_v (2uWY - ru^2 W^2) \\ & + 4\alpha_0 D_u (vY - 1)^2 + 2d_0 D_u\{(vY - 1)^2 - D_v\} + \nu D_v N^2, \qquad \nu = d_0(R - t)/R. \end{aligned} \tag{SE4}$$

This is (SD4), after the indicated exact centered subtraction and normalization, except that a positive part of the original Π_2 is kept inside the remainder. Indeed the original coefficient of $D_v N^2$ in Π_2/a is

$$\frac{4\beta + 2d}{a}\frac{R - t}{R} \;\geq\; 2\nu.$$

Thus subtracting only $\Pi_2/a - \nu D_v N^2$ remains legitimate, even when $\beta = 0$. The other three original terms are retained unchanged. All full first normals and every hard quadratic term are present in (SE4); there are no terms of higher normal degree.

A conductor that never divides by ε. Define the separated monomials and differences

$$A = \varepsilon wv, \quad B = v, \quad C = Euw, \quad D_0 = \varepsilon^2 j^3 E, \quad k = \varepsilon j,$$

$$x = B - C, \quad y = A - kC, \quad z = uB - kC, \quad h_0 = kC - D_0.$$

The two standard radical pairings are obtained by dividing the original four squared circuit monomials by δ: they are A, uB, kC, D_0, and their product is $(k^2BC)^2$. Their strict circuit is $T_4 = y^2 + z^2 + h_0^2 - 2yz + 2q_1^2$. The exact contact is $(u, w, v) = (k, j, \varepsilon j^2 E)$. Besides the three basic conductor identities,

$$C(u-k) = z - ux, \quad \varepsilon C(w-j) = y - \varepsilon wx, \quad C(v - \varepsilon j^2 E) = E(wz + jy - kwx), \tag{SE5}$$

use the additional, division-free identity

$$Cv(w-j) = Ew(w-j)z + Ejwy - Ekw^2x. \tag{SE6}$$

For a monic contact division $S = (u-k)T_u + (w-j)T_w + (v - \varepsilon j^2 E)T_v$, whenever $T_w = \varepsilon U + vV$, the bounded conductor is

$$\begin{aligned} CS = x[&-uT_u - \varepsilon wU - Ekw^2V - EkwT_v] \\ &+ y[U + EjwV + EjT_v] + z[T_u + Ew(w-j)V + EwT_v]. \end{aligned} \tag{SE7}$$

No division by $\varepsilon, j, k, r, a, t$, or a contact difference occurs.

To apply (SE7), first express the scalar in the four raw monomials. At zero normals let $V_0 = 1 - puvw$. Then

$$q_A = D_u wv, \quad q_B = D_u vV_0, \quad uW = C/E + u^2 hvV_0, \quad Y = C/E - hvV_0.$$

The scalar of (SE4) therefore has an exact raw allocation

$$f_A A^2 + f_{BB} B^2 + f_{BC} BC + f_{CC} C^2 + \rho D_0^2. \tag{SE8}$$

Expand only the nonconstant part of $(vY - 1)^2$ into B, C; its constant has $\alpha_0/D_0^2 = 2\widetilde{c}_\alpha/\widetilde{c}_3$. The coefficients in (SE8) have unit denominators independent of w. Every dependence on w in them comes through V_0, and hence has a factor ε^2; modulo ε they are independent of w and even in u, v. Both complete first normals are in (A, B, C) with bounded coefficients by the same exact formulas.

Write $\mathcal{L}_x(S), \mathcal{L}_y(S), \mathcal{L}_z(S)$ for the three coefficients produced by (SE7), using monic divisions in the order u, w, v. Substitution of $A = y + kC$, $B = x + C$, and $D_0 = kC - h_0$ leaves the scalar coefficient

$$S = k^2 f_A + f_{BB} + f_{BC} + f_{CC} + k^2 \rho.$$

It vanishes at contact, and $T_w(S) \in (\varepsilon^2)$, so (SE7) applies with $U = T_w(S)/\varepsilon$ and $V = 0$. Let

$$H = (2f_{BB} + f_{BC} + \mathcal{L}_x(S),\ 2kf_A + \mathcal{L}_y(S),\ \mathcal{L}_z(S),\ -2k\rho), \qquad J = H - 2k\rho_c(-2k, 1, 1, -1),$$

where the subscript c denotes exact contact evaluation. The native contact jets and the identity $kC(y + z - h_0 - 2kx) = q_1^2 - yz$ give $J_i|_c = 0$ for all four coefficients. The factor k needed for this identity comes from the original ρD_0^2 term. The contact Jacobian of (x, y, z) has determinant $-E^2\varepsilon^3 j^4 \neq 0$; its independence proves these vanishing statements without introducing its inverse into a coefficient.

It remains to check the second conductor. In this paragraph a bar means reduction modulo ε only. Write $\bar{S} = s(u^2, v^2)$ and put

$$p_s(U, V) = \frac{s(U, V) - s(0, V)}{U}, \qquad q_s(V) = \frac{s(0, V)}{V}.$$

These are polynomial quotients: parity gives the first, and contact gives $s(0, 0) = 0$. Direct substitution into (SE7) yields

$$\bar{J}_x = \overline{2f_{BB} + f_{BC}} - u^2 p_s(u^2, v^2), \quad \bar{J}_y = Ejvq_s(v^2), \quad \bar{J}_z = up_s(u^2, v^2) + Ewvq_s(v^2), \quad \bar{J}_h = 0.$$

The ordered w division first sets $u = k$, which is zero modulo ε. Consequently

$$(\overline{T_w(J_x)}, \overline{T_w(J_y)}, \overline{T_w(J_z)}, \overline{T_w(J_h)}) = (0, 0, Evq_s(v^2), 0).$$

Every second-stage T_w therefore belongs to (ε, v), as required by (SE7), with bounded polynomial coquotients. The complete normal raw coefficients are independent of w modulo ε; their single conductor has $T_w \in (\varepsilon)$. Thus both scalar divisions and both full normal lifts stay in the bounded coefficient ring. All denominators used here are the separated units already displayed in (SE3)–(SE4).

Finite residues of that full lift. Give $(\varepsilon, j, u, w, v, N, M_B)$ weights $(1, 1, 2, 1, 3, 5, 3)$; E, ℓ, τ have weight zero. Every positive-weight variable is infinitesimal, while all weight-zero coefficients are bounded. The contact substitutions preserve the grading; x has weight three and the other circuit roots weight five. Both (SE5) and (SE6) preserve weights, so after the scalar collection through weight ten and full first normals through weight five, every lifted remainder coefficient has positive weight. Restore the unit denominators afterward.

Set $H_0 = E + \ell$ and $V_* = 1 + \ell H_0$. Direct exact multiplication of (SE1)–(SE4), followed by (SE7), gives scalar residue

$$\frac{8(E+2)}{H_0 V_*} x^2 + 8T_4, \tag{SE9}$$

and complete first-normal residues

$$L_N = 0, \qquad L_M = \frac{16E(H_0+1)}{H_0 V_*}\, x.$$

The hard block has residue

$$\operatorname{diag}\left(\frac{4E}{H_0}, \frac{8(E^2+2E\ell+2\ell^2+2)}{V_*}\right). \tag{SE10}$$

These statements include coefficient corrections of positive weight; they are not asserted as literal equalities of the full rational scalar.

The only hard-soft coupled determinant is

$$\det G_{M,x} = \frac{64(E^2+2E\ell+4E+4\ell)}{H_0^2 V_*} > 0. \tag{SE11}$$

Together with $4E/H_0 > 0$ and the strict two-pairing circuit this gives the full bounded strict eight-root Gram. All original positive terms, the centered square, and the positive pure-coefficient factor a are then restored. All inverses and radicals remain separated between the two independent real closed fields.

B.4.19 Unit η and infinitesimal e: complementary tests

Lemma B.20. *Use the unchanged physical native C1 source with $c = 2$, $\lambda = n = 0$, positive original openings, and nonnegative original event coefficients. Suppose $k, \zeta, e = k\sqrt{\eta/\zeta}$ are positive infinitesimals, η is a positive bounded unit, and $r = 1 - a\eta > 0$. Then every independent projective test and state is tensor SOS except possibly the chart*

$$u, v, w > 0 \text{ infinitesimal}, \qquad Z = 1, \qquad w = (W + uY)/(1+u^2). \tag{SEO1}$$

There is no restriction on the positive relative scales of a, r, J or on k^2/r. This lemma supplies a join, not a certificate for the remaining chart (SEO1).

The small weak-event scale. Put $\kappa = k^2$, $K = 1 + \kappa$, $H = r + \eta$, $V_c = \kappa\eta\zeta$, $S = 1 - \kappa\eta + tV_c$, $R = 1 + \kappa r$, $T = K - \kappa\zeta t$, $M = R - t$, and $J = T - a$. Let P_4, P_d be the short native polynomials printed in the centered-A outer lemma. All of K, H, S, P_4, P_d are positive bounded units. Set

$$L = KHS, \quad N = KH(1+V_c) - J\eta P_d, \quad d_0 = 4\eta P_d, \quad A_0 = 2\eta^2 P_4, \quad D_0 = 2N.$$

After exact cancellation of the native r factors, the source is

$$d = \frac{ad_0}{L}, \qquad A = \frac{aA_0}{rL}, \qquad D = \frac{aD_0}{JL}, \qquad \frac{m}{ae^2} = \frac{2(1-t\zeta)(1+V_c)}{MS}. \tag{SEO2}$$

Here $\bar{N} = 1$. Native $\beta \geq 0$ gives the exact bound

$$\frac{t}{e^2} \leq \frac{2c_b}{\eta c_d} = \frac{K[1+(1-t)\zeta]H}{\eta P_d}.$$

Thus t is infinitesimal, $M \sim 1$, and $m/(ae^2) \sim 2$. In particular

$$\frac{k^2}{e^2} = \frac{\zeta}{\eta} \quad \text{is infinitesimal.} \tag{SEO3}$$

The ratios $d/a, A/a, D/a$ have positive real lower bounds, while d/a is a bounded unit. These statements allow arbitrary positive opening rates and literal $\beta = 0$.

Choose the critical parameter without a small denominator. There are two source cases. If a is a bounded unit, keep A, D, d and replace $c = 2$ by the critical number

$$c_* = \frac{d(d+2A)(d+2D)}{2(4AD - d^2)}. \tag{SEO4}$$

This is positive and bounded, and $c_* - 2 = O(k^2)$. Here is an exact short verification of the potentially delicate bound. Define

$$Q = 4A_0D_0 - rJd_0^2,$$

so $\bar{Q} = 16\bar{\eta}^2\bar{a}\bar{H} > 0$. Then

$$c_* - 2 = \frac{ad_0(d_0r + 2A_0)(d_0J + 2D_0) - 4QL}{2QL}. \tag{SEO5}$$

After $r = 1 - a\eta$ is inserted, its polynomial numerator vanishes identically at $\kappa = 0$, with ζ still free. It is therefore κ times a polynomial in bounded native variables; its denominator is a positive unit. No inverse of r or J occurs.

If a is infinitesimal, then $r, J \sim 1$. Divide the whole parent by a and replace the original A by

$$A_* = \frac{d^2(4 + d + 2D)}{2(8D - 2dD - d^2)}. \tag{SEO6}$$

The denominator is positive. In this case $A_*/a, D/a, d/a$ are positive bounded units, $c/a = 2/a$ is unbounded, and $(A_* - A)/a = O(k^2)$. To verify this without a rate comparison, put

$$B_0 = 8D_0L - a(2d_0D_0 + d_0^2J).$$

Its residue is $16\bar{H} > 0$, and

$$\frac{A_* - A}{a} = \frac{rd_0^2\{4JL + a(d_0J + 2D_0)\} - 2A_0B_0}{2rLB_0}. \tag{SEO7}$$

Again the numerator vanishes identically at $\kappa = 0$, so it has an exact polynomial factor κ, with a unit denominator.

Use $\sigma = 1$ in the first case and $\sigma = a$ in the second, and henceforth normalize all parent coefficients by σ. Let ϵ be the positive part of the critical increase in the chosen normalized coefficient. From (SEO2)–(SEO7),

$$\epsilon \text{ is infinitesimal}, \qquad \frac{\epsilon}{m/\sigma} = O(\zeta) \text{ is infinitesimal}. \tag{SEO8}$$

If the chosen critical value is smaller than the actual coefficient, retain the positive excess term instead. In either source case the chosen critical parent obeys

$$2c(4AD - d^2) = d(d + 2A)(d + 2D), \tag{SEO9}$$

with the letters in (SEO9) now denoting the normalized critical coefficients. In the first case c, d are units and A, D are bounded below by positive units; in the second A, D, d are units and c may be unbounded.

A critical-wall frame permitting the large coefficients. Write $D_u = 1 + u^2$, $D_v = 1 + v^2$ and

$$B = X + uZ, \quad U = uW, \quad V = vZ, \quad J_t = U - Y, \quad C = 4D + 2d,$$

and define the pure-coefficient numbers

$$y = \frac{d}{2A}, \quad b = \frac{4AD - d^2}{AC}, \quad s = 1 - b, \quad q = y + b.$$

The critical identity gives the useful exact relations

$$b = \frac{C}{4c + C}, \quad Cs = 4cb, \quad cb = \frac{d(2A + d)}{4A}. \tag{SEO10}$$

In both source cases, y, b, s, q are bounded and $b, s > 0$. The entire signed core plus the three critical terms is exactly

$$\begin{aligned} F_* = {} & 4cD_vJ_t^2 + 4AD_uD_vY^2 + C(J_t + uvB - D_uV)^2 + 2dD_vB^2 \\ & + 2dD_u\{v^2Y^2 - 2YV - V^2\}. \end{aligned} \tag{SEO11}$$

In the first source case use

$$H_A = \sqrt{A}(Y - yV), \qquad H_D = \sqrt{C}(J_t + uvB - (b + u^2)V). \tag{SEO12}$$

The inverse is $Y = yV + H_A/\sqrt{A}$ and $J_t = (b + u^2)V - uvB + H_D/\sqrt{C}$. All coefficients in (SEO11) after this substitution are bounded. Indeed the large A, C occur only in the normalized squares and in the bounded combinations Ay^2, $\sqrt{A}\,y$, Cs^2, $\sqrt{C}\,s$; use $Cs = 4cb$ and bounded c for the last two. At $u = v = 0$, the hard block on (B, H_A, H_D) is

$$\operatorname{diag}(2d, 4, 1 + 4c/C) \succ 0.$$

On its exact zero section the scalar coefficient of V^2 is

$$gu^2+\gamma v^2+4cu^4+g(y+2)u^2v^2+4cu^4v^2,\quad g=\frac{d(2A+d)}{A},\quad \gamma=\frac{Dd(2A+d)^2}{A^2(2D+d)}. \tag{SEO13}$$

Both g,γ are positive bounded units. Every hard-V entry has angular degree at least two, with bounded coefficients; this is an exact full-parent polynomial statement. Allocate it on uV, vV, leaving a factor u or v in its coefficient. The resulting exact five-root Gram on (B, H_A, H_D, uV, vV) therefore has strict residue.

In the second source case set $L_c = 4c + C$ and instead use

$$H_A=\sqrt{A}(Y-yV),\qquad H_c=\sqrt{L_c}(J_t-bV). \tag{SEO14}$$

Its coefficients are bounded because A, D, d are units and c/L_c, $cb/\sqrt{L_c}$ and cb^2 are bounded by (SEO10). Its hard block is $\mathrm{diag}(2d, 4, 1)$. The exact zero-section scalar is

$$gu^2+\gamma v^2+Cu^4+gy\,u^2v^2. \tag{SEO15}$$

The same complete hard-V allocation gives a strict five-root Gram on (B, H_A, H_c, uV, vV). Thus this is a full critical-wall certificate in both source cases, even with the stated unbounded coefficients. Every radical and denominator used in the roots is pure coefficient; no mixed inverse is used.

All unit-W states at the small-angle corner. Normalize $W = 1$, with Z, Y, X bounded, and take u, v infinitesimal. Retain the complete original α, β terms. Put

$$R_m=v-X+uvY-uZ=v(1+uY)-B,\qquad V=\frac{Z}{1+uY}(R_m+B). \tag{SEO16}$$

The displayed denominator is a positive pure-testing unit. In the first source case the only negative correction to the critical parent is $-4\epsilon D_v J_t^2$; (SEO12) and (SEO16) express $\sqrt{\epsilon}J_t$ as a bounded combination of the five critical roots and $\sqrt{m/\sigma}\,R_m$, with every coefficient having zero residue. In the second case the only negative correction is $-4\epsilon D_u D_v Y^2$; use $Y = yV + H_A/\sqrt{A}$ and (SEO16) for the identical conclusion. The only possibly delicate coefficient is a bounded testing multiple of $\sqrt{\epsilon/(m/\sigma)}$, and it is infinitesimal by (SEO8). The extra original m square contributes the strict diagonal 4. Hence the complete perturbed Gram remains strict. This includes unit or infinitesimal Z, literal $Z = 0$, and both signs of the bounded testing state. It uses the entire original weak m event, without a false positive-unit lower bound on m.

Finite outer angles without the weak event. The same critical parent also gives the two finite outer charts. In (SEO11) regard B, Y, J_t, V as independent roots. Exactly,

$$F_*(0,v)=F_*(0,0)+v^2\{2dB^2+(4A+2d)Y^2+4cJ_t^2\}. \tag{SEO17}$$

The zero-angle form has three positive hard pivots and null ray $(B, Y, J_t, V) = (0, y, b, 1)V$. On this null ray the additional coefficient is $\gamma v^2 > 0$. The normalized hard frames (SEO12) or (SEO14) keep every coefficient bounded. Thus when u is infinitesimal and v a bounded unit the full four-root residue is strict. The infinitesimal critical-parameter deficit is absorbed directly on these four roots; retain the entire m, α, β terms.

For bounded-unit u and infinitesimal v, set $V' = D_u V$. The other exact identity is

$$F_*(u,0)=F_*(0,0;V')+u^2\{4AY^2+2d(V')^2/D_u\}. \tag{SEO18}$$

Use the same normalized critical roots with V' in place of V. In the first source case include uvB in the H_D root so that the unbounded D event remains isolated for nonzero small v. The extra null mass in (SEO18) is positive in residue because d and u are units. The bounded full four-root Gram is strict, and the deficit is again a zero-residue correction. These two identities include every state, not merely the normal restriction.

For completeness the same argument covers both finite angles of nonzero residue, without a separate physical-source certificate. The exact determinant of (SEO11) on (B, Y, J_t, V) is

$$64cdD_u^2D_v^2\{A(2D+d)u^2+D(2A+d)v^2\}. \tag{SEO20}$$

The determinant in the normalized frame (SEO12) is this number divided by AC, namely

$$32cdD_u^2D_v^2\left\{u^2+\frac{D(2A+d)}{A(2D+d)}v^2\right\}.$$

It is a positive unit whenever at least one finite angle has nonzero residue. The hard restriction $V = 0$ is explicitly

$$\left(4D_uD_v+\frac{2dD_uv^2}{A}\right)H_A^2+H_D^2+4cD_v(H_D/\sqrt{C}-uvB)^2+2dD_vB^2,$$

which has a strict bounded residue. Hence its full residue Gram is positive definite by (SEO20). In frame (SEO14), the determinant is (SEO20) divided by $A(4c+C)$. Here A, D, d are units and $c/(4c+C) \sim 1/4$, so it too is a positive unit. Its hard restriction is

$$\left(4D_uD_v + \frac{2dD_uv^2}{A}\right)H_A^2 + \frac{4cD_v}{4c+C}H_c^2 + C(H_c/\sqrt{4c+C} + uvB)^2 + 2dD_vB^2,$$

again strict. All entries of both full transformed Grams are bounded as already proved. The infinitesimal deficit is a zero-residue correction on these four roots. This settles every finite angle pair outside the double-infinitesimal corner.

Projective infinity charts. For these charts normalize by a, clip the positive coefficients $c/a, A/a, D/a$ at suitable positive real lower bounds, and retain their positive original excess squares. Retain the whole original m, α, β terms. The normalized d/a is a unit. All the infinity-infinity and infinity-zero four-root identities in (BE2) and the subsequent outer routing apply without m. For $u = \infty$ in residue and v a bounded unit, the signed-core residue itself retains $2d\{(W+vX)^2 + (vY-Z)^2\}$; together with cW^2, AY^2 it is strict, so no $m(vY-Z)^2$ lower bound is needed. For $v = \infty$ and u a bounded unit the old signed-core and c, D root argument is unchanged. At the remaining $(u, 1/v) = (0,0)$ corner, replace the obsolete state-W mass argument by the exact root ideal $(X, Y, U = uW, Z)^2$ after retaining m. In $q^2F(u, 1/q)$ its complete selected residue is

$$2dX^2 + 4c(U-Y)^2 + (4A+2d)Y^2 + 4DZ^2 \succ 0. \tag{SEO19}$$

Every remaining occurrence of W carries u, so this is a full bounded Gram including all states. Literal infinity is included by these homogeneous identities. Literal zero axes have their unchanged original-event SOS. Finite tests with both noncommuting residues were included directly in (SEO20).

The residual state chart. Only u, v infinitesimal remain. Normalize the state by a maximal coordinate. On a negative pure-testing-core branch, X, Y have zero residue. The unit-W chart was just proved for every bounded Z. Otherwise Z is a unit, so normalize $Z = 1$ and W is infinitesimal. Then $w = (W + uY)/D_u$ is infinitesimal. The inherited exact testing Schur identity and sign gauges give (SEO1). Complex states split into their real and imaginary evaluations.

B.5 Nonvanishing $L = x\zeta$

For positive-unit or unbounded L, the reciprocal $z = 1/L$ is bounded. When η is unbounded, the uniform completion below combines two nonnegative source weights through their sum and covers every relative size of h and the infinitesimal $\sigma = 1/\eta$. The other source and testing domains retain their separate completions in complementary coordinates.

B.5.1 Unbounded L and unit η

This section proves this full original-parent subdomain, retaining every state normal and every independent source/testing rate. It makes no claim for infinitesimal σ. The common common remote theorem leaves u bounded and $w = 1/v$ infinitesimal; both u infinitesimal and u a bounded unit are covered below. All calculations use the exact NC source, with $d = 1$, $n = \lambda = 0$, and its original nonnegative D, β faces.

Source chart and bounded products. Put $x = k^2$, $z = q^2 = L^{-1}$, $r = bz$, $Q = 1-b$, $\theta = 1-h$, $K = 1+x$, $R = 1+xbz$, $M = R - xz\theta$, and

$$J = 1 + xh - \sigma(1-bz), \qquad E = 1 - xh/\sigma.$$

Here x, z are positive infinitesimals, $0 < b < 1$, and σ is a positive bounded unit. The NC source gives exactly

$$\begin{aligned}
P &= \sigma + K + \sigma z(1 + bx + 2xh) + \sigma bx^2hz^2 + x^2h\theta z,\\
V_0 &= z(1+xh)E + (1-bz)(1+xhz),\\
C &= cz, & c &= \frac{K(1+\sigma bz)E}{2(1-bz)P},\\
m &= \mu x^2 z, & \mu &= \frac{K(1+\sigma bz)h(1+\sigma z)}{2\sigma MP},\\
A &= a/z, & a &= \frac{V_0}{2bP},
\end{aligned}$$

$$\alpha=\ell/z,\qquad \ell=\frac{(1+\sigma bz)[(x/\sigma)Rh(1+\sigma z)+ME]}{2QP},$$
$$D=\frac{K(1+\sigma z)(1+\sigma bz)}{2JP}-\frac12,\qquad \beta=\frac{K(1+\sigma bz)(1+xhz)}{2\theta P}-\frac12.$$

In particular $c,\mu,ab,\ell Q,DJ,\beta\theta$ are bounded, $c,\mu,ab,\ell Q,DJ$ are positive units, and $ab-\ell Q\in z\mathcal{O}$. At $x=z=0$ their residues are

$$c_0=\frac{1}{2(1+\sigma)},\quad \mu_0=\frac{h}{2\sigma(1+\sigma)},\quad (ab)_0=(\ell Q)_0=\frac{1}{2(1+\sigma)},$$
$$(DJ)_0=\frac{\sigma^2}{2(1+\sigma)},\quad (\beta\theta)_0=\frac{(1+\sigma)h-\sigma}{2(1+\sigma)}.$$

Source nonnegativity gives $0<\sigma\le 1$ and $h\ge\sigma/(1+\sigma)$ at residue. Thus h is a positive unit. The openings b,Q,J,θ remain independent; $\beta=0$ is retained. The face $D=0$ cannot occur here since DJ is a positive unit.

Write $D_u=1+u^2$, $D_w=1+w^2$ and

$$M_w=W-wX+uY-uwZ,\quad R_w=uW-Y,\quad J_w=uwW+uX-wY-Z,\quad B_w=X+uZ,$$
$$K_w=J_w^2+D_u(Y-wZ)^2+D_wB_w^2-D_uD_wZ^2.$$

The complete reciprocal parent is

$$F_w=4mM_w^2+4czD_wR_w^2+4(a/z)D_uD_wY^2+4DJ_w^2+4(\ell/z)D_u(Y-wZ)^2+4\beta D_wB_w^2+2K_w.$$

Choose bounded positive-unit caps $a_*\le a$, $\ell_*\le\ell$, $D_*\le D$, and a bounded nonnegative cap $\beta_*\le\beta$. Keep as nonnegative squares the four centered excesses

$$4(a-a_*)D_uD_wN_y^2,\quad 4(\ell-\ell_*)D_uN_y^2,$$
$$4(D-D_*)(J_w+JZ)^2,\quad 4(\beta-\beta_*)D_wN_x^2,$$
$$N_x=X+uhZ,\qquad N_y=(Y-bwZ)/q.$$

These roots vanish at the exact source contact

$$u=k,\quad w=q\sqrt{\sigma},\quad (W,X,Y,Z)=(\sqrt{\sigma}/(kq),-kh,bq\sqrt{\sigma},1).$$

Set $d_J=(D-D_*)J$ and $b_\theta=\beta\theta$. Their coefficients and the centered excesses preserve the original parent.

Bounded full-normal remainder. Define $G=wZ/q$, $R_h=G-uqW$ and

$$c_N=8D_u[abD_w-\ell Q],\qquad c_G=4D_u[ab^2D_w+\ell Q^2].$$

The A,α remainder is $4D_u(a_*D_w+\ell_*)N_y^2+c_NN_yG+c_GG^2$. Here $c_N\in(z,w^2)\mathcal{O}$ and c_G is a positive unit. Keep $c_GR_h^2+c_NN_yR_h$ in the bounded Gram and use $G^2-R_h^2=2uwWZ-u^2zW^2$. The rest is the bounded polynomial

$$\begin{aligned}G_0=&\,4D_u(a_*D_w+\ell_*)N_y^2+c_NuqWN_y+c_G(2uwWZ-u^2zW^2)\\&+4mM_w^2+4czD_wR_w^2+4D_*J_w^2-8d_JZJ_w-4d_JJZ^2\\&+4\beta_*D_wB_w^2+8(b_\theta-\beta_*\theta)uD_wZB_w-4(b_\theta-\beta_*\theta)u^2\theta D_wZ^2+2K_w.\end{aligned}$$

All its coefficients are bounded. Put $N_z=Z-uwW/\sigma$ and substitute $Z=N_z+uwW/\sigma$, $X=N_x-uhZ$, $Y=qN_y+bwZ$. For $t=u^2$, $\xi=w^2$, define

$$T_0=1+th+b\xi,\quad M_0=1+t(h-Q)\xi/\sigma,\quad R_f=1-b\xi/\sigma,\quad J_f=1-T_0/\sigma.$$

In the formulas below c_N,c_G mean their even-polynomial expressions. The exact expansion is

$$G_0=\phi W^2+wL_xWN_x+uqL_yWN_y+uwL_zWN_z+H(N_x,N_y,N_z),$$

where

$$\begin{aligned}\phi=&\,4\mu x^2zM_0^2+4cz(1+\xi)tR_f^2+c_Gt(2\xi/\sigma-z)\\&+4D_*t\xi J_f^2-8d_Jt\xi J_f/\sigma-4d_JJt\xi/\sigma^2+4b_\theta\theta(1+\xi)t^2\xi/\sigma^2\\&+2[t\xi J_f^2+(1+t)tQ^2\xi^2/\sigma^2+(1+\xi)t^2\theta^2\xi/\sigma^2-(1+t)(1+\xi)t\xi/\sigma^2],\\L_x=&\,-8\mu x^2zM_0+t[(8D_*+4)J_f-8d_J/\sigma+(8b_\theta+4\theta)(1+\xi)/\sigma],\\L_y=&\,8\mu x^2zM_0-8cz(1+\xi)R_f+c_N-(8D_*+4)\xi J_f+8d_J\xi/\sigma-4(1+t)Q\xi/\sigma,\\L_z=&\,8\mu x^2z(h-Q)M_0-8cz(1+\xi)bR_f+2c_G-(8D_*+4)T_0J_f-8d_JJ_f+8d_J(T_0-J)/\sigma\\&+8b_\theta\theta(1+\xi)t/\sigma+4(1+t)Q^2\xi/\sigma+4(1+\xi)t\theta^2/\sigma-4(1+t)(1+\xi)/\sigma.\end{aligned}$$

For a compact explicit specification of the entire quadratic block, put

$$m_N = -wN_x + uqN_y + uw(h-Q)N_z, \quad r_N = -qN_y - bwN_z,$$
$$j_N = uN_x - wqN_y - T_0N_z, \quad b_N = N_x + u\theta N_z.$$

Then every quadratic normal coefficient is retained by the finite formula

$$\begin{aligned} H = {} & 4D_u(a_*D_w + \ell_*)N_y^2 + 4mm_N^2 + 4czD_wr_N^2 + (4D_* + 2)j_N^2 - 8d_JN_zj_N - 4d_JJN_z^2 \\ & + (4\beta_* + 2)D_wb_N^2 + 8(b_\theta - \beta_*\theta)uD_wN_zb_N - 4(b_\theta - \beta_*\theta)u^2\theta D_wN_z^2 \\ & + 2D_u(qN_y - QwN_z)^2 - 2D_uD_wN_z^2. \end{aligned}$$

The exact source substitutions give, at $(t,\xi) = (x,\sigma z)$,

$$\phi = \phi_t = \phi_\xi = L_x = L_y = L_z = 0.$$

Moreover $\phi(0,\xi) = 4\mu x^2 z$, $L_x(0,\xi) = -8\mu x^2 z$, and $\phi(t,\sigma z) \in z\mathcal{O}[t]$.

Exact six-root lift in the critical strip u infinitesimal. Monic division, first by $\xi - \sigma z$ and then by $t - x$, gives bounded polynomial coefficients satisfying

$$\begin{aligned} \phi &= z(t-x)^2A + t(t-x)(\xi-\sigma z)B + t(\xi-\sigma z)^2C, \\ L_x &= (t-x)(tA_x + zB_x) + t(\xi-\sigma z)C_x, \\ L_y &= (t-x)A_y + (\xi-\sigma z)B_y, \\ L_z &= (t-x)A_z + (\xi-\sigma z)B_z. \end{aligned}$$

For example, the constant coefficient of $L_x(t,\sigma z)/(t-x)$ equals $8\mu xz$, giving its (t,z) support. No small or mixed inverse enters these monic quotients. Also $A(0) = 4\mu$. Take the six finite separated roots

$$N_x, \quad N_y, \quad N_z, \quad R_h, \quad A_\theta = q(t-x)W, \quad V = uw(t-x)W.$$

The complete scalar lift follows from

$$\begin{aligned} z(t-x)^2W^2 &= A_\theta^2, \\ t(t-x)(\xi-\sigma z)W^2 &= u\sigma A_\theta R_h - \sigma VN_z, \\ t(\xi-\sigma z)^2W^2 &= \sigma^2(qR_h - wN_z)^2. \end{aligned}$$

All first normals lift by the following finite list, which includes their whole coefficient support:

$$\begin{aligned} wt(t-x)W &= uV, & wz(t-x)W &= qwA_\theta, \\ wt(\xi-\sigma z)W &= u\sigma qwR_h - u\sigma w^2N_z, & & \\ uq(t-x)W &= uA_\theta, & uq(\xi-\sigma z)W &= \sigma zR_h - \sigma qwN_z, \\ uw(t-x)W &= V, & uw(\xi-\sigma z)W &= \sigma qwR_h - \sigma w^2N_z. \end{aligned}$$

Together with the displayed H these give an exact bounded six-root Gram, not merely a restriction to the contact line. At residue its first five diagonal pivots are respectively

$$2 + 4\beta_*, \quad 4(a_* + \ell_*), \quad 4[D_* + d_J(1+\sigma)], \quad \frac{2}{1+\sigma}, \quad 4\mu_0.$$

They are positive units through all allowed openings. The only potentially nonzero remaining cross entry is N_zV. The exact relation

$$V^2 = \sigma tA_\theta^2 + u\sigma(t-x)A_\theta R_h - \sigma(t-x)VN_z$$

introduces an arbitrarily large finite positive V^2 pivot with only infinitesimal changes to all other entries. Its real Schur complement therefore makes the residue strictly positive definite. The exact bounded Gram is positive definite by the standard finite-dimensional local criterion. All excess squares are nonnegative with separated coefficients.

Away from the angular contact: u a bounded unit. The same source chart has a positive-unit angular gap at every such real residue. A convenient raw finite frame is $E_t = qW, F_t = Y/q, G_t = wZ/q, R_t = wW, X, Z$; products obey $E_tG_t = R_tZ$. The residue parent divided by $2/(1+\sigma)$ is exactly

$$u^2E_t^2 + D_u\{F_t^2/b + (F_t - G_t)^2/Q\} + \frac{(uR_t + uX - Z)^2}{J_0} + \frac{(X+uZ)^2}{\theta} - (1+\sigma)D_uZ^2, \quad J_0 = 1-\sigma.$$

Eliminate the positive F_t and X directions and allocate the common product between the remaining two quadratic blocks. Up to a common positive factor the two blocks have diagonal products AB, CD_0 and total off-diagonal magnitude E_0, with $U = u^2 > 0$ and

$$\delta = 1 - \sigma + \theta U, \qquad A = U, \quad B = 1+U, \quad C = U/\delta,$$
$$D_0 = (1+U)[(1+U)/\delta - (1+\sigma)], \quad E_0 = \sqrt{U}(1+U)/\delta.$$

Both diagonal products are strictly positive, since $1 + U - (1+\sigma)\delta = \sigma^2 + U[(1+\sigma)h - \sigma] > 0$. All square roots in this sentence are real-residue quantities used to choose real allocation constants, not mixed tensor radicals. The required strict inequality is $E_0 < \sqrt{AB} + \sqrt{CD_0}$. Set $N = U(2\sigma - \theta U)$. Directly,

$$E_0^2 - AB - CD_0 = (1+U)N/\delta, \quad N^2 - 4U^2[1 + U - (1+\sigma)\delta] = U^3(\theta^2 U - 4h).$$

If $N \le 0$ strictness is immediate. If $N > 0$, then $\theta^2 U < 2\sigma\theta \le 2h < 4h$, again giving strictness. The single-opening faces follow by their constrained limits. At the only double opening $J = \theta = 0$, so $\sigma = h = 1$, the exact limit is $E_0 - \sqrt{CD_0} \longrightarrow \sqrt{U}$, whereas $AB - U = U^2 > 0$. Here is a full-frame square completion which also justifies those cap limits. For clarity first take $u > 0$ and select a finite real ρ in the nonempty interval

$$\max\{-\sqrt{AB}, E_0 - \sqrt{CD_0}\} < \rho < \min\{\sqrt{AB}, E_0 + \sqrt{CD_0}\}.$$

Its nonemptiness follows from the strict allocation inequality above. Put

$$r_\rho = D_u/u - \rho\delta/U, \qquad s_\rho = D_u[2\rho/u - (1+\sigma)] - \rho^2\delta/U > 0.$$

The preceding complete residue form equals

$$UE_t^2 + D_uG_t^2 - 2\rho E_tG_t + \frac{D_u}{bQ}(F_t - bG_t)^2$$
$$+ \frac{\delta}{J_0\theta}\left[X + \frac{U\theta R_t + u(J_0 - \theta)Z}{\delta}\right]^2 + \frac{U}{\delta}(R_t - r_\rho Z)^2 + s_\rho Z^2.$$

The equality uses only $E_tG_t = R_tZ$ and is checked exactly. For $u < 0$ take the correspondingly signed ρ; all sign choices are real constants in this allocation. All triangular minimizer coefficients are bounded through every opening because U is a unit and $\delta = J_0 + U\theta$. Their positive hard weights stay bounded below. At the double face choose $|u| < |\rho| < \sqrt{U(1+U)}$, with matching sign; then s_ρ has a positive-unit limit. This proves uniform coercivity of the constrained residue, not just determinant positivity.

For completeness the exact centered cap rows in the raw frame are

$$F_t - bG_t, \quad uR_t + uX - qwF_t + (J-1)Z, \quad X + uhZ.$$

Only one of b, Q can tend to zero. The other two rows have hard parts on (R_t, X) with determinant u, a unit; each single row has a unit hard entry. Their simultaneous kernel is solved exactly by

$$F_t = bG_t, \qquad X = -uhZ, \qquad R_t = \{(u^2h - J + 1)Z + qwbG_t\}/u.$$

More generally, prescribed changes (e_F, e_D, e_β) in these three rows are produced by $\Delta F_t = e_F$, $\Delta X = e_\beta$, $\Delta R_t = (e_D - ue_\beta + qwe_F)/u$. This is a bounded separated right inverse: its only inversion is the pure-testing unit u. For the finite-cap existence argument fix u at its ordinary real residue, so the same inverse is an ordinary real finite-dimensional right inverse. No order on the tensor product is used. It lifts the limiting constrained kernel to the exact centered kernel while keeping every opening independent. The bounded residual form restricted to that kernel is the strictly positive constrained form just computed. An elementary block Schur complement then provides finite sufficiently large caps on every divergent row; bounded rows retain their full coefficients. This is the finite-cap lemma used in the UH proof. It yields a strictly positive bounded full-frame Gram. Exact coefficients differ from their residues infinitesimally, completing this strip. All divisions by δ and radicals in this paragraph belong only to the ordinary real residue calculation and its face limits. The exact tensor Gram uses the polynomial raw frame, real allocation constants, and finite source caps; it introduces no mixed inverse or radical.

Consequently this section closes the L unbounded, σ bounded-unit subdomain of C1.4 under the original parent, including all state normals, all independent opening rates, and $\beta = 0$. It does not close the σ infinitesimal subdomain or claim a whole-row count change.

B.5.2 Unit L and infinitesimal η: C1 substitution in LII

Theorem B.21. *The entire original C1 source subdomain*

$$x = k^2 \in \mathrm{Inf}_{>0}, \qquad L = x\zeta \text{ a positive bounded unit}, \qquad \eta \in \mathrm{Inf}_{>0}$$

has a full-parent tensor-SOS certificate for every independent projective testing pair. All original openings and the retained boundaries $D = 0$ or $\beta = 0$ remain allowed whenever they occur in this source subdomain.

The proof is an exact adapter to the strong-II local and remote certificates. Its additional event is retained with a centered square; it is not deleted from the parent.

The native source in the shared contact geometry. Keep the native notation of (NC1)–(NC4), but in this section denote its r, h, a by r_n, h_n, a_n to avoid collisions. Normalize $d = 1$. The exact native contact with $Z_c = 1$ is

$$(u_c, v_c; W_c, X_c, Y_c, Z_c) = \left(\sqrt{x}, \sqrt{x\eta\zeta}; \sqrt{\zeta/\eta}, -\sqrt{x}\, h_n, \sqrt{x}\, r_n \sqrt{\zeta/\eta}, 1\right).$$

It therefore has the same contact geometry used in (LII2), with

$$z = x, \quad y = L\eta, \quad B = L, \quad H = Lr_n = B - a_n y, \quad r = h_n = 1 - \theta, \quad t = a_n, \quad j = 1 + zr - t = J_n. \tag{LC1}$$

In particular z, y are positive infinitesimals. Here $r_n \sim 1$, $0 < Lr_n < 1$, and hence $b_0 := \operatorname{res} B$ satisfies $0 < b_0 \le 1$. We use b_0 only for this source residue.

Set, as in the local strong-II proof,

$$c = Cyt, \quad d_p = Dyj, \quad \mu = m/z^2, \quad a_A = A/y, \quad \sigma = \alpha(1 - H), \quad \tau = \beta(1 - r) = \beta\theta.$$

These are bounded source quantities, and c, d_p, μ, a_A, σ are positive units. Writing $r_0 = \operatorname{res} r$, exact substitution in (NC2)–(NC4), with opening factors canceled first, gives

$$\begin{aligned} c, d_p &\sim \frac{b_0}{2(1 + b_0)}, \qquad a_A \sim \frac{1}{2b_0(1 + b_0)}, \\ \mu &\sim \frac{r_0}{2b_0(1 + b_0)}, \qquad \sigma \sim \frac{b_0^2}{2(1 + b_0)}, \\ \operatorname{res}\tau &= \frac{b_0}{2(1 + b_0)} - \frac{1 - r_0}{2}. \end{aligned} \tag{LC2}$$

Since $\tau \ge 0$, the last identity implies $r_0 \ge 1/(1 + b_0) > 0$, proving the asserted unit lower bound for μ. Also $t + j = 1 + zr \sim 1$, and c, d_p being units proves C, D unbounded uniformly through either opening $t = 0$ or $j = 0$ in residue. Thus $D = 0$ is in fact absent from this specific subdomain by the exact source implications; it has not been excluded by an additional hypothesis. Literal $\beta = 0$ is included.

The source-harmonic quantity $yCD/(C + D)$ and the cap behavior of α have precisely the residues (RR1)–(RR2), with $h = b_0$. The remote proof (RR3)–(RR13) therefore applies unchanged, retaining the extra positive term $4\beta D_v(X + uZ)^2$ throughout. This covers every test outside the local region where u, v are both infinitesimal, including the original projective endpoints.

The additional event and its exact local remainder. For the local region use exactly the graph, roots, harmonic pair, and caps (LII2)–(LII4), with the source quantities (LC1)–(LC2). Put

$$R = v(U - z)W, \qquad E_\beta = \xi + u\zeta + R/B = (X + uZ) - \frac{z(1 - r)v}{B}W. \tag{LC3}$$

This root vanishes at the actual contact. Retain the entire positive term $4\beta D_v E_\beta^2$, in addition to the first, second and fourth terms of (LII5); there is no n term since $n = 0$. The exact remainder from the β event is

$$\begin{aligned} \Delta_\beta = {} & \frac{8\tau z}{B} D_v v W(\xi + u\zeta) \\ & + \frac{4\tau z}{B^2} D_v V\{2U - z(1 + r)\}W^2. \end{aligned} \tag{LC4}$$

All its coefficients are bounded even if β is unbounded: only $\tau = \beta(1 - r)$ occurs. The exact full remainder is (LII6) with its n line deleted, plus (LC4). Thus no state normal, boundary coefficient, or first contact jet has been omitted.

The R_hW gauge (LII7) is unchanged, because (LC4) contains no R_h. The coefficient representation (LII8), its exact contact and first-normal vanishings (LII9)–(LII10), and its bounded monic divisions (LII12)–(LII13) all remain valid. Explicitly, (LC4) adds $8\tau z(1+V)/B$ to P_ξ and P_ζ, adds zero to P_N, and adds its printed scalar line to ϕ. Hence the essential support $P_\xi \in (U, z)$ and $\deg_U \phi \le 2$ remain literal polynomial statements.

The only changed data in (LII11) are

$$\begin{aligned} d_s &= \frac{4z^2}{B^2}\left\{\mu(B+zr)^2 + \frac{r^2}{2} - \tau(1+r)\right\}, \\ e_s &= \frac{4z^2}{B^2}\left\{\frac{r^2}{2} - \tau(1+r)\right\}. \end{aligned} \tag{LC5}$$

Both quotients by z^2 are bounded; e_s need not be positive. Its contribution to the final Gram has an infinitesimal coefficient because it is multiplied by V. The decisive exact residue cancellation, using (LC2), is

$$\begin{aligned} \operatorname{res}\left\{\mu B^2 + \frac{r^2}{2} - \tau(1+r)\right\} &= \frac{b_0 r_0}{2(1+b_0)} + \frac{r_0^2}{2} - \left\{\frac{b_0}{2(1+b_0)} - \frac{1-r_0}{2}\right\}(1+r_0) \\ &= \frac{1}{2(1+b_0)}. \end{aligned} \tag{LC6}$$

Consequently

$$\operatorname{res}(d_s/z^2) = \frac{2}{b_0^2(1+b_0)}, \qquad \operatorname{res} A_\xi = \frac{4}{b_0(1+b_0)}. \tag{LC7}$$

The second equality is unchanged from (LII18): the new P_ξ term is a bounded multiple of z independent of U, so it has zero U-quotient residue. The four hard-normal residues (LII17) are unchanged as well.

Apply the exact allocations (LII15) to all the coefficients, including (LC4). They give the same bounded six-root Gram $(N, \xi, \zeta, R_h, R, T)$ as in (LII16). Its finite (ξ, R) block is exactly (LII19), with $h = b_0$, and has strictly positive Schur complement $2/[b_0(1+b_0)^2]$. The same exact conductor (LII20) supplies an arbitrarily large real T^2 pivot while changing every other residue entry by zero. The complete bounded Gram is strict. Restoring the four retained terms, including $4\beta D_v E_\beta^2$, proves the local tensor SOS for all independent rates and literal affine axes. The remote certificate above completes the theorem.

Source inequalities. Stationarity itself is the unchanged native source identity, not an added restriction. The result removes the whole L-unit, η-infinitesimal subdomain from C1.4. It does not assert the other L-unit source regimes or the entire C1.4 row.

B.5.3 Unit L, η: C1 substitution in LII

Theorem B.22. *For the original C1 source with $x \in \mathrm{Inf}_{>0}$ and both $L = x\zeta$ and η positive bounded units, the entire original parent is tensor-SOS for every independent projective testing pair. All original positive openings and literal $\beta = 0$ are included.*

The exact source formulas force D strong on this subdomain, so $D = 0$ is absent by source implication. No opening is removed by an extra hypothesis, and the proof does not invoke a table or atlas.

The source and the only testing strip left by four events. Use the exact shared geometry (LC1), and write $\rho = r_n$ for the native shape parameter and $r = 1 - \theta$ for the geometric one. Thus

$$B = L, \quad y = B\eta, \quad H = B\rho = B - yt, \quad t = (1-\rho)/\eta, \quad j = 1 + zr - t, \quad z = x \in \mathrm{Inf}_{>0}\,.$$

Here B, y, η are units. For the boundary coefficient formulas below set

$$P = 1 + B + y, \qquad P_4 = 1 + B(1-\rho), \qquad S = H + y = P - P_4. \tag{LU1}$$

All three quantities are positive units, uniformly through $\rho \to 0$, $1 - \rho \to 0$, $j \to 0$, and $1 - H \to 0$. The native formulas imply that D, C, A, α are strong. One forward cyclic relabeling sends these four consecutive events to C, A, α, m and sends testing coordinates to $(u', v') = (v, 1/u)$. The full-parent four-consecutive theorem (CE11)–(CE16) therefore covers every test except

$$u \in \mathrm{Inf}, \qquad v \text{ a bounded nonzero unit.} \tag{LU2}$$

The cycle is the inverse of (CE9)–(CE10), so its literal endpoint scope is the original homogeneous parent's scope.

It remains to prove (LU2). All roots and coefficients used below are valid for both testing signs, including literal $u = 0$.

A bounded full-parent remainder with every opening retained. Use the graph, harmonic roots and gauge (LII2)–(LII8), with $k = \sqrt{y}$ now a source unit. The bounded source data are

$$c = Cyt, \quad d_p = Dyj, \quad \mu = m/z^2, \quad \sigma = \alpha(1-H), \quad \tau = \beta(1-r), \quad A_H = AH.$$

They are bounded; c, d_p, μ, σ, A_H are units. In residue, with the boundary symbols of (LU1), their exact values are

$$c = \frac{S}{2P}, \quad d_p = \frac{BP_4}{2P}, \quad \mu = \frac{Sr(1+y)}{2B^2P}, \quad \sigma = \frac{BS}{2P}, \quad A_H = \frac{yP_4}{2P}, \quad \tau = \frac{S}{2P} - \frac{1-r}{2}. \tag{LU3}$$

These are residue identities, not substitutions of source and testing fields. They are obtained after multiplying away every possibly small opening in the exact native formulas. Since $\tau \geq 0$, $r \geq P_4/P$ in residue and is a positive unit. Also $t + j \sim 1$, so $\ell = y(C+D)$ has a positive lower bound, $q = D/(C+D)$ is bounded in $[0,1]$, and $g = yCD/(C+D)$ is a positive unit.

Choose positive bounded caps $\ell_* \leq \ell$, $A_* \leq A$, and $\alpha_* \leq \alpha$. Choose a bounded cap $0 \leq \beta_* \leq \beta$ as specified below. Retain the five contact-centered positive terms

$$\begin{gathered} 4(\ell-\ell_*)N^2, \qquad 4CV\eta_t^2, \qquad 4(A-A_*)D_uD_v\eta_t^2, \qquad 4(\alpha-\alpha_*)D_u(v\eta_t-\zeta)^2, \\ 4(\beta-\beta_*)D_v(\xi+u\zeta+R/B)^2, \qquad \eta_t = k(N+qR_h), \quad R = v(U-z)W. \end{gathered} \tag{LU4}$$

The exact remainder is (LII6), with its n line deleted, its A term replaced by

$$4A_*D_uD_v\eta_t^2 + \frac{8A_H}{B}D_uD_vuW\eta_t + \frac{4A_HH}{B^2}D_uD_vUW^2, \tag{LU5}$$

and with (LC4) plus $4\beta_*D_v(\xi+u\zeta+R/B)^2$ added. This specifies every coefficient of the full remainder. Only bounded source products occur, even at an unbounded original weight.

The same dependent-root gauge and exact contact equations give (LII9)–(LII13). The new constant scalar coefficients are

$$\frac{d_s + e_sV}{z^2} = \frac{4}{B^2}\left[\mu(B+zr)^2 + \left\{\frac{r^2}{2} - \tau(1+r) + \beta_*\right\}(1+V)\right]. \tag{LU6}$$

The support $P_\xi \in (U,z)$ and scalar degree at most two in U remain exact. Apply all of (LII15). As v is a unit, eliminate T by the exact pure-factor identity

$$T = (uk/v)R. \tag{LU7}$$

Its coefficient is a bounded infinitesimal tensor. Thus the entire remainder has a bounded Gram on (N, ζ, R_h, ξ, R). No source/testing ratio or mixed inverse has been introduced. We now verify its complete residue.

A three-square identity supplies all three state/angle normals. The following boundary computation is an exact rational identity before taking any opening limit. Define

$$\begin{aligned} C_0 &= \frac{\rho+\eta}{2(1-\rho)P}, & A_0 &= \frac{\eta P_4}{2\rho P}, \\ D_0 &= \frac{P_4}{2(\eta-1+\rho)P}, & \alpha_0 &= \frac{B^2(\rho+\eta)}{2(1-B\rho)P}. \end{aligned}$$

These may have unbounded openings; only the final capped identity below is used as a bounded Gram. For an indeterminate p, put

$$e = Y - \rho p, \qquad f = Z - vp/B, \qquad T_v = (V-y)p.$$

The entire leading three-state form is exactly

$$\begin{aligned} &4C_0D_v(p-Y)^2 + 4A_0D_vY^2 + (4D_0+2)(p-Y-vZ)^2 + (4\alpha_0+2)(vY-Z)^2 - 2D_vZ^2 \\ &\quad = \frac{4D_0}{B^2}(T_v + Be + Bvf)^2 + 4\alpha_0(f-ve)^2 + 4(C_0+A_0+\tfrac{1}{2})D_ve^2. \end{aligned} \tag{LU8}$$

This identity holds for every v, not only near the actual Bob angle. In the contact allocation at $z = u = 0$, $e = k(N+qR_h)$ and $T_v = -B(kR_h + v\zeta)$. Subtract the residue contributions of the retained terms (LU4). The harmonic identity then gives the complete three-normal block

$$\boxed{4\ell_*N^2 + 4gR_h^2 + 4(A_* + \tfrac{1}{2})D_vy(N+qR_h)^2 + 4\alpha_*\{\zeta - vk(N+qR_h)\}^2.} \tag{LU9}$$

It has strictly positive residue on (N, ζ, R_h) because ℓ_*, g, α_* are positive units. No normal has been suppressed. Every rational cancellation used to obtain (LU9) has bounded source products on both sides, so the identity extends continuously through every opening listed after (LU1).

The remaining two-root block and a finite cap. There is no finite residue coupling between these three roots and (ξ, R). Such terms carry u, z, or the coefficient in (LU7). For completeness, the exact first-normal support gives this block without dropping a term. To first order in (U, z) write $P_\xi = UA(V) + zZ(V)$; both coefficient functions are affine in V, and

$$Z(V) = \frac{D_v}{B}(-4r + 8\tau - 8\beta_*).$$

The exact contact equation gives $A(y) + Z(y) = 0$ in residue. After the first allocation of (LII15), the coefficient of ξR is therefore $A(V) - (V - y)(A' + Z') = -Z(V)$. Using (LU3) and (LU6), the complete block is

$$M_{\xi R} = \begin{pmatrix} 2D_v(1 + 2\beta_*) & \frac{2D_v}{B}(P_4/P + 2\beta_*) \\ \frac{2D_v}{B}(P_4/P + 2\beta_*) & \frac{2}{B^2}\{(P_4/P + 2\beta_*)D_v - (S/P)r(V - y)\} \end{pmatrix}. \tag{LU10}$$

Here all letters denote real residues when positivity is assessed.

If β is bounded, take $\beta_* = \beta$. The native relation $(1 - r)(2\beta + 1) = S/P$ follows from (LU3), and (LU10) has determinant

$$\frac{4SD_v}{B^2P}(1 + 2\beta)r(1 + y) > 0. \tag{LU11}$$

Its first diagonal entry is positive, including $\beta = 0$. If β is unbounded, take β_* to be a sufficiently large positive real cap. Increasing it adds the positive square $4\beta_* D_v(\xi + R/B)^2$. On its kernel $\xi = -R/B$, the uncapped block equals

$$\frac{2S}{B^2P}\{1 + ry + (1 - r)V\}R^2 > 0. \tag{LU12}$$

The elementary real two-by-two Schur argument therefore supplies a finite cap making (LU10) strictly positive. Unboundedness of the original β permits that cap.

Together (LU9)–(LU12) prove strict positive residue of the full five-root Gram. The bounded Gram lemma gives its tensor SOS; restoring (LU4) proves (LU2). The cyclic four-event complement completes every testing chart and original projective endpoint.

Verification scope. The exact source contact and monic contact divisions are the unchanged full-parent argument of (LII7)–(LII15), with the bounded modifications (LU4)–(LU6). This theorem closes the L-unit, η-unit subdomain of C1.4; the η-unbounded subdomain is not included.

B.5.4 Large η with bounded reciprocal $x\zeta$: one proof for every h/σ

Let K_c, K_t be separately labelled real closed extensions of $\mathbb{R}$, $T = K_c \otimes_{\mathbb{R}} K_t$, and ΣT^2 the finite sums of squares in T. Bounded coefficients belong to $B_c \otimes B_t$ and their real residue is taken in both factors. We use the bounded-Gram fact: a bounded symmetric matrix with positive-definite real residue is a Gram matrix of a finite sum of squares in T. Roots in the Gram need not be bounded. A positive source or testing factor can be absorbed into a square using the real closedness of its own field.

Theorem B.23. *Use the stationary C1 source with $d = 1$, $\lambda_{\mathrm{ev}} = n = 0$, $m, C, A, \alpha > 0$, $D, \beta \geq 0$, and the native strict gaps. Assume explicitly that*

$$x = k^2 \in \mathrm{Inf}_{c,>0}, \qquad \sigma = 1/\eta \in \mathrm{Inf}_{c,>0}, \qquad z = 1/(x\zeta) > 0 \text{ is bounded.}$$

There is no size restriction on $H = h/\sigma$. Then the complete reciprocal parent is in ΣT^2 for every independent pair $u, w \in \mathrm{Inf}_t$ and every real or complex state. There is no state-ratio or testing-ratio restriction. The literal faces $D = 0$ and $\beta = 0$ are included.

Equivalently, the theorem applies to the rational source below under $0 < b < 1$, $0 < r = bz < 1$, $0 < h < 1$, $E, M, J > 0$, $D, \beta \geq 0$, and the stated assumptions on x, σ, z.

The source and its uniform estimates. The rational identities and complete parent expansions below are algebraic for every positive z with the native gaps. Only their boundedness estimates and the theorem require z bounded. Put

$$r = bz, \quad Q = 1 - b, \quad 0 < b, r < 1, \quad \theta = 1 - h, \quad e = xh/\sigma = xH, \quad E = 1 - e, \quad K = 1 + x,$$

$$R_s = 1 + xr, \quad M = R_s - xz\theta > 0, \quad J = 1 + xh - \sigma(1 - r) > 0,$$

$$P = \sigma + K + \sigma z(1 + bx + 2xh) + \sigma b x^2 h z^2 + x^2 h\theta z.$$

Define the opening-cleared products

$$p = \frac{z(1+xh)E + (1-r)(1+xhz)}{2P},$$
$$l = \frac{(1+\sigma r)\{eR_s(1+\sigma z) + ME\}}{2P}, \qquad \kappa = \frac{K(1+\sigma r)(1+\sigma z)}{2MP}.$$

The event coefficients are

$$m = \kappa x^2 z H, \quad C = \frac{Kz(1+\sigma r)E}{2(1-r)P}, \quad A = p/r, \quad \alpha = l/(zQ),$$
$$D = \frac{K(1+\sigma z)(1+\sigma r) - JP}{2JP}, \qquad \beta = \frac{K(1+\sigma r)(1+xhz) - \theta P}{2\theta P}. \tag{P1}$$

Only the original nonnegative D, β sources are considered. Set $a = Az = p/b$, $\ell = \alpha z = l/Q$, and

$$H_0 = 1 + zQ, \quad B_s = P + xz(J - \sigma),$$
$$\mathcal{H} = R_s H_0 + z\{(1-r)xh(2+xr) - e[2 + x(1+h+r\theta)] + Ke^2\}.$$

Direct multiplication gives the two exact source identities

$$2\theta P\beta = hB_s - \sigma H_0, \tag{BE3}$$
$$2P(JD + x\theta\beta) = \sigma^2\mathcal{H}. \tag{BE4}$$

They contain the entire scale estimate required below. Initially $0 < e < 1$ follows from $E > 0$, so $\mathcal{H}$ is bounded; also $P, J, M, B_s \sim 1$. Eliminating β between the identities gives

$$eB_s = xH_0 + \sigma\mathcal{H} - 2PJD/\sigma \le xH_0 + \sigma\mathcal{H}. \tag{B.9}$$

Thus $e = O(x + \sigma)$ is infinitesimal. Consequently

$$\mathcal{H} - H_0 \in \mathrm{Inf}_c, \quad D = O(\sigma^2), \quad \beta\theta = O(h + \sigma), \quad p - H_0/2,\ l - 1/2,\ \kappa - 1/2 \in \mathrm{Inf}_c, \qquad H_0 = 1 + zQ \ge 1.$$

The quantities p, l, κ are bounded and positive, and $m = \kappa xze$ is infinitesimal. The individual positive coefficients $C, a = p/b, \ell = l/Q$ may be unbounded; they will stay in whole squares.

The following estimates do not assume that H is bounded. Set

$$\nu = (2D - 2\beta\theta + h)/\sigma, \quad \lambda = (8Dh + 4\beta\theta^2 + 2h^2)/\sigma - 4bp\sigma z, \quad K_A = 4\kappa H - \lambda,$$

and define

$$\lambda_0 = \lambda - 2H, \quad d_G = 4D/\sigma^2, \quad \rho = \lambda x/\sigma, \quad L_0 = \lambda\sigma.$$

The source identities imply

$$\nu = 2D/\sigma + [H_0 - ez(J - \sigma)]/P, \tag{B.10}$$
$$\lambda = 2H + 2e\theta z(J - \sigma)/P - 2\theta H_0/P + 8Dh/\sigma - 4bp\sigma z. \tag{B.11}$$

Hence ν, d_G, λ_0 are bounded, $\lambda_0 + 2\theta H_0$ is infinitesimal, and $L_0 - 2h$ is infinitesimal. For $K_A = 4\kappa H - \lambda$ one also has

$$K_A - \{2\theta H_0 - 2h(1-r)\} \in \mathrm{Inf}_c\,. \tag{B.12}$$

Here is a finite polynomial derivation of the only possible pole in this assertion. Put

$$N_\kappa = K(1+\sigma r)(1+\sigma z) - MP = xF_x + \sigma F_\sigma,$$

where $F_x = (N_\kappa - N_\kappa|_{x=0})/x$ and $F_\sigma = N_\kappa|_{x=0}/\sigma$ are polynomials in the bounded source arguments. Direct substitution gives $F_\sigma = r - 1 + \sigma rz$. Therefore $H(2\kappa - 1) = (eF_x + hF_\sigma)/(MP)$ has residue $-h(1-r)$, which proves (B.12). In particular K_A is bounded.

For later use write $R_\gamma = R_0 - 8pu^2$, where

$$R_0 = -4\{2(C+D)(r-1) + 2p + r - 1\}.$$

There is the uniform estimate

$$R_0 = O(x + \sigma + e), \qquad \lambda > 0 \implies R_0\sqrt{\lambda} \in \mathrm{Inf}_c\,. \tag{B.13}$$

To verify the first estimate without any source-pole division, temporarily treat e as an independent variable. The product $N = JPR_0$ is the polynomial

$$-4\{-JKz(1+\sigma r)(1-e)+(r-1)K(1+\sigma z)(1+\sigma r)+J[z(1+xh)(1-e)+(1-r)(1+xhz)]\}.$$

It vanishes identically at $x=\sigma=e=0$. Consecutive differences in x,σ,e express it as $xN_x+\sigma N_\sigma+eN_e$, with bounded polynomial quotients. For the second estimate use $\lambda = O(1+H)$ from (B.11) and $e = O(x+\sigma)$ from (B.9). The bound

$$R_0^2\lambda = O((x+\sigma)^2(1+H)), \qquad (x+\sigma)^2H = xe+2\sigma e+\sigma h$$

has an infinitesimal right side. This proves the square-root estimate for every possible H.

The same source formulas also give the identities used by the complementary testing charts. With $B = B_s$, put

$$d_0 = \sqrt{2PJD/(\sigma^2\mathcal{H})}, \qquad b_0 = \sqrt{2Px\theta\beta/(\sigma^2\mathcal{H})}.$$

The source identity gives

$$D = O(\sigma^2), \quad x\theta\beta = O(\sigma^2), \quad JD + x\theta\beta \asymp \sigma^2. \tag{BE5}$$

Then exactly

$$d_0, b_0 \ge 0, \quad d_0^2+b_0^2 = 1, \quad D = \frac{\sigma^2\mathcal{H}}{2PJ}d_0^2, \quad \beta = \frac{\sigma^2\mathcal{H}}{2Px\theta}b_0^2. \tag{BE6}$$

$$H = \frac{H_0}{B} + \frac{\sigma}{x}\frac{\mathcal{H}}{B}b_0^2. \tag{BE7}$$

$$m = m_0 + m_1 b_0^2, \quad m_0 = \kappa x^2 z H_0/B \asymp x^2 z, \quad m_1 = \kappa x\sigma z\mathcal{H}/B \asymp x\sigma z. \tag{BE8}$$

$$\frac{C(1-r)}{z} \sim \tfrac{1}{2}, \qquad Ar \sim H_0/2, \qquad \alpha Q z \sim \tfrac{1}{2}. \tag{BE9}$$

No inverse of a sphere coordinate is used. Another useful uniform source identity is $p-l \in zB_c$: with e temporarily independent, $2P(p-l)$ is a polynomial in bounded x,σ,z,b,h,e, and at $z=0$ its two source numerators are both one ($e+(1-e)=1$ for the second). Its finite quotient by z is therefore polynomial and bounded. Restoring $e = xh/\sigma$ proves the assertion through both radial openings.

The complete state polynomial. Write $t=u^2$, $\xi=w^2$, $D_u = 1+t$, $D_w = 1+\xi$, $q=\sqrt{z}$, $g = h-Q$. The original reciprocal parent is

$$\begin{aligned} F_w &= 4m(W-wX+uY-uwZ)^2 + 4CD_w(uW-Y)^2 + 4AD_uD_wY^2 \\ &\quad + (4D+2)(uwW+uX-wY-Z)^2 + (4\alpha+2)D_u(Y-wZ)^2 \\ &\quad + (4\beta+2)D_w(X+uZ)^2 - 2D_uD_wZ^2. \end{aligned} \tag{P4}$$

Use the homogeneous roots

$$\begin{gathered} N_x = X+uhZ, \quad N_y = (Y-bwZ)/q, \quad \Gamma = \sigma Z - uwW, \quad R = wZ/q - uqW, \\ A_\theta = q(t-x)W, \quad P_\beta = uZ-(x/\sigma)wW, \quad X_e = X+ewW, \quad V = uw(t-x)W, \\ J_C = N_y + bR, \qquad J_\ell = N_y - QR. \end{gathered}$$

The following definitions collect every remaining coefficient:

$$\begin{aligned} \nu &= (2D-2\beta\theta+h)/\sigma, \qquad \lambda = (8Dh+4\beta\theta^2+2h^2)/\sigma - 4bp\sigma z, \\ \lambda_0 &= \lambda - 2H = 4D(1+h)/\sigma - 2\theta\nu - 4bp\sigma z, \\ K_A &= 4\kappa H - \lambda, \qquad B_x = 8\kappa e, \\ A_y &= 8(p-l) + 4z\{(1+2m)g + 2Dh + 2p\sigma\}, \\ R_x &= -4\{2Db + (1+2m)g - 2\beta\theta\}, \\ R_\gamma &= -4\{2(C+D)(r-1) + 2p + r - 1 + 2pt\}, \\ \chi &= 8(pb+lQ) - 4h + 4eh(2D+1) - 8Dh \\ &\quad + 4z\{4Dbh + 4pb\sigma + (1+2m)g^2 + 2\beta\theta^2\}, \\ k_1 &= 2(1+2m)g^2 + 8Dhb + 8pb\sigma + 4\beta\theta^2, \end{aligned} \tag{SP3}$$

$$\begin{aligned} E_{yy} &= 2z\{(1+2m)t + (1+2D)\xi\}, \\ E_{RR} &= z\{8bD + 8bp\sigma + 4b\sigma(2C+2D+1)(r-1) + k_1 t + 2b^2(2D+1)\xi\}, \\ E_{yR} &= 4z\{t[2Dh + 2p\sigma + (1+2m)g] + b(2D+1)\xi + \sigma(2C+1)(r-1) + 2D[1+\sigma(r-1)] + 2p\sigma\}. \end{aligned}$$

Direct expansion of the original parent, using only the displayed source identities, gives the full four-state identity

$$\begin{aligned} F_w &= \mathcal{E}_+ + [2(1+\xi)+4m\xi]N_x^2 + [4a+2z(2C+1)]J_C^2 + 4\ell J_\ell^2 \\ &\quad + E_{yy}N_y^2 + E_{RR}R^2 + E_{yR}N_yR + (4D/\sigma^2)\Gamma^2 \\ &\quad - 4\nu N_x(u\Gamma) + R_\gamma(N_y+bR)(qw\Gamma) \\ &\quad - 4uqw(2m+2D+1)N_xN_y + uqwR_xN_xR + qwB_xN_xA_\theta + uA_yN_yA_\theta \\ &\quad + K_AA_\theta^2 + \chi uA_\theta R + \mathscr{S}, \\ \mathscr{S} &= \lambda\sigma P_\beta^2 + (\lambda x/\sigma)(t-x)(\xi-\sigma z)W^2, \end{aligned} \tag{SP6}$$

where all terms in the following remainder are retained tensor squares:

$$\begin{aligned} \mathcal{E}_+ &= 4\beta D_wN_x^2 + 2(2D+1)(uX_e)^2 + 4bpzt\Gamma^2 \\ &\quad + 4a(D_uD_w-1)J_C^2 + 4\ell tJ_\ell^2 + 4Cz\xi J_C^2. \end{aligned}$$

This identity includes all four state coordinates and both complete Gamma normal terms.

All E coefficients are infinitesimal. The state-normal cross coefficients outside $\mathscr{S}$, namely $\nu, R_\gamma, R_x, B_x, A_y, \chi$, are bounded; so are K_A, d_G. The possibly unbounded scalar coefficient λ is handled below. Indeed the only C poles outside positive squares occur through $C(r-1)$; β occurs there through the bounded product $\beta\theta$. Every non-Gamma cross coefficient has an explicit u or w factor and is infinitesimal. These conclusions use the uniform estimates just proved, not a bound on H.

Since $4a, 4\ell > 1$, clip these two whole positive radial squares at one and retain their source excesses. The resulting radial form is

$$J_C^2 + J_\ell^2 = (N_y+bR)^2 + (N_y-QR)^2,$$

whose real residue is positive definite because $b+Q=1$. The N_x coefficient has residue two. The exact testing sign gauges $(u,Y,Z)\mapsto(-u,-Y,-Z)$ and $(w,X,Z)\mapsto(-w,-X,-Z)$ allow $u,w\ge 0$; literal zero values are allowed.

A single construction for $\lambda > 0$. In this paragraph $\rho > 0$. Direct elimination of β in (BE3) gives the exact identity

$$\begin{aligned} S := d_G + \rho &= 2\mathcal{H}/P + d_G(1-J+2hx) + 2eH_0/P \\ &\quad - 2e^2z(J-\sigma)/P - 4bpxz. \end{aligned} \tag{B.14}$$

Thus $S-2H_0$ is infinitesimal. In particular S is a positive bounded source unit and d_G, ρ are both bounded. Also $q_0 := \lambda k$ is bounded, since $q_0^2 = L_0\rho$, and $\lambda x = \rho\sigma$ is infinitesimal.

Define the separated monomials, keeping the signs of W, Z,

$$B_1 = \sqrt{\lambda\sigma}\,uZ, \quad B_2 = \sqrt{\rho}\,uwW, \quad B_3 = \sqrt{\lambda}\,x\sqrt{z}W, \quad B_4 = \sqrt{\lambda x}\,wZ/\sqrt{z}.$$

Exactly

$$\mathscr{C} := \mathscr{S} + \lambda xR^2 = \sum_i B_i^2 - 4\lambda xuwWZ.$$

If $WZ > 0$, change the overall state sign so $W, Z > 0$, put $G = (B_1B_2B_3B_4)^{1/4}$ and $d_i = B_i - G$. Then

$$\mathscr{C} = \sum_i d_i^2 + \mathscr{C}_+,$$

$$\mathscr{C}_+ = 2G\{(\sqrt{B_1}-\sqrt{B_2})^2 + (\sqrt{B_3}-\sqrt{B_4})^2 + 2[(B_1B_2)^{1/4} - (B_3B_4)^{1/4}]^2\} \in \Sigma T^2.$$

All radicals separate into factors in K_c and K_t. If $WZ \le 0$, put $d_i = B_i$ and $\mathscr{C}_+ = -4\lambda xuwWZ \in \Sigma T^2$. In both cases the differences $d_i - d_j$ equal $B_i - B_j$.

Put $T_\theta = k\sqrt{z}(u-k)W$ and $\Gamma_* = \sqrt{d_G}\Gamma$. The following exact conductors require no inverse of D or sphere coordinate:

$$\sqrt{\rho}\,u\Gamma = k(d_1-d_2) - (u-k)(d_2-d_3) - \sqrt{\rho\sigma}\,T_\theta, \tag{B.15}$$

$$\begin{aligned} \sqrt{\rho}\sqrt{z}w\Gamma &= \sqrt{\rho}\sigma zR - (\sqrt{z}w + z\sqrt{\sigma})(d_2-d_3) \\ &\quad + \sqrt{\rho}(\sigma z/k + \sqrt{\sigma}\sqrt{z}w/k)T_\theta. \end{aligned} \tag{B.16}$$

Use $1 = (d_G+\rho)/S$ to write

$$u\Gamma = \frac{\sqrt{d_G}u}{S}\Gamma_* + \frac{\sqrt{\rho}}{S}(\sqrt{\rho}\,u\Gamma).$$

After (B.15), every coefficient on $(\Gamma_*, T_\theta, d_i)$ is infinitesimal. For the other complete normal use

$$\sqrt{z}wR_\gamma\Gamma = R_0\sqrt{z}w\Gamma - 8p\sqrt{z}wu(u\Gamma),$$

and apply the same identity $1 = (d_G + \rho)/S$ only to the first term. In (B.16) its only potentially singular T_θ coefficients reduce exactly to

$$\frac{R_0\rho\sigma z}{kS} = \frac{R_0 q_0 z}{S}, \qquad \frac{R_0\rho\sqrt{\sigma}\sqrt{z}w}{kS} = \frac{(R_0\sqrt{\lambda})\sqrt{\rho}\sqrt{z}w}{S}.$$

They are infinitesimal by (B.13); every other coefficient visibly is too. Thus both entire Gamma normal terms have infinitesimal coefficients on a single frame, including $D = 0$.

It remains to handle a possibly signed K_A. The same construction will work for every value of this source coefficient. Put

$$c = \max(0, 1 - K_A), \qquad a_0 = k/\sigma, \qquad J_A = \sqrt{z}(u - k)W, \quad F_A = u(Z - a_0 wW),$$

$$E_u = \sqrt{u}J_A, \quad E_k = \sqrt{k}J_A, \qquad U_* = \sqrt{q_0}E_u, \quad K_* = \sqrt{q_0}E_k.$$

One direct multiplication gives, for every state and all source sizes,

$$\mathscr{S} = L_0F_A^2 + 2q_0E_u^2 + q_0E_k^2 + 2q_0\sqrt{u}E_uR. \tag{B.17}$$

Average this with $\mathscr{S} = \mathscr{C} - \lambda xR^2$:

$$\mathscr{S} = \tfrac{1}{2}L_0F_A^2 + U_*^2 + \tfrac{1}{2}K_*^2 + \sqrt{q_0 u}\,U_*R + \tfrac{1}{2}\mathscr{C} - \tfrac{1}{2}\lambda xR^2.$$

The two cross/radial coefficients are infinitesimal. The term $L_0F_A^2/2$ is retained as a positive square.

If $c > 0$, (B.12) implies that h is a positive unit: in fact its residue is at least $1/4$. The inequality obtained from (B.9),

$$a_0^2(hB_s - \sigma H_0) \le \mathcal{H},$$

then proves a_0 bounded. Also $L_0 \sim 2h$ is a positive unit, so λ is unbounded. The exact angular certificate proved below is

$$\Delta_A := \varepsilon_C\mathscr{C} + u^2R^2 + \varepsilon_G\Gamma^2 - A_\theta^2 \in \Sigma T^2, \tag{B.18}$$

where

$$\varepsilon_C = 2/\lambda + u + k, \qquad \varepsilon_G = 2a_0u + (u + k)/L_0.$$

Both coefficients are infinitesimal whenever $c > 0$. Subtract $c(d_G/S)\Delta_A$ from the parent, keeping that nonnegative quantity separately. The angular part of the remainder is

$$(K_A + c)A_\theta^2 - c(\rho/S)A_\theta^2.$$

The first coefficient is a positive unit. The second square is an infinitesimal Gram correction on (U_*, K_*), by the exact relation

$$\rho A_\theta^2 = (\sqrt{a_0 u}\,U_* + \sqrt{a_0 k}\,K_*)^2. \tag{B.19}$$

The other subtractions have coefficients $cd_G\varepsilon_C/S$ on $\mathscr{C}$, cd_Gu^2/S on R^2, and $c\varepsilon_G/S$ on Γ_*^2; all are infinitesimal. If $c = 0$, none of these terms is used. This verifies the same construction in both situations without a new choice of roots or a source/test rate comparison.

Finally split the positive angular coefficient between A_θ and T_θ, retaining the exact nonnegative difference

$$A_\theta^2 - T_\theta^2 = z(u - k)^2W^2(u^2 + 2uk) \in \Sigma T^2.$$

The complete bounded Gram on

$$(N_x, N_y, R, \Gamma_*, A_\theta, T_\theta, d_1, \ldots, d_4, U_*, K_*)$$

has positive-definite residue: its radial block is $J_C^2 + J_\ell^2$, the N_x pivot is two, the Gamma pivot is one, the angular pivots are positive, and the circuit and last two pivots are respectively $1/2, \ldots, 1/2, 1, 1/2$. All other entries have zero residue. The remaining circuit coefficient $1/2 - cd_G\varepsilon_C/S$ has positive real residue $1/2$. The one-dimensional bounded-Gram fact makes it a tensor SOS, so its product with the retained $\mathscr{C}_+$ is also a tensor SOS. Restore the retained positive squares and apply the bounded-Gram fact.

Proof of the angular certificate, including state signs. For $U, V, Z_* > 0$ and positive separated a_* set

$$\Psi = 1 + U^2V^2 + U^2Z_*^2 + V^2Z_*^2 - 4UVZ_*, \quad T_* = U^2 - 1,$$
$$Q_* = VZ_* - U, \quad R_* = Z_* - UV, \quad G_* = U(V - Z_*).$$

The polynomial identity

$$\begin{aligned}&[2 + a_*(U+1)]\Psi + U^2Q_*^2 + [2U + (U+1)/a_*]R_*^2 - T_*^2\\ &= (T_* + UQ_*)^2 + 2Q_*^2 + (U+1)(\sqrt{a_*}G_* - R_*/\sqrt{a_*})^2\\ &\quad + a_*(U+1)(1 - VZ_*)^2 + 2a_*(U+1)VZ_*(U-1)^2\end{aligned}$$

is verified by multiplication. Expand each $U + 1$ factor before taking square roots. The right side is a finite separated SOS. For $W \neq 0$ substitute

$$U = u/k, \quad V = w/(\sqrt{\sigma}\sqrt{z}), \quad Z_* = \frac{\sqrt{\sigma}}{k\sqrt{z}}\frac{Z}{W}, \quad a_* = \lambda k, \quad M_* = k^2\sqrt{z}W.$$

Then $\mathscr{C} = \lambda M_*^2\Psi$, $A_\theta = M_*T_*$, $R = k\sqrt{z}WQ_*$ and $\Gamma = k\sqrt{\sigma}\sqrt{z}WR_*$. Multiplication by M_*^2 gives (B.18). These are legitimate separated factors even when an intermediate ratio is unbounded. If $Z_* < 0$, apply the displayed identity first to $|Z_*|$. Replacing $|Z_*|$ by Z_* increases Ψ by $8UV|Z_*|$ and each of Q_*^2, R_*^2 by $4UV|Z_*|$. The three resulting additional terms have positive separated coefficients and are SOS. Thus the same certificate holds for either state sign. At $Z_* = 0$ the original identity applies directly. If $W = 0$, $A_\theta = 0$, so (B.18) follows immediately from $\mathscr{C} \in \Sigma T^2$. No state coordinate is inverted there.

The case $\lambda \leq 0$. The definition of λ gives $H^2 \leq 2bpz$ and $\beta\theta^2/\sigma^2 \leq bpz$. Thus H is bounded, h is infinitesimal, $\theta \sim 1$, $M_\beta = \beta\theta/\sigma^2$ is bounded and $H - H_0, d_G - 2H_0$ are infinitesimal. In particular

$$L = \lambda/\sigma = 8(D/\sigma^2)h + 4M_\beta\theta + 2H^2 - 4bpz$$

is bounded. Direct multiplication gives

$$u^2\Gamma + V = \sigma u P_\beta,$$
$$A_\theta^2 + u\Gamma P_\beta + uA_\theta R = \sigma P_\beta^2 + (x/\sigma)(u^2 - x)(w^2 - \sigma z)W^2,$$

where $V = uw(u^2 - x)W$. Consequently

$$K_A A_\theta^2 + \mathscr{S} = 4\kappa H A_\theta^2 + Lu^2\Gamma^2 + L\Gamma V + \lambda u A_\theta R.$$

Add $1 + L^2$ times the exact zero

$$V^2 - \sigma u^2 A_\theta^2 - u\sigma(u^2 - x)A_\theta R + (u^2 - x)V\Gamma = 0.$$

The (Γ, V) real residue block is

$$\begin{pmatrix} 2\bar{H}_0 & \bar{L}/2 \\ \bar{L}/2 & 1 + \bar{L}^2 \end{pmatrix}, \qquad \det = 2\bar{H}_0 + (2\bar{H}_0 - 1/4)\bar{L}^2 > 0.$$

The N_x, radial and angular blocks are strict; every remaining cross coefficient, including both complete Gamma normal terms, is infinitesimal. The bounded-Gram fact proves this case, including $\lambda = 0$.

All constructions are literal on the testing axes. No inverse of D or β occurs, so both permitted source faces are included. The parent is real quadratic in the state; applying the proof to the real and imaginary parts proves the complex-state assertion. This completes the theorem.

B.5.5 A full-state completion for every Alice angle

Theorem B.24. *Retain the original stationary C14.1 source and all its intersections: $d = 1$, $\lambda_{\mathrm{ev}} = n = 0$, the native strict gaps, $m, C, A, \alpha > 0$, $D, \beta \geq 0$, and*

$$x = k^2, \qquad z = (x\zeta)^{-1}, \qquad \sigma = \eta^{-1} \quad \textit{positive source infinitesimals.}$$

There is no restriction on $H = h/\sigma$. The complete homogeneous parent is a finite tensor sum of squares at every independent projective testing pair and every real or complex state. The literal faces $D = 0$, $\beta = 0$ and all testing and state-coordinate zeros are included.

Proof. We first treat the corner in which the Bob half-angle is projectively unbounded. Put $w = 1/v$, including $w = 0$ at $v = \infty$. Thus w is infinitesimal or zero. If u is infinitesimal or zero, Theorem B.23 applies directly. For every other Alice test, $p = 1/u$ is bounded; $p = 0$ includes $u = \infty$. The exact angle gauges allow $p, w \geq 0$: changing the sign of p together with Y, Z, or changing the sign of w together with X, Z, preserves the homogeneous polynomial below.

The whole polynomial in the inverse-Alice chart. Write $D_p = 1+p^2$ and $D_w = 1+w^2$. The homogeneous parent is

$$\begin{aligned}\widetilde{F} &= 4m(pW - pwX + Y - wZ)^2 + 4CD_w(W - pY)^2 + 4AD_pD_wY^2 \\ &\quad + (4D+2)(wW + X - pwY - pZ)^2 + (4\alpha+2)D_p(Y - wZ)^2 \\ &\quad + (4\beta+2)D_w(pX + Z)^2 - 2D_pD_wZ^2.\end{aligned} \tag{IA1}$$

For $p \neq 0$ this is p^2 times the reciprocal parent (P4); the displayed polynomial defines its literal projective endpoint at $p = 0$. Set

$$q = \sqrt{z}, \quad Q = 1 - b, \quad r = bz, \quad H_0 = 1 + zQ,$$
$$E = qW, \quad F = Y/q, \quad G = wZ/q, \quad R = X + wW, \quad S = pX, \quad J = F - bG.$$

Only pure-source inverses occur in these roots. Direct expansion of the complete signed part of (P4) gives

$$K = R^2 + D_wS^2 + z(G - F)^2 + p^2D_wzF^2 - 2pE(G + w^2F) + 2qwS(G - F). \tag{IA2}$$

Thus (P4), in this chart, is $2K$ plus the six original positive event terms. This identity keeps all four state coordinates.

Three leading event weights complete the signed core. Define the positive source coefficients

$$C_0 = \frac{z}{2(1-r)}, \qquad A_0 = \frac{H_0}{2bz}, \qquad \alpha_0 = \frac{1}{2Qz}, \qquad B = \frac{H_0}{bQ}.$$

The opening-cleared estimates (BE9) imply

$$C/C_0 \sim 1, \qquad A/A_0 \sim 1, \qquad \alpha/\alpha_0 \sim 1.$$

In particular the pure-source number

$$f = \min\{1, C/C_0, A/A_0, \alpha/\alpha_0\}$$

is positive, $f \sim 1$, and each original event weight is at least f times its corresponding leading weight. This uses neither a bound on H nor positivity of D or β.

Completing the leading form gives the exact identity

$$\begin{aligned}&2z(G - F)^2 + 2p^2zF^2 - 4pEG + \frac{2}{1-r}(E - pzF)^2 \\ &\quad + \frac{2H_0}{b}(1+p^2)F^2 + \frac{2}{Q}(1+p^2)(F - G)^2 \\ &= \frac{2}{1-r}(E - pzJ - pG)^2 + 2B(1+p^2)J^2 + 2H_0G^2.\end{aligned} \tag{IA3}$$

Substituting $F = J + bG$ and $r = bz$ joins the negative product to the three event squares. The remaining coefficients can then be estimated together.

Put

$$J_* = \sqrt{B}J, \quad G_* = \sqrt{H_0}G, \quad a = \frac{z}{\sqrt{B}}, \quad c = \frac{1}{\sqrt{H_0}}, \quad L = aJ_* + cG_* = zJ + G.$$

The native inequality $r < 1$ gives the exact contraction

$$1 - a^2 - c^2 = \frac{zQ(1-r)}{H_0} \geq 0. \tag{IA4}$$

Consequently the source coefficients a, c are bounded, including arbitrarily small b or Q.

Restoring the exact event weights. The difference between the actual and retained weights contributes

$$\begin{aligned}\mathcal{X} &= 4m(pW - pwX + Y - wZ)^2 + 4D(wW + X - pwY - pZ)^2 + 4\beta D_w(pX + Z)^2 \\ &\quad + 4(C - fC_0)D_w(W - pY)^2 + 4(A - fA_0)D_pD_wY^2 \\ &\quad + 4(\alpha - f\alpha_0)D_p(Y - wZ)^2,\end{aligned}$$

and retain the nonnegative expression

$$\begin{aligned}\Pi &= \mathcal{X} + 2(1-f)\{R^2 + S^2 + z(G - F)^2 + p^2zF^2\} \\ &\quad + 2w^2S^2 + 2p^2w^2zF^2 + 4fC_0w^2(W - pY)^2 + 4fA_0D_pw^2Y^2.\end{aligned} \tag{IA5}$$

All its source coefficients are nonnegative; D_p, D_w and the remaining factors separate in the testing field. Each term is therefore a tensor sum of squares. Equations (IA1)–(IA3) give the full equality

$$\begin{aligned}\widetilde{F} &= f\mathcal{Q} + \Pi + \mathcal{E},\\ \mathcal{Q} &= 2R^2 + 2S^2 + \frac{2}{1-r}(E - pL)^2 + 2(1+p^2)J_*^2 + 2G_*^2,\\ \mathcal{E} &= -4(1-f)pEG - 4pw^2EF + 4qwS(G-F).\end{aligned} \tag{IA6}$$

A uniform reserve for every bounded inverse angle. Choose a real constant $P \geq 1$ with $p \leq P$ and put

$$t = p/P, \qquad E_* = E/P, \qquad g = (R, S, E_*, J_*, G_*)^{\mathsf{T}}.$$

Then $0 \leq t \leq 1$. Define

$$\mathcal{B} = 2R^2 + 2S^2 + 2(E_* - tL)^2 + 2(1+t^2)J_*^2 + 2G_*^2.$$

The difference $\mathcal{Q} - \mathcal{B}$ is exactly

$$2\left(\frac{P^2}{1-r} - 1\right)(E_* - tL)^2 + 2(P^2-1)t^2J_*^2 \in \Sigma T^2. \tag{IA7}$$

Moreover, the contraction (IA4) gives the explicit reserve

$$\begin{aligned}\mathcal{B} - \tfrac{1}{2}g^{\mathsf{T}}g = {}& \tfrac{3}{2}(R^2+S^2) + \tfrac{3}{2}(E_* - 4tL/3)^2 + \tfrac{3}{2}J_*^2 + \tfrac{5}{6}G_*^2\\ &+ \tfrac{2}{3}(1-t^2)G_*^2 + \tfrac{4}{3}t^2J_*^2\\ &+ \tfrac{2}{3}t^2\{(1-a^2-c^2)(J_*^2+G_*^2) + (cJ_* - aG_*)^2\}.\end{aligned} \tag{IA8}$$

Every radical needed on the right is in one labelled field. In particular no square root of a mixed coefficient is taken.

It remains to put the complete error in the same five roots. Its only nonzero upper-triangle matrix entries are

$$\begin{array}{c|c} (E_*, G_*) & -2(1-f)P^2t/\sqrt{H_0} - 2P^2btw^2/\sqrt{H_0}\\ (E_*, J_*) & -2P^2tw^2/\sqrt{B}\\ (S, J_*) & -2qw/\sqrt{B}\\ (S, G_*) & 2qQw/\sqrt{H_0}. \end{array} \tag{IA9}$$

The pure-source coefficient matrices at t, tw^2, w have absolute row bounds $2(1-f)P^2$, $4P^2$, and 4, respectively. Here use $1/\sqrt{B} \leq 1/2$, $b/\sqrt{H_0} \leq 1$, and

$$q^2/B = rQ/H_0 \leq 1, \qquad q^2Q^2/H_0 \leq 1.$$

The first bound is a source infinitesimal. In the other two terms, tw^2 and w are testing infinitesimals. Thus (IA9) defines a bounded tensor matrix with zero real residue, without imposing an order on the tensor product or comparing source and testing rates. The remaining Gram matrix is $fI/2$ plus this error; its residue is $I/2$. The strict bounded-Gram lemma proves its tensor sum of squares. Restoring (IA5), (IA7) and (IA8) proves the whole corner for every homogeneous real state. The roots themselves need not be bounded or independent.

The remaining Bob angles and the endpoints. On C14.1, (BE9) and z infinitesimal imply that A and α are unbounded. The adjacent-event theorem transported by (CE9)–(CE10) in Section B.2.4 therefore covers every Bob test that is not projectively unbounded, with all Alice tests. This and the corner just proved exhaust the projective testing pairs.

No inverse of D, β or a state coordinate occurs. The identities specialize literally at $p = 0$, $w = 0$ and zero state coordinates; $u = 0$ was covered separately at the start. Real and imaginary state parts give complex states. Finally restore the positive pure-field homogenization factors as in the original parent normalization. For unit states the inverse-Alice angular factor is $1/(D_pD_w)$, which remains positive at $p = 0$; arbitrary nonzero homogeneous states have the additional pure-testing state norm factor. This proves the stated full C14.1 domain. □

B.5.6 Unit L and unbounded η: all testing angles

Theorem B.25. *Consider the C1 source (P1), with $d = 1$, $x, \sigma = 1/\eta$ positive infinitesimal and $L = x\zeta$ a positive bounded unit. Assume the strict gaps $E, M, J > 0$, $0 < b, h, r < 1$ and $D, \beta \geq 0$. The complete parent is a tensor sum of squares at every independent projective angle pair and every real or complex state, including either permitted face $D = 0$ or $\beta = 0$.*

Proof. The source formulas (P1) and opening-safe estimates (BE3)–(BE9) give

$$C/C_0, A/A_0, \alpha/a_0 \sim 1, \qquad C_0 = \frac{1}{2L(1-r)}, \quad A_0 = \frac{1+L(1-r)}{2Lr}, \quad a_0 = \frac{L}{2(1-Lr)}.$$

Choose real $0 < \ell \le L \le U$ and a real $0 < \mu \le \min(1/4, C, A)$. Such a μ exists uniformly through the source openings because $C_0, A_0 \ge 1/(2U)$.

First let u be bounded. Put $c = (1+v^2)^{-1/2}$, $s = vc$, $P = uW$, $T = cZ$. The projective Bob endpoint has $c = 0$, $s = \pm 1$ and $w = 0$. When u, c are both infinitesimal, $w = 1/v$ is infinitesimal and Theorem B.23 applies, since $z = 1/L$ is bounded. This includes the literal corner. Otherwise put $(u_0, c_0, s_0) = \mathrm{res}(u, c, s)$ and choose real $M_* \ge 1$, $\tau > 0$ with $|u_0| \le M_*$ and $u_0^2 + c_0^2 \ge \tau^2$. The next reserve calculation is evaluated at $(u, c, s) = (u_0, c_0, s_0)$: every determinant, inverse and matrix inequality is then taken in the ordered source field K_c.

The leading selected quarter-parent in the four raw roots (P, X, Y, T) is

$$S_0 = C_0(P-Y)^2 + A_0(1+u^2)Y^2 + a_0(1+u^2)(sY-T)^2 + K_B/2,$$

$$K_B = X^2 + c^2(P-Y)^2 + s^2Y^2 - 2sPT + 2ucX(sP - sY + T) + u^2s^2X^2 + u^2(sY-T)^2.$$

Its $T = 0$ block is strictly positive. Its determinant is

$$\det G_0 = \frac{(1+u^2)^2(Lc^2 + u^2s^2)\{1+L(1-r)\}^2}{16L^2r(1-r)(1-Lr)} > 0.$$

All source-opening denominators cancel from the cofactor ratio:

$$\mathrm{tr}\, G_0^{-1} = \frac{2(L^2N_2 + LN_1 + N_0)}{(1+t)\{Lq + t(1-q)\}\{1+L(1-r)\}}, \quad t = u^2, \ q = c^2,$$

where, writing $e_r = 1 - r$,

$$N_2 = 1 + r^2 + e_rq\{1 + r(1-q)\} + e_r(2-qr)t + e_r^2(1-q)t^2,$$
$$N_1 = 2e_rq + \{2 - rq(3-q)\}t + e_r(1-q)t^2, \qquad N_0 = 1 + (2-q)t.$$

The numerator is bounded by a real constant depending only on U, M_*. Moreover $(1+t)\{Lq + t(1-q)\} \ge \min(\ell, 1)(q+t)$. Thus $G_0 \succeq \kappa I$ for some real $\kappa > 0$, uniformly through every source opening.

Set $d_s = |C/C_0 - 1| + |A/A_0 - 1| + |\alpha/a_0 - 1|$ and $f = 1 - d_s$. The selected form is the sum of $fS_0 + (1-f)K_B/2$ and the nonnegative event terms with coefficients $C - fC_0, A - fA_0, \alpha - fa_0$. The core at the real center has bounded coefficients, whereas d_s is source infinitesimal. This decomposition gives a positive real lower bound for the selected Gram with the original C, A, α.

Use the moving event coordinates $e = (P - Y, Y, sY - T, X)$. Both coordinate maps have squared Frobenius norm at most six. Clip C, A, α at a sufficiently large real N, retaining each excess as its whole original event. In these coordinates the core has norm at most a real K, independently of source openings. If the unmodified Gram has reserve $k > 0$, its complementary block remains at least k, its clipped large block is at least $N - K$, and the off-diagonal norm is at most K. Taking $N > K + 2K^2/k + 1$ preserves a positive real reserve, by $2Kab \le (2K^2/k)a^2 + (k/2)b^2$.

The clipped coefficients are bounded source elements. Keep the root column $e(u, c, s)$ unchanged and replace only its coefficient matrix by its value at (u_0, c_0, s_0). The difference belongs to $\mathrm{Sym}_4(B_c \otimes \mathfrak{m}_t)$, where $\mathfrak{m}_t$ denotes the testing infinitesimals. The source matrix at the center is bounded and has positive-definite real residue by the preceding estimate. The separated bounded-Gram lemma applies. Restore all clipped excesses and the complete original m, D, β terms.

It remains to treat unbounded u. Put $p = 1/u$ and use two Bob charts. For $|v| \le 1$ take (W, Y, B, R) with $B = pX + Z$, $R = vX$; the exact core is

$$K_p = B^2 - 2pvBW - 2vBY + (1+p^2)R^2 + 2(1+p^2)RW + (W - pY)^2 + (1+p^2)v^2Y^2.$$

Select only $C = A = \mu$. At $p = 0$ the selected quarter-parent is

$$\mu(1+v^2)(W^2 + Y^2) + \tfrac{1}{2}(W+R)^2 + \tfrac{1}{2}(B - vY)^2 \succeq (\mu/3)I.$$

For $|w| = |1/v| \le 1$ use (W, Y, X, H), $H = wZ$, and the exact core

$$K_{pw} = H^2 - 2pHW + 2pwHX - 2HY + w^2W^2 + 2wWX - 2pw^2WY + (1 + p^2 + p^2w^2)(X^2 + Y^2) - 2pwXY.$$

The corresponding selected value at $p = 0$ is

$$\mu(1 + w^2)(W^2 + Y^2) + \tfrac{1}{2}(X + wW)^2 + \tfrac{1}{2}(H - Y)^2 \succeq (\mu/3)I.$$

In each chart every remaining coefficient is bounded infinitesimal. The complete selected testing Gram is therefore positive definite and can be factored entirely in the testing field. Restore $C - \mu$, $A - \mu$ and all other original nonnegative event terms. The two charts cover every Bob angle, including their literal endpoints.

All normalizations are positive pure testing factors. The exact sign gauges are $u \mapsto -u$, $(Y, Z) \mapsto (-Y, -Z)$, and $v \mapsto -v$, $(X, Z) \mapsto (-X, -Z)$; they preserve the full parent. Real and imaginary splitting treats complex states. The event terms with coefficients D and β remain valid on their zero faces. □

B.6 Contact gaps when L vanishes

Here the coefficient gaps, rather than L alone, determine the remaining quadratic form. The gap j first separates a strict block from its singular limit. The completed J relation controls the remaining state normal when that block degenerates.

B.6.1 Small L: the source openings and complementary tests

Work in the original C1 source, with $d = 1$, $x \in \mathrm{Inf}_{>0}$, ζ noninfinitesimal, $L = x\zeta \in \mathrm{Inf}_{>0}$, and η bounded. Native $r = 1 - a\eta$, $h = 1 - \theta$ remain strictly positive. This section admits every original projective test except possibly u, v both infinitesimal. It does not admit the remaining two-soft local chart.

Exact source restrictions, including the total-scale intersection. Put $\rho = r + \eta$ and $j = \eta J = \eta(1 + xh) - 1 + r > 0$. Since $a < 1 + xh$, ρ is a positive bounded unit. The exact beta numerator is

$$2\beta\theta P_d = K\rho(xh + L) - \theta P_d. \tag{SB1}$$

It implies $\theta = O(x + L) = O(L)$, since $x/L = 1/\zeta$ is bounded. Consequently

$$h, M, E, Q, P_d, P_4 \sim 1, \qquad B_5 \sim 1. \tag{SB2}$$

For the last assertion, use the exact expression $B_5 = K(1 + L\eta)\rho - jP_d$: its residue is $r + \eta - (\eta - 1 + r) = 1$. Thus original $D = 0$ cannot occur on this source face; literal $\beta = 0$ remains included. The original formulas, with openings canceled before taking residues, give uniformly

$$\begin{gathered} C(1 - r) \sim \rho/2, \quad Ar/\eta \sim 1/2, \quad Dj \sim 1/2, \\ mL/x^2 \sim \rho/2, \quad \alpha/[L(L + x)] \sim \rho/2, \quad (2\beta + 1)\theta/(x + L) \sim \rho. \end{gathered} \tag{SB3}$$

In particular C, D are strong, m, α weak, and A is strong precisely when η is a unit. The four exact large-weight alternatives are

weight unbounded	equivalent opening
C	$1 - r \in \mathrm{Inf}$
D	$j \in \mathrm{Inf}$
A	$r \in \mathrm{Inf}$
β	$\theta/(x + L) \in \mathrm{Inf}$.

(SB4)

For the A equivalence, $r \in \mathrm{Inf}$ forces η to be a unit because $j > 0$; if r is a unit, η/r is bounded. All coefficients are bounded exactly when the four quantities in the right column of (SB4) are positive bounded units. That entire interior is proved by the positive-total-scale theorem. After intersection with $\delta_{\mathrm{tot}} \in \mathrm{Inf}_{>0}$, at least one of (SB4) therefore occurs. These are exhaustive necessary source alternatives, not assertions of their independent realizability.

The unit-η remote tests: two complete three-state blocks. Suppose first η is a unit, with real residue $e > 0$, and write $r_0 = \operatorname{res} r \in [0, 1]$. For an interior opening, the source residues are

$$C_0 = \frac{r_0 + e}{2(1 - r_0)}, \quad A_0 = \frac{e}{2r_0}, \quad D_0 = \frac{1}{2(e + r_0 - 1)}. \tag{SB5}$$

Retain the original m, α, β terms throughout. Bounded selected weights retain their actual coefficients; unbounded ones are capped at sufficiently large finite real numbers and their excesses retained as exact squares.

For u a bounded nonzero unit and v infinitesimal, use the full raw frame

$$p = uW, \quad Y, \quad R = vZ, \quad B_t = X + uZ.$$

The exact core is (CE2), and the D event is $J = p - Y + uvB_t - D_uR$. Every Gram coefficient is bounded after capping, and no state normal is omitted. Set $U = \operatorname{res} u^2 > 0$. The residue, divided by four, is $B_t^2/2$ plus

$$\begin{aligned} f_U = {} & C_0(p-Y)^2 + A_0(1+U)Y^2 + D_0\{p - Y - (1+U)R\}^2 \\ & + \tfrac{1}{2}\{(p-Y)^2 + (1+U)UR^2 - 2(1+U)pR\}. \end{aligned} \tag{SB6}$$

Its first two principal minors and determinant are, respectively,

$$\begin{aligned} \Delta_1 = \frac{e(e+r_0)}{2(1-r_0)j_0}, \qquad \Delta_2 = \frac{e^2(1+U)(e+r_0)}{4r_0(1-r_0)j_0}, \\ \Delta_3 = \frac{Ue(1+U)^2(e+r_0)^2}{8r_0(1-r_0)j_0} > 0, \qquad j_0 = e + r_0 - 1. \end{aligned} \tag{SB7}$$

For u infinitesimal and v a bounded nonzero unit, use the full frame $(p = uW, X, Y, R = vZ)$. The same exact parent has residue, divided by four, $(1+V)X^2/2$ plus

$$\begin{aligned} f_V = {} & C_0(1+V)(p-Y)^2 + A_0(1+V)Y^2 + D_0(p - Y - R)^2 \\ & + \tfrac{1}{2}\{(p-Y)^2 + VY^2 - 2pR\}, \qquad V = \operatorname{res} v^2 > 0. \end{aligned} \tag{SB8}$$

The leading two minors are positive, and

$$\det f_V = \frac{V(1+V)(e+r_0)^2}{8r_0(1-r_0)j_0} > 0. \tag{SB9}$$

These arguments include every opening in (SB4). To see explicitly why the finite cap remains valid, impose the kernels of the divergent events and inspect the remaining two-state Gram. The determinants are

opening	$\det(f_U\|_{\ker})$	$\det(f_V\|_{\ker})$
$r_0 = 0,\ e > 1$	$\dfrac{Ue^2(1+U)}{4(e-1)}$	$\dfrac{Ve}{4(e-1)}$
$r_0 = 1$	$\dfrac{U(1+U)^2(e+1)}{4}$	$\dfrac{V(e+1)}{4e}$
$j_0 = 0,\ 0 < r_0 < 1$	$\dfrac{U(1+U)^2}{4r_0}$	$\dfrac{V(1+V)}{4r_0(1-r_0)}$.

(SB10)

The respective kernels are $Y = 0$, $p = Y$, and $p - Y = (1+U)R$ for f_U or $p - Y = R$ for f_V. All first pivots are positive. At the only double opening, $r_0 = j_0 = 0, e = 1$, the two kernels leave the positive scalar $U(1+U)R^2/2$, respectively $VR^2/2$. There are no other multiple openings since $e > 0$. The elementary finite-penalty Schur lemma therefore supplies finite real caps simultaneously for every divergent selected event.

For completeness the remaining projective ends also use original homogeneous degree. If u is unbounded, (CE4)–(CE5) for bounded v, and (CE7)–(CE8) for unbounded v, are strict with any positive finite C, A caps. If v is unbounded and u bounded, put $w = 1/v$ and use $(p = uW, X, Y, Z)$. The exact D event is $w(p - Y) + uX - Z$. At $w = 0$ the core is $(1+u^2)(X^2 + Y^2)$, while the three caps give $C(p-Y)^2 + A(1+u^2)Y^2 + D(uX - Z)^2$; these are strict on all four roots, including $u = 0$. Each term suppressed in this residue has a bounded infinitesimal coefficient in this same full frame. Literal $w = 0$ is therefore included. The both-unit tests are covered by the common noncommuting-residue theorem. Together these statements exhaust the complement of $u, v \in \mathrm{Inf}$.

The infinitesimal-η remote tests. Here $r \sim 1$, C, D are unbounded, and $y = L\eta$ is infinitesimal. Define the pure-source harmonic coefficient $\gamma = yCD/(C + D)$. Uniformly through the a, J openings,

$$\gamma \sim L/2, \qquad A/y \sim 1/(2L), \qquad \gamma A/y \sim 1/4, \qquad y/\gamma \sim 2\eta \in \mathrm{Inf}\,. \tag{SB11}$$

The strong-II remote proof (RR3)–(RR6), which uses only C, D unbounded, covers every noninfinitesimal v and every u. For u noninfinitesimal and v infinitesimal, use (RR7)–(RR9) with zero alpha cap and $k = -1$. In its notation let $a = u/\sqrt{1+u^2}$, $b = 1/\sqrt{1+u^2}$, $\ell = vY$, $x_t = bX$, $R_0 = x_t + a(Z - \ell)$, and $q = \sqrt{y}Y$. Then

$$M_{-1} = \tfrac{1}{2}R_0^2 + \tfrac{1}{2}(b\ell)^2, \qquad v(Z - \ell) = avR_0 - bv(b\ell) - b\sqrt{y}\,V_h, \tag{SB12}$$

where $V_h = v(aX - bZ)/\sqrt{y}$. Normalize the harmonic pair as $P_h = \sqrt{\gamma}V_h$, $Q_h = \sqrt{A/y}\,q$. Its residue determinant is $1 - b_0^2 > 0$ by (SB11), where $b_0 = \operatorname{res} b < 1$; the off-diagonal entry is b_0. The sole negative term in (RR9), $-v^2(Z - \ell)^2/2$, is an infinitesimal Gram perturbation of $(R_0, b\ell, P_h)$ by (SB12), since $\sqrt{y/\gamma}$ is infinitesimal. Thus the full four-root Gram is strict. Retain the original positive $\alpha(Z - \ell)^2$, the C1 beta event, and every other unused term. The homogeneous endpoint $u = \infty$ has $a = \pm 1, b = 0$ and is included; literal $v = 0$ also causes no change.

B.6.2 The small-L native scalar and complete state normals

Normalize $d = 1$, use the native C1 source $\lambda = n = 0$, and keep all other original event weights, including literal $\beta = 0$. Put $U = u^2, V = v^2, D_u = 1 + U, D_v = 1 + V$, $\theta = 1 - h$, and

$$X = -huZ + \xi, \qquad Y = ruW + \nu.$$

Here ν denotes this state normal only. Direct expansion of the original event terms gives

$$\frac{F_g}{4} = A_w W^2 + A_z Z^2 - 2uv(b_0 + b_1 U)WZ,$$

with the complete graph coefficients

$$\begin{aligned}
A_w &= [(C + D + \tfrac{1}{2})(1 - r)^2 + Ar^2]U + Ar^2U^2 \\
&\quad + \{m + [2mr + C(1 - r)^2 + (A + \alpha + \tfrac{1}{2})r^2]U + r^2(m + A + \alpha + \tfrac{1}{2})U^2\}V, \\
A_z &= \alpha + [\alpha + (m + \beta + \tfrac{1}{2})\theta^2]U \\
&\quad + \{D + [2Dh + \tfrac{1}{2}h^2 + \beta\theta^2]U + (D + \tfrac{1}{2})h^2U^2\}V, \\
b_0 &= m\theta + D(1 - r) + \alpha r + \tfrac{1}{2}, \\
b_1 &= mr\theta + (D + \tfrac{1}{2})h(1 - r) + (\alpha + \tfrac{1}{2})r.
\end{aligned}$$

These identities are homogeneous in W, Z and need no state-ratio bound. The first-normal and hard-block formulas that follow are obtained by the same substitution, or by differentiating the original parent twice in X, Y. The entire original parent is exactly

$$\begin{aligned}
F/4 = F_g/4 &+ 2\xi(vWf_w + uZf_z) + 2\nu(uWk_w + vZk_z) \\
&+ H_{xx}\xi^2 + 2H_{xy}\xi\nu + H_{yy}\nu^2.
\end{aligned} \tag{SN1}$$

where

$$\begin{aligned}
f_w &= -m(1 + rU) + (D + \tfrac{1}{2})(1 - r)U, \\
f_z &= m\theta - (D + \tfrac{1}{2})V(1 + hU) + (\beta + \tfrac{1}{2})D_v\theta, \\
k_w &= mV(1 + rU) - CD_v(1 - r) + AD_uD_vr - (D + \tfrac{1}{2})(1 - r) + (\alpha + \tfrac{1}{2})D_urV, \\
k_z &= -m\theta U + (D + \tfrac{1}{2})(1 + hU) - (\alpha + \tfrac{1}{2})D_u, \\
H_{xx} &= m + \beta D_v + (D + \tfrac{1}{2})UV + D_v/2, \\
H_{yy} &= mUV + CD_v + AD_uD_v + D + \tfrac{1}{2} + (\alpha + \tfrac{1}{2})D_uV, \\
H_{xy} &= -(m + D + \tfrac{1}{2})uv.
\end{aligned} \tag{SN2}$$

There is no discarded normal or excess term in this identity. Its two exact first-normal contact equations are

$$\zeta f_w(x, y) + f_z(x, y) = 0, \qquad k_w(x, y) + \eta k_z(x, y) = 0, \qquad y = x\eta\zeta. \tag{SN3}$$

They follow by substituting (NC4), not by setting a testing angle equal to a source angle in the desired tensor assertion.

On the small-L, bounded-η source, with x, L infinitesimal and ζ noninfinitesimal, the beta-opening identity gives:

$$2\beta\theta P_d = K(r + \eta)(xh + L) - \theta P_d, \qquad \beta\theta = O(L), \quad \beta\theta^2 = O(L^2). \tag{SN4}$$

In particular, after the pure-source normalization $\widehat{\xi} = \sqrt{1 + \beta}\,\xi$, its cross coefficient is $\beta\theta/\sqrt{1 + \beta} = O(L)$, including every unbounded beta opening. The complete normal allocations are the retained RO/JO/HO and DG/LD/CJ terms.

B.6.3 The small-L scalar and finite Taylor quotients

Use the normalized native C1 parent, $d = 1$, with $x = k^2 > 0$ infinitesimal, and set

$$\rho = \eta + r, \quad y = x\eta\zeta, \quad L = x\zeta, \quad h = 1 - \theta, \quad s = \sqrt{\eta\zeta}, \quad \lambda = \rho/(2\eta).$$

The reference interior has

$$\eta, \zeta, r, 1 - r, j_0 = \eta + r - 1 \quad \text{positive bounded units.} \tag{IC1}$$

The formulas below are the exact scalar and full-normal dictionary. The opening-uniform RO proof supplies their normalized bounds when r or $1 - r$ tends to zero. The displayed reference X normal uses $\sqrt{\beta}$ when $\beta > 0$; RO replaces it throughout by $\sqrt{\beta + 1/2}$, which includes literal $\beta = 0$. The full scalar graph coefficients A_w, A_z, b_0, b_1 are printed explicitly in the native dictionary SN1–SN2. For IC4–IC21, first assume $Z \neq 0$ and normalize $Z = 1$ inside the testing field. Homogenizing the resulting state-linear roots restores every Z; the literal $Z = 0$ state has the direct core certificate. The reciprocal identities below are already homogeneous.

A bounded-ratio frame retaining all three state normals. First suppose the pure-testing ratio $t = v/u$ is bounded, possibly infinitesimal. Define

$$\begin{gathered} E = \eta uW - v, \quad b = \frac{r(1-r)}{\eta(\eta + 2r - 1)}, \quad N = Y - ruW - bE, \\ \xi = X + hu, \qquad N_x = \sqrt{\beta}\,\xi, \quad h = 1 - \theta. \end{gathered} \tag{IC4}$$

The state change is exact and uses only the pure-testing inverse u: $W = (v + E)/(\eta u)$, $Y = r(v + E)/\eta + bE + N$, $X = -hu + \xi$. Every normal vanishes at the actual source contact.

The exact scalar, with the negative defect absorbed. Set $U = u^2, V = v^2$, $s = \sqrt{\eta\zeta}$, $k = \sqrt{x}$, and introduce the positive separated roots

$$A_t = uv, \quad B_t = \frac{kv^2}{su}, \quad C_t = xs, \quad G = kv, \qquad F_v = V - y. \tag{IC6}$$

They satisfy $A_t B_t C_t = G^3$ and coincide at actual contact $(u, t) = (k, s)$. The pure-testing pole in B_t is inside the root; none of the coefficient arrays below has a negative power of t.

Use A_w, A_z, b_0, b_1 from (SN1)–(SN2), including the beta term. The scalar at $E = N = \xi = 0$ is exactly

$$\mathcal{S} = \frac{VA_w}{\eta^2 U} + A_z - \frac{2V}{\eta}(b_0 + b_1 U) = c_F F_v^2 + f_A A_t^2 + f_B B_t^2 + f_C C_t^2 - f_G G^2. \tag{IC7}$$

Here the complete coefficients are

$$\begin{aligned} c_F &= \frac{C(1-r)^2 + Ar^2 + (\alpha + \frac{1}{2})r^2 + 2mr}{\eta^2}, \\ a_{UV} &= \frac{Ar^2}{\eta^2} + \frac{h^2}{2} + 2Dh + \beta\theta^2 - \frac{2b_1}{\eta}, \\ a_{UUV} &= h^2(D + \tfrac{1}{2}), \qquad a_{UVV} = \frac{r^2(m + A + \alpha + \frac{1}{2})}{\eta^2}, \\ q_V &= \frac{C(1-r)^2 + Ar^2 + (D + \frac{1}{2})(1-r)^2}{\eta^2} + D - \frac{2b_0}{\eta}, \\ q_U &= \alpha + (m + \beta + \tfrac{1}{2})\theta^2, \\ f_A &= a_{UV} + a_{UUV}U + a_{UVV}V, \qquad f_B = \frac{m\zeta}{\eta x}, \\ f_C &= \frac{\alpha - c_F y^2 + q_U U}{\eta\zeta x^2}, \qquad f_G = -\frac{q_V + 2c_F y}{x}. \end{aligned} \tag{IC8}$$

Finite scalar Taylor arrays. Let $\epsilon_i = f_i - \lambda$ and $\epsilon_G = f_G - 3\lambda$, $h_c = \epsilon_A + \epsilon_B + \epsilon_C - \epsilon_G$. Equations (IC8) show that $h_c = h_{00} + aU + cV$ with bounded pure-source a, c. Exact contact value gives

$$h_c = a(u^2 - k^2) + c(u^2 t^2 - k^2 s^2). \tag{IC11}$$

For $d_u = u - k, d_t = t - s$, its finite second Taylor identity is

$$h_c = 2k(a + cs^2)d_u + 2k^2 sc\, d_t + (a + ct^2)d_u^2 + 2kc(t + s)d_u d_t + k^2 c\, d_t^2. \tag{IC12}$$

The coefficient of d_t^2 is infinitesimal. For any affine function $f = f_{00} + a_f U + c_f V$ among (IC8), the exact first difference is

$$f - f(k, s) = (u - k)(u + k)(a_f + c_f t^2) + (t - s)k^2(t + s)c_f. \tag{IC13}$$

These are the required bounded u arrays and infinitesimal t arrays; no appeal to unspecified Taylor membership is needed. The two exact first-jet balances are

$$2(\epsilon_{A0} - \epsilon_{C0}) + kh_{c,u0} = 0, \quad 2(\epsilon_{B0} - \epsilon_{C0}) + sh_{c,t0} = 0. \tag{IC14}$$

They follow from the original stationary contact, and are also checked directly using (NC4). The extra square $c_F F_v^2$ has zero value and first jets there. Apply the accompanying three-circuit contact-lift lemma with the exact conductors

$$G(u - k) = k(A_t - G), \qquad G(t - s) = s(B_t - G). \tag{IC15}$$

Explicit finite allocations of every first normal. Use f_w, f_z, k_w, k_z in (SN2), and define the polynomials

$$\begin{aligned} P_Y(U,V) &= 2(k_w/\eta + k_z),\\ P_E(U,V) &= \frac{2}{\eta^2}(A_w/U - mV/U) - \frac{2}{\eta}(b_0 + b_1 U) + bP_Y(U,V). \end{aligned} \tag{IC16}$$

The quotient $A_w/U - mV/U$ is polynomial, as its displayed numerator shows. For either polynomial P use the exact divisions

$$P = P_{00} + UP_U + VP_V, \quad P_U = \frac{P - P(0,V)}{U}, \quad P_V = \frac{P(0,V) - P(0,0)}{V}. \tag{IC17}$$

The normalized bounds on these exact arrays are supplied by RO. The complete first normals of the original $F/4$ in the frame (IC4) are exactly

$$\begin{aligned} \mathcal{L}_N &= vP_Y,\\ \mathcal{L}_E &= 2mvt^2/\eta^2 + vP_E,\\ \mathcal{L}_x &= \frac{1}{\sqrt{\beta}}\left[2u\theta(m+\beta+\tfrac{1}{2}) - \frac{2m}{\eta}ut^2(1+rU) + 2uVZ_X\right],\\ Z_X &= (\beta+\tfrac{1}{2})\theta - (D+\tfrac{1}{2})(1+hU) + (D+\tfrac{1}{2})(1-r)/\eta. \end{aligned} \tag{IC18}$$

These identities follow by differentiating the entire original parent after the full state substitution (IC4).

Their actual root allocations are

$$\begin{aligned} vP &= \frac{P_{00}}{k}G + u(P_U + t^2P_V)A_t,\\ \frac{2m}{\eta^2}vt^2 &= \frac{2mst}{\eta^2 k}B_t,\\ \mathcal{L}_x &= \frac{2u\theta(m+\beta+\frac{1}{2})}{\sqrt{\beta}\,xs}C_t - \frac{2ms(1+rU)}{\eta k\sqrt{\beta}}B_t + \frac{2vZ_X}{\sqrt{\beta}}A_t. \end{aligned} \tag{IC19}$$

The exact original contact makes each complete normal zero at $(u,t) = (k,s)$. For a normal $\ell = \sum b_i A_i + b_G G$, put $B = \sum b_i + b_G$, and use the explicit monic divisions

$$B = (u-k)\frac{B(u,t) - B(k,t)}{u-k} + (t-s)\frac{B(k,t) - B(k,s)}{t-s}. \tag{IC21}$$

Reciprocal testing-ratio identities. Return to homogeneous state coordinates and set

$$p = uW,\ q = vZ,\ E = \eta p - q, \quad Y = rp + bE + N, \quad B = X + uZ = \xi + \theta uZ.$$

Use the complete raw root list

$$E,\ N,\ N_x = \sqrt{\beta}\xi, \quad P = \sqrt{m}\,vW, \quad Q = \sqrt{\alpha}Z, \quad R_t = u^2W, \quad S_t = uvW. \tag{IC22}$$

$$\sqrt{x}\,p = \sqrt{x/m}\,(u/v)P, \quad Y = rp + bE + N, \quad q = \eta p - E, \tag{IC23}$$

For completeness, with $B = 0$ the exact higher-angle part of the C, A, D terms and half core, after the quadratic base is removed, is

$$\begin{aligned} &U\{AY^2 - 2Dq(p - Y - q) + q^2/2 - pq\}\\ &\quad + V\{C(p-Y)^2 + (A+\tfrac{1}{2})Y^2\} + (A+\tfrac{1}{2})UVY^2 + (D+\tfrac{1}{2})U^2q^2. \end{aligned} \tag{IC24}$$

The terms involving B are equally explicit. Relative to $B = 0$, the D term adds $2DuvB(p - Y - D_u q) + DUVB^2$, and the half core adds $uvB(p - Y - D_u q) + (D_v + UV)B^2/2$. Here

$$B = \frac{N_x}{\sqrt{\beta}} + \frac{\theta u}{\sqrt{\alpha}}Q, \quad uvp = uS_t,$$

The opening-specific proof bounds their coefficients. Write $M = vW - X + uvY - uZ$. The three remaining original terms keep their entire roots:

$$\begin{aligned} \sqrt{m}\,M &= P + \sqrt{m}\,uvY - \sqrt{m}\,B,\\ \sqrt{\alpha}\sqrt{D_u}(vY - Z) &= \sqrt{D_u}\{\sqrt{\alpha}(rS_t + v(bE+N)) - Q\},\\ \sqrt{\beta}\sqrt{D_v}B &= \sqrt{D_v}\{N_x + \sqrt{\beta\theta^2/\alpha}\,uQ\}. \end{aligned} \tag{IC26}$$

B.6.4 Unit positive j: all r openings

Theorem B.26. *Use the original physical native C1 source, normalized by $d = 1$, and retain all original intersections. Suppose that $x = k^2$ is positive infinitesimal and*

$$\eta, \quad \zeta, \quad j_0 = \eta + r - 1 \quad \text{are positive bounded units.} \tag{RO1}$$

Then the complete original parent is tensor SOS at every independent projective test and every state. The original $0 < r < 1$ may approach either endpoint at any positive rate, and the literal $\beta = 0$ face is included. No lower positive real bound on r, $1 - r$, or β is imposed. This extends the covered IC1 domain; it does not cover an infinitesimal j_0, infinitesimal η, or unbounded ζ.

All scalar and Gram identities in this section use the normalized parent $F/4$; multiplication of every square root by 2 restores F.

The proof reuses the unchanged exact scalar and full-normal identities (IC7)–(IC8), (IC16)–(IC19), and the three-circuit lemma. Its new content is a uniform weighting of the two divergent event openings, and a different X normal that retains bounded and zero beta.

Opening-cleared source arrays. Put $\rho = \eta + r$, $y = x\eta\zeta$, $h = 1 - \theta$, $c = C(1 - r)$ and $a = Ar$. The beta numerator (SB1) gives $\theta = x\pi$ with $\pi > 0$ bounded. We do not require π infinitesimal. The original source formulas give, uniformly for all $0 < r < 1$,

$$\begin{gathered} c = \rho/2 + O(x), \quad a = \eta/2 + O(x), \quad D = 1/(2j_0) + O(x), \quad h = 1 + O(x), \\ m/x \sim \rho/(2\zeta), \qquad \alpha/x^2 \sim \rho\zeta(1+\zeta)/2, \qquad B_x\theta/x = \frac{K\rho(h+\zeta)}{2P_d}, \quad B_x = \beta + \tfrac{1}{2} \geq \tfrac{1}{2}. \end{gathered} \tag{RO2}$$

These are finite canceled numerator identities, followed by residue evaluation. Every canceled denominator at $x = 0$ is a product of positive units from $\eta, \zeta, j_0, j_0 + r$ and ρ, with no factor r, $1 - r$, or π. In particular $D > 0$ on this domain; the original $D = 0$ face has empty intersection here, as already proved in (SB2). Both $\theta/\sqrt{\alpha}$ and $B_x\theta^2/\alpha$ are bounded, not necessarily infinitesimal.

The exact source formulas for the two cleared strong weights are

$$c = \frac{K\rho E}{2P_d}, \qquad a = \frac{\eta P_4}{2P_d}.$$

Their order-x differences in (RO2), and the following arrays, have opening-free finite canceled formulas:

$$\frac{P_{Y,00}}{x}, \quad \frac{P_{E,00}}{x}, \quad \frac{\theta(m + B_x)}{x}, \quad \frac{\alpha + (m + B_x)\theta^2}{x^2}, \quad \frac{q_V}{x}. \tag{RO3}$$

Here P_Y, P_E, q_V are precisely (IC16) and (IC8). To check why no unbounded event remains in these arrays, replace every occurrence of $C(1 - r)^2$ by $c(1 - r)$, every Ar^2 by ar, every $C(1 - r)$ by c, and every Ar by a before division. No other occurrence of C or A survives in the scalar or first-normal polynomials. This is a finite symbolic criterion for boundedness on the stated field domain, not a numerical endpoint approximation.

The normalized hard frame. Define the pure-source coefficients

$$b = \frac{r(1-r)}{\eta(j_0 + r)}, \quad H_0 = \frac{\eta(j_0 + r)}{2r(1-r)j_0}, \quad \gamma = \frac{\rho}{2\eta(j_0 + r)}.$$

Both $1/\sqrt{H_0}$ and $1/\sqrt{B_x}$ are bounded; H_0 itself may be unbounded. Put

$$E_t = \eta uW - vZ, \quad N_y = \sqrt{H_0}(Y - ruW - bE_t), \quad N_x = \sqrt{B_x}(X + huZ). \tag{RO4}$$

All three normals vanish at the actual source contact. For the comparison weights

$$C_0 = \frac{\rho}{2(1-r)}, \quad A_0 = \frac{\eta}{2r}, \quad D_0 = \frac{1}{2j_0},$$

the exact quadratic base is

$$C_0(p - Y)^2 + A_0Y^2 + D_0(p - Y - q)^2 + \tfrac{1}{2}(p - Y)^2 - pq = H_0(Y - rp - b(\eta p - q))^2 + \gamma(\eta p - q)^2. \tag{RO5}$$

Thus the divergent normal has already been normalized before taking residue. The cancellation factors needed for its entire hard block are

$$\frac{C_0}{H_0} = \frac{r\rho j_0}{\eta(j_0 + r)}, \quad \frac{A_0}{H_0} = \frac{(1-r)j_0}{j_0 + r}, \quad \frac{D_0}{H_0} = \frac{r(1-r)}{\eta(j_0 + r)}. \tag{RO6}$$

For example, after $p = (v + E_t)/\eta$ and $Z = 1$, the E_t coefficients in $p - Y$ and Y are respectively

$$\frac{1-r}{\eta} - b = \frac{(1-r)j_0}{\eta(j_0+r)}, \qquad \frac{r}{\eta} + b = \frac{r\rho}{\eta(j_0+r)}. \tag{RO7}$$

The strong C, A factors therefore cancel even on those complete first-state directions. For instance an error $\delta_c(p-Y)^2/(1-r)$, with $\delta_c = O(x)$, has a N_y^2 coefficient $\delta_c/((1-r)H_0) = O(x)$, an E_t^2 coefficient $O(x)(1-r)$, and an E_tN_y coefficient $O(x)/\sqrt{H_0}$. The A error has the same properties using rH_0 and the second formula in (RO7). The original higher-angle factors have the same cleared coefficients, multiplied by $U = u^2$ or $V = v^2$.

Suppose first that $u, v > 0$ are infinitesimal and the pure-testing ratio $t = v/u$ is bounded. After the state substitution (RO4), the complete hard quadratic on (E_t, N_y, N_x) consequently has bounded entries and residue

$$\operatorname{diag}(\gamma, 1, 1). \tag{RO8}$$

Here γ is a positive bounded unit even when either r opening is infinitesimal. The additional mt^2 contribution is infinitesimal. The beta and half-core coefficients jointly give the third pivot because m/B_x is infinitesimal. These estimates concern the full hard block, not only its restriction to the contact graph.

The complete scalar and all first normals. Normalize $Z = 1$, homogenizing at the end as in IC1. Use the unchanged positive separated roots

$$A_t = uv, \quad B_t = \frac{kv^2}{su}, \quad C_t = xs, \quad G = kv, \quad F_v = V - y, \qquad s = \sqrt{\eta\zeta}.$$

The scalar at all three normals zero is exactly (IC7). Its coefficient residues are still

$$c_F \sim \frac{\rho}{2\eta^2}, \quad f_A, f_B, f_C \sim \lambda = \frac{\rho}{2\eta}, \quad f_G \sim 3\lambda. \tag{RO9}$$

They do not depend on the residue of π. Every coefficient of U, V in (IC8) is bounded after the clearing described above; for the additional term in f_C use $B_x\theta^2/x^2 = (B_x\theta/x)\pi$. The exact contact value and the two first-jet identities (IC11)–(IC14) are unchanged. Their finite arrays therefore obey precisely the hypotheses of the three-circuit lemma, with s a unit. It gives the same strict finite circuit Gram and the strict F_v pivot. In particular the negative soft defect has been allocated by the entire scalar (IC7), not dropped by comparing only its quadratic base.

The complete first normals are (IC18), with the N line divided by $\sqrt{H_0}$ and the X line divided by $\sqrt{B_x}$ instead of $\sqrt{\beta}$. Their exact root allocations are (IC19) with those same replacements. Every coefficient remains infinitesimal: the N_y factors are bounded source factors times the old cleared polynomials; the X coefficient on C_t is

$$\frac{2u\theta(m + B_x)}{\sqrt{B_x}\, xs},$$

which is infinitesimal by (RO3), and its coefficients on B_t, A_t are respectively $O(k)$ and $O(v)$. In the latter coefficient $Z_X = B_x\theta - (D + \frac{1}{2})(1 + hU) + (D + \frac{1}{2})(1 - r)/\eta$ is bounded. The E_t line uses $m/x, P_{E,00}/x$ and bounded polynomial arrays, as in IC1. There is no negative power of t in any coefficient.

Each coefficient is polynomial in u, t with bounded pure-source arrays; every term depending on t carries u, k or $m/k = O(k)$. The exact monic differences (IC21) hence have bounded u quotient and infinitesimal t quotient after evaluation at $u = k$. The unchanged exact source contact cancels the bare G term in each complete normal. The conductors $G(u - k) = k(A_t - G)$ and $G(t - s) = s(B_t - G)$ therefore allocate all normals with infinitesimal coefficients on the strict circuit roots. Joining this allocation to (RO8) proves the entire parent on the bounded-ratio chart.

The reciprocal ratio chart. For v/u unbounded use the homogeneous root list

$$E_t, \quad N_y, \quad N_x, \quad P_m = \sqrt{m}\, vW, \quad Q_\alpha = \sqrt{\alpha} Z, \quad R_t = u^2 W, \quad S_t = uvW. \tag{RO10}$$

The error of the actual quadratic base relative to (RO5) is an $O(x)$ quadratic in (p, E_t, N_y) after the opening cancellations (RO6)–(RO7). Every $\sqrt{x}\, p$ factor is allocated exactly by

$$\sqrt{x}\, p = \sqrt{x/m}\, (u/v) P_m,$$

with a bounded source unit x/m and an infinitesimal pure-testing ratio u/v. Thus the entire error has zero Gram residue on (RO10). All coefficients in the exact higher-angle identity (IC24) remain bounded after

$Y = rp + bE_t + N_y/\sqrt{H_0}$. This follows term by term from (RO6), $b = O(r(1-r))$, $Ar = a$, and $C(1-r) = c$. Its only nonzero scalar residues are the two positive diagonal entries

$$\kappa_R = \rho\eta/2, \qquad \kappa_S = \rho/2. \tag{RO11}$$

No opening has been substituted before cancellation.

It remains essential to combine beta with its half-core part. Set $B_t^{\text{state}} = X + uZ$ and $J_0 = p - Y - D_u vZ$. Relative to $B_t^{\text{state}} = 0$, the exact D, beta and half-core increment is

$$2(D + \tfrac{1}{2})uvB_t^{\text{state}}J_0 + (D + \tfrac{1}{2})UV(B_t^{\text{state}})^2 + B_x D_v (B_t^{\text{state}})^2. \tag{RO12}$$

This whole-polynomial identity is checked in Use

$$B_t^{\text{state}} = \frac{N_x}{\sqrt{B_x}} + \frac{\theta u}{\sqrt{\alpha}} Q_\alpha.$$

The first two terms of (RO12) have zero Gram residue on (RO10); D is bounded, $1/\sqrt{B_x}$ and $\theta/\sqrt{\alpha}$ are bounded, and every coefficient carries u or v. The three remaining positive terms have their entire roots

$$\begin{aligned} \sqrt{m}\, R_m &= P_m + \sqrt{m}\, uvY - \sqrt{m}\, B_t^{\text{state}}, \\ \sqrt{\alpha}\sqrt{D_u}(vY - Z) &= \sqrt{D_u}\{\sqrt{\alpha}[rS_t + v(bE_t + N_y/\sqrt{H_0})] - Q_\alpha\}, \\ \sqrt{B_x}\sqrt{D_v}B_t^{\text{state}} &= \sqrt{D_v}\{N_x + \sqrt{B_x\theta^2/\alpha}\, uQ_\alpha\}. \end{aligned} \tag{RO13}$$

Each is its leading root plus an infinitesimal-coefficient combination of (RO10). The square root in the last line is pure source and bounded; its factor u supplies the needed infinitesimal even when beta is zero or bounded. The full parent therefore has a bounded Gram with strict diagonal residue

$$(\gamma, 1, 1, 1, 1, \kappa_R, \kappa_S). \tag{RO14}$$

This proves the remaining ratio chart.

All-tests join and exact scope. The earlier small-L, bounded-η remote theorem includes (RO1) with every original opening, so it covers the complement of u, v both infinitesimal. Literal axes have the elementary full-core identities recorded in IC1. Its exact sign gauges apply unchanged. Homogenizing the $Z = 1$ identity gives polynomial state identities, including $Z = 0$; applying them to real and imaginary parts covers complex states. All inverse and radical factors above are pure source or pure testing. This completes the claimed full-parent all-tests certificate.

The newly removed source region is the portion of (RO1) outside IC1, namely either r or $1 - r$ infinitesimal, with all total-scale and native source inequalities retained. Remaining C14.3 source configurations still include infinitesimal η, unbounded ζ, and nonunit or nonpositive j_0 while the actual $j = \eta(1 + xh) - 1 + r$ remains positive. No conclusion on those regions or on C14.4 follows from this theorem.

A local algebraic extension below beta zero. The preceding two local ratio-chart arguments have a stronger precise scope that is useful when another event is restored later. Assume the exact native stationary coefficient formulas (NC1)–(NC4), $x > 0$ infinitesimal, (RO1), $0 < r, h < 1$, and

$$B_x = \beta + \tfrac{1}{2} \geq b_* > 0 \quad \text{for some real } b_*. \tag{RO15}$$

Original source physicality and $\beta \geq 0$ are not required for this local assertion. Then the full C1-form parent is tensor SOS at all independent u, v both infinitesimal, including their literal zero axes. In particular $-1/2 < \beta < 0$ is allowed whenever $\beta + 1/2$ is a positive bounded unit.

Indeed $P_d \sim 1$ before any h estimate, since all its source arguments are bounded. The exact formula $B_x\theta/x = K\rho(h + \zeta)/(2P_d)$ has bounded right side, and (RO15) therefore implies θ/x bounded. All of (RO2)–(RO14) follows, with $1/\sqrt{B_x} \leq 1/\sqrt{b_*}$. In particular $m, \alpha > 0$, $C, A > 0$, and $D \sim 1/(2j_0) > 0$ on this domain. No remaining use of $\beta \geq 0$ occurs in the local scalar, first-normal or hard-block proof: its complete contribution is combined with the half-core as (RO12). At $u = 0$ the signed core contributes $D_v(X^2 + Y^2)/2$, so its X^2 part plus the beta term is $B_x D_v X^2$. At $v = 0$ the same combination is $B_x(X + uZ)^2$ and the remaining half-core is $(uW - Y)^2/2$. Thus both axes are also exact SOS under (RO15). This proves the local and literal-axis assertions under (RO15).

B.6.5 The positive j opening

Theorem B.27. *Use the original physical native C1 source with $d = 1$, $x = k^2 > 0$ infinitesimal, and*

$$\eta,\ \zeta,\ r,\ 1-r \quad \textit{positive bounded units}, \qquad j = \eta(1+xh) - 1 + r \quad \textit{positive infinitesimal}. \tag{JO1}$$

Then the complete original parent is tensor SOS for every independent projective test and every state. Literal $\beta = 0$ is retained. There is no assumption that $j_0 = \eta + r - 1$ is positive: $j_0 = 0$ and a slightly negative j_0 are included. Simultaneous infinitesimal r and j are outside this theorem.

All scalar and Gram identities in this section use the normalized parent $F/4$; multiplication of every square root by 2 restores F.

The divergence $D \asymp 1/j$ must be centered by its full event, including its testing-angle dependence. A normal centered only by its quadratic limit would lose control when $j \ll x$. The following exact frame removes that problem before any residue is taken.

Source bounds and contact-exact coordinates. Write $\theta = 1 - h = x\pi$, where $\pi > 0$ is bounded by (SB1), $B_x = \beta + 1/2 \geq 1/2$, and $y = x\eta\zeta$. Put

$$R_0 = 1 + j - \eta = r + \eta x h, \qquad g(U) = R_0 - \eta h U, \qquad U = u^2,\ V = v^2. \tag{JO2}$$

Thus $g(x) = r$ exactly. The native source formulas, with $r = 1 + j - \eta(1 + xh)$, imply

$$\begin{gathered} C, A = O(1), \quad Dj = \tfrac{1}{2} + O(x), \quad m/x \sim 1/(2\zeta), \quad \alpha/x^2 \sim \zeta(1+\zeta)/2, \\ B_x\theta/x = K(\eta + r)(h + \zeta)/(2P_d), \qquad \theta/\sqrt{\alpha}, \quad B_x\theta^2/\alpha = O(1). \end{gathered} \tag{JO3}$$

In particular $D > 0$, so its square root and inverse are pure source and legitimate. Set

$$\begin{gathered} p = uW, \quad q = vZ, \quad E_t = \eta p - q, \qquad N_d = \sqrt{D}\,(J_t + jp), \\ J_t = uW + uvX - Y - vZ, \qquad N_x = \sqrt{B_x}(X + huZ), \quad n = N_d/\sqrt{D}, \quad \xi = N_x/\sqrt{B_x}. \end{gathered} \tag{JO4}$$

The exact inverse state formulas are polynomial and homogeneous:

$$Y_0 = (1+j)p - q - n, \qquad Y = Y_0 - hUq + uv\xi, \qquad X = -huZ + \xi, \qquad J_t = n - jp. \tag{JO5}$$

All three normals vanish at the actual source contact. At their zero graph, $q = \eta p$ and $Y = g(U)p$, so the original native contact graph is recovered at $U = x$ without changing any source contact jet.

For later use introduce positive reference weights

$$\widetilde{C} = \frac{1+j}{2(\eta - j)}, \quad \widetilde{A} = \frac{\eta}{2R_0}, \quad \widetilde{D} = \frac{1}{2j}.$$

Since $R_0, 1 - R_0, \eta$ are units, exact source numerator division gives

$$C - \widetilde{C} = O(x), \qquad A - \widetilde{A} = O(x), \qquad Dj - \tfrac{1}{2} = O(x). \tag{JO6}$$

The last assertion is deliberately about Dj, not D. The canceled denominators of (JO3),(JO6), at $x = j = 0$, contain only the unit factors $\eta, \zeta, 1 - \eta$ and constants. No comparison of j/x has been made.

The complete bounded hard block. The reference quadratic base has the exact identity

$$\begin{aligned} &\widetilde{C}(p - Y_0)^2 + \widetilde{A}Y_0^2 + \widetilde{D}(p - Y_0 - q)^2 + \tfrac{1}{2}(p - Y_0)^2 - pq \\ &\quad = \widetilde{\gamma}E_t^2 + \widetilde{H}\{(1 - \widetilde{b})E_t - n\}^2, \\ &\widetilde{H} = \frac{\eta(j + R_0)}{2R_0(1 - R_0)j}, \quad \widetilde{b} = \frac{R_0(1 - R_0)}{\eta(j + R_0)}, \quad \widetilde{\gamma} = \frac{1+j}{2\eta(j + R_0)}. \end{aligned} \tag{JO7}$$

No p term survives. Furthermore

$$1 - \widetilde{b} = \frac{j(1+j)}{\eta(j + R_0)}, \qquad \widetilde{H}/D \sim 1. \tag{JO8}$$

On (E_t, N_d) this is a bounded hard Gram with diagonal residue $(1/(2\eta(1-\eta)), 1)$ and vanishing off-diagonal residue. The actual-base error has coefficients $O(x)$ in (p, E_t, N_d). For its apparently singular part use

$$(D - \widetilde{D})(n - jp)^2, \quad \frac{D - \widetilde{D}}{D} = O(x), \quad \frac{(D - \widetilde{D})j}{\sqrt{D}} = O(x\sqrt{j}), \quad (D - \widetilde{D})j^2 = O(xj). \tag{JO9}$$

These are consequences of (JO3),(JO6) and hold for every relative source rate. All other weights in the base are bounded.

For bounded $t = v/u$ at $u, v > 0$ both infinitesimal, put $Z = 1$. Then $p = (v + E_t)/\eta$ in (JO5). The higher-angle terms have bounded infinitesimal hard coefficients, and the m hard coefficient mt^2 is infinitesimal. Combining beta with its half-core part gives the third hard pivot on N_x. Thus the full hard residue is

$$\operatorname{diag}(1/(2\eta(1-\eta)), 1, 1). \tag{JO10}$$

The exact higher-angle identity given below also proves these hard bounds directly; no state normal is suppressed here.

An exact scalar three-circuit with bounded polynomial coefficients. Use $s = \sqrt{\eta\zeta}$ and the same separated roots

$$A_t = uv, \quad B_t = \frac{kv^2}{su}, \quad C_t = xs, \quad G = kv, \quad F_v = V - y.$$

On the zero graph of (JO4), the exact scalar is

$$\mathcal{S} = c_F F_v^2 + f_A(U,V)A_t^2 + f_B B_t^2 + f_C(U)C_t^2 - f_G G^2, \tag{JO11}$$

where, with $g = g(U)$,

$$\begin{aligned}
c_F &= \frac{2mR_0 + C(1-R_0)^2 + (A + \alpha + \frac{1}{2})R_0^2}{\eta^2},\\
q_U &= \alpha + (m + B_x)\theta^2,\\
q_V &= \frac{C(1-R_0)^2 + AR_0^2 + (D + \frac{1}{2})j^2}{\eta^2} - \frac{2m\theta}{\eta} - \frac{(2\alpha+1)R_0}{\eta} - \tfrac{1}{2},\\
f_B &= \frac{m\zeta}{\eta x}, \quad f_C = \frac{\alpha - c_F y^2 + q_U U}{\eta\zeta x^2}, \quad f_G = -\frac{q_V + 2c_F y}{x},\\
f_A &= \frac{mV(g^2 - 2\eta h)}{\eta^2} - \frac{2m\theta g}{\eta} + C(1+V)\{2h(1-R_0)/\eta + h^2 U\}\\
&\quad + \{A(1+V) + (\alpha + \tfrac{1}{2})V\}\{g^2/\eta^2 - 2R_0 h/\eta + h^2 U\}\\
&\quad - (2\alpha+1)(g/\eta - h) + B_x\theta^2 - \tfrac{1}{2}.
\end{aligned} \tag{JO12}$$

In particular f_A is an explicit bounded polynomial of degree at most two in U and one in V; the divergent D is absent. The only D factor elsewhere is $Dj^2 = O(j)$. These formulas follow by substituting $W = v/(\eta u)$, $X = -hu$, $Y = g(U)v/\eta$ in the entire parent;

The source bounds and (JO12) give

$$c_F \sim 1/(2\eta^2), \qquad f_A, f_B, f_C \sim \lambda = 1/(2\eta).$$

Exact scalar contact gives $f_G = f_A(x,y) + f_B + f_C(x)$, proving that f_G is bounded and $f_G \sim 3\lambda$ without estimating a singular quotient q_V/x. Every nonconstant coefficient in f_A, f_C is bounded; for q_U/x^2 use (JO3).

For clarity the required finite Taylor step works for these higher polynomial coefficients too. Let $H(u,t) = f_A(u^2, u^2t^2) + f_B + f_C(u^2) - f_G$. It is a finite polynomial in u, t with bounded source arrays and $H(k,s) = 0$. Write $\Delta_u = u - k$, $\Delta_t = t - s$. A finite identity, obtained by monic polynomial division, is

$$H = H_u(k,s)\Delta_u + H_t(k,s)\Delta_t + H_{uu}\Delta_u^2 + H_{ut}\Delta_u\Delta_t + H_{tt}\Delta_t^2, \tag{JO13}$$

where one may take

$$\begin{aligned}
H_{uu} &= \frac{H(u,t) - H(k,t) - (u-k)H_u(k,t)}{(u-k)^2},\\
H_{ut} &= \frac{H_u(k,t) - H_u(k,s)}{t-s},\\
H_{tt} &= \frac{H(k,t) - H(k,s) - (t-s)H_t(k,s)}{(t-s)^2}.
\end{aligned}$$

These displayed divisions mean their finite polynomial quotients, not mixed inverses. The first two arrays are bounded and H_{tt} is infinitesimal: every t-dependent monomial of $H(k,t)$ contains k^2. The coefficient first differences similarly have a bounded u quotient and infinitesimal t quotient. Exact first angular stationarity supplies the two unchanged contact-jet balances of the three-circuit lemma. The chain rule is legitimate here because $g(x) = r$ and the original first state derivatives vanish at contact. Consequently (JO11) has the strict finite three-circuit Gram and the strict F_v pivot, including the complete soft defect.

Every complete first normal. The following bounded polynomials are explicit:

$$\begin{aligned}
P_Y/2 &= m\{V(1+gU)/\eta - \theta U\} - CD_v(1-g)/\eta + AD_uD_vg/\eta \\
&\quad + (D+\tfrac12)j/\eta + (\alpha+\tfrac12)D_u(gV/\eta - 1), \\
P_E &= \frac{2mV}{\eta^2}\{g+1+j+g(1+j)U\} - \frac{2m\theta}{\eta}\{1+(1+j)U\} \\
&\quad - \frac{2CD_vj(1-g)}{\eta^2} + \frac{2AD_uD_vg(1+j)}{\eta^2} + \frac{2(D+\frac12)j^2}{\eta^2} \\
&\quad + \frac{2(\alpha+\frac12)D_u(1+j)}{\eta}(gV/\eta - 1), \\
Z_X &= B_x\theta + P_Y/2 - (D+\tfrac12)j/\eta.
\end{aligned} \tag{JO14}$$

Their only D factors are Dj and Dj^2; D cancels entirely from Z_X. Direct full-parent differentiation gives

$$\begin{aligned}
\mathcal{L}_d &= -vP_Y/\sqrt{D}, \\
\mathcal{L}_E &= 2mvt^2/\eta^2 + vP_E, \\
\mathcal{L}_x &= \frac{1}{\sqrt{B_x}}\{2u\theta(m+B_x) - 2mut^2(1+gU)/\eta + 2uVZ_X\}.
\end{aligned} \tag{JO15}$$

No normal is missing. For $P = P_Y, P_E$ use the exact finite division $P = P_{00} + UP_U + VP_V$. Contact implies respectively

$$\begin{aligned}
P_{Y,00}/x &= -P_{Y,U}(x,y) - \eta\zeta P_{Y,V}(y), \\
P_{E,00}/x &= -2(m/x)\eta\zeta/\eta^2 - P_{E,U}(x,y) - \eta\zeta P_{E,V}(y).
\end{aligned} \tag{JO16}$$

Thus these coquotients are bounded by finite formulas with no j inverse. Allocate vP by $vP = (P_{00}/k)G + u(P_U + t^2P_V)A_t$; allocate the additional $2mvt^2/\eta^2$ on B_t with coefficient $2mst/(\eta^2k)$. The X line allocates exactly on (C_t, B_t, A_t) with coefficients

$$\frac{2u\theta(m+B_x)}{\sqrt{B_x}xs}, \quad -\frac{2ms(1+gU)}{\eta k\sqrt{B_x}}, \quad \frac{2vZ_X}{\sqrt{B_x}}. \tag{JO17}$$

Every displayed coefficient is infinitesimal; for the first use $\theta(m+B_x)/x = O(1)$, and for the others use $m/x = O(1)$. The N_d row has the additional bounded factor $1/\sqrt{D}$. Each sum of coefficient functions vanishes at contact. Its finite u difference is bounded and its t difference at $u = k$ is infinitesimal, because every t dependence carries u or k. The two unchanged conductors therefore place every whole normal on the strict circuit roots with infinitesimal coefficients. Joining (JO10) proves the full parent on the bounded-ratio chart.

The reciprocal testing ratio: a complete polynomial remainder. For v/u unbounded, return to homogeneous state coordinates and use

$$E_t,\ N_d,\ N_x,\ P_m = \sqrt{m}\,vW,\ Q_\alpha = \sqrt{\alpha}Z,\ R_t = u^2W,\ S_t = uvW. \tag{JO18}$$

The entire base error from (JO9) has zero Gram residue on this list: its p^2 coefficient is $O(x)$ and $\sqrt{x}\,p = \sqrt{x/m}(u/v)P_m$. Every coefficient is a product of a bounded source factor and a bounded testing factor; u/v is pure testing.

Here is the full higher-angle identity used to control the other terms. Write $T_s = -hUq + uv\xi$, so $Y = Y_0 + T_s$, with Y_0 in (JO5). After removing the actual quadratic base and the three entire positive terms mR_m^2, $\alpha D_u(vY - Z)^2$ and $B_xD_v(X + uZ)^2$, the exact remainder is

$$\begin{aligned}
\mathcal{R} &= C\{V(p-Y_0)^2 - 2D_v(p-Y_0)T_s + D_vT_s^2\} \\
&\quad + A\{(D_uD_v - 1)Y_0^2 + 2D_uD_vY_0T_s + D_uD_vT_s^2\} \\
&\quad + \tfrac12 D_uVY^2 - UqY + (h - \tfrac12)Uq^2 - uvq\xi.
\end{aligned} \tag{JO19}$$

This is a quadratic in the full state, with bounded coefficients C, A, h and no D. It is checked as a whole polynomial. Insert $q = \eta p - E_t$, $n = N_d/\sqrt{D}$, $\xi = N_x/\sqrt{B_x}$. Every term contains U, V or uv. Cross terms with p have infinitesimal coefficients on (JO18), using $Up = uR_t$, $Vp = vS_t$ and $uvp = uS_t$. Hard quadratic terms have zero residue. Terms $U^aV^bp^2$ are allocated on R_t^2 when $a > 0$, and on S_t^2 when $a = 0, b > 0$. Their two only nonzero residues, verified from (JO19), are

$$\kappa_R = \eta/2, \qquad \kappa_S = 1/2. \tag{JO20}$$

Every remaining such coefficient contains another U or V. These identities explicitly exhaust the scalar and every normal coupling of (JO19).

The three retained positive terms have leading roots $P_m, -Q_\alpha, N_x$. Their remaining root coefficients are infinitesimal because m, α are infinitesimal and $\theta/\sqrt{\alpha}$, $B_x\theta^2/\alpha$ are bounded:

$$\sqrt{m}\,R_m = P_m + \sqrt{m}\,uvY - \sqrt{m}(\xi + \theta uZ), \quad \sqrt{B_x D_v}(X + uZ) = \sqrt{D_v}\{N_x + \sqrt{B_x\theta^2/\alpha}\,uQ_\alpha\}.$$

The alpha term is $\sqrt{D_u}(\sqrt{\alpha}\,vY - Q_\alpha)$. Thus the complete parent has a bounded Gram on (JO18) with strictly positive diagonal residue

$$(1/(2\eta(1-\eta)), 1, 1, 1, 1, \eta/2, 1/2). \tag{JO21}$$

This proves the reciprocal-ratio chart and completes the local proof.

All-tests join and receipt. The common small-L, bounded-η remote theorem applies to (JO1), including the exact original source openings and literal beta zero. It covers the complement of both-infinitesimal testing angles. The original exact sign gauges, literal axes, homogeneous state extension and real/imaginary decomposition apply as in IC1 and RO. All coefficients and radicals used above are pure-field quantities; no mixed inverse or comparison of source and testing rates occurs.

They verify the original full-parent substitution, every complete first normal, the scalar identity, all newly used source denominator factors, the reference hard cancellation, the whole reciprocal higher-angle identity and both strict soft pivots. The finite polynomial bounds and full-rates circuit allocation are the proof above, rather than an inference from those checks. This admits precisely (JO1) and its exact symmetry images. It does not remove the simultaneous r, j opening or any unbounded-ζ or infinitesimal-η source.

B.6.6 The simultaneous positive r, j opening

Theorem B.28. *Use the original physical native C1 source, with $d = 1$ and $x = k^2 > 0$ infinitesimal. Suppose η, ζ are positive bounded units and r and the actual $j = \eta(1 + xh) - 1 + r$ are both positive infinitesimals. Then the complete original parent is tensor SOS for every independent projective test and every state. Literal $\beta = 0$ is retained. No relative rate among x, r, j or between source and testing parameters is imposed.*

All scalar and Gram identities in this section use the normalized parent $F/4$; multiplication of every square root by 2 restores F.

The essential change is to combine the whole positive A, D events before choosing the scalar graph. The weighted uvX term is retained inside a hard root. Solving for $p = uW$, rather than for $q = vZ$, then gives a polynomial inverse with a pure-source unit denominator.

Source bounds and the harmonic frame. Write $U = u^2$, $V = v^2$, $D_u = 1 + U$, $D_v = 1 + V$, $\theta = 1 - h = x\pi$, $B_x = \beta + 1/2$, and

$$T = 1 + xh, \qquad e = \eta T = 1 - r + j, \qquad y = x\eta\zeta. \tag{HO1}$$

The bounded-η source bound gives $\pi = O(1)$. Here $e, \eta, h, 1 - r \sim 1$. Exact native numerator cancellation gives

$$\begin{gathered} C - C_* = O(x), \qquad Ar - e/2 = O(x), \qquad Dj - 1/2 = O(x), \\ C_* = \frac{1+j}{2(1-r)}, \qquad A_* = \frac{e}{2r}, \qquad D_* = \frac{1}{2j}, \\ m/x \sim 1/(2\zeta), \qquad \alpha/x^2 \sim \zeta(1+\zeta)/2, \\ B_x\theta/x = O(1), \qquad q_U/x^2 = O(1), \quad q_U = \alpha + (m + B_x)\theta^2. \end{gathered} \tag{HO2}$$

This proves boundedness at arbitrary simultaneous source rates, rather than just nonvanishing at a selected boundary. In particular $A, D > 0$ and both are unbounded. Also $\theta/\sqrt{\alpha}$ and $B_x\theta^2/\alpha$ are bounded, including $\beta = 0$.

Define pure-source positive quantities

$$\mathsf{H} = A + D, \qquad \rho = \frac{D}{\mathsf{H}}, \qquad \mathsf{K} = \frac{AD}{\mathsf{H}}, \qquad \delta = \frac{Ar + Dj}{\mathsf{H}}. \tag{HO3}$$

Both H and K are unbounded, $0 < \rho < 1$, and $\mathsf{K}^{-1} = A^{-1} + D^{-1} = O(r + j)$. Put $p = uW$, $q = vZ$, $J_0 = p + uvX - q$, and use the full-event roots

$$\begin{aligned} E &= \sqrt{\mathsf{K}}\{J_0 - (r - j)p\}, \\ N &= \sqrt{\mathsf{H}}\{Y - \rho J_0 - \delta p\}, \\ N_x &= \sqrt{B_x}(X + huZ), \qquad \ell = E/\sqrt{\mathsf{K}}, \quad n = N/\sqrt{\mathsf{H}}, \quad \xi = N_x/\sqrt{B_x}. \end{aligned} \tag{HO4}$$

All three vanish at the actual source contact. Their exact inverse is

$$\begin{aligned} p &= \frac{(1+hU)q - uv\xi + \ell}{e}, \quad Y = rp + \rho\ell + n, \qquad X = -huZ + \xi, \\ J_t &= -jp + (1-\rho)\ell - n, \qquad J_t = p + uvX - Y - q. \end{aligned} \tag{HO5}$$

The denominator e is pure source. In particular no inverse of $1 + hU$ occurs, even if the testing and source rates are unrelated. The useful exact weighted identities are

$$\frac{A\rho^2}{\mathsf{K}} = \rho, \qquad \frac{A}{\mathsf{H}} = 1 - \rho, \qquad \frac{D(1-\rho)^2}{\mathsf{K}} = 1 - \rho, \qquad \frac{D}{\mathsf{H}} = \rho. \tag{HO6}$$

The complete hard block and its reference cancellation. The A, D base has the exact identity

$$\begin{aligned} AY^2 + DJ_t^2 = N^2 + E^2 + 2p\left\{\frac{Ar + Dj}{\sqrt{\mathsf{H}}}N + \sqrt{\mathsf{K}}(r-j)E\right\} \\ + (Ar^2 + Dj^2)p^2. \end{aligned} \tag{HO7}$$

Both displayed cross coefficients and the last coefficient are infinitesimal: the squared E cross coefficient is $O(r+j)$ and the squared N cross coefficient is $O(\mathsf{H}^{-1})$. For bounded testing ratio $t = v/u$, (HO5) has bounded coefficients on the hard roots, and the only potentially large m coefficient is mt^2/K, which is infinitesimal. The higher A term is controlled by (HO6); for example

$$\frac{Ar^2}{\mathsf{K}} = r^2 + \frac{Ar}{Dj}rj = O((r+j)^2), \qquad \frac{Ar\rho}{\mathsf{K}} = r. \tag{HO8}$$

Consequently its hard coefficients carry an additional U or V. The bounded C and half-core terms have infinitesimal coefficients on E, N, and combining the beta term with its half-core part gives the third hard pivot. The full hard residue on (E, N, N_x) is $\mathrm{diag}(1, 1, 1)$.

A stronger exact base statement is needed for unbounded t. Set $\widetilde{q} = q - uvX = ep - \ell$, so that $J_t = p - Y - \widetilde{q}$. The reference quadratic form

$$\mathcal{B}_* = C_*(p-Y)^2 + A_*Y^2 + D_*(p - Y - \widetilde{q})^2 + \tfrac{1}{2}(p-Y)^2 - p\widetilde{q} \tag{HO9}$$

has no soft p coordinate after substituting $Y = rp + \rho\ell + n$ and $\widetilde{q} = ep - \ell$:

$$\mathcal{B}_* = (A_* + C_* + \tfrac{1}{2})(\rho\ell + n)^2 + D_*\{(1-\rho)\ell - n\}^2. \tag{HO10}$$

This is an exact contact cancellation, checked before taking residues. Since $A_*/A = 1 + O(x)$ and $D_*/D = 1 + O(x)$, (HO6) shows that its bounded hard Gram on (E, N) has residue $\mathrm{diag}(1, 1)$. Replacing the reference weights by the actual C, A, D creates a quadratic error with coefficients $O(x)$ in (p, E, N). For the singular weights this follows from $(A - A_*)Y^2 = O(x)\,AY^2$ and $(D - D_*)J_t^2 = O(x)\,DJ_t^2$, using (HO6),(HO7); neither an x/r nor an x/j estimate is used.

The exact bounded-ratio scalar circuit. At $E = N = N_x = 0$ and $Z = 1$, let

$$w(U) = \frac{1 + hU}{e}, \qquad p = w(U)v, \quad Y = rw(U)v, \quad X = -hu. \tag{HO11}$$

Since $w(x) = 1/\eta$, this graph agrees exactly with the native source contact. The source contact value and all first state and angular jets are therefore common by the chain rule.

For compact scalar coefficients set $a = C(1-r)^2$, $b = Ar^2$, $d = (D + 1/2)j^2$, $c = (\alpha + 1/2)r^2$, and

$$L_A(U) = 1 + 2h + (2h + h^2)U + h^2U^2.$$

The complete scalar is exactly

$$\mathcal{S} = c_F(V - y)^2 + f_A(U, V)A_t^2 + f_BB_t^2 + f_C(U)C_t^2 - f_GG^2, \tag{HO12}$$

on the separated roots $A_t = uv$, $B_t = kv^2/(su)$, $C_t = xs$, $G = kv$, where $s = \sqrt{\eta\zeta}$. The coefficients are

$$\begin{aligned}
c_F &= \frac{2m(h+r)+a+b+c}{e^2},\\
q_V &= \frac{a+b+d}{e^2} - \frac{2m\theta}{e} - \frac{(2\alpha+1)r}{e} - \tfrac{1}{2},\\
f_B &= \frac{m\eta\zeta}{e^2x}, \qquad f_C = \frac{\alpha - c_F y^2 + q_U U}{\eta\zeta x^2}, \qquad f_G = -\frac{q_V + 2c_F y}{x},\\
f_A &= \frac{mV}{e^2}\{(h+r)^2 + 2hr + 2hr(h+r)U + h^2r^2U^2\} - \frac{2m\theta}{e}(h+r+hrU)\\
&\quad + \frac{a}{e^2}(2h+h^2U)(1+V) + \frac{b}{e^2}L_A(U)(1+V) + \frac{d}{e^2}(2h+h^2U) + \frac{c}{e^2}L_A(U)V\\
&\quad - \frac{(2\alpha+1)r}{e}(1+h+hU) + B_x\theta^2 - \tfrac{1}{2}.
\end{aligned} \tag{HO13}$$

The only opening weights here are Ar^2 and Dj^2, both infinitesimal. Thus f_A is a bounded polynomial of degree at most two in U and one in V, and all nonconstant coefficients of f_C are bounded by (HO2). Direct source reduction gives

$$c_F \sim \tfrac{1}{2}, \qquad f_A, f_B, f_C \sim \tfrac{1}{2}. \tag{HO14}$$

Exact scalar contact gives $f_G = f_A(x,y) + f_B + f_C(x) \sim 3/2$, without separately estimating q_V/x.

Apply the finite three-circuit lemma with $\lambda = 1/2$. The polynomial Taylor arrays are the monic quotients in (JO13), now applied to $f_A(u^2, u^2t^2) + f_B + f_C(u^2) - f_G$. Their u and mixed arrays are bounded; the t,t array and the coefficient t first differences are infinitesimal, since every t dependence at $u = k$ contains k^2. The two exact angular contact jets give the two circuit balances. This supplies the strict three-circuit Gram and the strict $V - y$ pivot for the entire scalar, including its negative soft term.

All complete first normals. The following formulas come from differentiating the entire original parent after (HO5), with $Z = 1$. Put $a_E = r/e + \rho$, $b_E = (1-r)/e - \rho$, $c_E = 1 - \rho - j/e$. The bounded polynomials are

$$\begin{aligned}
P_N/2 &= m\{wV(1+rU) - \theta U\} - CD_v(1-r)w + AD_uD_vrw\\
&\quad + (D+\tfrac{1}{2})jw + (\alpha+\tfrac{1}{2})D_u(rwV-1),\\
P_E/2 &= \frac{mV}{e^2}(h+r+hrU) - \frac{m\theta}{e} + ma_E\{wV(1+rU) - \theta U\}\\
&\quad + CD_v(1-r)wb_E + AD_uD_vrwa_E - (D+\tfrac{1}{2})jwc_E + (\alpha+\tfrac{1}{2})D_u(rwV-1)a_E,\\
Z_X &= \frac{m\theta(1+rU)}{e} + B_x\theta\\
&\quad - \frac{CD_v(1-r)^2w + AD_uD_vr^2w + (D+\tfrac{1}{2})j^2w + (\alpha+\tfrac{1}{2})D_ur(rwV-1)}{e}.
\end{aligned} \tag{HO15}$$

Their only opening-weight factors are Ar, Dj, Ar^2, Dj^2. Every coefficient is bounded uniformly in the source rates and $0 < \rho < 1$. The three complete first normals are

$$\begin{aligned}
\mathcal{L}_N &= \frac{vP_N}{\sqrt{\mathsf{H}}},\\
\mathcal{L}_E &= \frac{2mvt^2/e^2 + vP_E}{\sqrt{\mathsf{K}}},\\
\mathcal{L}_x &= \frac{1}{\sqrt{B_x}}\left\{2u\theta(m+B_x) - 2mwut^2(1+rU)\left(1 + \frac{V(1+rU)}{e}\right) + 2uVZ_X\right\}.
\end{aligned} \tag{HO16}$$

For either $P = P_N, P_E$, use its exact finite monic decomposition $P = P_{00} + UP_U + VP_V$. State contact gives

$$\begin{aligned}
P_{N,00}/x &= -P_{N,U}(x,y) - \eta\zeta P_{N,V}(y),\\
P_{E,00}/x &= -2(m/x)\eta\zeta/e^2 - P_{E,U}(x,y) - \eta\zeta P_{E,V}(y).
\end{aligned} \tag{HO17}$$

Thus these coquotients are bounded without any new opening inverse. Allocate $vP = (P_{00}/k)G + u(P_U + t^2P_V)A_t$. The extra E term allocates on B_t with coefficient $2mst/(e^2k\sqrt{\mathsf{K}})$. The three X terms allocate respectively on C_t, B_t, A_t, with coefficients

$$\frac{2u\theta(m+B_x)}{xs\sqrt{B_x}}, \qquad \frac{-2msw(1+rU)\{1+V(1+rU)/e\}}{k\sqrt{B_x}}, \qquad \frac{2vZ_X}{\sqrt{B_x}}. \tag{HO18}$$

Every displayed allocation coefficient is infinitesimal. The coefficient sums vanish at contact by the exact state jets; their finite u first differences are bounded and their t first differences at $u = k$ are infinitesimal. This follows directly from the bounded polynomial arrays in (HO15): every t dependence carries u or k. The two unchanged circuit conductors therefore place every whole normal on the strict circuit roots with infinitesimal coefficients. Together with the complete hard block, this proves all bounded t, including infinitesimal t, without a missing state normal.

The reciprocal ratio: retain both soft directions. If v/u is unbounded, work homogeneously and use seven roots

$$E, \quad N, \quad N_x, \quad P_m = \sqrt{m}\, vW, \quad Q_\alpha = \sqrt{\alpha} Z, \quad R_t = uvZ, \quad S_t = uvW. \tag{HO19}$$

The base error following (HO10) has zero Gram residue on this list: use the exact separated identity $\sqrt{x}\, p = \sqrt{x/m}(u/v)P_m$. This absorbs every $O(x)$ coefficient in (p, E, N), since x/m is a source unit and u/v is a pure-testing infinitesimal.

After removing the actual quadratic base (HO9), with actual C, A, D, and the three entire positive terms mM_t^2, $\alpha D_u(vY - Z)^2$, $B_x D_v(X + uZ)^2$, the full remaining quadratic is exactly

$$\begin{aligned} \mathcal{R} = {} & CV(p - Y)^2 + A(U + V + UV)Y^2 + \tfrac{1}{2} D_u V Y^2 - \theta U q Y \\ & + (h - \tfrac{1}{2})Uq^2 + \tfrac{1}{2} h^2 U^2 q^2 - uv\xi\{Y + (1 + hU)q\} + \tfrac{1}{2} UV\xi^2. \end{aligned} \tag{HO20}$$

This identity retains the whole D event in the quadratic base, so no coefficient containing a bare divergent D remains. It is checked as one complete homogeneous polynomial.

Here is a finite allocation of every term. The exact identity

$$up = \frac{(1 + hU)R_t - u^2 v\xi + uE/\sqrt{\mathsf{K}}}{e} \tag{HO21}$$

has bounded coefficients and uses no mixed inverse. The C term has leading root $\sqrt{C}(1-r)S_t$; its other coefficients are $-\sqrt{C}\,\rho v/\sqrt{\mathsf{K}}$ and $-\sqrt{C}\, v/\sqrt{\mathsf{H}}$ on E, N, both infinitesimal. All three A higher terms have zero residue. For example,

$$\sqrt{A}\, uY = \sqrt{A}\, r(up) + \sqrt{\rho}\, uE + \sqrt{1 - \rho}\, uN, \qquad \sqrt{A}\, vY = \sqrt{A}\, rS_t + \sqrt{\rho}\, vE + \sqrt{1 - \rho}\, vN. \tag{HO22}$$

Here $\sqrt{A}\, r = O(\sqrt{r})$ and (HO8),(HO21) control every coefficient; the uvY term is v times the first. The half-VY^2 term is smaller by the same identities. The $-\theta UqY$ term is $-\theta R_t(uY)$ and has zero residue. The next two terms are $(h - 1/2)R_t^2$ and $h^2 U R_t^2/2$. Finally, the cross term is $-\xi v(uY) - \xi v(1 + hU)R_t$ and all its coefficients, as well as those of $UV\xi^2/2$, are infinitesimal. Thus this allocation exhausts the complete quadratic (HO20); its only nonzero soft residues are $1/2$ on each of R_t, S_t.

The three retained positive terms have leading roots $P_m, -Q_\alpha, N_x$:

$$\begin{aligned} \sqrt{m}\, M_t &= P_m + \sqrt{m}\, uvY - \sqrt{m}\, \xi - \sqrt{m/\alpha}\, \theta u Q_\alpha, \\ \sqrt{\alpha D_u}(vY - Z) &= \sqrt{D_u}(\sqrt{\alpha}\, vY - Q_\alpha), \\ \sqrt{B_x D_v}(X + uZ) &= \sqrt{D_v}\{N_x + \sqrt{B_x \theta^2/\alpha}\, uQ_\alpha\}. \end{aligned} \tag{HO23}$$

Every other coefficient here is infinitesimal by (HO2),(HO21), because $\theta\sqrt{m/\alpha} = O(\sqrt{x})$. The complete bounded Gram on (HO19) therefore has strictly positive diagonal residue

$$\operatorname{diag}(1, 1, 1, 1, 1, 1/2, 1/2). \tag{HO24}$$

This proves the reciprocal chart for the full parent and every state.

All-tests join and receipt. The common small-L, bounded-η remote theorem covers the complement of both-infinitesimal testing angles on this original physical source domain. Its exact source openings and literal beta zero remain available. The original exact sign gauges, literal axes, homogeneous state extension and real/imaginary decomposition join as in IC1, RO and JO. All roots, radicals and inverses above are pure-field or separated; every bound is a product of a source bound and a testing bound.

Those checks support the finite bounds and circuit allocation proved above; they are not being used as a substitute for them. This theorem removes precisely the simultaneous positive infinitesimal r, j opening with unit η, ζ.

Corollary B.29 (The entire unit-η, ζ small-L region)**.** *Within the original physical native small-L C1.4 source, suppose $x > 0$ is infinitesimal and η, ζ are positive bounded units. Then the full parent is tensor SOS at every independent projective test and every state, with all original source intersections retained.*

Proof. The positive actual gap $j = \eta(1 + xh) - 1 + r$ is bounded. If j is a unit, then $j_0 = j - \eta xh$ is a positive unit, so RO applies, including either r opening. If j is infinitesimal, then $1 - r \sim \eta$ is a unit. A unit r is exactly the JO domain; an infinitesimal r is exactly the HO domain. These alternatives exhaust the stated source, require no comparison of j with x, and each supplies its complete local, remote and endpoint join. □

B.6.7 The simultaneous small-η parent identity

Polynomial identity. This is an independently reconstructed exact identity for the complete original C1 parent, not a certificate admitting C14.3b. It retains the same stationary source, $\Lambda = n = 0$, $\beta \geq 0$, all j and $1 - r$ openings, independent source/testing fields, and the remaining local v/u-infinitesimal testing chart. No mixed inverse or comparison of source and testing rates is used below.

Write $\mathcal{P} = F/4$, initially at $Z = 1$. Set

$$a = 1/\zeta, \quad t = \eta, \quad y = tx/a, \quad U = u^2, \quad V = v^2, \quad B = \beta + \tfrac{1}{2}, \quad D_p = D + \tfrac{1}{2},$$

$$E = tuW - v, \quad Q_t = Y - rv/t, \quad P = E/t - Q_t,$$
$$\xi = X + hu, \quad R_D = v(U - x), \quad T_D = R_D - uv\xi/h.$$

Thus $Q_t = rE/t + \nu$ in the R15 notation. Put

$$q = \alpha + (m + B)(1 - h)^2, \quad \mu = m - a^2 q, \quad \lambda = \mu/(tax), \quad H = avW - u - a\xi,$$

$$F_0 = V - y, \quad F_b = F_0 + vE, \quad F_t = F_0 + (t/r)vQ_t, \quad T_0 = A + \alpha + \tfrac{1}{2}, \quad T_m = T_0 + m, \quad e_4 = r^2 T_m/t^2.$$

Here $F_t = \widetilde{F}$ and $F_b = \overline{F}$. Define pure-source coefficients

$$b = Ar/t - m(1 - h) - \alpha - \tfrac{1}{2}, \qquad b' = b + e_4 yt/(2r),$$

$$\kappa = 2[B(1 - h) - D_p j/t - mr/t], \qquad \gamma = 2b'/h + 2D_p j/r - 2tB(1 - h)/r.$$

They satisfy $b' = O(x)$, $t\kappa \sim -1$ and $\gamma \sim 1$.

Complete parent, with no omitted hard row. Let

$$L = UV + (ax/t)V^2/U + tx^2/a - 3xV,$$

$$\mathscr{C} = \lambda L + \frac{\mu V}{t^2 U}E^2 + 2\left(\frac{\mu v^3}{t^2 U} - \lambda xv\right)E.$$

The exact full-parent identity is

$$\begin{aligned}\mathcal{P} = {} & qH^2 + \mathscr{C} + CP^2 + AQ_t^2 + D_p(P - hT_D)^2 + 2b'Q_tT_D \\ & + \mathscr{R} + e_4UF_t^2 - e_4yu(H + a\xi)F_t \\ & + AUQ_t^2 + B(1 + V)\xi^2 - 2(\mu/a)H\xi - \mu\xi^2 \\ & + \kappa uF_t\xi + \gamma uvQ_t\xi, \end{aligned} \tag{JF1}$$

where the entire joint radial row is

$$\mathscr{R} = \frac{C}{t^2}(F_b - rF_t)^2 + \frac{r^2T_0}{t^2}F_t^2 + \left(\frac{2mr}{t^2} + e_4x\right)F_bF_t. \tag{JF2}$$

In particular all $mVEQ_t/t$ terms are included in JF2; they are not separate errors. The $mUVQ_t^2$ contribution is included in the complete $e_4UF_t^2$ row, so it cannot also be retained independently.

A short independent way of certifying JF1 is useful. Without imposing any source formulas, the difference between the original parent and its displayed right side is exactly a linear combination of only

$$vE,\ vQ_t,\ u\xi,\ UV,\ V,\ 1.$$

The displayed expression has zero value and all five first derivatives at the original contact $U = x, V = y, E = Q_t = \xi = 0$. Its six coefficients therefore vanish by the six original stationary contact balances. This support calculation and the six contact identities prove the identity for the complete original parent, including every state normal.

One strict radial pivot, with its exact companion. Set $\rho = m + rT_m x/2$. Then JF2 has the pure-source completion

$$\mathscr{R} = \frac{C}{t^2}[F_b - r(1-\rho/C)F_t]^2 + \frac{r^2}{t^2}[T_0 + 2\rho - \rho^2/C]F_t^2. \tag{JF3}$$

Here $\rho = O(x)$ and $T_0 + 2\rho - \rho^2/C \sim 1/2$; all denominators are pure source and this remains uniform at both source openings. Both F_t/t and F_b/t have bounded allocations on JF3. This does not provide an independent strict F_0/t pivot. The non-small remaining coefficient is precisely $\gamma uvQ_t\xi$. In particular a source split making $x/\sqrt{t}$ bounded does not remove it, because $\gamma \sim 1$.

The extra circuit row participates in the leading cancellation. Use $c^2 = a/t$, $k^2 = x$, $p = ckv/u$, and homogenize now in Z:

$$J_0 = uv - x/c, \quad R_c = kv(tcW - Z), \quad R = \sqrt{\lambda}\, tk(cv - u)W, \quad \tau = tcW - Z.$$

The generic exact circuit completion is

$$\mathscr{C} = \lambda J_0^2 Z^2 + R^2 + \underbrace{\frac{2\lambda k}{p}R_c^2}_{\mathcal{T}^2} - \lambda x E^2.$$

The full-D conductor is

$$R_D = uJ_0 Z + kR_c - (x/c)(H + a\xi), \qquad H + a\xi = avW - uZ. \tag{JF4}$$

The additional exact polarization is

$$E(H + a\xi) = uv\tau^2 - t(u - cv)^2 WZ. \tag{JF5}$$

Consequently the extra circuit row and the principal cross left after using JF4 satisfy

$$\mathcal{T}^2 + \kappa h(x/c)E(H + a\xi) = \frac{x}{c}(2\lambda + \kappa h)uv\tau^2 - \kappa ht\frac{x}{c}(u - cv)^2 WZ. \tag{JF6}$$

This cancels the previously non-small E–angular Gram coupling. JF6 is used as an exact identity, rather than as a separate positive term. In particular, its first coefficient has not been allocated to a strict positive row. The sign of WZ must be retained. The second summand is a literal SOS on the pure-testing sign chart $WZ \geq 0$ because $-\kappa > 0$, but not on the opposite sign chart.

An exact allocation of the negative small extra-row fraction. There is nevertheless a useful conductor:

$$x\frac{k}{p}R_c^2 = UR_c^2 - kuJ_0 R_c\tau. \tag{JF7}$$

If the testing state is normalized with Z a unit and W bounded, then $\tau/Z = tc(W/Z) - 1$ is bounded. Thus the right side is an infinitesimal Gram on $\sqrt{\lambda}R_c$ and $\sqrt{\lambda}J_0 Z$, with coefficients U and $ku(\tau/Z)$. Since $\sqrt{\lambda}R_c = R + \sqrt{\lambda}kE$, its additional coefficient on $E/\sqrt{t}$ is $\sqrt{\lambda t}\, k = O(k)$. Therefore a negative $O(x)$ fraction of $\mathcal{T}^2$ has a bounded infinitesimal allocation on this state chart. No bound on p is used. The companion identity

$$F_b/t = (W/Z)J_0 Z + (x/a)\tau \tag{JF8}$$

may supply a strict xZ radial pivot in a simultaneous state-chart reassembly, since $\tau/Z \sim -1$ there. All divisions in JF7–JF8 are pure testing. Neither identity is an authorization to assume a strict $E/\sqrt{t}$ pivot after the leading source block has become semidefinite. That final simultaneous allocation is still required.

A direct conductor on the normalized Z-unit state chart. There is a simpler route on this state chart that does not perform JF4–JF6. Put $e_E = E/\sqrt{t}$ and retain the original strict source completion of the first line of JF1. Since $Q_t = E/t - P$ and $vZ = tuW - E$,

$$uvQ_t\xi = \sqrt{t}\, U(W/Z)e_E\xi - u(\xi/Z)e_E^2 - uvP\xi. \tag{JF9}$$

When the testing state is normalized with Z a unit, W/Z and X/Z are bounded and $\xi/Z = X/Z + hu$ is bounded. Thus all three coefficients in JF9 are infinitesimal in the bounded tensor ring. The last uses the bounded pure-source allocation of P on the completed source roots. The first line of JF1 has source Schur pivot $tS \sim 1/2$, and $E/\sqrt{t}$ has a bounded allocation there, including after the full D_p shift. Consequently JF9 removes the remaining γ coupling on this state chart without producing a semidefinite source frame. The accompanying complete state-chart certificate, if used, must still list the retained positive rows and the strict angular block; JF9 alone does not cover states with normalized Z infinitesimal.

Use of the full identity. The complete JF1 identity is the input to DG3/LD9 and the completed J-row conductor in Section B.6.13. Those arguments allocate the remaining normal and retain the full angular and radial terms.

B.6.8 The small-η angular cancellation

Use the small-η, unit-ζ source: $x, \eta > 0$ infinitesimal, ζ a positive bounded unit, and $\beta \geq 0$. Put

$$a = 1/\zeta, \quad c^2 = a/\eta, \quad k^2 = x, \quad B_x = \beta + \tfrac{1}{2}, \quad q = \alpha + (m + B_x)(1-h)^2, \quad \mu = m - a^2 q, \quad \lambda = \mu/(\eta a x).$$

For homogeneous testing states use the exact roots

$$\xi = X + huZ, \quad E = \eta uW - vZ, \quad D_0 = u - cv, \quad J = (uv - x/c)Z,$$
$$R_D = v(u^2 - x)Z, \quad R_c = kv(\eta cW - Z), \quad \mathcal{Q} = \sqrt{q}(avW - uZ - a\xi).$$

Direct multiplication gives

$$R_D - uJ = (x/c)D_0 Z, \qquad \mathcal{Q}/\sqrt{q} = -D_0 Z + (c/k)R_c - a\xi.$$

Hence the complete principal ξ row cancels exactly to

$$-2\eta\lambda ckR_c\xi + 2\eta\lambda c(R_D - uJ)\xi = -2\eta\lambda x\mathcal{Q}\xi/\sqrt{q} - 2\eta\lambda xa\xi^2. \tag{SC5}$$

Together with the singular rank's $+\mu\xi^2$, the last term leaves $-\mu\xi^2$, because $\mu = \eta\lambda ax$. The latter coefficient is infinitesimal relative to $B_x\xi^2$. The complete angular block on $(\mathcal{Q}, \sqrt{B_x}\xi)$ has strictly positive residue: the original source expressions give

$$B_x q/x^2 \sim \frac{B_x\zeta(1+\zeta)}{2} + \frac{(1+\zeta)^2}{4} > \frac{1}{4} \sim (\eta\lambda)^2.$$

For unbounded B_x the normalized off-diagonal tends to zero. For bounded $B_x \geq 1/2$, the displayed difference is bounded below by $\zeta(3+2\zeta)/4 > 0$. The source substitutions are the regular DG/LD dictionary; the direct derivative and margin calculation is also printed in the refinement proof. All inverses and radicals in these coefficient expressions are pure source. The full radial and remaining normal allocations are the retained JF/DG/LD/CJ terms; SC5 supplies their angular block.

B.6.9 Small L, unbounded ζ and unit η

Theorem B.30. *Retain the original physical C1 source and every original contact, strict-gap, nonnegative-weight, total-scale and earlier-exclusion intersection. Suppose $x = k^2, L = x\zeta$ are positive infinitesimals, ζ is unbounded and η is a positive bounded unit. Then the complete original parent is finite tensor SOS for every independent projective test and every real or complex state. No positivity or noninfinitesimal lower bound is imposed on $j_0 = \eta + r - 1$: only the actual original gap $j = \eta(1 + xh) - 1 + r > 0$ is used. Both r openings, simultaneous r, j openings and literal $\beta = 0$ are included.*

This theorem treats the whole η-unit source portion of C14.3a. Every identity below uses the normalized quarter-parent $\mathcal{P} = F/4$. The complete parent identity JF1–JF3 from R16 is algebraic in the original stationary source; none of its identities assumes unit ζ. We retain that identity and change its source normalization.

Opening-cleared source bounds. Put

$$a = 1/\zeta, \quad x = aL, \quad y = \eta L, \quad \theta = 1 - h = L\pi, \quad B = \beta + \tfrac{1}{2}, \quad D_p = D + \tfrac{1}{2}, \quad \rho_0 = r + \eta.$$

The original beta numerator gives bounded $\pi > 0$; $h, M, E, Q, P_d, P_4 \sim 1$ as in SB1–SB3. In particular ρ_0 is a positive bounded unit. All formulas here preserve actual $j > 0$, rather than replacing it by j_0. Define, exactly as in JF1,

$$\begin{gathered} q = \alpha + (m+B)\theta^2, \quad \mu = m - a^2 q, \quad \lambda = \mu/(\eta a^2 L), \\ T_0 = A + \alpha + \tfrac{1}{2}, \quad T_m = T_0 + m, \quad e_4 = r^2 T_m/\eta^2, \\ b' = Ar/\eta - m\theta - \alpha - \tfrac{1}{2} + rT_m L/2, \\ \kappa = 2[B\theta - D_p j/\eta - mr/\eta], \qquad \gamma = 2b'/h + 2D_p j/r - 2\eta B\theta/r. \end{gathered} \tag{EU1}$$

Finite canceled source division gives

$$\begin{gathered} C(1-r) \sim \rho_0/2, \quad Ar \sim \eta/2, \quad D_p j \sim \rho_0/2, \quad m/(a^2 L) \sim \rho_0/2, \\ \alpha/L^2 \sim \rho_0/2, \quad B\theta/L \sim \rho_0/2, \quad q/L^2 - \rho_0(1+\pi)/2 \in \mathrm{Inf}, \\ \mu/(a^2 L) \sim \rho_0/2, \quad rT_m \sim \rho_0/2, \quad b'/L \sim -\rho_0/4, \quad \eta\kappa \sim -\rho_0. \end{gathered} \tag{EU2}$$

The canceled denominators at $a = L = 0$ are only $1, 2, 4$; there is no r, $1 - r$, j, θ or j_0 factor. Thus these are uniform through every opening in the theorem. In particular C, A, D_p are bounded below by positive real constants, although any of them may be unbounded. Also q/L^2 and λ are positive bounded units, $\mu > 0$, and $b' = O(L)$. The positive lower bound for D_p uses bounded positive actual j and the third estimate, not positivity of j_0.

Use the complete homogeneous parent. For nonnegative testing angles u, v put $U = u^2, V = v^2$ and define

$$\begin{gathered} E_t = \eta u W - v Z, \quad Q_t = Y - r v Z/\eta, \quad P = E_t/\eta - Q_t, \quad \xi = X + h u Z, \\ R_D = v(U - x)Z, \quad T_D = R_D - uv\xi/h, \quad H = avW - uZ - a\xi, \\ F_b = (V - y)Z + vE_t, \qquad F_t = (V - y)Z + (\eta/r)vQ_t. \end{gathered} \tag{EU3}$$

All inverses in these roots are pure source. The exact full parent is

$$\begin{aligned} \mathcal{P} = {} & qH^2 + \mathscr{C} + CP^2 + AQ_t^2 + D_p(P - hT_D)^2 + 2b'Q_tT_D + \mathscr{R} \\ & + e_4UF_t^2 - e_4yu(H + a\xi)F_t + AUQ_t^2 + B(1 + V)\xi^2 - 2(\mu/a)H\xi - \mu\xi^2 \\ & + \kappa u F_t\xi + \gamma uvQ_t\xi. \end{aligned} \tag{EU4}$$

This is the complete JF1 identity, including every hard row and every first normal. Set

$$\rho = m + rT_m x/2 = O(x), \qquad K_r = T_0 + 2\rho - \rho^2/C.$$

Then $K_r/T_m \sim 1$ and K_r is bounded below by a positive real constant. The complete radial row is exactly

$$\mathscr{R} = R_b^2 + R_t^2, \qquad R_b = \frac{\sqrt{C}}{\eta}[F_b - r(1 - \rho/C)F_t], \qquad R_t = \frac{r\sqrt{K_r}}{\eta}F_t. \tag{EU5}$$

It follows that

$$\frac{F_b}{\eta} = \frac{R_b}{\sqrt{C}} + \frac{1 - \rho/C}{\sqrt{K_r}}R_t \tag{EU6}$$

is a bounded pure-source allocation even when r is infinitesimal. No independent bare $(V - y)Z$ pivot has been inserted.

A strict source frame without either gap inverse bound. Use the three original weighted triangle roots

$$N_C = \sqrt{C}\,P, \quad N_A = \sqrt{A}\,Q_t, \quad N_D = \sqrt{D_p}(P - hT_D). \tag{EU7}$$

Their positive squares are exactly the three original source-triangle terms in EU4. They give the bounded pure-source allocations

$$P = \frac{N_C}{\sqrt{C}}, \quad Q_t = \frac{N_A}{\sqrt{A}}, \quad T_D = \frac{N_C/\sqrt{C} - N_D/\sqrt{D_p}}{h}, \quad E_t = \eta(N_C/\sqrt{C} + N_A/\sqrt{A}). \tag{EU8}$$

Consequently the entire $2b'Q_tT_D$ term is an infinitesimal Gram on this strict diagonal frame. This does not cap or remove a divergent original event and does not assert boundedness of its coefficient.

The complete two-normal cancellation. The source definitions in EU1 give the exact identity

$$\frac{\eta\kappa}{r} + \gamma = 2b'/h - 2m.$$

Since $F_t = F_b - vE_t + (\eta/r)vQ_t$, both complete normal rows in EU4 therefore satisfy

$$\boxed{\kappa u F_t\xi + \gamma uvQ_t\xi = \kappa u F_b\xi + uv\{(2b'/h - 2m)Q_t - \kappa E_t\}\xi.} \tag{EU9}$$

This is the crucial simultaneous allocation. It cancels the potentially unbounded $1/r$ coefficients before any estimate is made. It neither estimates the two original rows separately nor uses their contact vanishing as a substitute for the identity.

Set $N_x = \sqrt{B}\xi$ and $\mathcal{Q} = \sqrt{q}H$. By EU2, EU6 and EU8, every coefficient of EU9 on $(N_C, N_A, R_b, R_t, N_x)$ is infinitesimal: the first uses $\eta\kappa = O(1)$ and an extra u; the second uses $b', m \in \mathrm{Inf}$ and an extra uv; the third uses $\eta\kappa = O(1)$ and an extra uv. The factors $1/\sqrt{B}$, $1/\sqrt{A}$, $1/\sqrt{C}$ and $1/\sqrt{K_r}$ are bounded. No testing-state normalization occurs.

Every remaining scalar and hard term. The complete angular block on $(\mathcal{Q}, N_x)$ is

$$\mathcal{Q}^2 - \frac{2\mu}{a\sqrt{qB}}\mathcal{Q}N_x + (1-\mu/B)N_x^2.$$

Its off-diagonal coefficient is $O(a)$ and μ/B is infinitesimal:

$$\frac{\mu}{a\sqrt{qB}} = a\,\frac{\mu/(a^2L)}{\sqrt{q/L^2}\sqrt{B}}.$$

Thus its residue is the identity, including literal $\beta = 0$. The radial-angular cross has the exact normalized expression

$$-e_4yu(H+a\xi)F_t = -\frac{rT_mL}{\sqrt{K_r}}u\left(\frac{\mathcal{Q}}{\sqrt{q}} + \frac{aN_x}{\sqrt{B}}\right)R_t. \tag{EU10}$$

Both coefficients are infinitesimal because $rT_m = O(1)$, $L/\sqrt{q} = O(1)$ and u is infinitesimal.

The coupled circuit $\mathscr{C}$ is exactly the sum of the following three whole positive squares, followed by $-\lambda x E_t^2$:

$$\sqrt{\lambda}(uv - x/c)Z, \qquad \sqrt{\lambda}\,\eta k(cv-u)W, \qquad \sqrt{2\lambda/c}\,k\sqrt{uv}(\eta cW - Z), \qquad c^2 = a/\eta, \quad k^2 = x. \tag{EU11}$$

The last negative term is an infinitesimal Gram on EU8 since λ is bounded and x infinitesimal. All radicals in EU11 are separated pure-source and pure-testing radicals.

Retain EU11's three squares, $e_4UF_t^2$, AUQ_t^2 and $BV\xi^2$ as whole positive terms. Every other term of EU4 has now been assigned explicitly to the seven-root frame

$$(N_C, N_A, N_D, R_b, R_t, \mathcal{Q}, N_x). \tag{EU12}$$

The Gram on this frame has identity residue and bounded coefficients. The bounded-Gram lemma proves finite tensor SOS for the complete parent whenever both testing angles are infinitesimal. Every estimate is a product of a source bound and a testing bound; there is no source/testing order comparison and no inverse of a mixed tensor.

All tests, endpoints and states. The roots and the whole-parent identity are homogeneous in all four state coordinates. In particular no Z inverse or state restriction was used. The positive circuit rows EU11 are defined on literal $u = 0$ or $v = 0$ and their squared identity is polynomial there. The common exact sign gauges give all local sign charts. The unchanged small-L, bounded-η remote theorem covers the complement of both infinitesimal testing angles on this exact source domain, retaining all its projective endpoints and openings. Applying the real homogeneous identity to real and imaginary state parts proves the complex-state assertion. All of the source unit-η portion of C14.3a is therefore all-tests covered.

The complementary source. The remaining C14.3a sources have η infinitesimal, hence $r \sim 1$, with all original $1-r, j$ openings retained. Their complete parent is treated by the LD identities and the completed J-row conductor in Sections B.6.11 and B.6.13.

B.6.10 Small η and unit ζ: the damped positive block

Theorem B.31. *Normalize $d = 1$ and retain the unchanged physical C14.3b source, with $x, \eta > 0$ infinitesimal, ζ a positive bounded unit, every original j and $1-r$ opening, $\beta \geq 0$ including literal $\beta = 0$, and all common intersections. On the additional* pure-source *subchart*

$$x/\eta \text{ bounded}, \tag{DG1}$$

the entire original parent is finite tensor SOS for every independent projective angle and state test. No relation between a source rate and a testing rate is assumed. The proof below treats the whole local corner; the common remote theorem gives its complement. The source subchart x/η unbounded is not covered by this theorem.

The sign reduction and the complete frame. Use the sign gauges to take $u, v \geq 0$. For real states with $WZ \leq 0$ the independent half-core sign lemma proved in this revision gives a direct tensor-SOS decomposition of the original signed half core; all remaining original event terms are positive source multiples of squares. Thus only $WZ > 0$ needs the following construction. No normalization or inverse of a state coordinate is used here.

Use all homogeneous JF1 definitions, and abbreviate $Q = Q_t$, $T = T_D$, $B = B_x$, $a = 1/\zeta$, $x = k^2$, $a = \eta c^2$. In particular

$$E = \eta uW - vZ = \eta(P+Q), \quad H = avW - uZ, \quad \tau = \eta cW - Z, \quad D_0 = u - cv,$$
$$J = (uv - x/c)Z, \quad R = -\sqrt{\lambda}\,\eta k D_0 W, \quad f_t = F_t/\eta, \quad \mathcal{Q} = \sqrt{q}(H - a\xi), \quad N_x = \sqrt{B}\xi.$$

The full radial term $\mathscr{R}$ is JF2, with the strict two-root completion JF3. Its roots will be denoted R_b, f_t; both normalized pivots have positive unit residue. The angular term is exactly

$$\mathscr{A} = \mathcal{Q}^2 - \frac{2\mu}{a\sqrt{qB}}\mathcal{Q}N_x + (1-\mu/B)N_x^2. \tag{DG2}$$

Its normalized Schur pivot is strict by the unchanged R15/R16 source margin, including $\beta = 0$ and unbounded B. Those previously proved angular and radial blocks are used once, as their complete terms.

Damp the remaining normal and complete both circuit companions. Set the positive source unit

$$G_0 = 2\eta\lambda - 2\eta \sim 1, \qquad \epsilon = \gamma - G_0/h,$$

$$\widehat{R} = R + \frac{G_0 k}{2\sqrt{\lambda}}Q, \qquad \widehat{J} = \sqrt{\lambda}J + \frac{G_0 u}{2\sqrt{\lambda}}Q, \qquad c_U = A - \frac{G_0^2}{4\lambda}.$$

Use the exact full-D conductor and polarization displayed in Chapter 6 to conduct only $(G_0/h)uvQ\xi$, leaving $\epsilon uvQ\xi$ unaltered. Complete *both* R and J, incorporating the entire native AUQ^2 row into the latter completion. Since $G_0/(2\lambda) = \eta - \eta/\lambda$, the complete original parent has the following exact identity:

$$\begin{aligned} F/4 = {} & \mathscr{A} + \mathscr{R} + S_0 - \lambda x(\eta P + \eta Q/\lambda)^2 + \widehat{R}^2 + \widehat{J}^2 \\ & + c_U UQ^2 + e_4 UF_t^2 + BV\xi^2 + \frac{2x}{c}uv\tau^2 + \frac{G_0 x}{c}D_0^2 WZ \\ & - e_4 yuHF_t + \kappa uF_t\xi + \frac{G_0 x}{c}PH + \epsilon uvQ\xi, \end{aligned} \tag{DG3}$$

where

$$S_0 = CP^2 + AQ^2 + D_p(P - hT)^2 + 2b_0 QT, \qquad b_0 = b' - G_0/2. \tag{DG4}$$

This is a simultaneous completion of the entire remaining normal and the coupled circuit. It retains every full-D, full-Y and higher normal of JF1. There is no second bare radial pivot and no extracted uncentered m event. The coefficient of the extra $uv\tau^2$ row is exactly $2x/c > 0$, because $2\lambda - G_0/\eta = 2$.

Two strictly positive source deficits replace the failed soft pivot. Put $K = CD_p/(C + D_p)$, $s = x + \eta$, and

$$\Delta_0 = h^2 AK - b_0^2.$$

The original source formulas imply exact finite divisions

$$\Delta_0 = xd_x + \eta d_\eta, \qquad \frac{c_U}{\eta} = xe_x + \eta e_\eta, \tag{DG5}$$

with bounded source coefficients having the following residues:

$$\begin{array}{c|c} d_x & \pi/2 + 1/4 + 1/(4a) \\ d_\eta & \sigma/2 + 3/4 \\ e_x & \pi + 1 + 1/(2a) \\ e_\eta & \sigma + 3/2 \end{array} \qquad h = 1 - x\pi, \quad r = 1 - \eta\sigma. \tag{DG6}$$

All four residues are strictly positive. Here $\pi, \sigma \geq 0$ are bounded by the unchanged physical source estimates. Thus

$$\Delta_0/s \text{ is a positive bounded unit}, \qquad c_U/(\eta s) \text{ is a positive bounded unit}. \tag{DG7}$$

These assertions compare source quantities only and do not use DG1.

For clarity, the divisions in DG5 are opening-safe. Write $C_n = (1-r)C$, $D_j = jD_p$ and $\Theta_B = B(1-h)$, all given by their unchanged unit-denominator source formulas. The identity $\eta\Theta_B - D_j = -C_n$ gives $b' - \gamma h/2 = -hC_n/r$. The gap-cleared harmonic coefficient is

$$\eta K = \frac{C_n D_j}{C_n(1 + xh - \sigma) + D_j\sigma}.$$

Its denominator is a positive unit, even at either opening. The quantities $\eta\lambda$, b', A/η and γh also have unit-denominator expressions after their displayed monomial factors are canceled. Both functions on the left of DG5 vanish at $x = \eta = 0$. Define their η coefficients by their exact $x = 0$ values divided by η, and then divide the remainder by x. This gives DG5 with only source-unit denominators; differentiation at the origin gives DG6. No assumption on x/η enters these exact finite divisions.

The source roots control the complete soft direction. There is the exact pure-source completion

$$S_0 = N_s^2 + N_T^2 + N_Q^2,$$

$$N_s = \sqrt{C + D_p}\left(P - \frac{hD_p}{C + D_p}T\right), \quad N_T = h\sqrt{K}\left(T + \frac{b_0}{h^2K}Q\right), \quad N_Q = \sqrt{\frac{\Delta_0}{h^2K}}Q. \tag{DG8}$$

As $\eta K \sim 1/2$, DG7 gives $N_Q/\{\sqrt{\eta s}\,Q\}$ a positive source unit. The inverse identities show

$$T = \frac{N_T}{h\sqrt{K}} - \frac{b_0}{h^2K}Q, \qquad P = \frac{N_s}{\sqrt{C + D_p}} + \frac{hD_p}{C + D_p}T. \tag{DG9}$$

Consequently P has bounded coefficients on (N_s, N_T, N_Q): the first two coefficients are $O(\sqrt{\eta})$ and the last is $O(\sqrt{\eta/s})$, which is bounded because $\eta \le s$. This is the complete P control after the full D shift; it is not an assumption that the old $E/\sqrt{\eta}$ pivot remains strict.

The negative square in DG3 is an infinitesimal Gram on these three roots. Its root is

$$\sqrt{\lambda x}(\eta P + \eta Q/\lambda) = \eta\sqrt{\lambda x}\,P + \eta\sqrt{x/\lambda}\,Q.$$

Since $\eta\lambda \sim 1/2$, the first coefficient is $O(\sqrt{\eta x})$ on the bounded allocation of P; the second, after normalizing Q by N_Q, is $O(\eta\sqrt{x/s})$. Both are infinitesimal, without DG1.

Every remaining complete normal is small on this frame. The angular-radial term in DG3 is

$$-e_4 y\eta u\left(\mathcal{Q}/\sqrt{q} + aN_x/\sqrt{B}\right) f_t.$$

Its coefficients are $O(u)$ and $O(xu)$, using $e_4\eta^2 \sim 1/2$ and q/x^2 a positive bounded unit. The term $\kappa u F_t \xi$ has coefficient $(\eta\kappa)u/\sqrt{B}$, again infinitesimal. The new PH term has coefficients

$$\frac{G_0 x}{c\sqrt{q}} \quad \text{and} \quad \frac{G_0 a x}{c\sqrt{B}}$$

on $P\mathcal{Q}$ and PN_x; these are respectively $O(\sqrt{\eta})$ and $O(x\sqrt{\eta})$, so DG9 allocates the whole term infinitesimally.

Finally $\epsilon = O(s)$ by the same gap-free source expressions. The complete, still unconducted normal has coefficient

$$\frac{\epsilon u v}{\sqrt{\eta s}\sqrt{B}} = O\left(\sqrt{s/\eta}\,uv\right) \tag{DG10}$$

on $N_Q N_x$. DG1 is used *exactly here*: it makes s/η bounded, and hence the coefficient infinitesimal for independent infinitesimal u, v. No testing comparison is imposed.

Thus the remainder on

$$(N_s, N_T, N_Q, R_b, f_t, \mathcal{Q}, N_x)$$

has a bounded Gram with strictly positive residue. The bounded-Gram lemma makes it finite tensor SOS. Add the whole positive rows $\widehat{R}^2$, $\widehat{J}^2$, $c_U U Q^2$, $e_4 U F_t^2$, $BV\xi^2$, $(2x/c)uv\tau^2$ and $(G_0 x/c)D_0^2 WZ$. The last is a literal tensor square on the current pure-testing sign chart; all radicals are source-only or testing-only. This proves the complete local parent, not merely its scalar restriction.

Equality, endpoints and the global join. At original contact $P, Q, T, J, R, \tau, F_t, \xi$ vanish, so all newly completed rows vanish too. DG3 preserves all contact normals exactly. The row formulas have no angle ratio or state-coordinate inverse; $\sqrt{uv}$ is defined as zero on a literal axis. Thus they include both literal local axes. The common gauges cover every angle sign. For a complex state, split the real-coefficient Hermitian parent into its real and imaginary evaluations and apply either the $WZ \le 0$ core lemma or this $WZ > 0$ proof separately to each real state. There is no assumption that both real and imaginary state components have a unit Z or W. Combine the local proof with the unchanged remote all-state theorem to obtain the all-tests assertion on DG1.

The hypothesis DG1 is used only in the final allocation DG10. The exact DG3 identity and preceding source bounds also apply without it; the completed J-row conductor in Section B.6.13 supplies the remaining allocation for arbitrary x/η.

B.6.11 Small η and unbounded ζ: the damped positive block

Theorem B.32. *Normalize $d = 1$ and retain the unchanged physical C14.3a source, its actual stationary contact, every strict source gap, $\beta \geq 0$ including literal $\beta = 0$, the original total-scale intersection and every earlier exclusion. Suppose $x, L = x\zeta, \eta$ are positive infinitesimals, ζ is unbounded, and*

$$L/\eta \text{ is bounded.} \tag{LD1}$$

Then the entire original parent is finite tensor SOS at every independent projective angle and state test, real or complex. This is a whole-state, all-tests source-subchart certificate. No condition is imposed on the testing angle ratio or on the source ratio a/η, where $a = 1/\zeta$. The source subchart with η infinitesimal and L/η unbounded is not covered here.

The proof uses the exact complete damped identity DG3, but its source scale is $L + \eta$ rather than $x + \eta$. This change must be proved: one cannot invoke the unit-ζ estimates by substituting an infinitesimal a into their displayed $1/a$ coefficients. All formulas below use the actual physical source and the original contact.

Source variables and opening-cleared coefficients. Put

$$a = 1/\zeta, \quad x = aL, \quad y = \eta L, \quad h = 1 - L\pi, \quad r = 1 - \eta\sigma, \quad B = \beta + \tfrac{1}{2}.$$

The original source estimates give bounded $\pi, \sigma \geq 0$, and actual $j > 0$ says $0 < \sigma < 1 + xh$, $j = \eta(1 + xh - \sigma) > 0$. Neither opening has a positive-residue lower bound. The exact native source formulas displayed below give

$$\frac{m}{a^2 L} \sim \frac{1}{2}, \quad \frac{\alpha}{L^2} \sim \frac{1}{2}, \quad \frac{q}{L^2} - \frac{1+\pi}{2} \in \mathrm{Inf}, \quad \frac{\mu}{a^2 L} \sim \frac{1}{2},$$

$$\eta\lambda \sim \frac{1}{2}, \quad \eta K \sim \frac{1}{2}, \quad \eta\kappa \sim -1, \quad \eta^2 e_4 \sim \frac{1}{2}, \quad b' = O(L), \tag{LD2}$$

where $q = \alpha + (m + B)(1 - h)^2$, $\mu = m - a^2 q$, $\lambda = \mu/(\eta a^2 L)$ and $K = CD_p/(C + D_p)$.

Here is a completely regular specification of the new source expressions. Write

$$K_x = 1 + x, \ R_s = 1 + xr, \ T_s = 1 + xh, \ M_s = R_s - aL\pi, \ E_s = 1 - x\eta h, \ Q_s = 1 - Lr,$$

$$P_d = 1 + 2xh + rx^2 h + rxL + L + \eta\{K_x L + x^2 hL\pi\}, \qquad P_4 = T_s E_s + (1 - r)(xh + L).$$

Define the opening-cleared quantities

$$C_n = \frac{K_x(r+\eta)E_s}{2P_d}, \qquad D_j = \frac{K_x(1+y)(r+\eta)}{2P_d}, \qquad A_\eta = \frac{P_4}{2rP_d},$$

$$m_n = \frac{K_x(r+\eta)h(1+y)}{2M_s P_d}, \quad \alpha_n = \frac{(r+\eta)\{aR_s h(1+y) + M_s E_s\}}{2Q_s P_d}, \quad B_n = \frac{K_x(r+\eta)(1+ah)}{2P_d}. \tag{LD3}$$

Thus $C_n = (1 - r)C$, $D_j = jD_p$, $A_\eta = A/\eta$, $m_n = m/(a^2 L)$, $\alpha_n = \alpha/L^2$ and $B_n = B(1 - h)/L$. In particular

$$q_n := q/L^2 = \alpha_n + m\pi^2 + B_n\pi, \qquad \Lambda := \eta\lambda = m_n - Lq_n,$$

$$\eta K = \frac{C_n D_j}{C_n(T_s - \sigma) + D_j\sigma}. \tag{LD4}$$

There is no a^{-1} in (LD3)–(LD4). Their denominators at $a = L = \eta = 0$ have residues 1, 2 or 1/2, independently of π, σ. The remaining definitions are

$$T_m = A + \alpha + \tfrac{1}{2} + m, \quad b' = A_\eta r - mL\pi - \alpha - \tfrac{1}{2} + rT_m L/2,$$

$$G_f := \gamma h = 2b' + 2D_j h/r - 2\eta L B_n h/r, \quad G_0 = 2\Lambda - 2\eta \sim 1, \quad b_0 = b' - G_0/2. \tag{LD5}$$

These are the actual native-source expressions used in the completion.

Positive source deficits at the new scale. Set $s = L + \eta$ and retain DG's definitions

$$\Delta_0 = h^2AK - b_0^2, \qquad c_U = A - \frac{G_0^2}{4\lambda}, \qquad \epsilon = \gamma - G_0/h.$$

The regular source formulas above give exact finite divisions

$$\Delta_0 = Ld_L + \eta d_\eta, \qquad \frac{c_U}{\eta} = Le_L + \eta e_\eta, \qquad G_f - G_0 = Lf_L + \eta f_\eta, \tag{LD6}$$

with bounded coefficients and residues

	L coefficient	η coefficient
Δ_0	$\pi/2 + 1/4$	$\sigma/2 + 3/4$
c_U/η	$\pi + 1/2$	$\sigma + 3/2$
$G_f - G_0$	$\pi + 1/2$	$\sigma + 2$

(LD7)

(as usual the bounded π, σ on the right can be replaced by their real residues). To justify the finite divisions explicitly, each expression on the left of (LD6) vanishes at $L = \eta = 0$ *for arbitrary* a. Subtract its exact value at $L = 0$, divide that difference by L, and divide the $L = 0$ value by η. Clearing the source-unit denominators proves each divisibility in a polynomial ring. Evaluation of these quotients at $a = L = \eta = 0$ gives (LD7). There is consequently no hidden pure-a remainder.

All four coefficients for the first two rows have strictly positive residue. Hence

$$\Delta_0/s \text{ and } c_U/(\eta s) \text{ are positive bounded source units}, \qquad \epsilon = O(s). \tag{LD8}$$

This source calculation itself does not use (LD1).

The same complete damped identity, without a state chart. Use DG's unshifted $H = avW - uZ$; the strict angular root is $\mathcal{Q} = \sqrt{q}(H - a\xi)$, so this H differs from the shifted letter H in EU3. Retain every other homogeneous JF/DG definition. In particular

$$E = \eta(P + Q), \quad N_x = \sqrt{B}\xi, \quad \tau = \eta cW - Z, \quad D_0 = u - cv, \quad c^2 = a/\eta.$$

After the pure-testing sign gauges, the half-core lemma supplies all $WZ \le 0$ states. On $WZ > 0$, the exact identity DG3 is valid for this source without any change:

$$\begin{aligned} F/4 = {} & \mathscr{A} + \mathscr{R} + S_0 - \lambda x(\eta P + \eta Q/\lambda)^2 + \widehat{R}^2 + \widehat{J}^2 \\ & + c_U UQ^2 + e_4 UF_t^2 + BV\xi^2 + \frac{2x}{c}uv\tau^2 + \frac{G_0 x}{c}D_0^2 WZ \\ & - e_4 yuHF_t + \kappa uF_t\xi + \frac{G_0 x}{c}PH + \epsilon uvQ\xi. \end{aligned} \tag{LD9}$$

Here $S_0 = CP^2 + AQ^2 + D_p(P - hT)^2 + 2b_0QT$, $\widehat{R} = R + G_0\sqrt{x}Q/(2\sqrt{\lambda})$ and $\widehat{J} = \sqrt{\lambda}J + G_0uQ/(2\sqrt{\lambda})$. The identity conducts only $G_0uvQ\xi/h$ and completes both circuit companions exactly, retaining every first normal and hard coupling. Its derivation uses only $x = k^2$, $c^2 = a/\eta$ and the homogeneous JF conductor, not a bounded reciprocal of a or c.

Strict source, radial and angular frames. Use the exact source roots

$$N_s = \sqrt{C + D_p}\left(P - \frac{hD_p}{C + D_p}T\right), \quad N_T = h\sqrt{K}\left(T + \frac{b_0}{h^2K}Q\right), \quad N_Q = \sqrt{\frac{\Delta_0}{h^2K}}Q. \tag{LD10}$$

Their squares sum to S_0, and $N_Q/[\sqrt{\eta s}\,Q]$ is a positive source unit. The inverse source identities give bounded coefficients for P on this frame: the coefficients of N_s, N_T are $O(\sqrt{\eta})$ and that of N_Q is $O(\sqrt{\eta/s})$. This uses only $\eta \le s$, so no old $E/\sqrt{\eta}$ pivot is imported.

The complete radial term has its common strict roots R_b, f_t, with $f_t = F_t/\eta$. Indeed $\rho = m + rT_m x/2 = O(x)$ and $K_r \sim 1/2$. The complete angular block is

$$\mathscr{A} = \mathcal{Q}^2 - \frac{2\mu}{a\sqrt{qB}}\mathcal{Q}N_x + (1 - \mu/B)N_x^2.$$

It has identity residue, since

$$\frac{\mu}{a\sqrt{qB}} = \frac{a[\mu/(a^2L)]}{\sqrt{q/L^2}\sqrt{B}} = O(a), \qquad \mu/B \in \mathrm{Inf}\,. \tag{LD11}$$

This includes unbounded B and literal $\beta = 0$.

The entire remainder has infinitesimal coefficients. The negative-square root in (LD9) has, on the source frame, the same coefficient bounds as DG:

$$\eta\sqrt{\lambda x}\,P + \eta\sqrt{x/\lambda}\,Q, \qquad O(\sqrt{\eta x}), \quad O(\eta\sqrt{x/s}). \tag{LD12}$$

They are infinitesimal; here $x = aL$ and no comparison of a with η is needed. For the radial-angular term use $H = \mathcal{Q}/\sqrt{q} + aN_x/\sqrt{B}$ and $F_t = \eta f_t$. Since $e_4 y\eta = (e_4\eta^2)L = O(L)$ and $\sqrt{q} \asymp L$, its two coefficients are $O(u)$ and $O(xu)$. The κ term has coefficient $(\eta\kappa)u/\sqrt{B}$, which is also infinitesimal.

The new PH term improves at this source end:

$$\frac{G_0 x}{c\sqrt{q}} = O(\sqrt{a\eta}), \qquad \frac{G_0 a x}{c\sqrt{B}} = O(x\sqrt{a\eta}). \tag{LD13}$$

Thus the bounded P allocation makes both of its angular coefficients infinitesimal. Finally, the complete unprocessed normal is

$$\frac{\epsilon uv}{\sqrt{\eta s}\sqrt{B}} N_Q N_x \quad \text{up to a bounded positive source-unit factor.}$$

Its coefficient is

$$O\Big(\sqrt{1 + L/\eta}\,uv\Big), \tag{LD14}$$

which is infinitesimal under the bounded-source-ratio hypothesis (LD1). This is the only use of that hypothesis.

Consequently the complete remainder on $(N_s, N_T, N_Q, R_b, f_t, \mathcal{Q}, N_x)$ has a bounded strict-residue Gram. The bounded-Gram lemma gives its finite tensor SOS. Every other term in (LD9) is a whole positive square: in particular $c_U > 0$, $G_0 > 0$, and uv, WZ are nonnegative pure-testing factors on this sign chart. No mixed inverse or radical is taken.

All tests and exact remaining source end. No state normalization, state-coordinate inverse or testing ratio occurs in this proof. The division-free circuit rows and $\sqrt{uv} = 0$ include both literal local axes. The half-core exit and the exact angle sign gauges cover every real state; splitting the real-coefficient Hermitian form covers complex states. The unchanged small-L, bounded-η remote theorem covers all other projective testing pairs, including literal infinities. This proves the theorem with every original source opening retained.

The restriction LD1 is used only in LD14. The complete LD9 identity, source bounds and all preceding normal allocations are inputs to the completed J-row conductor in Section B.6.13, which treats the full infinitesimal-η source at arbitrary L/η. The unit-η source is the EU theorem of Section B.6.9.

B.6.12 The direct large-X/Z state bound

In any local chart with u infinitesimal, suppose $Z \neq 0$ and the pure-testing ratio $|X/Z|$ is unbounded. Then

$$\frac{1+v^2}{2}\{(X+uZ)^2 - (1+u^2)Z^2\} > 0. \tag{DS3}$$

Indeed, after division by $Z^2 > 0$, the square $(X/Z + u)^2$ is unbounded and $1 + u^2$ is bounded. The factor $1 + v^2$ is positive. This expression is the complete X square and the negative Z term of the signed quarter-core. Its other two terms are $(uW + uvX - Y - vZ)^2/2$ and $(1 + u^2)(vY - Z)^2/2$, both nonnegative. Thus the signed core itself is a nonnegative pure-testing scalar and has a square root in the testing field. Every original event remains a square with its nonnegative source coefficient. This proves a direct separated SOS certificate for the complete parent, with arbitrary W, Y.

The state chart $Z = 0$ and the sign chart $WZ \leq 0$ have the direct core certificates already stated in the geometric reduction. Consequently the C14.3 local certificate only needs the remaining case $Z \neq 0$, $WZ > 0$, X/Z bounded. The completed-J conductor CJ treats that whole case without bounding W/Z or Y/Z. Apply the appropriate real chart to each real and imaginary part to include complex states.

B.6.13 The completed J relation for the entire remaining normal

Scope and unchanged inputs. Retain the complete actual-parent identity DG3, equivalently LD9, with $d = 1$, every original source opening and residual intersection, and $\beta \geq 0$ including literal $\beta = 0$. On C14.3b use $s = x + \eta$ and the unit $a = 1/\zeta$; on C14.3a use $s = L + \eta$ with $x = aL$ and a infinitesimal. In both cases s and η are positive infinitesimals. The exact DG5–DG9 and LD3–LD10 source identities give

$$\Lambda = \eta\lambda \sim \tfrac{1}{2}, \qquad \epsilon = O(s), \qquad N_Q = \sqrt{\frac{\Delta_0}{h^2 K}}\,Q, \qquad Q = \frac{\omega}{\sqrt{\eta s}} N_Q, \tag{CJ1}$$

where ω is a bounded positive source unit. The complete strict frame is

$$\mathcal{F} = (N_s, N_T, N_Q, R_b, f_t, \mathcal{Q}, N_x), \qquad N_x = \sqrt{B}\,\xi, \quad B = \beta + \tfrac{1}{2}. \tag{CJ2}$$

All DG3/LD9 residuals other than $\epsilon uvQ\xi$ already have infinitesimal Gram coefficients on this frame without any bound on s/η. The two residuals in the exact split CJ6a below have that same property whenever $Z \neq 0$ and X/Z is bounded. Thus the only expression to be allocated on that state chart is

$$N = \epsilon U(W/Z)E\xi, \qquad E = \eta(P+Q) = \eta uW - vZ. \tag{CJ3}$$

There is no restriction on Y/Z or W/Z below.

An exact identity from the already retained completed row. Let $\phi = \xi/Z = X/Z + hu$, and retain exactly DG's completed square

$$\widehat{J} = \sqrt{\lambda}\,(uv - x/c)Z + \frac{G_0 u}{2\sqrt{\lambda}}Q, \qquad G_0 = 2\eta\lambda - 2\eta, \qquad c^2 = a/\eta.$$

The defining identity $vZ = \eta uW - \eta(P+Q)$ gives

$$\boxed{\frac{\widehat{J}}{\sqrt{\lambda}} = \eta UW - \frac{x}{c}Z - \eta u\left(P + \frac{Q}{\lambda}\right).} \tag{CJ4}$$

Multiply CJ4 by $\epsilon\phi E/\eta$. The complete remaining normal has the exact conductor

$$\boxed{N = \frac{\epsilon\phi}{\eta\sqrt{\lambda}}E\widehat{J} + \epsilon\phi uE\left(P + \frac{Q}{\lambda}\right) + \frac{\epsilon x}{\eta c}E\xi.} \tag{CJ5}$$

The only state inverse here is the pure-testing Z^{-1}. No inverse of u, v, W, or a mixed expression occurs. No radial row, source pivot, or completed square is duplicated.

Bounded allocations and infinitesimal coefficients. Define two dependent roots

$$\mathcal{E} = \sqrt{s/\eta}\,E, \qquad \mathcal{P} = P + Q/\lambda.$$

Both have bounded allocations on the existing source roots (N_s, N_T, N_Q). Indeed the exact DG9/LD10 inverse identities give coefficients $O(\sqrt{\eta})$, $O(\sqrt{\eta})$ and $O(\sqrt{\eta/s})$ for P. Hence the coefficients for $\sqrt{\eta s}\,P$ are bounded (in fact infinitesimal), whereas

$$\mathcal{E} = \sqrt{\eta s}\,P + \omega N_Q, \qquad Q/\lambda = \frac{\omega}{\Lambda}\sqrt{\eta/s}\,N_Q. \tag{CJ6}$$

Here $\eta/s \le 1$ is a pure-source inequality. These are bounded allocations of dependent roots, not new positive pivots. For completeness, the other two terms of the same normal have the particularly simple allocations

$$\epsilon uvQ\xi - N = -\frac{\epsilon}{s}u\phi\mathcal{E}^2 - \frac{\epsilon uv}{\sqrt{B}}PN_x. \tag{CJ6a}$$

Their coefficients are infinitesimal, since u, v are local infinitesimals and P has a bounded source-root allocation.

In these roots CJ5 is exactly

$$\begin{aligned} N = {} & \frac{\epsilon}{s}\frac{\sqrt{s}}{\sqrt{\Lambda}}\phi\mathcal{E}\widehat{J} \\ & + \epsilon\sqrt{\eta/s}\,u\phi\mathcal{E}\mathcal{P} + \frac{\epsilon}{s}\frac{x\sqrt{s}}{\sqrt{a}\sqrt{B}}\mathcal{E}N_x. \end{aligned} \tag{CJ7}$$

All three coefficients are infinitesimal bounded tensor elements. For the first two use ϵ/s bounded, s infinitesimal, $\eta/s \le 1$, and bounded $\phi = X/Z + hu$. The last coefficient is also infinitesimal:

$$\frac{x\sqrt{s}}{\sqrt{a}} = \begin{cases} O(x\sqrt{s}), & a \text{ a bounded positive unit,} \\ \sqrt{a}\,L\sqrt{s}, & x = aL,\ a \text{ infinitesimal.} \end{cases} \tag{CJ8}$$

Also $1/\sqrt{B}$ is bounded, even at literal $\beta = 0$ and when B is unbounded. The apparent inverse $\sqrt{a}$ is canceled in the large-ζ case before taking bounds. Every coefficient is a finite separated expression; ϕ is expanded as $X/Z + hu$. No source rate is compared with any testing rate.

Full local reassembly. Add the already present row $\widehat{J}^2$ to the strict frame CJ2 and allocate CJ7 to it and the source roots. This produces an infinitesimal Gram perturbation of the previous strict-residue matrix plus a unit $\widehat{J}$ diagonal entry. The separated bounded-Gram lemma therefore applies to the full remainder. Retain each other positive row from DG3/LD9 exactly once:

$$\widehat{R}^2, \quad c_U U Q^2, \quad e_4 U F_t^2, \quad BV\xi^2, \quad (2x/c)uv\tau^2, \quad (G_0 x/c)D_0^2 WZ.$$

The last two are literal tensor squares on the unchanged sign chart $u, v \geq 0$, $WZ > 0$. The complete angular and radial terms remain inside CJ2; all full-D, full-Y and higher normals are those of the exact original-parent identity, not truncated expansions.

Consequently the entire local original parent is tensor SOS when $Z \neq 0$ and X/Z is bounded, with no condition on s/η, u^2W/Z, or Y/Z. The proof is valid on literal local angle axes because it uses no angle inverse. At original contact E, P, Q, ξ and $\widehat{J}$ vanish, so this reassembly preserves equality and all original contact normals exactly.

The local state and complementary testing cases. The reviewed pure-testing half-core exit covers $WZ \leq 0$, including $Z = 0$. When $Z \neq 0$ and X/Z is unbounded, DS3 in Section B.6.12 is a direct original-parent tensor-SOS exit. The above argument covers every remaining real state on the local angle chart. Applying these real certificates to real and imaginary state parts gives the full complex parent. Thus the shared local C14.3a/b obligation is closed, with all source gaps and $\beta = 0$ retained. The SB remote theorem in Section B.6.1 and the unit-η theorem in Section B.6.9 supply the other testing and source charts. The complete source partition is printed in Section B.1.2.

B.7 Vanishing L with unbounded η

Both L and $\sigma = 1/\eta$ now vanish. With $S = L + \sigma$, the three source ratios $\kappa = x/S$, $\nu = \sigma/S$ and $\ell = L/S$ satisfy $\nu + \ell = 1$. Their exact relations organize the remaining contacts, including the endpoints at which two of these ratios vanish.

B.7.1 The small-L, unbounded-η source sphere

This is a source theorem, not an additional all-tests certificate. Use the original physical C1 source with $x, L = x\zeta > 0$ infinitesimal, ζ noninfinitesimal and η unbounded. Put $\sigma = 1/\eta$, $S = \sigma + L$, and retain the original $0 < r, h < 1$, $\theta = 1 - h$, $M > 0$, and $D, \beta \geq 0$. The native gap gives $E = 1 - xh/\sigma \geq 1/18$. Set

$$e = \frac{xh}{\sigma} = 1 - E, \quad P = \sigma P_d, \quad J = 1 + xh - \sigma(1 - r), \quad H_0 = 1 + L(1 - r), \quad B = P + x(J - \sigma),$$

$$\mathcal{H} = RH_0 + (1 - r)xh(2 + xr) - e\{2 + x(1 + h + r\theta)\} + Ke^2, \quad R = 1 + xr, \quad K = 1 + x. \tag{ES1}$$

No $E \sim 1$ assumption is made here. The original source formulas give the two exact identities

$$2\theta P\beta = hB - \sigma H_0, \qquad 2P(JD + x\theta\beta) = \sigma^2\mathcal{H}. \tag{ES2}$$

They are the beta-shared identities with the reciprocal-L factors cleared. In particular their use does not require bounded $1/L$.

The positive denominator is exactly

$$P = \sigma(1 + L + 2xh + rxL + rx^2h) + KL + x^2h\theta. \tag{ES3}$$

Since $x/L = 1/\zeta$ is bounded and $x^2h = x\sigma e$, all terms beyond $\sigma + L$ in (ES3), divided by S, are infinitesimal. Consequently

$$\frac{P}{S} \sim 1, \qquad J \sim 1, \qquad \frac{B}{S} - (1 + x/S) \in \mathrm{Inf}, \qquad H_0 \sim 1, \qquad \mathcal{H} - E^2 \in \mathrm{Inf}\,. \tag{ES4}$$

Here x/S is bounded. Both B/S and $\mathcal{H}$ are positive bounded units, uniformly through all original openings; for the latter use the actual bound $E \geq 1/18$.

Define nonnegative pure-source sphere coordinates

$$d_0^2 = \frac{2PJD}{\sigma^2\mathcal{H}}, \qquad b_0^2 = \frac{2Px\theta\beta}{\sigma^2\mathcal{H}}.$$

Then

$$d_0^2 + b_0^2 = 1, \qquad D = \frac{\sigma^2\mathcal{H}}{2PJ}d_0^2, \qquad x\theta\beta = \frac{\sigma^2\mathcal{H}}{2P}b_0^2. \tag{ES5}$$

Thus $D = O(\sigma^2/S) = O(\sigma)$ and $x\theta\beta = O(\sigma^2/S)$, including either literal coefficient boundary. The first identity of (ES2) becomes

$$eB = xH_0 + \sigma\mathcal{H}b_0^2. \tag{ES6}$$

Put $\kappa = x/S$ and $\nu = \sigma/S$. Multiplying $e(B/S) = \kappa H_0 + \nu\mathcal{H}b_0^2$ by e and using $e\nu = \kappa h$ gives

$$e^2(B/S) = \kappa\{eH_0 + h\mathcal{H}b_0^2\}. \tag{ES7}$$

All factors in braces are bounded and nonnegative. Hence $e^2 = O(x/S)$, and (ES6) also gives $x/S = O(e)$. In particular ζ unbounded implies e infinitesimal, since $x/S \le 1/\zeta$. The unit-ζ cases may have e a positive bounded unit.

The other original coefficients have the opening-cleared bounds

$$\frac{CS(1-r)}{E} \sim \tfrac{1}{2}, \qquad \frac{ASr}{E} \sim \tfrac{1}{2}, \qquad m = \frac{K(1+\sigma r)eS}{2M\zeta P}, \tag{ES8}$$

$$\alpha = \frac{(1+\sigma r)\{LReS + L^2ME\}}{2QP} = O(L), \qquad Q = 1 - Lr \sim 1. \tag{ES9}$$

For A use $P_4 = (1+xh)E + (1-r)(xh+L) \sim E$. The bound on α follows because $0 < M < R$ is bounded and $L/S \le 1$. Thus C, A are unbounded while D, α are infinitesimal on the entire stated source domain. No bound on m is asserted through $M = 0$, and no bound on beta is asserted through $\theta = 0$.

Finally put $\mu = K(1+\sigma r)S/(2M\zeta P) > 0$. The original m term has the exact shared-sphere split

$$m = m_0 + m_1 b_0^2, \qquad m_0 = \frac{\mu x H_0}{B}, \qquad m_1 = \frac{\mu\sigma\mathcal{H}}{B}. \tag{ES10}$$

The remaining source opening M is displayed explicitly. These identities provide the weights and boundedness estimates for C14.4. Section B.1.2 assembles the full-parent certificates for the remaining infinitesimal-u tests and the complementary testing charts. The inequalities above use only the order in each individual field.

B.7.2 The unmatched Bob residue

This is a partial-domain theorem. It retains the original physical C14.4 source, all strict gaps, nonnegative event weights, the original total-scale and native source inequalities, and independent source and testing fields. Its source has $x, L > 0$ infinitesimal, $\zeta = L/x$ noninfinitesimal and η unbounded. Write $\sigma = 1/\eta$, $S = \sigma + L$ and

$$e = \frac{xh}{\sigma}, \quad E = 1 - e, \quad \kappa = \frac{x}{S}, \quad \nu = \frac{\sigma}{S}, \quad \ell = \frac{L}{S}, \quad \nu + \ell = 1. \tag{SE1}$$

The letter κ in this section denotes x/S, not the different coefficient denoted by κ in the R19 reciprocal-L chart. All source inequalities are taken in the coefficient field. Real residues are compared only after taking residues separately in the two fields.

The exact source input. Use ES1–ES10 from the retained R13 source proof. In particular $1/18 \le E < 1$, $0 < h, r < 1$, $\theta = 1 - h$, and

$$\begin{gathered} J = 1 + xh - \sigma(1-r), \qquad M = 1 + xr - \theta/\zeta > 0, \\ Q = 1 - Lr, \quad H_0 = 1 + L(1-r), \quad B = P + x(J - \sigma), \\ P = \sigma(1 + L + 2xh + rxL + rx^2h) + (1+x)L + x^2h\theta, \\ \mathcal{H} = (1+xr)H_0 + (1-r)xh(2+xr) - e\{2 + x(1 + h + r\theta)\} + (1+x)e^2. \end{gathered} \tag{SE2}$$

The opening-safe sphere identities are

$$\begin{gathered} 2\theta P\beta = hB - \sigma H_0, \qquad d_0^2 + b_0^2 = 1, \quad d_0, b_0 \ge 0, \\ D = \frac{\sigma^2\mathcal{H}}{2PJ}d_0^2, \qquad x\theta\beta = \frac{\sigma^2\mathcal{H}}{2P}b_0^2, \\ e\frac{B}{S} = \kappa H_0 + \nu\mathcal{H}b_0^2, \qquad e^2\frac{B}{S} = \kappa(eH_0 + h\mathcal{H}b_0^2). \end{gathered} \tag{SE3}$$

Here $P/S \sim 1$, $J, Q, H_0 \sim 1$, $B/S - (1+\kappa)$ is infinitesimal, and $\mathcal{H} - E^2$ is infinitesimal. In particular B/S and $\mathcal{H}$ are positive bounded units. These estimates are the proved ES bounds; they do not use bounded η.

Source dichotomy. On the entire C14.4 source,

$$e \text{ is infinitesimal} \quad \Longleftrightarrow \quad \kappa = x/S \text{ is infinitesimal.} \tag{SE4}$$

Indeed the first nonnegative sum in SE3 gives $\kappa = O(e)$, and the second gives $e^2 = O(\kappa)$. Both implications are ordered-field statements entirely inside the source field.

Suppose e is infinitesimal. If ζ is unbounded, then θ/ζ is infinitesimal and $M \sim 1$ directly. If ζ is a positive bounded unit, then the exact relation

$$\ell = \zeta\kappa \tag{SE5}$$

gives ℓ infinitesimal and $\nu \sim 1$. Consequently $B/\sigma \sim 1$. Since $\beta \geq 0$, the first identity of SE3 implies $hB \geq \sigma H_0$, whence $h \sim 1$ and θ is infinitesimal. Thus $M \sim 1$ in this case as well. We have proved, uniformly on the whole small-e source,

$$E \sim 1, \quad M \sim 1, \quad m \in \mathrm{Inf}_{>0}\,. \tag{SE6}$$

For the last statement use the exact original coefficient

$$m = \frac{(1+x)(1+\sigma r)eS}{2M\zeta P}; \tag{SE7}$$

$1/\zeta$ is bounded and every other coefficient apart from e is bounded, including the reciprocal of M just proved to be a unit. No assumption on the ratio of a source infinitesimal to a testing infinitesimal occurs.

The two nonnegative terms in the first sum in SE3 imply νb_0^2 infinitesimal. Hence the exact sphere formula and the exact formula for α yield

$$\frac{D}{S} - \frac{\nu^2}{2} \in \mathrm{Inf}, \qquad \frac{\alpha}{S} - \frac{\ell^2}{2} \in \mathrm{Inf}, \qquad SC(1-r) \sim \tfrac{1}{2}, \qquad SAr \sim \tfrac{1}{2}. \tag{SE8}$$

For clarity the second formula is obtained without dividing by ℓ:

$$\frac{\alpha}{S} = \frac{(1+\sigma r)\{\ell(1+xr)e + \ell^2 ME\}}{2Q(P/S)}. \tag{SE9}$$

These are additive bounded residue estimates, including $D = 0$ whenever that boundary has points in the stated domain. They do not assert a relative estimate at a vanishing pivot.

Theorem: every unmatched Bob residue on the small-e source. Let e be infinitesimal. In the testing field choose normalized homogeneous Bob coordinates (c, s) with $c^2 + s^2 = 1$. For every independent projective test and every real or complex state satisfying

$$\overline{s^2} \neq \bar{\ell}, \tag{SE10}$$

the complete original C14.4 parent is a finite sum of squares in $K_c \otimes_{\mathbb{R}} K_t$.

Proof with the complete parent retained. Noninfinitesimal Alice tests use the unchanged adjacent-C, A theorem: SE8 and S infinitesimal imply that both C and A are unbounded. For infinitesimal or zero u, apply the exact EB3–EB7 homogeneous frame from the retained R18 proof. Those formulas are algebraic identities; their only source-limit inputs to the bounded-Gram proof are precisely m infinitesimal and SE8, now established on the larger source domain. To make the imported implication explicit, put

$$H = S(C+A), \quad \rho = \frac{C}{C+A}, \quad a_h = \frac{SCA}{C+A}, \qquad p = \frac{uW}{\sqrt{S}},\ q = \sqrt{S}Z,\ y = \frac{Y}{\sqrt{S}}.$$

Then $1/\sqrt{H}$ is bounded, $0 < \rho < 1$, $a_h \sim 1/2$, and the harmonic normal $N_y = \sqrt{H}(y - \rho p)$ can be substituted by $y = \rho p + N_y/\sqrt{H}$ with bounded source coefficients. The remaining normals are the EB4 N_m, N_b, N_r, with only pure-field radicals and the positive testing denominators $1 + u^2$ and $1 + u^2 s^2$. The exact full quarter-parent is EB7,

$$N_y^2 + a_h p^2 + ASu^2y^2 + M_B^2 + D_B^2 + A_B^2 + \beta(X + uZ)^2 + \tfrac{1}{2}K_B. \tag{SE11}$$

All EB5 core terms and all EB6 event amplitudes are retained. In particular no state graph is imposed and neither first-normal coupling is removed. The two positive rows ASu^2y^2 and $\beta(X + uZ)^2$ are left whole. On the other six roots $(N_y, N_m, N_b, N_r, p, q)$ the bounded Gram has residue equal to $\mathrm{diag}(1, 1, 1/2, 1/2)$ plus the scalar block

$$\frac{1}{2}\begin{pmatrix} 1 & -\bar{s}\bar{c} \\ -\bar{s}\bar{c} & \bar{\nu}^2\bar{s}^2 + \bar{\ell}^2\bar{c}^2 \end{pmatrix}. \tag{SE12}$$

Its determinant is $(\overline{s^2} - \bar{\ell})^2/4$, so SE10 makes this finite coefficient Gram strictly positive definite. The separated bounded-Gram lemma supplies tensor SOS; restoring the two whole rows gives the complete parent. The exact homogeneous formulas allow $c = 0$, $s = 0$ and $u = 0$. Signs and real/imaginary state parts are handled exactly as in EB, and the original positive source and testing normalizations are pure-field factors. This proves the stated domain.

New part, and exact retained complement. Unbounded ζ already forces $\kappa \le 1/\zeta$ infinitesimal, so the new theorem contains the entire R18 unmatched-residue certificate. Its additional domain is ζ a positive bounded unit, e infinitesimal, and Bob sine square with nonzero residue; this includes Bob infinity. In this added domain ℓ has zero residue, so the remaining Bob chart is exactly finite infinitesimal v (including zero, which may be handled by independent literal-axis exits).

After the unmatched-residue certificates, the complementary C14.4 domain is contained in the union of (i) e infinitesimal and matched Bob residue, and (ii) ζ a positive bounded unit and e a positive bounded unit. All original intersections remain in both. In case (ii), SE7 shows that m has a positive real lower bound, while its upper bound may diverge through $M = 0$. This is a source estimate. The complete certificates for both complementary domains, with their exact source partition, are joined in Section B.1.2.

B.7.3 The homogeneous unmatched-Bob quadratic

Precise certificate. Retain the entire original C14.4 source: $x, L > 0$ infinitesimal, η unbounded, $\zeta = L/x$ noninfinitesimal, all original physical intersections, every positive opening, and either permitted literal $D = 0$ or $\beta = 0$. This section additionally assumes ζ unbounded. Put

$$\sigma = 1/\eta, \qquad S = \sigma + L, \qquad \nu = \sigma/S, \qquad \ell = L/S = 1 - \nu.$$

For a real projective Bob test choose its normalized homogeneous coordinates (c, s), so $c^2 + s^2 = 1$; on the finite chart these are $c = (1 + v^2)^{-1/2}$, $s = vc$. The complete original parent is tensor SOS at every independent test and state satisfying

$$\overline{s^2} \neq \overline{\ell}. \tag{EB1}$$

Bars mean real residues of bounded elements in their respective fields. This compares real residues only, not independent infinitesimal rates. The adjacent-C, A theorem already covers noninfinitesimal Alice tests. Hence only u infinitesimal is proved below. No all-tests closure of C14.4 is asserted: the matched residue and bounded-unit ζ source remain.

Source limits, including both sphere endpoints. Use ES1–ES10 without changing their source entry. Since $x/S \le x/L = 1/\zeta$ is infinitesimal, ES7 gives $e = xh/\sigma$ infinitesimal, and hence $E \sim 1$. Moreover $M = 1 + xr - \theta/\zeta \sim 1$. ES8 then gives m infinitesimal. The exact equation ES6, divided by S, is

$$e(B/S) = (x/S)H_0 + \nu\mathcal{H}b_0^2.$$

Both summands are nonnegative; B/S and $\mathcal{H}$ tend to one. Consequently νb_0^2 is infinitesimal. Therefore ES5 yields

$$\frac{D}{S} - \frac{\nu^2}{2} \in \mathrm{Inf}, \qquad \frac{\alpha}{S} - \frac{\ell^2}{2} \in \mathrm{Inf}, \qquad m \in \mathrm{Inf}_{>0}\,. \tag{EB2}$$

For the second formula, divide the exact ES9 expression by S:

$$\frac{\alpha}{S} = \frac{(1 + \sigma r)\{\ell Re + \ell^2 ME\}}{2Q(P/S)}.$$

All displayed denominator factors tend to one. These are additive residue assertions, valid also when ν or ℓ is infinitesimal. In particular no division by either sphere coordinate is used.

The opening-cleared ES8 bounds give

$$c_* = SC(1 - r) \sim \tfrac{1}{2}, \qquad a_* = SAr \sim \tfrac{1}{2}.$$

Define the pure-source harmonic quantities

$$H = S(C + A), \qquad \rho = C/(C + A), \qquad a_h = SCA/(C + A) = \frac{c_* a_*}{c_* r + a_*(1 - r)}. \tag{EB3}$$

Then $1/\sqrt{H}$ is bounded, $0 < \rho < 1$, and $a_h \sim 1/2$, uniformly through both r openings. No upper bound on C or A has been assumed.

An exact all-state frame for every Bob chart. Let $D_u = 1 + u^2$ and $A_u = 1 + u^2 s^2$, both positive testing units. Use the separated roots

$$p = \frac{uW}{\sqrt{S}}, \quad q = \sqrt{S}Z, \quad y = \frac{Y}{\sqrt{S}}, \quad N_y = \sqrt{H}(y - \rho p), \quad N_m = \sqrt{m}\, sW,$$

$$N_b = \sqrt{A_u}\left(X + \frac{uc^2}{A_u}Z\right), \qquad N_r = \frac{\sqrt{D_u}\,usc}{\sqrt{A_u}}Z. \tag{EB4}$$

Only pure-source and pure-testing radicals or inverses occur. In particular no inverse of c, s, u, or a state coordinate is needed. The inverse harmonic allocation $y = \rho p + N_y/\sqrt{H}$ has bounded coefficients.

Write $b = X + uZ$. The exact original signed core K/D_v in finite Bob coordinates equals the following homogeneous expression:

$$\begin{aligned} K_B = {} & N_b^2 + N_r^2 - 2D_u sc\,pq + Sc^2(p-y)^2 + SD_u s^2 y^2 \\ & + \frac{2\sqrt{S}\,usc}{\sqrt{A_u}}(p-y)N_b + \frac{2\sqrt{S}\,us^2\sqrt{D_u}}{\sqrt{A_u}}(p-y)N_r. \end{aligned} \tag{EB5}$$

Here $D_u s$ denotes $D_u\, s$, and similarly in the other products. One derivation starts with CE2 divided by D_v, sets $R = usZ$, and completes its entire (b, R) block:

$$A_u b^2 - 2D_u s\, bR + D_u R^2 = N_b^2 + N_r^2.$$

The cross term from $[\sqrt{S}\,c(p-y) + usb]^2$ is allocated by $cb = cN_b/\sqrt{A_u} + \sqrt{D_u}sN_r/\sqrt{A_u}$. This proves EB5 without deleting a first state normal.

The original positive event roots have the exact bounded expressions

$$\begin{aligned} M_B &= N_m - \frac{\sqrt{m}\,c}{\sqrt{A_u}}N_b - \frac{\sqrt{mD_u}\,s}{\sqrt{A_u}}N_r + \sqrt{mS}\,us\,y, \\ D_B &= \sqrt{DS}\,c(p-y) + \frac{\sqrt{D}\,us}{\sqrt{A_u}}N_b - \sqrt{D/S}\,\frac{D_u s}{A_u}q, \\ A_B &= \sqrt{\alpha D_u S}\,sy - \sqrt{\alpha D_u/S}\,cq. \end{aligned} \tag{EB6}$$

They are, respectively, the normalized original m, D, α event amplitudes. The formulas are valid at $D = 0$ as well. In the D row, the identity $usb - D_u sZ = usN_b/\sqrt{A_u} - D_u sZ/A_u$ avoids an apparent reciprocal-c pole.

Consequently the full original quarter-parent, divided by D_v on the finite chart, is exactly

$$\mathcal{P}_B = N_y^2 + a_h p^2 + ASu^2y^2 + M_B^2 + D_B^2 + A_B^2 + \beta b^2 + \tfrac{1}{2}K_B. \tag{EB7}$$

The two explicit positive rows ASu^2y^2 and βb^2 can be retained unchanged. All other rows in EB7 use EB4 once. The formula is a full four-state identity and is not a restriction to a state ray.

Strictness precisely off the matched residue. Every coefficient of EB5 except the printed pq term is either its fixed positive pivot or infinitesimal after EB3. In EB6, the m row is N_m plus infinitesimal coefficients; the D and α rows have only their q coefficients at residue. Use EB2, $A_u, D_u \sim 1$, and EB3. The resulting residue Gram on $(N_y, N_m, N_b, N_r, p, q)$ is the direct sum of $\mathrm{diag}(1, 1, 1/2, 1/2)$ and

$$\frac{1}{2}\begin{pmatrix} 1 & -\bar{s}\bar{c} \\ -\bar{s}\bar{c} & \bar{\nu}^2\bar{s}^2 + \bar{\ell}^2\bar{c}^2 \end{pmatrix}. \tag{EB8}$$

The determinant of this last block is

$$\frac{1}{4}(\bar{s}^2 - \bar{\ell})^2. \tag{EB9}$$

Thus EB1 makes the entire bounded finite Gram strictly positive. The separated bounded-Gram lemma gives tensor SOS, and restoring the two retained positive rows proves the complete parent. The roots are allowed to be dependent or even identically zero at an endpoint; the claim concerns the coefficient Gram, so this does not affect the argument.

The same homogeneous formulas allow $c = 0$ and $s = 0$ literally. They include every Bob projective endpoint, every sign, and every local Alice axis. Nonlocal Alice charts use the common adjacent C, A theorem. The real and imaginary state parts are treated separately. Every normalization is pure source or pure testing.

The exact remaining obstacle. The leading scalar has the useful two-square identity

$$\tfrac{1}{2}p^2 - sc\,pq + \tfrac{1}{2}(\nu^2 s^2 + \ell^2 c^2)q^2 = \tfrac{1}{2}(p - scq)^2 + \tfrac{1}{2}(s^2 - \ell)^2 q^2. \tag{EB10}$$

It does not complete the matched-residue chart: there the second pivot collapses, and the actual source corrections must be allocated through exact contact conductors. Arbitrary small positive D or β openings do not supply a missing fixed pivot. Unit ζ also requires new source limits because e can have nonzero residue. These two obligations are retained, not declared empty.

B.7.4 The common finite matched-Bob angular relation

This exact auxiliary identity supplies the angular helper budget in FM. Work in the separated tensor ring $K_c \otimes_{\mathbb{R}} K_t$. Let S, κ, ν, ℓ be positive source quantities, with $\nu + \ell = 1$, and put

$$x = \kappa S, \qquad j = \sqrt{\nu\ell}, \qquad a_B = \sqrt{\ell/\nu}. \tag{MB1}$$

For independent testing variables u, v, W, Z define

$$\begin{gathered} T = \kappa W, \quad U = uZ, \quad P = uW/\sqrt{S}, \quad Q = \sqrt{S}Z, \\ N = T - jU, \quad G = P - jQ, \quad \delta = v - a_B, \\ B = \delta Q, \quad B_u = \delta U. \end{gathered} \tag{MB2}$$

The monomial relation $TU = \kappa PQ$ is exact. It gives

$$\delta^2 N^2 + 2j\kappa\delta GB + 2j^2\kappa B^2 - j^2 B_u^2 = (\delta T)^2. \tag{MB3}$$

In particular, if ν, ℓ are positive bounded units, κ is infinitesimal and δ has zero real residue, then any fixed real positive diagonal on B_u can be allocationd at an infinitesimal perturbation of the Gram matrix on (N, G, B). Indeed, for any fixed $c > 0$ the exact SOS remainder is

$$\frac{c}{j^2}\delta^2 N^2 + \frac{2c\kappa}{j}\delta GB + 2c\kappa B^2 - cB_u^2 = \frac{c}{j^2}(\delta T)^2. \tag{MB4}$$

Every displayed coefficient is a finite sum of bounded separated factors, with zero real residue except the constant c in the desired B_u pivot. The only inverses are of positive source units. The factor δ is a difference of a pure testing and a pure source quantity; no inverse or radical of that difference is taken. The right side has the displayed separated SOS root $\sqrt{c}\,\delta T/j$.

Two exact radial conductors and one cross conductor are

$$\begin{aligned} (u^2 - x)W/\sqrt{S} &= uG - \sqrt{S}N, \\ (u^2 - x)Z &= (\kappa\sqrt{S}/j)G - (u/j)N, \\ (u^2 - x)\delta Z^2 &= (\kappa/j)GB - (1/j)NB_u. \end{aligned} \tag{MB5}$$

These identities require no bound on the state coordinates or on any ratio of source and testing infinitesimals. In the local matched chart with u infinitesimal their first two right sides have only infinitesimal coefficients on (N, G).

B.7.5 Unit κ, ν, ℓ: complementary Bob charts

Exact scope and conclusion. Retain the original physical C14.4 source, its strict source gaps, and its native source inequalities. Assume

$$S = \sigma + L \in \mathrm{Inf}_{>0}, \qquad \kappa = x/S, \quad \nu = \sigma/S, \quad \ell = L/S \quad \text{are positive bounded units,} \qquad \nu + \ell = 1. \tag{FU1}$$

Together with the matched theorem UKR1–UKR21, the argument below proves that the complete original parent is a finite separated tensor SOS for every projective test and every state on FU1. It includes arbitrary openings of $M, \theta, r, 1 - r$, literal $D = 0$ and $\beta = 0$, and independent source and testing infinitesimals. It makes no claim at the ν- or ℓ-infinitesimal source endpoints.

All normalizations are pure-source or pure-testing operations. All real Schur complements below prove strictness of a real residue matrix; no such inverse is used as a tensor-ring coefficient. The strict bounded-Gram lemma is the one in the surviving Lemma 3.2.

The full parent and opening-safe source bounds. Write $D_u = 1 + u^2, D_v = 1 + v^2$, and divide the parent by 4:

$$\begin{aligned} F/4 = {} & m(vW - X + uvY - uZ)^2 + CD_v(uW - Y)^2 + AD_uD_vY^2 \\ & + (D + \tfrac{1}{2})(uW - Y + uvX - vZ)^2 + (\alpha + \tfrac{1}{2})D_u(vY - Z)^2 \\ & + (\beta + \tfrac{1}{2})D_v(X + uZ)^2 - \tfrac{1}{2}D_uD_vZ^2. \end{aligned} \tag{FU2}$$

Keep the exact FM2–FM3 source formulas, with $c_0 = SC(1 - r)$, $a_0 = SAr$, $d_0 = D/S$, and $a_1 = \alpha/S$. These four coefficients and $\beta\theta$ are bounded, with no inverse of the possibly small M or θ in their normalized formulas. The D/S numerator has its factor S canceled exactly.

The following letters refer only to real residues:

$$A_* = \nu - \kappa h = \nu E, \quad R = 1 - A_*, \quad \nu = A_* + \kappa h, \quad \ell = 1 - A_* - \kappa h. \tag{FU3}$$

The exact source yields

$$\begin{gathered} E \ge 1/18, \qquad h - A_* \ge 0, \qquad A_*^2 - \kappa(h - A_*) \ge 0, \qquad R - \kappa \ge 0, \\ c_0 = a_0 = E/2, \qquad d_0 = \{A_*^2 - \kappa(h - A_*)\}/2, \\ a_1 = \frac{\ell\{\kappa h + (R - \kappa)A_*\}}{2\nu}, \qquad 2\beta\theta = h - A_*. \end{gathered} \tag{FU4}$$

In particular $h \ge \nu/(1+\kappa)$, so h, E, a_1, c_0, a_0 have positive real lower bounds. The actual source has m bounded below by a positive real, while C, A are unbounded. Here m can be unbounded, and so can β. They cannot both be unbounded: m unbounded forces $R = \kappa$, whence $\ell = \kappa\theta$ makes θ a unit; β unbounded forces $\theta = 0$, whence $R - \kappa = \ell$ makes M a unit. For bounded m, β their real values are

$$m = \frac{\kappa^2 h}{2\nu(R - \kappa)}, \qquad \beta = \frac{h - A_*}{2(1 - h)}. \tag{FU5}$$

These formulas are used only when the indicated real denominators are positive. Every estimate below for the actual parent uses bounded normalized source coefficients before taking residues.

A complete finite-Bob frame, including the state normals. Assume u is infinitesimal and v is finite. We may cap m and β by finite pure-source values $m_c > 0, \beta_c \ge 0$ and retain the complete excesses in their original event squares. When a weight is bounded, keep it unchanged. When it is unbounded, the sufficiently large real cap is chosen below. Define

$$B_0 = X + uZ, \quad N_y = \frac{Y - ruW}{\sqrt{Sr(1-r)}}, \qquad T = \kappa W,\ U = uZ,\ P = \frac{uW}{\sqrt{S}},\ Q = \sqrt{S}Z. \tag{FU6}$$

All six rows are legal separated expressions, and $TU = \kappa PQ$ holds exactly. No restriction is placed on their values or on state ratios. Set $a = u^2$, $s_y = \sqrt{Sr(1-r)}$, and

$$\begin{aligned} M_0 &= (v/\kappa)(1 + ra)T - B_0, \\ A_P &= D_v\{c_0(1-r) + a_0 r(1+a)\} \\ &\quad + S\{(D + \tfrac12)(1-r)^2 + (\alpha + \tfrac12)(1+a)r^2v^2\}. \end{aligned} \tag{FU7}$$

The entire capped parent is $F_0 + 2L_yN_y + H_yN_y^2$, where

$$\begin{aligned} F_0 &= m_cM_0^2 + A_PP^2 + (1+a)\{d_0(1+a)v^2 + a_1\}Q^2 + \tfrac12(1+a)v^2U^2 \\ &\quad - 2(D + \tfrac12)(1+a)v^2B_0U - (1+a)v\{1 + 2D(1-r) + 2\alpha r\}PQ \\ &\quad + \{(\beta_c + \tfrac12)D_v + (D + \tfrac12)av^2\}B_0^2 + 2(D + \tfrac12)(1-r)uv\sqrt{S}\,B_0P, \end{aligned} \tag{FU8}$$

$$\begin{aligned} L_y &= m_cuvs_yM_0 + \sqrt{r(1-r)}D_v\{-c_0 + a_0(1+a)\}P \\ &\quad + S\sqrt{r(1-r)}\{-(D + \tfrac12)(1-r) + (\alpha + \tfrac12)(1+a)rv^2\}P \\ &\quad - (D + \tfrac12)uvs_yB_0 + \sqrt{r(1-r)}v(1+a)(D - \alpha)Q, \\ H_y &= rc_0D_v + (1-r)a_0(1+a)D_v \\ &\quad + Sr(1-r)\{m_cu^2v^2 + D + \tfrac12 + (\alpha + \tfrac12)(1+a)v^2\}. \end{aligned} \tag{FU9}$$

These are exact identities in the two unrestricted original state coordinates X, Y; B_0 is not set to zero. Their coefficients are bounded across all source openings. In L_y every coefficient has zero residue, since c_0, a_0 have the same residue, and the other terms contain u or S. The N_y pivot has residue $ED_v/2 > 0$. Thus it suffices to make the remaining five-row real residue strict.

Finite nonzero unmatched Bob residue. Take $v > 0$ with a positive finite residue different from $b = \sqrt{\ell/\nu}$. The complete real residue of FU8–FU9 is

$$\begin{aligned} &m_c\{(v/\kappa)T - B_0\}^2 + qD_vB_0^2 - v^2B_0U + \tfrac12v^2U^2 - vPQ \\ &\quad + \tfrac{E}{2}D_vP^2 + (d_0v^2 + a_1)Q^2 + \tfrac{E}{2}D_vN_y^2, \qquad q = \beta_c + \tfrac12. \end{aligned} \tag{FU10}$$

For the moment use the uncapped finite values m, β from FU5. Eliminating B_0 in this real matrix, and using $TU = \kappa PQ$ only to name the adjusted cross coefficient, gives

$$a_T T^2 + a_U U^2 + a_P P^2 + a_Q Q^2 - 2fPQ, \tag{FU11}$$

where, for $c = v^2$ and $H = m + q(1 + c)$,

$$a_T = \frac{mq(1+c)c}{\kappa^2 H}, \quad a_U = \frac{c(2H-c)}{4H}, \quad a_P = E(1+c)/2, \quad a_Q = d_0 c + a_1, \\ f = \frac{v}{2}(1 + mc/H). \tag{FU12}$$

The replacement of the real Schur cross term TU by κPQ is implemented in the actual bounded Gram by a real multiple of the exact zero $TU - \kappa PQ$. The elimination of B_0 itself is only a real strictness test. Every diagonal is positive, including when D or β is zero. The reduced determinant in a formal positive real variable y is

$$\Delta(y, c) = (a_T \kappa^2 + a_P y)(a_U y + a_Q) - f^2 y. \tag{FU13}$$

It obeys $4H\Delta/(1 + c) = \mathcal{P}(y, c)$ with

$$\mathcal{P} = p_{22} y^2 c^2 + p_{21} y^2 c + p_{12} y c^2 + p_{11} y c + p_{10} y + p_{02} c^2 + p_{01} c, \tag{FU14}$$

$$\begin{array}{ll} p_{22} = E\beta, & p_{21} = E(m+q), \\ p_{12} = 2m\beta + 2qEd_0, & p_{11} = -m - q + 2E\{(m+q)d_0 + qa_1\}, \\ p_{10} = 2(m+q)Ea_1, & p_{02} = 4mqd_0, \quad p_{01} = 4mqa_1. \end{array} \tag{FU15}$$

All coefficients except possibly p_{11} are nonnegative, and p_{21}, p_{10}, p_{01} are strictly positive. Direct substitution of FU3–FU5 gives the exact three contact identities

$$\mathcal{P}(\kappa, \ell/\nu) = \mathcal{P}_y(\kappa, \ell/\nu) = \mathcal{P}_c(\kappa, \ell/\nu) = 0. \tag{FU16}$$

These finite rational identities can also be checked without radicals. For positive real y, c,

$$\frac{\mathcal{P}(y, c)}{yc} = p_{22} yc + p_{21} y + p_{12} c + p_{11} + p_{10}/c + p_{02} c/y + p_{01}/y. \tag{FU17}$$

As a real function of $(\log y, \log c)$, this is strictly convex: each nonconstant summand has nonnegative Hessian, and $p_{21} y + p_{10}/c$ alone has positive definite Hessian. FU16 says its value and gradient vanish at the contact. It is consequently positive at every other positive point. This real argument is only a way to determine a finite residue matrix, not a use of logarithms or mixed inverses in the tensor ring.

Since $c \neq \ell/\nu$, FU13 is positive for all $y > 0$. Minimizing its quotient by y gives

$$f < \kappa\sqrt{a_T a_U} + \sqrt{a_P a_Q}. \tag{FU18}$$

Indeed the minimum of $a_P a_U y + a_T \kappa^2 a_Q / y$ is $2\kappa\sqrt{a_T a_U a_P a_Q}$, attained at a positive y; equality in FU18 would give a zero of FU13. Choose a real λ with $|\lambda| < \sqrt{a_T a_U}$ and $|f - \kappa\lambda| < \sqrt{a_P a_Q}$. Adding the exact zero $-2\lambda(TU - \kappa PQ)$ makes both 2×2 blocks in FU11 strictly positive. Restoring the positive real B_0 pivot and the N_y pivot proves strictness of the complete six-row residue. All actual Gram entries were bounded in FU8–FU9, so the strict bounded-Gram lemma gives the tensor SOS.

Unbounded m or β in the unmatched chart. The preceding strictness persists under a sufficiently large finite cap in the one possible unbounded weight. This assertion has an opening-safe direct verification. If $m \to \infty$, divide FU14–FU15 by m before taking the real limit. Then

$$(p_{22}, p_{21}, p_{12}, p_{11}, p_{10}, p_{02}, p_{01})/m \longrightarrow (0, E, 2\beta, -1 + 2Ed_0, 2Ea_1, 4qd_0, 4qa_1). \tag{FU19}$$

At $R = \kappa$, the three contact identities FU16 remain exact for this limit polynomial. The strict convexity proof FU17 still applies because its y and $1/c$ coefficients are positive. The Schur coefficients tend to

$$a_T \to q(1+c)c/\kappa^2, \qquad a_U \to c/2, \qquad f \to v(1+c)/2, \tag{FU20}$$

with a_P, a_Q unchanged. Therefore FU18 is strict at the limit, and continuity makes it strict for a sufficiently large finite real m_c.

If $\beta \to \infty$, divide instead by q before taking the limit at $h = 1$. The corresponding coefficient vector is

$$(E, E, 2m + 2Ed_0, -1 + 2E(d_0 + a_1), 2Ea_1, 4md_0, 4ma_1), \tag{FU21}$$

again with the three exact contact identities and the two strictly positive convexity coefficients. Now $a_T \to mc/\kappa^2$, $a_U \to c/2$, $f \to v/2$. A sufficiently large finite β_c therefore works. All the other normalized actual coefficients have their real residues with bounded unit denominators; no $O(S)$ term was divided by M or θ. Restore the complete excesses $(m - m_c)(vW - X + uvY - uZ)^2$ and $(\beta - \beta_c)D_v(X + uZ)^2$. These caps are used only on the unmatched chart and need not vanish at the actual contact.

Infinitesimal or zero Bob coordinate. If v is infinitesimal, use FU8–FU9 with the six rows

$$(B_0, T_v, U_v, P, Q, N_y), \qquad T_v = vT, \quad U_v = vU. \tag{FU22}$$

Cap m at any sufficiently small fixed positive real $m_c \le m$, and discard no excess. One may set $\beta_c = 0$ and retain its whole event. The exact coefficients remain bounded: the B_0U coefficient is now a multiple of vB_0U_v, and M_0 becomes $(1 + ra)T_v/\kappa - B_0$. Their real residue is

$$m_c(T_v/\kappa - B_0)^2 + \tfrac{1}{2}B_0^2 + \tfrac{1}{2}U_v^2 + \tfrac{E}{2}(P^2 + N_y^2) + a_1Q^2 > 0. \tag{FU23}$$

This also covers literal $v = 0$. The rows need not be independent coordinates on the state space.

Infinitesimal reciprocal Bob coordinate. Put $w = 1/v$ and use the homogeneous parent w^2F at $w = 0$ as well as for nonzero infinitesimal w. The complete signed core in the quarter-parent is $K_w/2$, with

$$\begin{aligned} K_w = {} & u^2H^2 - 2u^2HY - 2uHW + 2uwHX + u^2w^2W^2 \\ & + 2u^2wWX - 2uw^2WY + (1 + u^2 + w^2)(X^2 + Y^2) - 2uwXY, \qquad H = wZ. \end{aligned} \tag{FU24}$$

This is the exact CE6 full core, so all state normals are present. FU4 gives the real alternative

$$d_0 + \kappa\beta\theta = (\nu E)^2/2 > 0. \tag{FU25}$$

Thus either D/S has a positive real lower bound, or β has one. Both cases allow independent rates for u, w, S.

If β is strong, use the four rows (W, X, Y, U), $U = uZ$, in FU24 by replacing uH with wU. Every coefficient is bounded and the core residue is $X^2 + Y^2$. The exact m event is $(W - wX + uY - wU)^2$, and the β event is $D_w(X + U)^2$. Positive finite caps give the strict residue

$$m_cW^2 + \tfrac{1}{2}(X^2 + Y^2) + \beta_c(X + U)^2 > 0. \tag{FU26}$$

All other event terms and both excesses remain whole.

If D/S is strong, choose positive real c, d, m_c with $c \le c_0, a_0$, $d \le D/S$, and $m_c \le m$. Select exactly

$$C_c = c/[S(1 - r)], \qquad A_c = c/(Sr), \qquad D_c = dS. \tag{FU27}$$

Retain all excesses, including the whole $A_cu^2D_wY^2$. Writing $Y = ruW + \sqrt{Sr(1 - r)}N_y$, the other two selected C, A terms have the exact harmonic identity

$$C_cD_w(uW - Y)^2 + A_cD_wY^2 = cD_w(P^2 + N_y^2). \tag{FU28}$$

Use six rows (W, V, X, P, N_y, Q) with $V = uwZ$ and the exact monomial identity $VW = wPQ$. Replacing uH by V in FU24, the potentially non-small cross term $-2VW$ is exactly $-2wPQ$. Every other off-diagonal core coefficient is infinitesimal after $Y = \sqrt{S}\{rP + \sqrt{r(1 - r)}N_y\}$; its residue is $V^2 + X^2$. The selected m and D events are the complete squares

$$m_c(W - wX + uY - V)^2, \qquad d\{uw\sqrt{S}W + u\sqrt{S}X - w\sqrt{S}Y - Q\}^2. \tag{FU29}$$

The full selected residue is strictly positive:

$$m_c(W - V)^2 + \tfrac{1}{2}(V^2 + X^2) + c(P^2 + N_y^2) + dQ^2 > 0. \tag{FU30}$$

The strict bounded-Gram lemma applies to the exact complete forms, and all unselected positive terms are restored. Both reciprocal proofs include literal $w = 0$ and $u = 0$.

Exhaustion of tests and exact joins. If u is noninfinitesimal, C, A are unbounded by FU4. The two-adjacent-events theorem CE1–CE10 in Section B.2.4 therefore applies to the original full parent and all states. Its hypotheses require only nonnegative event weights and these two unbounded coefficients; no small-κ source statement is imported.

For infinitesimal u, every positive projective v is either infinitesimal, a finite unit, or has infinitesimal reciprocal. FU22–FU30 cover the two end cases. A finite unit has either the matched residue, covered by UKR1–UKR21, or a different residue, covered by FU10–FU21. The transformations

$$(u, Y, Z) \mapsto (-u, -Y, -Z), \qquad (v, X, Z) \mapsto (-v, -X, -Z) \tag{FU31}$$

preserve FU2 exactly and handle both signs. Literal axes and projective endpoints were included in the displayed raw frames. Complex states split into real and imaginary parts. No proof in this exhaustion bounds, normalizes, or relates the testing-state ratios W/Z, X/Z, or Y/Z.

Consequently every testing chart is covered on exactly FU1, with every original source predicate retained. The remaining source endpoints are separate from this conclusion.

B.7.6 Unit κ, ν, ℓ: the matched Bob residue

Scope. This theorem retains the entire original physical C14.4 source, including its strict gaps and native source inequalities. Assume $S \in \mathrm{Inf}_{>0}$ and $\kappa = x/S, \nu = \sigma/S, \ell = L/S$ are positive bounded units, with $\nu + \ell = 1$. The testing chart is $u \in \mathrm{Inf}$ and finite v whose absolute value has the same residue as $b = \sqrt{\ell/\nu}$. All real or complex states are allowed. The unmatched Bob tests on this source are proved in Section B.7.5; the source endpoints are separate cells in Section B.1.2. The present proof is confined to its stated matched-Bob chart and uses no mixed inverse.

The dictionary FM2–FM3 remains an exact source identity on this larger domain, but its small-κ estimates FM4 are replaced below. Set $j = \sqrt{\nu\ell}$ and retain the rows

$$N_x = X + huZ, \quad N_y = (Y - ruW)/\sqrt{Sr(1-r)}, \qquad T = \kappa W,\ U = uZ,\ P = uW/\sqrt{S},\ Q = \sqrt{S}Z,$$
$$N = T - jU, \quad G = P - jQ, \quad \delta = v - b, \quad B = \delta Q, \quad B_u = \delta U, \quad R_Z = (u^2 - x)Z. \tag{UKR1}$$

The radial and cross identities FM8 hold unchanged. Their coefficients on R_Z remain infinitesimal because S, u are infinitesimal and $\kappa, 1/j$ are bounded.

A sharper exact angular budget. For any fixed real $0 < \rho < 1$, the monomial identity $TU = \kappa PQ$ gives

$$\frac{\delta^2}{j^2(1-\rho^2)}N^2 + \frac{2\kappa\delta}{j(1+\rho)}GB + \frac{2\kappa}{1+\rho}B^2 - B_u^2 = \frac{(\delta T - \rho j B_u)^2}{j^2(1-\rho^2)}. \tag{UKR2}$$

Its first two coefficients have zero residue. Thus a fixed cB_u^2 pivot costs $q\kappa cB^2$ in residue, where $q = 2/(1+\rho) > 1$ can be chosen arbitrarily close to one. Every inverse in UKR2 is a pure source unit or a positive real. There is no inverse of δ.

The actual source through both coefficient openings. The source has $1/18 \le E = 1 - \kappa h/\nu < 1$ and κ is bounded because $1/\zeta$ is bounded. The exact $\beta\theta$ identity gives a positive real lower bound on h. Consequently m has a positive real lower bound, although it may be unbounded. Let

$$R_s = 1 + xr, \quad p = \frac{\kappa\theta}{\ell R_s} \in (0,1), \qquad \mu_c = m(1-p) = \frac{\kappa^2 h(1+x)(1+\nu Sr)}{2\nu\ell P_0 R_s}. \tag{UKR3}$$

The quantity μ_c is a positive bounded unit through $M \to 0$. All of $SC(1-r), SAr, D/S, \alpha/S, \beta\theta$ are bounded without inverses of $r, 1-r, \theta, D, \beta$; this follows directly from FM3 after the explicit factor S in the D/S numerator is canceled. Their limits are evaluated below.

Only in the real residue calculation use the abbreviations

$$A = \nu - \kappa h = \nu E, \quad R = 1 - A, \quad d = h - A. \tag{UKR4}$$

Here $A, R, h, \nu, \ell, \kappa$ are positive units, and the original nonnegative weights and positive M imply

$$d \ge 0, \quad A^2 - \kappa d \ge 0, \quad R - \kappa \ge 0, \qquad \nu = A + \kappa h, \quad \ell = 1 - A - \kappa h. \tag{UKR5}$$

Indeed in residue $2\beta\theta = d$, $2D/S = A^2 - \kappa d$, and $\ell M = R - \kappa$. These are additive residue identities; no relative error estimate is asserted at a vanishing opening.

Retain unbounded weights in actual contact-zero rows. Define

$$V = v(W + uY), \quad B_0 = X + uZ, \quad N_m = B_0 - pV. \tag{UKR6}$$

At the actual source contact $u = \sqrt{x}$, $v = b$, $W/Z = j\sqrt{S/\kappa}$, $X = -h\sqrt{x}Z$, $Y = r\sqrt{x}W$, both N_m and N_x vanish. This is an identity at the source contact, not an equation imposed on independent tests.

If m is bounded, set $m_c = m$. If it is unbounded, choose a positive real m_c sufficiently large, as justified below. Similarly keep $\beta_c = \beta$ when β is bounded, and choose a sufficiently large positive real β_c when it is unbounded. An unbounded positive field element exceeds either fixed real cap. Retain the entire positive excesses

$$(m - m_c)N_m^2 + (\beta - \beta_c)D_v N_x^2. \tag{UKR7}$$

The remaining parent is exact, with its m part replaced by

$$m_0 B_0^2 - 2m_1 B_0 V + m_2 V^2, \qquad m_0 = m_c, \quad m_1 = \mu_c + m_c p, \quad m_2 = \mu_c(1+p) + m_c p^2. \tag{UKR8}$$

This follows from the identity $m(V - B_0)^2 - (m - m_c)(B_0 - pV)^2$ and $\mu_c = m(1-p)$. Every coefficient of UKR8 is bounded. The residual β terms are exactly

$$\beta_c D_v N_x^2 + 2(\beta\theta)D_v N_x uZ + (\beta\theta)\theta D_v u^2 Z^2. \tag{UKR9}$$

All coefficients of UKR9 are bounded. The complete cross terms are retained, and both original excesses in UKR7 are restored once at the end.

The complete bounded scalar and normal frame. Apply the FM9–FM19 polynomial construction to this capped parent with the following explicit replacements: in A_T replace m/κ^2 by m_2/κ^2; in A_U replace its $m\theta^2$ term by $m_0\theta^2$; in K replace its $m\theta(1+ra)$ term by $m_1\theta(1+ra)$. In the complete first-normal formulas replace

entry	replacement
H_x first term	m_0
H_y term mu^2v^2	$m_2u^2v^2$
H_{xy} term m	m_1
L_x coefficient $-m/\kappa$	$-m_1/\kappa$
L_U terms $m\theta-(mj/\kappa)v(1+ra)$	$m_0\theta-(m_1j/\kappa)v(1+ra)$
Y_P term $mv^2(1+ra)$	$m_2v^2(1+ra)$
Y_Q term $-m\theta a$	$-m_1\theta a$.

(UKR10)

Include β_cD_v in H_x and omit the separately retained $\beta D_vN_x^2$ term of FM15. All other entries are unchanged. These replacements follow by expanding UKR8–UKR9 in the two unrestricted state normals. They are not a scalar-only operation.

The graph scalar and both complete first normals still vanish to the orders required by FM11 and FM19: the subtracted rows in UKR7 vanish at the actual source contact. Their squares therefore change neither the value nor any first derivative there. The same finite monic divisions supply the exact bounded FM13 scalar and FM18–FM19 full normal conductors. The Y normal has zero-residue cross coefficients, since $SC(1-r)$ and SAr both have residue $E/2$ and the remaining Y coefficients contain S or u^2. Its diagonal residue is $E/(2\nu)>0$.

The only nontrivial real residue uses (N_x,N,G,B,B_u). Writing

$$H_*=h-\tfrac12+\tfrac{\theta}{2}(\ell+\kappa h),\qquad r_*=H_*-j^2m_2/\kappa^2, \tag{UKR11}$$

its exact entries are

$$\begin{aligned} H_c&=\begin{pmatrix} m_0+(\beta_c+1/2)/\nu & -(m_1/\kappa)b\\ -(m_1/\kappa)b & (m_2/\kappa^2)b^2\end{pmatrix},\qquad A_G=E/(2\nu),\\ z&=\begin{pmatrix}-j(m_1/\kappa+E)\\ -r_*/\nu\end{pmatrix},\qquad g_*=E\ell-m_1\theta-\tfrac12+2m_2\ell/\kappa+\kappa r_*/\nu,\\ &\qquad H_B=\nu(m_1\theta+\tfrac12).\end{aligned} \tag{UKR12}$$

Thus the residue form is $(N_x,N)H_c(N_x,N)^t+A_GG^2+2g_*GB+H_BB^2+2z^t(N_x,N)^tB_u$ before the UKR2 pivot is added. In particular the $\kappa r_*/\nu$ contribution to g_* is retained; it is not infinitesimal in the present source sector.

The exact real contact curvature and all source faces. First take an interior real source, $\theta>0$ and $R>\kappa$, without capping, so

$$m=\frac{\kappa^2h}{2\nu(R-\kappa)},\qquad \beta=\frac{h-A}{2\theta},\qquad m_0=m_1=m_2=m. \tag{UKR13}$$

The two hard pivots and the angular pivot obey the exact identities

$$\begin{aligned}(N_x,N)H_c(N_x,N)^t&=\frac{m}{\kappa^2}(bN-\kappa N_x)^2+\frac{R}{2\nu\theta}N_x^2,\\ J_*&=H_B-\frac{g_*^2}{A_G}-\kappa z^tH_c^{-1}z\\ &=\frac{AR+\kappa(h-A)}{2AhR}\{4A^2hR-\kappa(Rh-A)^2\}.\end{aligned} \tag{UKR14}$$

All inverses in UKR14 are used only in a real finite matrix calculation to prove strictness. They are not inverses in the tensor ring. The crucial bracket has the positive decomposition

$$\begin{aligned}4A^2hR-\kappa(Rh-A)^2&=A^3R(4-A)+A^4(R-\kappa)\\ &\quad+A^2R(h-A)(3+A+2\kappa)\\ &\quad+R^2(h-A)\{A^2-\kappa(h-A)\}.\end{aligned} \tag{UKR15}$$

Its first term is a positive unit and all remaining terms are nonnegative by UKR5. This includes the literal $D=0$ and $\beta=0$ faces and both limiting openings.

For completeness the strictness is uniform toward those openings. Put $\mu=m/\kappa^2$ and retain H_* from UKR11 with uncapped coefficients. Then the two entries of $H_c^{-1}z$ are

$$z_x=-\frac{2\nu\theta}{R}\left(jE+\frac{\kappa H_*}{\nu b}\right),\qquad z_N=\frac{\kappa}{b}z_x+\nu-\frac{H_*}{\mu\ell}. \tag{UKR16}$$

They stay bounded. Also

$$g_* = -\frac{2A^2 + Ah\kappa + A\kappa - A - h\kappa}{2\nu} \tag{UKR17}$$

stays bounded, while A_G and J_* have positive real lower bounds near any fixed allowed residue point. UKR14 gives a positive real lower bound on H_c because μ and $R/(2\nu\theta)$ have positive lower bounds and b is a unit. Completing the real source quadratic form with the bounded coefficients UKR16–UKR17 therefore gives a uniform positive lower bound on

$$Q_{\mathrm{phys}} = (N_x, N)H_c(N_x, N)^t + A_G G^2 + 2g_* GB + H_B B^2 + 2\sqrt{\kappa}\, z^t (N_x, N)^t B. \tag{UKR18}$$

This is an auxiliary real matrix used to choose a diagonal budget. It does not impose $B_u = \sqrt{\kappa}B$, or any relation between source and testing angles, in the tensor certificate.

Why finite caps preserve a strict real residue. When m, β are bounded no cap is needed and UKR18 is already strict. If m is unbounded, $R - \kappa = 0$ in residue; since ℓ is a unit and $\ell = R - \kappa h$, θ is then a unit, so β is bounded. If β is unbounded, $\theta = 0$ in residue, and $R - \kappa = \ell > 0$, so m is bounded. Thus at most one unbounded penalty is present.

For the m opening, the real contact-zero linear form in UKR18 is

$$n_m = N_x - (pb/\kappa)N - (pj/\sqrt{\kappa})B, \tag{UKR19}$$

and $p \to 1$. Its N coefficient is a unit. UKR8 shows that subtracting the whole mn_m^2 leaves a bounded coefficient matrix with a well-defined limit at the opening. That limit is strict on $\ker n_m$: approach the residue point by real $\kappa < R$ with A, h fixed, adjust any kernel vector continuously to the exact kernel using the unit N coefficient, and apply the uniform lower bound from UKR18. The parameter inequalities UKR5 are preserved when κ is decreased. A real symmetric matrix strict on the kernel of a nonzero linear form becomes strictly positive definite after adding a sufficiently large fixed real multiple of the form's square; this follows by a real block Schur complement. Choose m_c that large in UKR7.

For the β opening the same argument uses the exact penalty $(\beta/\nu)N_x^2$ and its kernel $N_x = 0$. Approach $h = 1$ from below with A, κ fixed, which also preserves UKR5, and choose β_c sufficiently large. This is a calculation on real residue matrices, not a mixed-field limiting certificate. The actual capped coefficients have those residues because UKR3, UKR8 and UKR9 have bounded denominators through the original independently varying S, M, θ openings. In particular no estimate divides an $O(S)$ error by M or θ.

We have proved that the actual caps can be chosen so that the capped matrix UKR18 is strictly positive definite. It follows that $H_c \succ 0$, $A_G > 0$, and

$$H_B - g_*^2/A_G - \kappa L > 0, \qquad L = z^t H_c^{-1} z \geq 0. \tag{UKR20}$$

Complete separated SOS allocation. Choose $\rho < 1$ sufficiently close to one and then a positive real c so that

$$L < c < \frac{H_B - g_*^2/A_G}{q\kappa}, \qquad q = 2/(1+\rho). \tag{UKR21}$$

Such choices exist by UKR20. Use c times UKR2 to introduce the cB_u^2 pivot. The full real residue on (N_x, N, G, B, B_u) is strictly positive: eliminate H_c to leave $(c - L)B_u^2$, and eliminate A_G to leave $(H_B - q\kappa c - g_*^2/A_G)B^2$. The N_y pivot is $E/(2\nu) > 0$ and its cross coefficients have zero residue. Every other full normal and radial term has already been allocated by the exact bounded FM constructions and UKR10. The strict separated bounded-Gram lemma supplies a finite tensor SOS. Restore the square on the right of UKR2 and both complete excess terms UKR7. This is the full original parent, not its restriction to a state graph.

The original sign transformations of FM5 handle negative u, v, and the proof includes $u = 0$. Complex states split into real and imaginary parts. Arbitrary state ratios and independent source and testing rates are allowed throughout. Unmatched unit-κ Bob tests and the ν- or ℓ-infinitesimal source endpoints remain outside this theorem.

B.7.7 Unit κ, ℓ and infinitesimal ν

Scope. Retain every original physical C14.4 source predicate, strict gap, nonnegative event coefficient, and original native source inequalities. This theorem assumes

$$S = \sigma + L \in \mathrm{Inf}_{>0}, \quad \kappa = x/S \text{ a positive bounded unit}, \quad \nu = \sigma/S \in \mathrm{Inf}_{>0}, \quad \ell = L/S = 1 - \nu. \tag{NU1}$$

It proves a finite separated tensor SOS for the complete original parent at every projective test and every real or complex state. The sector with both κ, ν infinitesimal is not included. No relative source/testing rate, state-ratio bound, or mixed inverse is used. The original literal $D = 0$ and $\beta = 0$ faces are included.

Opening-safe source parameters and the two sphere masses. Use the exact FM2–FM3 native dictionary, and write

$$\begin{aligned} q = \nu/\ell, \quad \zeta = \ell/\kappa = L/x, \quad e = \kappa h/\nu, \quad E = 1 - e, \\ c_0 = SC(1-r), \quad a_0 = SAr, \quad d_0 = D/S, \quad a_1 = \alpha/S. \end{aligned} \tag{NU2}$$

In NU1, q is infinitesimal and ζ is a positive bounded unit. The exact source sphere SE3 gives $e(B/S) = \kappa H_0 + \nu\mathcal{H}b_0^2$. Since $B/S \sim 1+\kappa$, $H_0 \sim 1$, and $\mathcal{H} \sim E^2$, its real residues satisfy

$$e = \frac{\kappa}{1+\kappa}, \quad E = \frac{1}{1+\kappa}, \quad h \in \mathrm{Inf}, \quad \theta \sim 1, \quad c_0 = a_0 = a_1 = E/2. \tag{NU3}$$

The two equalities for e, E in this display are residue equalities; the definition $e = \kappa h/\nu$ remains exact. In particular e, E, c_0, a_0, a_1 are positive bounded units.

Define actual nonnegative source coefficients

$$d_s = \frac{D}{Sq^2}, \qquad n_s = \frac{\beta}{q^2}. \tag{NU4}$$

These are bounded even when either is zero. More explicitly the SE3 sphere identities give

$$d_s = \frac{\ell^2\mathcal{H}}{2P_0 J} d_0^{\mathrm{sph}\,2}, \qquad n_s = \frac{\ell^2\mathcal{H}}{2\kappa P_0\theta} b_0^{\mathrm{sph}\,2}, \qquad Jd_s + \kappa\theta n_s = \frac{\ell^2\mathcal{H}}{2P_0}. \tag{NU5}$$

Here $(d_0^{\mathrm{sph}})^2 + (b_0^{\mathrm{sph}})^2 = 1$; the superscript distinguishes sphere coordinates from D/S. Every denominator in NU5 is a positive unit. At least one of d_s, n_s is a positive unit. No sphere coordinate is inverted.

The sole possibly unbounded event coefficient needed locally is m. Remove its pole by its actual contact-zero event, not by dividing an error by M. With $R_s = 1 + xr$ set

$$p = \frac{\kappa\theta}{\ell R_s} \in (0,1), \qquad \mu_c = m(1-p) = \frac{\kappa e(1+x)(1+\nu Sr)}{2\ell P_0 R_s}. \tag{NU6}$$

Thus μ_c is a positive bounded unit and $m \geq \mu_c$. Choose a fixed positive real $\epsilon < m$. On the reciprocal Bob chart define $V = W + uY$ and $B_c = w(X + uZ)$. Retain the whole positive term $(m-\epsilon)(B_c - pV)^2$. The remaining m quadratic is exactly

$$m_0 B_c^2 - 2m_1 B_c V + m_2 V^2, \qquad m_0 = \epsilon, \quad m_1 = \mu_c + \epsilon p, \quad m_2 = \mu_c(1+p) + \epsilon p^2. \tag{NU7}$$

All three coefficients are bounded through $M \to 0$ with arbitrary rate. Their residues are

$$\mu_c = \frac{\kappa^2}{2(1+\kappa)}, \quad p = \kappa, \qquad m_2 = \kappa^2(\epsilon + \tfrac{1}{2}) > 0. \tag{NU8}$$

The source $M > 0$ also implies $0 < \bar{\kappa} \leq 1$.

Local reciprocal chart and exact angular budgets. Assume u, w are infinitesimal, including literal zero. Use the homogeneous reciprocal parent $F_w/4$ from P4, with its m term replaced by NU7 and all other terms intact. Put

$$\begin{aligned} X_e = X + ewW, \quad N_y = (Y - ruW)/\sqrt{Sr(1-r)}, \quad a = u^2, \quad c = w^2, \quad \delta = c - q, \\ P = uW/\sqrt{S}, \quad Q = \sqrt{S}\, wZ, \quad N = W - \zeta uwZ, \quad G = P - \ell Q, \\ B = \sqrt{S}\,\delta Z, \quad B_u = u\delta Z, \quad R = w(a-x)Z = \frac{\sqrt{S}}{\zeta} G - \frac{u}{\zeta} N. \end{aligned} \tag{NU9}$$

Every row is a finite separated expression. No state coordinate is divided by another. Since $W(uwZ) = PQ$, two exact identities are

$$\begin{aligned} \delta^2 N^2 + 2\zeta\delta wGB + 2\zeta\ell w^2 B^2 - \zeta^2 (wB_u)^2 = (\delta W)^2, \\ \delta^2 G^2 + 2\ell\delta wNB_u + 2\ell\zeta w^2 B_u^2 - \ell^2 (wB)^2 = (\delta P)^2. \end{aligned} \tag{NU10}$$

Consequently, from either positive sphere pivot in NU5, any fixed positive diagonal budgets on both wB and wB_u can be obtained at infinitesimal bounded-Gram cost. To see this without a hidden division, if d_s is a unit, replace B by $\sqrt{d_s}B/\sqrt{d_s}$ in the first identity, divide by ζ^2, and use $wB = w(\sqrt{d_s}B)/\sqrt{d_s}$ for the other budget. All cost coefficients are infinitesimal. If n_s is a unit, use the second identity divided by ℓ^2 and $wB_u = w(\sqrt{n_s}B_u)/\sqrt{n_s}$. The factor ζw^2 is infinitesimal because ζ is a unit in NU1. This is the precise restriction preventing a silent extension to infinitesimal κ.

The entire scalar graph before normal restoration. Write $D = Sd_0$, $\alpha = Sa_1$, and define bounded coefficient polynomials in the independent testing squares a, c:

$$\begin{aligned}
A_P &= (1+c)\{c_0(1-r) + a_0 r(1+a)\} \\
&\quad + S\{(D+\tfrac{1}{2})(1-r)^2 c + (\alpha+\tfrac{1}{2})(1+a)r^2\}, \\
A_N &= m_2(1+ra)^2 + 2m_1 ec(1+ra) + m_0 e^2 c^2 + (\beta+\tfrac{1}{2})e^2 c(1+c) \\
&\quad - (D+\tfrac{1}{2})e\{2(1-r) - e\}ac, \\
K &= m_1(1+ra) + m_0 ec + (D+\tfrac{1}{2})(1-r-e) \\
&\quad + (\alpha+\tfrac{1}{2})(1+a)r + (\beta+\tfrac{1}{2})e(1+c), \\
F_Z &= D + \alpha c + \beta a + (m_0 + \alpha + \beta + \tfrac{1}{2})ac, \\
J_1 &= K - \ell A_P, \qquad L_1 = J_1 - \zeta A_N, \\
H &= F_Z - \ell^2 S c A_P + ac(\zeta^2 A_N - 2\zeta J_1).
\end{aligned} \tag{NU11}$$

For $X = -ewW$, $Y = ruW$, the exact graph scalar is

$$F_0 = A_N N^2 + A_P G^2 - 2uwZL_1 N + HZ^2. \tag{NU12}$$

This graph is used to compute coefficients; the unrestricted normals are restored explicitly below. The actual source contact has $a = x, c = q, W = \zeta uwZ$, so $N = G = X_e = N_y = 0$. The retained row $B_c - pV$ also vanishes there. Direct source substitution in NU11 yields

$$L_1(x,q) = 0, \qquad H(x,q) = H_a(x,q) = H_c(x,q) = 0, \qquad H(a,0) = D + \beta a. \tag{NU13}$$

These are exact rational source identities; no independent testing angle is identified with a source angle in the certificate.

All divisions in the next display are finite monic polynomial divisions at the bounded pure-source contacts x, q. They define bounded coefficient polynomials $L_a, L_c, H_{20}, H_{11}, H_{02}$:

$$\begin{aligned}
L_1(a,c) &= (a-x)L_a(a,c) + (c-q)L_c(c), \\
H(a,c) &= H(x,c) + (a-x)H_a(x,c) + c(a-x)^2 H_{20}(a,c), \\
H_a(x,c) &= (c-q)\{-\beta/q + cH_{11}(c)\}, \\
H(x,c) &= S(c-q)^2\{d_s + \kappa n_s + cH_{02}(c)\}.
\end{aligned} \tag{NU14}$$

Here the factor c in the quadratic a remainder follows from $H(a,0) = D + \beta a$. The two constant terms follow by putting $c = 0$: $H_a(x,0) = \beta$ and $H(x,0) = D + \beta x$. In particular the displayed apparent divisions by q are bounded: $\beta/q = n_s q$. Finally $H(x,c)/S$ is a bounded coefficient polynomial directly from NU11, because $x = \kappa S$, $D/S = d_0$, and $\alpha/S = a_1$ are bounded. Thus dividing it by the monic $(c-q)^2$ introduces no unbounded coefficient. This verifies all asserted coefficient bounds without using relative orders of S, q, u, w.

The useful mixed identity is

$$c(a-x)(c-q)Z^2 = \frac{1}{\zeta}\{G(wB) - N(wB_u)\}. \tag{NU15}$$

Substituting NU14 into NU12 gives the complete exact scalar allocation

$$\begin{aligned}
F_0 &= A_N N^2 + A_P G^2 + d_s B^2 + n_s B_u^2 + H_{02}(wB)^2 + H_{20}R^2 \\
&\quad + \frac{H_{11} - n_s}{\zeta}\{G(wB) - N(wB_u)\} - 2uL_a NR - 2L_c N(wB_u).
\end{aligned} \tag{NU16}$$

The residues of A_N, A_P are respectively $\kappa^2(\epsilon + 1/2)$ and $E/2$, both positive. The row R has infinitesimal coefficients on (N, G) by NU9. Every remaining signed term is a bounded quadratic term on N, G, wB, wB_u. The angular rows are retained as the complete nonnegative squares $(\sqrt{d_s}B)^2$ and $(\sqrt{n_s}B_u)^2$; neither is prematurely discarded.

Both complete state normals. Put $s_y = \sqrt{Sr(1-r)}$ and $r_y = \sqrt{r(1-r)}$. Define

$$\begin{aligned}
A_x &= -m_0 ec - m_1(1+ra) + (D+\tfrac{1}{2})a(1-r-e) - e(\beta+\tfrac{1}{2})(1+c), \\
T_x &= \zeta c A_x + \beta - D + c(m_0 + \beta + \tfrac{1}{2}), \\
Y_P &= (1+c)\{-c_0 + a_0(1+a)\} \\
&\quad + S\{m_2(1+ra) + m_1 ec - (D+\tfrac{1}{2})c(1-r-e) + (\alpha+\tfrac{1}{2})(1+a)r\}, \\
Y_Q &= -m_1 a + (D+\tfrac{1}{2}) - (\alpha+\tfrac{1}{2})(1+a), \qquad Y_C = \ell Y_P + Y_Q.
\end{aligned} \tag{NU17}$$

The entire capped quarter-parent, before any allocation, is

$$F_0 + H_x X_e^2 + H_y N_y^2 + 2H_{xy} X_e N_y + 2L_x X_e + 2L_y N_y, \tag{NU18}$$

where

$$\begin{aligned} H_x &= m_0 c + (D + \tfrac{1}{2})a + (\beta + \tfrac{1}{2})(1 + c), \\ H_y &= r c_0 (1 + c) + (1 - r) a_0 (1 + a)(1 + c) \\ &\quad + S r (1 - r)\{m_2 a + (D + \tfrac{1}{2})c + (\alpha + \tfrac{1}{2})(1 + a)\}, \\ H_{xy} &= -uw s_y (m_1 + D + \tfrac{1}{2}), \\ L_x &= w A_x N + u Z T_x, \qquad L_y = r_y \{Y_P G + Y_C Q\}. \end{aligned} \tag{NU19}$$

These formulas retain every quadratic state coefficient and both first normals. Their real hard residues are $H_x = 1/2$, $H_y = E/2$, $H_{xy} = 0$.

Exact source substitution gives $T_x(x, q) = Y_C(x, q) = 0$. Moreover $T_x(a, 0) = \beta - D$, and its a dependence has a factor c. Thus bounded monic divisions give

$$\begin{aligned} T_x(a, c) &= c(a - x)X_a + (c - q)\{(D - \beta)/q + c X_c(c)\}, \\ Y_C(a, c) &= (a - x)Y_a(a, c) + (c - q)Y_c(c). \end{aligned} \tag{NU20}$$

No coefficient here needs an inverse of a sphere coordinate. Using NU4 and NU9, the full normal conductors become

$$\begin{aligned} L_x &= w A_x N + uw X_a R + d_s \sqrt{S}\, quB - n_s q B_u + w X_c(w B_u), \\ L_y &= r_y \{Y_P G + \sqrt{S}\, Y_a R + Y_c(wB)\}. \end{aligned} \tag{NU21}$$

The two potentially weak terms are face-safe: $d_s \sqrt{S}\, quB = (\sqrt{d_s}\sqrt{S}\, qu)(\sqrt{d_s} B)$ and $n_s q B_u = (\sqrt{n_s} q)(\sqrt{n_s} B_u)$. Their coefficients are infinitesimal, even when either sphere mass is zero. Also Y_P has zero residue by NU3 and NU17. All other terms in NU21 have infinitesimal coefficients on N, G, R or bounded coefficients on the two angular helper rows.

Strict complete Gram and restoration. Begin with the six rows

$$(N, G, X_e, N_y, \sqrt{d_s} B, \sqrt{n_s} B_u). \tag{NU22}$$

Their real diagonal residue in NU16–NU21 is

$$\operatorname{diag}\big(\kappa^2(\epsilon + \tfrac{1}{2}), E/2, 1/2, E/2, 1, 1\big) > 0. \tag{NU23}$$

Every cross coefficient among these rows is infinitesimal after the displayed R substitution. Literal zero rows are harmless: the coefficient matrix may be strict even when the chosen rows are linearly dependent or vanish.

Use NU10 to add sufficiently large fixed real pivots for the two further rows wB, wB_u, at infinitesimal cost to NU22. The signed scalar terms in NU16 and the remaining normal terms in NU21 have bounded coefficients on these helper rows. A sufficiently large fixed pair of diagonals therefore gives a strictly positive real eight-row Gram, by an ordinary real Schur complement. All entries are finite sums of bounded separated factors. The separated bounded-Gram lemma supplies a finite SOS. Restore the exact positive square remainders in NU10 and the whole excess $(m - \epsilon)(B_c - pV)^2$ in NU7. This gives the entire original parent, with both unrestricted state normals.

All projective testing joins on NU1. Noninfinitesimal Alice tests use CE1–CE10 because C, A are unbounded. For infinitesimal u, the local proof above covers every infinitesimal reciprocal w, including zero. It remains to check finite Bob coordinates v.

For v infinitesimal use the exact FU22–FU23 frame; its source inputs here are only m strong, c_0, a_0 with common positive residue $E/2$, and a_1 positive, all proved in NU3–NU6. For a finite positive unit v use the complete FU6–FU12 bounded parent and real Schur calculation, capping the original m event at a finite real only if m is unbounded. In the present source residue $\beta = d_0 = 0$, $a_1 = E/2$, and, if $\kappa < 1$,

$$m = \frac{\kappa^2}{2(1 - \kappa^2)}, \qquad E = \frac{1}{1 + \kappa}. \tag{NU24}$$

The determinant polynomial FU14 becomes exactly

$$\mathcal{P}(y, c) = p_{21} c (y - \kappa)^2 + p_{10} y > 0, \qquad p_{21} = E(m + \tfrac{1}{2}), \quad p_{10} = (m + \tfrac{1}{2})E^2, \quad y, c > 0. \tag{NU25}$$

It gives the strict four-term circuit inequality FU18. If $\bar{\kappa} = 1$, then m is unbounded. The normalized limit is

$$\mathcal{P}(y, c)/m \longrightarrow \tfrac{1}{2}c(y-1)^2 + \tfrac{1}{4}y > 0. \tag{NU26}$$

The Schur coefficients have finite positive limits, so a sufficiently large finite cap preserves strictness, exactly as in FU19–FU20. The whole positive original event excess is restored. These finite-Bob arguments use no lower bound on ν, and do not import the source scope of the FU theorem.

The exact sign transformations FU31 cover negative angles; all displayed frames include literal axes and projective endpoints. Complex states split into real and imaginary parts. No chart normalizes a state coordinate or bounds a state ratio. This completes the all-tests theorem on precisely NU1.

B.7.8 Infinitesimal κ and unit ν, ℓ

Scope. This theorem treats the source sector stated in FM1: S, κ are positive infinitesimals and ν, ℓ are positive bounded units. The source endpoints $\nu \in \mathrm{Inf}$ and $\ell \in \mathrm{Inf}$, and the unit-κ source, are separate cells in the complete partition of Section B.1.2. All original physical C14.4 source conditions, strict gaps, nonnegative event coefficients, total-scale conditions and native source inequalities are retained.

Theorem. Let the original native C14.4 source have

$$S = \sigma + L > 0, \qquad x = \kappa S, \qquad \nu = \sigma/S, \quad \ell = L/S, \quad \nu + \ell = 1, \qquad S, \kappa \in \mathrm{Inf}_{>0}, \tag{FM1}$$

with ν, ℓ positive bounded units. Then the complete original parent is a finite SOS in $K_c \otimes_{\mathbb{R}} K_t$ for every independent projective test and every real or complex state. The argument below supplies the missing finite matched-Bob chart; the other tests use the explicitly scoped SE theorem.

Exact opening-safe source dictionary. Write $\theta = 1 - h$, $0 < r, h < 1$, $e = \kappa h/\nu$, $E = 1 - e$, and set

$$\begin{aligned} J &= 1 + xh - \nu S(1-r), \quad M = 1 + xr - \kappa\theta/\ell, \quad Q_s = 1 - \ell S r, \\ P_0 &= 1 + S\{\nu\ell + 2\nu\kappa h + \kappa\ell + \kappa^2 h\theta\} + S^2\{\nu r\kappa\ell + \nu r\kappa^2 h\}. \end{aligned} \tag{FM2}$$

These are the actual native source quantities, with $P = SP_0$. The converted P1 coefficients, without applying P1's different unit-L asymptotic hypotheses, are exactly

$$\begin{aligned} c_0 = SC(1-r) &= \frac{(1+x)(1+\nu S r)E}{2P_0}, \\ a_0 = SAr &= \frac{(1+xh)E + (1-r)(\ell S + xh)}{2P_0}, \\ d_0 = D/S &= \frac{(1+x)(1+\nu S r) - JP_0}{2JSP_0}, \\ a_1 = \alpha/S &= \frac{(1+\nu S r)\{\ell(1+xr)e + \ell^2 ME\}}{2Q_s P_0}, \\ b_1 = \beta\theta &= \frac{(1+x)(1+\nu S r)(\ell + \kappa h) - \theta P_0}{2P_0}, \\ \mu = m/\kappa^2 &= \frac{(1+x)(1+\nu S r)h}{2M\ell\nu P_0}. \end{aligned} \tag{FM3}$$

The numerator in the d_0 formula has a factor S, so its displayed S denominator cancels in the source polynomial ring. Thus all six coefficients belong to the bounded source algebra generated by $S, \kappa, h, r, \nu, \ell$ and inverses of $J, M, Q_s, P_0, \nu, \ell$. None of these inverses is of an opening $r, 1-r, \theta, D$ or β. The original source gives

$$\begin{aligned} &J, M, Q_s, P_0, E \sim 1, \qquad c_0, a_0 \sim \tfrac{1}{2}, \qquad d_0 \sim \nu^2/2, \quad a_1 \sim \ell^2/2, \\ &b_1 - (h-\nu)/2 \in \mathrm{Inf}, \qquad \mu - h/(2\nu\ell) \in \mathrm{Inf}, \qquad \bar{h} \geq \bar{\nu} > 0. \end{aligned} \tag{FM4}$$

The last inequality follows from $b_1 = \beta\theta \geq 0$. Hence μ is a positive bounded unit. These estimates are uniform through $r, 1-r, \theta \to 0$; no comparison with a testing quantity has been used. In this sector D/S is a positive unit, so $D = 0$ has empty intersection; $\beta = 0$ is allowed.

Exact full parent and unrestricted normals. On the finite Bob chart write $D_u = 1 + u^2$, $D_v = 1 + v^2$. The complete quarter-parent is

$$\begin{aligned} F/4 = {} & m(vW - X + uvY - uZ)^2 + CD_v(uW - Y)^2 + AD_uD_vY^2 \\ & + (D + \tfrac{1}{2})(uW - Y + uvX - vZ)^2 + (\alpha + \tfrac{1}{2})D_u(vY - Z)^2 \\ & + (\beta + \tfrac{1}{2})D_v(X + uZ)^2 - \tfrac{1}{2}D_uD_vZ^2. \end{aligned} \tag{FM5}$$

No state restriction is made. Put

$$N_x = X + huZ, \qquad N_y = \frac{Y - ruW}{\sqrt{Sr(1-r)}}. \tag{FM6}$$

The source square root is nonzero even when its size is arbitrarily small. It is used only to define a permitted tensor root; every coefficient below is bounded without its inverse.

After the unmatched-residue and adjacent-C, A exits it is enough to take u infinitesimal and v a bounded positive unit with the same residue as $b = \sqrt{\ell/\nu}$. Set

$$\begin{gathered} a = u^2, \quad j = \sqrt{\nu\ell}, \quad T = \kappa W, \quad U = uZ, \\ P = uW/\sqrt{S}, \quad Q = \sqrt{S}Z, \quad N = T - jU, \quad G = P - jQ, \\ \delta = v - b, \quad B = \delta Q, \quad B_u = \delta U. \end{gathered} \tag{FM7}$$

Every state coordinate remains arbitrary, including zero and unbounded coordinates. The rows satisfy the exact MB identities

$$\begin{gathered} R_Z = (a-x)Z = (\kappa\sqrt{S}/j)G - (u/j)N, \\ (a-x)\delta Z^2 = (\kappa/j)GB - (1/j)NB_u, \\ \delta^2 N^2 + 2j\kappa\delta GB + 2j^2\kappa B^2 - j^2 B_u^2 = (\delta T)^2. \end{gathered} \tag{FM8}$$

These polynomial identities hold in the separated tensor ring.

The full graph scalar. Let F_0 denote FM5 with $N_x = N_y = 0$; it is used only as the constant coefficient of the full normal expansion. Define bounded source/testing polynomials

$$\begin{aligned} A_T &= \mu v^2(1+ra)^2, \\ A_P &= (1+v^2)\{c_0(1-r) + a_0 r(1+a)\} \\ &\quad + S\{(D+\tfrac12)(1-r)^2 + (\alpha+\tfrac12)(1+a)r^2v^2\}, \\ A_Q &= d_0 v^2 + a_1, \\ A_U &= m\theta^2 + \alpha + b_1\theta + \theta^2/2 \\ &\quad + v^2\{2Dh + h - \tfrac12 + b_1\theta + \theta^2/2\} + (D+\tfrac12)h^2v^2a, \\ K &= m\theta(1+ra) + (D+\tfrac12)(1-r)(1+ha) + (\alpha+\tfrac12)(1+a)r, \\ H &= vK - \kappa j A_T, \quad R_U = A_U - j^2 A_T, \\ g &= jA_P - H, \quad R_Q = A_Q + j^2 A_P - 2jH, \\ J_Z &= SR_Q + aR_U. \end{aligned} \tag{FM9}$$

Direct expansion of every term of FM5 gives

$$F_0 = A_T N^2 + A_P G^2 + 2gGQ + J_Z Z^2. \tag{FM10}$$

For example the only relation used to obtain FM10 from the five monomials T^2, U^2, P^2, Q^2, PQ is $TU = \kappa PQ$.

The actual source stationarity identities, also verified directly from FM2–FM3, are

$$g(x,b) = 0, \qquad J_Z(x,b) = \partial_a J_Z(x,b) = \partial_v J_Z(x,b) = 0. \tag{FM11}$$

These are coefficient-field identities at the actual source contact; they impose no equation on independent tests.

Finite polynomial division. All following quotients mean monic polynomial division in formal a or v, performed before evaluation in the tensor ring. No mixed inverse is introduced. There are bounded coefficient polynomials $g_a, g_v, H_{20}, H_{11}, H_{02}$ defined exactly by

$$\begin{aligned} g(a,v) &= (a-x)g_a(a,v) + (v-b)g_v(v), \\ J_Z(a,v) &= (a-x)^2 H_{20}(a,v) + (a-x)(v-b)H_{11}(v) + S(v-b)^2 H_{02}(v). \end{aligned} \tag{FM12}$$

For clarity, first divide $g(a,v) - g(x,v)$ by $a-x$ and then $g(x,v)$ by $v-b$. For the second line, first subtract the constant and linear Taylor terms in $a-x$ and divide the remainder by $(a-x)^2$. Divide $\partial_a J_Z(x,v)$ by $v-b$ for H_{11}. Finally divide the bounded polynomial $R_Q(x,v) + \kappa R_U(x,v)$ by $(v-b)^2$ for H_{02}. FM11 makes every division exact. In particular H_{02} is bounded without dividing a bounded error by S.

Substituting FM8 into FM10 now yields the exact scalar identity

$$\begin{aligned} F_0 = {} & A_T N^2 + A_P G^2 + 2g_v GB + H_{02}B^2 + H_{11}\{(\kappa/j)GB - (1/j)NB_u\} \\ & + H_{20}R_Z^2 + 2\sqrt{S}\, g_a G R_Z. \end{aligned} \tag{FM13}$$

Thus every coefficient is bounded and all remaining radial contributions are infinitesimal matrix entries on (N, G).

Here and below a displayed residue formula uses the real residues of its bounded source parameters. FM3 gives explicitly

$$\bar{A}_T = \frac{h}{2\nu^2}, \quad \bar{A}_P = \frac{1}{2\nu}, \quad \bar{g}_v = \frac{\ell - \nu}{2}, \quad \bar{H}_{02} = \frac{\nu}{2}, \quad \bar{H}_{11} = -\theta j. \tag{FM14}$$

For example at $S = \kappa = 0$ and $a = 0$ the three relevant polynomials are $g = j(1+v^2)/2 - v/2$, $R_Q = \nu(v-b)^2/2$, and $R_U = \theta(\ell - \nu v^2)/2$. These establish FM14 by the finite divisions just specified.

Both complete first-normal terms. Write the full expansion of FM5 as

$$F/4 = F_0 + H_x N_x^2 + H_y N_y^2 + 2H_{xy}N_x N_y + 2N_x L_x + 2N_y L_y + \beta D_v N_x^2. \tag{FM15}$$

The exact bounded diagonal and cross coefficients are

$$\begin{aligned} H_x &= m + (D + \tfrac{1}{2})u^2v^2 + D_v/2, \\ H_y &= rc_0 D_v + (1-r)a_0 D_u D_v \\ &\quad + Sr(1-r)\{mu^2v^2 + D + \tfrac{1}{2} + (\alpha + \tfrac{1}{2})D_u v^2\}, \\ H_{xy} &= -uv\sqrt{Sr(1-r)}(m + D + \tfrac{1}{2}). \end{aligned} \tag{FM16}$$

In particular $\bar{H}_x = \bar{H}_y = 1/(2\nu)$ and $\bar{H}_{xy} = 0$. The last term of FM15 is retained whole; it allows β to be arbitrarily large or zero.

Define

$$\begin{aligned} L_U &= m\theta - (mj/\kappa)v(1+ra) - (D+\tfrac{1}{2})v^2(1+ha) + (b_1 + \theta/2)(1+v^2) \\ &\quad + (D+\tfrac{1}{2})(1-r)vjS, \\ Y_P &= (1+v^2)\{-c_0 + a_0(1+a)\} + S\{mv^2(1+ra) - (D+\tfrac{1}{2})(1-r) + (\alpha+\tfrac{1}{2})(1+a)rv^2\}, \\ Y_Q &= -m\theta a + (D+\tfrac{1}{2})(1+ha) - (\alpha+\tfrac{1}{2})(1+a), \qquad Y_C = jY_P + vY_Q. \end{aligned} \tag{FM17}$$

The expression $mj/\kappa = \kappa\mu j$ is bounded and infinitesimal. The complete original normal terms are exactly

$$\begin{aligned} L_x &= -\kappa\mu v(1+ra)N + (D+\tfrac{1}{2})(1-r)vu\sqrt{S}\, G + L_U U, \\ L_y &= \sqrt{r(1-r)}\{Y_P G + Y_C Q\}. \end{aligned} \tag{FM18}$$

The actual source gives $L_U(x,b) = Y_C(x,b) = 0$. Perform the same finite monic divisions to write

$$L_U = (a-x)l_a + (v-b)l_v, \qquad Y_C = (a-x)y_a + (v-b)y_v. \tag{FM19}$$

Then $L_U U = u l_a R_Z + l_v B_u$ and $Y_C Q = \sqrt{S}\, y_a R_Z + y_v B$. All their coefficients are bounded. Moreover $\bar{l}_v = -j$, while Y_P and y_v have zero residue. This follows directly from the source-limit identities $L_U = (\ell - \nu v^2)/2$ at $a = 0$, and $Y_P = Y_Q = 0$ at $a = 0$. Consequently $L_x + jB_u$ is an infinitesimal bounded combination of (N, G, B_u) and L_y is an infinitesimal bounded combination of (N, G, B). This accounts for both unrestricted first-normal terms, including all $\beta\theta$ terms and the independent $r, 1-r, \theta$ openings.

A strict residue for the full six-row Gram. Use FM8 with a fixed real $c = 4$ to introduce a cB_u^2 pivot at infinitesimal Gram cost. More explicitly subtract

$$(c/j^2)\delta^2 N^2 + (2c\kappa/j)\delta GB + 2c\kappa B^2 \tag{FM20}$$

from the bounded part of FM15, add cB_u^2 to that part, and retain the exact positive square $(\sqrt{c}\,\delta T/j)^2$. Their sum is unchanged by FM8. The residue on (N_x, N_y, N, G, B, B_u) is therefore

$$\frac{N_x^2 + N_y^2}{2\nu} + \frac{h}{2\nu^2}N^2 + \frac{G^2}{2\nu} + (\ell - \nu)GB + \frac{\nu}{2}B^2 + \theta N B_u - 2jN_x B_u + 4B_u^2. \tag{FM21}$$

The (G, B) block has determinant $\nu\ell > 0$ and positive diagonal. The remaining block has positive diagonal pivots $1/(2\nu), 1/(2\nu), h/(2\nu^2)$, and its final scalar Schur pivot is

$$4 - 2\nu j^2 - \frac{\theta^2\nu^2}{2h} > 0. \tag{FM22}$$

Indeed $j^2 \le 1/4$, $0 < \nu \le 1$, $h \ge \nu$ and $0 \le \theta \le 1$ in residue make the two subtracted terms at most $1/2$ each. All entries before taking residues are finite sums of bounded separated factors. The strict bounded-Gram lemma therefore supplies a finite tensor SOS. Restoring the single square from FM20 and the whole positive $\beta D_v N_x^2$ row gives the entire FM5. No mixed matrix inverse or mixed square root has been used.

Testing, sign, and boundary joins. The argument includes $u = 0$ and every real state without a state normalization or a sign condition. Negative u is changed to $-u$ together with $(Y, Z) \mapsto (-Y, -Z)$; negative v is changed to $-v$ together with $(X, Z) \mapsto (-X, -Z)$. FM5 is exactly invariant under these transformations. On FM1, $e = \kappa h/\nu$ is infinitesimal and C, A are unbounded. The SE10–SE12 theorem covers every unmatched homogeneous Bob residue, including Bob zero and infinity, and invokes the unchanged adjacent-C, A theorem for noninfinitesimal Alice tests. The only remaining Bob residue has finite nonzero v and $\bar{v}^2 = \bar{\ell}/\bar{\nu}$, which is the matched chart just proved. Thus the union covers all projective tests on FM1. Real and imaginary state parts are treated separately for complex states. All divisions and normalizations used in these joins are pure source or pure testing operations; all original source predicates remain.

Dependencies and limits. The analytical inputs are the native source formulas P1 converted as FM2–FM3, the full original parent, the strict separated bounded-Gram lemma in Lemma 3.2, and Section B.7.2 with its preserved EB and adjacent-C, A dependencies. The exact budget is independently printed in Section B.7.4. The source endpoint and unit-κ sectors require separate complete arguments; this theorem makes no claim about them.

B.7.9 Infinitesimal κ, ℓ and unit ν

Theorem and exact scope. Retain the original native C14.4 physical source, all strict gaps, nonnegative event weights, total-scale conditions and native source inequalities. Let

$$S = \sigma + L, \quad x = \kappa S, \quad \nu = \sigma/S, \quad \ell = L/S, \quad \nu + \ell = 1, \quad \zeta = L/x = \ell/\kappa.$$

Suppose $S, \ell > 0$ are infinitesimal and ζ is noninfinitesimal. Then the complete original parent is a finite sum of squares in the separated ring $K_c \otimes_{\mathbb{R}} K_t$ for every independently chosen projective test and every real or complex state. The proof includes $\beta = 0$ and unrestricted openings of $r, 1 - r, \theta$. The intersection with $D = 0$ is empty by the source estimate below. No earlier missing R21 endpoint proof is used.

Source estimates at this endpoint. Use the exact actual-source formulas FM2–FM3 and SE1–SE3, with $0 < r, h < 1$, $\theta = 1 - h$, $e = \kappa h/\nu$, $E = 1 - e \ge 1/18$, $U_s = (1 + x)(1 + \nu S r)$, and

$$J = 1 + xh - \nu S(1 - r), \quad M = 1 + xr - \kappa\theta/\ell > 0, \quad Q_s = 1 - \ell S r,$$
$$a_0 = SAr, \qquad c_0 = SC(1 - r), \qquad P = SP_0.$$

Put $g = \kappa/\ell = 1/\zeta$, which is bounded. Then $\nu \sim 1$ and $\kappa \in \mathrm{Inf}$. Since

$$0 \le \beta\theta = \frac{U_s(\ell + \kappa h) - \theta P_0}{2P_0},$$

we have $\theta = O(\ell + \kappa) = O(\ell)$ and $\beta\theta = O(\ell)$, hence $h \sim 1$, $\beta\theta^2 = O(\ell^2)$ and $M \sim 1$. The exact source formulas give

$$\begin{aligned} J, Q_s, P_0, E \sim 1, \quad c_0, a_0 \sim \tfrac{1}{2}, \quad D/S \sim \tfrac{1}{2}, \quad m \in \mathrm{Inf}_{>0}, \\ \frac{qm}{\kappa^2} = \frac{U_s h}{2M\nu^2 P_0} \sim \tfrac{1}{2}, \qquad q = \ell/\nu, \\ \frac{\alpha}{S\ell^2} = \frac{(1 + \nu S r)\{(1 + xr)e/\ell + ME\}}{2Q_s P_0}, \qquad \frac{\alpha}{S\ell^2} - \frac{1 + g}{2} \in \mathrm{Inf}\,. \end{aligned} \tag{EL0a}$$

The last difference is uniform for bounded g, because $e/\ell = gh/\nu$. All statements are inside the source field; none compares a testing rate to a source rate. In particular D/S is a positive unit, while β may be zero, bounded, or unbounded.

The full parent and local testing chart. Noninfinitesimal Alice tests and unmatched Bob residues will be joined at the end. The remaining chart has $u, v \ge 0$ infinitesimal, including literal zero axes. Write $a = u^2$, $c = v^2$, $b = \sqrt{q}$, $j = \nu b$, and $d = D + 1/2$. The complete original quarter-parent is

$$\begin{aligned} F/4 = {} & m(vW - X + uvY - uZ)^2 + C(1 + c)(uW - Y)^2 \\ & + A(1 + a)(1 + c)Y^2 + d(uW - Y + uvX - vZ)^2 \\ & + (\alpha + \tfrac{1}{2})(1 + a)(vY - Z)^2 + (\beta + \tfrac{1}{2})(1 + c)(X + uZ)^2 \\ & - \tfrac{1}{2}(1 + a)(1 + c)Z^2. \end{aligned} \tag{EL0b}$$

All state coordinates are arbitrary elements of the testing field. No state coordinate is divided by or bounded below. Put $R_s = 1 + xr$, $p = \kappa\theta/(\ell R_s)$, $\mu = m(1-p)$, and

$$V = v(W + uY), \qquad N = X + uZ - pV, \qquad Y = ruW + \sqrt{Sr(1-r)}N_y. \tag{EL1}$$

These are pure-source rescalings of polynomial state normals. The original m event is $(1-p)V - N$, the beta event is $N + pV$, and the core event is $uW - Y + puvV - (1+a)vZ + uvN$; thus no term has been dropped. On $N = N_y = 0$ define

$$\begin{aligned} L_0 &= \mu(1-p) + (\beta + \tfrac{1}{2})p^2, \qquad L_1 = (\beta + \tfrac{1}{2})p^2, \\ B_c &= 1 - r + pc(1 + ra), \\ \lambda(a,c) &= \frac{q}{\kappa^2}(1+ra)^2(L_0 + L_1 c), \\ A_P &= (1+c)[c_0(1-r) + a_0 r(1+a)] + S[dB_c^2 + (\alpha + \tfrac{1}{2})(1+a)r^2 c], \\ K &= (1+a)[dB_c + (\alpha + \tfrac{1}{2})r], \\ F_Z &= (1+a)[\alpha + Dc + dac]. \end{aligned} \tag{EL2}$$

Here $c_0 = SC(1-r)$ and $a_0 = SAr$. The complete graph scalar is

$$F_0 = c(1+ra)^2(L_0 + L_1 c)W^2 + A_P P^2 - 2vKPQ + F_Z Z^2, \quad P = uW/\sqrt{S}, \quad Q = \sqrt{S}Z. \tag{EL3}$$

Let $T = \kappa W, U = uZ$ and use the exact four-term circuit

$$\Psi = (\nu/\ell)cT^2 + \nu^2 cU^2 + \kappa\nu\ell Q^2 + \kappa P^2 - 4\kappa\nu vPQ. \tag{EL4}$$

Its four positive monomial square roots are $A_1 = (\kappa/b)vW$, $A_2 = \nu v u Z$, $A_3 = \nu b\sqrt{x}Z$, and $A_4 = \sqrt{\kappa}P$. Their product is a square. The usual two paired AM–GM identities make Ψ a separated SOS, including $WZ < 0$ by sign excess. The multiplier λ itself is a sum of separated monomial squares with positive pure-source coefficients; no mixed radical of λ is taken.

Set $\Gamma = P - \nu v Q$, $\Lambda = vP - \ell Q$, and

$$\widehat{A} = A_P - \kappa\lambda, \quad \widehat{K} = K - 2\kappa\nu\lambda, \quad \widehat{F} = F_Z - \lambda(\nu^2 ac + \kappa\nu\ell S).$$

Then the following equality is purely polynomial and exact:

$$F_0 - \lambda\Psi = \widehat{A}\Gamma^2 - 2vJ\Gamma Q + HZ^2, \quad J = \widehat{K} - \nu\widehat{A}, \quad H = \widehat{F} + \nu^2 Sc\widehat{A} - 2\nu Sc\widehat{K}. \tag{EL5}$$

The exact coefficient bounds are $\lambda_0 = qL_0/\kappa^2 \sim 1/2$ and $\lambda_1 = qL_1/\kappa^2 = O(\ell)$. Indeed the exact opening-safe formulas are

$$\lambda_1 = \frac{\theta(2\beta\theta + \theta)}{2\ell\nu R_s^2}, \qquad \lambda_0 = \frac{U_s h M}{2\nu^2 P_0 R_s^2} + \lambda_1.$$

Here θ/ℓ and $(\beta\theta)/\ell$ are bounded, so the formula for λ_1/ℓ has only bounded source factors. No comparison between S and ℓ is made. The mixed term arising from $-\nu^2\lambda_1 c^2 a$ is allocated in EL18 below using an actual row of the same circuit.

Exact helper rows and the complete two-normal conductor

The following exact budgets and complete normal conductors use a positive real fraction of Ψ and strict pivots on (Γ, Λ). The scalar calculation below supplies precisely these pivots; it does not use the state normals to obtain them. Put $\delta = v - b$, $G_0 = P - jQ$, $B_o = \delta Q$, and

$$\begin{aligned} E_{12} = A_1 - A_2 = (v/b)(T - jU), \quad V_h = v\delta U, \\ R_v = v(a-x)Z, \quad B = \sqrt{S}(c-q)Z, \quad R_0 = \ell\sqrt{S}(a-x)Z. \end{aligned} \tag{EL6}$$

The exact angular relations are

$$(v+b)G_0 = \Lambda + b\Gamma, \qquad \nu(v+b)B_o = \Lambda - v\Gamma = \nu B. \tag{EL7}$$

In particular B has a bounded linear expression in the two strict angular rows. Multiply MB3 by v^2/j^2 and use $\eta = \kappa(v+b)/j \in \mathrm{Inf}$. The exact SOS inequality is

$$V_h^2 \preceq_{\mathrm{SOS}} \frac{\delta^2}{\nu^2}E_{12}^2 + \eta(\Lambda + b\Gamma)^2 + \frac{\eta + 2\kappa}{\nu^2}(\Lambda - v\Gamma)^2. \tag{EL8}$$

For a direct verification, set $R_1 = vG_0, R_2 = vB_o$. The mixed-term replacement uses the legal SOS

$$\eta(R_1^2 + R_2^2) - 2\kappa\delta R_1 R_2/j = \frac{\kappa v}{j}(R_1 - R_2)^2 + \frac{\kappa b}{j}(R_1 + R_2)^2.$$

Replacing v^2 by $(v+b)^2$ leaves $(2vb+b^2)G_0^2$ and $(2vb+b^2)B_o^2$, with positive separated multipliers. The remaining MB3 square is $(v\delta T/j)^2$. Every cost in EL8 is infinitesimal, since $\kappa/j = (\kappa/\ell)b$ is infinitesimal.

Two further exact relations are

$$R_v = \frac{\kappa v\sqrt{S}}{\nu b}\Gamma - \frac{u}{\nu}E_{12} + \frac{xv(v-b)}{b}Z, \tag{EL9}$$

$$\frac{x^2}{qS}B^2 - \left(\frac{xv(v-b)}{b}Z\right)^2 = \frac{x^2}{q}(2vb+b^2)(v-b)^2Z^2. \tag{EL10}$$

Thus R_v is free at infinitesimal Gram cost because $x^2/(qS) = \kappa S(\kappa/q)$ is infinitesimal. Finally

$$2\nu^2 ScR_v^2 + 2\nu^2(a-x)^2B^2 - R_0^2 = [\nu\sqrt{S}(2c-q)(a-x)Z]^2 \tag{EL11}$$

supplies R_0 with infinitesimal costs. None of EL6–11 bounds or divides a state coordinate or a mixed source/testing sum.

For the exact complete N half derivative set

$$\begin{aligned} B_0(c) &= [-\mu + (\beta + \tfrac{1}{2})p(1+c)]/\kappa, \\ F_a(a,c) &= r\kappa B_0(c) + d(1-r) + dpc(1+ra), \\ D_t(a,c) &= d(1+a) - \nu S F_a(a,c). \end{aligned} \tag{EL12}$$

All their coefficients are bounded, including at $\beta = 0$ or unbounded β: μ/κ and $(\beta+1/2)p/\kappa = (\beta\theta+\theta/2)/(\ell R_s)$ are bounded. Write the monic polynomial differences $D_t - D_t(x,q) = (a-x)D_a + (c-q)D_c$ and $B_0(c) - B_0(q) = (c-q)B_{01}$. Source contact gives $D_t(x,q) = \nu B_0(q)$. The entire half derivative is

$$\begin{aligned} L_N = {} & bB_0(c)E_{12} + uv\sqrt{S}F_a\Gamma - uvD_aR_v \\ & + [-D_t(x,q) + \nu bB_{01}(v+b) - v(v+b)D_c]V_h. \end{aligned} \tag{EL13}$$

Hence EL8–10 allocate every N cross term without a state bound. The normal $\widehat{N} = \sqrt{1+\beta}N$ has diagonal at least $(\beta+1/2)/(1+\beta) \geq 1/2$; dividing the cross coefficients by $\sqrt{1+\beta}$ only improves the displayed bounds.

The complete other half derivative is

$$L_y = \sqrt{r(1-r)}[Y_P\Gamma + vY_KQ], \tag{EL14}$$

where

$$\begin{aligned} Y_P = {} & (1+c)[-c_0 + a_0(1+a)] \\ & + S\{c(1+ra)(L_0 + L_1c) + d(-1+pac)B_c + (\alpha+\tfrac{1}{2})(1+a)rc\}, \\ Y_K = {} & \nu Y_P + (1+a)(D - \alpha - dpac). \end{aligned}$$

The polynomial coefficients are bounded, Y_P is infinitesimal, and actual source contact gives $Y_K(x,q) = 0$. Monic division gives $Y_K = (a-x)Y_a + (c-q)Y_c$ with bounded coefficients, so

$$L_y = \sqrt{r(1-r)}[Y_P\Gamma + \sqrt{S}Y_aR_v + vY_cB]. \tag{EL15}$$

The exact quadratic normal coefficients are

$$\begin{aligned} H_N &= m + dac + (\beta+\tfrac{1}{2})(1+c), \\ H_y &= rc_0(1+c) + (1-r)a_0(1+a)(1+c) \\ &\quad + Sr(1-r)[d(-1+pac)^2 + (\alpha+\tfrac{1}{2})(1+a)c + ac(L_0+L_1c)], \\ H_{Ny} &= \sqrt{Sr(1-r)}uv[\kappa B_0(c) - d + dpac]. \end{aligned} \tag{EL16}$$

After the source-only normalization of N, the first two diagonal residues are respectively in $[1/2, 1]$ and $1/2$, and the cross coefficient is infinitesimal. Equations EL13–16 include all terms of the full original parent; they require no W/Z or testing/source rate restriction. The scalar allocation and exact strict-Gram assembly are given next.

Complete scalar allocation, including the mixed radial term

Both $\widehat{A}$ and the coefficients of J, H in EL5 are bounded polynomials in a, c: their source coefficients are built from bounded $p, d, \alpha, c_0, a_0, \lambda_0, \lambda_1, \kappa, S, \nu$. In fact $p = O(\kappa)$, $\mu/\kappa = O(\kappa/\ell)$, and $(\beta + 1/2)p^2 = O(\kappa^2)$. Consequently $\widehat{A} \sim 1/2$ on the local chart. Actual source contact, checked in the original parent, gives

$$J(x, q) = 0, \qquad H(x, q) = H_a(x, q) = H_c(x, q) = 0. \tag{EL17}$$

These are exact identities, not residue equalities. They also follow by evaluating the actual contact state in EL1: all four circuit monomials agree, $\Gamma = 0$, and both graph normals vanish. The vanishing first derivatives of the original contact parent then give the three scalar jets.

The additional actual circuit row

$$E_{23} = A_2 - A_3 = \nu(uv - \sqrt{xq})Z$$

resolves the otherwise unbounded mixed coefficient. Direct expansion using $x = \kappa S$ gives the exact identity

$$c(a - x)(c - q)Z^2 = \frac{c - q}{\nu^2}E_{23}^2 + \frac{2\sqrt{\kappa q}}{\nu}E_{23}B - \kappa B^2. \tag{EL18}$$

Every coefficient on the right is infinitesimal. To see the equality without division by a testing coordinate, use

$$(ac - xq)Z^2 = E_{23}^2/\nu^2 + 2\sqrt{xq}\, ZE_{23}/\nu, \qquad c(a - x) = ac - xq - x(c - q).$$

All square roots in this calculation are pure-source roots.

For completeness, the single circuit supplies positive fixed pivots on both E_{12} and E_{23}. If $WZ \geq 0$, use the two AM–GM pairings having respectively these differences and average them; then $\Psi - \frac{1}{4}(E_{12}^2 + E_{23}^2)$ is separated SOS. If W, Z are both negative, first change the common state sign; the parent and all squared differences are unchanged. If $WZ < 0$, the negative product term of Ψ is nonnegative, and

$$\sum_{i=1}^{4} A_i^2 - \tfrac{1}{4}[(A_1 - A_2)^2 + (A_2 - A_3)^2]$$

is a real positive-definite quadratic form in the A_i. For example its leading 3×3 block has diagonal $(3/4, 1/2, 3/4)$ and successive leading minors $3/4, 5/16, 3/16$; the fourth diagonal is 1. This supplies a finite real Gram factorization. The remaining $-4\kappa\nu uvWZ$ is a separated square on this pure-testing sign chart. The zero-state boundaries use the same identities. Since $\lambda_0 \sim 1/2$ and $\lambda_1 \geq 0$, $\lambda - 1/4$ is a separated SOS multiplier:

$$\lambda - \tfrac{1}{4} = (\lambda_0 - \tfrac{1}{4}) + \lambda_0(2ra + r^2a^2) + \lambda_1 c(1 + ra)^2.$$

Thus $\lambda\Psi$ leaves strict real diagonal $1/16$ on each of E_{12}, E_{23} and an explicitly retained separated SOS remainder. No mixed square root of λ has been used.

Perform monic polynomial division in a and then in c. By EL17 there are bounded coefficient polynomials such that

$$\begin{aligned} J &= (a - x)J_a + (c - q)J_c, \\ H &= (a - x)^2 H_{20}(a, c) + (a - x)(c - q)H_{11}(c) + (c - q)^2 H_{02}(c). \end{aligned} \tag{EL19}$$

The division is by monic polynomials; it introduces no inverse of $a - x$, $c - q$, or any mixed expression. The following four particular coefficient identities are exact:

$$\begin{aligned} H_{20}(a, 0) &= -\kappa\nu\ell S\lambda_0 r^2, \\ H_{11}(0) &= -\alpha/q + 2\kappa\nu^2 S\lambda_0 r(1 + rx), \\ H_{02}(0)/S &= \nu^2\left[(1 + x)\frac{\alpha}{S\ell^2} - g\nu\lambda_0(1 + rx)^2\right] \sim \tfrac{1}{2}, \\ H_{02}(c) &= H_{02}(0) + \nu^2 S^2 dp^2(1 + rx)^2 c. \end{aligned} \tag{EL20}$$

The first three also follow immediately by evaluating $H(a, 0) = \alpha(1 + a) - \kappa\nu\ell S\lambda_0(1 + ra)^2$ and using EL17; the last follows from the c^3 coefficient. In particular $H_{11}(0)/(S\ell)$ is bounded and $H_{02}(c)/S \sim 1/2$. The g terms cancel in the real residue of the third line by EL0a. This cancellation is why the independent ratio S/ℓ never enters the estimate.

Write $H_{20} = H_{20}(a,0) + c\widetilde{H}_{20}$ and $H_{11} = H_{11}(0) + c\widetilde{H}_{11}$, with bounded coefficients. Substitution into EL5 gives the exact scalar assembly

$$\begin{aligned} F_0 = {} & \lambda\Psi + \widehat{A}\Gamma^2 + \frac{H_{02}(c)}{S}B^2 \\ & - 2\sqrt{S}J_a\Gamma R_v - 2vJ_c\Gamma B - g\nu\lambda_0 r^2 R_0^2 + \widetilde{H}_{20}R_v^2 \\ & + \frac{H_{11}(0)}{S\ell}R_0 B \\ & + \widetilde{H}_{11}\left[\frac{c-q}{\nu^2}E_{23}^2 + \frac{2\sqrt{\kappa q}}{\nu}E_{23}B - \kappa B^2\right]. \end{aligned} \tag{EL21}$$

This includes every scalar term, including the one involving $\lambda_1 c^2 a$. It supplies strict real pivots on $(E_{12}, E_{23}, \Gamma, B)$; the two radial rows have arbitrary fixed positive pivots available at infinitesimal cost by EL9–11. There is no discarded source-dependent signed term.

Strict bounded Gram and all-test completion

Use the main rows

$$E_{12},\ E_{23},\ \Gamma,\ B,\ \widehat{N} = \sqrt{1+\beta}N,\ N_y$$

and the three auxiliary rows V_h, R_v, R_0. EL21 and EL13–16 express the entire EL0b as a bounded quadratic form in these nine rows, together with the retained circuit SOS remainder. The real diagonal of the main block is bounded below by $(1/16, 1/16, 1/2, 1/2, 1/2, 1/2)$. Its off-diagonal entries are infinitesimal except for couplings to the auxiliary rows. These are bounded, as are the possibly signed auxiliary diagonal entries.

Choose a sufficiently large fixed real positive diagonal on each auxiliary row, and allocation it using EL8–11. This is an exact SOS operation: if $Q - R^2$ is a displayed SOS budget, replace the quadratic form by its difference with a real multiple of $Q - R^2$ and retain that multiple as a separate SOS. EL8 and EL9–10 cost only infinitesimal entries on the main rows; EL11 is allocationd last and costs an infinitesimal entry on R_v as well as on B. The real auxiliary diagonals can be chosen large enough that the real Schur complement is positive definite. This argument uses only a finite real matrix, never a mixed-field matrix inverse. The resulting bounded coefficient Gram has strictly positive real residue. The common strict separated bounded-Gram lemma in Lemma 3.2 makes the complete form a finite SOS in $K_c \otimes_{\mathbb{R}} K_t$. Restoring the budget and circuit remainders proves the original full parent.

There is no restriction on W, Z, their ratio, or either normal; β was never discarded or divided by. The actual retained normal diagonal handles an arbitrarily large β before any cross-term estimate. The only inverse involving an arbitrarily small opening is the pure-source scaling used to define N_y; its inverse never appears in a Gram coefficient. All helper roots are products of source and testing factors or finite sums thereof.

The local proof includes $u = 0$ and $v = 0$, since it never divides by either. Negative u is reduced by $(u, Y, Z) \mapsto (-u, -Y, -Z)$ and negative v by $(v, X, Z) \mapsto (-v, -X, -Z)$; EL0b is invariant under each map. Real and imaginary state parts are handled separately. By EL0a both C, A are unbounded, so the adjacent-C, A proof covers noninfinitesimal Alice tests. Since κ and e are infinitesimal, the SE10–SE12 theorem covers every homogeneous Bob sine-square residue different from $\bar{\ell} = 0$, including Bob infinity. The only remaining Bob chart is exactly finite infinitesimal v, proved above. Therefore all projective tests and all states on the original ℓ-infinitesimal source intersection are covered. All original source predicates are retained; no numerical sampling or completion marker is an input to this conclusion.

B.7.10 Exact quotients at the doubly infinitesimal source

Keep the original C14.4 source, its full native contact, and all original coefficient signs. The common doubly small endpoint has

$$S = \sigma + L \in \mathrm{Inf}_{>0}, \quad \kappa = x/S \in \mathrm{Inf}_{>0}, \quad \nu = \sigma/S \in \mathrm{Inf}_{>0}, \quad \ell = L/S = 1 - \nu, \quad q = \nu/\ell, \quad n_s = \beta/q^2. \tag{SN1}$$

Here n_s may be bounded or unbounded. The formulas below are exact identities; the complete BN and UN proofs supply their distinct normalized bounds on the bounded and unbounded τ domains. Set

$$c_0 = SC(1-r), \quad a_0 = SAr, \quad d_s = D/(Sq^2), \quad a_1 = \alpha/S, \quad \mu = m/\kappa^2, \quad e = \kappa h/\nu, \quad \tau = h/q. \tag{SN2}$$

The actual source-sphere identities are

$$d_s = \frac{\ell^2\mathcal{H}d_0^2}{2(P/S)J}, \qquad n_s = \frac{\ell^2\mathcal{H}b_0^2}{2\kappa(P/S)\theta}, \qquad Jd_s + \kappa\theta n_s = \frac{\ell^2\mathcal{H}}{2(P/S)}. \tag{SN3}$$

The source symbols in SN3 have the definitions in FM2–FM3 and SE1–SE3. In particular $e = \kappa\tau/\ell$ exactly.

The complete reciprocal parent. For infinitesimal Alice coordinate u and reciprocal Bob coordinate w, including either literal zero, use $D_u = 1 + u^2, D_w = 1 + w^2$ and

$$\begin{aligned} F_w/4 = {} & m(W - wX + uY - uwZ)^2 + CD_w(uW - Y)^2 + AD_uD_wY^2 \\ & + (D + \tfrac{1}{2})(uwW + uX - wY - Z)^2 + (\alpha + \tfrac{1}{2})D_u(Y - wZ)^2 \\ & + (\beta + \tfrac{1}{2})D_w(X + uZ)^2 - \tfrac{1}{2}D_uD_wZ^2. \end{aligned} \tag{SN5}$$

Keep both unrestricted state normals

$$X_e = X + ewW, \qquad N_y = (Y - ruW)/\sqrt{Sr(1-r)}. \tag{SN6}$$

Every inverse and radical in these expressions is pure-source or pure-testing. Write $a = u^2, c = w^2, x = \kappa S, \delta = c - q$ and set

$$\begin{gathered} N = \kappa W - \ell uwZ, \quad G = uW/\sqrt{S} - \ell\sqrt{S}\, wZ, \\ B = \sqrt{S}\,\delta Z, \quad B_u = u\delta Z, \quad R = w(a-x)Z = \frac{\kappa\sqrt{S}\,G - uN}{\ell}. \end{gathered} \tag{SN7}$$

The last equality is the exact homogeneous radial conductor.

Two free helper pivots. The exact polynomial identity

$$\delta^2N^2 + 2\kappa\ell\delta wGB + 2\kappa\ell^2w^2B^2 - \ell^2(wB_u)^2 = (\kappa\delta W)^2 \tag{SN8}$$

is an exact SOS identity. Whenever a current strict frame controls N, G, B, it supplies a prescribed real pivot on wB_u at infinitesimal Gram cost. BN and UN give the required ordered helper constructions even when the original d_sB^2 pivot is absent.

Exact scalar dictionary and the $1/\kappa$ cancellations. Put $d = D + 1/2$ and define the following polynomials in a, c:

$$\begin{aligned} A_P &= (1+c)\{c_0(1-r) + a_0r(1+a)\} + S\{d(1-r-e)^2c + (\alpha+\tfrac{1}{2})(1+a)r^2\}, \\ A_N &= m(1+ra+ec)^2 + (\beta+\tfrac{1}{2})e^2c(1+c), \\ K &= m(1+ra+ec) + d(1-r-e) + (\alpha+\tfrac{1}{2})(1+a)r + (\beta+\tfrac{1}{2})e(1+c), \\ F_Z &= D + \alpha c + \beta a + (m+\alpha+\beta+\tfrac{1}{2})ac, \\ J_1 &= K - \ell A_P, \quad L_1 = J_1 - (\ell/\kappa)A_N, \\ H &= F_Z - \ell^2ScA_P + ac\{(\ell/\kappa)^2A_N - 2(\ell/\kappa)J_1\}. \end{aligned} \tag{SN9}$$

These are the exact graph coefficients at $X = -ewW, Y = ruW$; the negative $dac, e\{2(1-r) - e\}$ contribution has been placed in A_P, not left in A_N. The displayed combination is used intact before taking any normalized bounds.

The actual source satisfies $L_1(x,q) = H(x,q) = H_a(x,q) = H_c(x,q) = 0$ and $H(a,0) = D + \beta a$. Finite monic division therefore gives

$$\begin{gathered} L_1 = (a-x)L_a + \delta L_c, \\ H = S\delta^2(d_s + \kappa n_s + cH_{02}) + (a-x)\delta(-qn_s + cH_{11}) + c(a-x)^2H_{20}. \end{gathered} \tag{SN10}$$

The quotients mean literal polynomial differences at $a = x, c = q$, not division by a mixed tensor. For example L_a is $(L_1(a,c) - L_1(x,c))/(a-x)$, and L_c is the quotient of $L_1(x,c) - L_1(x,q)$ by $c - q$.

Individually $H_{20}, H_{11}, L_c/\kappa$ need not be bounded. The complete scalar uses the following combined coefficients:

$$C_R = H_{20} + 2\ell L_a/\kappa, \quad C_G = \kappa(H_{11} - n_s)/\ell, \quad C_N = (n_s - H_{11})/\ell - 2L_c/\kappa. \tag{SN11}$$

Substitution of $e = \kappa\tau/\ell$ gives the exact cancellations

$$\begin{aligned} C_R &= -\ell\mu r(a\ell r + 2c\kappa\tau + 2\ell), \\ C_N &= n_s/\ell - Sa_0\ell r + (\beta + m + \tfrac{1}{2})\tau^2(c/\ell + 1/\ell^2) + 2\kappa\mu\tau - m\tau^2/\ell, \\ H_{02} &= \kappa\tau^2(\beta + m + \tfrac{1}{2}). \end{aligned} \tag{SN12}$$

For further pole cancellation, write $A_P = P_0(a) + cP_1(a)$. Direct differentiation in SN9 gives

$$\begin{aligned} \kappa H_{11} = {} & 2\ell^2P_1(x) + \ell^2xa_0r + 2\ell^2\kappa\mu e(1+2rx) + \frac{\ell^2(\beta+1/2)e^2}{\kappa} - 2\ell(m+\beta+\tfrac{1}{2})e \\ & + (c+q)\frac{\ell^2(m+\beta+1/2)e^2}{\kappa}. \end{aligned} \tag{SN13}$$

These are identities of the complete coefficient combinations. Their respective normalized estimates are proved in BN and UN.

The entire scalar graph is now

$$\begin{aligned} F_0 = {} & (A_N/\kappa^2)N^2 + A_P G^2 + d_s B^2 + n_s B_u^2 + H_{02}(wB)^2 + C_R R^2 - 2\sqrt{S}\, L_a G R \\ & + C_G G(wB) + C_N N(wB_u). \end{aligned} \tag{SN14}$$

For verification the mixed identity is $c(a-x)\delta Z^2 = \{\kappa G(wB) - N(wB_u)\}/\ell$. Combining this with $uN = \kappa\sqrt{S}\, G - \ell R$ proves SN14 directly from SN9–SN10.

Both complete state normals. Define

$$\begin{aligned} A_X &= -m(1+ra+ec) + da(1-r-e) - e(\beta+\tfrac12)(1+c), \\ T_X &= (\ell/\kappa)cA_X + \beta - D + c(m+\beta+\tfrac12), \\ Y_P &= (1+c)\{-c_0 + a_0(1+a)\} + S\{m(1+ra+ec) - dc(1-r-e) + (\alpha+\tfrac12)(1+a)r\}, \\ Y_C &= \ell Y_P - ma + d - (\alpha+\tfrac12)(1+a). \end{aligned} \tag{SN15}$$

Their actual contact values satisfy $T_X(x,q) = Y_C(x,q) = 0$. Use finite polynomial quotients

$$T_X = c(a-x)X_a + \delta\{(D-\beta)/q + cX_c\}, \qquad Y_C = (a-x)Y_a + \delta Y_c. \tag{SN16}$$

Although $X_a, A_X/\kappa$ separately may be unbounded, their complete conductor has

$$\begin{aligned} \widehat{A}_X &= A_X/\kappa - aX_a/\ell = -\kappa\mu - \frac{\tau}{\ell}\{(\beta+\tfrac12)(1+c) + mc\}, \\ \widehat{X}_a &= \kappa X_a = \ell\{d(1-r-e) - mr\}, \qquad X_c = -\tau(m+\beta+\tfrac12). \end{aligned} \tag{SN17}$$

The exact quarter-parent is $F_w/4 = F_0 + H_X X_e^2 + H_Y N_y^2 + 2H_{XY} X_e N_y + 2L_X X_e + 2L_Y N_y$, with

$$\begin{aligned} H_X &= mc + da + (\beta+\tfrac12)(1+c), \\ H_Y &= rc_0(1+c) + (1-r)a_0(1+a)(1+c) + Sr(1-r)\{ma + dc + (\alpha+\tfrac12)(1+a)\}, \\ H_{XY} &= -uw\sqrt{Sr(1-r)}(m+d), \\ L_X &= w\widehat{A}_X N + \frac{uw\sqrt{S}}{\ell}\widehat{X}_a G + d_s\sqrt{S}\, quB - n_s q B_u + wX_c(wB_u), \\ L_Y &= \sqrt{r(1-r)}\{Y_P G + \sqrt{S}\, Y_a R + Y_c(wB)\}. \end{aligned} \tag{SN18}$$

Every formula retains the original r and $1-r$ openings and both complete state normals. In particular, the term $n_s q B_u$ may be written $q\sqrt{n_s}(\sqrt{n_s}B_u)$ when that row is used; literal $n_s = 0$ needs no division. BN and UN allocate these actual rows, with their own weights, instead of treating separate uncanceled coefficients as bounded.

Explicit finite divisions. All quotient symbols above can be defined directly in the polynomial ring in a, c over the source field. In addition to L_a, L_c specified after SN10, use

$$\begin{aligned} H_{20} &= \frac{H(a,c) - H(x,c) - (a-x)H_a(x,c)}{c(a-x)^2}, \\ H_{11} &= \frac{H_a(x,c)/(c-q) + qn_s}{c}, \\ H_{02} &= \frac{H(x,c)/(S(c-q)^2) - d_s - \kappa n_s}{c}, \\ X_a &= \frac{T_X(a,c) - T_X(x,c)}{c(a-x)}, \qquad X_c = \frac{T_X(x,c)/(c-q) - (D-\beta)/q}{c}, \\ Y_a &= \frac{Y_C(a,c) - Y_C(x,c)}{a-x}, \qquad Y_c = \frac{Y_C(x,c) - Y_C(x,q)}{c-q}. \end{aligned}$$

These displays specify polynomial division, not evaluation of an inverse of $a-x$, $c-q$ or c. Divisibility follows from $H(x,q) = H_a(x,q) = H_c(x,q) = 0$, $H(a,0) = D + \beta a$, $T_X(x,q) = Y_C(x,q) = 0$, and the explicitly affine dependence of T_X on a. Thus all exact coefficients are defined without an external program. Pure-source divisions such as $1/S$ or $1/q$ are retained until their cancellations in the printed combinations.

B.7.11 The doubly infinitesimal source with bounded τ

Source scope and result. Retain the original entered physical C14.4 source, all strict gaps and all native source inequalities. Assume

$$S = \sigma + L,\ \kappa = x/S,\ \nu = \sigma/S \in \mathrm{Inf}_{>0}, \qquad \ell = 1 - \nu, \quad q = \nu/\ell, \quad \tau = h/q \text{ is bounded.} \tag{BN1}$$

The complete original parent is a finite separated tensor SOS for every test and every real or complex state on BN1. The quantity β/q^2 is unrestricted, including zero and unbounded values. The proof below is a circuit completion of SN, not an import of a missing endpoint proof.

Exact source estimates. Use the definitions of SN2, except that $n_s = \beta/q^2$ need not be bounded. The exact sphere SN3 remains valid. Put $b = \beta/q$ and $z = \kappa\beta/q^2$. Then

$$\begin{gathered} h = q\tau \in \mathrm{Inf}, \quad \theta \sim 1, \quad e = \kappa\tau/\ell \in \mathrm{Inf}, \quad M \sim 1, \\ b = \frac{\tau(B/S) - \ell H_0}{2\theta(P/S)}, \qquad \mu\ell^2 = \frac{\tau(1+x)(1+\sigma r)}{2M(P/S)}, \\ Jd_s + \theta z = \frac{\ell^2\mathcal{H}}{2(P/S)} \sim \tfrac{1}{2}. \end{gathered} \tag{BN2}$$

In particular b, z, d_s, μ, τ are bounded, τ and μ have positive real lower bounds, and at least one of d_s, z is a positive bounded unit. The exact inequality $\beta \geq 0$ supplies the lower bound on τ. Also $c_0, a_0, a_1 \sim 1/2$, D, α, m, β are infinitesimal. Although the two terms below need not have small difference before subtraction, their displayed bounded formulas in BN2 give

$$A_{00} := \mu\ell^2 - b \sim \tfrac{1}{2}. \tag{BN3}$$

This argument takes residues only of bounded source coefficients and never uses an estimate divided by q. It allows arbitrary relative rates of S, κ, q.

A retained circuit and two actual positive rows. Use the full reciprocal parent SN5, the two unrestricted normals SN6, and the shifted exact scalar dictionary SN9. Let $\zeta = \ell/\kappa$, $a = u^2, c = w^2$, $x = \kappa S$, and assume $u, w \geq 0$ by the original sign gauges. Both may be zero. For every state define

$$\begin{aligned} A_1 &= \sqrt{\beta}\, uZ, & A_2 &= \sqrt{\beta}\, W/(\zeta\sqrt{q}), \\ A_3 &= \sqrt{\beta}\sqrt{x/q}\, wZ, & A_4 &= \sqrt{\beta}\, uwW/(\zeta q\sqrt{x}), \end{aligned} \tag{BN4}$$

$$\Psi = \sum_{i=1}^{4} A_i^2 - \frac{4\beta}{\zeta q} uwWZ, \qquad E_1 = A_1 - A_2, \quad E_2 = A_3 - A_4. \tag{BN5}$$

Every monomial and its radical is a product of a pure-source and a pure-testing element. This remains true if x/q is unbounded. The exact inequality meant here is a separated SOS certificate:

$$\Psi - \tfrac{1}{2}(E_1^2 + E_2^2) \in \Sigma T^2. \tag{BN6}$$

Indeed, if $WZ \geq 0$, then $A_1A_2, A_3A_4 \geq 0$ and

$$\Psi = E_1^2 + E_2^2 + 2(\sqrt{A_1A_2} - \sqrt{A_3A_4})^2.$$

The radicals displayed here split into pure-source and pure-testing radicals. If $WZ < 0$, use instead

$$\Psi - \tfrac{1}{2}(E_1^2 + E_2^2) = \tfrac{1}{2}(A_1 + A_2)^2 + \tfrac{1}{2}(A_3 + A_4)^2 + \frac{4\beta}{\zeta q}|uwWZ|.$$

The last positive separated monomial is a square. Thus BN6 is valid for all states, including $WZ = 0$. At the actual source contact every A_i agrees, so Ψ and its first contact derivatives vanish.

The exact scalar after circuit subtraction. Write $N_0 = W/\zeta - uwZ = N/\ell$, where N is SN7, and keep G, B, B_u, R from SN7. Thus

$$R = \frac{\sqrt{S}}{\zeta} G - uN_0. \tag{BN7}$$

Subtract the entire circuit BN5 from the exact scalar graph in SN9. Define

$$\begin{gathered} A = \zeta^2 A_N - b, \qquad A'_P = A_P - zc/\ell^2, \\ K' = K - 2\beta/(\zeta q), \quad J' = K' - \ell A'_P, \quad L' = J' - A/\zeta, \\ H' = F_Z - \beta a - \beta xc/q - \ell^2 ScA'_P + ac(A - 2\zeta J'). \end{gathered} \tag{BN8}$$

Then $F_0 - \Psi = AN_0^2 + A'_P G^2 - 2\zeta uwZL'N_0 + H'Z^2$. All coefficients of A and A'_P are bounded:

$$A = \mu\ell^2(1 + ra + ec)^2 + (\beta + \tfrac{1}{2})\tau^2 c(1 + c) - b, \qquad \overline{A} = \overline{A'_P} = \tfrac{1}{2}. \tag{BN9}$$

Here and below the residue is taken at infinitesimal u, w.

The circuit subtraction has especially simple exact effects on the SN10 quotient dictionary:

$$\begin{aligned} &L'_a = L_a, \qquad L'_c = L_c + \kappa n_s/\ell, \qquad H'_{20} = H_{20}, \quad H'_{02} = H_{02}, \quad H'_{11} = H_{11} - 2n_s, \\ &H' = S\delta^2(d_s + cH_{02}) + c(a - x)\delta H'_{11} + c(a - x)^2 H_{20}. \end{aligned} \tag{BN10}$$

In particular no coefficient n_s remains on its own in the final conductors. With C_R and L_a from SN11–SN13 put

$$C'_G = \kappa(H_{11} - 2n_s)/\ell, \qquad C'_N = -H_{11} - 2\ell L_c/\kappa. \tag{BN11}$$

These are bounded. The cancellation for the second coefficient is explicit:

$$C'_N = -Sa_0\ell^2 r + (\beta + m + \tfrac{1}{2})\tau^2(c + 1/\ell) + 2\ell\kappa\mu\tau - m\tau^2. \tag{BN12}$$

For C'_G use the bounded formula SN13 and $\kappa n_s = z$. The exact full scalar is consequently

$$\begin{aligned} F_0 = \Psi + AN_0^2 + A'_P G^2 + d_s B^2 + H_{02}(wB)^2 + C_R R^2 - 2\sqrt{S}\, L_a GR \\ + C'_G G(wB) + C'_N N_0(wB_u). \end{aligned} \tag{BN13}$$

This is an identity in unrestricted state coordinates, not a restriction to a contact graph in the full parent.

Free helper rows across the whole source sphere. First reserve the positive rows E_1, E_2 from BN6. If d_s is a unit, wB has infinitesimal cost on $\sqrt{d_s}B$. Otherwise z is a positive bounded unit. The exact identity

$$E_2 = -\sqrt{z}\,\{(w/\ell)G + \sqrt{S}\, w(w - \sqrt{q})Z\}$$

then gives

$$wB = -\frac{w + \sqrt{q}}{\sqrt{z}} E_2 - \frac{w(w + \sqrt{q})}{\ell} G. \tag{BN14}$$

Every coefficient in BN14 is infinitesimal, so any fixed real pivot on wB is free at infinitesimal Gram cost. Only the chosen pure-source unit $\sqrt{z}$ is inverted. Literal $\beta = 0$ uses the first branch.

After this pivot has been allocationd, the normalized SN8 identity is

$$\delta^2 N_0^2 + \frac{2\kappa\delta}{\ell} G(wB) + 2\kappa(wB)^2 - (wB_u)^2 = (\delta W/\zeta)^2. \tag{BN15}$$

It supplies any fixed pivot on wB_u at infinitesimal cost. The order of these two allocations is deliberate; no weak B diagonal is needed in the second step.

Both full normals and their only additional row. Use the complete $H_X, H_Y, H_{XY}, Y_P, Y_a, Y_c$ from SN18 and its bounded $\widehat{A}_X, \widehat{X}_a, X_c$ from SN17. These formulas only require τ, μ, β bounded, not n_s bounded. The entire first-normal part is

$$\begin{aligned} L_X &= \ell w \widehat{A}_X N_0 + \frac{uw\sqrt{S}}{\ell} \widehat{X}_a G + d_s\sqrt{S}\, quB - bB_u + wX_c(wB_u), \\ L_Y &= \sqrt{r(1 - r)}\{Y_P G + \sqrt{S}\, Y_a R + Y_c(wB)\}. \end{aligned} \tag{BN16}$$

The apparent extra B_u coupling is paid by the actual circuit row E_1, using the exact, zero-face-safe identity

$$bB_u = -(w + \sqrt{q})\sqrt{b}\, E_1 - b(w + \sqrt{q})N_0. \tag{BN17}$$

All coefficients on the right have zero residue because b is bounded. No inverse of b or β is used. In the B term in L_X, write $d_s B = \sqrt{d_s}(\sqrt{d_s}B)$; this is likewise face-safe when $D = 0$. All direct normal coefficients are infinitesimal, and $H_X, H_Y \sim 1/2$, H_{XY} is infinitesimal. There is no missing X or Y normal and no imposed state ratio.

Allocation and all-tests conclusion. Keep a fixed positive fraction of each circuit row in BN6. Use the strict base rows $(N_0, G, X_e, N_y, \sqrt{d_s}B, E_1, E_2)$, omitting any zero row. Their main four diagonal residues are $1/2$; the last three have fixed positive pivots wherever their rows are nonzero. Terms containing R are infinitesimal matrix perturbations by BN7. Purchase sufficiently large fixed pivots on wB, wB_u by BN14–BN15, and use them for the bounded cross terms in BN13 and BN16. Formula BN17 pays the remaining first-normal row at infinitesimal cost. Every allocation is taken from an explicitly retained positive square or from the exact SOS remainder BN6. The resulting bounded Gram has strictly positive definite real residue. The strict bounded-Gram lemma therefore gives finite tensor SOS for SN5 with both unrestricted normals restored.

This proves all infinitesimal u, w, including the literal axes. For noninfinitesimal Alice use the unchanged adjacent-C, A theorem CE1–CE10; C, A are unbounded on BN1. For the other Bob charts use the small-e unmatched theorem SE10–SE12, since $\overline{\ell} = 1$. The sign gauges and real/imaginary state split are the original exact ones. The local argument already handles either sign of WZ, so no state normalization or state-chart exclusion is required. Both literal $D = 0$ and $\beta = 0$ are covered whenever they intersect BN1.

The other source cells. The unbounded-h/q source and the ℓ-infinitesimal endpoint are proved in Sections B.7.12 and B.7.9. Their source partition with the present bounded-h/q theorem is printed in Section B.1.2.

B.7.12 The doubly infinitesimal source with unbounded τ

Exact scope. Retain the full original entered C14.4 source, all its strict gaps and native source inequalities. Suppose S, κ, ν are positive infinitesimals, $\ell = 1 - \nu$, $q = \nu/\ell$, and

$$\tau = h/q \text{ is unbounded}, \qquad \theta = 1 - h > 0, \quad e = \kappa h/\nu \in \mathrm{Inf}, \quad \zeta = \ell/\kappa. \tag{UN1}$$

This section proves finite separated tensor SOS for the complete parent at every projective test and every state on UN1. The opening $\theta \in \mathrm{Inf}$ is included. There is no state-ratio restriction or source/testing relative-rate assumption.

Normalized source estimates and a canceled radial mass. Keep $c_0 = SC(1-r)$, $a_0 = SAr$, $d_s = D/(Sq^2)$, $a_1 = \alpha/S$, $P_0 = P/S$, and the exact SE2–SE3 source sphere. Define

$$d = D + \tfrac{1}{2}, \quad n = \beta\theta^2 + 2Dh + \tfrac{1}{2}h^2 > 0, \quad z_n = \kappa n/q^2, \quad A_{00} = m\zeta^2 - n/q. \tag{UN2}$$

Here n is a scalar coefficient, not an omitted event weight. It is bounded even when β is unbounded. The exact source gives

$$\begin{gathered} m\zeta^2 = \frac{\tau(1+x)(1+\sigma r)}{2MP_0}, \qquad m\zeta \in \mathrm{Inf}, \quad D, \alpha, m \in \mathrm{Inf}, \quad c_0, a_0, a_1 \sim \tfrac{1}{2}, \\ e\tau = \frac{\ell\{eH_0 + h\mathcal{H}b_0^2\}}{B/S} \text{ is bounded}, \\ z_n = \frac{\ell^2\mathcal{H}\theta b_0^2}{2P_0} + \frac{\ell^2(eH_0 + h\mathcal{H}b_0^2)}{2(B/S)} + 2\kappa S d_s h. \end{gathered} \tag{UN3}$$

Consequently d_s, z_n are bounded and $d_s + z_n \sim 1/2$; at least one is a positive bounded unit. This includes literal $D = 0$. Literal $\beta = 0$ has no points on UN1, because $\beta \geq 0$ has the exact boundary equation $\tau(B/S) = \ell H_0$ when $\beta = 0$.

For completeness, the following finite source division proves the bound on A_{00} without dividing an error by θ or q. Put $U = (1+x)(1+\sigma r)$ and $M_* = \ell(1+xr) - \kappa\theta = \ell M$, and form the bounded source polynomial

$$T = \ell JU + JM_*P_0 - JM_*\kappa\theta(J - \sigma) - 2M_*U.$$

In this expression P_0, J, U, M_* are their exact native polynomial formulas in κ, ν, S, h, r, with $\ell = 1 - \nu$. Thus

$$\begin{gathered} T = \kappa T_\kappa + \ell\nu S T_\nu, \qquad T_\nu = r - 1 - \nu - \nu S(1-r)(r+\ell), \\ T_\kappa = \{T - T|_{\kappa=0}\}/\kappa \quad \text{is a bounded source polynomial}, \\ A_{00} = \frac{\theta\ell H_0}{2P_0} + \frac{eT_\kappa + h\ell S T_\nu}{2JMP_0}. \end{gathered} \tag{UN4}$$

The quotient is a finite monic polynomial division in a pure source variable. All denominator factors are positive bounded units. Hence A_{00} is bounded and $A_{00} - \theta/2$ is infinitesimal. Moreover $n/q = m\zeta^2 - A_{00} \sim \tau/2$ is positive unbounded. If θ is infinitesimal, then $n \sim 1/2$ and $\kappa/q = e\ell/h$ is infinitesimal. These facts supply the two source subcases used below.

The actual circuit and three reserved positive rows. Use the complete original reciprocal quarter-parent SN5. For infinitesimal $u, w \geq 0$, including zero, retain the unrestricted normals

$$N_x = X + huZ, \qquad N_y = (Y - ruW)/\sqrt{Sr(1-r)}. \tag{UN5}$$

Put $a = u^2, c = w^2, x = \kappa S, \delta = c - q$ and

$$\begin{gathered} N_0 = W/\zeta - uwZ, \quad G = uW/\sqrt{S} - \ell\sqrt{S}\,wZ, \\ B = \sqrt{S}\delta Z, \quad B_u = u\delta Z, \quad R = w(a-x)Z = \sqrt{S}\,G/\zeta - uN_0, \quad G_h = h(a - xc/q)Z. \end{gathered} \tag{UN6}$$

Define A_1, A_2, A_3, A_4 by BN4 with β replaced by the positive n, and set

$$\Psi_n = \sum_i A_i^2 - 4nuwWZ/(\zeta q), \qquad E_1 = A_1 - A_2, \quad E_2 = A_3 - A_4, \quad E_3 = A_1 - A_3. \tag{UN7}$$

The explicit separated AM–GM square decompositions in BN6 give

$$\Psi_n - \tfrac{1}{4}(E_1^2 + E_2^2 + E_3^2) \in \Sigma T^2. \tag{UN8}$$

To verify the simultaneous reserve, average BN6 with the pairing $\Psi_n - (A_1 - A_3)^2 - (A_2 - A_4)^2 \in \Sigma T^2$. For the latter pairing the products A_1A_3, A_2A_4 are nonnegative; if $WZ < 0$, the additional signed cross term is a positive separated monomial. Thus UN8 holds for both signs of WZ, as well as its zero face. No radical of a general mixed sum is used.

Exact scalar subtraction and radial cancellation. The complete scalar graph at $N_x = N_y = 0$ is

$$F_0 = m(1+ra)^2W^2 + A_P P^2 + F_Z Z^2 - 2KuwWZ, \quad P = uW/\sqrt{S}, \tag{UN9}$$

where

$$\begin{aligned} A_P &= (1+c)\{c_0(1-r) + a_0 r(1+a)\} + S\{d(1-r)^2 c + (\alpha + \tfrac{1}{2})(1+a)r^2\}, \\ K &= m\theta(1+ra) + d(1-r)(1+ha) + (\alpha + \tfrac{1}{2})(1+a)r, \\ F_Z &= D + \alpha c + na + \{m\theta^2 + \alpha + (\beta + \tfrac{1}{2})\theta^2\}ac + dh^2a^2. \end{aligned} \tag{UN10}$$

Subtract Ψ_n, put

$$\begin{gathered} A = m\zeta^2(1+ra)^2 - n/q, \quad A'_P = A_P - z_n c/\ell^2, \\ K' = K - 2n/(\zeta q), \quad J' = K' - \ell A'_P, \quad L' = J' - A/\zeta, \\ H' = F_Z - na - nxc/q - \ell^2 S c A'_P + ac(A - 2\zeta J'). \end{gathered} \tag{UN11}$$

The exact residual graph is $AN_0^2 + A'_P G^2 - 2\zeta uwZL'N_0 + H'Z^2$. At the actual source contact $L'(x,q) = H'(x,q) = H'_a(x,q) = H'_c(x,q) = 0$. Also $H'(a,0) = D + dh^2a^2$. Finite monic contact division gives

$$\begin{gathered} L' = (a-x)L_a + \delta L_c, \\ H' = (D + dh^2x^2)\delta^2/q^2 + (a-x)\delta(-2dh^2x/q + cH_{11}) + (a-x)^2(dh^2 + cH_{20}). \end{gathered} \tag{UN12}$$

The otherwise possible third-c quotient is exactly zero. Define $P_1(a)$ to be the coefficient of c in A_P. The complete combined coefficients have short bounded formulas:

$$\begin{aligned} L_a &= m\theta r + dh(1-r) + (\alpha + \tfrac{1}{2})r - \ell\{(1+c)a_0 r + S(\alpha + \tfrac{1}{2})r^2\} \\ &\quad - m\zeta r\{2 + r(a+x)\}, \\ C_R &= H_{20} + 2\zeta L_a = -m\zeta^2 r(2+ra), \\ C_G &= H_{11}/\zeta = 2\ell P_1(x) + \ell x a_0 r - 2z_n/\ell, \\ C_N &= -H_{11} - 2\zeta L_c = -Sa_0\ell^2 r. \end{aligned} \tag{UN13}$$

Only C_R may be unbounded separately. Its entire contribution must be combined before estimating: $A + aC_R = A_{00}$ exactly. Using $R = \sqrt{S}\,G/\zeta - uN_0$ therefore gives

$$\begin{aligned} F_0 &= \Psi_n + A_{00}N_0^2 + \widetilde{A}_P G^2 + 2u\sqrt{S}\,(L_a - C_R/\zeta)N_0 G \\ &\quad + d_s B^2 + dG_h^2 + C_G G(wB) + C_N N_0(wB_u), \\ \widetilde{A}_P &= A'_P + SC_R/\zeta^2 - 2SL_a/\zeta \sim \tfrac{1}{2}. \end{aligned} \tag{UN14}$$

Every coefficient in UN14 is bounded after this combination: $C_R/\zeta = -m\zeta r(2+ra)$ is infinitesimal and $C_R/\zeta^2 = -mr(2+ra)$ is bounded. The N_0G coefficient is infinitesimal. No estimate of an unbounded isolated radial coefficient is used.

Complete X conductor and its exact positive completion. The full N_x hard coefficient is $H_X = mc + da + (\beta + 1/2)(1 + c)$. Its complete half-linear coefficient, after using the actual contact identity, is

$$\begin{aligned} L_X &= -m\zeta w N_0 + d\tau a(wN_0 + B_u) + uw\sqrt{S}\{d(1 - r - e) - mr\}G + t_0 B_u, \\ t_0 &= \frac{D + h/2 - \beta\theta}{q} = D/q + \frac{\ell H_0 - \kappa\tau(J - \sigma)}{2P_0} \sim \tfrac{1}{2}. \end{aligned} \tag{UN15}$$

The exact source identity in the second line cancels the potential $1/q$ pole. Here $D/q = Sq\, d_s$ is infinitesimal. The single unbounded τ term is removed with the actual square dG_h^2, not estimated. Namely

$$\begin{aligned} G_h &= -\tau u(wN_0 + B_u) + ew\sqrt{S}\, G, \quad N_x + \tau(wN_0 + B_u) = X + ewW, \\ &dG_h^2 + daN_x^2 + 2d\tau a(wN_0 + B_u)N_x \\ &\quad = da(X + ewW)^2 - 2de\tau uw\sqrt{S}\,(wN_0 + B_u)G + de^2 Sw^2 G^2. \end{aligned} \tag{UN16}$$

The remainder has bounded infinitesimal coefficients on N_0, G, wB_u because $e\tau$ is bounded. Retain the entire positive square on the right and also the whole term $\beta(1+c)N_x^2$. The remaining N_x diagonal is $mc+(1+c)/2 \sim 1/2$. The remaining L_X is UN15 with its $d\tau a$ term removed; all its direct coefficients are infinitesimal except the bounded coefficient t_0 on B_u. The construction thus includes arbitrarily large β.

The unrestricted Y normal and cross coefficient. Put

$$\begin{aligned} Y_P &= (1 + c)\{-c_0 + a_0(1 + a)\} + S\{m(1 + ra) - d(1 - r)c + (\alpha + \tfrac{1}{2})(1 + a)r\}, \\ Y_C &= \ell Y_P - m\theta a + d(1 + ha) - (\alpha + \tfrac{1}{2})(1 + a). \end{aligned} \tag{UN17}$$

The actual source has $Y_C(x, q) = 0$, so finite monic division gives $Y_C = (a - x)Y_a + \delta Y_c$ with bounded polynomial coefficients. The complete remaining entries are

$$\begin{aligned} L_Y &= \sqrt{r(1 - r)}\{Y_P G + \sqrt{S}\, Y_a R + Y_c(wB)\}, \\ H_Y &= rc_0(1 + c) + (1 - r)a_0(1 + a)(1 + c) \\ &\quad + Sr(1 - r)\{ma + dc + (\alpha + \tfrac{1}{2})(1 + a)\} \sim \tfrac{1}{2}, \\ H_{XY} &= -uw\sqrt{Sr(1 - r)}(m + d) \in \mathrm{Inf}\,. \end{aligned} \tag{UN18}$$

In particular Y_P is infinitesimal. Every coefficient is bounded through $r = 0$ and $r = 1$ openings; no inverse of r or $1 - r$ remains after the displayed pure-source normal scaling. UN14–UN18 account for all terms of SN5, including both independent state normals.

Free pivots, including the θ-infinitesimal opening. Reserve the three actual circuit rows in UN8. In either sphere chart, wB is free at infinitesimal cost: use $\sqrt{d_s}B$ when d_s is a unit, and otherwise use the exact identity

$$wB = -\frac{w + \sqrt{q}}{\sqrt{z_n}}E_2 - \frac{w(w + \sqrt{q})}{\ell}G. \tag{UN19}$$

If θ is a unit, A_{00} already supplies a strict N_0 pivot. Since q/n is infinitesimal, the exact identity

$$B_u = -(w + \sqrt{q})\sqrt{q/n}\, E_1 - (w + \sqrt{q})N_0 \tag{UN20}$$

supplies any fixed B_u pivot at infinitesimal cost. Then wB_u is free by multiplication by the pure-testing infinitesimal w.

If θ is infinitesimal, first supply the missing N_0 pivot from the actual circuit, using

$$N_0 = \sqrt{q/n}(E_2 - E_1) - \frac{w - \sqrt{q}}{\sqrt{n}}E_3 + \frac{\sqrt{\kappa/q}}{\ell}\, wG. \tag{UN21}$$

All coefficients are infinitesimal: $n \sim 1/2$ and κ/q is infinitesimal by UN4. Identity UN21 holds without requiring z_n to be a unit; the apparent source factor in UN19 has canceled exactly before estimating. After this allocation, UN20 supplies B_u, and multiplication by w supplies wB_u. Thus the order of allocations involves no circular use of a missing pivot. The bounded, possibly signed A_{00} in this source subcase has zero residue and is absorbed after the strict N_0 pivot is obtained. No inverse of θ is introduced.

Final allocation and unchanged testing joins. Use the strict rows G, N_x, N_y, the applicable N_0 pivot, and fixed retained fractions of $E_1, E_2, E_3, \sqrt{d_s}B$ wherever nonzero. Purchase sufficiently large real pivots on B_u, wB, wB_u by UN19–UN21. Every allocation has infinitesimal Gram cost on the preceding strict rows. All signed terms in UN14–UN18 are then either infinitesimal matrix perturbations or bounded couplings to those freely allocationd helper pivots. A fixed sufficiently large choice makes the complete residue matrix strictly positive definite. The separated strict bounded-Gram lemma proves finite SOS. Restore the positive terms retained in UN8 and UN16, including $\beta(1+c)N_x^2$; nothing from the original parent is omitted.

This proves all infinitesimal u, w, their literal zero faces and arbitrary states. Noninfinitesimal Alice tests use CE1–CE10 because C, A are unbounded. The other Bob tests use the SE10–SE12 unmatched certificate, since e is infinitesimal and $\overline{\ell} = 1$. The original sign gauges and real/imaginary state split are exact; the circuit argument treats both signs of WZ. These joins give the claimed all-tests theorem on UN1.

The theorem just proved has precisely the source domain UN1. Its join with the bounded-h/q theorem and the other C14.4 source cells is printed in Section B.1.2.

B.8 Completion of the C1 certificate

The source partition in Section B.1.2 is exhausted. For unbounded η and bounded $z = 1/(x\zeta)$, Theorem B.23 gives the local certificate for every $H = h/\sigma$; the unit-Alice and complementary projective charts cover the other tests. When L is infinitesimal, the positive-j completions, the completed J relation and the five normalized source domains give the remaining certificates. Each represents the full state quadratic form, with its positive event terms restored.

The real statement applies separately to the real and imaginary state parts because the parent is a real quadratic form. Source and testing homogenization factors are pure-field squares. Restoring the original positive d therefore proves the complete projective C1 assertion.

Chapter C

The C2 cases

All source coefficients lie in the real closed field K_c, and all testing measurements and states lie in the independently labelled real closed field K_t. We use $K_c \otimes_{\mathbb{R}} K_t$ without an algebraic-disjointness assumption. A retained positive square may have unbounded pure-source coefficients. Only the matrix coefficients to which the bounded strict-Gram lemma is applied must be bounded. The native physical entry, bounded-coefficient theorem, and arbitrary-scale noncommuting-residue theorem are the common preceding results. Normalize $d = 1$ by a pure-source square and write $P = F/4$. The common four-state parent is

$$\begin{aligned} F = 4P &= 4nD_uD_vX^2 + 4mM^2 + 4CD_v(uW - Y)^2 + 4AD_uD_vY^2 \\ &\quad + 4DJ^2 + 4\alpha D_u(vY - Z)^2 \\ &\quad + 2\{J^2 + D_u(vY - Z)^2 + D_v(X + uZ)^2 - D_uD_vZ^2\}, \\ M &= vW - X + uvY - uZ, \quad J = uW + uvX - Y - vZ, \quad D_u = 1 + u^2, \quad D_v = 1 + v^2. \end{aligned} \tag{C.1}$$

The finite coefficient-extraction and monic-division operations are those in Section B.1.1; they are field-independent. For every displayed state substitution $s = s_0 + A\xi$, every scalar, complete first normal and hard coefficient below means respectively $[1]\mathcal{Q}$, $[\xi_i]\mathcal{Q}$, and $\frac{1}{2}\partial_i\partial_j\mathcal{Q}$, where $\mathcal{Q} = (F - \Pi)(s_0 + A\xi)$ and Π is the displayed sum of retained positive squares. Quadratic state degree makes this the exact whole polynomial. Coefficients need not be expanded to define them. All support, divisibility, source-bound and residue verifications remain part of the relevant calculation below.

The proof follows the source geometry. We first determine the stationary coefficients and treat two weak complementary events by a weighted radial completion. The remaining families are organized by the scales of the actual angles. For one small angle the state parameter separates bounded and unbounded source scales; for two small angles the source ratios define R1–R10. Within each family the scalar calculation is completed by the state-normal and complementary-testing arguments.

C.1 Stationary sources and their classification

C.1.1 Source domains

The native entry theorem supplies an actual physical stationary source with $d = 1$, $\lambda_{\text{event}} = \beta = 0$, $n \geq 0$, and positive m, C, A, D, α. In the native coordinates $u_c = c$, $v_c = ac$ its strict gaps are

$$c, a, b, t > 0, \quad 0 < r < 1, \quad h = b - a^2t > 0, \quad g = (1 + c^2)b - (1 - r + a^2c^2t) > 0,$$

$$j = 1 + c^2r - t > 0, \qquad 1 - c^2h > 0, \qquad b - a^2c^2r > 0.$$

These are the original source equations. If $(1 + n + m + C + A + D + \alpha)^{-1}$ is a unit, the bounded-coefficient theorem applies. Henceforth that reciprocal is infinitesimal. Call an event weak when its coefficient is infinitesimal and strong otherwise; strong coefficients may be unbounded.

The exact reflection

$$(u, v; W, X, Y, Z; n, m, C, A, D, \alpha) \longmapsto (1/v, 1/u; Z, X, Y, W; n, D, \alpha, A, m, C) \tag{C.2}$$

satisfies $P = u^2v^2P'$ and is a homogeneous polynomial identity at the literal endpoints. With $z = c^2$ its source map is

$$c' = (ac)^{-1}, \ a' = a, \ b' = a^2/b, \ r' = a^2zr/b, \quad t' = (1 - zh)/b,$$

$$h' = a^2zh/b, \quad g' = a^2j/b, \quad j' = g/b, \quad \Delta' = \Delta/(b^2z^2), \quad N' = a^2N/b^2.$$

The remaining three strict gaps are positive by the original gaps. This map preserves the total coefficient scale. In particular it orients an exactly-one-weak source to C weak and α strong.

Write I, U, O for positive infinitesimal, bounded-unit, and unbounded finite actual angles. The complete cases

are

(u_c, v_c)	C, α strong	C weak, α strong
II	II calculation	$\varnothing$
IU	one small angle	$\varnothing$
IO	strong IH	weak IH
UI	strong UI	$\varnothing$
UU	five positive events	$\varnothing$
UO	reflected IU	reflected IU
OI	strong HI	$\varnothing$
OU	reflected UI	$\varnothing$
OO	reflected II	reflected weak II.

Each source domain is treated below for all independent testing angles and states. The following bounds establish that these domains exhaust the source.

If C, α are both weak, the four event-cycle physicality tests force n, m, A, D unbounded. One test is $4D\varepsilon^2 - 2\varepsilon + 4(2+\varepsilon)C + 8(1+\varepsilon)\alpha$: a sufficiently small positive real ε excludes bounded D. The matrix-square core bound gives $H \geq -2u_c v_c \|\psi_c\|^2$. Thus each matching event amplitude divided by $\sqrt{u_c v_c}$ is infinitesimal. Compact source angles and the reflection imply c bounded, $a, b, t \sim 1$, and $r, h, g, j \in \mathrm{I}$. For $c \in \mathrm{I}$ the relative matching equations give $v_c/u_c \sim 1$ and $W_c/Z_c \sim 1$ before scaled stationarity is used. The radial calculations give $N \asymp r + h + g + j$ and $4\alpha/(c^4 N) \sim 1$; they retain all four state amplitudes and both state charts. For c a unit, $\alpha/C \asymp 1$. The outer calculations in both cases use $1/(Cn), 1/(Cm), 1/(CA), 1/(CD) = O(1)$, with their reflected counterparts when c is a unit. This covers all two-weak sources.

The empty cells follow from the original source inequalities. For II, $L < y(1 + zr)$ and the positive source numerator show that weak C forces weak α. For UI the same numerator and $n \geq 0$ would force r infinitesimal and h a unit, contradicting weak C. For IU, bounded b obeys

$$r_0\{b_0 + k_0^2(b_0+1)\} \leq b_0, \quad r_0 \geq 1 - b_0, \quad b_0 \geq \frac{k_0}{\sqrt{1+k_0^2}} > 0, \qquad k = v_c,$$

and gives strong C. For unbounded b, $C \asymp L^{-1}$ with L bounded, again giving strong C. For UU the positive numerator and unit source denominator make C, α weak or strong together. Finally, for OI or OU put $q = u_c^{-2} \in \mathrm{I}$ and $y = v_c^2$ bounded. The exact n numerator is $-qP/(2rQ)$, $Q > 0$, and

$$P \geq r^2(1+y)(H+yr) - Tqr(y+2) - q^2(T^2+T), \quad 0 < T < H + y.$$

If q/r were infinitesimal, $P > 0$ would contradict $n \geq 0$. Thus r/q is bounded, $H \sim 1$, and the canceled C numerator is a unit with bounded positive denominator. These arguments include equality $n = 0$.

For IU, bounded b is a unit by the preceding bound. Its exhaustive opening cases are g a unit; g infinitesimal and r a unit; and g, r both infinitesimal. The three one-angle calculations retain, respectively, the ordinary hard normal, the full m normal, and both centered n, m squares. If b is unbounded, put $x = u_c^2$, $B = xb$, $H = xh$, $L = x(b-h)$. The gaps imply $0 < H < 1$, $0 < L < k^2(1+xr)$ and $B = H + L$ bounded. Exact cancellation gives

$$1 - r \asymp 1, \quad m \asymp b^{-2}, \quad C \asymp L^{-1}, \quad A \asymp H^{-1}, \quad D \asymp B/j, \quad \alpha \asymp B/(1-H).$$

For infinitesimal B use the homogeneous one-angle calculation; for unit B use the strong-event calculation, which includes unbounded C or α. The reflection supplies every UO source.

On strong II the source equations give B a unit, $H, 1 - r$ units, and

$$m/z^2, \quad A/y, \quad CL, \quad Dyj, \quad \alpha(1-H) \asymp 1, \qquad nr = O(1).$$

These are the coefficient assumptions of the shared II local and outer calculations. For IO the weak IH source has B unbounded and $\rho = B/y$ with residue in $[0, 1]$; its interior, two endpoints and open-axis calculations exhaust that interval. The strong IH source instead has B a unit, $r = O(1/y)$ and $Dy^2 \asymp 1$ and uses the same polynomial relations with its own coefficient bounds. Strong UI has bounded $nr, m\mu, \alpha\pi, C\rho k^2, D\sigma k^2, Ct - Dj$, which are the inputs of its three-normal calculation. On OI, direct source cancellation gives $R = r/q$ bounded, $H \sim 1$ and

$$mqK, \quad Cyt, \quad Dyj, \quad \alpha qE, \quad A/(qy) \asymp 1.$$

The HI circuit calculation includes every $n \geq 0$, including $R_0 = 0$; its multiplier $2(1+R_0)^2/(3+R_0)$ lies in $[2/3, 2]$. On strong UU, the source equations force m, A, D strong, so the five-event proof uses the positive real lower bound of $\min(1, m, C, A, D, \alpha)$. The remaining reflected entries follow from (C.2). The directed weak OO entry is the following weak-II calculation.

The directed weak II source

Use $z = u_c^2$, $y = v_c^2$, $H = v_c Y_c$, $L = v_c(u_c W_c - Y_c)$, $B = H + L$, $r = -X_c/u_c$. The source equations are (C.25), with $\delta = B\Delta$ and the positive numerator

$$M_s = B^2 M = (1+z)(B+zr)H + yG + yzj + zrL(1+y), \qquad G = B + z(H + r - 1).$$

Since $L < y(1+zr)$, L is infinitesimal and B bounded. If B is infinitesimal, the exact bound

$$0 < \alpha \le \frac{(B+zr)(1+z)(H+y)}{2(1-H)} \in \mathrm{I}$$

uses $\delta \ge B + z$. If B is a unit, $H = B - L$, δ and the positive numerator are units, giving strong α. Thus weak α is exactly $B \in \mathrm{I}$ in this chart.

Set $a = z/B$, $t = y/B$, $h = H/B$, $l = L/B = 1 - h$, $e = 1 - r$. The exact gaps give

$$g = 1 - ae + hz > 0, \quad j = 1 + zr - l/t > 0, \quad ae < 1 + hz.$$

Hence unit e forces bounded a; unbounded a forces infinitesimal e; infinitesimal a gives $g \sim 1$; and infinitesimal g gives $ae \sim 1$. The source formula gives

$$r(2n+1) \asymp 1, \quad AH/y \asymp 1, \qquad m = \frac{(1+y)(1+zr)S_c}{2}\frac{z}{\delta}\frac{1 + zH/B - g}{g}.$$

Thus m is bounded for unit g, and $m \asymp 1/g$ for infinitesimal g. The positive source numerator also gives the unbounded-coefficient equivalences

coefficient	n	m	A	C	D
condition	$r \in \mathrm{I}$	$g \in \mathrm{I}$	$H/y \in \mathrm{I}$	$L/(H+y) \in \mathrm{I}$	$t \in \mathrm{I}$ or $j \in \mathrm{I}$.

For t unbounded, $j \sim 1$ and $D = O(1/t)$. These estimates concern the original source.

Split t first. Infinitesimal t is R1 and unbounded t is R2, proved below with their complete scalar and both-normal calculations. For unit t and bounded a, Proposition C.27 proves all cases at once. For unit t and unbounded a, necessarily e is infinitesimal, $r \sim 1$. The remaining exhaustive cases are

h unit, $l \in \mathrm{I}$	R3, the EN calculation
h, l units, $j \in \mathrm{I}$	R4, the EN calculation
h, l, j units, $g \in \mathrm{I}$	R6, RI or RB
$h, g \in \mathrm{I}$	R7, RU
$h, j \in \mathrm{I}$, g unit	R8, the joint-D calculation
$h \in \mathrm{I}$, g, j units	R10, the weighted/thin calculation.

Indeed if h, l, j, g are all units, all original coefficients are bounded by the displayed identities, contradicting the assumed total scale. Infinitesimal r is impossible when a is unbounded. No other ordered alternative remains.

The R6 source inequality forces $b = a^2 B$ bounded, and its two cases b unit or infinitesimal are RB and RI. RU treats both b cases for R7. In R8 and R10, split n/e^2 into noninfinitesimal, positive infinitesimal, and zero. The first uses the weighted source Schur bound; the second uses the explicitly paired positive coefficients and the full contact-normal correction; equality uses the exact OP4 native C1 map. OP4 preserves the physical parent and the actual contact, so only the native C1 small-e or centered-A theorem is used at $n = 0$. The positive-thin calculation never declares its formal negative-β tuple physical. These source splits and the finite and homogeneous outer calculations below exhaust every independent testing pair and state.

C.1.2 Stationary coefficients and contact

In this appendix, c denotes the actual half-angle and C the event weight previously denoted c. Set

$$z = c^2, \quad A_c = 1 + zr, \quad h = b - a^2 t, \quad g = (1+z)b - (1 - r + a^2 zt), \quad j = 1 + zr - t,$$

$$\begin{aligned}\Delta &= (1+z)zb^2 + 2z(1+z)br + (1+z)(1+a^2 z)b \\ &\quad + z^2(1-a^2)r^2 + 2z(1+a^2 z)r + 1 + a^2 z, \\ N &= (1+z)(b+r)(b+a^2(1-t)) - a^2(1+a^2 z)(1-r)t.\end{aligned}$$

The unchanged $d = 1$, $\lambda = \beta = 0$ source is

$$\begin{aligned} n &= \frac{(b+r)A_c(A_c+a^2zt) - r\Delta}{2r\Delta}, & m &= \frac{(1+a^2z)(1-r)A_c(A_c+a^2zt)}{2g\Delta}, \\ C &= \frac{A_cN}{2a^2t\Delta}, & A &= \frac{a^2A_c(b+r)(A_c+a^2zt)}{2h\Delta}, \\ D &= \frac{(1+z)(b+r)(b-a^2zr)(A_c+a^2zt)}{2a^2j\Delta}, & \alpha &= \frac{z^2(b+r)N}{2(1-zh)\Delta}. \end{aligned} \tag{C.3}$$

Its exact contact is

$$(u_c, v_c; W_c, X_c, Y_c, Z_c) = \left(c, ac; \frac{b}{a}, -cr, c\left(\frac{b}{a} - at\right), 1\right). \tag{C.4}$$

The positive chart requires $a, c, b, t > 0$, $0 < r < 1$, $h, g, j > 0$, $1 - zh > 0$, $b - a^2zr > 0$, and nonnegative original event weights. The source-entry proof uses this positive chart without a shape-unit assumption.

The numerator has the useful positive identity

$$N = (1+c^2)(b+r)h + a^2\{g + j + rt(1+a^2c^2)\} > 0.$$

It follows by substituting the three source-gap definitions; every term is positive in the domain used by the contact-entry theorem. For the determinant calculation use the aliases

$$R_2 = R_m, \quad R_3 = uW - Y, \quad R_5 = R_D, \quad R_\alpha = vY - Z, \qquad (e_1, e_2, e_3, e_4, e_5) = (n, m, C, A, D).$$

Uniqueness of the stationary coefficients

The six stationary coefficients are unique, as follows. Let M have rows indexed by (W, X, Y, Z, u, v) and columns equal to the derivatives of

$$D_uD_vX^2, \quad R_2^2, \quad D_vR_3^2, \quad D_uD_vY^2, \quad R_5^2, \quad D_uR_\alpha^2,$$

evaluated at $(u, v; W, X, Y, Z) = (c, ac; b/a, -cr, c(b/a - at), 1)$. Write $z = c^2$, $D = 1 + z$, $E = 1 + a^2z$, $h = b - a^2t$, $g = Db - (1 - r + a^2zt)$, and $j = 1 + zr - t$. Factoring the six columns by

$$2cr, \quad 2cg, \quad 2act, \quad -2ch/a, \quad 2acj, \quad 2(1-zh)$$

and the first row by c leaves the matrix

$$B = \begin{pmatrix} 0 & a & E & 0 & -1 & 0 \\ -DE & -1 & 0 & 0 & -az & 0 \\ 0 & az & -E & -DE & 1 & -acD \\ 0 & -c & 0 & 0 & ac & D \\ zrE & zh-1 & (b/a)E & -zhE/a & azr - b/a & c(1-zh) \\ azrD & b/a + zh/a & a^2zt & -zDh & 1+zr & -cDh/a \end{pmatrix}.$$

Use the pivots in row 2, column 1; row 1, column 3; and row 4, column 6. Eliminate those three columns, multiply the remaining second and third rows by D and E, and factor $-DE$ from their middle column. The determinant reduces to $\det B = D^2E^2 \det T$, where

$$T = \begin{pmatrix} a & 1 & a^2z \\ -1 - b - z(r + a^2t) & zh/a & az(r - 1 + zh) \\ b/a + az(h - r) & zh & E(1+zh) + z(r + a^2t) \end{pmatrix}.$$

Subtract (zh/a) times the first row from the second and zh times the first row from the third. Expanding along the middle column gives

$$\det T = [1 + b + z(b+r)][1 + z(a^2 + b + r)] - z(1-r)(b - a^2zr) = \Delta.$$

The last equality follows by expanding the two products and is exactly the source definition of Δ. Restoring the factors proves

$$\det M = -64ac^6rgthj(1-zh)D^2E^2\Delta. \tag{C.5}$$

Each source coefficient occurs with a factor 4 in the parent, so the stationary linear system in $(e_1, e_2, e_3, e_4, e_5, \alpha)$ has determinant $4^6 \det M$. On the positive source domain this is nonzero. Since the six source formulas above satisfy all six stationary equations, they are the unique stationary coefficients.

C.2 Two weak complementary coefficients

C.2.1 Entry from two weak coefficients

Here weak means infinitesimal after division by the original positive signed coefficient d. All source calculations take place in one real closed coefficient field. Testing fields remain independent.

Theorem C.1 (C2 entry without a contact-chart hypothesis)**.** *Let the physical canonical parent have $\lambda = \beta = 0$, $d > 0$, and nonnegative original event coefficients. Suppose it has an actual finite noncommuting stationary zero contact, and*

$$C/d,\ \alpha/d \quad \text{are infinitesimal.}$$

Then its full parent is tensor SOS at every independent real or complex projective test. No hypothesis on the other four event weights, on the contact-angle ratio, or on the signs of the source gaps is imposed.

More precisely, normalize $d = 1$. After local real sign gauges and at most the exact C2 reflection, the actual contact enters the positive rational C2 source with

$$c \text{ bounded}, \qquad a, b, t \longrightarrow 1, \qquad r, h, g, j > 0 \text{ infinitesimal.}$$

In particular all four matching weights are unbounded and both complementary weights are strictly positive on this finite-contact domain.

The proof has three ingredients: a finite physical witness makes the matching weights large; the quadratic CHSH excess controls the contact amplitudes at the scale of the actual angles; stationarity fixes their signs. This last step establishes the source chart rather than assuming it.

The physical bound at the correct angular scale

For $u, v > 0$, put $D_u = 1 + u^2, D_v = 1 + v^2$, and let $M(u, v)$ be the real symmetric matrix of the signed core

$$H = 2\{R_D^2 + D_u(vY - Z)^2 + D_v(X + uZ)^2 - D_u D_v Z^2\},$$

where

$$R_m = vW - X + uvY - uZ, \qquad R_D = uW + uvX - Y - vZ.$$

Write $L = D_u D_v$. Direct matrix multiplication gives

$$(M - LI)^2 = L^2 I + 4uvLJ, \qquad J = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \\ 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \qquad J^2 = I. \tag{C.6}$$

For a unit real state ψ, Cauchy–Schwarz therefore gives

$$\psi^T M \psi \geq L - \sqrt{L^2 + 4uvL} \geq -2uv.$$

The last inequality follows by squaring the positive quantities $L + 2uv$ and $\sqrt{L^2 + 4uvL}$; their squared difference is $4u^2v^2$. Homogeneity proves

$$H \geq -2uv\,\|\psi\|^2. \tag{C.7}$$

This is the quadratic CHSH bound in the present normalization.

At a normalized zero contact, every event term is nonnegative. Define the four matching amplitudes, in the fixed order (n, A, m, D), by

$$\mathcal{L} = (\sqrt{D_u D_v}X, \sqrt{D_u D_v}Y, R_m, R_D).$$

Equation (C.7) gives, for each matching event,

$$e_i \mathcal{L}_i^2 \leq uv/2. \tag{C.8}$$

All square roots and inequalities in this paragraph are coefficient-side.

Matching weights and the compact contact geometry

First every matching weight is unbounded. To see the mechanism for D, use the coefficient-field experiment

$$u = 1, \quad v = \sqrt{1 + \delta}, \quad (W, X, Y, Z) = (1, 0, 0, \sqrt{1 + \delta}), \qquad \delta > 0.$$

The n, A, m positive square terms vanish, and the full eight-event parent is exactly

$$4D\delta^2 - 2\delta + 8\lambda + 4(2 + \delta)C + 8(1 + \delta)\alpha + 4(2 + \delta)(1 + \delta)\beta.$$

If $D \le B$ for a positive real B, choose the positive real $\delta = 1/[4(B+1)]$. The first two terms have a strictly negative real upper bound, whereas the complementary terms are infinitesimal. This contradicts physicality. The exact even rotations of the event cycle give the same argument for n, A, m, preserving the sum of the four complementary weights. Hence $1/n, 1/A, 1/m, 1/D$ are infinitesimal.

Normalize the actual state to norm one and use positive actual half-angles, obtainable by local real basis gauges. The weaker bound $H/(D_u D_v) \ge -2$ shows that all normalized matching probabilities vanish in residue. In particular $X, Y \to 0$. Put

$$c_u = (1+u^2)^{-1/2},\ s_u = uc_u, \qquad c_v = (1+v^2)^{-1/2},\ s_v = vc_v.$$

If $\sigma = \mathrm{res}(c_u s_v)$ and $\tau = \mathrm{res}(s_u c_v)$, the two remaining matching equations read

$$\sigma\overline{W} - \tau\overline{Z} = 0, \qquad \tau\overline{W} - \sigma\overline{Z} = 0.$$

At least one of $\overline{W}, \overline{Z}$ is nonzero. Thus $\sigma = \tau \ge 0$. If positive, the two actual half-angles are bounded units with the same residue. If zero, both are infinitesimal or both unbounded, by the two unit-circle identities. The exact C2 reflection sends (u, v) to $(1/v, 1/u)$ and exchanges (C, α) and (m, D), preserving the hypotheses. We may therefore assume either ordinary bounded-unit angles or two infinitesimal angles.

Relative coalescence when both angles vanish

Suppose u, v are infinitesimal. From (C.8), all four $\mathcal{L}_i/\sqrt{uv}$ are infinitesimal. In particular $X/\sqrt{uv}, Y/\sqrt{uv} \to 0$. Set $a = v/u > 0$. Dividing the two other matching amplitudes by $\sqrt{uv}$ gives

$$\sqrt{a}\,W - \frac{Z}{\sqrt{a}} \longrightarrow 0, \qquad \frac{W}{\sqrt{a}} - \sqrt{a}\,Z \longrightarrow 0.$$

The discarded terms are $X/\sqrt{uv}, Y/\sqrt{uv}$ and $\sqrt{uv}X, \sqrt{uv}Y$, all infinitesimal. If a were unbounded, these equations would force both W, Z infinitesimal; the same follows if a is infinitesimal. Both contradict normalization. Thus a is a bounded unit. Taking residues now yields $a \to 1$ and $W/Z \to 1$, with both W, Z units. Consequently

$$X/u,\ Y/u,\ R_m/u,\ R_D/u \longrightarrow 0.$$

Normalize $W = 1$ by a pure coefficient factor. Write

$$x = X/u, \quad y = Y/u, \quad q = v/u, \qquad (u, x, y, Z, q) \longrightarrow (0, 0, 0, 1, 1).$$

The core divided by u^2 is the polynomial

$$\begin{aligned}\mathcal{H} := H/u^2 = 2\{&(1 - y - qZ + u^2 qx)^2 + (1+u^2q^2)(x+Z)^2 \\ &+ (1+u^2)(u^2q^2y^2 - 2qyZ - q^2Z^2)\}.\end{aligned} \tag{C.9}$$

It has zero residue at this point. The nonnegative alpha positive square term and the zero-contact equation give

$$0 \le 4\frac{\alpha}{u^2}(1+u^2)(u^2qy - Z)^2 \le -\mathcal{H}.$$

Its square factor is a positive unit. Hence

$$\alpha/u^2 \longrightarrow 0. \tag{C.10}$$

This estimate is derived before the scaled stationary equations are used.

Stationarity fixes the source signs

In the small-angle case introduce the four exact local coordinates

$$\ell = \left(\sqrt{D_u D_v}\,x, \sqrt{D_u D_v}\,y, q - x + u^2qy - Z,\ 1 + u^2qx - y - qZ\right).$$

Their Jacobian with respect to (x, y, Z, q) has residue

$$J_0 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & -1 & 1 \\ 0 & -1 & -1 & -1 \end{pmatrix}, \qquad \det J_0 = 2.$$

Thus its inverse and the transformed derivatives are bounded at the actual contact. Only this pointwise coefficient-field inverse is used. Direct differentiation of (C.9) gives

$$\mathrm{res}(\nabla_\ell \mathcal{H}) = 2(1, -1, -1, 1).$$

The full parent divided by u^2 is exactly

$$4\sum_i e_i\ell_i^2 + 4C(1+u^2q^2)(1-y)^2 + 4(\alpha/u^2)(1+u^2)(u^2qy-Z)^2 + \mathcal{H}.$$

The two complementary-gradient terms have zero residue, by weak C and (C.10). The original stationary equations imply stationarity in ℓ, with u fixed. These equations first show that every $e_i\ell_i$ is bounded, so its residue is defined. Consequently

$$\operatorname{res}(n\ell_n, A\ell_A, m\ell_m, D\ell_D) = \tfrac{1}{4}(-1,1,1,-1). \tag{C.11}$$

Every matching weight is positive, so $X<0, Y>0, R_m>0, R_D<0$ as exact source inequalities.

For ordinary angles the argument is shorter. Normalize $W=1$; the actual contact has residue $(X,Y,Z,v) = (0,0,1,u)$, where u is a bounded unit. Use the unscaled $\mathcal{L}$ coordinates. Their Jacobian determinant at that residue is $2u(1+u^2)^2$, and

$$\nabla_{\mathcal{L}} H = 2u(1,-1,-1,1)$$

there. All complementary gradients are bounded times C,α, hence infinitesimal. Stationarity gives $\operatorname{res}(e_i\mathcal{L}_i) = \overline{u}\,(-1,1,1,-1)/4$, with the same exact signs.

The original source and all independent tests

In either case $W/Z\to 1$, $v/u\to 1$, and $X/u, Y/u\to 0$. Set, using the original actual contact,

$$c=u,\quad a=v/u,\quad b=aW/Z,\quad r=-X/(uZ),\quad t=(uW-Y)/(vZ).$$

Then $a,b,t\to 1$ and

$$h=b-a^2t=\frac{aY}{uZ}>0,\qquad g=\frac{R_m}{uZ}>0,\qquad j=-\frac{R_D}{vZ}>0.$$

All four r,h,g,j are infinitesimal, so $0<r<1$. The remaining chart boundaries are positive units:

$$1-c^2h=\frac{Z-vY}{Z}>0,\qquad b-a^2c^2r=\frac{a(W+vX)}{Z}>0.$$

These definitions give exactly

$$h=b-a^2t,\qquad g=(1+c^2)b-(1-r+a^2c^2t),\qquad j=1+c^2r-t.$$

Finally the stationary coefficient matrix has the nonzero determinant

$$4^6[-64ac^6rgt\,hj(1-c^2h)(1+c^2)^2(1+a^2c^2)^2\Delta].$$

The preceding three-by-three reduction gives this determinant. For $z=c^2$, its positive expansion is

$$\Delta = 1+(1+z)b+z(1+z)b^2+2z(1+z)br+z^2r^2+2zr+a^2z\{(1+z)b+1+zr(2-r)\}\geq 1.$$

The original coefficients therefore equal the unique positive rational C2 source formulas. In particular their positive numerator identity forces $C,\alpha>0$. If c is infinitesimal, apply the small-angle radial theorem and its outer-test corollary; if c is a unit, apply the ordinary-angle radial corollary. Reflect back when necessary and restore d by a pure-source square root. This proves Theorem C.1.

This covers the finite-contact two-weak branch. Weak C alone need not imply weak α; the one-weak family below supplies the complementary case.

C.2.2 Matching the weak coefficients to the strong events

A coefficient condition symmetric in the two weak positive square terms closes both commuting angle corners at once. It also gives a second exact radial source family whose contact angles have ordinary positive residue.

Theorem C.2 (Two matched weak positive square terms)**.** *Let the original physical C2 parent have $d=1$, $\lambda=\beta=0$, and event weights (n,m,C,A,D,α). Suppose $C,\alpha>0$ are infinitesimal, all four other weights are positive, and each of*

$$Cn,\ Cm,\ CA,\ CD,\qquad \alpha n,\ \alpha m,\ \alpha A,\ \alpha D$$

has bounded reciprocal in the coefficient field. Then the full parent is tensor SOS at every independent projective test and every real or complex state.

Proof. Lemma C.5 applies directly to every commuting residue test except the finite zero-zero corner. For that corner use the exact reflection

$$(u,v;W,X,Y,Z;n,m,C,A,D,\alpha)\longmapsto(1/v,1/u;Z,X,Y,W;n,D,\alpha,A,m,C). \tag{C.12}$$

More explicitly, if F denotes the full parent with its indicated arguments, then

$$F(u,v;W,X,Y,Z;n,m,C,A,D,\alpha)=u^2v^2F(1/v,1/u;Z,X,Y,W;n,D,\alpha,A,m,C). \tag{C.13}$$

This is a homogeneous polynomial identity, so its interpretation in reciprocal charts includes literal zero and infinity endpoints.

For a short verification, the reflected R_2,R_5 amplitudes, after multiplication by uv, are $-R_5,-R_2$. The reflected C and α positive square terms exchange, while the n,A positive square terms are fixed. If $\mathcal{H}$ is the signed core, its common expanded form is

$$\begin{aligned}\mathcal{H}/2 &= u^2(W^2+Z^2)+(1+v^2+u^2v^2)(X^2+Y^2)\\ &\quad+2u^2v(WX-YZ)+2u(XZ-WY)-2uv(WZ+XY).\end{aligned}$$

Multiplying its reflected expression by u^2v^2 fixes the first, second and last terms, and exchanges the middle two. This proves the whole-parent identity, including its signed part.

The reflected weak weight is α, its four strong weights are n,D,A,m, and its other nonnegative weak positive square term is C. Thus all hypotheses of Lemma C.5 hold for the reflected parent. Its infinity-infinity chart is precisely the original zero-zero chart. Restore the homogeneous pure-testing factors to obtain the certificate there as well.

For noncommuting residue tests apply the physical-source theorem, using positive pure-source normalization when a coefficient is unbounded. Restore that factor and split real and imaginary state parts. □

Corollary C.3 (Radial C2 source with ordinary contact angles)**.** *Use the same rational source* (C.3)*, now with source half-angle c a positive bounded unit. Suppose $h=b-a^2t,r,g,j$ are positive infinitesimals and $a,1-r,t$ are positive bounded units. Then the full parent is tensor SOS at every independent projective test and every real or complex state.*

Proof. Put $z=c^2$ and continue to write $C=e_3$ for the weak event weight. The exact source identities

$$b-1=-r+g-zh,\qquad t-1=zr-j,\qquad a-1=\frac{g+j-(1+z)(r+h)}{t(a+1)}$$

remain valid for this ordinary source angle. Thus a,b,t have residue one. The positive source identity

$$N=(1+z)(b+r)h+a^2\{g+j+rt(1+a^2z)\}$$

shows that $N>0$ is infinitesimal and $h/N,r/N,g/N,j/N$ are bounded: each of the four positive coefficients in this sum is a bounded unit.

Here is the entire normalized coefficient calculation. Write $z_0=\overline{z}>0$. Substitution of $a=b=t=1,r=0$ in the bounded source numerators and unit denominators gives

$$\begin{aligned}\overline{\Delta}&=2(1+z_0)^2,\qquad \overline{C/N}=\frac{1}{4(1+z_0)^2},\qquad \overline{\alpha/C}=z_0^2,\\ \overline{rn}&=\overline{hA}=\frac{1}{4(1+z_0)},\qquad \overline{gm}=\overline{jD}=\frac14.\end{aligned} \tag{C.14}$$

For example, the exact quotients used in the first line are

$$\frac{C}{N}=\frac{1+zr}{2a^2t\Delta},\qquad \frac{\alpha}{C}=\frac{z^2(b+r)a^2t}{(1-zh)(1+zr)}.$$

The four strong normalized products use the numerators of (C.3) after multiplication by r,h,g,j, respectively; no vanishing source gap is inverted in this calculation. All six weights are therefore positive, C,α are infinitesimal, and their normalized quotients displayed above are bounded units. For instance

$$\frac{1}{Cn}=\frac{r/N}{(C/N)(rn)},\qquad \frac{1}{\alpha n}=\frac{C/\alpha}{Cn};$$

the other six product bounds follow in the same way from $g/N,h/N,j/N$. These are exactly the hypotheses of Theorem C.2.

For completeness, the source is physical by the unchanged exact contact argument. The rational stationary identities (C.4) are valid at this value of c as well. At that noncommuting contact the state matrix has a nonzero kernel vector. Its restriction to $Z = 0$ is positive definite: the signed core is nonnegative there and the positive event positive square terms have zero common kernel. Hence the other three eigenvalues are positive. The exact contact and angular stationarity give the stationary zero of the Laurent determinant; its constant-plus-positive-exponentials form makes this its global minimum. The determinant criterion and homogeneous continuity prove physicality, exactly as for the small-angle source. Theorem C.2 now applies.

The corollary applies to the stated radial subfamily of the ordinary-angle cell. □

C.2.3 Weighted radial completion

Normalize the original positive d by a pure-source square. The two-weak source reduction is applied after this normalization; it has already given $c \in \mathrm{I}$, $a, b, t \sim 1$ and $r, h, g, j \in \mathrm{I}$, where $h = b - a^2 t$. Use the exact native coefficients in Section C.1.2, with $(e_1, e_2, e_3, e_4, e_5) = (n, m, C, A, D)$ and

$$(u_c, v_c; W_c, X_c, Y_c, Z_c) = (c, ac; b/a, -cr, c(b/a - at), 1).$$

The two source identities needed for all relative opening rates are

$$N = (1 + c^2)(b + r)h + a^2\{g + j + rt(1 + a^2c^2)\} > 0, \qquad N \asymp h + r + g + j. \tag{C.15}$$

Consequently

$$U_1 = re_1, \quad U_2 = ge_2, \quad U_4 = he_4, \quad U_5 = je_5 \sim \tfrac{1}{4}, \qquad e_3/N \sim \tfrac{1}{4}, \qquad \frac{4\alpha}{c^4 N} \sim 1.$$

These follow directly from the native fractions after their respective opening factors are canceled. In particular all four e_i above are unbounded, and $h/N, r/N, g/N, j/N$ are bounded. They also imply $1/(Cn), 1/(Cm), 1/(CA), 1/(CD) = O(1)$, the assumptions of the complete radial outer calculation.

The same source reduction forces n, m, A, D unbounded, so $n = 0$ is impossible in this case. No zero-coefficient face is being discarded.

Proposition C.4. *On this source domain the full parent is tensor SOS at every independent projective testing pair and every real or complex state.*

First let the independent tests u, v be infinitesimal or zero and take the pure-testing state chart $Z = 1$, $W \geq 0$ bounded. Define

$$R_m = vW - X + uvY - u, \qquad J = uW + uvX - Y - v,$$
$$H_1 = X + ru, \quad H_2 = R_m - gu, \quad H_4 = Y - (h/a)u, \quad H_5 = J + jv.$$

Retain the four whole positive squares

$$4e_1 D_u D_v H_1^2 + 4e_2 H_2^2 + 4e_4 D_u D_v H_4^2 + 4e_5 H_5^2.$$

Their differences from the original terms are, respectively,

$$-8U_1 D_u D_v u H_1 + 4rU_1 D_u D_v u^2, \quad 8U_2 u H_2 + 4gU_2 u^2,$$
$$\frac{8U_4}{a} D_u D_v u H_4 + \frac{4hU_4}{a^2} D_u D_v u^2, \quad -8U_5 v H_5 + 4jU_5 v^2.$$

Thus the entire remaining polynomial has bounded coefficients, including all its state normals. With $n_1 = H_1$, $n_5 = H_5$ the exact inverse is

$$X = -ru + n_1, \qquad Y = uW - (1 - j + ru^2)v + uvn_1 - n_5.$$

Define its scalar f, linear coefficients L_1, L_5, and quadratic normal matrix by coefficient extraction in (n_1, n_5). Quadratic state degree proves that this is the whole remainder.

Put

$$R = au - v, \qquad S = u(W - W_c), \qquad T = u^2 - c^2.$$

Then $f \in (R, S, T)^2$ and $L_1, L_5 \in (R, S, T)$ with bounded coefficients. Here is the finite division proof. Parity makes $f(u, au, W) = f_0(u^2, W)$ even and $f_v(u, au, W), L_i(u, au, W)$ odd. At zero angles the value is 4α, independent of W, so both nonconstant W coefficients of f_0 are divisible by u^2. Write $f_0 = A(\xi)(W - W_c)^2 + B(\xi)(W - W_c) + C(\xi)$. The exact source contact gives

$$A = \xi A_1, \quad B = \xi(\xi - c^2)B_1, \quad C = (\xi - c^2)^2 C_1.$$

Monic division gives these polynomials without a factor $1/c$. Likewise $f_v(u, au, W) = S\ell_1 + uT\ell_2$. Finite Taylor division in $v-au$ proves the scalar assertion. The identical odd-parity argument proves the linear-normal assertions. All higher normal terms are already quadratic by construction.

The exact current identity removes S. Put $K = 1 - j + ru^2$ and

$$Q(\xi) = aW_c - a^2(1-j) - a^2r(\xi + c^2).$$

The identities $H_2 = G - (1-u^2v^2)H_1 - uvH_5$ and $G = -[D_uW - Ku(v+au)]R + aD_uS + uTQ(u^2)$ give

$$S = \frac{H_2 + (1-u^2v^2)H_1 + uvH_5 + [D_uW - Ku(v+au)]R - uTQ(u^2)}{aD_u}. \tag{C.16}$$

Its denominator is a product of pure-source and pure-testing units. After restoring the four retained squares this is an exact bounded Gram on $K_i = \sqrt{e_i}H_i$ for $i = 1, 2, 4, 5$, together with R, T.

There is one further exact linear relation:

$$\begin{aligned}\zeta = {} & aD_uH_4 - H_2 - (1 - u^2v^2 + aD_uuv)H_1 + (aD_u - uv)H_5 \\ & - [D_uW + (a - uv)K]R + u[Q(u^2) + a^2D_ur]T = 0.\end{aligned} \tag{C.17}$$

Add the exact zero

$$\frac{8U_4}{a^3}\zeta R + \zeta^2 = 0. \tag{C.18}$$

The hard and R residue can be computed directly on the present all-small domain. Modulo (h, r, g, j), followed by the joint real residue $c = u = v = 0$, the degree-two scalar is

$$2(u-v)(uW - v) = 2R(R + S).$$

Equation (C.16) gives $S = WR$ on that residue, while $\zeta = -(1+W)R$ and $U_4 = 1/4$. The displayed zero changes $2(1+W)R^2$ into $(1+W)^2R^2$. The four scaled hard diagonals are 4 and their other couplings vanish.

It remains to divide the radial column by $\sqrt{N}$. Let $I = (h, r, g, j)$ in the bounded source coefficient ring. The exact identities

$$b - 1 = -r + g - c^2h, \quad t - 1 = c^2r - j, \quad a - 1 = \frac{g + j - (1+c^2)(r+h)}{t(a+1)}$$

show that subtraction of the $I = 0$ specialization of any coefficient leaves I with bounded coquotients. Its denominators are units after the four opening factors have been absorbed in U_i. Finite monic division and the displayed current substitution preserve this fact.

The exact simplifications $Q(u^2) = h - a^2ru^2$ and $Q(u^2) + a^2D_ur = h + a^2r$ belong to I. The only radial cross coefficient requiring a further correction is the RT coefficient. For an explicit calculation before taking the testing residue, set $D_c = 1 + c^2$, $Y_* = uW - v$, $G_* = D_uvW - D_vu$. At $I = 0$ the whole scalar remainder is

$$f_* = 2uG_* + \frac{2}{D_c}D_uD_vuY_* + 2\{D_u(vY_* - 1)^2 + D_vu^2 - D_uD_v\}.$$

It satisfies

$$f_*(u,u,1) = 0, \quad (f_*)_W(u,u,1) = \frac{2u^2(u^2-c^2)D_u}{D_c}, \quad (f_*)_v(u,u,1) = \frac{-2u(u^2-c^2)D_u}{D_c}.$$

The monic division order just specified therefore gives, after eliminating S, the RT coefficient $2u(D_uW + 1 - uv)/D_c$. The earlier zero adds none at this specialization. The R coefficient of ζ is $-(D_uW + 1 - uv)$, so add the further exact zero

$$(2u/D_c)\zeta T = 0.$$

The resulting RT coefficient belongs to uI.

Now use $T_N = \sqrt{N}T$. Every hard–T_N coefficient is u times bounded factors divided by $\sqrt{Ne_i}$, which is bounded because $1/\sqrt{Ne_i} = \sqrt{(r,g,h,j)_i/N}/\sqrt{U_i}$. Its residue is zero. The corrected RT_N coefficient belongs to $uI/\sqrt{N}$, again bounded with zero residue. The T^2 coefficient belongs to I because $f_*(u,u,1) = 0$; every contribution other than its value at $u = 0$ has a further factor u^2. Its value at $u = 0$ is exactly $4\alpha/c^4$. Therefore its quotient by N is bounded and has residue 1. The complete weighted residue is

$$4K_1^2 + 4K_2^2 + 4K_4^2 + 4K_5^2 + (1+W_0)^2R^2 + T_N^2 \succ 0.$$

The separated strict-Gram lemma proves the full chart.

For $W = 1$, Z infinitesimal, retain the original four squares in X, R_m, Y, J and the whole α square. The exact testing identities

$$v = \frac{R_m + (1 - u^2v^2)X + uD_vZ + uvJ}{D_u},$$

$$u = \frac{D_uY + [Z(1 - u^2v^2) - D_uuv]X + (D_u + uvZ)J + ZR_m}{D_u - D_vZ^2}$$

have pure-testing unit denominators. They express the entire remaining quadratic form on the four scaled retained amplitudes. To check this without discarding a state term, the half-core has the exact completion

$$H = 2\{C_*^2 + D_vQ_*^2 + D_uv^2Y^2 + D_uu^2v^2(1 - u^2v^2)Z^2/D_v\} - 4D_uuvZ,$$

where $C_* = u + uv(X + uZ) - Y$ and $Q_* = X + uZ - D_uuv^2Z/D_v$. Substitute the two identities in every term. All coefficients are bounded and the residue on the four scaled amplitudes is $4I$; thus the same lemma applies.

The testing sign gauges allow $u, v \geq 0$. The X, Y block of half the core has eigenvalues $L \pm uv > 0$, $L = 1 + v^2 + u^2v^2$, and Schur complement

$$\frac{uvD_uD_v\{uv(W^2 + Z^2) - 2LWZ\}}{L^2 - u^2v^2}.$$

For $WZ \leq 0$ the core is a nonnegative pure-testing value and all events are already tensor squares. For $WZ > 0$, the two preceding state charts are exhaustive. Homogeneous restoration and real/imaginary splitting prove every state. Section C.2.4 then proves all other independent projective testing pairs.

C.2.4 Complementary testing angles

The radial corner has four strong event positive square terms and one weak positive square term. The weak positive square term must be retained at its exact scale at the opposite commuting corner. An elementary conductor identity makes that scale sufficient; it does not compare coefficient and testing infinitesimals. Throughout this section C denotes the event weight e_3, whereas c continues to denote the source half-angle in (C.3).

Lemma C.5 (Four strong positive square terms and a weak opposite-corner pivot)**.** *Consider the full parent* (C.1)*, with*

$$(e_1, e_2, e_3, e_4, e_5) = (n, m, C, A, D), \qquad \alpha \geq 0.$$

Suppose $C > 0$ is infinitesimal and the four positive pure coefficient quantities Cn, Cm, CA, CD have bounded reciprocals. Then the full parent is tensor SOS at every commuting residue test outside the finite zero-zero corner, for every real or complex projective state. The four strong coefficients and the roots in the proof may be unbounded.

Proof. The hypotheses imply that $1/n, 1/m, 1/A, 1/D$ are infinitesimal. First consider the outer axes other than the infinity-infinity corner. Choose $\mu = \min(1, n, m, A, D)$, a positive bounded unit. Retain the nonnegative excess of each strong positive square term, together with the whole C and α positive square terms. The remaining parent consists of the signed core and the four positive square terms with common weight μ.

Here are the four kernel calculations, with positive factors $1 + u^2$ or $1 + v^2$ omitted only from the positive square term amplitudes. At $u = 0$, finite nonzero v, the four amplitudes are $(X, vW - X, Y, -Y - vZ)$; at $v = 0$, finite nonzero u, they are $(X, -X - uZ, Y, uW - Y)$. Their common kernels are zero, and half the signed core is, respectively,

$$D_v(X^2 + Y^2), \qquad (uW - Y)^2 + (X + uZ)^2.$$

At $u = \infty$, finite v, the four homogenized amplitudes are $(X, vY - Z, Y, W + vX)$; at $v = \infty$, finite u, they are $(X, W + uY, Y, uX - Z)$. These common kernels are zero even when the finite angle is zero. Half the homogenized signed core is, respectively,

$$(W + vX)^2 + (vY - Z)^2, \qquad D_u(X^2 + Y^2).$$

Thus the selected state Gram has strict positive residue. Its coefficients are bounded in each indicated local testing chart, so the strict-residue Gram lemma proves the whole chart. This includes the mixed zero-infinity endpoints and either sign of every finite angle.

It remains to treat $u = v = \infty$ at residue. Put $p = 1/u$, $q = 1/v$, using the homogeneous reciprocal coordinates at exact infinity, and write $D_p = 1 + p^2$, $D_q = 1 + q^2$. Thus p, q are infinitesimal or zero. The homogeneous full parent in this chart is exactly

$$\begin{aligned}
\widehat{F} &= 4nD_pD_qX^2 + 4mM^2 + 4CD_q(W - pY)^2 + 4AD_pD_qY^2 + 4DJ^2 + 4\alpha D_p(Y - qZ)^2 + H,\\
M &= pW - pqX + Y - qZ, \qquad J = qW + X - pqY - pZ,\\
H &= 2\{J^2 + D_p(Y - qZ)^2 + D_q(pX + Z)^2 - D_pD_qZ^2\}.
\end{aligned} \tag{C.19}$$

For finite nonzero u, v this is p^2q^2F; homogeneous interpretation also gives the exact endpoint polynomial.

Set $L = 1 + p^2 + p^2q^2$. The needed conductor identity is

$$\frac{H}{2} = qWJ - qZM + L(X^2 + Y^2) - 2pqXY + q(W + pqZ)X - q(pqW + Z)Y. \tag{C.20}$$

For a direct check, put $M_0 = pW - qZ$, $J_0 = qW - pZ$. The scalar part of $H/2$ is

$$J_0^2 + (q^2 - p^2)Z^2 = qWJ_0 - qZM_0.$$

Its linear X, Y part is $2q(W+pqZ)X-2q(pqW+Z)Y$, and its quadratic part is $L(X^2+Y^2)-2pqXY$. Substitution of $M = M_0 - pqX + Y$, $J = J_0 + X - pqY$ gives (C.20).

First normalize $W = 1$ and assume Z bounded. No bound is required on X or Y. Retain the α positive square term and use the five roots

$$K_1 = \sqrt{n}X, \quad K_2 = \sqrt{m}M, \quad K_4 = \sqrt{A}Y, \quad K_5 = \sqrt{D}J, \quad T_C = \sqrt{C}\,W.$$

Here $T_C = \sqrt{C}$ in this state chart. Equations (C.19)–(C.20) give the following *exact* Gram expression for the remainder:

$$\begin{aligned}\widehat{F} - 4\alpha D_p(Y - qZ)^2 = {} & 4D_pD_q(K_1^2 + K_4^2) + 4K_2^2 + 4K_5^2 + 4D_q\left(T_C - p\sqrt{C/A}\,K_4\right)^2 \\ & + 2L\left(\frac{K_1^2}{n} + \frac{K_4^2}{A}\right) - \frac{4pq}{\sqrt{nA}}K_1K_4 \\ & + 2qT_C\left\{\frac{K_5}{\sqrt{CD}} - \frac{ZK_2}{\sqrt{Cm}} + \frac{(1+pqZ)K_1}{\sqrt{Cn}} - \frac{(pq+Z)K_4}{\sqrt{CA}}\right\}.\end{aligned} \tag{C.21}$$

Every denominator is pure coefficient. The four inverse square roots $1/\sqrt{Ce_i}$ are bounded by hypothesis, and $\sqrt{C/A} = C/\sqrt{CA}$ is infinitesimal. All coefficients of this Gram are therefore in the bounded tensor ring. Every displayed correction has zero residue, whereas its five positive square term pivots have residue 4. Thus its residue is exactly

$$4(K_1^2 + K_2^2 + K_4^2 + K_5^2 + T_C^2) \succ 0.$$

The strict-residue Gram lemma applies without any bound on the roots. In particular it makes no comparison between C and p or q.

The other state chart is $Z = 1$, W infinitesimal or zero. Here the exact pure-testing conductor is a unit:

$$q = \Gamma := \frac{M + WJ + (pq - W)X + (Wpq - 1)Y}{W^2 - 1}. \tag{C.22}$$

Indeed $M + WJ = q(W^2 - 1) + (W - pq)X + (1 - Wpq)Y$. Substitution in (C.20) gives

$$\frac{H}{2} = L(X^2 + Y^2) - 2pqXY + \Gamma\{WJ - M + (W + pq)X - (pqW + 1)Y\}. \tag{C.23}$$

This is an exact quadratic form on X, M, Y, J, with bounded pure-testing coefficients. Retain the whole C and α positive square terms separately. Scale the four remaining roots as before. Every coefficient of the scaled core has zero residue, because each contains two inverse strong positive square term square roots. Adding the four strong positive square terms consequently produces the strict residue $4(K_1^2 + K_2^2 + K_4^2 + K_5^2)$. Again only the Gram coefficients, not X, Y or the positive square term roots, needed boundedness.

These two state charts are exhaustive when $(W, Z) \neq (0, 0)$: if $W \neq 0$ and Z/W is bounded use the first; otherwise $Z \neq 0$ and W/Z is infinitesimal, giving the second. Normalization and restoration use pure-testing factors. When $W = Z = 0$, (C.20) is just the bounded quadratic form $H/2 = L(X^2 + Y^2) - 2pqXY$; after the same four-root scaling its coefficients have zero residue, so the four retained strong positive square terms again give a strict Gram. This includes the zero state. Splitting a complex state into real and imaginary parts completes the proof. □

Corollary C.6 (The radial source has no missing outer commuting chart)**.** *Use the unchanged exact source* (C.3) *under the hypotheses of Theorem C.4: h, r, g, j are positive infinitesimals and $a, 1 - r, t$ are positive bounded units. Then the full parent is tensor SOS at every commuting residue test and every real or complex state. Together with the physical-source noncommuting-residue theorem, this proves all tests for this exact radial source family.*

Proof. The source identities and positive decomposition of N proved in Section C.2.3 give

$$N \asymp h + r + g + j, \qquad U_1 = re_1,\ U_2 = ge_2,\ U_4 = he_4,\ U_5 = je_5 > 0 \quad \text{bounded units},$$

where $N/(h+r+g+j)$ has residue 1. Moreover, directly from (C.3),

$$\frac{C}{N} = \frac{A_c}{2a^2 t\Delta} > 0 \quad \text{is a bounded unit.}$$

For example,

$$\frac{1}{Ce_1} = \frac{r/N}{(C/N)U_1},$$

and the three analogous formulas use $g/N, h/N, j/N$. Thus all four product reciprocals in Lemma C.5 are bounded, while C is positive infinitesimal. That lemma covers every outer commuting chart, and Theorem C.4 covers the zero-zero chart.

The final noncommuting join uses only the already-established physicality of the exact source and the physical-source noncommuting-residue theorem. In particular the exact contact and Laurent-determinant physicality argument used for the unit-N source requires $N > 0$, not that N have positive residue. All five event weights are positive. On the three-dimensional state subspace $Z = 0$ the signed core is nonnegative, and the positive event positive square terms have zero common kernel at the noncommuting contact; hence the state restriction is positive definite. The exact stationary state is a nonzero kernel vector, so the other three state eigenvalues are positive. Determinant stationarity remains exact, and the same Laurent-determinant argument proves physicality. No coefficient-entry claim is made here for C2 sources outside the stated radial family. □

C.3 Strong sources with two small actual angles

The following full-parent polynomial identities and normal divisions are also used by the C1 coefficient domains. Their use there requires the C1 source bounds and coefficient identities proved in the C1 chapter; it does not use a C2 source conclusion with altered coefficients.

C.3.1 Local strong-II Gram

Theorem C.7 (Strong-II local allocation)**.** *Let the positive C2 source satisfy $z, y \in \mathrm{Inf}_{>0}$ and let $B = H + L$ be a positive unit. Keep the exact source coefficients of (GE1), with $n \geq 0$ and the remaining displayed weights positive. Then the original C2 parent belongs to $\sum (K_c \otimes_{\mathbb{R}} K_t)^2$ whenever both independent affine testing coordinates u, v are infinitesimal, including either literal testing axis. There is no comparison hypothesis between u^2, z, between v^2, y, or between the two source rates. All openings of r, L/y, $1 + zr - L/y$, and $1 - H$ are retained, as is literal $n = 0$.*

This is the local full-parent calculation. Section C.3.2 proves the other testing charts and the literal projective endpoints. All identities below are polynomial in testing coordinates. Inverses are taken only in the pure source field, and the final step uses the bounded Gram strict-real-residue lemma already used in the completed local UI certificate. In particular no ordering or inverse of the mixed tensor is used.

Source units and notation. Write $U = u^2, V = v^2, k = \sqrt{y}$ and

$$a_c = 1 + zr, \quad t = L/y, \quad j = a_c - t, \quad s = a_c/B, \quad c = CL, \quad d = Dyj.$$

The source entry gives bounded t, j, r, positive units $B, c, d, m/z^2, A/y$, bounded nr and $\alpha(1-H)$, and strong α. These facts include the arbitrary openings $t \to 0$, $j \to 0$, $r \to 0$, and $1 - H \to 0$. They follow directly from the positive source formulas and were proved in (DI7). Let $h = \mathrm{st}\, B$; then $0 < h \leq 1$ and, with $r_0 = \mathrm{st}\, r$,

$$\begin{gathered} c_0 = d_0^{\mathrm{pair}} = \frac{h}{2(1+h)}, \quad \mathrm{st}(m/z^2) = \frac{1 - r_0}{2h^2(1+h)}, \quad \mathrm{st}(A/y) = \frac{1}{2h(1+h)}, \\ \mathrm{st}(nr) = \frac{1}{2(1+h)} - \frac{r_0}{2}, \qquad \mathrm{st}\{\alpha(1-H)\} = \frac{h^2}{2(1+h)}. \end{gathered} \tag{LII1}$$

These residues follow directly from (GE1): $L \to 0$, $H \to h$, $a_c \to 1$, $G \to h$, $\delta \to h(1+h)$, and the source numerator $M \to h^2$. Substitution in the six rational coefficient formulas gives (LII1), including the identity $\mathrm{st}(nr) = 1/[2(1+h)] - r_0/2$ without division by n. Define the source quantities (with q, g bounded)

$$\ell = y(C+D) = c/t + d/j, \qquad q = \frac{D}{C+D} = \frac{dt}{cj+dt}, \qquad g = \frac{yCD}{C+D} = \frac{cd}{cj+dt}.$$

Here $0 < q < 1$, g is a positive unit with residue $h/[2(1+h)]$, and ℓ has a positive real lower bound, although it may be unbounded. Indeed $cj + dt$ is a unit because $t + j = a_c$ is a unit. Choose positive real $\ell_* \leq \ell$ and $\alpha_* \leq \alpha$.

The state roots and the exact C,D completion. Use the source-fixed parity graph

$$X=-\frac{zrv}{B}W+\xi,\qquad Y=\frac{Hu}{B}W+\eta,\qquad Z=\frac{uv}{B}W+\zeta. \tag{LII2}$$

It agrees with the actual kernel at $U=z, V=y$ and is a change of all four homogeneous state coordinates with determinant one. Set

$$Q_n=u\xi-\zeta,\quad K=ukW,\quad R_h=\frac{vQ_n-su(V-y)W}{k},\qquad N=\frac{\eta}{k}-qR_h. \tag{LII3}$$

Thus $\eta=k(N+qR_h)$. Every root in (LII3) vanishes at the actual contact. If $Q=uW-Y$ and $J=Q+v(uX-Z)$, then

$$Q/k=(t/B)K-N-qR_h,\qquad J/k=(1-q)R_h-N-(j/B)K.$$

Consequently the exact weighted completion is

$$\begin{aligned}4CQ^2+4DJ^2&=4\ell N^2+4gR_h^2+\frac{8(d-c)}{B}KN-8gsKR_h+\frac{4(ct+dj)}{B^2}K^2,\\ 4CVQ^2&=4CV\eta^2+\frac{4ct}{B^2}VK^2-\frac{8c}{B}VK(N+qR_h).\end{aligned} \tag{LII4}$$

There is no estimate in these identities. Retain the following four nonnegative, contact-centered positive square terms outside the remainder:

$$4(\ell-\ell_*)N^2,\qquad 4CV\eta^2,\qquad 4n(1+U)(1+V)\xi^2,\qquad 4(\alpha-\alpha_*)(1+U)(v\eta-\zeta)^2. \tag{LII5}$$

Call the exact remainder G. Its coefficients are bounded after writing $\eta=k(N+qR_h)$ and using (LII4). For completeness, its entire formula is

$$\begin{aligned}G&=4\ell_*N^2+4gR_h^2+\frac{8(d-c)}{B}KN-8gsKR_h\\ &\quad+\frac{4(ct+dj)}{B^2}K^2+\frac{4ct}{B^2}VK^2-\frac{8c}{B}VK(N+qR_h)\\ &\quad+4n(1+U)(1+V)\left\{\frac{z^2r^2V}{B^2}W^2-\frac{2zrv}{B}W\xi\right\}\\ &\quad+4\alpha_*(1+U)(v\eta-\zeta)^2+4\alpha(1+U)\left\{\frac{(1-H)^2UV}{B^2}W^2-\frac{2(1-H)uv}{B}W(v\eta-\zeta)\right\}\\ &\quad+4mM^2+4A(1+U)(1+V)Y^2\\ &\quad+2\{J^2+(1+U)(vY-Z)^2+(1+V)(X+uZ)^2-(1+U)(1+V)Z^2\},\end{aligned} \tag{LII6}$$

where $M=vW-X+uvY-uZ$ and (LII2) is used in the last line. Every potentially unbounded original coefficient occurs here only through nr, $\alpha(1-H)$, or the bounded quantities in (LII1). No division by a testing coordinate has been introduced.

One exact gauge removes the dependent-root first normal. In the formal bounded polynomial (LII6), the coefficient of R_hW is $uk\rho(U,V)$ for a bounded source polynomial ρ. Replace this entire term using the exact identity

$$KR_h=uvWQ_n-sU(V-y)W^2. \tag{LII7}$$

Call the resulting polynomial $\widetilde{G}$. It is equal to G on the roots (LII3), and has no R_hW term. This gauge is useful because R_h is a dependent root: it permits the state and angular contact equations to be read as formal coefficient vanishings.

More precisely, by parity write uniquely

$$\widetilde{G}=Q_0(N,\xi,\zeta,R_h)+W\{vP_\xi(U,V)\xi+uvP_\zeta(U,V)\zeta+ukP_N(U,V)N\}+\phi(U,V)W^2, \tag{LII8}$$

where all coefficients are bounded source polynomials and Q_0 is quadratic in the four displayed roots. Formula (LII6), followed by the one replacement (LII7), is an exact coefficient-extraction specification for every entry in (LII8).

The retained positive square terms (LII5) vanish together with their first jets at the actual contact. The exact original source state equations thus give

$$P_\xi(z,y)=P_\zeta(z,y)=P_N(z,y)=0. \tag{LII9}$$

Indeed the physical η derivative first forces the N coefficient to vanish; the physical ξ, ζ derivatives then force their formal coefficients to vanish because the $R_h W$ coefficient was removed. The exact original contact and the two angular derivatives now give

$$\phi(z, y) = \phi_U(z, y) = \phi_V(z, y) = 0. \tag{LII10}$$

This uses the six exact original source substitutions already checked in R6. It imposes no new relation between source and testing fields.

Bounded division and the complete angular allocation. There are two additional elementary support facts in (LII8):

$$P_\xi \in (U, z), \qquad \deg_U \phi \le 2, \qquad \phi(0, V) = V(d_s + e_s V), \tag{LII11}$$

where

$$d_s = \frac{4}{B^2}\{m(B + zr)^2 + (n + \tfrac{1}{2})z^2 r^2\}, \qquad e_s = \frac{4}{B^2}(n + \tfrac{1}{2})z^2 r^2.$$

Both d_s/z^2 and e_s/z^2 are bounded. The first support assertion is literal bounded-coefficient membership: setting $U = z = 0$ in the extracted polynomial gives zero. It is also clear before expansion: all ξW terms not supplied by the gauge contain U, m, nzr, or zr; those supplied by the gauge contain U.

Monic division using (LII9) therefore gives bounded polynomials

$$\begin{aligned} P_\xi &= (U - z)A_\xi + z(V - y)B_\xi, \\ P_\zeta &= (U - z)A_\zeta + (V - y)B_\zeta, \\ P_N &= (U - z)A_N + (V - y)B_N. \end{aligned} \tag{LII12}$$

For the first line, evaluate P_ξ at $U = z$: the support assertion supplies the factor z with bounded quotient, and its value at $V = y$ is zero. All other divisions are by the monic polynomials $U - z$ or $V - y$. In particular none creates an inverse of a small source or a testing difference.

The scalar has the exact bounded decomposition

$$\phi = \frac{d_s + e_s V}{z^2} V(U - z)^2 + U(U - z)(V - y)A_s(V) + U(V - y)^2 C_s(V). \tag{LII13}$$

Here is a constructive proof, including boundedness. Write $\phi = a(V)U^2 + b(V)U + V(d_s + e_s V)$ and subtract the first term of (LII13). The remaining U^2, U coefficients a_1, b_1 vanish at $V = y$ by the value and U derivative in (LII10). Divide them by $V - y$ to obtain A_s, B_s. The V derivative gives $zA_s(y) + B_s(y) = 0$, so set $C_s = (B_s + zA_s)/(V - y)$. All three divisions are monic divisions of bounded polynomials. This proves (LII13) exactly, not asymptotically.

Introduce the two legal angular roots

$$R = v(U - z)W, \qquad T = uk(U - z)W. \tag{LII14}$$

The following identities allocate every term in (LII12)–(LII13):

$$\begin{aligned} zv(V - y)W &= \frac{uVQ_n - uvkR_h}{s} - (V - y)R, \\ uv(V - y)W &= \frac{VQ_n - vkR_h}{s}, \\ uk(V - y)W &= \frac{vkQ_n - yR_h}{s}, \\ U(U - z)(V - y)W^2 &= -\frac{R_h T - uRQ_n}{s}, \\ U(V - y)^2 W^2 &= \frac{VQ_n^2 - 2vkR_h Q_n + yR_h^2}{s^2}. \end{aligned} \tag{LII15}$$

Together with $v(U - z)W = R$, $uv(U - z)W = uR$, and $uk(U - z)W = T$, these are an explicit complete allocation of the full remainder as a bounded symmetric Gram in

$$(N, \xi, \zeta, R_h, R, T). \tag{LII16}$$

The original first state normals are all present. The scalar restriction has not been used as a substitute for them.

The finite block and the missing pivot. The residue of Q_0 on the first four roots is

$$\operatorname{diag}(4\ell_*, 2, 4\alpha_*, 4g_0). \tag{LII17}$$

For example the signed core at $W = 0$ is exactly $2(\xi + u\zeta - uv\eta)^2 + 2(1+U)V\xi^2 + 2(1+V)\eta^2$; all its other entries vanish in the present residue. The extra terms in the first three lines of (LII15) have infinitesimal Gram coefficients except for $A_\xi \xi R$ and the allowed $A_N NT$. Likewise the last line has only infinitesimal coefficients, and the fourth line has one potentially finite $R_h T$ entry. Thus, apart from entries in the last row and column T, the sole finite off-diagonal entry is between ξ and R.

The exact coefficient extraction yields

$$\operatorname{st} A_\xi = \frac{4}{h(1+h)}, \qquad \operatorname{st}(d_s/z^2) = \frac{2}{h^2(1+h)}. \tag{LII18}$$

For a short independent check of the first value, the unscaled original first-normal coefficient on the $V = y$ slice is

$$-4nzr(1+y) + 4m(1-H) - (4D+2)yj + 2(1+y) \longrightarrow 2 - 4d_0^{\text{pair}} = \frac{2}{1+h}.$$

Its full multiplier is $2/B$. Passing from the original normal to (LII8) changes this U-quotient only by a bounded multiple of U or z, so its residue is unchanged. The second value follows from (LII1):

$$\operatorname{st}\{m(B+zr)^2/z^2 + (n+\tfrac{1}{2})r^2\} = \frac{1-r_0}{2(1+h)} + \frac{r_0}{2(1+h)} = \frac{1}{2(1+h)}.$$

Hence the finite (ξ, R) Gram block is exactly

$$\begin{pmatrix} 2 & \dfrac{2}{h(1+h)} \\ \dfrac{2}{h(1+h)} & \dfrac{2}{h^2(1+h)} \end{pmatrix}, \qquad \text{Schur complement} = \frac{2}{h(1+h)^2} > 0. \tag{LII19}$$

This calculation retains $r_0 = 0$, $n = 0$, and $h = 1$.

Finally the exact toric conductor is the zero identity

$$\boxed{T^2 - UR^2 - \frac{U-z}{s}R_hT + \frac{u(U-z)}{s}RQ_n = 0.} \tag{LII20}$$

Add any positive real multiple Λ of this zero identity to the Gram. Its only nonzero residue contribution is ΛT^2. The five-by-five block excluding T is strictly positive by (LII17)–(LII19); all entries incident to T are bounded. Choose a sufficiently large real Λ by the ordinary real Schur complement. The entire six-by-six residue Gram is then strictly positive. The bounded Gram lemma gives a tensor SOS for G, and adding the four positive square terms (LII5) proves the theorem.

Quantifiers and endpoint discipline. No proof step divides by u, v, $U - z$, or $V - y$: parity quotients and monic polynomial divisions are coefficient extractions whose final identities are polynomial. Thus the two literal affine axes are included. The source inverses $1/B$, $1/k$, and the source positive square term square roots are legal even when their values are large; all coefficients of the final Gram are bounded. The source ratios $t, j, r, 1-H$ have not been bounded away from zero. The new roots raise testing degree, so the local certificate is not homogenized to assert the original literal projective endpoints. Those are joined by the separate direct-parent remote certificate.

Source input. Source contact and stationarity follow from the native-entry identification in Chapter 5 and the exact source formulas in Section C.1.2. The scaled bounds used above are the II specialization of GE1–GE2 in Section C.6.4, recorded there as DI7.

C.3.2 Complementary strong-II tests

Use the positive C2 source with z, y infinitesimal and $B = H + L$ a positive unit. We prove the full parent for every testing pair with at least one noninfinitesimal angle, including literal axes and projective infinities. The double-small testing chart was treated above.

Source constants, including all openings. Put $t = L/y$, $j = 1+zr-t$, and write $h = \operatorname{res} H$. Then $0 < h \le 1$, $t + j = 1 + zr \sim 1$, and the exact source formulas give

$$Cyt \sim Dyj \sim \frac{h}{2(1+h)}, \qquad \frac{A}{y} \sim \frac{1}{2h(1+h)}.$$

Consequently C, D are unbounded, and the positive pure-source quantity

$$\gamma = \frac{yCD}{C+D} = \frac{(Cyt)(Dyj)}{(Cyt)j + (Dyj)t} \sim \frac{h}{2(1+h)}. \tag{RR1}$$

These statements are uniform when either t or j tends to zero. If $h < 1$, then

$$\alpha \sim \frac{h^2}{2(1-h^2)};$$

if $h = 1$, then α is unbounded. Thus, for every real $0 < h_0 < h$, the positive real number

$$\beta_0 = \frac{h_0^2}{2(1-h_0^2)} < \alpha \tag{RR2}$$

is an admissible cap. No lower bound on $r, j, t, 1-H$ is imposed, and $n = 0$ is retained.

Homogeneous testing coordinates. For finite u, v define

$$a = \frac{u}{\sqrt{1+u^2}}, \quad b = \frac{1}{\sqrt{1+u^2}}, \qquad c = \frac{v}{\sqrt{1+v^2}}, \quad d = \frac{1}{\sqrt{1+v^2}}.$$

Use the same homogeneous formulas at infinity, with $(a, b) = (1, 0)$ or $(c, d) = (1, 0)$. These are the original parent's homogeneous endpoint values. In particular no raised-degree graph is being specialized at an endpoint. Put

$$E = aW - bY, \quad K = aX - bZ, \quad L = bX + aZ, \quad J = dE + cK, \quad Q = cY - dZ.$$

The original parent divided by $4(1+u^2)(1+v^2)$ is exactly

$$nX^2 + m\{bcW - bdX + acY - adZ\}^2 + CE^2 + AY^2 + (D + \tfrac{1}{2})J^2 + (\alpha + \tfrac{1}{2})Q^2 + \tfrac{1}{2}(L^2 - Z^2). \tag{RR3}$$

At projective endpoints this equation means the original homogeneous form, not a division by an infinite number. All radicals and normalizations used here belong to a single testing field.

Proposition C.8 (All tests with v not infinitesimal)**.** *For every such v, including $v = \infty$, and every u, including $u = 0, \infty$, the original full parent is tensor-SOS.*

Proof. Here c is a unit. Set $P = bL$, $R_1 = \sqrt{C}E$, and $R_2 = \sqrt{D}J$. Orthogonality gives

$$L^2 - Z^2 = P^2 + 2aKP - b^2K^2, \qquad K = \frac{R_2}{c\sqrt{D}} - \frac{dR_1}{c\sqrt{C}}. \tag{RR4}$$

Both coefficients on the right are infinitesimal bounded tensors: $1/c$ is bounded in the testing field, while $1/\sqrt{C}$ and $1/\sqrt{D}$ are infinitesimal in the source field. Therefore

$$R_1^2 + R_2^2 + \tfrac{1}{2}P^2 + aKP - \tfrac{1}{2}b^2K^2$$

has a bounded Gram with residue $\operatorname{diag}(1, 1, 1/2)$. It is tensor-SOS by the strict bounded-Gram lemma. Every other term of (RR3), namely the retained n, m, A, α positive square terms and $J^2/2 + Q^2/2$, is already SOS. This uses all four state coordinates. □

An exact harmonic identity for the remaining remote strip. Now let v be infinitesimal, possibly zero, and let u not be infinitesimal, possibly infinity. Thus a is a unit. Write $D_v = 1 + v^2$, and put

$$s = C + D, \quad \theta = D/s, \quad N = \sqrt{s}(E + \theta vK), \quad V = \frac{vK}{\sqrt{y}}, \quad q = \sqrt{y}Y.$$

No comparison between v and $\sqrt{y}$ is made. The roots may be unbounded; their Gram coefficients below are bounded. The exact pure-source harmonic identity is

$$CD_vE^2 + D(E + vK)^2 = N^2 + \gamma V^2 + Cv^2E^2. \tag{RR5}$$

Also set $x = bX$, $z_0 = Z$, and $\ell = vY$. The single toric relation needed below is the polynomial identity

$$ax\ell - b^2 z_0 \ell = bVq. \tag{RR6}$$

For a real cap $0 < \beta_0 < \alpha$, the original parent divided by $4(1+u^2)$ is the sum of the explicitly positive square terms

$$\begin{aligned} nD_v X^2 + m\{b(vW - X) + a(vY - Z)\}^2 + N^2 + Cv^2E^2 \\ + \tfrac{1}{2}(E + vK)^2 + (\alpha - \beta_0)(\ell - z_0)^2 \end{aligned} \tag{RR7}$$

and the following exact bounded quadratic form:

$$G = \gamma V^2 + \frac{A}{y} D_v q^2 + (\beta_0 + \tfrac{1}{2})(\ell - z_0)^2 + \frac{D_v}{2}(x^2 + 2axz_0 - b^2 z_0^2). \tag{RR8}$$

This decomposition is valid also at $u = \infty$, by the original homogeneous convention. It has used no state graph or state inverse.

Proposition C.9 (All tests with u not infinitesimal and v infinitesimal)**.** *The full original parent is tensor-SOS for all these tests, including $v = 0$ and $u = \infty$.*

Proof. For a real constant k define

$$M_k = (\beta_0 + \tfrac{1}{2})(\ell - z_0)^2 + \tfrac{1}{2}(x^2 + 2axz_0 - b^2 z_0^2) + k(ax\ell - b^2 z_0 \ell).$$

By (RR6), the exact form (RR8) equals

$$\boxed{G = D_v M_k + \gamma V^2 + \frac{A}{y} D_v q^2 - k D_v bVq - v^2(\beta_0 + \tfrac{1}{2})(\ell - z_0)^2.} \tag{RR9}$$

This is a full-parent identity; every state-normal term is present.

First suppose b has positive residue b_0. Since a is a unit, $0 < b_0 < 1$. Choose $0 < h_0 < h$ sufficiently close to h that

$$b_0(1 + h - h_0) < 1,$$

and take the admissible cap (RR2). There exists a positive real κ satisfying

$$\frac{1}{1 + b_0 h_0} < \kappa < \min\left\{\frac{1}{1 - b_0 h_0}, \frac{1}{b_0(1+h)}\right\}. \tag{RR10}$$

Set $k = -\kappa$. The Gram of M_k on (x, z_0, ℓ) has first pivot $1/2$, second Schur pivot β_0, and determinant

$$\det M_k = \frac{2\beta_0 b^2 k^2 - (2\beta_0 + 1)(k+1)^2}{8}. \tag{RR11}$$

Its residue is strictly positive by the first two inequalities in (RR10), because $h_0^2 = 2\beta_0/(2\beta_0 + 1)$. The Gram on (V, q) in (RR9) has positive diagonal residues and determinant residue

$$\frac{1}{4}\left\{\frac{1}{(1+h)^2} - k^2 b_0^2\right\} > 0. \tag{RR12}$$

Thus both blocks have strictly positive residue. The final term in (RR9) is a bounded infinitesimal perturbation on (x, z_0, ℓ), so the combined bounded Gram remains strict and is tensor-SOS.

If b is infinitesimal, including $b = 0$, choose any positive real cap $\beta_0 < \alpha$ and take $k = -1$. There is now an exact three-square identity

$$M_{-1} = \tfrac{1}{2}\{x + a(z_0 - \ell)\}^2 + \beta_0(z_0 - \ell)^2 + \tfrac{1}{2}(b\ell)^2. \tag{RR13}$$

On these three roots and (V, q), (RR9) has positive diagonal residue; the only off-block term is $D_v bVq$, whose coefficient is infinitesimal. The negative perturbation changes only the $(z_0 - \ell)^2$ coefficient by an infinitesimal. Its Gram is again strict. Formula (RR7) restores all retained positive square terms. The argument does not divide by u, v, b, or any state coordinate, so the asserted literal boundaries are included. □

Scope and join. The two propositions cover precisely the complement of the region where both testing angles are infinitesimal, for every strong-II source with B a unit. They preserve independent source and testing fields, arbitrary source openings, testing signs, and literal $n = 0$. Combined with a separately verified full-parent case of that local region, they close the strong II/HH orbit by the exact C2 reflection. They do not close the B-infinitesimal weak face.

C.4 One small actual angle: bounded state scale

C.4.1 A lower bound on the state parameter

This short reduction concerns the same exact rational source, before assuming that b is a unit. Suppose $x > 0$ is infinitesimal, k is a positive bounded unit, $b > 0$ is bounded, $0 < r < 1$, $0 < h < b$, $g = b - 1 + r + xh > 0$, and $n \geq 0$. No further source physicality is needed for the following necessary bound.

All quantities in

$$(b + r)A_c S_c \geq r\Delta$$

are bounded, so taking residues gives

$$r_0\{b_0 + k_0^2(b_0 + 1)\} \leq b_0.$$

The brace is strictly positive even if $b_0 = 0$. On the other hand $g > 0$ gives $r_0 \geq 1 - b_0$. If $b_0 \leq 1$, combining the two inequalities yields

$$(1 - b_0)\{b_0 + k_0^2(b_0 + 1)\} \leq b_0, \quad \text{or equivalently} \quad b_0^2 \geq \frac{k_0^2}{1 + k_0^2}.$$

For $b_0 \geq 1$ this lower bound is automatic. Hence in every case

$$b_0 \geq \frac{k_0}{\sqrt{1 + k_0^2}} > 0.$$

Thus bounded b is necessarily a unit. In particular there is no additional b-infinitesimal source domain in this actual-contact scaling. Combining the two new single-angle certificates, the uncaptured part of this bounded-b entry domain has exactly the joint necessary degeneration $r_0 = g_0 = 0$, hence $b_0 = 1$; the other remaining scale is unbounded b. This is a necessary source reduction, not a certificate for the joint degeneration.

C.4.2 A bounded-shape family at all tests

This calculation gives a source-domain extension of the fixed rational family. Its scalar uniqueness proof is an exact six-term Laurent identity and a three-cycle arithmetic–geometric mean inequality. The full-normal and outer-chart arguments include literal $n = 0$.

Put $x > 0$ infinitesimal in the coefficient field and $\rho = \sqrt{x}$. Choose positive bounded coefficient units k, b, r, h such that

$$1 - r, \quad b - h, \quad g = b - 1 + r + xh, \quad b - k^2 r$$

are positive bounded units. Set

$$c = \rho, \quad a = k/\rho, \quad t = x(b - h)/k^2, \quad j = 1 + xr - t, \quad a_2 = k^2/x, \quad A_c = 1 + xr, \quad S_c = A_c + a_2 xt.$$

Thus $h = b - a_2 t$, $g = (1 + x)b - (1 - r + a_2 xt)$, and $j = 1 + xr - t$. Use the exact original rational source

$$\begin{aligned}
\Delta &= (1 + x)xb^2 + 2x(1 + x)br + (1 + x)(1 + a_2 x)b + x^2(1 - a_2)r^2 + 2x(1 + a_2 x)r + 1 + a_2 x,\\
N &= (1 + x)(b + r)(b + a_2(1 - t)) - a_2(1 + a_2 x)(1 - r)t,\\
n &= \frac{(b + r)A_c S_c - r\Delta}{2r\Delta},\\
m &= \frac{(1 + a_2 x)(1 - r)A_c S_c}{2g\Delta},\\
C &= \frac{A_c N}{2a_2 t\Delta},\\
A &= \frac{a_2 A_c (b + r)S_c}{2h\Delta},\\
D &= \frac{(1 + x)(b + r)(b - a_2 xr)S_c}{2a_2 j\Delta},\\
\alpha &= \frac{x^2(b + r)N}{2(1 - xh)\Delta}.
\end{aligned}$$

Assume $n \geq 0$ and physicality of this canonical source. The latter is the same inherited source-physicality hypothesis used for the general noncommuting-residue exit; no conclusion below is predicated on a new source enumeration. Literal $n = 0$ is allowed.

The parent, with $d = 1$ and $\lambda = \beta = 0$, is

$$F = 4nD_uD_vX^2 + 4mM^2 + 4CD_v(uW - Y)^2 + 4AD_uD_vY^2 + 4DJ^2 + 4\alpha D_u(vY - Z)^2 + H,$$
$$M = vW - X + uvY - uZ, \qquad J = uW + uvX - Y - vZ,$$
$$H/2 = J^2 + D_u(vY - Z)^2 + D_v(X + uZ)^2 - D_uD_vZ^2, \qquad D_u = 1 + u^2, \quad D_v = 1 + v^2.$$

Its exact stationary contact is

$$(u, v; W, X, Y, Z) = (\rho, k; \rho b/k, -\rho r, xh/k, 1).$$

Writing $\kappa = b/k$ and using a subscript zero for residues gives

$$\Delta_0 = (1 + k_0^2)(b_0 + 1), \qquad n_0 = \frac{b_0 + r_0 - r_0\Delta_0}{2r_0\Delta_0} \geq 0,$$
$$m_0 = \frac{1 - r_0}{2(b_0 - 1 + r_0)(b_0 + 1)} > 0,$$
$$C_* := (xC)_0 = \frac{k_0^2(b_0 + r_0)}{2(b_0 - h_0)\Delta_0}, \qquad A_* := (xA)_0 = \frac{k_0^2(b_0 + r_0)}{2h_0\Delta_0},$$
$$D_* := (D/x)_0 = \frac{(b_0 + r_0)(b_0 - k_0^2 r_0)}{2k_0^2\Delta_0}, \qquad E_* := (\alpha/x)_0 = \frac{k_0^2(b_0 + r_0)^2}{2\Delta_0}.$$

All starred quantities are positive. Thus n is bounded, m is a positive unit, C, A have scale x^{-1}, and D, α have scale x.

Theorem C.10 (General bounded-shape single-angle source)**.** *Every source in the displayed domain has tensor SOS for the full original parent at all independent real or complex projective testing states and all projective testing angles. No comparison between a coefficient infinitesimal and a testing infinitesimal is needed. The conclusion includes $n = 0$. Applying the exact C2 reflection gives the corresponding original orientation with weak C and strong α; this is a sufficient source domain, not a global one-weak C2 classification.*

The finite $u = 0$ axis and its full normals. Set $a_0 = 4m_0$, $b_0' = a_0 + 2$, and

$$H_0(v) = (4n_0 + 2)(1 + v^2) + a_0, \quad \eta = h/b, \quad \gamma = -r - \frac{a_0 b - b_0'}{H_0(k)}.$$

Here γ is infinitesimal and $\eta_0 = C_*/(C_* + A_*)$. Use the exact contact graph

$$X_g = \frac{a_0 vW - b_0' uZ}{H_0(v)} + \gamma uZ, \qquad Y_g = \eta uW.$$

The denominator $H_0(v)$ is a pure testing unit and $H_0(k)$ is a pure coefficient unit. Let $N_X = X - X_g$, $N_Y = (Y - Y_g)/\rho$. Writing $f = F|_g = f_{ww}W^2 + f_{wz}WZ + f_{zz}Z^2$, exact monic divisions give

$$f = aP^2 + b_sB^2 + c_sV^2 + eE^2 + g_sBV, \qquad (P, B, V, E) = (uW/\rho, uZ, vW, \rho Z),$$

where

$$a = x(f_{ww} - f_{ww}|_{u=0})/u^2, \qquad b_s = (f_{zz} - f_{zz}|_{u=0})/u^2,$$
$$c_s = f_{ww}|_{u=0}/v^2, \qquad e = f_{zz}|_{u=0}/x, \qquad g_s = f_{wz}/(uv).$$

These displayed quotients are polynomial divisions, not inverses of testing infinitesimals. Parity gives the divisions by u^2, v^2, uv; $f_{zz}|_{u=0} = 4Dv^2 + 4\alpha$ gives the x factor. Only C, A can produce coefficient poles, and they occur in f_{ww} with a factor u^2; its prefactor x removes them.

Put $Q = 4C_*A_*/(C_* + A_*) > 0$. The bounded coefficient residues are

$$q(v) = a_0^{\mathrm{sc}} = QD_v, \qquad b(v) = b_0'\frac{H_0 - a_0 - 2}{H_0},$$
$$\bar{c}(v) = a_0\frac{H_0 - a_0}{H_0}, \qquad g(v) = -2b_0'\frac{H_0 - a_0}{H_0}, \qquad e(v) = 4D_*v^2 + 4E_*.$$

Thus $c(v) := v^2\bar{c}(v)$ and $d(v) := -vg(v)$ give the weighted leading scalar

$$q(v)u^2W^2/x + b(v)u^2Z^2 + c(v)W^2 - d(v)uWZ + e(v)xZ^2.$$

The full expansion is

$$F = f + N_X L_X + N_Y L_Y + H_{XX} N_X^2 + 2H_{XY} N_X N_Y + H_{YY} N_Y^2.$$

Its hard residue is $\operatorname{diag}(H_0(v), 4D_v(C_* + A_*))$. Direct differentiation gives

$$L_X = V\ell_w + B\ell_z, \qquad L_Y = P(B_2 + xe_w) + B(\rho u e_z) + Ee_*,$$

where

$$B_2 = 8D_v\{(xA)\eta D_u - (xC)(1-\eta)\}, \qquad e_* = 8v(D-\alpha).$$

All ℓ_w, ℓ_z, e_w, e_z are bounded; the residues of ℓ_w, ℓ_z, B_2, e_* vanish identically in finite v. For the second identity, after subtracting uWB_2/x from $F_Y|_g$, the remaining W coefficient is ue_w and the remaining Z coefficient is $u^2 e_z + e_*$. The constant signed-core derivatives cancel. This proves full normal restoration wherever the scalar Gram on these four roots is strict.

Global scalar strictness by an exact cycle identity. Let $\nu = 2n_0 + 1 \geq 1$, $\mu = 2m_0 + 1 > 1$, $z = v^2$, $\mathcal{H} = \nu(1+z) + \mu - 1$, $e_2 = 4D_*$ and $e_0 = 4E_*$. Then

$$b = \frac{2\mu(\nu(1+z)-1)}{\mathcal{H}}, \quad c = \frac{2(\mu-1)z\nu(1+z)}{\mathcal{H}}, \quad d = \frac{4\mu v\nu(1+z)}{\mathcal{H}}, \quad e = e_2 z + e_0.$$

For $s, z > 0$ define $\mathcal{D}(s,z) = (qs+c)(bs+e) - d^2 s/4$. The following exact identity is the key:

$$\begin{aligned}\frac{\mathcal{H}\mathcal{D}(s,z)}{(1+z)sz} &= 2Q\mu(\nu-1)s/z + 2Q\mu\nu s + 2(\mu-1)\nu e_2 z/s + 2(\mu-1)\nu e_0/s \\ &\quad + Qe_0(\nu+\mu-1)/z + Qe_2\nu z \\ &\quad + Q\{e_2(\nu+\mu-1) + e_0\nu\} - 4\mu\nu.\end{aligned}$$

Every nonconstant coefficient is positive, except the first may be zero when $n_0 = 0$. Exact contact and stationarity, followed by taking residues, give a stationary zero at $(s,z) = (1, k_0^2)$. Here is an algebraic proof that it is the unique zero on the whole positive quadrant. Rescale $z = k_0^2 t$ and write the Laurent expression as

$$\mathcal{L} = a_\ell s/t + b_\ell s + c_\ell t/s + d_\ell/s + e_\ell/t + f_\ell t - \Sigma, \qquad \Sigma = a_\ell + b_\ell + c_\ell + d_\ell + e_\ell + f_\ell.$$

The constant term has this form by its zero value at $(1,1)$. Stationarity gives $a_\ell - c_\ell = d_\ell - b_\ell = f_\ell - e_\ell =: \tau$. Put $p_\ell = \min(a_\ell, c_\ell) \geq 0$, $q_\ell = \min(b_\ell, d_\ell) > 0$, $r_\ell = \min(e_\ell, f_\ell) > 0$. Exactly,

$$\begin{aligned}\mathcal{L} &= p_\ell(s/t + t/s - 2) + q_\ell(s + 1/s - 2) + r_\ell(t + 1/t - 2) \\ &\quad + \begin{cases}\tau(s/t + 1/s + t - 3), & \tau \geq 0, \\ (-\tau)(t/s + s + 1/t - 3), & \tau \leq 0.\end{cases}\end{aligned}$$

Each bracket is nonnegative by the two- or three-term arithmetic–geometric mean inequality. The two strictly positive pair terms force $s = t = 1$ at equality. They also give a positive definite Hessian there: the other brackets are nonnegative and vanish at that point. Thus both remote uniqueness and local nondegeneracy are proved by the same finite algebraic identity, over any real closed residue field. No inference from a merely local Hessian is being used.

Consequently, for finite nonzero v with $v^2 \neq k_0^2$,

$$|d(v)| < 2\sqrt{b(v)c(v)} + 2\sqrt{q(v)e(v)}.$$

Indeed the minimum of $\mathcal{D}(s,z)/s$ over $s > 0$ is $(\sqrt{bc} + \sqrt{qe})^2 - d^2/4$ and is attained, so positivity is strict. Split the exact cross term using $BV = vPE$ and the pure testing function

$$\theta = \frac{|v|\sqrt{b(v)\bar{c}(v)}}{|v|\sqrt{b(v)\bar{c}(v)} + \sqrt{q(v)e(v)}}.$$

The resulting residue blocks on (B,V) and (P,E) are strictly positive. The denominator is a pure testing unit. At $v_0 = 0$ the same conclusion holds when $n_0 > 0$, with zero cross residues. The case $n_0 = 0, v_0 = 0$ is treated separately below.

The matched finite-angle residue. For $v_0 = k_0$, normalize $Z = 1$ when $Z \neq 0$. If W is infinitesimal, put

$$R = (uW - \kappa x)/\rho, \quad S = u - \kappa^{-1}W, \quad U = \rho(v - k), \quad T = u(v - k).$$

The exact conductors are

$$u^2 - x = uS + \kappa^{-1}\rho R, \qquad W^2 - \kappa^2 x = -\kappa WS + \kappa\rho R, \qquad uW - \kappa x = \rho R.$$

Write $\xi = u^2, \omega = W^2, p = uW$ and

$$f = p^2 Q_b/x + B_b, \qquad Q_b = 4D_v\{(xC)(1 - \eta)^2 + (xA)\eta^2 D_u\}.$$

Here B_b is bounded and polynomial in these three invariants with only the separate unit denominators already specified. Put $G = \kappa^2 x Q_b + B_b$ and take successive monic differences

$$G = G_0(v) + (\xi - x)G_\xi + (\omega - \kappa^2 x)G_\omega + (p - \kappa x)G_p.$$

Exactly,

$$f = R^2 Q_b + \kappa^2 S^2 G_\omega + \rho R\mathcal{A} + uS\mathcal{B} + G_0,$$

where $\mathcal{A} = 2\kappa Q_b + \kappa^{-1}G_\xi + \kappa G_\omega + G_p$ and $\mathcal{B} = G_\xi - \kappa^2 G_\omega$. The contact Jacobian of (R, S) with respect to (u, W) has determinant -2. Exact stationarity therefore gives $\mathcal{A}_c = \mathcal{B}_c = 0$. Cancelling ρ here proves vanishing and does not introduce a Gram coefficient inverse. Monic contact division of these two functions and the conductors above give a bounded Gram on (R, S, U, T). Also $G_0 = xg_0(v)$ with bounded g_0, and exact contact gives a double factor $(v - k)^2$, producing its bounded U^2 coefficient.

The full normals have zero-residue lifts on these roots. For L_X, write $L_X = uA_1 + WB_1 = -\kappa B_1 S + u(A_1 + \kappa B_1)$. Both coefficient functions have identically zero leading part for every v; exact normal contact makes $A_1 + \kappa B_1$ vanish at contact. Monic division gives coefficients with factors $u\rho$, u^2, or uW, and a $(v - k)$ quotient of an identically zero leading function. All have zero residue. For L_Y, write $L_Y = RB_2 + \rho E_2$. Its bounded even E_2 vanishes exactly at contact, and both B_2, E_2 have identically zero leading part. Monic division gives factors $x, \rho u, \rho W$, or the same zero-leading-function quotient. Thus the entire normal restoration is bounded and has zero residue.

The soft residue has the form

$$q(k_0)R^2 + \kappa_0^2 c(k_0)S^2 + A_1' RU + B_1' ST + E_1' U^2.$$

It satisfies

$$E_1' > \frac{(A_1')^2}{4q(k_0)} + \frac{(B_1')^2}{4\kappa_0^2 c(k_0)}.$$

To prove this strict inequality, the leading scalar at $u = \rho s, W = \rho w$ is $\Phi = qs^2 w^2 + bs^2 + cw^2 - dsw + e$. Complete its w square:

$$\Phi = (qs^2 + c)\left(w - \frac{ds}{2(qs^2 + c)}\right)^2 + \frac{\mathcal{D}(s^2, v^2)}{qs^2 + c}.$$

The global Laurent calculation gives a positive definite Hessian of the second term in (s, v) at $(1, k_0)$; both coordinate derivatives are nonzero there. The first square supplies the missing w direction. Hence the full three-variable Hessian is positive definite. On first variations $T = U$; the displayed inequality is exactly its Schur complement after the invertible (R, S) change. Choose a real δ strictly between the second summand and $E_1' - (A_1')^2/(4q(k_0))$. Add δ times the exact zero

$$T^2 - U^2 - ST(v - k) - \kappa^{-1}RU(v - k) = 0.$$

The two blocks (R, U) and (S, T) are now strictly positive, and so is the full six-root residue Gram.

If the ratio W/Z is not infinitesimal, instead normalize $W = 1$; then Z is bounded. The earlier scalar is aP^2 plus a term of positive residue $c(k_0)$; its normal lifts have zero residue on $(P, 1)$. This is strict. The case $Z = 0$ is a literal sum of the event squares and the three positive core squares. The simultaneous sign change $(u, v, X, Y) \mapsto (-u, -v, -X, -Y)$ covers $v_0 = -k_0$.

The double-zero corner when $n_0 = 0$. Use instead the pure coefficient quantities

$$H_c = 4n + 4m + 2, \quad \xi_c = 4m/H_c, \quad \zeta = (4m + 2)/H_c, \quad \eta_c = C/(C + A),$$

and graph $X_g = \xi_c vW - \zeta uZ$, $Y_g = \eta_c uW$. Put $\epsilon = 4\alpha + 4n(4m + 2)/H_c > 0$. Then ϵ is infinitesimal, while n/ϵ and x/ϵ are bounded. Use the hard roots as before and scalar roots

$$P = uW/\rho, \quad V = vW, \quad E = \rho Z, \quad B_\epsilon = \sqrt{\epsilon}\, uZ, \quad T = uvZ.$$

Direct positive square term expansion gives exactly

$$f_{zz} = 4\alpha + \epsilon u^2 + 4n\zeta^2 u^4 + 4Dv^2 + u^2 v^2 B_b,$$

where B_b is bounded with residue 2. Its first four terms give coefficients $4\alpha/x$, $1 + (4n\zeta^2/\epsilon)u^2$, and $(4D/x)v^2$ on E^2, B_ϵ^2, E^2 respectively. The W^2 contribution has bounded coefficients on P^2, V^2 with positive residues Q and $4m_0/(2m_0+1)$. The WZ contribution is $uvWZ$ times a bounded function; use $uvWZ = vPE$ to give it zero cross residue. Thus the five scalar roots have strict diagonal residue $(Q, 4m_0/(2m_0+1), 4E_*, 1, 2)$.

For completeness, L_X has a zero-residue bounded coefficient on V. At $v = 0$ its Z coefficient is exactly $-8n\zeta u^3$; on B_ϵ this is $-8n\zeta u^2/\sqrt{\epsilon}$, which is infinitesimal because n/ϵ is bounded. The remaining Z coefficient has a factor uv^2, giving a zero-residue multiple of T. For L_Y, the large-positive square term coefficient is now exactly $B_2 = 8D_v(xA)\eta_c u^2$; the remaining terms lift as

$$L_Y = P(B_2 + xe_w) + T(\rho u e_z) + E\,8v(D - \alpha),$$

with bounded e_w, e_z. All coefficients have zero residue. The hard residue is $\mathrm{diag}(4m_0 + 2, 4(C_* + A_*))$. This includes literal $n = 0$ without deleting a coordinate.

The remaining commuting charts. The following exact core identities retain all unused event positive square terms. Every retained coefficient is nonnegative; in particular the n positive square term can be retained even when $n = 0$. They prove strict bounded Grams without dividing by a testing infinitesimal.

If u is a finite unit and $v \to 0$, take

$$R = (uW - Y)/\rho, \quad L = Y/\rho, \quad B = X + uZ, \quad V = vZ.$$

Then $M = -B + (\rho v/u)(R + D_u L)$, and exactly

$$\begin{aligned} H/2 = {} & (1 + D_u v^2)B^2 + u^2 D_u V^2 - 2uvD_u BV \\ & + 2\rho R(uvB - D_u V) + xR^2 - 2D_u \rho LV + D_u x v^2 L^2. \end{aligned}$$

Retain n, D, α. The residue on (B, V, R, L) is diagonal with entries $(4m_0 + 2, 2u_0^2 D_{u,0}, 4C_*, 4A_* D_{u,0})$.

If $u \to \infty$ and v is finite, put $p = 1/u$ and use $p^2 F$. Take $R = (W - pY)/\rho$, $L = Y/\rho$, $B = Z + pX$, $V = vX$. The homogenized m amplitude is $-B + \rho p v R + \rho v D_p L$, and the exact core is

$$\begin{aligned} \widehat{H}/2 = {} & B^2 + D_p V^2 + 2\rho D_p RV - 2\rho p v RB + xR^2 \\ & + 2D_p \rho p LV - 2D_p \rho v LB + D_p x v^2 L^2. \end{aligned}$$

Retain n, D, α. The residue on (B, V, R, L) is diagonal with entries $(4m_0 + 2, 2, 4C_* D_{v,0}, 4A_* D_{v,0})$.

If $v \to \infty$ and u is bounded, put $q = 1/v$ and use $q^2 F$. Take $R = (uW - Y)/\rho$, $L = Y/\rho$, $T = uqZ$, $N_1 = \rho Z$. Exactly,

$$\begin{aligned} \widehat{H}/2 &= (D_u + q^2)X^2 + D_u x L^2 + T^2 + 2qXT + q^2 xR^2 + 2uq\rho XR - 2qRN_1 - 2D_u qLN_1, \\ 4D\widehat{J}^2 &= 4(D/x)(-N_1 + \rho uX + xqR)^2. \end{aligned}$$

Retain n, m, α. The residue on (X, R, L, T, N_1) is diagonal with entries

$$(2D_{u,0}, 4C_*, 4A_* D_{u,0}, 2, 4D_*).$$

Finally if both angles are infinite, use $p = 1/u, q = 1/v$ and $p^2 q^2 F$. Set $R = (W - pY)/\rho$, $L = Y/\rho$, $T = qZ$. Then

$$\widehat{H}/2 = (1 + p^2 D_q)X^2 + xq^2 R^2 + xD_p L^2 + T^2 + 2\rho qXR + 2pqXT - 2\rho pRT - 2\rho D_p LT.$$

Retain n, m, D, α. The residue on (X, R, L, T) is diagonal with entries $(2, 4C_*, 4A_*, 2)$.

Exhaustion and independence. These charts cover every commuting angle residue: the whole finite $u = 0$ axis, the finite-unit $v = 0$ axis, either infinite axis, and their intersections. When both angle residues are noncommuting, apply the inherited physical noncommuting-residue theorem at arbitrary coefficient scales. Complex states require no additional chart: if $Z \neq 0$, multiply by a phase and normalize $Z = 1$; the real parent is the sum of its real-state value and its imaginary-state value, the latter having $Z = 0$ and hence being a literal square sum. If $Z = 0$ this is already true of both parts. Homogeneity covers all projective state charts. Roots may be unbounded throughout. All inverses or radicals used in coefficients are pure coefficient or pure testing operations on units, except $\sqrt{\epsilon}$ in roots and its controlled coefficient ratio $n/\sqrt{\epsilon}$. No mixed inverse and no comparison of independent infinitesimals occurs. Applying the strict bounded-tensor Gram lemma completes the proof.

C.4.3 Three arbitrarily small source openings

This theorem removes three unit assumptions from the source of Theorem C.10. Both positive openings h and $b-h$ may approach zero, and r may approach zero, making n arbitrarily large. The conclusions concern the entire original parent and every independent test. Source physicality remains an explicit hypothesis; no new source classification is claimed.

Theorem C.11 (Three-opening single-angle source)**.** *Use exactly the rational C2 source and contact of Section C.4.2. Suppose $x>0$ is infinitesimal, k,b are positive bounded units,*

$$0<r<1, \qquad 0<h<b,$$

and $g=b-1+r+xh$ is a positive bounded unit. Assume $n\geq 0$ and physicality of the original source. There is no lower bound on $r,h,b-h$. Then the entire original parent has tensor SOS at all independent projective measurement pairs and all real or complex projective states. Literal $n=0$ is included.

Write $\rho=\sqrt{x}$, $\eta=h/b$, and retain the source quantities A_c,S_c,Δ from that section.

Entry and the two complementary openings. First the former assumptions that $1-r$ and $b-k^2r$ are units follow from $n\geq 0$. Indeed all quantities in

$$(b+r)A_cS_c\geq r\Delta$$

are bounded, and $\Delta_0=(1+k_0^2)(b_0+1)$. Taking residues yields

$$r_0\leq \frac{b_0}{b_0+k_0^2(b_0+1)}<1, \qquad b_0-k_0^2r_0\geq \frac{b_0^2(1+k_0^2)}{b_0+k_0^2(b_0+1)}>0.$$

Thus m is a positive bounded unit, and $D/x,\alpha/x$ are positive bounded units, regardless of whether r has zero residue.

Set

$$T_C=xC(b-h), \qquad T_A=xAh, \qquad H=x(C+A).$$

The original source formulas, with their apparent opening denominators cancelled, show that T_C,T_A are bounded positive units and have the same residue

$$T_0=\frac{k_0^2(b_0+r_0)}{2\Delta_0}>0.$$

Consequently H is bounded below by a positive real, although it may be unbounded. The four identities

$$\begin{aligned} xC(1-\eta)&=T_C/b, & xA\eta&=T_A/b,\\ xC(1-\eta)^2&=T_C(b-h)/b^2, & xA\eta^2&=T_Ah/b^2 \end{aligned} \tag{CO1}$$

are the reason no new scalar singularity appears when either opening collapses. In particular their last two terms add to a bounded unit, with residue T_0/b_0. The effective scalar coefficient is

$$Q=4T_0/b_0>0. \tag{CO2}$$

It is independent of the limiting value of h.

A bounded n: restoration with the normalized hard root. If r is a unit, the exact formula for n makes n bounded. Use the contact graph $X_g,Y_g=\eta uW$ of the prior theorem. All of its scalar coefficients remain bounded: the only potentially unbounded weights are C,A, and on the graph they occur solely through the last two products in (CO1), divided by x and multiplied by u^2W^2. Its leading scalar is exactly the scalar in that theorem with Q replaced by (CO2). The quantities n_0,m_0,D_*,E_* have their former values. Hence the entire six-term Laurent identity, its balanced AM–GM decomposition, global equality characterization, and positive contact Hessian apply without change.

Replace the old hard root $N_Y=(Y-Y_g)/\rho$ by

$$\widehat{N}_Y=\sqrt{H}\,N_Y=\sqrt{C+A}(Y-Y_g). \tag{CO3}$$

Its exact hard coefficient from the C,A positive square terms is

$$4D_v\frac{C+AD_u}{C+A}=4D_v\left(1+\frac{A}{C+A}u^2\right), \tag{CO4}$$

which is bounded and has residue $4D_{v,0}$ on the finite $u=0$ residue axis. The other positive square term contributions to this coefficient and to the mixed hard entry are bounded: $1/(C+A)=x/H$ and $1/\sqrt{C+A}=\rho/\sqrt{H}$ are bounded and infinitesimal.

For clarity the complete first-normal verification is also retained. The previous unnormalized Y derivative has exactly the form

$$L_Y = P(B_2 + xe_w) + B(\rho u e_z) + Ee_*, \qquad B_2 = 8D_v\{(xA)\eta D_u - (xC)(1-\eta)\}, \quad e_* = 8v(D-\alpha). \tag{CO5}$$

Here $P=uW/\rho$, $B=uZ$, $E=\rho Z$. The functions e_w, e_z come only from the bounded n, m, D, α positive square terms and the signed core, so remain bounded. Equation (CO1) makes B_2 bounded with identically zero residue for every finite v on $u=0$; e_* has the same property. The coefficient of $\widehat{N}_Y$ is exactly $L_Y/\sqrt{H}$, so its bounded zero-residue lifts are preserved. The X derivative contains neither C nor A and is unchanged. Thus all four-root remote certificates restore both full normals.

For matched finite-angle residues the scalar has exactly the old form $f = p^2Q_b/x + B_b$, with

$$Q_b = 4D_v\{(xC)(1-\eta)^2 + (xA)\eta^2 D_u\}.$$

Both Q_b and B_b have bounded coefficients by (CO1). The monic divisions in u^2-x, $W^2-\kappa^2 x$, $uW-\kappa x$, and $v-k$, where $\kappa=b/k$, therefore give precisely the same bounded Gram on R, S, U, T. Its residue is unchanged. The old first-normal contact divisions use only bounded B_2, e_w, e_z and exact stationary contact; multiplication by the pure coefficient $1/\sqrt{H}$ preserves their zero-residue lifts. Hence the complete six-root Gram has strictly positive residue.

If $n_0=0$ at the double-zero test, use the previous exact graph with $\eta_c = C/(C+A)$. Now

$$xC(1-\eta_c) = xA\eta_c = \frac{xCA}{C+A}, \qquad \left(\frac{xCA}{C+A}\right)_0 = T_0/b_0. \tag{CO6}$$

These quantities are bounded units even when H is unbounded. Thus the exact ϵ reserve and five scalar roots in that proof remain valid. Its Y normal is again divided by $\sqrt{H}$, and its hard Y residue is now 4 rather than $4(C_*+A_*)$. This proves the literal $n=0$ case as well as its infinitesimal neighborhood without deleting a state coordinate.

An infinitesimal r: the opposite event can be unbounded. Suppose now r is infinitesimal. Let $N=4n+2$. The exact source formula gives

$$Nr = \frac{2(b+r)A_cS_c}{\Delta}. \tag{CO7}$$

Thus Nr is a bounded positive unit, N is unbounded, and both nr and nr^2 are bounded. Use the exact-contact graph

$$X_g = -ruZ, \qquad Y_g = \eta uW, \qquad \widehat{N}_X = \sqrt{N}(X + ruZ), \quad \widehat{N}_Y = \sqrt{C+A}(Y - \eta uW). \tag{CO8}$$

At the original contact $u=\rho, v=k, W=\rho b/k, Z=1$ this graph has exactly the required X, Y values. The potentially large n positive square term contributes $4nr^2D_uD_vu^2Z^2$ to the graph scalar, and $-8nrD_uD_vuZ$ to its unscaled X derivative. Equation (CO7) therefore controls the entire scalar and every full first normal, not only their values at contact.

The hard residue on $(\widehat{N}_X, \widehat{N}_Y)$ is

$$\operatorname{diag}(D_{v,0}, 4D_{v,0}). \tag{CO9}$$

Indeed $H_{XX}/N = D_uD_v + O(1/N)$, since the remaining source weights that enter H_{XX} are m, D; (CO4) controls H_{YY}. The mixed entry is bounded times $1/(\sqrt{N}\sqrt{C+A})$ and has zero residue.

The five exact scalar monic divisions on P, B, V, E from the previous proof still apply. Their residues, with $z=v^2$, are now

$$q = Q(1+z), \quad b_s = 2\mu, \quad c_s = 2(\mu-1)z, \quad d_s = 4\mu v, \quad e = E_2z + E_0, \tag{CO10}$$

where $\mu = 2m_0+1 > 1$, $E_2 = 4(D/x)_0 > 0$, $E_0 = 4(\alpha/x)_0 > 0$, and Q is given by (CO2). Define $\mathcal{D}(s,z) = (qs+c_s)(b_ss+e) - d_s^2 s/4$. Exactly,

$$\begin{aligned}\frac{\mathcal{D}(s,z)}{sz} &= 2Q\mu s/z + 2Q\mu s + 2(\mu-1)E_2z/s + 2(\mu-1)E_0/s \\ &\quad + QE_0/z + QE_2z + Q(E_0+E_2) - 4\mu.\end{aligned} \tag{CO11}$$

All six nonconstant Laurent coefficients are positive. Exact stationary contact, applied after the bounded graph substitution $u=\rho s_1, W=\rho w_1$, gives a stationary zero at $(s,z)=(1,k_0^2)$. Alternatively this follows by direct substitution of

$$m_0 = \frac{1}{2(b_0-1)(b_0+1)}, \quad Q = \frac{2k_0^2}{\Delta_0}, \quad E_2 = \frac{2b_0^2}{k_0^2\Delta_0}, \quad E_0 = \frac{2k_0^2b_0^2}{\Delta_0}.$$

Here $b_0 > 1$ follows from $g_0 = b_0 - 1 > 0$. Rescale $z = k_0^2 t$ and use exactly the balanced reciprocal-pair and three-cycle identity of the previous theorem. Its $s + 1/s - 2$ and $t + 1/t - 2$ coefficients are strictly positive. Thus (CO11) has a unique zero in the positive quadrant and a positive definite Hessian there. This proves remote scalar strictness and the matched-contact Schur inequality, with no appeal to numerical evidence.

For the full X normal, direct differentiation and parity give

$$F_X|_g = V A_X + B B_X,$$

where the complete coefficients are

$$\begin{aligned} A_X &= -8m(1+\eta u^2) + (8D+4)(1-\eta)u^2, \\ B_X &= -8nrD_uD_v + 8m(1-r) - (8D+4)v^2(1+ru^2) + 4D_v(1-r). \end{aligned}$$

These are bounded polynomials in the bounded testing coordinates. Consequently the coefficient of $\widehat{N}_X$ has bounded zero-residue coefficients on V, B because $1/\sqrt{N}$ is infinitesimal. The full Y normal is (CO5), with the bounded graph (CO8); (CO1) gives all of its required bounds and zero residues. Its complete residual coefficients are

$$\begin{aligned} e_w &= 8mv^2(1+\eta u^2) - (8D+4)(1-\eta) + (8\alpha+4)D_u v^2 \eta, \\ e_z &= v\{-8m(1-r) + (8D+4)r - (8\alpha+4)\}. \end{aligned}$$

This restores the remote scalar certificates, including $v_0 = 0$, where (CO10) has strictly positive coefficients on P, B, V, E and zero cross residues. No special $n_0 = 0$ corner occurs in this case.

At a matched residue write, exactly,

$$\frac{F_X|_g}{\sqrt{N}} = uA_1 + WB_1 = -\kappa B_1 S + u(A_1 + \kappa B_1).$$

Both A_1, B_1 have identically zero residue for every finite v; their finite monic coquotients also retain the factor $1/\sqrt{N}$. Exact normal contact makes $A_1 + \kappa B_1$ vanish at the source contact. The same four conductors R, S, U, T therefore give bounded zero-residue coefficients, exactly as in the bounded-n argument. For the Y normal the previous division $L_Y = RB_2 + \rho E_2'$ has bounded B_2, E_2' with identically zero leading functions; division by $\sqrt{H}$ preserves those properties. All quadratic normal coefficients are already bounded by (CO4),(CO7),(CO9). Thus the matched-contact Gram for the entire parent is strictly positive in residue. The other state charts and the simultaneous sign copy use precisely the previous proof.

Every other commuting test. Choose the pure coefficient

$$\sigma = \min(1, xC, xA).$$

It is a positive bounded unit, since T_C, T_A are units while $h, b - h$ are bounded. Retain the nonnegative excess positive square terms

$$4(C - \sigma/x)D_v(uW - Y)^2, \qquad 4(A - \sigma/x)D_u D_v Y^2,$$

and use the four outer core identities in the previous theorem with $C = A = \sigma/x$. These identities retain the entire n positive square term; they never use boundedness of n. Their displayed hard entries $4C_*, 4A_*$ are simply replaced by $4\sigma_0$, while $m_0 > 0$ and $(D/x)_0 > 0$ are unchanged. Every residue Gram stays strictly positive. This covers the finite-unit u axis, both reciprocal axes and their double corners, including literal projective ends.

The finite $u = 0$ axis, those outer axes, and the inherited theorem for noncommuting-residue tests exhaust all independent measurements. Restore the retained positive square terms and positive homogeneous factors, and split complex states into real and imaginary parts. This proves Theorem C.11. In particular the newly proved source scales permit either A or C to be arbitrarily larger than x^{-1}, independently of an arbitrarily large n. They cannot both have unbounded ratios to this scale because $h + (b - h) = b$ is a unit. There is no prescribed comparison between these coefficient scales and any testing infinitesimal.

C.4.4 The remaining opening with bounded opposite event

Theorem C.12 (The unbounded-m opening with bounded n)**.** *In Theorem C.11, replace the assumption that $g = b - 1 + r + xh$ is a positive bounded unit by $g > 0$ and the assumption that at least one of r, g is a bounded unit. The all-tests tensor-SOS conclusion still holds.*

The only new case is infinitesimal g; it is automatically bounded. Our hypothesis now makes r a unit. Since $b_0 + r_0 = 1$ and b is a positive unit, $r_0 < 1$. Thus n is bounded, $0 < b_0 < 1$, and $1 - r$ is a unit. The source formulas imply

$$mg = \frac{(1+k^2)(1-r)A_c S_c}{2\Delta}, \tag{CG1}$$

which is a positive bounded unit. Thus m is unbounded. The two-opening bounds (CO1), positivity of $D/x, \alpha/x$, and the consequence that $b - k^2 r$ is a unit remain valid.

Use the original strong amplitude as the hard normal. Put $\eta = h/b$ and use the exact contact graph

$$Y_g = \eta u W, \qquad X_g = v(1 + \eta u^2)W - (1 + g)uZ. \tag{CG2}$$

At contact this gives $X_g = \rho[b + xh - 1 - g] = -\rho r$. The original m amplitude satisfies exactly $M|_g = guZ$. Normalize the full hard directions by

$$E_M = \sqrt{m}\{X - X_g - uv(Y - Y_g)\}, \qquad E_Y = \sqrt{C + A}(Y - Y_g). \tag{CG3}$$

In these variables

$$M = guZ - E_M/\sqrt{m}, \quad X = X_g + E_M/\sqrt{m} + uvE_Y/\sqrt{C + A}.$$

Consequently the unbounded positive square term has no E_Y contribution at all. This triangular change is essential: scaling X alone would leave the uncontrolled coefficient $\sqrt{m/(C + A)}uv$. The entire scalar contribution of that positive square term is $4mg^2u^2Z^2$, which is bounded by (CG1). Its full X derivative contributes $-8mg\,uZ$, also bounded, before division by $\sqrt{m}$.

All other source weights entering the X row are n, D and the unit signed weight, so the full X derivative on (CG2) has bounded coefficients on $V = vW, B = uZ$. Hence the E_M first normal has zero-residue bounded coefficients on those roots. The E_Y first normal is

$$\frac{(F_Y + uvF_X)|_g}{\sqrt{C + A}}. \tag{CG4}$$

The m positive square term cancels identically in $F_Y + uvF_X$, before any graph substitution. The remaining large terms are exactly the C, A terms controlled by (CO1); all other coefficients are bounded. In particular the old (CO5) form on P, B, E holds with bounded coquotients and identically zero leading corrections. The complete quadratic hard block is bounded and has residue

$$\operatorname{diag}(4, 4D_{v,0}) \tag{CG5}$$

on the finite $u = 0$ axis. The m positive square term gives precisely its first diagonal entry, the C, A positive square terms give the second, and all mixed entries from bounded positive square terms contain $1/(\sqrt{m}\sqrt{C + A})$.

The remaining five Laurent terms already force strictness. The exact scalar divisions on P, B, V, E are bounded for the same reason: C, A enter through (CO1), and m only through mg^2. Their leading weighted scalar is

$$q\,u^2W^2/x + b_s u^2Z^2 + c_s W^2 - d_s uWZ + exZ^2,$$

where, for $z = v^2$ and $\nu = 2n_0 + 1 \geq 1$,

$$\begin{aligned} &q = Q(1 + z), \quad b_s = 2\nu(1 + z) - 2, \quad c_s = 2z\nu(1 + z), \quad d_s = 4v\nu(1 + z), \\ &e = E_2 z + E_0, \qquad Q = 4T_0/b_0, \quad E_2 = 4(D/x)_0 > 0, \quad E_0 = 4(\alpha/x)_0 > 0. \end{aligned} \tag{CG6}$$

Set $\mathcal{D}(s, z) = (qs + c_s)(b_s s + e) - d_s^2 s/4$. Its exact Laurent identity is

$$\begin{aligned} \frac{\mathcal{D}(s, z)}{(1 + z)sz} = {} & 2Q(\nu - 1)s/z + 2Q\nu s + 2\nu E_2 z/s + 2\nu E_0/s \\ & + QE_0/z + QE_2 - 4\nu. \end{aligned} \tag{CG7}$$

Contact and stationarity give a zero at $(s, z) = (1, k_0^2)$. After $z = k_0^2 t$, write the five coefficients in order as a, b, c, d, e. Here $a \geq 0$ and $b, c, d, e > 0$. Stationarity gives $c = a + e$ and $b = d + e$. The zero value fixes the constant, so exactly

$$\mathcal{L} = a(s/t + t/s - 2) + d(s + 1/s - 2) + e(t/s + s + 1/t - 3). \tag{CG8}$$

Every summand is nonnegative by AM–GM. The last bracket alone forces $s = t = 1$ at equality: its three positive terms have product one, and equality makes all three equal. Its exponent vectors $(-1, 1), (1, 0), (0, -1)$ span two dimensions; equivalently its Hessian in $\log s, \log t$ at $(1, 1)$ is $\left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$. Hence it already has a positive definite contact Hessian. This proves global uniqueness and local nondegeneracy even when $n_0 = 0$ and the first pair is absent.

For remote finite nonzero v the same scalar cross-term split as before gives a strict Gram. The case $v_0 = 0, n_0 > 0$ is also strict on P, B, V, E. The full normals are restored by (CG3)–(CG5). At matched finite-angle residues the

whole scalar still has the form $p^2Q_b/x + B_b$ with bounded Q_b, B_b. The four monic conductors R, S, U, T of the prior theorem therefore give a bounded Gram, and (CG8) proves its strict Schur inequality.

The full first normals have the required same conductor lifts. For E_M, their coefficient is $(uA_1 + WB_1)/\sqrt{m}$ with bounded polynomial A_1, B_1. Monic contact division retains the factor $1/\sqrt{m}$ in every coquotient. Exact contact makes the residual coefficient $A_1 + \kappa B_1$ vanish, giving the old four-root lift with zero residue. For E_Y, the cancellation in (CG4) leaves precisely the old large-positive square term term B_2 plus bounded residual terms; its exact contact and identically zero leading part give the same $RB_2 + \rho E_2'$ division, then the bounded factor $1/\sqrt{H}$. These are full normal identities, not just Hessian checks. Quadratic state degree and (CG5) finish the complete matched-contact certificate.

The double-zero test when $n_0 = 0$. The graph (CG2) is replaced in this remote corner. Put

$$H_c = 4m + 4n + 2, \quad \xi = 4m/H_c, \quad \zeta = (4m + 2)/H_c, \quad \eta_c = C/(C + A).$$

Use

$$Y_g = \eta_c uW, \quad X_g = \xi v(1 + \eta_c u^2)W - \zeta uZ,$$

and the triangular normals

$$E_X = \sqrt{H_c}\{X - X_g - uv(Y - Y_g)\}, \qquad E_Y = \sqrt{C + A}(Y - Y_g). \tag{CG9}$$

The entire graph amplitude is

$$M|_g = (1 - \xi)v(1 + \eta_c u^2)W + (\zeta - 1)uZ.$$

Both $m(1 - \xi)$ and $m(\zeta - 1)$ are bounded because n is bounded. Thus the scalar and full first normals have bounded coquotients after the same x scaling as before.

Set $\epsilon = 4\alpha + 4n\zeta > 0$. It is infinitesimal, with $n/\epsilon, x/\epsilon$ bounded. The complete scalar Z^2 coefficient is exactly

$$f_{zz} = 4\alpha + \epsilon u^2 + 4n\zeta^2 u^4 + 4Dv^2 + u^2v^2B_b, \tag{CG10}$$

where B_b is bounded with residue 2. This identity follows directly from the positive square terms and the exact relation $4n\zeta^2 + (4m + 2)(1 - \zeta)^2 = 4n\zeta$. The W^2 coefficients on $P = uW/\rho, V = vW$ have positive residues $Q, 2$. The cross term is $uvWZ$ times a bounded function and can be written as a zero-residue multiple of PE using $uvWZ = vPE$. Thus the five scalar roots

$$P, \quad V, \quad E = \rho Z, \quad B_\epsilon = \sqrt{\epsilon}\, uZ, \quad T = uvZ$$

have strict diagonal residue $(Q, 2, E_0, 1, 2)$.

For the full X derivative, at $v = 0$ the Z coefficient is exactly $-8n\zeta u^3$. After division by $\sqrt{H_c}$ its coefficient on B_ϵ is infinitesimal because $n/\sqrt{\epsilon} = (n/\epsilon)\sqrt{\epsilon}$ is infinitesimal. The remaining Z coefficient has the factor uv^2 and bounded coquotient; its coefficient on T has zero residue. The W coefficient has a factor v and bounded coquotient, multiplied by $1/\sqrt{H_c}$. For E_Y the exact m cancellation in $F_Y + uvF_X$ again applies; (CO6) gives $B_2 = 8D_v(xA)\eta_c u^2$, and the remaining lift is

$$P(B_2 + xe_w) + T(\rho u e_z) + E\, 8v(D - \alpha),$$

divided by $\sqrt{H}$, with bounded e_w, e_z. Every coefficient has zero residue. The hard residue of (CG9) is $\mathrm{diag}(1, 4)$, and all mixed hard entries vanish in residue. This proves the full-parent corner including literal $n = 0$.

All other tests. For the remaining commuting axes use the earlier outer identities, cap C, A at σ/x as before, and also cap m at $\min(m, 1)$. Retain the entire nonnegative excess m positive square term. The selected m weight is a unit and all previous displayed strict residues apply. The noncommuting-residue theorem covers the other tests. Restore all positive factors and split complex states as before. This proves the theorem for arbitrary positive g, including the new unbounded-m regime.

The remaining assumptions in this source family are now only that k, b are units, x is positive infinitesimal, all original contact openings are positive, at least one of r, g is a unit, $n \geq 0$, and the source is physical. The joint degeneration $r, g \to 0$, where both n, m can be unbounded, is not covered. The result does not yet treat unbounded b or a degenerate residue of the finite actual angle k. Section C.4.1 excludes infinitesimal b under these entry hypotheses.

C.4.5 Joint opening of the two event coefficients

Theorem C.13 (Both event openings may close)**.** *Use the exact rational C2 source and original parent of Section C.4.2. Suppose $x > 0$ is infinitesimal, k, b are positive bounded units, $0 < r < 1$, $0 < h < b$, and $g = b - 1 + r + xh > 0$. Suppose r, g are infinitesimal, $n \geq 0$, and the source is physical. Then the entire original parent is tensor SOS at every independent real or complex projective testing state and every independent projective pair of testing angles. No lower bound on either h or $b - h$ is required.*

Together with the three-opening and last-opening theorems this covers the bounded-k, b rational source family without requiring either r or g to be a unit. Nonunit k and unbounded b remain separate source domains.

Source bounds, including both complementary openings. Write $\rho = \sqrt{x}$, $\eta = h/b$, $\kappa = b/k$, $K = k_0 > 0$, and $D_K = 1 + K^2$. Since r, g are infinitesimal, $b_0 = 1$. The source formulas give

$$(nr)_0 = \frac{1}{4D_K}, \qquad (mg)_0 = \frac{1}{4}, \qquad \Delta_0 = 2D_K.$$

Thus n, m are positive unbounded pure coefficient quantities, nr, mg are bounded units, and nr^2, mg^2 are infinitesimal. In particular $m - 1 > 0$. The existing opening cancellation identities, with $T_C = xC(b - h)$ and $T_A = xAh$, give bounded positive units

$$(T_C)_0 = (T_A)_0 = \frac{K^2}{4D_K}, \qquad (D/x)_0 = \frac{1}{4K^2 D_K}, \quad (\alpha/x)_0 = \frac{K^2}{4D_K}.$$

They hold even if h or $b - h$ is infinitesimal. Specifically,

$$xC(1 - \eta) = T_C/b, \quad xA\eta = T_A/b, \quad xC(1 - \eta)^2 = T_C(b - h)/b^2, \quad xA\eta^2 = T_A h/b^2. \tag{JC1}$$

Put $H_c = x(C + A)$. This quantity is bounded below by a positive real, although it can be unbounded. Its reciprocal and reciprocal square root are bounded pure coefficient elements.

Center the excess positive square terms before taking normals. Retain exactly the original amplitudes

$$M = vW - X + uvY - uZ, \qquad J = uW + uvX - Y - vZ, \qquad D_u = 1 + u^2, \quad D_v = 1 + v^2.$$

Let F denote the full original parent, including its signed core H. Define G by the following exact polynomial identity:

$$F = G + 4nD_u D_v (X + ruZ)^2 + 4(m - 1)(M - guZ)^2. \tag{JC2}$$

An explicit expression, requiring no elimination or mixed inverse, is

$$\begin{aligned} G = {} & 4M^2 + 4CD_v(uW - Y)^2 + 4AD_u D_v Y^2 + 4DJ^2 + 4\alpha D_u(vY - Z)^2 + H \\ & - 8nrD_u D_v uZX - 4nr^2 D_u D_v u^2 Z^2 \\ & + 8(m - 1)g\, uZM - 4(m - 1)g^2 u^2 Z^2. \end{aligned} \tag{JC3}$$

Every coefficient in (JC3) except C, A is bounded. At the exact original contact

$$(u, v; W, X, Y, Z) = (\rho, k; \kappa\rho, -r\rho, xh/k, 1)$$

both subtracted squares and all of their first derivatives vanish. Consequently G has exactly the same zero and full stationary contact as F. This statement is essential: merely capping the uncentered original positive square terms would not preserve the contact.

On the finite $u = 0$ residue axis use

$$\begin{aligned} X_g = -ruZ, \qquad Y_g = \eta uW, \qquad E_X = \sqrt{n}(X + ruZ), \\ E_Y = \sqrt{C + A}(Y - \eta uW), \qquad E_M = \sqrt{m - 1}(M - guZ). \end{aligned} \tag{JC4}$$

Expand only G in E_X, E_Y and retain $4E_M^2$ as written. The root E_M is allowed to be redundant. No substitution for W is made. In particular, the positive C positive square term introduces no coefficient of the form Cu^2/n or Cu^2/m.

The complete graph scalar and its global leading inequality. Write $f = G|_g = f_{ww}W^2 + f_{wz}WZ + f_{zz}Z^2$. The same exact monic divisions as in the prior single-angle proof give

$$f = aP^2 + b_sB^2 + c_sV^2 + eE^2 + g_sBV, \qquad (P, B, V, E) = (uW/\rho, uZ, vW, \rho Z), \tag{JC5}$$

where

$$a = x\frac{f_{ww} - f_{ww}|_{u=0}}{u^2}, \quad b_s = \frac{f_{zz} - f_{zz}|_{u=0}}{u^2}, \quad c_s = \frac{f_{ww}|_{u=0}}{v^2}, \quad e = \frac{f_{zz}|_{u=0}}{x}, \quad g_s = \frac{f_{wz}}{uv}.$$

These are polynomial coquotients, not inverses of testing infinitesimals. Their coefficients are bounded by (JC1)–(JC3). The exact constant $f_{zz}|_{u=0} = 4Dv^2 + 4\alpha$ supplies the x factor. For finite v their residues are

$$q(v) = \frac{K^2(1+v^2)}{D_K}, \qquad b_s = 4, \quad c_s = 4, \quad g_s = -10, \qquad e(v) = \frac{v^2/K^2 + K^2}{D_K}. \tag{JC6}$$

Thus the leading scalar, using the exact identity $BV = vPE$, is

$$qP^2 - 2vPE + eE^2 + 4(V - B)^2. \tag{JC7}$$

The exact identity on the bounded full state graph is

$$q(v)e(v) - v^2 = \frac{(v^2 - K^2)^2}{D_K^2}. \tag{JC8}$$

If v is a finite nonzero residue with $v^2 \neq K^2$, then $10|v| < 8|v| + 2\sqrt{qe}$. Splitting the exact $BV = vPE$ cross term with the pure testing function

$$\theta = \frac{4|v|}{4|v| + \sqrt{q(v)e(v)}}$$

gives strictly positive residue blocks on (B, V) and (P, E). At $v_0 = 0$, allocate the entire cross term using $BV = vPE$; (JC5) then has strictly positive diagonal residue $(q(0), 4, 4, e(0))$ and zero cross residues. There is no exceptional $n_0 = 0$ corner in the joint regime.

Here is also a global equality and Hessian check needed at the matched angles. For $z = v^2 > 0$ and $s > 0$, set

$$\mathcal{D}(s, z) = (qs + 4z)(4s + e) - 25zs.$$

Exactly,

$$\begin{aligned} \frac{\mathcal{D}(s,z)}{s} &= 4qs + 4ez/s + qe - 9z \\ &= 4\big(\sqrt{qs} - \sqrt{ez/s}\big)^2 + (\sqrt{qe} - \sqrt{z})(\sqrt{qe} + 9\sqrt{z}). \end{aligned} \tag{JC9}$$

The second term is nonnegative by (JC8); its equality forces $z = K^2$. The first then forces $s = 1$. Moreover the second term is

$$\frac{(z - K^2)^2}{D_K^2}\frac{\sqrt{qe} + 9\sqrt{z}}{\sqrt{qe} + \sqrt{z}},$$

whose quadratic coefficient at contact is strictly positive. The first square supplies a nonzero s derivative there. Hence $\mathcal{D}$ has a positive definite Hessian at $(1, K^2)$. These radicals are used only in the residue-field inequality; no radical of an element mixing coefficient and testing fields is used.

Restore every normal, with bounded coefficients. From (JC2)–(JC4) the entire parent is exactly

$$F = f + E_XL_X + E_YL_Y + H_{XX}E_X^2 + 2H_{XY}E_XE_Y + H_{YY}E_Y^2 + 4E_M^2. \tag{JC10}$$

The first normals are $L_X = G_X|_g/\sqrt{n}$ and $L_Y = G_Y|_g/\sqrt{C + A}$. The only unbounded weights C, A do not enter the X row. Thus

$$L_X = V\ell_w + B\ell_z$$

with bounded coefficient functions carrying the infinitesimal pure coefficient factor $1/\sqrt{n}$. Both have identically zero residue on the entire finite $u = 0$ axis, including $v = 0$.

For the second normal the exact form is

$$L_Y = \frac{1}{\sqrt{H_c}}\left\{P(B_2 + xe_w) + B(\rho ue_z) + Ee_*\right\}, \tag{JC11}$$

where

$$B_2 = 8D_v\{(xA)\eta D_u - (xC)(1 - \eta)\}, \qquad e_* = 8v(D - \alpha).$$

All e_w, e_z are bounded: all new terms in (JC3) have bounded coefficients, and their Y derivative has an additional u^2Z factor. Equation (JC1) makes B_2 bounded with identically zero residue, even at either closing complementary opening. Thus every coefficient in (JC11) has zero residue.

The entire hard block is bounded and has residue

$$\operatorname{diag}(4D_{v,0}, 4D_{v,0}, 4) \quad \text{on } (E_X, E_Y, E_M). \tag{JC12}$$

The first entry comes from the retained centered n positive square term; the bounded G contribution is divided by n. The second is

$$4D_v \frac{C + AD_u}{C + A} = 4D_v \left(1 + \frac{A}{C + A} u^2\right)$$

plus bounded terms divided by $C + A$. The mixed entry contains $1/(\sqrt{n}\sqrt{C + A})$ times a bounded function. The third entry is the unexpanded centered m positive square term. This proves the full-parent assertion at all remote finite-axis tests, by the strict Gram in (JC5) and the zero-residue first normals.

The matched finite-angle residues. At $v_0 = K$, normalize $Z = 1$ and suppose W is infinitesimal. Use exactly the prior contact conductors

$$R = (uW - \kappa x)/\rho, \quad S = u - \kappa^{-1}W, \quad U = \rho(v - k), \quad T = u(v - k),$$

with exact identities

$$u^2 - x = uS + \kappa^{-1}\rho R, \quad W^2 - \kappa^2 x = -\kappa WS + \kappa\rho R, \quad uW - \kappa x = \rho R. \tag{JC13}$$

Here $f = p^2 Q_b/x + B_b$, with bounded B_b and

$$Q_b = 4D_v\{(xC)(1 - \eta)^2 + (xA)\eta^2 D_u\}.$$

Boundedness follows from (JC1); the remaining terms use only the bounded coefficients of (JC3). The full exact stationary contact of G proved after (JC3) supplies every vanishing required by the prior monic-contact-division argument. Successive monic differences in $u^2 - x, W^2 - \kappa^2 x, uW - \kappa x, v - k$ therefore give a bounded Gram on R, S, U, T.

For precision, its strict residue follows from the same explicit Schur-complement step in that argument, now with

$$\Phi = q(v)s^2w^2 + 4s^2 + 4v^2w^2 - 10vsw + e(v).$$

Completing the w square gives

$$\Phi = (qs^2 + 4v^2)\left(w - \frac{5vs}{qs^2 + 4v^2}\right)^2 + \frac{\mathcal{D}(s^2, v^2)}{qs^2 + 4v^2}.$$

Equation (JC9) proves a positive definite three-variable Hessian at $(s, w, v) = (1, 1/K, K)$. Consequently the residue before the conductor-zero correction has form

$$q(K)R^2 + 4S^2 + A_1RU + B_1ST + E_1U^2, \qquad E_1 > A_1^2/(4q(K)) + B_1^2/16.$$

Choose a real δ strictly between $B_1^2/16$ and $E_1 - A_1^2/(4q(K))$, and add δ times the exact zero

$$T^2 - U^2 - ST(v - k) - \kappa^{-1}RU(v - k) = 0.$$

The two residue blocks (R, U) and (S, T) are strictly positive.

An explicit equivalent residue check can be used in place of this Schur choice. The capped current has the exact conductor lift

$$(M - gu)|_g = -v\kappa S + \kappa T + \eta uv\rho R + \eta u\kappa\rho U.$$

Retaining its T^2 term, rather than first moving that term to U^2, gives the scalar residue

$$K^2R^2 + \frac{2(K^2 - 1)}{D_K} RU + \frac{U^2}{K^2} + 4(S - T/K)^2.$$

Adding $2/D_K^2$ times the same conductor zero leaves the (R, U) Schur margin $2/D_K^2 > 0$, and leaves $4(S - T/K)^2 + (2/D_K^2)T^2$ on (S, T), also strictly positive.

The full-normal lifts are also the ones actually required for (JC10), rather than scalar-only lifts. Write $L_X = uA_1' + WB_1' = -\kappa B_1' S + u(A_1' + \kappa B_1')$. Both functions and all their bounded monic coquotients retain $1/\sqrt{n}$. Exact $G_X = 0$ at contact supplies vanishing of $A_1' + \kappa B_1'$. Using (JC13) gives zero-residue coefficients on

R, S, U, T. For (JC11), write its numerator as $RB_2 + \rho E_2$. Both B_2, E_2 have identically zero leading functions and E_2 vanishes at contact. The same monic divisions give factors $x, \rho u, \rho W$, or a zero-leading-function $(v - k)$ coquotient; division by $\sqrt{H_c}$ preserves boundedness and zero residue. Thus the complete seven-root Gram on $(R, S, U, T, E_X, E_Y, E_M)$ is strict in residue.

If W/Z is not infinitesimal, normalize $W = 1$ and keep Z bounded. Then the scalar has a strict residue on $(P, 1)$: its positive constant contribution is $4K^2$. All normalized first normals have zero residue as above. For $Z = 0$, the original parent is already the literal sum of its event positive square terms and three positive core squares. The simultaneous sign copy gives $v_0 = -K$.

Every other independent test. The outer-chart proof need not use the centered decomposition. Use the original parent, cap C, A at σ/x, where $\sigma = \min(1, xC, xA)$ is a positive bounded unit, and cap m at 1. Retain the original nonnegative excess positive square terms and the entire original n positive square term. The four exact outer-core identities in the prior single-angle theorem apply without change: their strict residues require only $\sigma_0 > 0$, the selected $m = 1$, and $(D/x)_0 > 0$. They never bound n. This covers the finite unit u axis, both reciprocal axes, all their intersections, and literal projective ends. The inherited physical noncommuting-residue theorem covers the remaining tests.

These charts and the finite $u = 0$ axis exhaust the projective measurement pairs. Restore the retained squares and homogeneous factors, then apply the real identities to the real and imaginary state parts.

C.5 One small actual angle: unbounded state scale

C.5.1 Source reduction at unbounded state scale

Use the original positive rational source with $d = 1$, actual $u_c = \sqrt{x}$ positive infinitesimal and $v_c = k$ a positive bounded unit. The global entry theorem supplies its positive chart inequalities. The preceding rational theorems, including the joint-opening theorem, now cover every bounded b: the inherited $n \geq 0$ entry inequality makes bounded b a unit. We classify the remaining b unbounded without changing any original coefficient.

Put

$$B = xb, \qquad H = xh, \qquad L = x(b - h) = B - H, \qquad a^2 = k^2/x, \quad t = L/k^2.$$

The original gaps give

$$0 < H < 1, \qquad 0 < L < k^2(1 + xr), \qquad B = H + L.$$

Thus B, H, L are bounded and $x/B = 1/b$ is infinitesimal. Set $A_c = 1 + xr$ and $S = A_c + L$, both positive bounded units. Also

$$g/b = 1 + (r - 1)/b + H/b \longrightarrow 1, \qquad j = 1 + xr - L/k^2 > 0, \quad j \leq 1 + x.$$

The following exact rearrangements of the original short polynomials control every event scale:

$$\begin{aligned} x^2 N &= (1 + x)(B + xr)(H + k^2) - x(1 + k^2)(1 - r)L, \\ \Delta/b &= (1 + x)B + 2x(1 + x)r + (1 + x)(1 + k^2) \\ &\quad + (x^2 - k^2 x)r^2/b + 2x(1 + k^2)r/b + (1 + k^2)/b. \end{aligned}$$

Consequently $xN/b = x^2 N/B$ and Δ/b are positive bounded units, with respective residues

$$H_0 + k_0^2, \qquad B_0 + 1 + k_0^2.$$

Here $H, L \leq B$ controls the divided corrections, and $x/B \to 0$.

The exact physical condition $n \geq 0$ is

$$r(\Delta/b) \leq (1 + r/b)A_c S.$$

It implies

$$r_0 \leq \frac{1 + L_0}{1 + k_0^2 + B_0}, \qquad 1 - r_0 \geq \frac{k_0^2 + H_0}{1 + k_0^2 + B_0} > 0.$$

Thus $1 - r$ is a positive bounded unit without any lower bound on r. The original formulas now give the exact comparabilities

$$m \asymp b^{-2}, \qquad C \asymp L^{-1}, \qquad A \asymp H^{-1}, \qquad D \asymp B/j, \qquad \alpha \asymp B/(1 - H). \tag{SR1}$$

Every omitted factor in $\asymp$ is a positive bounded unit with bounded reciprocal. For instance,

$$CL = \frac{A_c(xN/b)}{2\Delta/b}, \qquad AH = \frac{k^2 A_c(1 + r/b)S}{2\Delta/b}$$

are positive units; the other identities follow using $g/b \to 1$ and $(b - k^2 r)/b \to 1$.

There are exactly two unbounded-b scale leaves:

1. If $B = xb$ is infinitesimal, so are H, L, while $j \to 1$ and $1 - H \to 1$. Thus C, A are unbounded and m, D, α are weak, with $m \asymp b^{-2}$ and $D \asymp \alpha \asymp xb$. This is the one-weak-complementary-event coefficient domain, with weak α and strong C in this actual-angle orientation.

2. If B is a positive bounded unit, then C, A, D, α are all strong and m is weak. The new strong-complement pitchfork applies to this entire leaf at all upper event scales. In particular it includes C unbounded when L is infinitesimal and α unbounded when $1 - H$ is infinitesimal.

Unbounded B is impossible by the original chart inequalities. The homogeneous all-tests theorem of Section C.5.3 closes the remaining infinitesimal-B leaf without assuming a bounded actual state ratio. Thus both unbounded-b leaves, as well as all bounded-b openings, are now covered. Literal $n = 0$ is preserved.

C.5.2 Homogeneous completion of the scalar graph

This lemma is a scalar lifting statement. It neither supplies the full state normals nor proves positivity of its coefficient Gram by itself. Its purpose is to avoid any bound on the contact state ratio.

Let $B, \sigma > 0$ be pure coefficient infinitesimals, let $k > 0$ be a pure coefficient bounded unit, and put $x = \sigma B$, $\xi = u^2$. Suppose

$$f = a(\xi, v)W^2 + 2ud(\xi, v)WZ + c(\xi, v)Z^2, \quad a = \frac{\xi}{B}Q + \sigma^2 a_0, \qquad c = Be + \xi b, \tag{HL1}$$

where Q, a_0, d, e, b have bounded polynomial coefficients. Bounded pure testing unit denominators may be cleared beforehand. Assume the value and all first derivatives of f vanish at

$$u = \sqrt{x}, \quad v = k, \quad W = \frac{1}{k}\sqrt{B/\sigma}, \quad Z = 1. \tag{HL2}$$

No bound on B/σ is assumed.

Define the homogeneous roots

$$R = \frac{uW - (B/k)Z}{\sqrt{B}}, \qquad S = uZ - \sigma k W, \qquad U = \sqrt{B}(v - k)Z, \qquad T = u(v - k)Z. \tag{HL3}$$

They satisfy the exact conductors

$$(\xi - x)W = u\sqrt{B}R + (B/k)S, \qquad (\xi - x)Z = uS + \sigma k\sqrt{B}R. \tag{HL4}$$

Then f has an exact symmetric Gram on (R, S, U, T) whose entries are bounded polynomials. In particular no inverse of σ, B, a testing infinitesimal, or a mixed expression is required in these Gram entries. The roots themselves need not be bounded.

First set $v = k$. In this paragraph every scalar coefficient is evaluated at $v = k$. State stationarity and u stationarity give

$$\begin{aligned} a_c = -\sigma k d_c, \qquad c_c = -Bd_c/k, \\ B(a_\xi)_c + kd_c + 2kx(d_\xi)_c + \sigma k^2 (c_\xi)_c = 0, \end{aligned} \tag{HL5}$$

where c on a subscript means evaluation at $\xi = x$. These are identities in the pure coefficient field; deriving them does not insert an inverse into a tensor Gram.

Set, by monic polynomial division in ξ,

$$s = \frac{c + Bd/k}{\xi - x}, \qquad q = -kd - \sigma k^2 s, \qquad h_2 = \frac{Ba + k\xi d + \sigma k^2 c + \sigma Bkd}{(\xi - x)^2}. \tag{HL6}$$

All three quotients have bounded polynomial coefficients. The first numerator vanishes at x by (HL5), and the last numerator and its ξ derivative vanish there by the same three identities. Moreover $Ba = \xi Q + B\sigma^2 a_0$ is bounded. Division by a monic polynomial preserves bounded coefficients and is not evaluation of an inverse of $\xi - x$.

Direct multiplication gives the exact identity

$$f(\xi, k) = qR^2 + sS^2 + h_2 \left(uR + \frac{\sqrt{B}}{k} S \right)^2. \tag{HL7}$$

For later residue computations, if $E_0 = e + d/k$, then

$$s = b + B\frac{E_0(\xi) - E_0(x)}{\xi - x} + x\frac{b(\xi) - b(x)}{\xi - x}. \tag{HL8}$$

Thus $s_0 = b_0$, $q_0 = -k_0 d_0$, and the h_2 correction has zero residue in every Gram entry. Here subscripts 0 denote residues, not evaluation at the contact polynomial variable.

The complete angular coquotient. Write $\tau = v - k$ and, for each of Q, a_0, d, e, b, form its bounded monic difference in v, denoted respectively by $Q_1, a_{01}, d_1, e_1, b_1$:

$$Q(\xi, v) - Q(\xi, k) = \tau Q_1(\xi, v),$$

and similarly for the other four functions. Put

$$H_1 = Q_1/k^2 + 2d_1/k + e_1, \qquad L_1 = a_{01}/k^2 + b_1, \qquad J_1 = H_1 + \sigma L_1.$$

Angular stationarity at (HL2) gives exactly $J_1(x, k) = 0$. Consequently the successive monic differences

$$J_1 = (\xi - x)J_\xi + \tau J_v, \quad J_\xi = \frac{J_1(\xi, v) - J_1(x, v)}{\xi - x}, \qquad J_v = \frac{J_1(x, v)}{v - k} \tag{HL9}$$

are bounded polynomials. Let $D_1 = L_1 + BJ_\xi$. The full exact angular correction is

$$\begin{aligned} f(\xi, v) - f(\xi, k) = \tau Q_1 R^2 + \frac{\tau a_{01}}{k^2} S^2 \\ + \{2(Q_1/k + d_1) + \sigma k D_1\} RU \\ + \left\{D_1 - 2a_{01}/k^2\right\} ST + J_v U^2. \end{aligned} \tag{HL10}$$

To check this without any division, use $uW/\sqrt{B} = R + \sqrt{B}Z/k$ and $\sigma W = (uZ - S)/k$. The only term not already among the displayed root products becomes

$$\tau Z^2\{BH_1 + \xi L_1\} = D_1\tau(\xi - x)Z^2 + BJ_v\tau^2 Z^2 = D_1(ST + \sigma kRU) + J_v U^2,$$

by (HL4). Equations (HL7) and (HL10) prove the asserted homogeneous Gram lift for every state W, Z simultaneously. There is no state normalization and no discarded endpoint $Z = 0$.

The usual exact zero, now with the new source scale, is

$$T^2 - \sigma U^2 - ST(v - k) - \sigma kRU(v - k) = 0. \tag{HL11}$$

This is the appropriate identity for redistributing the angular diagonal. Its U^2 coefficient is σ, which is infinitesimal; replacing it by 1 would erase the smaller radial scale. Any positivity argument must use the actual residue coefficients of (HL7)–(HL10), rather than importing the bounded-state Schur test without this scale correction.

Strict scalar corollary for the small-B source. Suppose additionally that the coefficientwise leading functions on the finite $u = 0$ axis are

$$Q_0(v) = \frac{2K^2(1 + v^2)}{1 + K^2}, \qquad d_0(v) = -2v, \qquad e_0(v) = \frac{2(v^2/K^2 + K^2)}{1 + K^2}, \qquad b_0(K) > 0,$$

where $K = k_0$. Then the (R, U) residue block produced above is

$$2K^2R^2 + \frac{4(K^2 - 1)}{1 + K^2}RU + \frac{2}{K^2}U^2,$$

with Schur margin $8/(1 + K^2)^2 > 0$. The S^2 coefficient has residue $b_0(K) > 0$, the RS residue is zero, and the only other possible residue entry is a bounded ST coefficient, say ℓ. Add a positive real multiple δ of (HL11), with $\delta > \ell^2/(4b_0(K))$. Its correction to the U^2 entry has zero residue because σ is infinitesimal, whereas it supplies the positive T^2 diagonal δ. Both residue blocks are then strictly positive. This proves the scalar strict bounded Gram for all homogeneous states in the matched-angle chart. Full normals and the other testing charts remain separate requirements.

C.5.3 The case b unbounded and xb infinitesimal

Theorem C.14 (The remaining unbounded-state rational C2 source)**.** *Use the exact original rational C2 source. Suppose $x > 0$ is infinitesimal, k is a positive bounded unit, b is positive unbounded, $B = xb$ is infinitesimal, $0 < r < 1$, $0 < h < b$, and all original positive-chart inequalities hold. Assume $n \geq 0$ and physicality of the source. Then the full original parent has tensor SOS for every independent real or complex projective state and every independent projective pair of measurement angles.*

The conclusion includes $n = 0$, arbitrarily large n, arbitrary complementary openings, and every value of the pure coefficient ratio B/σ, where $\sigma = 1/b$. In particular, the actual contact ratio $W/Z = k^{-1}\sqrt{B/\sigma}$ need not be bounded. This proof uses the homogeneous scalar lemma in Section C.5.2; it restores both full state normals and exhausts the other testing charts.

Exact source scales. Put

$$\sigma = 1/b, \quad x = \sigma B, \quad \eta = h/b, \quad y = B\eta, \quad z = B(1-\eta), \quad A_c = 1 + \sigma B r, \quad S_c = A_c + z.$$

The positive chart gives $0 < \eta < 1$; both y, z are infinitesimal. Let $\widehat{\Delta} = \sigma\Delta$, $\widehat{N} = \sigma^2 B N_{\mathrm{src}}$, $\widehat{g} = \sigma g = 1 + \sigma(r - 1 + B\eta)$ and $j = A_c - z/k^2$. The cancelled exact formulas are

$$\begin{aligned}
nr &= \frac{(1+\sigma r)A_c S_c - r\widehat{\Delta}}{2\widehat{\Delta}},\\
m &= \frac{\sigma^2(1+k^2)(1-r)A_c S_c}{2\widehat{g}\widehat{\Delta}},\\
C &= \frac{A_c\widehat{N}}{2z\widehat{\Delta}}, \qquad A = \frac{k^2 A_c(1+\sigma r)S_c}{2y\widehat{\Delta}},\\
D &= \frac{B(1+\sigma B)(1+\sigma r)(1-\sigma k^2 r)S_c}{2k^2 j\widehat{\Delta}},\\
\alpha &= \frac{B(1+\sigma r)\widehat{N}}{2(1-y)\widehat{\Delta}}.
\end{aligned} \tag{UB1}$$

Here $\widehat{\Delta}, \widehat{N}, \widehat{g}, j, 1-y$ are bounded positive units. Write $K = k_0 > 0$ and $D_K = 1 + K^2$. Their relevant residues imply

$$\begin{gathered}
(nr)_0 = \tfrac{1}{2}(D_K^{-1} - r_0), \quad (m/\sigma^2)_0 = (1 - r_0)/2,\\
(Cz)_0 = (Ay)_0 = K^2/(2D_K), \quad (D/B)_0 = 1/(2K^2 D_K), \quad (\alpha/B)_0 = K^2/(2D_K).
\end{gathered} \tag{UB2}$$

Since $n \geq 0$, $r_0 \leq D_K^{-1} < 1$, so $1 - r$ is a unit. Set $N = 4n + 2 \geq 2$. Exactly,

$$Nr = \frac{2(1+\sigma r)A_c S_c}{\widehat{\Delta}}, \tag{UB3}$$

which is a bounded positive unit. Thus nr, nr^2 are bounded even when n is unbounded, while $1/N$ and $1/\sqrt{N}$ are bounded.

All opening cancellations needed below are uniform:

$$BC(1-\eta) = Cz, \quad BA\eta = Ay, \quad BC(1-\eta)^2 = Cz(1-\eta), \quad BA\eta^2 = Ay\eta. \tag{UB4}$$

The quantity $H_B = B(C + A)$ is bounded below by a positive real; its reciprocal and reciprocal square root are bounded. No assumption that η or $1 - \eta$ is a unit is made.

The whole finite $u = 0$ axis. Use the original parent and the exact-contact graph

$$X_g = -ruZ, \qquad Y_g = \eta u W, \qquad E_X = \sqrt{N}(X + ruZ), \quad E_Y = \sqrt{C + A}(Y - \eta u W). \tag{UB5}$$

The actual contact is

$$u = \sqrt{\sigma B}, \quad v = k, \quad W = k^{-1}\sqrt{B/\sigma}, \quad Z = 1.$$

Let $\xi = u^2$ and $f = F|_g$. Exact polynomial coquotients give

$$f = a(\xi, v)W^2 + 2ud(\xi, v)WZ + c(\xi, v)Z^2, \quad a = \frac{\xi}{B}Q + \sigma^2 a_0, \quad c = Be + \xi b. \tag{UB6}$$

Every coefficient of Q, a_0, d, e, b is bounded by (UB3)–(UB4). For example $a_0 = 4(m/\sigma^2)v^2$, $e = 4(D/B)v^2 + 4\alpha/B$, and $Q = B(a - a|_{u=0})/u^2$. These are monic polynomial divisions. The full leading functions at $u = 0$ are

$$\begin{aligned} q(v) &= 2K^2(1+v^2)/D_K, & d_0(v) &= -2v, \\ e_0(v) &= 2(v^2/K^2 + K^2)/D_K, & (a_0)_0(v) &= 2(1-r_0)v^2, \\ b_0(v) &= 2r_0(1+v^2)/D_K - 4r_0 + 2. \end{aligned} \tag{UB7}$$

For remote tests use

$$P = uW/\sqrt{B}, \quad B_u = uZ, \quad V = \sigma v W, \quad E = \sqrt{B}Z.$$

Then (UB6) has the exact bounded Gram

$$QP^2 + bB_u^2 + (a_0/v^2)V^2 + eE^2 + 2dPE, \tag{UB8}$$

where a_0/v^2 is a polynomial coquotient. The (P, E) residue block is strictly positive whenever $v_0^2 \neq K^2$, since

$$q(v)e_0(v) - 4v^2 = 4(v^2 - K^2)^2/D_K^2. \tag{UB9}$$

This is a global identity for every finite residue v, not a local-contact positivity inference.

Both complete state normals. The exact quadratic normal expansion is

$$F = f + E_X L_X + E_Y L_Y + H_{XX}E_X^2 + 2H_{XY}E_X E_Y + H_{YY}E_Y^2. \tag{UB10}$$

Its entire hard block is bounded with residue $\mathrm{diag}(D_{v,0}, 4D_{v,0})$. Indeed the X pivot is $ND_uD_v + O(1)$ before division by N, with the remaining terms either infinitesimal in u or of order m; the Y pivot from the strong positive square terms is exactly $4D_v(1 + [A/(C + A)]u^2)$. All mixed entries have zero residue and contain no C, A contribution.

The complete unnormalized X derivative has the form

$$F_X|_g = uZA_X + W(\xi B_1 + \sigma^2 B_0), \tag{UB11}$$

with bounded polynomial coefficients, where

$$\begin{aligned} B_0 &= -8(m/\sigma^2)v(1+\eta\xi), \\ B_1 &= v(8D+4)(1-\eta), \\ A_X &= -8nrD_uD_v + 8m(1-r) - (8D+4)v^2(1+r\xi) + 4D_v(1-r). \end{aligned}$$

Thus in the remote roots its P, V coefficients have zero residue; its only possible nonzero residue coupling is to B_u. The exact residue Schur expression for the (B_u, E_X) block is

$$(Nr^2D_v - 4r + 2) - \frac{(4 - 2NrD_v)^2}{4ND_v} = 2 - \frac{4}{ND_v}. \tag{UB12}$$

It is strictly positive for every finite nonzero residue v, because $N \geq 2$. At $v_0 = 0$ it is strict if n is unbounded or has positive residue. The sole zero case $n_0 = v_0 = 0$ is treated below.

For the other complete first normal,

$$L_Y = \frac{1}{\sqrt{H_B}}\{P(B_2 + Be_w) + B_u(\sqrt{B}\,ue_z) + Ee_*\}, \tag{UB13}$$

where e_w, e_z are bounded,

$$B_2 = 8D_v\{(BA)\eta D_u - (BC)(1-\eta)\}, \qquad e_* = 8v(D - \alpha).$$

All coefficients have zero residue, by (UB4), $B \to 0$, and $u \to 0$. This verifies both full normals and proves the full parent at every remote finite-axis test covered by (UB12).

Matched residues without any state-ratio restriction. For $v_0 = K$, apply the homogeneous scalar lemma to (UB6). Its contact and full stationarity hypotheses hold exactly because the graph (UB5) passes through the original stationary contact. Use the homogeneous roots

$$R = \frac{uW - (B/k)Z}{\sqrt{B}}, \quad S = uZ - \sigma kW, \quad U = \sqrt{B}(v-k)Z, \quad T = u(v-k)Z. \tag{UB14}$$

They satisfy

$$(\xi - x)W = u\sqrt{B}R + (B/k)S, \qquad (\xi - x)Z = uS + \sigma k\sqrt{B}R. \tag{UB15}$$

The lemma constructs an exact bounded scalar Gram, valid for every W, Z simultaneously. Its (R, U) residue is

$$2K^2R^2 + \frac{4(K^2-1)}{D_K}RU + \frac{2}{K^2}U^2,$$

whose Schur margin is $8/D_K^2 > 0$. Its radial S^2 residue is $s_0 = 2(1 - r_0) > 0$; its only other possible residue coupling is ℓST, with a bounded real residue ℓ.

To lift the full X normal, set

$$L = A_X + (B/k)B_1 + (\sigma/k)B_0.$$

Equations (UB11) and (UB15) give exactly

$$F_X|_g = u\sqrt{B}B_1R - (\sigma/k)B_0S + uZL.$$

At the source contact $L(x, k) = 0$, because $F_X = 0$ there. Write its bounded monic differences as $L = (\xi - x)L_\xi + (v - k)L_v$. Then

$$F_X|_g = u\sqrt{B}(B_1 + \sigma kL_\xi)R + \{\xi L_\xi - (\sigma/k)B_0\}S + L_vT. \tag{UB16}$$

After division by $\sqrt{N}$, all coefficients are bounded. The R, S coefficients have zero residue; the T coefficient may have a finite residue γ (it is $-8K/(D_K\sqrt{N_0})$ when N is bounded). This nonzero full-normal coupling is retained, not silently discarded.

For (UB13), put $A_Y = B_2 + Be_w$ and $L'_Y = A_Y/k + \xi e_z + e_*$. Its numerator is exactly $RA_Y + \sqrt{B}ZL'_Y$. Exact normal contact gives $L'_Y(x, k) = 0$. Monic differences and (UB15) yield bounded coefficients on R, S, U, all of zero residue: the R, S corrections carry factors σB or $u\sqrt{B}$, and the U coefficient is the angular coquotient of an identically zero leading function. Division by $\sqrt{H_B}$ preserves these bounds.

Finally add δ times the exact zero

$$T^2 - \sigma U^2 - ST(v - k) - \sigma kRU(v - k) = 0. \tag{UB17}$$

Choose a real $\delta > \ell^2/(4s_0) + \gamma^2/(4D_K)$. This supplies a T^2 residue δ at zero residue cost to the U^2 coefficient, since σ is infinitesimal. The block on (S, E_X, T) has strictly positive Schur complement; the (R, U) block and E_Y diagonal are already strict. Thus the complete six-root Gram on (R, S, U, T, E_X, E_Y) is strictly positive in residue. The calculation is homogeneous in all four state coordinates and needs no bound on W/Z, no comparison of B with σ, and no state endpoint exclusion. The exact simultaneous sign copy treats $v_0 = -K$.

The double-zero test, including literal $n = 0$. Suppose now $n_0 = v_0 = 0$. Use the pure coefficient quantities

$$H_c = 4n + 4m + 2, \quad \zeta = (4m + 2)/H_c, \quad \xi_c = 4m/H_c, \quad \eta_c = C/(C + A),$$

and graph $X_g = \xi_c vW - \zeta uZ$, $Y_g = \eta_c uW$. Normalize its X normal by $\sqrt{H_c}$ and its Y normal by $\sqrt{C + A}$. Put

$$\epsilon = 4\alpha + 4n(4m + 2)/H_c > 0.$$

Then ϵ is infinitesimal, n/ϵ and B/ϵ are bounded, and the harmonic product $BCA/(C + A)$ is a unit. Use the five scalar roots

$$P = uW/\sqrt{B}, \quad V = \sigma vW, \quad E = \sqrt{B}Z, \quad B_\epsilon = \sqrt{\epsilon}\,uZ, \quad T_0 = uvZ.$$

The exact inherited identity, with its original coefficients, is

$$f_{zz} = 4\alpha + \epsilon u^2 + 4n\zeta^2u^4 + 4Dv^2 + u^2v^2B_b, \qquad (B_b)_0 = 2.$$

The W^2 coefficient at $u = 0$ is exactly

$$\frac{4m(2n + 1)}{2n + 2m + 1}v^2 + (4n + 2)\xi_c^2v^4.$$

Therefore its coefficient on V^2 has positive residue $4(m/\sigma^2)_0 = 2(1 - r_0)$. The scalar residue on the five roots is strictly diagonal, with entries

$$\left(\frac{2K^2}{D_K},\ 2(1 - r_0),\ \frac{2K^2}{D_K},\ 1,\ 2\right).$$

The cross term is allocated using $uvWZ = vPE$, so it has zero residue.

Both full normals are restored, including when $n = 0$ exactly. At $v = 0$ the X derivative is $-8n\zeta u^3 Z$, whose coefficient on B_ϵ is infinitesimal since n/ϵ is bounded. Its remaining Z coefficient is uv^2 times a bounded function, so it has zero residue on T_0. Its W coefficient is a sum of u^2vW times a bounded function and mvW times a bounded function divisible by v^2 at $u = 0$. Allocate the former to P with coefficient $u\sqrt{B}\,v$, and the latter to V with coefficient m/σ; both have zero residue. In particular one must not divide the former by σ. The Y derivative is the inherited harmonic-opening identity: $B_2 = 8D_v(BA)\eta_c u^2$, with bounded residual coquotients. Its coefficients on P, T_0, E all have zero residue. The hard residue is $\mathrm{diag}(1, 4)$. This proves the full-parent corner without deleting a state coordinate.

All remaining projective tests. For the other commuting charts use the four original outer-core identities with ρ replaced by $\sqrt{B}$, and cap C, A at s/B, where $s = \min(1, BC, BA)$ is a positive bounded unit. Retain both original nonnegative excess positive square terms and the entire original n positive square term. No positive lower bound on m is needed: the first two outer pivots are $4m_0 + 2 = 2$, while the other two charts retain the whole m positive square term and do not use it for strictness. The reciprocal-v chart uses $(D/B)_0 > 0$ exactly as before. Thus the finite unit u axis, both reciprocal axes, all double corners, and literal projective endpoints have strict bounded Grams. The inherited physical noncommuting-residue theorem covers all remaining testing pairs.

The finite-axis identities are homogeneous in the state and include unbounded actual state ratios. Every mixed quotient used above is a monic polynomial quotient; coefficient inverses and radicals remain in one labelled field. Restore the retained squares and apply the strict bounded-Gram lemma to each real and imaginary state part.

C.6 One small actual angle: strong complementary events

C.6.1 Two unbounded retained events

Theorem C.15 (The n, D-unbounded C2 pitchfork)**.** *Let the physical canonical C2 source over a real closed coefficient field have $\lambda = \beta = 0$, with d, C, α positive bounded units, $A \geq 0$ bounded, and $n, m, D \geq 0$ arbitrary. Suppose there is an actual finite noncommuting stationary zero with $u_c = k > 0$ infinitesimal and $v_c = a > 0$ a bounded unit. Then the entire original parent is tensor SOS at every independent projective measurement pair and every real or complex projective state. There is no bound on n, D, or on their ratio. In fact physicality and contact force A to be a positive unit, D to be bounded below by a positive real, and m/k^4 to be a positive bounded unit.*

This extends the preceding pitchfork by removing the upper bound on D. A bound on m need not be assumed: it follows from physicality. The original D positive square term supplies a state normal, so an unbounded weight is not put into a bounded Gram coefficient.

Write the unchanged parent as

$$\begin{aligned} F &= dH + 4nD_uD_vX^2 + 4mR_2^2 + 4CD_vr^2 + 4AD_uD_vY^2 + 4DJ^2 + 4\alpha D_up^2,\\ H &= 2\{J^2 + D_up^2 + D_vB^2 - D_uD_vZ^2\},\\ R_2 &= vW - X + uvY - uZ, \quad r = uW - Y, \quad J = uW + uvX - Y - vZ,\\ p &= vY - Z, \quad B = X + uZ, \quad D_u = 1 + u^2, \quad D_v = 1 + v^2. \end{aligned} \tag{TD1}$$

Entry and bounded contact coordinates. At $u = 0, v = a$, the state block on (X, Y, Z) of $dH + 4CD_vr^2 + 4\alpha D_up^2$ has positive definite residue. Its W column is $O(k)$ and its W^2 entry is $O(k^2)$. Completing this ordinary bounded hard block gives, for positive real constants b, B_0, the coefficient-field inequality

$$dH + 4CD_vr^2 + 4\alpha D_up^2 \geq b(X^2 + Y^2 + Z^2) - B_0k^2W^2$$

at the actual angles. All omitted positive square terms are nonnegative. Thus an actual zero has $W \neq 0$ and, on setting $W = 1$, $X, Y, Z = O(k)$. Since $R_2 = a + O(k)$ is a unit, the same inequality and its retained m positive square term give $m = O(k^2)$.

The state equation $F_Y = 0$ gives $DJ = O(k)$: every other term is $O(k)$, using bounded A, C, α, d, $m = O(k^2)$ and the bounds just obtained. The equation $F_X = 0$ then gives $(d + 2n)X = O(k^2)$, since the J contribution is $O(kDJ)$ and the m contribution is $O(m)$, while $Z = O(k)$. Consequently, with

$$N = d + 2n, \qquad x = X_c/k^2, \quad z = Z_c/k, \quad j = J_c/k,$$

we have

$$Nx,\ z,\ j,\ Dj \quad \text{bounded}, \qquad N^{-1} \text{ bounded}. \tag{TD2}$$

The bound on j follows directly from $X, Y, Z = O(k)$, even before a lower bound for D has been obtained. No inverse of D has yet been used.

For the scale entry, put $U=k^2, V=a^2$, $L=d+2D$, $\theta=n/N$ and use

$$t_2=d+2(\alpha+m+A),\quad t_3=d+2(C+D),\quad t_4=d+2(\alpha+m),$$
$$t_6=d+2(\alpha+C+A),\quad t_8=d+2(C+A+D).$$

The original Laurent-determinant stationary equations, divided only by N, are

$$\begin{aligned}U^2\{4\theta A t_3t_4+4CA(1+2D/N)t_4V\}&=mV\{4\alpha t_8+4Dt_6V\},\\ 4\alpha A(1+2m/N)t_3+4\theta A t_3t_4U&=\{4mDt_6/U+4CDt_2\}V^2.\end{aligned}\tag{TD3}$$

Here $t_2,t_4,t_6,t_3/L,t_8/L$ are positive bounded units. After division by L, the braces in the first equation are A times a positive unit on the left and a positive unit on the right. To check the lower bound uniformly when both n,D diverge, observe

$$\theta+\frac{1+2D/N}{L}=\frac{n+1}{N}+\frac{2n}{NL},\tag{TD4}$$

which is a positive bounded unit; all its neighboring factors in (TD3) have positive upper and lower real bounds. Thus m/U^2 is A times a positive unit. The second equation, again divided by L, has the form

$$A\cdot(\text{positive unit})+O(AU)=(D/L)\cdot(\text{positive unit})+O(AU).\tag{TD5}$$

If A were infinitesimal, then D/L and hence D would be infinitesimal, as would m. This is incompatible with physicality: on the test state $(W,X,Y,Z)=(1,0,u,zu)$ the n positive square term vanishes exactly, and at the residue source $m=A=D=0$ choose

$$z=\frac{v(2\bar\alpha+\bar d)}{2\bar\alpha}.$$

The coefficient of u^2 in the full polynomial is then $-\bar d v^2(2\bar\alpha+\bar d)/\bar\alpha<0$ for real $v\neq0$. The bounded residue polynomial cannot be nonnegative for all real small u. Therefore A is a unit. Equations (TD3)–(TD5) now prove that D/L and m/k^4 are positive bounded units. In particular $D>0$ and D^{-1} is bounded. Set $\gamma=4ma^2/k^4$.

A graph that keeps both large positive square terms bounded. For arbitrary independent testing angles use the exact graph

$$X_g=xu^2W,\qquad Z_g=zuW,\qquad J_g=juW,$$

which fixes

$$Y_g=u(1-j-vz+xu^2v)W.\tag{TD6}$$

It matches the entire actual contact, including its Y coordinate. The three exact state normals are

$$E_X=\sqrt{N}(X-xu^2W),\qquad E_J=\sqrt{D}(J-juW),\qquad E_Z=Z-zuW.\tag{TD7}$$

The inverse uses only pure coefficient inverses:

$$X=xu^2W+E_X/\sqrt{N},\quad Z=zuW+E_Z,\quad Y=u(1-j)W+uvX-vZ-E_J/\sqrt{D}.\tag{TD8}$$

The n positive square term on the graph is $4nx^2u^4D_uD_vW^2$, and the D positive square term is $4Dj^2u^2W^2$. Both have bounded coefficients by (TD2). Their linear coefficients are bounded too, since nx and $\sqrt{D}\,j=(Dj)/\sqrt{D}$ are bounded. The exact quadratic normal block is bounded: its only large positive square terms become $4(n/N)D_uD_vE_X^2$ and $4E_J^2$.

Set $W=1$, $K=k^2$, and

$$R=u^2-(K/a)v,\qquad T=u(v-a).$$

The entire graph scalar has the exact shape

$$F|_g=4mv^2+u^2H_0(u^2,v),\qquad H_0\in B_c[u^2,v].\tag{TD9}$$

The three contact scalar jets give exactly the same monic division as for the preceding pitchfork:

$$F|_g=\gamma R^2+u^2\{(u^2-K)^2H_{20}+(u^2-K)(v-a)H_{11}+(v-a)^2H_{02}\},\tag{TD10}$$

with bounded polynomial coquotients. After $u^2-K=R+(K/a)(v-a)$ this is a bounded Gram on (R,T). Its R residue pivot is $\bar\gamma>0$.

The full first normals, before the indicated scale divisions, have exact shapes

$$(F_X+uvF_Y)|_g=-8mv+u^2L_X(u^2,v),\quad(-F_Y)|_g=uL_J(u^2,v),\quad(F_Z-vF_Y)|_g=uL_Z(u^2,v).\tag{TD11}$$

The coefficient of E_X is the first expression divided by $\sqrt{N}$; that of E_J is the second divided by $\sqrt{D}$. These scaled polynomials have bounded coefficients, as shown above. Contact makes $L_X(K,a) = 8ma/K$, and makes the scaled J, Z functions vanish at (K,a). Monic division by $u^2 - K, v - a$ therefore puts every first normal in (R,T) with bounded coquotients. Every coefficient coupling E_X to R,T is infinitesimal; the R coefficients in the E_J, E_Z rows are infinitesimal. For example, the X row is exactly

$$\frac{1}{\sqrt{N}}\left[(8ma/K + u^2 L_{X1})R + u\{(K/a)L_{X1} + L_{X2}\}T\right], \tag{TD12}$$

and the other two rows have form $uL_{i1}R + \{(K/a)L_{i1} + L_{i2}\}T$ after their scale division. State degree is two, so there are no omitted higher-normal terms. Equations (TD7)–(TD12) give an exact bounded full-parent Gram on (E_X, E_J, E_Z, R, T).

The residue Gram is strict. The X pivot has residue $2(1+\bar{a}^2)$ and the R pivot has residue γ. For the remaining hard pair the unscaled state form is

$$B(v)[Y,Z] = \{4(C+A) + 2d\}(1+v^2)Y^2 + 4D(Y+vZ)^2 + 4\alpha(vY - Z)^2. \tag{TD13}$$

In the scaled coordinates $Y = -vE_Z - E_J/\sqrt{D}$, $Z = E_Z$ its coefficients are bounded and its residue is positive definite. If D is bounded, this follows from the invertible residue change and the positive form (TD13). If D is unbounded, its residue is

$$4E_J^2 + \left(\{4(\bar{C}+\bar{A}) + 2\bar{d}\}(1+v^2)v^2 + 4\bar{\alpha}(1+v^2)^2\right)E_Z^2 \succ 0.$$

The exact minimized boundary coefficient of u^2, before residue, is

$$f(v) = \frac{8(1+v^2)}{\det B(v)/L}\left(\frac{Q_{10}}{L} + \frac{Q_0}{L}v^2 + \frac{Q_{12}}{L}v^4\right), \tag{TD14}$$

where

$$Q_{10} = 4\alpha A[d + 2(C+D)], \qquad Q_{12} = 4CD[d + 2(\alpha + A)].$$

The middle coefficient Q_0 is defined by the exact Schur numerator; its explicit value is unnecessary. All quantities in (TD14) are bounded, $\det B/L$ is a positive unit for bounded real v, and $Q_{10}/L, Q_{12}/L$ are positive units. One may verify boundedness of Q_0/L directly from the determinant expansion, which is affine in D. This is only a residue calculation of a real hard Schur complement, not inversion of a mixed tensor element.

The full contact normal and scalar equations imply that the residue of (TD6) at $v = \bar{a}$ minimizes the hard form, and that $\bar{f}(\bar{a}) = \bar{f}'(\bar{a}) = 0$. Thus the real quartic bracket in (TD14) equals

$$\overline{Q_{12}/L}\,(v^2 - \bar{a}^2)^2.$$

Its second derivative at $\bar{a}$ is strictly positive. This is exactly the Schur complement on T of the remaining residue block (E_J, E_Z, T). Hence the whole five-root residue Gram is positive definite. The bounded strict-residue Gram lemma proves the local certificate at $\bar{u} = 0, \bar{v} = \bar{a}$. The Bob sign copy covers $\bar{v} = -\bar{a}$.

All other independent tests. For u infinitesimal and v bounded with $\bar{v} \neq \pm\bar{a}$, retain the original m, n positive square terms. Set $K_J = \sqrt{D}J$ and substitute

$$Y = uW + uvX - vZ - K_J/\sqrt{D}.$$

The exact remainder is a bounded quadratic Gram on (X, Z, K_J, uW). Its hard residue is the block just considered with X pivot $2\bar{d}(1+\bar{v}^2)$; its soft Schur complement is $\bar{f}(\bar{v}) > 0$. This includes $v = 0$ and all infinitesimal v. For $u, q = 1/v$ infinitesimal use the homogeneous parent $q^2F(u, 1/q)$. Here $J_h = quW + uX - qY - Z$. Retain m, n and substitute $Z = quW + uX - qY - K_J/\sqrt{D}$ with $K_J = \sqrt{D}J_h$. The remainder is a bounded Gram on (X, Y, K_J, uW). Its soft residue is $\lim_{q\to 0} q^2\bar{f}(1/q) > 0$, since $\overline{Q_{12}/L} > 0$ and $\overline{\det B_\infty/L} > 0$. These are polynomial homogeneous expressions, so literal projective endpoints are included.

At every other commuting test the capping proof from the preceding all-tests corollary applies verbatim: choose $\mu = \min(1, C/d, A/d, D/d, \alpha/d)$, a positive unit, retain every excess and the whole m, n positive square terms, and use its displayed open-axis state forms and two corner quadratic forms. That proof used lower bounds, not upper bounds, on these four source weights. Noncommuting residue tests use the retained arbitrary-physical-source theorem. Finally homogenize (TD6)–(TD12) in W, replacing R, T by RW, TW; restore pure testing projective factors and split complex states into real and imaginary parts. This proves the theorem for all states, including $W = 0$, without comparing k, n, D with independent testing infinitesimals.

Exact scope. This is an all-tests theorem for the stated actual-contact domain. It does not bound or remove the residual regimes where C/d or α/d is unbounded, A/d is unbounded, or the actual contact has a different angle-residue type. The C2 reflection $(m, C, D, \alpha; u_c, v_c) \mapsto (D, \alpha, m, C; 1/v_c, 1/u_c)$ supplies its corresponding reflected domain. No exhaustive C2 join follows from these two domains alone.

C.6.2 Three arbitrary event weights

Theorem C.16 (All scales of the three retained events)**.** *Let K_c, K_t be independent real closed extensions of $\mathbb{R}$. Let a physical canonical C2 source over K_c have $\lambda = \beta = 0$, with d, C, α positive bounded units and $n, m, A, D \geq 0$ arbitrary. Suppose an actual finite noncommuting stationary zero has $u_c = k > 0$ infinitesimal and $v_c = a > 0$ a bounded unit. Then the entire original parent is tensor SOS at every independent projective measurement pair and every real or complex projective state. No ratios among n, A, D are restricted.*

The bounded-A result is a special case. The mechanism is to keep all three possibly large event amplitudes as state coordinates. The exact parent and its amplitudes are (TD1); in particular $J = uW + uvX - Y - vZ$.

Source entry with three large weights. The initial coercivity estimate in the preceding proof uses only d, C, α, not A. Thus after $W_c = 1$,

$$X_c, Y_c, Z_c = O(k), \qquad m = O(k^2).$$

Now use $F_Z = 0$ first. The A and n positive square terms do not enter this state equation. Every term other than $-8DvJ$ is $O(k)$, because $v_c = a$ is a unit and the just displayed bounds hold. Therefore $DJ_c = O(k)$. The Y equation then gives $AY_c = O(k)$, and the X equation gives $(d + 2n)X_c = O(k^2)$. Write

$$N = d + 2n, \quad T_A = d + 2A, \quad L = d + 2D, \qquad x = X_c/k^2, \quad y = Y_c/k, \quad j = J_c/k.$$

Consequently

$$x, y, j, Nx, Ay, Dj \text{ are bounded}, \qquad N^{-1} \text{ bounded}. \tag{TA1}$$

These estimates do not yet invert A or D.

Use the exact stationary equations (TD3) and divide both by $T_A L$. Now $t_2/T_A, t_6/T_A, t_3/L, t_4$ are positive bounded units. In the first equation the coefficient multiplying U^2 is (A/T_A) times a positive bounded unit: its remaining bracket is

$$4t_4 \left\{ \theta \frac{t_3}{L} + CV \frac{1 + 2D/N}{L} \right\},$$

whose lower bound follows from (TD4). The coefficient multiplying m is also a positive bounded unit:

$$4V \left\{ \alpha \frac{t_8}{T_A L} + \frac{D}{L} \frac{t_6}{T_A} V \right\}. \tag{TA2}$$

Indeed, if D/L is bounded below, the second summand is a unit. Otherwise D is infinitesimal and L is a unit, while $t_8/T_A = 1 + 2(C + D)/T_A$ is a unit. All displayed factors have bounded upper bounds in either case. Hence

$$\frac{m}{U^2} = \frac{A}{T_A} \cdot (\text{positive bounded unit}). \tag{TA3}$$

The second stationary equation becomes

$$\frac{A}{T_A} \cdot (\text{positive unit}) + O\left(\frac{A}{T_A} U\right) = \frac{D}{L} \cdot (\text{positive unit}) + O\left(\frac{A}{T_A} U\right). \tag{TA4}$$

Thus A/T_A and D/L are comparable. If they were infinitesimal, A, D, m would all be infinitesimal, contradicting exactly the physicality test printed after (TD5). Therefore

$$A/T_A, \ D/L, \ m/k^4 \quad \text{are positive bounded units.} \tag{TA5}$$

In particular A, D have bounded reciprocals, but may be arbitrarily large. Set $\gamma = 4ma^2/k^4$, also a positive bounded unit.

The three-positive square term graph. On the local testing domain $\bar{v} = \bar{a} \neq 0$, put $b = 1 - j - y$ and use

$$X_g = xu^2W, \qquad Y_g = yuW, \qquad J_g = juW, \qquad Z_g = u\left(\frac{b}{v} + xu^2\right)W. \tag{TA6}$$

This graph matches the whole actual contact: the definition of J_c gives $Z_c/k = b/a + xk^2$. Its three hard normals are

$$E_X = \sqrt{N}(X - xu^2W), \quad E_Y = \sqrt{A}(Y - yuW), \quad E_J = \sqrt{D}(J - juW). \tag{TA7}$$

The exact inverse is

$$X = xu^2W + E_X/\sqrt{N}, \quad Y = yuW + E_Y/\sqrt{A}, \quad Z = \frac{uW + uvX - Y - juW - E_J/\sqrt{D}}{v}. \tag{TA8}$$

Every denominator is either pure coefficient or the pure testing unit v. All graph coefficients and all scaled normal coefficients are bounded. For the three large positive square terms this follows explicitly from

$$nx^2,\ nx,\ n/N, \quad Ay^2,\ \sqrt{A}\,y, \quad Dj^2,\ \sqrt{D}\,j \quad \text{bounded}, \tag{TA9}$$

using (TA1),(TA5). Their quadratic normal contributions are exactly $4(n/N)D_uD_vE_X^2$, $4D_uD_vE_Y^2$, and $4E_J^2$. Every other original source coefficient is bounded, including m.

For clarity, the finite Taylor division used below works in the bounded Laurent polynomial ring in v: for positive integer r,

$$v^{-r} - a^{-r} = -(v - a)\frac{\sum_{i=0}^{r-1} v^i a^{r-1-i}}{a^r v^r}.$$

Thus a Laurent polynomial vanishing at $(U, v) = (K, a)$ belongs to $(U - K, v - a)$ with bounded Laurent coquotients; repeating the same identity gives the second-order division. There is no localization at a difference of the two fields.

Set $K = k^2$, $R = u^2 - (K/a)v$ and $T = u(v - a)$. The entire graph scalar is again $4mv^2 + u^2H_0(u^2, v)$, now with bounded Laurent coefficients in v. Its three scalar contact jets give (TD10), and hence an exact bounded Gram on (R, T) with R residue pivot $\bar{\gamma}$. The full first normals before their pure source scale divisions are

$$\begin{aligned} (F_X + uF_Z)|_g &= -8mv + u^2L_X(u^2, v), \\ (F_Y - F_Z/v)|_g &= uL_Y(u^2, v), \\ (-F_Z/v)|_g &= uL_J(u^2, v). \end{aligned} \tag{TA10}$$

Divide the rows by $\sqrt{N}, \sqrt{A}, \sqrt{D}$, respectively. The scaled coefficients are bounded by (TA9). At contact they vanish. Laurent Taylor division gives exactly the root membership and infinitesimal R couplings proved in (TD11)–(TD12). The X row has infinitesimal coefficients on both R, T because $m/K = O(K)$; the other rows have infinitesimal coefficients on R. The exact quadratic state remainder is bounded by (TA8)–(TA9), so the entire original parent has a bounded five-root Gram on (E_X, E_Y, E_J, R, T).

Uniform strictness without a largest-event choice. The X and R residue pivots are $2(1 + \bar{a}^2)$ and $\bar{\gamma}$. The remaining hard state form is still (TD13). At $u = 0$, (TA8) gives

$$Y = E_Y/\sqrt{A}, \qquad Z = -\frac{E_Y}{v\sqrt{A}} - \frac{E_J}{v\sqrt{D}}.$$

In these coordinates (TD13) is the sum of the two explicit pivots

$$4(1 + v^2)E_Y^2 + 4E_J^2$$

and the nonnegative bounded form $(4C + 2d)(1 + v^2)Y^2 + 4\alpha(vY - Z)^2$. Its residue is therefore positive definite for every combination of bounded or unbounded A, D.

The boundary Schur function is the exact expression

$$f(v) = \frac{8(1 + v^2)}{\det B(v)/(T_AL)}\left(\frac{Q_{10}}{T_AL} + \frac{Q_0}{T_AL}v^2 + \frac{Q_{12}}{T_AL}v^4\right), \tag{TA11}$$

with Q_{10}, Q_{12} as in (TD14). For v a bounded unit all its coefficients and denominator are bounded, and

$$\frac{Q_{10}}{T_AL} = 4\alpha\frac{A}{T_A}\frac{t_3}{L}, \qquad \frac{Q_{12}}{T_AL} = 4C\frac{D}{L}\frac{t_2 - 2m}{T_A}$$

are positive units. The determinant has the positive term $16AD(1+v^2)v^2$, so its quotient by T_AL has a positive lower bound. Its other terms are bounded after that division. The exact middle coefficient is polynomial of degree at most one in each of A, D, so $Q_0/(T_AL)$ is bounded too.

The contact normal equations and scalar jets give $\bar{f}(\bar{a}) = \bar{f}'(\bar{a}) = 0$ as before. Therefore the real quartic bracket equals $\overline{Q_{12}/(T_AL)}(v^2-\bar{a}^2)^2$. Its strict second derivative gives the positive soft Schur pivot on T. The whole five-root residue Gram is strict. This proves the local all-state certificate; the Bob sign identity gives its negative sign copy. Homogenization in W includes $W = 0$.

Outer tests when both A and D are large. For u infinitesimal and v a bounded unit with $\bar{v} \neq \pm\bar{a}$, retain the entire m, n positive square terms and use the four roots

$$X, \quad K_A = \sqrt{A}Y, \quad K_D = \sqrt{D}J, \quad uW.$$

Substitute $Y = K_A/\sqrt{A}$ and $Z = (uW + uvX - K_A/\sqrt{A} - K_D/\sqrt{D})/v$. The exact remainder is a bounded quadratic Gram. Its hard residue has the positive X pivot $2\bar{d}(1+\bar{v}^2)$ and the two explicit pivots just displayed; its soft Schur complement is $\bar{f}(\bar{v}) > 0$ by (TA11).

At u, v both infinitesimal, and at $u, 1/v$ both infinitesimal, one need not divide by v. Cap C, A, D, α at the positive unit $\mu = \min(1, C/d, A/d, D/d, \alpha/d)$ after division by d, and retain all excesses and the full m, n positive square terms. The bounded selected parent has an exact Gram on (X, Y, Z, uW) in the finite chart and on the same four roots in the homogeneous reciprocal chart. At $(u, v) = (0, 0)$ its soft Schur complement is $8Q_{10,\mu}/\det B_\mu(0) > 0$; at $(u, 1/v) = (0, 0)$ it is $8Q_{12,\mu}/\det B_{\mu,\infty} > 0$. The hard blocks are positive by (TD13) and its homogeneous limit. Hence both entire endpoint neighborhoods have strict-residue Grams, including literal zeros and infinities. These expressions use only the bounded selected parent; no residue of an unbounded A or D is taken.

For every other commuting test, the open-axis and corner forms of the preceding proof apply with the four lower bounds (TA5). Noncommuting residue tests use the physical-source theorem. Restore the positive pure factors and split complex states into real and imaginary parts.

Coverage gained and remaining restriction. The theorem removes every relative-scale restriction among n, A, D from the finite pitchfork. Its reflected form covers the corresponding ordinary–infinite actual contacts. It still assumes C/d and α/d are bounded units and this actual-angle residue type. Unbounded C/d or α/d, other actual-angle types, and the exhaustive C2 join remain outside this theorem.

C.6.3 Arbitrary complementary event scales

Theorem C.17 (No upper bound on the two complementary events)**.** *Let $K_c, K_t \supset \mathbb{R}$ be independent real closed fields. Let the original canonical C2 parent be physical over K_c, with $\lambda = \beta = 0$, d a positive bounded unit, $C, \alpha > 0$ having bounded reciprocals, and $n, m, A, D \geq 0$ arbitrary. Suppose it has an actual finite noncommuting stationary zero with $u_c = k > 0$ infinitesimal and $v_c = a > 0$ a bounded unit. Then the entire original parent is tensor SOS at every independent projective measurement pair and every real or complex projective state. There is no upper bound or relative-scale condition on any event weight.*

The case of bounded C, α is the existing three-large-event theorem. We prove both remaining cases, without modifying the parent or identifying the source and testing fields. We use

$$\begin{gathered} D_u = 1+u^2, \quad D_v = 1+v^2, \quad r = uW - Y, \quad p = vY - Z, \\ J = uW + uvX - Y - vZ, \quad R_2 = vW - X + up, \quad B = X + uZ, \\ H = 2\{J^2 + D_u p^2 + D_v B^2 - D_u D_v Z^2\}, \\ F = dH + 4nD_uD_vX^2 + 4mR_2^2 + 4CD_vr^2 + 4AD_uD_vY^2 + 4DJ^2 + 4\alpha D_u p^2. \end{gathered} \tag{SC1}$$

Every square in this formula retains its original coefficient.

Necessary reductions from the full parent. Cap only C, α at positive bounded units in the initial hard state estimate. At the actual angles the bounded selected form is bounded below by $b(X^2+Y^2+Z^2) - B_0k^2W^2$, with positive real b, B_0. Every retained positive square term is nonnegative. Hence the contact has $W_c \neq 0$; on setting $W_c = 1$,

$$X_c, Y_c, Z_c = O(k), \qquad m = O(k^2).$$

Each positive event positive square term is $O(k^2)$ as well. Thus, writing $y = Y_c/k$, $z = Z_c/k$, $x_1 = X_c/k$, $j = J_c/k$,

$$Ay^2,\ Dj^2,\ C(1-y)^2,\ \alpha(ay-z)^2 = O(1), \qquad j = 1 - y - az + aX_c. \tag{SC2}$$

If C is unbounded then $\bar{y}=1$, so A is bounded. If D were also unbounded, j,z would be infinitesimal, and the residue of

$$H_c/(2k^2)=j^2+(1+k^2)(ay-z)^2+(1+a^2)(x_1+Z_c)^2-(1+k^2)(1+a^2)z^2$$

would be $\bar{a}^2+(1+\bar{a}^2)\bar{x}_1^2>0$. This contradicts $H_c\le 0$ at the zero of (SC1). Therefore C unbounded forces both A,D bounded. If α is unbounded and A were unbounded, (SC2) would make y,z infinitesimal and $\bar{j}=1$; the same residue would instead be $1+(1+\bar{a}^2)\bar{x}_1^2>0$. Thus α unbounded forces A bounded. It remains to prove precisely

$$\text{(I) } C \text{ unbounded, } A,D \text{ bounded;} \qquad \text{(II) } C \text{ bounded, } \alpha \text{ unbounded, } A \text{ bounded.} \tag{SC3}$$

Uniform stationary scale entry. Set $U=k^2$, $V=a^2$, $N=d+2n$, $\theta=n/N$, and

$$t_2=d+2(\alpha+m+A),\quad T=t_3=d+2(C+D),\quad t_4=d+2(\alpha+m),$$
$$t_6=d+2(\alpha+C+A),\quad t_8=d+2(C+A+D).$$

The exact original Laurent stationary equations are

$$\begin{aligned} U^2At_4\{\theta T+C(1+2D/N)V\}&=mV\{\alpha t_8+Dt_6V\},\\ \alpha A(1+2m/N)T+\theta ATt_4U&=\{mDt_6/U+CDt_2\}V^2.\end{aligned} \tag{SC4}$$

Define only pure coefficient quantities

$$\delta=CD/T,\quad e=(D/T)(t_6/\alpha),\quad b=t_2/\alpha,\quad \mathcal{B}=\theta+(C/T)(1+2D/N)V,\quad \mathcal{G}=t_8/T+eV.$$

Because one of C,D is bounded in either case (SC3), $\delta\le\min(C,D)/2$ is bounded. The identities

$$e=(D/T)\{2+(d+2A)/\alpha\}+2\delta/\alpha,\quad \mathcal{G}=1+2A/T+eV,\quad \mathcal{B}=\theta+(C/T)V+(2\delta/N)V$$

show boundedness of all these quantities. The quantities $b,\mathcal{G},t_4/\alpha$ are positive bounded units. So is $\mathcal{B}$: since C,N have bounded reciprocals and d is bounded,

$$\frac{T}{C(N+2D)}\le\frac{d}{CN}+\frac{2}{N}+\frac{1}{C}=O(1),$$

and consequently

$$\theta+(C/T)(1+2D/N)=\frac{n+C(N+2D)/T}{d+2n}\ge\frac{n+c_0}{d+2n}$$

for a positive real c_0. The bounded-unit factor V preserves this lower bound. Dividing (SC4) by αT gives exactly

$$\frac{m}{U^2}=A\rho,\qquad A\mathcal{H}=\delta bV^2,\qquad \rho=\frac{(t_4/\alpha)\mathcal{B}}{V\mathcal{G}},\quad \mathcal{H}=1+2m/N+\theta(t_4/\alpha)U-U\rho eV^2. \tag{SC5}$$

Here ρ is a positive bounded unit and $\mathcal{H}=1+O(U)$. These identities were obtained before inverting A or D. They show $A\asymp\delta$. Since C has a bounded reciprocal, δ infinitesimal implies D infinitesimal. If A were infinitesimal, therefore, so would D,m be. But the exact physical test

$$u=v=1,\qquad (W,X,Y,Z)=(1,0,1,1)$$

has $X=r=p=0$, $R_2=1$, $J=-1$, $H=-2$, and hence $F=-2d+4m+16A+4D$. Its residue would be negative. We conclude

$$A\asymp 1,\qquad D^{-1}=O(1),\qquad m/k^4\asymp 1,\qquad \gamma=4ma^2/k^4\asymp 1. \tag{SC6}$$

In case (I), D is a bounded unit; in case (II), it may be arbitrarily large. In both cases δ is a positive bounded unit.

Full state equations and bounded graph data. For later use, the three entire state derivatives of (SC1) are

$$\begin{aligned} F_X/4&=D_v[d+2nD_u]X+(d+2D)uvJ+duD_vZ-2mR_2,\\ F_Y/4&=-(d+2D)J+(d+2\alpha)vD_up-2CD_vr+2AD_uD_vY+2muvR_2,\\ F_Z/4&=-(d+2D)vJ-(d+2\alpha)D_up+duD_vB-dD_uD_vZ-2muR_2.\end{aligned} \tag{SC7}$$

In case (I), the Z equation first yields $\alpha p_c=O(k)$, because D is bounded and every core coordinate is $O(k)$. Then the Y equation yields $Cr_c=O(k)$. In case (II), C,A are bounded, so the Y,Z equations are a system with bounded right side $O(k)$ and unit coefficient matrix

$$\begin{pmatrix}-1 & a\\ -a & -1\end{pmatrix}$$

on $((d+2D)J_c,(d+2\alpha)(1+k^2)p_c)$. They yield $DJ_c,\alpha p_c=O(k)$. Finally the X equation in either case yields $NX_c=O(k^2)$. Thus put $x=X_c/k^2$ and

$$\begin{array}{ll}\text{(I)} & r_0=r_c/k,\ p_0=p_c/k, \qquad Nx,Cr_0,\alpha p_0=O(1),\\ \text{(II)} & j_0=J_c/k,\ p_0=p_c/k, \qquad Nx,Dj_0,\alpha p_0=O(1).\end{array} \tag{SC8}$$

All unweighted graph constants are bounded as well. The reciprocals of N,C,D,α used below are bounded pure coefficient elements.

Two exact graphs retaining the large positive square terms. For independent finite testing angles, case (I) uses

$$X_g=xu^2W,\quad r_g=r_0uW,\quad p_g=p_0uW,\quad Y_g=(1-r_0)uW,\quad Z_g=\{v(1-r_0)-p_0\}uW. \tag{SC9}$$

Its exact normals and inverse are

$$\begin{gathered}E_X=\sqrt{N}(X-xu^2W),\quad E_r=\sqrt{C}(r-r_0uW),\quad E_p=\sqrt{\alpha}(p-p_0uW),\\ X=xu^2W+E_X/\sqrt{N},\quad Y=(1-r_0)uW-E_r/\sqrt{C},\quad Z=vY-p_0uW-E_p/\sqrt{\alpha}.\end{gathered} \tag{SC10}$$

Case (II) uses

$$X_g=xu^2W,\quad J_g=j_0uW,\quad p_g=p_0uW,\quad Y_g=\frac{u(1-j_0+vp_0+xu^2v)}{D_v}W,\quad Z_g=vY_g-p_0uW. \tag{SC11}$$

Here the exact normals and inverse are

$$\begin{gathered}E_X=\sqrt{N}(X-xu^2W),\quad E_J=\sqrt{D}(J-j_0uW),\quad E_p=\sqrt{\alpha}(p-p_0uW),\\ X=xu^2W+E_X/\sqrt{N},\quad J=j_0uW+E_J/\sqrt{D},\quad p=p_0uW+E_p/\sqrt{\alpha},\\ Y=(uW+uvX+vp-J)/D_v,\qquad Z=vY-p.\end{gathered} \tag{SC12}$$

Both graphs match the entire actual contact. All denominators in these formulas are pure coefficient factors or the pure testing unit D_v. No inverse of a mixed tensor element occurs. The possibly unbounded positive square terms become respectively

$$\begin{array}{ll}\text{(I)} & 4(n/N)D_uD_vE_X^2,quad4D_vE_r^2,quad4D_uE_p^2,\\ \text{(II)} & 4(n/N)D_uD_vE_X^2,quad4E_J^2,quad4D_uE_p^2\end{array}$$

in the quadratic normal remainder. Their graph and first-normal coefficients are bounded by (SC8), for example $nx,nx^2,Cr_0^2,\sqrt{C}r_0,\alpha p_0^2,\sqrt{\alpha}p_0$ in case (I), and $Dj_0^2,\sqrt{D}j_0$ in case (II). Every remaining source coefficient is bounded.

Entire scalar and entire first-normal division. Set $W=1$, $K=k^2$, and $R=u^2-(K/a)v$, $T_s=u(v-a)$. Since $R_2|_g=v+(p_0-x)u^2$ for both graphs, the entire scalar has the exact shape

$$\Phi=F|_g=4mv^2+u^2H_0(u^2,v), \tag{SC13}$$

with bounded polynomial coefficients in case (I), and bounded coefficients in $B_c[v,D_v^{-1}]$ in case (II). The three scalar contact jets give $H_0(K,a)=-\gamma K$, $H_{0,U}(K,a)=\gamma$, and $H_{0,v}(K,a)=-2\gamma K/a$. In particular the explicit expression

$$\Psi=H_0-\gamma u^2+2\gamma(K/a)v$$

satisfies $\Phi-\gamma R^2=u^2\Psi$, and its value and both first derivatives vanish at $(u^2,v)=(K,a)$. Finite Taylor division of Ψ therefore gives

$$\Phi=\gamma R^2+u^2\{(u^2-K)^2H_{20}+(u^2-K)(v-a)H_{11}+(v-a)^2H_{02}\}, \tag{SC14}$$

with bounded coquotients. This is an exact bounded Gram on (R,T_s), with separate residue pivot $\bar\gamma$ on R. For the rational case, Taylor division is legitimate because

$$D_v^{-r}-D_a^{-r}=-(v-a)(v+a)\frac{\sum_{i=0}^{r-1}D_v^iD_a^{r-1-i}}{D_a^rD_v^r}.$$

Iterating this identity gives the second-order division; only pure factors have been inverted.

Here are the full first normals, rather than initial forms. In case (I) they are

$$\mathcal{N}_X = F_X|_g, \quad \mathcal{N}_r = -(F_Y + vF_Z)|_g, \quad \mathcal{N}_p = -F_Z|_g, \tag{SC15}$$

and in case (II) they are

$$\mathcal{N}_X = \{F_X + (uv/D_v)(F_Y + vF_Z)\}|_g, \quad \mathcal{N}_J = -(F_Y + vF_Z)|_g/D_v, \quad \mathcal{N}_p = (vF_Y - F_Z)|_g/D_v. \tag{SC16}$$

The derivatives in (SC7) specify every term. In particular the following exact positive square term expressions expose the cancellations:

$$\begin{aligned} -(F_Y + vF_Z)/(4D_v) &= (d + 2D)J + 2Cr - 2AD_uY - duvB + dvD_uZ, \\ (vF_Y - F_Z)/(4D_v) &= (d + 2\alpha)D_up - 2Cvr + 2AvD_uY + 2muR_2 - duB + dD_uZ. \end{aligned} \tag{SC17}$$

The actual coefficients of the normal roots are obtained by dividing these rows by $\sqrt{N}, \sqrt{C}, \sqrt{\alpha}$ in case (I), and by $\sqrt{N}, \sqrt{D}, \sqrt{\alpha}$ in case (II). Using (SC8), their exact forms after those divisions are

$$\mathcal{N}_X/\sqrt{N} = -8mv/\sqrt{N} + u^2 L_X(u^2, v), \qquad \mathcal{N}_i/\sqrt{w_i} = uL_i(u^2, v), \tag{SC18}$$

where all L_i have bounded coefficients in the indicated ring. The contact equations give $L_X(K, a) = 8ma/(K\sqrt{N})$ and $L_i(K, a) = 0$. Writing $L_X = L_X(K, a) + (u^2 - K)L_{X1} + (v - a)L_{X2}$, and similarly for L_i, gives the full exact row identities

$$\begin{aligned} \mathcal{N}_X/\sqrt{N} &= (8ma/(K\sqrt{N}) + u^2 L_{X1})R + u\{(K/a)L_{X1} + L_{X2}\}T_s, \\ \mathcal{N}_i/\sqrt{w_i} &= uL_{i1}R + \{(K/a)L_{i1} + L_{i2}\}T_s. \end{aligned} \tag{SC19}$$

Thus the X row has infinitesimal couplings to both soft roots, and the other rows have infinitesimal couplings to R. The exact state degree is two, so the bounded quadratic remainder already accounts for every higher-normal term. Equations (SC10)–(SC19) give an exact full-parent bounded five-root Gram.

A common exact real Schur certificate. The X and R residue pivots are $2(1 + \bar{a}^2)$ and $\bar{\gamma}$. The unscaled boundary hard form on Y, Z is

$$\mathsf{B}(v)[Y, Z] = sD_vY^2 + 4D(Y + vZ)^2 + 4\alpha(vY - Z)^2, \qquad s = 4(C + A) + 2d. \tag{SC20}$$

In case (I), $Y = -E_r/\sqrt{C}$, $Z = -vE_r/\sqrt{C} - E_p/\sqrt{\alpha}$ at $u = 0$. The form has the explicit pivots $4D_vE_r^2 + 4E_p^2$ plus a bounded nonnegative form. In case (II), $Y = (vE_p/\sqrt{\alpha} - E_J/\sqrt{D})/D_v$ and $Z = (-E_p/\sqrt{\alpha} - vE_J/\sqrt{D})/D_v$. The explicit pivots are $4E_J^2 + 4E_p^2$, again with a bounded nonnegative remainder. The hard residue is strict in both cases.

The determinant and the exact boundary Schur numerator are

$$\begin{aligned} \det \mathsf{B}(v) &= 4(1 + v^2)\{s(D\,v^2 + \alpha) + 4D\alpha(1 + v^2)\}, \\ Q_{10} &= 4\alpha A[d + 2(C + D)], \qquad Q_{12} = 4CD[d + 2(\alpha + A)], \\ Q_0 &= 8ACD + 8AC\alpha - 4ADd - 2Ad^2 - 2Cd^2 - 4Cd\alpha - 2Dd^2 - 4Dd\alpha - d^3 - 2d^2\alpha. \end{aligned} \tag{SC21}$$

Put $S = \alpha T$. Every coefficient of $\det \mathsf{B}/S$ and of $Q_{10}/S, Q_0/S, Q_{12}/S$ is bounded. This follows directly from (SC21), bounded A, bounded $C/T, D/T, \delta$, and bounded $1/\alpha$. Moreover

$$Q_{10}/S = 4A \asymp 1, \qquad Q_{12}/S = 4\delta[d + 2(\alpha + A)]/\alpha \asymp 1. \tag{SC22}$$

The normalized determinant has a positive lower bound for every bounded unit real v. Indeed it equals C/T times the determinant of the scaled hard form in case (I), and equals $(D/T)D_v^2$ times that determinant in case (II). The factors C/T and D/T, respectively, are positive units in their cases.

We now take residues of these bounded coefficients and use an ordinary real variable v. The exact real hard Schur function is

$$\bar{f}(v) = \frac{8(1 + v^2)}{\overline{\det \mathsf{B}/S}(v)} \left\{ \overline{Q_{10}/S} + \overline{Q_0/S}v^2 + \overline{Q_{12}/S}v^4 \right\}. \tag{SC23}$$

This formula is an ordinary real Schur calculation, never an inverse of $\det \mathsf{B}(v)$ in $K_c \otimes K_t$. The contact first-normal identities identify the graph at $\bar{a}$ with the unique minimizer of the scaled real hard form. The scalar contact jets give $\bar{f}(\bar{a}) = \bar{f}'(\bar{a}) = 0$. Because $\bar{a} \neq 0$, the quartic bracket in (SC23) is exactly

$$\overline{Q_{12}/S}(v^2 - \bar{a}^2)^2.$$

Its strict second derivative supplies the soft Schur pivot

$$\tfrac{1}{2}\bar{f}''(\bar{a}) = \frac{32(1+\bar{a}^2)\bar{a}^2\overline{Q_{12}/S}}{\overline{\det \mathsf{B}/S}(\bar{a})} > 0.$$

The entire five-root Gram therefore has strictly positive residue when u is infinitesimal or zero and $\bar{v} = \bar{a}$. The strict-residue Gram lemma proves tensor SOS. The exact identity

$$F(u,-v;W,-X,Y,-Z) = F(u,v;W,X,Y,Z)$$

supplies $\bar{v} = -\bar{a}$.

All other independent tests, including literal endpoints. If u is infinitesimal and v is a bounded unit with $\bar{v} \neq \pm\bar{a}$, retain the full original m, n positive square terms. In case (I) use the roots

$$X, \quad K_C = \sqrt{C}r, \quad K_\alpha = \sqrt{\alpha}p, \quad uW$$

and the exact inverse $Y = uW - K_C/\sqrt{C}$, $Z = vY - K_\alpha/\sqrt{\alpha}$. In case (II) use

$$X, \quad K_D = \sqrt{D}J, \quad K_\alpha = \sqrt{\alpha}p, \quad uW$$

and the inverse in (SC12), with unshifted J, p. The entire remaining parent is a bounded quadratic Gram on these four roots. Its separate X residue pivot is $2\bar{d}(1+\bar{v}^2)$; its hard pair is the strict form (SC20) in the scaled coordinates just displayed, and its soft Schur complement is $\bar{f}(\bar{v}) > 0$ by (SC23).

For the two endpoint neighborhoods $u, v \sim 0$ and $u, q = 1/v \sim 0$, divide by d, choose

$$\mu = \min(1, C/d, A/d, D/d, \alpha/d) > 0,$$

a bounded unit by (SC6), and retain all excesses together with the full m, n positive square terms. The selected bounded parent has signed weight one, $C = A = D = \alpha = \mu$, and $m = n = 0$. In the finite chart it is exactly a Gram on (X, Y, Z, uW). At $(u, v) = (0, 0)$ its hard block is positive and its soft Schur complement is $8Q_{10,\mu}/\det \mathsf{B}_\mu(0) > 0$. For the reciprocal chart use the polynomial homogeneous parent $q^2F_\mu(u, 1/q)$, on the same four roots. At $(u, q) = (0, 0)$ its hard block is $\mathrm{diag}(2, 4(C + A + \alpha) + 2, 4D)$ in the (X, Y, Z) variables, evaluated at the capped weights, and its soft Schur complement is $8Q_{12,\mu}/\det \mathsf{B}_{\mu,\infty} > 0$. These are bounded polynomial Grams, so they include the full infinitesimal neighborhoods and literal $v = 0$ or $q = 0$.

For completeness, the other open commuting axes use the same cap; here the other angle has a nonzero finite residue. At $v = 0$ the core is $2\{(uW - Y)^2 + (X + uZ)^2\}$ and the A, α positive square terms force $Y = Z = 0$; finite nonzero u then forces $W = X = 0$. At $u = \infty$ the homogeneous core is $2\{(W + vX)^2 + (vY - Z)^2\}$; the C, A positive square terms force $W = Y = 0$, and finite nonzero v forces $X = Z = 0$. At $v = \infty$ the core is $2(1 + u^2)(X^2 + Y^2)$; the C, D positive square terms force $W = Z = 0$ for finite nonzero u. Thus these complete state matrices are strict.

Only the two corners with $u = \infty$ remain. Put $p_u = 1/u$. At $(p_u, v) = (0, 0)$ the selected state form is $(2+8\mu)W^2 + 4\mu Y^2 + (2+4\mu)Z^2$. Its sole zero ray is X. In the chart $X = 1$, the five small variables (p_u, v, W, Y, Z) have exact quadratic part

$$(2 + 4\mu)(W + v)^2 + 2(Z + p_u)^2 + 4\mu(W^2 + Y^2 + Z^2) \succ 0.$$

At $(p_u, q) = (0, 0)$ the selected state form is $4\mu W^2 + (2 + 4\mu)X^2 + (2 + 8\mu)Y^2$. In its sole zero-ray chart $Z = 1$, the variables (p_u, q, W, X, Y) have exact quadratic part

$$2X^2 + (2 + 4\mu)(Y - q)^2 + 4\mu\{W^2 + Y^2 + (X - p_u)^2\} \succ 0.$$

There are no lower-order terms. Allocating every higher monomial to two coordinate roots leaves an infinitesimal factor, giving a strict-residue Gram for the entire selected polynomial at each corner. Retain the original excess positive square terms throughout. After pure testing projective normalization, states outside each displayed zero ray have positive residue value and are covered by the scalar strict-residue lemma.

Noncommuting-residue tests use the inherited arbitrary-physical-source theorem. This exhausts all independent projective tests. Finally homogenize the local graph identities in W, replacing R, T_s by RW, T_sW; they then include $W = 0$. Restore all pure positive coefficient and testing factors, and split complex states into real and imaginary parts. No source infinitesimal was compared with an independent testing infinitesimal anywhere in the argument.

Domain. This proves the finite pitchfork when $C/d, \alpha/d$ have positive real lower bounds, with arbitrary upper event scales, and its C2 reflection. Other actual-angle types and vanishing complementary lower bounds remain separate cases.

C.6.4 Source bounds from an infinitesimal first angle

Let $z = u_c^2$ be infinitesimal, $y = v_c^2 > 0$ arbitrary, and use the exact positive C2 source. Set

$$H = zh \in (0,1), \quad L = yt > 0, \quad B = zb = H + L,$$
$$A_c = 1 + zr, \quad S = A_c + L, \quad j = A_c - L/y > 0.$$

Here $0 < r < 1$. Define

$$\begin{aligned}\delta &= z + (1+z)B + (1+z)B^2 + 2z(1+z)Br + z^3r^2 + 2z^2r \\ &\quad + y\{(1+z)B + z + z^2r(2-r)\}, \\ G &= B + z(H + r - 1), \\ M &= (1+z)(B+zr)(H+y) - z(1+y)(1-r)L \\ &= (1+z)(B+zr)H + yG + yzj + zrL(1+y) > 0.\end{aligned}$$

The unchanged original coefficient formulas are

$$\begin{aligned} n &= \frac{(B+zr)A_cS - r\delta}{2r\delta}, & m &= \frac{z^2(1+y)(1-r)A_cS}{2G\delta}, \\ C &= \frac{A_cM}{2L\delta}, & A &= \frac{yA_c(B+zr)S}{2H\delta}, \\ D &= \frac{(1+z)(B+zr)(B-zyr)S}{2yj\delta}, & \alpha &= \frac{(B+zr)M}{2(1-H)\delta}. \end{aligned} \tag{GE1}$$

Assume C and α are strong, meaning each has a positive real lower bound in the coefficient field. Then B is a bounded positive unit. Indeed

$$\delta \geq (1+y)(B+z), \qquad M \leq (1+z)(B+z)(1+y)$$

give $CL \leq A_c(1+z)/2$, so L and B are bounded. If B were infinitesimal then so would H, while $\alpha(1-H) \leq (B+zr)(1+z)/2$ would force α infinitesimal, a contradiction.

Next, $n \geq 0$ gives $r\delta \leq (B+zr)A_cS$ and therefore $yr \leq A_cS$, since $\delta \geq y(B+zr)$. Thus G, $B - zyr$ and $\delta/(1+y)$ are positive units. Since $L < yA_c$, the ratio $(H+y)/(1+y)$ is a positive unit: if y is infinitesimal then $H = B - L$ is a unit, and otherwise $y/(1+y)$ is a unit. The positive second expression for M now shows that $M/(1+y)$ is a positive unit.

Finally $1-r$ is a unit. To see this, let $E(r) = (B+zr)A_cS - r\delta$, holding B, H, y, z fixed. Its coefficients divided by $1+y$ are bounded, while direct expansion at $r = 1$ gives

$$E(1) = -(H+y)(B+z)(1+z).$$

The right side divided by $1+y$ is negative with nonzero residue. If $1-r$ were infinitesimal, $E(r)/(1+y)$ would have the same negative residue, contradicting $n \geq 0$.

Consequently the following exact scaled quantities are positive units:

$$\frac{m}{z^2}, \quad CL, \quad \alpha(1-H), \quad \frac{AH(1+y)}{y}, \quad Dyj(1+y). \tag{GE2}$$

For the IH source subcase y is unbounded, hence $r = O(1/y)$ is infinitesimal, j is a unit, and Dy^2 is a positive unit. Put $\rho = B/y$, $s = r/\rho$, and $c_s = (1+L)/B$. Then s is bounded and, with $N = 4n+2$ and $\nu = Nr$,

$$\frac{\nu}{\rho} \longrightarrow 2c_s, \quad \frac{nr}{\rho} \longrightarrow \frac{c_s - s}{2}, \quad \frac{D}{\rho^2} \longrightarrow \frac{c_s}{2}, \quad CL \longrightarrow \frac{1}{2}, \quad \frac{Bm}{z^2} \longrightarrow \frac{c_s}{2},$$

$$AH \longrightarrow \frac{1+L}{2}, \qquad \alpha(1-H) \longrightarrow \frac{B}{2}. \tag{GE3}$$

Here arrows mean equality of residues of bounded expressions, with B, H, L, s replaced on the right by their residues. The expressions remain valid at $H_0 = 0$, $H_0 = 1$, or $L_0 = 0$ whenever allowed by $B_0 > 0$. The reason is that the scaled rational functions have denominators bounded away from zero after the displayed opening factors have been absorbed; no unscaled $1/H$, $1/(1-H)$ or $1/L$ is asserted bounded.

The II source corollary. For the original positive C2 source of GE1, suppose z, y are infinitesimal, without an initial strength assumption on C or α. Then α is strong if and only if $B = H + L$ is a positive unit. On this face C is unbounded, H and $1 - r$ are positive units, and

$$\frac{m}{z^2}, \quad \frac{A}{y}, \quad CL, \quad Dyj, \quad \alpha(1 - H) \asymp 1, \qquad nr = O(1). \tag{DI7}$$

Here every displayed comparison is in the coefficient field. Indeed $0 < L < yA_c$ makes L infinitesimal and B bounded. If B is infinitesimal, then H is infinitesimal and GE1 gives

$$0 < \alpha \le \frac{(B + zr)(1 + z)(H + y)}{2(1 - H)} \in \mathrm{Inf}_c,$$

using $\delta \ge B + z$ and $M \le (1 + z)(B + zr)(H + y)$. If B is a unit, then $H = B - L$ is a unit. The positive expressions in GE1 show that G, δ, M are units and hence that CL and $\alpha(1 - H)$ are positive units. Thus C is unbounded and α is strong, so the hypotheses used to prove GE2 hold. That proof also gives $1 - r$ a positive unit, from the strictly negative value of $E(1)$ and $E(r) \ge 0$. Now GE2 yields the five bounds in DI7 because H and $1 + y$ are units. Finally the exact GE1 identity

$$nr = \frac{(B + zr)A_cS - r\delta}{2\delta}$$

is bounded. No lower bound on r, j, or $1 - H$ was imposed, and the argument includes $n = 0$.

C.7 One small and one unbounded actual angle

C.7.1 Source bounds for weak C

Write $z = u_c^2$, $y = v_c^2$, and use the exact contact quantities

$$H = v_cY_c = zh, \quad L = v_c(u_cW_c - Y_c) = yt, \quad B = H + L = u_cv_cW_c = zb, \quad R = -u_cX_c = zr.$$

Thus $0 < H < 1$, $0 < R < z$, $L > 0$, and

$$L < y(1 + R), \quad B > yR, \quad G = B + R - z(1 - H) > 0.$$

Set $S = 1 + R + L$ and

$$\begin{aligned}\widetilde{\Delta} &= (1 + z)B^2 + 2(1 + z)BR + (1 + z)(1 + y)B \\ &\quad + (z - y)R^2 + 2z(1 + y)R + z(1 + y), \\ \widehat{N} &= (1 + z)(B + R)(H + y) - (1 + y)(z - R)L.\end{aligned}$$

These are exactly $z\Delta, z^2N_{\mathrm{src}}$. In particular,

$$\widehat{N} = (1 + z)(B + R)H + yG + z\{y(1 + R) - L\} + RL(1 + y) > 0.$$

The original coefficients, without an approximation, are

$$\begin{aligned} C &= \frac{(1 + R)\widehat{N}}{2L\widetilde{\Delta}}, & \alpha &= \frac{(B + R)\widehat{N}}{2(1 - H)\widetilde{\Delta}}, \\ A &= \frac{y(1 + R)(B + R)S}{2H\widetilde{\Delta}}, & D &= \frac{(1 + z)(B + R)(B - yR)S}{2\{y(1 + R) - L\}\widetilde{\Delta}}, \\ m &= \frac{z(1 + y)(z - R)(1 + R)S}{2G\widetilde{\Delta}}, & n + \frac{1}{2} &= \frac{z(B + R)(1 + R)S}{2R\widetilde{\Delta}}. \end{aligned} \tag{WI1}$$

Uniform source reduction for IH. Suppose z is positive infinitesimal, y is positive unbounded, and $n \ge 0$. Then C is weak if and only if B is unbounded. To prove the nontrivial direction, suppose B is bounded. Since $R < z$ and $G > 0$, one has $z \le 4(B + R)$: use $B \ge 1/2$ or, otherwise, $B + R > z(1 - H) > z/2$. Direct division in (WI1) gives bounded $\widetilde{\Delta}/[y(B + R)]$. Moreover,

$$\frac{\widehat{N}}{y(B + R)} = (1 + z)(1 + H/y) - (1 + 1/y)\frac{(z - R)L}{B + R}$$

has residue 1, since $L \le B$ and z is infinitesimal. Consequently $C \ge c/L$ for a positive real c, which is strong because L is bounded.

Now let B be unbounded. Put

$$\rho = B/y, \qquad e = 1 + R - L/y > 0.$$

Here $0 < \rho < 1 + R + H/y$, so ρ is bounded and its residue lies in $[0, 1]$. The following exact normalized quantities have residue 1:

$$\frac{\widetilde{\Delta}}{B(B+y)}, \qquad \frac{\widehat{N}}{yB}, \qquad \frac{L}{B}, \qquad \frac{S}{B}, \qquad \frac{G}{B}. \tag{WI2}$$

They give

$$n + \tfrac{1}{2} = \frac{\rho}{2r(1+\rho)}(1 + o(1)).$$

The inequality $n \ge 0$ therefore gives r/ρ bounded and $r \le \rho/[1+\rho](1+o(1)) \le 1/2 + o(1)$. In particular $1 - r$ is a positive unit, and $R/\rho = z(r/\rho)$ is infinitesimal. The normalized leading relations, uniform at arbitrary openings $H, 1 - H, e$, are

$$\begin{aligned} C &= \frac{1 + o(1)}{2B(1+\rho)}, & A &= \frac{B(1+o(1))}{2H(1+\rho)}, \\ \alpha &= \frac{B(1+o(1))}{2(1-H)(1+\rho)}, & m &= \frac{z^2(1-r)(1+o(1))}{2B(1+\rho)}, \\ D &= \frac{\rho^2(1+o(1))}{2e(1+\rho)}. \end{aligned} \tag{WI3}$$

Here each $1 + o(1)$ means a positive coefficient unit of residue one; it never denotes a comparison with a testing infinitesimal. In particular A, α are unbounded, m, C are infinitesimal, and Cm is infinitesimal for every weak-C IH source. Thus the four reciprocal-product exit can never admit this entire directed leaf. If ρ is infinitesimal then e is a unit and D is weak; if ρ is a unit then D is strong. Literal $n = 0$ remains allowed throughout the reduction. No lower bound on r was used.

C.7.2 Interior double corner

This proof covers the stated local testing chart; the remote commuting charts are treated below.

Theorem C.18 (Full-parent IH corner, interior shape)**.** *Use the exact positive C2 source with $d = 1$, the omitted original λ, β coefficients zero, and $n \ge 0$. Suppose its actual angles have type IH, and C is weak. Use the source notation of §C.7.1; thus z is infinitesimal and B unbounded. Write $\rho = B/y$ (a shape parameter, not an event coefficient). Assume*

$$\rho, \quad 1 - \rho \quad \textit{are positive bounded units.} \tag{IC1}$$

At every independent testing pair for which u and the homogeneous reciprocal coordinate $w = 1/v$ are infinitesimal or zero, the entire original parent has a separated-field tensor-SOS certificate, for all real or complex state coordinates. The source openings $H, 1 - H$ and the positive radial parameter r are arbitrary; n may be unbounded. Literal testing $u = 0$ and $v = \infty$, either testing sign, and literal source $n = 0$ are included. No state coordinate or certificate root is required to be bounded.

Proof. Put $N = 4n + 2$ and $\nu = Nr$. Source reduction (WI1)–(WI3) shows that ν and $1 - r$ are bounded positive units, even when N is unbounded. In residue,

$$\nu_0 = \frac{2\rho_0}{1+\rho_0}, \qquad 2r_0 \le \nu_0, \qquad D_0 = \frac{\rho_0^2}{2(1-\rho_0^2)}. \tag{IC2}$$

All assumptions and divisions in this sentence concern the coefficient field alone. The uniform coefficient estimates needed below are

$$BC \asymp 1, \quad \frac{Bm}{z^2} \asymp 1, \quad \frac{AH}{B} \asymp 1, \quad \frac{\alpha(1-H)}{B} \asymp 1, \quad D \asymp 1, \quad nr \le \nu/4.$$

In particular C, m are infinitesimal, nr, nr^2 are bounded, $mB/z \asymp z$, and $m\sqrt{B}/z \asymp z/\sqrt{B}$. These estimates are uniform in $r, H, 1 - H$ and do not assert that A/B or α/B is bounded at an opening.

The homogeneous testing parent is exactly

$$\begin{aligned} \widehat{F} &= 4nD_uD_wX^2 + 4mM^2 + 4CD_w(uW - Y)^2 + 4AD_uD_wY^2 + 4DJ^2 + 4\alpha D_u(Y - wZ)^2 \\ &\quad + 2\{J^2 + D_u(Y - wZ)^2 + D_w(X + uZ)^2 - D_uD_wZ^2\}, \\ M &= W - wX + uY - uwZ, \qquad J = uwW + uX - wY - Z, \qquad D_u = 1 + u^2, \quad D_w = 1 + w^2. \end{aligned} \tag{IC3}$$

For $w \neq 0$ it is $w^2F(u, 1/w)$; (IC3) defines the literal projective endpoint as well. Let G be (IC3) restricted to the graph

$$X = -ruZ, \qquad Y = HwZ. \tag{IC4}$$

Use the following five scalar roots, all defined using only pure coefficient radicals and inverses:

$$\begin{aligned} &R_0 = (uW - BwZ)/\sqrt{B}, && R_1 = (zW - BuwZ)/\sqrt{B}, \\ &R_2 = uwW - \rho Z, && T = (u^2 - z)Z, && E = \sqrt{BH}\,(w^2 - 1/y)Z. \end{aligned} \tag{IC5}$$

They vanish at every signed exact source contact. Define the pure coefficient quantities

$$\begin{aligned} K &= 4AH^2 + 8DHr + 4\alpha(1-H)^2 + 4m(H + r - 1)^2 + 2(H + r - 1)^2 + 4nr^2, \\ k &= K - 4B^2m/z^2 + 8znr^2 + 8AH^2/y. \end{aligned} \tag{IC6}$$

The exact source formulas and state/angular stationarity give the following finite polynomial Gram identity:

$$\begin{aligned} G = &\left(4BC - \frac{kz}{B}\right) R_0^2 + \frac{4Bm}{z^2} R_1^2 + 2(2C + 2D + 1)R_2^2 \\ &+ r^2\{4nw^2 + 2(2D + 2n + 1)\}T^2 + \frac{H}{B}\{4Au^2 + 2(2A + 2C + 2D + 1)\}E^2 \\ &+ \frac{ku}{B} R_0R_1 + \left\{-4r(2D + 1) + \frac{k}{B}\right\} R_2T - \frac{4\sqrt{H}(2C + 2D + 1)}{\sqrt{B}} R_2E. \end{aligned} \tag{IC7}$$

For clarity, the only non-obvious algebraic step is the exact toric conductor

$$B(u^2 - z)(w^2 - 1/y)Z^2 = (u^2 - z)ZR_2 - zR_0^2 + uR_0R_1. \tag{IC8}$$

Before (IC8), the last mixed radial term is $k(u^2 - z)(w^2 - 1/y)Z^2$. Expanding (IC7), its difference from G has only the four monomials $uwWZ, u^2Z^2, w^2Z^2, Z^2$; the unchanged original source sets all four coefficients exactly to zero.

All coefficients of (IC7) are bounded: BC, Bm/z^2, AH/B, $\alpha(1 - H)/B$, nr, D and k/B are bounded. No inverse of r is used, and $1/\sqrt{N}$ is bounded because $N \geq 2$. Moreover,

$$(k/B)_0 = \frac{2r_0}{1 + \rho_0}.$$

Add to the right side of (IC7) the following exact zero, obtained by one more toric conductor:

$$\rho T^2 + (u^2 - z)TR_2 - u(zR_0 - uR_1)(uR_0 - R_1) = 0. \tag{IC8a}$$

The new T^2 term has positive unit residue ρ_0. Every other coefficient introduced by (IC8a) is infinitesimal. This step is what makes the proof uniform when r is infinitesimal and n unbounded.

Suppress residue bars temporarily. The R_0, R_1, E diagonal residues are respectively

$$\frac{2}{1 + \rho}, \qquad \frac{2(1 - r)}{1 + \rho}, \qquad \frac{2}{1 + \rho},$$

which are positive. The only surviving off-diagonal block is

$$(R_2, T): \qquad \begin{pmatrix} \dfrac{2}{1 - \rho^2} & -\dfrac{r}{1 - \rho} \\ -\dfrac{r}{1 - \rho} & \rho + r\nu + 4Dr^2 \end{pmatrix}. \tag{IC9}$$

Using (IC2), its determinant is exactly

$$\frac{2\rho}{1 - \rho^2} + \frac{r\{4\rho - r(3\rho + 1)\}}{(1 - \rho)(1 + \rho)^2} > 0.$$

Indeed $0 \leq r \leq \rho/(1 + \rho)$, and consequently $4\rho - r(3\rho + 1) \geq \rho(3 + \rho)/(1 + \rho) > 0$. The first determinant summand is already strictly positive, including $r_0 = 0$. Thus (IC7) after the exact zero (IC8a) has strictly positive definite residue, including literal $n = 0$ and unbounded n.

It remains to restore the full state normals; the graph certificate alone is not the claimed conclusion. Put

$$E_X = \sqrt{N}(X + ruZ), \qquad E_Y = \sqrt{B/[H(1 - H)]}(Y - HwZ).$$

We give exact factorizations of both linear normal terms. Let $U = u^2, V = w^2, s = 1/y$ and define L_X, L_Y by $4L_X = \partial_X \widehat{F}|_{\text{graph}}$ and $4L_Y = \partial_Y \widehat{F}|_{\text{graph}}$. Direct expansion gives

$$\begin{aligned} L_X &= (2D+1)uR_2 - \frac{2m\sqrt{B}}{z} wR_1 + uZP_X(U,V), \\ P_X &= (2D+1)\rho - \frac{2mB}{z} V - 2nr(1+U)(1+V) - r(2D+1)U - 2D - r \\ &\quad + V\{1 - H - r - 2DH - 2m(H+r-1)\}. \end{aligned} \tag{IC10}$$

Every coefficient of P_X is bounded, and source stationarity gives $P_X(z,s) = 0$. Write its exact monic division as

$$P_X = (U-z)f_X(V) + (V-s)g_X.$$

Both f_X, g_X have bounded coefficients. Explicitly,

$$\begin{aligned} f_X(V) &= -2nr(1+V) - r(2D+1), \\ g_X &= -2mB/z - 2nr(1+z) + 1 - H - r - 2DH - 2m(H+r-1). \end{aligned}$$

Thus no inverse of r or of either opening is introduced. We obtain

$$L_X = \left(2D + 1 + \frac{g_X}{B}\right) uR_2 - \frac{2m\sqrt{B}}{z} wR_1 + uf_X(V)T - \frac{uwg_X}{\sqrt{B}} R_0. \tag{IC11}$$

Here we used the exact identity $B(V-s)Z = R_2 - \sqrt{B}\, wR_0$; this avoids dividing by $\sqrt{H}$ when H is infinitesimal. Every displayed multiplier is infinitesimal in the bounded tensor ring; in particular $m\sqrt{B}/z \asymp z/\sqrt{B}$.

For the other normal, put

$$\begin{aligned} f(V) &= 2m - 2C(1+V) - (2D+1)V, \\ P_Y(U,V) &= Bf(V) + 2AH(1+U)(1+V) + 2CH(1+V) + 2DHV + 2DrU + 2D \\ &\quad - 2\alpha(1-H)(1+U) + 2m(H+r-1)U + (H+r-1)U + H(V+1). \end{aligned}$$

Then

$$L_Y = \sqrt{B} f(V) R_0 + wZP_Y(U,V), \qquad P_Y(z,s) = 0.$$

The coefficients of P_Y/B are bounded. Write $P_Y = (U-z)f_Y(V) + (V-s)g_Y$. The two quotient coefficients are explicitly

$$\begin{aligned} f_Y(V) &= 2AH(1+V) + 2Dr - 2\alpha(1-H) + (2m+1)(H+r-1), \\ g_Y &= -B(2C+2D+1) + 2AH(1+z) + 2CH + 2DH + H. \end{aligned}$$

The displayed source estimates show directly that f_Y/B and g_Y/B have bounded coefficients, including both opening limits. The elementary exact identity

$$(U-z)\sqrt{B}\, wZ = zR_0 - uR_1$$

therefore gives

$$\frac{L_Y}{\sqrt{B/[H(1-H)]}} = \sqrt{H(1-H)} \left\{ f(V) + \frac{zf_Y(V)}{B} \right\} R_0 - \frac{\sqrt{H(1-H)}uf_Y(V)}{B} R_1 + \frac{\sqrt{1-H}wg_Y}{B} E. \tag{IC12}$$

Again every displayed multiplier is infinitesimal. This argument retains the full X, Y linear terms; none is estimated on the graph and then discarded.

Finally the normal quadratic block, in (E_X, E_Y), is bounded. Its residue is diagonal with entries

$$1, \qquad \frac{2}{1+\rho_0} > 0.$$

Indeed the X^2 coefficient before scaling is $4nD_uD_w + 4mw^2 + (4D+2)u^2 + 2D_w$; the Y^2 coefficient is $4CD_w + 4AD_uD_w + (4D+2)w^2 + (4\alpha+2)D_u + 4mu^2$. Their mixed coefficient is $-4uw(2m+2D+1)$. For the Y normalization, if $L = B/[H(1-H)]$, then A/L and α/L are bounded and

$$\frac{4(A+\alpha)}{L} = 4\left\{ \frac{AH}{B}(1-H) + \frac{\alpha(1-H)}{B} H \right\} \longrightarrow \frac{2}{1+\rho_0}.$$

Thus the stated normal residue remains strict even when $H_0 = 0$ or $H_0 = 1$; no denominator involving a vanishing opening was dropped. Equations (IC11)–(IC12) show that all cross terms from these normals to the five scalar roots have zero residue. Thus the exact seven-root Gram for the entire (IC3) is bounded and has positive definite residue. The bounded positive-residue Gram lemma yields tensor SOS. Its roots may be unbounded; it uses no mixed inverse or radical. The homogeneous chart and real/imaginary splitting give all endpoints and states stated in the theorem. □

This calculation proves the interior $0 < \rho_0 < 1$ testing corner. Section C.7.3 treats the source endpoints; Sections C.7.5 and C.7.4 prove the other testing charts.

C.7.3 The two source endpoints

We use the exact parent, graph, roots, coefficients K, k, and both normal factorizations (IC3)–(IC12) in the preserved interior-corner proof. Those identities are exact for every positive canonical C2 source; their validity does not require the interior-shape assumption. Here z and testing u, w are infinitesimal (zero testing coordinates are allowed), B is unbounded, $H \in (0,1)$, $\rho = B/y$, and $N = 4n+2$, $\nu = Nr$. All source and testing rates are independent.

The endpoint $\rho_0 = 0$. Put $s = r/\rho$ and $\tau = 1/B$. Nonnegativity of n and the exact source formulas imply that s is bounded. The following normalized source quantities have the indicated residues:

$$\nu/\rho \to 2, \quad nr/\rho \to (1-s)/2, \quad D/\rho^2 \to 1/2, \quad BC, Bm/z^2, AH/B, \alpha(1-H)/B \to 1/2,$$
$$k/(B\rho) \to 2s, \qquad 0 \le s_0 \le 1. \tag{WIE1}$$

These are uniform opening identities: after substituting $B = 1/\tau$, $y = 1/(\tau\rho)$, $r = s\rho$, their normalized rational functions have a nonzero denominator at $z = \tau = \rho = 0$, including $H = 0, 1$. Thus they concern simultaneous, not iterated, infinitesimal limits.

Replace T by $T_\rho = \rho T$ in (IC7). Its coefficients are bounded; for example $r^2 n/\rho^2 = s(nr/\rho)$ is bounded. Add the exact zero

$$T_\rho^2 + (u^2 - z)T_\rho R_2 - \rho u(zR_0 - uR_1)(uR_0 - R_1) = 0. \tag{WIE2}$$

This is (IC8a) multiplied by ρ, and uses no mixed inverse. The residue Gram has R_0, R_1, E diagonal entries $2, 2, 2$, and

$$(R_2, T_\rho): \quad \begin{pmatrix} 2 & -s_0 \\ -s_0 & 1+2s_0 \end{pmatrix}.$$

Its determinant $2 + 4s_0 - s_0^2 \ge 2$ is strictly positive on $[0,1]$.

The entire normal restoration survives this rescaling. In (IC11), replace $f_X T$ by $(f_X/\rho)T_\rho$, where

$$f_X/\rho = -2(nr/\rho)(1+w^2) - s(2D+1)$$

is bounded. Every multiplier still contains testing u or w, and $1/\sqrt{N}$ is bounded. The remaining coefficients in (IC11) are unchanged and infinitesimal. Formula (IC12) uses the root R_1 rather than T and remains bounded and infinitesimal unchanged. The normal quadratic residue is $\mathrm{diag}(1,2)$. Hence the full seven-root Gram is strict. This includes arbitrary positive $r, H, 1-H$ and literal $n = 0$.

The endpoint $\rho_0 = 1$. Set $e = 1 + zr - \rho + H/y = j > 0$, an infinitesimal, and write

$$\rho = \frac{1 + zr - e}{1 - H\tau}, \qquad \tau = 1/B.$$

The exact source formulas give the simultaneous normalized residues

$$De \to 1/4, \quad \nu \to 1, \quad nr \to (1-2r)/4, \quad BC \to 1/4, \quad Bm/z^2 \to (1-r)/4,$$
$$AH/B, \alpha(1-H)/B \to 1/4, \quad 0 \le r_0 \le 1/2. \tag{WIE3}$$

Put $\kappa = k - 8DHr$; then $\kappa/B \to r$. The scaled rational functions have regular nonzero denominators at $z = \tau = e = 0$ after the displayed powers of e are absorbed.

The large coefficient D must be retained in a square, not treated as a bounded Gram coefficient. Introduce the full-state root

$$F_D = \sqrt{D}\,(J + eZ), \qquad a = \sqrt{H/B}, \qquad Q = R_2 - rT - aE.$$

On the graph, $Q = J + eZ$. Subtracting its square from (IC7) gives the exact identity

$$\begin{aligned} G - 4DQ^2 = {} & (4BC - \kappa z/B)R_0^2 + 4BmR_1^2/z^2 + 2(2C+1)R_2^2 \\ & + r^2(4nw^2 + 4n + 2)T^2 + \frac{H}{B}\{4A(1+u^2) + 4C + 2\}E^2 \\ & + \kappa u R_0 R_1/B + (-4r + \kappa/B)R_2 T - 4a(2C+1)R_2 E. \end{aligned} \tag{WIE4}$$

Add the exact zero (IC8a). Restore the full normals

$$E_X = \sqrt{N}(X + ruZ), \qquad E_Y = \sqrt{B/[H(1-H)]}(Y - HwZ).$$

Use the exact relation

$$R_2 = F_D/\sqrt{D} + rT + aE - uE_X/\sqrt{N} + wE_Y/\sqrt{B/[H(1-H)]}. \tag{WIE5}$$

The inverse $1/\sqrt{D}$ is purely in the source field and infinitesimal. All coefficients after this substitution are bounded.

For completeness, the first-normal quotients of $\widehat{F} - 4F_D^2$ are precisely the D-removed versions of (IC11)–(IC12). Explicitly,

$$\begin{aligned}
\bar{f}_X &= -2nr(1+V) - r,\\
\bar{g}_X &= -2mB/z - 2nr(1+z) + 1 - H - r - 2m(H+r-1),\\
\bar{f} &= 2m - 2C(1+V) - V,\\
\bar{f}_Y &= 2AH(1+V) - 2\alpha(1-H) + (2m+1)(H+r-1),\\
\bar{g}_Y &= -B(2C+1) + 2AH(1+z) + 2CH + H.
\end{aligned}$$

The constant term in the Y monic polynomial acquires $2De$, but its monic quotient coefficients are unchanged; the X polynomial has the constant correction $-2De$. Exact source stationarity still sets the constant remainders to zero because $J + eZ$ vanishes at the contact. Thus the displayed quotient formulas reconstruct both linear normals. Their normalized coefficients are infinitesimal by (WIE3); in particular De is bounded and no bare D survives. The normal quadratic block of the D-removed expression has residue $\mathrm{diag}(1,1)$.

In the order $(R_0, R_1, T, E, F_D, E_X, E_Y)$, the complete residue Gram is diagonal:

$$\mathrm{diag}\{1, 1-r_0, 1+r_0-r_0^2, 1, 4, 1, 1\}. \tag{WIE6}$$

For instance the T^2 entry comes from substituting $R_2 = r_0 T$ in (WIE4): it is $r_0\nu_0 - r_0^2$, plus the unit contribution from (IC8a). Every entry is positive for $0 \le r_0 \le 1/2$. This proves full-parent tensor SOS at the corner also when $1-\rho$ is infinitesimal. Along with the preserved interior proof and the $\rho_0 = 0$ argument, every allowed source shape is covered at the independent testing corner u, w infinitesimal.

C.7.4 The open testing axis

Domain and source input. The source and testing real closed fields are independent. Let B_c, B_t be their rings of bounded elements and use $B = B_c \otimes_{\mathbb{R}} B_t$ for Gram coefficients. Source square roots and testing square roots are always taken in their respective fields. In the weak-C IH source chart, write $z = u_c^2$ infinitesimal, $y = v_c^2$ unbounded, $H = zh \in (0,1)$, $B_* = zb$ unbounded and $\rho = B_*/y$ bounded. The source entry calculation gives

$$0 \le \rho_0 \le 1, \qquad (B_*C)_0 = \left(\frac{AH}{B_*}\right)_0 = \left(\frac{\alpha(1-H)}{B_*}\right)_0 = c := \frac{1}{2(1+\rho_0)}. \tag{OA1}$$

In particular C is infinitesimal and A, α are unbounded. For $0 < \rho_0 < 1$,

$$D_0 = \frac{\rho_0^2}{2(1-\rho_0^2)}; \tag{OA2}$$

for $\rho_0 = 0$, D is positive infinitesimal; for $\rho_0 = 1$, D is unbounded. These statements follow from $D \sim \rho^2/[2e(1+\rho)]$ and $e_0 = 1 - \rho_0$. No lower bound on $n \ge 0$ is imposed.

Statement. For every testing u with nonzero finite residue and every infinitesimal $w = 1/v$, including $w = 0$, the complete reciprocal-v canonical C2 parent is a tensor sum of squares for every state. The same assertion holds for complex states by applying the real quadratic form to real and imaginary parts separately.

Exact parent and retained squares. Put $D_u = 1 + u^2$, $D_w = 1 + w^2$ and

$$M = W - wX + uY - uwZ, \qquad J = uwW + uX - wY - Z.$$

The parent is

$$\begin{aligned}
\widehat{F} = {} & 4nD_uD_wX^2 + 4mM^2 + 4CD_w(uW - Y)^2 + 4AD_uD_wY^2 + 4DJ^2 + 4\alpha D_u(Y - wZ)^2\\
& + 2\{J^2 + D_u(Y - wZ)^2 + D_w(X + uZ)^2 - D_uD_wZ^2\}.
\end{aligned}$$

Retain the entire n, m positive square terms. When $\rho_0 = 0$, retain the entire D positive square term as well and set $d = 0$. When $0 < \rho_0 < 1$, set $d = D$. When $\rho_0 = 1$, choose a sufficiently large positive real d below D and retain $4(D-d)J^2$. Denote the remainder by F_*. All retained terms are tensor squares or sums of tensor squares.

Use the graph $X = 0$, $Y = HwZ$, and define

$$P = \frac{uW}{\sqrt{B_*}}, \quad E = \sqrt{B_*}\, wZ, \quad V = uwW, \quad E_Y = \sqrt{\frac{B_*}{H(1-H)}}(Y - HwZ). \tag{OA3}$$

Crucially, the exact root relation is

$$PE = VZ. \tag{OA4}$$

On the graph, the C, A, α, d positive square terms are respectively

$$\begin{aligned} &4B_*CD_w(P - HE/B_*)^2, \\ &\frac{4D_u}{B_*}\{AH^2D_w + \alpha(1-H)^2\}E^2, \\ &4d\{V - (1 + Hw^2)Z\}^2. \end{aligned}$$

The remaining core is exactly

$$2V^2 - 4(1 + Hw^2)PE + \frac{2}{B_*}\{H^2 + u^2(1-H)^2 + H^2w^2\}E^2. \tag{OA5}$$

Thus every graph coefficient is bounded. The last E^2 coefficient is infinitesimal, uniformly for all $0 < H < 1$.

Restoring the Y normal. Let $L = B_*/[H(1-H)]$ and $Y_* = HwZ$. The first normal, divided by four, is

$$\begin{aligned} L_Y := \frac{1}{4}\partial_Y F_*\big|_{X=0,Y=Y_*} &= 2CD_w(Y_* - uW) + 2D_u\{AHD_w - \alpha(1-H)\}wZ \\ &\quad - 2dw\{V - (1 + Hw^2)Z\} - w\{V - (1 + Hw^2)Z\} + D_u(H-1)wZ. \end{aligned} \tag{OA6}$$

Every coefficient of $L_Y/\sqrt{L}$ in (P, E, V, Z) is infinitesimal when d is bounded, including $d = 0$. Indeed the only potentially finite contribution is

$$2D_u\frac{AH - \alpha(1-H)}{B_*}\sqrt{H(1-H)}E,$$

and it has zero residue by (OA1); the additional w^2 contribution is infinitesimal. The other terms contain C, $w/\sqrt{L}$ or $1/B_*$. The quadratic E_Y coefficient has residue $4cD_{u,0} > 0$, and the XE_Y coefficient has zero residue. These statements remain valid when H or $1 - H$ is infinitesimal. No inverse of d, H or $1 - H$ enters a Gram coefficient unaccompanied by its compensating factor.

The strict residue when $0 < \rho_0 < 1$. Write $s = \sqrt{1 + u_0^2} > 1$ and $r = \rho_0$. Using (OA4), split the core cross term as $-4PE = -4aPE - 4(1-a)VZ$, where a is any real number satisfying

$$\frac{s}{s+r} < a < \min\left\{\frac{s}{1+r}, \frac{s}{s-r}\right\}. \tag{OA7}$$

The interval is nonempty: both upper endpoints exceed $s/(s+r)$. The residue Gram splits into the positive E_Y pivot, a (P, E) block

$$G_{PE} = \begin{pmatrix} 2/(1+r) & -2a \\ -2a & 2s^2/(1+r) \end{pmatrix},$$

and a (V, Z, X) block

$$G_{VZX} = \begin{pmatrix} 2 + 4d_0 & -4d_0 - 2(1-a) & 2u_0(1+2d_0) \\ -4d_0 - 2(1-a) & 4d_0 & -4d_0u_0 \\ 2u_0(1+2d_0) & -4d_0u_0 & 2s^2 + 4d_0u_0^2 \end{pmatrix}. \tag{OA8}$$

The (V, X) principal block is positive definite. Its Schur complement is positive exactly when

$$\frac{s}{s+r} < a < \frac{s}{s-r},$$

as direct substitution of $d_0 = r^2/[2(1-r^2)]$ shows. Hence (OA7) makes both blocks strictly positive. This checks the entire X normal, not merely the scalar graph.

The endpoint $r = 1$. For finite $d > 0$, positivity of the (V, Z, X) block is equivalent to $a_- < a < a_+$, where

$$a_\pm(d) = \frac{(2d+1)s^2 \pm s\sqrt{2d(2d+1)}}{(2d+1)u_0^2 + 1}.$$

As $d \to \infty$, the lower endpoint tends to $s/(s+1)$ and the upper endpoint to $s/(s-1)$. Choose first a real a strictly between $s/(s+1)$ and $\min\{s/2, s/(s-1)\}$, then a sufficiently large finite d. The interval is nonempty because $s > 1$. Since source D is unbounded this cap is admissible, and the full residue is strict.

The endpoint $r = 0$. Take $a = 1$, $d = 0$, having retained the whole D positive square term, and omit Z from the root list. The residue is the direct sum of

$$2(P-E)^2 + 2u_0^2E^2, \qquad 2(V+u_0X)^2 + 2X^2, \qquad 2D_{u,0}E_Y^2. \tag{OA9}$$

It is strictly positive. The exact rewriting has bounded coefficients: (OA5) uses only P, E, V, and the first X normal from the core is $uV + u(1-H)w^2Z = uV + u(1-H)wE/\sqrt{B_*}$. Thus no unscaled Z or inverse of D remains.

Conclusion. In all three cases the original remainder has a bounded Gram matrix with strictly positive real residue. The bounded-Gram lifting lemma therefore makes it a tensor sum of squares. Adding the retained source-positive positive square terms proves the assertion. The argument is valid for arbitrary $n \geq 0$, including $n = 0$, and literal projective $w = 0$; no comparison between source and testing infinitesimals was used.

C.7.5 Complementary tests for weak C

Assume the weak-C IH source entry, with no restriction on $\rho_0 \in [0, 1]$, on positive $r, H, 1-H$, or on $n \geq 0$. The uniform source inputs are B unbounded, C, m infinitesimal, $BC, AH/B, \alpha(1-H)/B$ positive units with common residue $c = 1/[2(1+\rho_0)]$, and A, α unbounded. Every square retained below has nonnegative pure-source coefficient; products $1+u^2$ and $1+v^2$ are themselves testing sums of squares.

Finite v, infinitesimal u. Write $Y_A = \sqrt{A}Y$ and $Z_\alpha = \sqrt{\alpha}(vY - Z)$, so $Y = Y_A/\sqrt{A}$ and $Z = vY_A/\sqrt{A} - Z_\alpha/\sqrt{\alpha}$. Retain the complete n, C, D positive square terms. Use the five roots

$$U = uW, \quad X, \quad V = \sqrt{m}\,vW, \quad Y_A, \quad Z_\alpha.$$

The m positive square term is exactly $4(V - \sqrt{m}X + u\sqrt{m/\alpha}Z_\alpha)^2$. All coefficients in the remaining core are bounded after substituting Y, Z, and all coefficients involving Y or Z are infinitesimal. The full residue Gram is diagonal:

$$2U^2 + 2(1+v_0^2)X^2 + 4V^2 + 4(1+v_0^2)Y_A^2 + 4Z_\alpha^2. \tag{IR1}$$

This includes literal $u = 0$ or $v = 0$; redundant zero roots cause no problem for a positive Gram representation.

Finite v, unit u. Retain n, m, C, D. Use (W, X, Y_A, Z_α). The residue of the core and selected A, α positive square terms is

$$2\{u_0^2(W+v_0X)^2 + (1+v_0^2)X^2\} + 4(1+u_0^2)(1+v_0^2)Y_A^2 + 4(1+u_0^2)Z_\alpha^2, \tag{IR2}$$

which is strictly positive because $u_0 \neq 0$.

Finite v, unbounded u. Use the homogeneous coordinate $p = 1/u$, infinitesimal or zero. Retain n, m, D and use

$$P = pX, \quad Q = W + vX, \quad S = \sqrt{C}W, \quad Y_A, \quad Z_\alpha.$$

The C positive square term becomes $4(1+v^2)(S - p\sqrt{C/A}Y_A)^2$. The entire selected expression has bounded coefficients, and residue

$$2Q^2 + 2(1+v_0^2)P^2 + 4(1+v_0^2)S^2 + 4(1+v_0^2)Y_A^2 + 4Z_\alpha^2. \tag{IR3}$$

Thus every finite-v chart is complete, independently of n.

Both testing angles unbounded. Let $p = 1/u$, $w = 1/v$ be infinitesimal or zero and put $D_p = 1 + p^2$, $D_w = 1 + w^2$. Retain the entire n, m positive square terms. Set $d = \min(1, D) > 0$ in the source field and retain $4(D - d)J^2$. The remainder is exactly

$$\begin{aligned} F_* &= 4CD_w(W - pY)^2 + 4AD_pD_wY^2 + 4dJ^2 + 4\alpha D_p(Y - wZ)^2 \\ &\quad + 2\{J^2 + D_p(Y - wZ)^2 + D_w(pX + Z)^2 - D_pD_wZ^2\}, \\ J &= wW + X - pwY - pZ. \end{aligned}$$

Use the graph $X = -wW$, $Y = HwZ$, the normals $E_X = X + wW$, $E_Y = \sqrt{L}(Y - HwZ)$ with $L = B/[H(1-H)]$, and scalar roots

$$S = \sqrt{C}W, \quad E = \sqrt{B}\,wZ, \quad U = pwW, \quad P = \sqrt{d}\,pZ. \tag{IR4}$$

On the graph the exact expression is

$$\begin{aligned} F_*|_{\text{graph}} &= 4D_w(S - pH\sqrt{C/B}E)^2 + \frac{4D_p}{B}\{AH^2D_w + \alpha(1 - H)^2\}E^2 \\ &\quad + 4(1 + Hw^2)^2P^2 + 2D_wU^2 - \frac{4pD_w}{\sqrt{BC}}SE + \frac{2}{B}\{(1 - H)^2 + H^2p^2D_w\}E^2. \end{aligned} \tag{IR5}$$

All coefficients are bounded and its residue is $4S^2 + 4cE^2 + 2U^2 + 4P^2$.

The first X normal, divided by four, is exactly

$$L_X = -pD_wU + \frac{p(1 - H)w}{\sqrt{B}}E - 2\sqrt{d}(1 + Hw^2)P. \tag{IR6}$$

The first Y normal is

$$\begin{aligned} L_Y = &-2p\sqrt{C}D_wS + \frac{E}{\sqrt{B}}\{2Cp^2HD_w + 2D_p(AHD_w - \alpha(1 - H)) \\ &\quad + (2d + 1)p^2(1 + Hw^2) - D_p(1 - H)\}. \end{aligned} \tag{IR7}$$

After dividing by $\sqrt{L}$, all its coefficients are infinitesimal; the only possible nonzero term is proportional to $(AH - \alpha(1 - H))/B$, whose residue is zero. The quadratic normal coefficients have residues $2 + 4d_0$ and $4c$, and their mixed coefficient has zero residue. The only nonzero scalar-normal coupling is the (P, E_X) block

$$\begin{pmatrix} 4 & -4\sqrt{d_0} \\ -4\sqrt{d_0} & 2 + 4d_0 \end{pmatrix}, \tag{IR8}$$

whose determinant is exactly 8. Thus the full Gram is strict even when D is infinitesimal and even when $n = 0$. The normalization uses $\sqrt{d}$ but never $1/\sqrt{d}$ in a Gram coefficient.

Each displayed Gram is a representation of the complete selected state quadratic form, not just its graph restriction. The bounded positive-residue lemma and the retained squares prove tensor SOS. These charts, the open-axis theorem, and the three local source-shape proofs cover every independent testing pair.

C.7.6 Local completion for strong C

Assume the strong IH source entry of §C.6.4: z is infinitesimal, y unbounded, $B = H + L$ a bounded positive unit, $0 < H < 1$, $L > 0$, and C, α strong. The source parameter $\rho = B/y$ is infinitesimal, $s = r/\rho$ bounded, and $c_s = (1 + L)/B$ is a positive unit. Testing $u, w = 1/v$ are infinitesimal or zero. We use the exact graph, five roots and both full normal identities (IC3)–(IC12) of the preserved corner calculation; none of those algebraic identities assumes C weak.

The source quantities obey (GE3), and direct rational simplification gives additionally

$$\frac{k}{B\rho} \longrightarrow \frac{2(B^2s + BHs + BH - Bs - H^2 + H)}{B^2}. \tag{SL1}$$

No opening denominator occurs in this normalized expression.

Set $T_\rho = \rho T$ and $\eta = \sqrt{H(1 - H)/B}$, so $Y = HwZ + \eta E_Y$. The full Gram's entire explicit dependence on C is exactly

$$\begin{aligned} &4C(U_1^2 + U_2^2), \quad U_1 = uW - Y - LwZ = \sqrt{B}R_0 - \eta E_Y, \\ &U_2 = w(uW - Y) - (L/y)Z = R_2 - \sqrt{H/B}E - w\eta E_Y. \end{aligned} \tag{SL2}$$

The scalar and normal identities verifying this assertion use

$$\sqrt{H}R_2 - \sqrt{BH}\,wR_0 - \sqrt{B}E = 0. \tag{SL3}$$

Choose a positive real cap $c_* \le C$ and retain $4(C-c_*)(U_1^2+U_2^2)$. In the remaining exact Gram replace explicit C by c_*, leaving k and all other original source quantities unchanged. The distinction matters: this is an exact square subtraction from the complete Gram, not a perturbation of the source. Add λ times the exact zero (WIE2), where $\lambda > 0$ is a real number to be chosen. All resulting coefficients are bounded, using (GE3), (SL1), and the opening-normalized roots.

We display the entire residue. The roots R_1 and E_X have isolated positive pivots $2c_s$ and 1. The (R_0, E_Y) block is

$$\begin{pmatrix} 4Bc_* & -4c_*\sqrt{H(1-H)} \\ -4c_*\sqrt{H(1-H)} & (4c_*+4A+4\alpha+2)H(1-H)/B \end{pmatrix}. \tag{SL4}$$

Here and below all expressions in a residue matrix mean their bounded residues; products such as AH are evaluated by (GE3), even when A itself is unbounded. Its determinant is exactly $8c_*(1+L) > 0$. Put $a_* = 2(2c_*+1)$. The (R_2, E) block is

$$\begin{pmatrix} a_* & -a_*\sqrt{H/B} \\ -a_*\sqrt{H/B} & H(4A+a_*)/B \end{pmatrix}, \tag{SL5}$$

with determinant $2a_*c_s > 0$. The only remaining coupling is between R_2 and T_ρ, whose Gram entry is $c_s(H/B - s)$. The T_ρ diagonal is $\lambda + 2c_s s$. Choose λ larger than the finite Schur-complement threshold of the positive block (SL5). This produces a strictly positive residue matrix. Since this choice is made after taking a real residue, it is a real scalar and introduces no mixed inverse.

We spell out why no normal term is hidden in this block description. In (IC11), f_X/ρ is bounded by (GE3) and every scalar-root multiplier contains u or w. In (IC12), the only nonzero limiting term is the explicit C contribution to R_0E_Y, already included in (SL2)–(SL4). After the cap, all other normal multipliers contain z, u, or w. The unbounded factors A and α always occur as AH and $\alpha(1-H)$ after normalization, hence remain bounded also at $H_0 = 0, 1$. The normal X coefficient divided by $N = 4n+2$ has residue one, and its coupling to E_Y vanishes. The remaining Y coefficient is exactly the lower-right entry of (SL4). These facts follow directly from the displayed exact normal factorizations, not from scalar-graph positivity.

The bounded positive-residue Gram lemma now gives tensor SOS for the complete original parent, after restoring the retained positive $C - c_*$ squares. This holds for all states and all independent local tests, for both opening endpoints, for arbitrary $n \ge 0$, and at literal $u = 0$ or $w = 0$.

C.7.7 Complementary tests for strong C

Use the source entry of §C.6.4. In particular $B = H + L$ is a bounded positive unit, m, D are infinitesimal, C, A, α are strong, and the bounded products satisfy

$$CL \to \tfrac{1}{2}, \qquad AH \to (1+L)/2, \qquad \alpha(1-H) \to B/2. \tag{SR1}$$

Whenever an event coefficient is unbounded, a positive finite real cap may be used and its excess retained in its original positive square. When a coefficient is bounded it may be used unchanged.

Finite-penalty principle. Suppose a real symmetric quadratic form is strictly positive on the common kernel of finitely many real linear forms. Adding sufficiently large positive multiples of their squares makes the complete form strictly positive. To verify this without a limiting assertion, choose an orthogonal splitting into their common kernel and its complement. The upper block has a positive lower eigenvalue c, and the penalty matrix on the complementary block has a positive lower eigenvalue e. If the remaining lower block has norm at most b and the cross block norm is at most a, any penalty M with $Me > b + a^2/c$ makes the Schur complement strictly positive. All choices are real scalars. This principle makes the opening cases below uniform; no unbounded coefficient is inserted into a bounded Gram.

Testing u infinitesimal, v finite. Retain the n, D positive square terms and excess caps of C, A, α. Write their finite selected coefficients as $c, a, b > 0$. Use roots $U = uW, Y, Z, X, V = \sqrt{m}\,vW$. The m positive square term is exactly

$$4(V - \sqrt{m}X + \sqrt{m}\,uvY - \sqrt{m}\,uZ)^2.$$

Together with the selected positive square terms and the original core this is an exact bounded Gram expression. Its residue has isolated X, V pivots $2(1+v^2), 4$ and the following (U, Y, Z) block (residue notation suppressed):

$$\begin{pmatrix} 2+4cD_v & -2-4cD_v & -2v \\ -2-4cD_v & 2D_v+4cD_v+4aD_v+4bv^2 & -4bv \\ -2v & -4bv & 4b \end{pmatrix}. \tag{SR2}$$

In the interior, use $c = 1/(2L)$, $a = (1+L)/(2H)$ and $b = B/[2(1-H)]$. Its leading principal minors are

$$\frac{2(L+D_v)}{L}, \quad \frac{4(L+1)(D_v-H)(L+D_v)}{HL(1-H)}, \quad \frac{8B(L+1)^2D_v}{HL(1-H)}, \tag{SR3}$$

all positive. At an opening, apply the finite-penalty principle:

- $H_0 = 0$: the large A positive square term forces $Y = 0$; the restricted two-dimensional determinant is $4(L+1) > 0$.
- $H_0 = 1$: the large α positive square term forces $Z = vY$; the restricted determinant is $4(L+1)^2D_v/L > 0$.
- $L_0 = 0$: the large C positive square term forces $U = Y$; the restricted determinant is $4/(1-H) > 0$.
- $L_0 = 0$, $H_0 = 1$: both constraints leave the strictly positive form $2Y^2$.

The case $L_0 = H_0 = 0$ is excluded by $B_0 > 0$. Thus (SR2) can always be made strictly positive with bounded selected coefficients.

Testing u a unit, $w = 1/v$ infinitesimal. Retain n, m and the excess caps of C, A, α; retain the full D positive square term using its natural root. Put $U = uW$, $E = wZ$, $P_Z = \sqrt{D}Z$. In roots (U, Y, E, X, P_Z) the exact selected expression is

$$\begin{aligned} &4cD_w(U-Y)^2 + 4aD_uD_wY^2 + 4bD_u(Y-E)^2 \\ &\quad + 4\{\sqrt{D}(wU + uX - wY) - P_Z\}^2 \\ &\quad + 2\{(wU + uX - wY)^2 + D_u(Y-E)^2 + D_wX^2 - E^2 - 2UE + 2YE + 2uwXE\}. \end{aligned} \tag{SR4}$$

Its isolated X, P_Z residue pivots are $2D_u, 4$. The (U, Y, E) block is

$$\begin{pmatrix} 4c & -4c & -2 \\ -4c & 4c + (4a+4b+2)D_u & -4bD_u - 2u^2 \\ -2 & -4bD_u - 2u^2 & 4bD_u + 2u^2 \end{pmatrix}. \tag{SR5}$$

For interior openings the leading minors are

$$2/L, \qquad 4(1+L)D_u/[HL(1-H)], \qquad 8u^2(1+L)^2D_u/[HL(1-H)], \tag{SR6}$$

strict because $u_0 \neq 0$. The opening constraints and restricted positive determinants are respectively

$$\begin{array}{c|c|c} H_0 = 0 & Y = 0 & 4u^2(1+1/L) \\ H_0 = 1 & Y = E & 4u^2(1+L)/L \\ L_0 = 0 & U = Y & 4u^2D_u/[H(1-H)] \end{array}$$

and at $L_0 = 0, H_0 = 1$ the remaining form is $2u^2Y^2$. Finite penalties therefore give a strict full Gram at every opening.

Both testing angles unbounded. Use (IR4)–(IR7), replacing C by a finite positive cap c_* and setting $d = D$ (which is infinitesimal in the strong IH source). Retain the excess C positive square term and the entire n, m positive square terms. The roots are

$$\begin{aligned} &S = \sqrt{c_*}W, \quad E = \sqrt{B}\,wZ, \quad U = pwW, \quad P = \sqrt{D}\,pZ, \\ &E_X = X + wW, \qquad E_Y = \sqrt{B/[H(1-H)]}(Y - HwZ). \end{aligned}$$

The exact graph identity (IR5) remains valid. The E^2 coefficient has residue $2c_s$, $c_s = (1+L)/B$, because its core contribution $2(1-H)^2/B$ must be retained when B is a unit. In (IR7), the potentially nonzero first-normal coefficient is now

$$2\{AH - \alpha(1-H)\} - (1-H),$$

whose residue is zero by (SR1). The normal E_Y pivot is also $2c_s$. All scalar-normal cross coefficients are infinitesimal, since $\sqrt{D}$ is infinitesimal. Thus the full diagonal residue is

$$\operatorname{diag}(4, 2c_s, 2, 4, 2, 2c_s) > 0. \tag{SR7}$$

The remaining remote charts. When u is unbounded and v is a unit, the homogeneous core has residue $2\{(W+vX)^2+(vY-Z)^2\}$. Positive finite caps on the C and A positive square terms supply W^2 and Y^2, respectively, so the residue is strict on (W,X,Y,Z). When u is unbounded and v infinitesimal, put $p=1/u$ and use the five roots $(P,Q,W,Y,Z)=(pX,W+vX,W,Y,Z)$. Positive caps c,a,b give the residue

$$2Q^2+2(P+Z)^2+4cW^2+4aY^2+4bZ^2>0. \tag{SR8}$$

When u is a unit and v infinitesimal, the core has residue $2\{(uW-Y)^2+(X+uZ)^2\}$; the positive A and α caps give Y^2 and Z^2, making the residue strict. In these expressions all excess event coefficients are retained in their positive squares. When both testing angles are units, the inherited noncommuting-test theorem applies independently of source scales.

Exhaustive join. Classify each testing angle in the projective line as infinitesimal, a bounded unit, or unbounded, with literal zero and infinity included in the corresponding homogeneous chart. The infinitesimal/unbounded cell is the local theorem of §C.7.6. The infinitesimal/finite cells follow from (SR2); the unit/unbounded cell from (SR4); the unbounded/unbounded cell from (SR7); the unbounded/finite and unit/infinitesimal cells from (SR8) and its preceding argument; and the unit/unit cell from the inherited theorem. These nine cells are exhaustive. Both exact sign gauges $(u,W,X)\mapsto(-u,-W,-X)$ and $(v,W,Y)\mapsto(-v,-W,-Y)$ preserve the full parent, so every signed testing chart is included. Every certificate retains the full state normals and the omitted original λ,β remain zero. Therefore the entire strong IH source leaf has all-tests tensor-SOS case, including $n=0$ and every projective endpoint.

C.8 One ordinary and one small actual angle

C.8.1 Source reduction at an ordinary–small contact

This section records the original source, every parent positive square term, and the bounded products used in the following reciprocal and local certificates. It is an exact algebraic reduction, not an all-tests case. The coefficient and testing fields remain independent real closed extensions of $\mathbb{R}$.

Source definitions and physical domain. Take the actual source angles $u_c=a$ and $v_c=k$, where $a>0$ is a bounded unit and $k>0$ is infinitesimal. Set $z=a^2$ and $\varepsilon=k^2/z$. In the original canonical C2 chart put

$$h=b-\varepsilon t,\quad A_c=1+zr,\quad j=A_c-t,\quad g=(1+z)b-1+r-\varepsilon zt,\quad S=A_c+\varepsilon zt. \tag{UR1}$$

The native positive gaps are $r,1-r,t,h,g,j,1-zh$; assume all original event signs, including $n\ge 0$, and assume C,α are bounded below by positive real numbers. No positive lower bound on a native opening is imposed. Define the two complete source polynomials

$$\begin{aligned}\Delta&=1+(1+z)b+z(1+z)b^2+2z(1+z)br+z^2r^2+2zr\\&\quad+\varepsilon z\{(1+z)b+1+zr(2-r)\},\\N&=(1+z)(b+r)\{b+\varepsilon(1-t)\}-\varepsilon(1+\varepsilon z)(1-r)t.\end{aligned} \tag{UR2}$$

With $d=1$ and $\lambda=\beta=0$, the unchanged original coefficients are

$$\begin{aligned}n&=\frac{(b+r)A_cS-r\Delta}{2r\Delta},&m&=\frac{(1+k^2)(1-r)A_cS}{2g\Delta},\\C&=\frac{A_cN}{2\varepsilon t\Delta},&A&=\frac{\varepsilon A_c(b+r)S}{2h\Delta},\\D&=\frac{(1+z)(b+r)(b-k^2r)S}{2\varepsilon j\Delta},&\alpha&=\frac{z^2(b+r)N}{2(1-zh)\Delta}.\end{aligned} \tag{UR3}$$

These are the original rational C2 source formulas, with its angle-square parameter replaced by z and its squared angle-ratio parameter by ε; all quantities formerly supplied by a source-file reference are displayed in (UR1)–(UR3).

The source bounds used below follow directly from these definitions. First $t<A_c$, $h<1/z$, and $b=h+\varepsilon t$ give bounded t,h,b. The inequality $(1+z)b+r>1$ gives a positive lower bound on $b+r$; (UR2) makes Δ a positive bounded unit. The exact positive identity

$$N=(1+z)(b+r)h+\varepsilon\{g+j+rt(1+\varepsilon z)\} \tag{UR4}$$

shows that infinitesimal h would make b,N infinitesimal and $1-zh$ a unit, contradicting strong α. Thus h,b,N are units. The numerator of $2rn\Delta$ evaluated at $r=1$ is exactly

$$-z(\varepsilon+h)(b+1)(z+1).$$

It has negative unit residue, so $n \geq 0$ forces $1 - r$ to be a unit. Consequently

$$mg, \quad A/\varepsilon, \quad \varepsilon Ct, \quad \varepsilon Dj, \quad \alpha(1 - zh) \quad \text{are positive bounded units,} \qquad nr \text{ is bounded.} \tag{UR5}$$

In particular m is strong, A is infinitesimal, and C, D may have arbitrarily large positive sizes. Literal $n = 0$ is retained.

The entire parent and exact contact. For independent testing angles and a homogeneous state define

$$D_u = 1 + u^2, \quad D_v = 1 + v^2, \quad r_{\mathrm{amp}} = uW - Y, \quad Q = uX - Z,$$
$$J = r_{\mathrm{amp}} + vQ = uW + uvX - Y - vZ,$$
$$M = vW - X + uvY - uZ, \quad p = vY - Z, \quad B = X + uZ.$$

The signed core and the full original parent are

$$\mathcal{K} = 2\{J^2 + D_u p^2 + D_v B^2 - D_u D_v Z^2\},$$
$$F = 4nD_u D_v X^2 + 4mM^2 + 4CD_v r_{\mathrm{amp}}^2 + 4AD_u D_v Y^2 + 4DJ^2 + 4\alpha D_u p^2 + \mathcal{K}. \tag{UR6}$$

Normalize the actual contact by $W_c = 1$. Introduce the bounded pure-source quantities

$$x = -r/b, \quad \rho = t/(ab), \quad \sigma = -j/(ab), \quad s = \sigma - \rho = -A_c/(ab), \quad \mu = g/b, \quad \pi = (zh - 1)/(ab). \tag{UR7}$$

Here s is a nonzero bounded unit. The exact stationary contact is

$$(u_c, v_c; W_c, X_c, Y_c, Z_c) = \left(a, k; 1, kx, a - \rho k^2, \frac{k}{ab}\right), \tag{UR8}$$

and its amplitudes are

$$r_{\mathrm{amp},c} = \rho k^2, \quad J_c = \sigma k^2, \quad M_c = \mu k, \quad p_c = \pi k, \quad Q_c = sk.$$

Substitution of (UR3) gives $F = 0$ and all four state and both angle first derivatives zero at (UR8). These are the unchanged native contact identities; no equality of independent testing rates is used.

Products with all possible opening poles cancelled. Define two positive bounded units

$$c_1 = \frac{zA_c N}{2\Delta}, \qquad c_2 = \frac{z(1+z)(b+r)(b-k^2 r)S}{2\Delta}.$$

The exact products required by the local cap argument are

$$\begin{aligned} nx &= -\frac{(b+r)A_c S - r\Delta}{2b\Delta}, & m\mu &= \frac{(1+k^2)(1-r)A_c S}{2b\Delta}, \\ \alpha\pi &= -\frac{z^2(b+r)N}{2ab\Delta}, & C\rho k^2 &= \frac{c_1}{ab}, \qquad D\sigma k^2 = -\frac{c_2}{ab}. \end{aligned} \tag{UR9}$$

Their products with the bounded factors x, μ, π also show $nx^2, m\mu^2, \alpha\pi^2$ bounded. Crucially, cancellation before estimating gives the further exact identity

$$Ct - Dj = \frac{(1+z)(b+r)\{A_c j - zt(b - \varepsilon z r)\} - A_c(1+\varepsilon z)(1-r)t}{2\Delta}. \tag{UR10}$$

Thus $C\rho + D\sigma = (Ct - Dj)/(ab)$ is bounded for every positive rate of t, j, including infinitesimal openings.

Set

$$L = C + D, \qquad q = D/L, \qquad H = CD/L, \qquad \tau = (C\rho + D\sigma)/L, \qquad \gamma = k^2 H.$$

Here $q, 1/L, \tau, L\tau$ are bounded, and

$$\gamma = \frac{c_1 c_2}{c_1 j + c_2 t} \asymp 1, \qquad t + j = A_c \asymp 1. \tag{UR11}$$

The denominator in (UR11) is a positive unit even if either opening is infinitesimal. At finite source residues it gives precisely H_0 in (RC3). Equations (UR9)–(UR11) also show that the bounded finite part extends continuously to the allowed boundary residues: no inverse of $r, g, 1 - zh, t, j$ remains in these products.

Proposition C.19 (Exact bounded UI remainder)**.** *For bounded source caps $0 \le n_* \le n$ and positive bounded-unit caps $m_* \le m$, $\alpha_* \le \alpha$, the parent (UR6) is the sum of explicit positive centered positive square terms and a bounded polynomial remainder G. Every centered positive square term vanishes with its full first jet at (UR8), so G has exact zero value and all first derivatives there.*

Proof. The exact original C, D combination is

$$CD_v r_{\mathrm{amp}}^2 + DJ^2 = L(r_{\mathrm{amp}} + qvQ)^2 + Hv^2Q^2 + Cv^2 r_{\mathrm{amp}}^2.$$

Retain the three nonnegative positive square terms

$$\begin{aligned} P_L &= 4L(r_{\mathrm{amp}} + qvQ - \tau k^2 W)^2, \\ P_H &= 4H(vQ - sk^2 W)^2, \\ P_C &= 4Cv^2(r_{\mathrm{amp}} - \rho k^2 W)^2. \end{aligned} \tag{UR12}$$

Their subtraction leaves exactly

$$\begin{aligned} G_{CD} = {} & 8L\tau k^2 W(r_{\mathrm{amp}} + qvQ) - 4L\tau^2 k^4 W^2 \\ & + 8Hsk^2 vWQ - 4Hs^2 k^4 W^2 \\ & + 8C\rho k^2 v^2 W r_{\mathrm{amp}} - 4C\rho^2 k^4 v^2 W^2. \end{aligned} \tag{UR13}$$

All its coefficients are bounded by (UR9)–(UR11), including $L\tau^2 = (L\tau)^2/L$. Centering at $k^2 W$ is essential to this boundedness assertion.

Also retain

$$\begin{aligned} P_n &= 4(n - n_*)D_u D_v (X - xvW)^2, \\ P_m &= 4(m - m_*)(M - \mu vW)^2, \\ P_\alpha &= 4(\alpha - \alpha_*)D_u (p - \pi vW)^2. \end{aligned} \tag{UR14}$$

The exact corresponding remainders are

$$\begin{aligned} G_n &= 4n_* D_u D_v X^2 + 8(n - n_*)xvW D_u D_v X - 4(n - n_*)x^2 v^2 W^2 D_u D_v, \\ G_m &= 4m_* M^2 + 8(m - m_*)\mu vWM - 4(m - m_*)\mu^2 v^2 W^2, \\ G_\alpha &= 4\alpha_* D_u p^2 + 8(\alpha - \alpha_*)\pi vW D_u p - 4(\alpha - \alpha_*)\pi^2 v^2 W^2 D_u. \end{aligned} \tag{UR15}$$

These coefficients are bounded by (UR9) and the bounded caps. With

$$G = G_{CD} + G_n + G_m + G_\alpha + 4AD_u D_v Y^2 + \mathcal{K},$$

the entire exact identity is

$$F = P_L + P_H + P_C + P_n + P_m + P_\alpha + G. \tag{UR16}$$

It includes the signed core and every original state-normal term. All six retained roots vanish at (UR8), proving the contact-jet assertion. This proposition does not claim positivity of G. □

For the local theorem retain $P_L, P_C, P_n, P_m, P_\alpha$ outside the bounded Gram and combine G with

$$P_H = 4\gamma R_h^2, \qquad R_h = (vQ - sk^2 W)/k. \tag{UR17}$$

The factor $1/k$ is pure coefficient. The exact graph normals are

$$E_X = X - xvW, \quad E_Z = Z - (ux - s)vW, \quad E_r = r_{\mathrm{amp}} - \rho k^2 W.$$

Their inverse is polynomial:

$$X = E_X + xvW, \quad Z = E_Z + (ux - s)vW, \quad Y = uW - \rho k^2 W - E_r, \quad Q = svW + uE_X - E_Z.$$

Thus no state is discarded, including $W = 0$. The following reciprocal lemma identifies the leading relative-angle completion; the local certificate supplies the complete normal lift, cap choice, and strict Gram. The outer-angle and other commuting-axis join remains a separate obligation, so this reduction adds no all-tests source case.

C.8.2 Reciprocal completion of a singular angle

The following algebraic identity applies at the indicated singular angle.

Lemma C.20 (Reciprocal contact completion)**.** *Let A_0, A_2 be symmetric matrices over a real closed field, let q_0 be a column vector, and consider*

$$f(q,V,W) = q^T(A_0 + V^2A_2)q + 2VWb^Tq + (c_0 + c_2V^2)W^2.$$

Suppose $(q,V,W) = (q_0,1,1)$ is a zero with all first derivatives zero. Then

$$b = -(A_0+A_2)q_0, \qquad c_0 = q_0^TA_2q_0, \qquad c_2 = q_0^TA_0q_0,$$

and the entire polynomial has the exact decomposition

$$\boxed{f = (q - Vq_0W)^TA_0(q - Vq_0W) + (Vq - q_0W)^TA_2(Vq - q_0W).} \tag{RC1}$$

Consequently $A_0, A_2 \succeq 0$ imply a sum of squares. If $A_0 \succ 0$ and $q_0^TA_2q_0 > 0$, its nonzero zero states have $V = \pm 1$ and $q = Vq_0W$.

Proof. The q equation gives the displayed value of b. The V equation then gives $c_2 = q_0^TA_0q_0$, and the W equation gives $c_0 = q_0^TA_2q_0$. Expanding the right side of (RC1) proves the identity. The zero statement follows first from the strictly positive A_0 term and then from $(V^2-1)^2W^2q_0^TA_2q_0$. □

Homogeneous form without a mixed inverse. For a positive pure-coefficient k, substitute $q = Q/k$ and $V = v/k$, and multiply (RC1) by k^2. It becomes

$$(Q - vq_0W)^TA_0(Q - vq_0W) + \frac{1}{k^2}(vQ - k^2q_0W)^TA_2(vQ - k^2q_0W). \tag{RC2}$$

Thus its roots use only the pure-coefficient factor $1/k$. There is no inverse or radical of a mixed tensor element and no comparison between a testing rate and k.

The bounded-source C2 UI leading model. Here $u_c = a$ is a positive bounded unit and $v_c = k$ is infinitesimal, with $d = 1$. To state a residue model, first restrict to the subdomain where the original n, m, α are bounded and the source parameters $r, g, 1 - a^2b$ have nonzero residues. In the following formulas a, b, r denote those residues, $z = a^2$, and

$$A_c = 1 + zr, \quad g = (1+z)b - 1 + r, \quad \Delta = 1 + (1+z)b + z(1+z)b^2 + 2z(1+z)br + z^2r^2 + 2zr.$$

The source formulas give

$$\begin{aligned} n &= \frac{(b+r)A_c^2 - r\Delta}{2r\Delta}, \qquad m = \frac{(1-r)A_c^2}{2g\Delta}, \qquad \alpha = \frac{z^2(1+z)b(b+r)^2}{2(1-zb)\Delta}, \\ H_0 &= \overline{k^2CD/(C+D)} = \frac{z(1+z)(b+r)b}{2\Delta}, \qquad A_0^{\mathrm{weak}} = \overline{A/k^2} = \frac{A_c^2(b+r)}{2zb\Delta}. \end{aligned} \tag{RC3}$$

Normalize the actual contact by $W = 1$. With $v = kV$, $X = k\xi$, $Z = k\zeta$ and the leading constraint $Y = aW$, the leading polynomial of F/k^2 is

$$\begin{aligned} f = {} & 4n(1+a^2)\xi^2 + 4m\{V(1+a^2)W - \xi - a\zeta\}^2 + 4H_0V^2(a\xi - \zeta)^2 \\ & + 4A_0^{\mathrm{weak}}(1+a^2)a^2W^2 + (4\alpha+2)(1+a^2)(VaW - \zeta)^2 \\ & + 2(\xi + a\zeta)^2 - 2(1+a^2)\zeta^2. \end{aligned} \tag{RC4}$$

It has the form of the lemma, with $q = (\xi,\zeta)^T$,

$$q_0 = (-r/b, 1/(ab))^T,$$

$$q^TA_0q = 4n(1+a^2)\xi^2 + (4m+2)(\xi+a\zeta)^2 + 4\alpha(1+a^2)\zeta^2, \qquad q^TA_2q = 4H_0(a\xi - \zeta)^2.$$

Here $A_0 \succ 0$ and $H_0 > 0$. Put $s_0 = aq_{0,1} - q_{0,2} = -(1+a^2r)/(ab) \neq 0$. The exact source identities (RC3) verify all three coefficient relations in the lemma. Hence the apparently problematic relative normal and scale difference combine into the single root

$$\frac{v(aX - Z) - s_0k^2W}{k}. \tag{RC5}$$

The ordinary centered normals $(X,Z) - vq_0W$ supply the remaining positive directions. Requiring separate positive pivots for the relative normal and the scale difference would obscure this completion.

This proves an exact leading-model identity on its stated bounded subdomain. It does not discard the additional large source scales: those are handled algebraically by the centered-positive square term reduction in Section C.8.1. The complete local lifted Gram is proved in Section C.8.3; the outer projective join remains a separate obligation.

C.8.3 Local contact certificate

This section uses the exact full-parent reduction and source notation in Section C.8.1. It proves a local independent-test statement. The other commuting axis and the outer projective join are not included.

Theorem C.21 (Local ordinary–small C2 certificate)**.** *Use the unchanged physical C2 source with $d = 1$, actual contact $u_c = a$ a positive bounded unit and $v_c = k$ a positive infinitesimal, and C, α bounded below by positive real numbers. Then the full parent is tensor SOS for every independent test with v infinitesimal and $u - a$ infinitesimal, and every projective state. The original $n = 0$ boundary and all positive source opening rates are included. No upper bound on n, m, C, D, α is imposed.*

Here $u - a$ is the difference of the corresponding tensor factors. The residue condition means that their real residues agree; it does not identify the source and testing fields or compare their rates. The exact sign gauges give the corresponding negative-angle charts.

Bounded full remainder and its exact graph. The positive square term reduction leaves a bounded polynomial remainder G and the positive relative square

$$4\gamma R_h^2, \qquad \gamma = k^2CD/(C+D) \asymp 1, \qquad R_h = \frac{v(uX - Z) - sk^2W}{k}.$$

Every other retained positive square term is an exact positive square vanishing at the full actual contact. For positive bounded caps m_*, α_* and $0 \le n_* \le n$, set

$$E_X = X - xvW, \quad E_Z = Z - (ux - s)vW, \quad E_r = uW - Y - \rho k^2 W, \qquad Q_N = uE_X - E_Z.$$

The full inverse is polynomial. All terms of G, including its complete first-normal rows and quadratic normal block, are retained. There are no higher state-normal terms.

Put $w = u - a$, $\xi = v^2$, and use the additional separated roots

$$S = vwW, \qquad T = kwW, \qquad R = (v^2 - k^2)W/k.$$

The exact identities

$$kR = \frac{kR_h - vQ_N}{s}, \qquad RT = \frac{R_hT - SQ_N}{s}, \qquad S^2 - T^2 = \frac{w}{s}(R_hT - SQ_N) \tag{UI1}$$

use only the bounded pure-source inverse $1/s$.

Complete scalar and normal lifting. On the graph $E_X = E_Z = E_r = 0$, write $G = \phi(u, \xi)W^2$. Here $r_{\mathrm{amp}} = \rho k^2 W$, $Q = svW$, and M, p, B are bounded polynomials times vW. In (UR13)–(UR16), every scalar term therefore contains k^2 or $\xi = v^2$ with a bounded coquotient: the canceled products (UR9)–(UR11) bound the C, D terms, and A/k^2 is bounded by (UR5). Thus $\phi \in (k^2, \xi)$ in the bounded source polynomial ring. The exact contact identities give $\phi = \phi_u = \phi_\xi = 0$ at $(u, \xi) = (a, k^2)$. Successive monic Taylor division gives

$$\phi = w^2(k^2A_0 + \xi A_1) + w(\xi - k^2)B + (\xi - k^2)^2C_0, \tag{UI2}$$

with bounded coefficients. To see the asserted first factor directly, divide first in $u - a$; the second u coquotient stays in (k^2, ξ). The constant and first u terms are then divided in $\xi - k^2$. No source infinitesimal is inverted by these operations. Equations (UI1) give the allocation explicitly:

$$\phi W^2 = A_0T^2 + A_1S^2 + \frac{B}{s}(R_hT - SQ_N) + \frac{C_0}{s^2}(kR_h - vQ_N)^2.$$

Every coefficient is bounded, and the last term contributes only zero-residue coefficients to the Gram on (R_h, E_X, E_Z).

The full first E_X, E_Z coefficients are $vL_X(u, \xi)$ and $vL_Z(u, \xi)$. Exact contact gives $L_i(a, k^2) = 0$, so they are

$$v\{wL_{i0} + (\xi - k^2)L_{i1}\}.$$

The first summand is allocated on S, and the second uses $vkR = v(kR_h - vQ_N)/s$. The entire E_r coefficient belongs to (k^2, ξ) with bounded coquotients and vanishes at contact. Indeed, differentiating (UR13) gives bounded multiples of k^2 and v^2; the derivatives of M, p contain v, and the A term contains k^2. For the signed core, $\frac{1}{2}\partial_{E_r}\mathcal{K} = 2J - 2D_uvp$ on this graph, which has the same factors. The same division writes the coefficient as

$$w(k^2L_{r0} + \xi L_{r1}) + (\xi - k^2)L_{r2}.$$

It consequently uses kT, vS, kR. All these E_r couplings have zero residue. Thus (UI1)–(UI2) produce an exact bounded Gram on

$$(E_r, E_X, E_Z, R_h, S, T). \tag{UI3}$$

This is a full-parent lift, not a restriction to a state kernel.

The strict matched Hessian. First take finite positive residue source weights. The leading model before setting the ordinary test angle equal to a is (RC4), with every testing occurrence of a replaced by u; its source coefficients remain those of (RC3). Let G denote one half of its angular Hessian after eliminating its two state directions, in variables (V,u) at the exact contact. Direct differentiation and two-by-two Schur elimination give

$$\mathsf{G}_{uu}=\frac{8(1-r)(1+zr)^2}{zb\Delta}>0, \tag{UI4}$$

$$\det\mathsf{G}=\frac{16(b+r)^2(1+zr)^3\{3z+4-(z+2)^2r\}}{b^2\Delta^2}. \tag{UI5}$$

The other two entries, making boundedness explicit, are

$$\mathsf{G}_{Vu}=\frac{4(1+zr)^2\{zb+zr+2r-2\}}{ab\Delta},$$
$$\mathsf{G}_{VV}=\frac{8(1+zr)\{z^2b^2+z^2br+zb^2+2zbr-zb-r+1\}}{b\Delta}.$$

For a short check of strictness, the exact leading n numerator is bQ_n, where

$$Q_n=1+r(z-1)-z(z+2)r^2-z(z+1)br=(1-(z+1)r)(1+zr)-zrg.$$

Since $Q_n\ge 0$ and $g\ge 0$, one has $r\le 1/(z+1)$. Consequently

$$3z+4-(z+2)^2r\ \ge\ \frac{z(2z+3)}{z+1}>0. \tag{UI6}$$

All denominators in (UI4)–(UI6) are positive bounded units. These formulas therefore give a strict, uniformly bounded-away-from-zero angular Schur form on the source residue domain, including $n=0$. They contain no inverse of r, g, or $1-zb$.

Large weights and finite real caps. We use the elementary finite-dimensional penalty lemma: if a real symmetric form B is strictly positive on the common kernel of the rows of a matrix V, then $B+MV^TV$ is positive definite for some finite real $M>0$. A proof is to split into $\ker V$ and its orthogonal complement. The first diagonal block is strict; taking M sufficiently large makes its Schur complement strict too.

The bounded finite part obtained by the centered positive square terms is continuous at all source boundary residues. This assertion is algebraic: its only large-weight products are

$$nx,\ nx^2,\ m\mu,\ m\mu^2,\ \alpha\pi,\ \alpha\pi^2,\quad C\rho k^2,\ D\sigma k^2,\ k^2H,$$

and their bounded cap analogues. Substitution of the original source formulas cancels the possible $r,g,1-zh,t,j$ poles in those products; their denominators are the displayed positive units. For example,

$$m\mu=\frac{(1+k^2)(1-r)A_cS}{2b\Delta},\qquad \alpha\pi=-\frac{z^2(b+r)N}{2ab\Delta}.$$

The other cancellations are

$$nx=-\frac{(b+r)A_cS-r\Delta}{2b\Delta},\quad nx^2=x(nx),\quad m\mu^2=\mu(m\mu),\quad \alpha\pi^2=\pi(\alpha\pi),$$

$$C\rho k^2=\frac{c_1}{ab},\qquad D\sigma k^2=-\frac{c_2}{ab},\qquad c_1=\frac{zA_cN}{2\Delta},\quad c_2=\frac{z(1+z)(b+r)(b-k^2r)S}{2\Delta}.$$

Finally $C\rho+D\sigma=(Ct-Dj)/(ab)$, using the exact canceled formula in the full-parent reduction. Thus even this difference has no hidden small-opening denominator. For k^2H, write $k^2C=c_1/t$, $k^2D=c_2/j$, with c_1,c_2 positive bounded units. Then $k^2H=c_1c_2/(c_1j+c_2t)$, whose denominator is a unit because $t+j=A_c$ is a unit. Its residue is exactly H_0 in (RC3).

At most two of n,m,α can diverge. Indeed m can diverge only at $g=0$, while α can diverge only at $1-zb=0$; these two equalities would give $g=1/z+r>0$. The hard-state parts of the three possible diverging centered rows are, up to nonzero unit factors,

$$E_X,\qquad E_X+aE_Z,\qquad E_Z.$$

Every possible pair is independent with a unit determinant.

Here is why their constrained limiting finite form is strict. Approach the boundary by the positive leading-source domain. The state block is bounded below by a positive real form because m and α are strong. Its minimizing hard vector for a fixed angular vector is bounded: solve the at most two diverging positive square term equations

in the hard variables, using the unit determinants just displayed; this supplies a bounded competitor, and all remaining coefficients stay bounded. If h_0 is that competitor and $h_{\min}$ the state minimizer for angular vector q, the exact identity

$$(h_0 - h_{\min})^T P(h_0 - h_{\min}) = F(h_0, q) - q^T \mathsf{G} q$$

and $P \succeq cI$ make boundedness of $h_{\min}$ explicit. Together with (UI4)–(UI6), the state Schur factorization then bounds the full matched Hessian below by a positive real multiple of the Euclidean norm. For a vector in the common limiting positive square term kernel, adjust only its hard coordinates to lie in the exact nearby positive square term kernels. The adjustment tends to zero and annihilates the large weights exactly. Taking the limit proves strict positivity of the bounded finite part on that common kernel. The penalty lemma now chooses finite real caps for every diverging weight. Bounded source weights can be retained in full. All caps are below their original weights, so the remaining centered excesses are positive squares.

This limiting argument uses only real finite-dimensional forms to choose caps. No inverse of a tensor matrix and no mixed root are used in the full-parent identity. The boundary residues are limits of the positive leading-source domain: g and $1 - zb$ specify the interval for b, and decreasing a positive r if necessary makes $Q_n \geq 0$ strict. Thus the argument also covers a literal original $n = 0$ and all combinations of the allowed boundary residues.

Separate angular roots and full strictness. The residue of the hard block in (UI3) is

$$2E_r^2 + 4n_*(1 + a^2)E_X^2 + (4m_* + 2)(E_X + aE_Z)^2 + 4\alpha_*(1 + a^2)E_Z^2 + 4\gamma R_h^2,$$

which is strict. In the residue Gram, S couples only to the two ordinary hard normals, and T only to R_h; the E_r couplings vanish. Setting $S = T$ gives the matched leading Hessian under an invertible bounded change of coordinates. Indeed at $V = 1$ the ordinary normals are its two state deviations, R_h supplies the relative-angle deviation with nonzero coefficient $2s$, and $S = T$ supplies the ordinary-angle deviation. The chosen caps therefore make the sum of the two angular Schur diagonals strictly positive.

Add an appropriate real multiple of the last zero identity in (UI1). It redistributes mass between S^2 and T^2; all accompanying coefficients contain w and have zero residue. Since their Schur diagonal sum is positive, choose this multiple so that both Schur diagonals are positive. The entire six-root residue Gram is now positive definite. The strict-residue Gram lemma proves the exact bounded remainder plus relative positive square term is tensor SOS.

Restore the centered excesses and the other C/D squares. The construction is homogeneous in W with a polynomial state inverse, so it includes $W = 0$ and every state. Real and imaginary parts give complex states. The complementary angle charts are proved next.

C.8.4 Complementary angle charts

Theorem C.22 (Strong C2 UI, all independent tests)**.** *Use the original C2 source (UR1)–(UR3), with $d = 1$, actual source angles $u_c = a > 0$ a bounded unit and $v_c = k > 0$ infinitesimal, and C, α strong. Then the entire parent (UR6) is a sum of squares in the tensor product of the independent source and testing fields, for every testing angle, including projective endpoints, and every homogeneous state. The statement retains the literal boundary $n = 0$ and arbitrary positive rates of every source opening.*

Section C.8.3 covers v infinitesimal and u with the same residue as a. Its complement follows from a rank-one allocation and positive frames at the testing endpoints.

A rank-one allocation lemma. Let P be a positive definite real symmetric matrix, let ℓ, b be real columns, and let $H, c_0 > 0$. Set

$$d = \ell^T P^{-1} \ell, \qquad e = \ell^T P^{-1} b, \qquad h = c_2 - b^T P^{-1} b.$$

Suppose

$$p(T) = c_0 + (h + 4Hdc_0)T + 4H(dh + e^2)T^2 > 0 \quad (T \geq 0), \qquad dh + e^2 > 0. \tag{UA1}$$

There exists a real λ such that

$$|\lambda| < \sqrt{4Hc_0}, \qquad h + 2e\lambda - d\lambda^2 > 0. \tag{UA2}$$

Indeed put $s = \sqrt{4Hc_0}$. If $d > 0$ and $|e| < ds$, take $\lambda = e/d$. If $|e| \geq ds$, the value at the appropriate endpoint is $h + 2|e|s - ds^2$. Were this nonpositive, writing $x = |e| - ds \geq 0$ would give $h + ds^2 \leq -2sx$ and $dh + e^2 \leq x^2$; the quadratic in (UA1) would then have a nonpositive value at a positive argument. Thus a nearby interior λ works. This includes equality $|e| = ds$. The scalar interval argument also holds in a limiting case $d = 0$; then $e \neq 0$ follows from $dh + e^2 > 0$. (For a finite positive P itself, $d = 0$ would force $\ell = e = 0$, so this case occurs only when using the continuous scalar extension at a boundary.)

Consequently the two real matrices

$$\begin{pmatrix} P & b - \lambda\ell \\ (b - \lambda\ell)^T & c_2 \end{pmatrix}, \qquad \begin{pmatrix} 4H & \lambda \\ \lambda & c_0 \end{pmatrix} \tag{UA3}$$

are positive definite. This is an elementary real lemma. Its inverse and square root are never taken in the tensor ring.

The finite ordinary angle away from contact. Assume v infinitesimal and u a bounded unit whose real residue has square different from a_0^2. Keep the centered positive square terms $P_L, P_C, P_n, P_m, P_\alpha$ of (UR12), (UR14) outside the Gram, and combine P_H with the exact remainder. Use the six separated roots

$$r_0 = uW - Y, \quad X, \quad Z, \quad V_0 = vW, \quad K_0 = kW, \quad R_0 = \frac{v(uX - Z)}{k}. \tag{UA4}$$

They satisfy the exact zero relation

$$R_0K_0 - (uX - Z)V_0 = 0. \tag{UA5}$$

All divisions in (UA4) are pure source divisions.

For completeness the boundedness of the C, D remainder in this frame is explicit. Write $\ell_0 = L\tau$, $c_R = C\rho k^2$, and $Q = uX - Z$. Then (UR13) gives exactly

$$\begin{aligned} G_{CD} + P_H = {} & 4\gamma R_0^2 + 8\ell_0 kK_0 r_0 + 8\ell_0 qk^2 V_0 Q - 4\ell_0 \tau k^2 K_0^2 \\ & + 8c_R v V_0 r_0 - 4c_R \rho k^2 V_0^2. \end{aligned} \tag{UA6}$$

All displayed coefficients are bounded by (UR9)–(UR11); all terms except the first have zero residue. The A positive square term is exactly $4(A/k^2)D_uD_v(uK_0 - kr_0)^2$. Substituting $Y = uW - r_0$ in (UR15) and the signed core gives a bounded Gram on (UA4), with no discarded normal term. Its residue has an isolated $2r_0^2$ term. First suppose the leading source weights are finite. The remaining residue is

$$q_s^T P q_s + 2V_0 b_*^T q_s + c_2 V_0^2 + 4HR_0^2 + c_0 K_0^2, \quad q_s = (X, Z)^T, \tag{UA7}$$

where $D_u = 1 + u^2$, $H = H_0$, $A_w = A_0^{\mathrm{weak}}$ are (RC3), and

$$\begin{aligned} P &= \begin{pmatrix} 4nD_u + 4m + 2 & (4m+2)u \\ (4m+2)u & (4m+2)u^2 + 4\alpha D_u \end{pmatrix}, \quad \ell = (u, -1)^T, \\ b_* &= \begin{pmatrix} -4mD_u \\ -D_u u(4m + 4\alpha + 2) \end{pmatrix}, \quad c_0 = 4A_w D_u u^2, \quad c_2 = 4mD_u^2 + (4\alpha + 2)D_u u^2. \end{aligned} \tag{UA8}$$

The matrix P is strictly positive because m, α are strong. The last sign-changing part of (UA7) is handled by adding 2λ times (UA5). This gives precisely the two blocks (UA3).

Here is a direct proof of (UA1) for every ordinary angle away from contact. Put $N_* = 2n + 1$, $T_* = 2m + 2\alpha + 1$, $U = u^2$, and $T = V^2$. The leading state polynomial (RC4), with its testing a replaced by u but its source unchanged, has determinant

$$\det f = 8D_u^2 u^2 V^2 \, \mathcal{L}(U, T), \tag{UA9}$$

where

$$\begin{aligned} \mathcal{L} = {} & \frac{4\alpha m N_*}{U} + 4A_w H T_* U + \frac{4A_w \alpha(2m + N_*)}{T} + \frac{4A_w n T_* U}{T} \\ & + 2HN_*T_*T + \frac{4HmN_*T}{U} + 4A_w H(2m + N_*) - N_*T_*. \end{aligned}$$

After division by N_*T_* all possible opening poles cancel. Define

$$i_N = N_*^{-1}, \quad t_m = m/T_*, \quad t_{am} = \alpha m/T_*, \quad q_\alpha = \frac{\alpha(2m + N_*)}{N_*T_*}, \quad i_\alpha = 1/\alpha.$$

Writing $z = a^2$, $A_c = 1 + zr$, and using the leading Δ of (RC3), these quantities are exactly

$$\begin{aligned} i_N &= \frac{r\Delta}{(b + r)A_c^2}, & t_m &= \frac{(1 - r)(1 - zb)}{2b(b + r)(z + 1)}, \\ t_{am} &= \frac{z^2(b + r)(1 - r)}{4\Delta}, & q_\alpha &= \frac{bz^2\{(z + 1)b + (z + 2)r - 1\}}{2A_c^2}, \\ i_\alpha &= \frac{2(1 - zb)\Delta}{z^2(z + 1)b(b + r)^2}. \end{aligned} \tag{UA10}$$

All denominators here are units. The normalized Laurent polynomial is

$$\frac{\mathcal{L}}{N_*T_*} = \frac{4t_{am}}{U} + 4A_w H i_N U + \frac{4A_w q_\alpha}{T} + 2A_w(1 - i_N)\frac{U}{T} + 2HT + 4Ht_m\frac{T}{U} + 4A_w H q_\alpha i_\alpha - 1. \tag{UA11}$$

Its value and its two first derivatives vanish at $(U,T)=(z,1)$, as direct substitution verifies. In logarithmic coordinates the nonconstant terms have nonnegative coefficients. The coefficients of U^{-1} and T are positive units. If $i_N>0$, the coefficients of U and T^{-1} are positive; if $i_N=0$, that of U/T is positive. The function is therefore strictly convex in logarithmic coordinates and has its unique minimum at $(z,1)$. This proves strict positivity at fixed $U\neq z$ for every $T>0$. The constant and leading coefficients of the equivalent quadratic (UA1) are strictly positive: $c_0>0$, and the positive T coefficient in (UA11) gives the second assertion. Thus (UA2)–(UA3) prove the required strict residue Gram.

Opening boundaries and divergent source weights. The preceding proof extends to all allowed opening residues, rather than assuming n,m,α bounded. The following formulas make the needed uniformity checkable. Put

$$J_0=q_\alpha+\frac{u^2(1-i_N)}{2}.$$

Then J_0 is a positive unit, and direct two-by-two inversion gives

$$\begin{aligned}\frac{\det P}{N_*T_*}&=8D_uJ_0,\\ d=\frac{u^2i_N+q_\alpha i_\alpha}{4J_0},\qquad e&=\frac{uD_u}{2J_0},\qquad h=\frac{D_u(4t_{am}-u^2)}{J_0},\\ (P^{-1}b_*)_1=\frac{i_N(u^2-4t_{am})}{2J_0},\quad (P^{-1}b_*)_2&=-\frac{u\{4t_{am}i_N+(1-i_N)u^2+1\}}{2J_0}.\end{aligned}\tag{UA12}$$

To check the unit assertion, if $i_N=0$ then $J_0=u^2/2>0$. If $i_N>0$, the weight N_* is finite and the strong lower bounds on m,α give $q_\alpha>0$: this follows directly from $\alpha(2m+N_*)/[N_*(2m+2\alpha+1)]$, whether both weights are finite or one diverges. Thus (UA12) has only unit denominators on the entire source residue domain.

The strictly positive leading coefficient is also explicit:

$$dh+e^2=\frac{D_u(u^2+2t_m)}{2J_0}>0.$$

Equations (UA10)–(UA12) prove continuity of d,e,h and of the minimizing hard vector. The convexity proof for (UA11) remains strict at every allowed boundary. Choose λ from the strict interval (UA2) at that boundary; the same strict inequalities hold nearby. Moreover P is bounded below by a positive real matrix, while $P^{-1}b_*$ and $P^{-1}\ell$ are bounded. Its Schur factorization therefore bounds the first block in (UA3) below by a positive real multiple of the identity, uniformly in the nearby source. The second block is a strictly positive real matrix as well.

The finite leading sources used in this limiting argument are valid real interior sources. This point must be established separately: the actual source has $b=h+(k^2/z)t$, and it is not legitimate to substitute that actual b into a leading formula with denominator $1-zb$ when this denominator has an infinitesimal sign change. For the limiting real parameters, put

$$f(b,r)=1+r(z-1)-brz(z+1)-r^2z(z+2).$$

Their domain is $b>0$, $0\le r<1$, $g=(1+z)b-1+r\ge 0$, $1-zb\ge 0$, and $f(b,r)\ge 0$. For $0<s<1$, let

$$b_s=sb+(1-s)/(1+z),\qquad r_s=sr.$$

Direct expansion gives

$$g_s=sg,\qquad 1-zb_s>0,\qquad f(b_s,r_s)=(1-s)(1+s-sr)+s^2f(b,r)>0.$$

Increase b_s by a sufficiently small $\epsilon>0$. This makes g_s strict, decreases the last two strict margins by respectively $z\epsilon$ and $r_sz(z+1)\epsilon$, and therefore preserves them. If $r=0$, then increase r_s by a sufficiently small positive amount; g increases and f remains positive by continuity. Letting these perturbations tend to zero as $s\to 1$ supplies valid finite real leading sources approaching every allowed residue. Their t may be chosen as $(1+zr)/2$, since the leading harmonic weight is independent of t. The normalized leading identity $2rn\Delta=bf(b,r)$ verifies the nonnegative-n requirement. The same construction supplies the real-interior approximation used in the finite-cap step of the local UI certificate.

At most two of n,m,α diverge: m can diverge only at $g=0$, and α only at $1-zb=0$, which cannot occur together. Their centered excess rows have hard-state parts proportional to

$$X,\qquad X+uZ,\qquad Z.$$

Any possible pair is independent because u is a unit. For a vector in their common limiting kernel, change only (X,Z) by a vanishing amount to annihilate the corresponding nearby exact rows. Their large penalties then

vanish exactly. Taking the limit in the uniform strict lower bound proves that the bounded finite part is strictly positive on that limiting kernel. The elementary finite-dimensional penalty lemma used in the local theorem now supplies finite real caps n_*, m_*, α_*. Their centered excesses are precisely the retained positive square terms (UR14). The exact bounded remainder with those caps, after adding (UA5), has a strictly positive residue Gram on (UA4). This proves the finite ordinary-angle complement for all source openings and literal $n = 0$.

Endpoint frames. Choose positive real μ, m_*, α_* below, respectively, k^2C, k^2D, m, and α. The products k^2C, k^2D are strong by (UR5), and $A_w = A/k^2$ is a positive unit. Retain n and all positive excesses in full. The following formulas concern the parent with $C = D = \mu/k^2$, $m = m_*$, $\alpha = \alpha_*$ and $A = A_w k^2$. They therefore suffice. The root coordinates need not themselves be bounded; only their displayed Gram coefficients must be bounded.

At u, v infinitesimal use

$$E = (uW - Y)/k, \quad R = v(uX - Z)/k, \quad V = vW, \quad K = kuW, \quad X, Z.$$

The exact amplitudes are

$$J = k(E + R), \quad M = D_u V - X - uZ - uvkE, \quad vY - Z = uV - Z - vkE, \quad kY = K - k^2E.$$

Their full residue form is

$$4\mu\{E^2 + (E + R)^2\} + 4m_*(V - X)^2 + 4\alpha_* Z^2 + 4\overline{A_w}K^2 + 2X^2, \tag{UA13}$$

which is strict. Literal zero testing angles are included without any division by a testing angle.

For bounded u and finite unit v, put $E = (uW - Y)/k$, $F = J/k$. Then

$$Y = uW - kE, \quad Z = uX + k(E - F)/v.$$

The roots (E, F, W, X) have strict residue

$$4\mu D_v E^2 + 4\mu F^2 + \{4m_* D_u^2 + (4\alpha_* + 2)D_u u^2\}(vW - X)^2 + 2D_v D_u X^2. \tag{UA14}$$

For bounded u and $v = \infty$ write $q = 1/v$ and homogenize by q^2. The same roots are defined using the homogeneous J, and

$$Y = uW - kE, \qquad Z = uX + kqE - kF.$$

The strict residue is (UA14) with $D_v = 1$ and $(vW - X)^2$ replaced by W^2. This includes the literal endpoint $q = 0$.

For $u = \infty$, put $p = 1/u$, $D_p = 1 + p^2$, and homogenize by p^2. Write $r_p = W - pY$, $Q_p = X - pZ$, and $P = vY - Z$. Then

$$J = r_p + vQ_p, \qquad M = D_p P - pQ_p + pvr_p.$$

For v infinitesimal use the six roots

$$E = r_p/k, \quad F = vQ_p/k, \quad P, \quad R = pQ_p, \quad K = kY, \quad S = pZ.$$

The signed core is exactly

$$2k^2(E + F)^2 + 2D_p P^2 + 2D_v R^2 - 4D_p D_v RP + 4D_p D_v pFK + 2D_p D_v S^2.$$

Its cross-term reduction uses $RvY = pFK$, an exact identity. Adding every capped positive square term gives the strict residue

$$4\mu\{E^2 + (E + F)^2\} + (4m_* + 2)(P - R)^2 + 4\alpha_* P^2 + 4\overline{A_w}K^2 + 2S^2. \tag{UA15}$$

For finite unit v, instead use $E = r_p/k$, $F = J/k$, P, K, S. Since $Q_p = k(F - E)/v$, the strict residue is

$$4\mu D_v E^2 + 4\mu F^2 + (4m_* + 4\alpha_* + 2)P^2 + 4\overline{A_w}D_v K^2 + 2D_v S^2. \tag{UA16}$$

Finally, at both infinite endpoints put $q = 1/v$ and homogenize by p^2q^2. Use

$$E = (W - pY)/k, \quad F = (qW + X - pqY - pZ)/k, \quad P = Y - qZ, \quad K = kY, \quad S = pZ.$$

Here $M = D_p P + pkD_q E - pqkF$ and the entire signed core is

$$2k^2F^2 + 2D_p P^2 + 2D_q p^2 k^2 (F - qE)^2 + 4D_q D_p k(F - qE)S + 2D_q D_p S^2.$$

The strict residue is (UA16) with $D_v = 1$.

Exhaustion and restoration. A testing angle is infinitesimal, a bounded unit, or projectively infinite. Equations (UA13)–(UA16) cover every pair except ordinary u with infinitesimal v. The local theorem covers the equal-residue contact, and (UA4)–(UA12) cover its complement. Exact sign gauges are

$$(u, W, X) \mapsto (-u, -W, -X), \qquad (v, W, Y) \mapsto (-v, -W, -Y);$$

all unspecified variables are fixed. Both preserve the full parent, so negative ordinary contact residues are included. All endpoint frames have been written after homogenization, including literal projective endpoints. Each proof has an exact bounded Gram with strict real residue; the strict-residue Gram lemma gives tensor SOS. Restoring all retained positive square terms completes the original parent. Real and imaginary parts treat complex states. This closes the entire strong C2 UI case and its exact symmetry image HU.

C.9 An unbounded first angle and a small second angle

C.9.1 Source entry at the HI corner

Use the original positive rational C2 source with $d = 1$, actual u_c unbounded and v_c infinitesimal. Put

$$q = u_c^{-2}, \quad y = v_c^2, \quad H = u_c^2 h \in (0, 1), \quad R = r/q, \quad B = b/q, \quad L = yt = B - H, \quad T = B - yR.$$

The already proved source identities (WE1)–(WE3) show that R is bounded using only $n \geq 0$, q infinitesimal and y bounded. Here is their short contradiction argument, to make this use explicit. The positive source gaps imply

$$0 < T < H + y, \quad g = H + (1 + y)r - 1 + qT > 0,$$

and $n = -qP/(2rQ)$ with $Q > 0$ and

$$P \geq r^2(1 + y)(H + yr) - Tqr(y + 2) - q^2(T^2 + T).$$

If q/r were infinitesimal, then $T/(H + yr)$ and $(T^2 + T)/(H + yr)$ would be bounded. For unit r this follows from $T < H + y \leq (H + yr)/r$; for infinitesimal r, the inequality $g > 0$ forces H to have residue one. Division by $r^2(H + yr)$ would give $P > 0$, contrary to $n \geq 0$. Thus R is bounded. (The exact positive remainder in this lower bound is printed in (WE3), so this argument uses no asymptotic sign substitution.)

Now $r = qR$ is infinitesimal and $g > 0$ gives $H_0 = 1$. Define the positive opening

$$E = (1 - H)/q.$$

The same gap implies $E < (1 + y)R + T$, so E is bounded. Since $j = 1 + R - t > 0$, t is bounded; consequently

$$H = 1 - qE, \quad B = H + yt \sim 1, \quad j = 1 + R - t, \quad K = B + R - E = g/q > 0.$$

All of E, t, j, K are positive bounded, while $t + j = 1 + R$ and $E + K = B + R$ are positive units.

Let Δ and N denote the unchanged original source polynomials, and set $\mathcal{N} = N/q$. Their exact expressions are

$$\begin{aligned}\Delta &= (1 + q)\{B^2 + 2BR + (1 + y)B\} + (1 - yq)R^2 + 2(1 + y)R + 1 + y,\\ \mathcal{N} &= (1 + q)(B + R)(H + y) - (1 + y)(1 - qR)yt.\end{aligned}$$

Thus Δ is a positive bounded unit and

$$\bar{\Delta} = (1 + \bar{R})(3 + \bar{R}), \qquad \overline{\mathcal{N}} = 1 + \bar{R} > 0.$$

Set $S = 1 + R + yt$, also a positive unit. Direct substitution into the original coefficients gives the following exact equalities:

$$\begin{aligned} mqK &= \frac{(1 + y)(1 - qR)(1 + R)S}{2\Delta}, & Cyt &= \frac{(1 + R)\mathcal{N}}{2\Delta},\\ \frac{A}{qy} &= \frac{(1 + R)(B + R)S}{2H\Delta}, & Dyj &= \frac{(1 + q)(B + R)(B - yR)S}{2\Delta},\\ \alpha qE &= \frac{(B + R)\mathcal{N}}{2\Delta}, & n &= \frac{(B + R)(1 + R)S - R\Delta}{2R\Delta}. \end{aligned} \tag{HI-entry}$$

Every one of the first five right-hand sides is a positive bounded unit; in particular $B - yR$ has residue one. The last numerator has residue $(1 + \bar{R})(1 - \bar{R})$, so $n \geq 0$ implies $0 \leq \bar{R} \leq 1$. This does not imply $R \leq 1$ as a source inequality, and no such stronger claim is used.

Since q, y are infinitesimal and E, t, j, K bounded, the positive numbers qK, yt, yj, qE are infinitesimal. Dividing their unit products in (HI-entry) proves

$$m, C, D, \alpha \quad \text{are all unbounded}, \qquad A \asymp qy.$$

This conclusion follows from the original source formulas and $n \geq 0$; no additional strong-weight assumption is needed for an actual HI source.

C.9.2 The HI contact for every $n \ge 0$

Use the HI source estimates of Section C.9.1. Let $q = u_c^{-2}$ and $y = v_c^2$ be positive infinitesimals, and put $p = 1/u$ for the testing angle. The full target here is $\widehat{F} = p^2 F(1/p, v)$, with its polynomial value at $p = 0$. Here the independent testing p, v are infinitesimal or zero. The source numbers $R = r/q$, H, and the opening quantities t, j, K, E are those of the source calculation; R is bounded, $H \sim 1$, and

$$S_c = 1 + R + yt = j + t(1 + y) = K + E(1 + q).$$

Set $d_p = 1 + p^2$, $d_v = 1 + v^2$, $\delta_p = p^2 - q$, $\delta_v = v^2 - y$, and use the bounded positive source products

$$c = Cyt, \quad d = Dyj, \quad \mu = mqK, \quad a_* = \alpha q E.$$

The original n is included from the start. Define

$$\nu = 2nR = \frac{(B_{\mathrm{src}} + R)(1 + R)S_c - R\Delta}{\Delta}, \qquad \zeta = R + \nu, \quad h_* = H - \nu.$$

The exact source identities are

$$\frac{2AH}{qy} = \zeta, \qquad c - d = \frac{y}{2}\{\zeta(1 + q) - t\}, \qquad \mu - a_* = \frac{q}{2}\{E - \zeta(1 + y)\}.$$

They cancel the apparent opening denominators before any normal is estimated. At $R_0 = r_0 \in [0, 1]$ they give

$$\nu_0 = \frac{1 - r_0}{3 + r_0}, \qquad c_0 = d_0 = \mu_0 = (a_*)_0 = \frac{1 + r_0}{2(3 + r_0)}.$$

Thus these products remain units at $r_0 = 0$, even when n is unbounded. Literal $n = 0$ is retained in every formula.

Define the state coordinates

$$A_0 = W - pY, \quad B_0 = vY - Z, \quad S = X - vW - pZ + pvY, \quad s = -S/S_c, \quad Q = \sqrt{qy}\, Y/H,$$

$$P = p/\sqrt{q}, \quad V = v/\sqrt{y}, \quad T = PV, \quad N_c = (A_0 - tvs)/\sqrt{y}, \quad N_m = (B_0 + Eps)/\sqrt{q},$$

$$N_D = d_v N_c + t\delta_v V s, \qquad N_M = d_p N_m - E\delta_p P s.$$

The exact source has $P = V = 1$, $s = Q = \sqrt{q}$ and all four N's zero. Retain the positive squares

$$\mathcal{E} = \frac{4cjd_v}{tS_c}N_c^2 + \frac{4dt(1 + y)}{jS_c}N_D^2 + \frac{4\mu E(1 + q)}{KS_c}N_M^2 + \frac{4a_* K d_p}{ES_c}N_m^2$$

and $\mathcal{N} = 4nd_p d_v x^2$, where

$$x = X + (R/H)pvY = -gs + (H + R)TQ + v\sqrt{y}N_c - p\sqrt{q}N_m, \qquad g = S_c - tv^2 - Ep^2.$$

The complete difference of the original n term and this square is

$$\begin{aligned} 4nd_p d_v(X^2 - x^2) = {} & 4\nu d_p d_v g\, TsQ - 2\nu(2H + R)d_p d_v T^2 Q^2 \\ & - 4\nu d_p d_v TQ(v\sqrt{y}N_c - p\sqrt{q}N_m). \end{aligned}$$

All its coefficients are bounded, regardless of the size of n.

The exact quadratic-state expansion is

$$\begin{aligned} \widehat{F} - \mathcal{E} - \mathcal{N} = {} & h_c N_c^2 + h_m N_m^2 - 4\sqrt{qy}\, pvd_p d_v N_c N_m \\ & + N_c(C_s V s + \beta_c TQ) + N_m(M_s P s + \beta_m TQ) + \mathcal{G}, \end{aligned}$$

where

$$h_c = \frac{4d_v}{S_c}\{c(1 + y) + dd_v\} + 2yd_v(1 + v^2 + p^2v^2), \quad h_m = \frac{4d_p}{S_c}\{\mu d_p + a_*(1 + q)\} + 2qd_p(1 + p^2 + p^2v^2),$$

$$C_s = 4d_v[y\{\zeta(1 + q) - d_p g\} + 2dt\delta_v/S_c], \quad M_s = 4d_p[q\{d_v g - \zeta(1 + y)\} - 2\mu E\delta_p/S_c],$$

$$\beta_c = 4h_*\sqrt{y}\, vd_p d_v, \qquad \beta_m = -4h_*\sqrt{q}\, pd_p d_v,$$

and

$$\mathcal{G} = f_A P^2 s^2 + f_B V^2 s^2 + f_C T^2 Q^2 + f_D Q^2 - eTsQ,$$

$$f_A = 4[\mu K + a_* E d_p - 2\mu E\delta_p + \mu E^2\delta_p^2/S_c] + 2qd_v(E\delta_p - K + tv^2)^2 - 2E^2qv^2d_p,$$
$$f_B = 4[ctd_v + dj - 2dt\delta_v + dt^2\delta_v^2/S_c] + 2y(t\delta_v - j)^2,$$
$$f_C = 2\{H^2 - \nu(2H + R)\}d_pd_v, \quad f_D = 2H(R + \nu)d_pd_v, \quad e = 4h_*d_pd_vg.$$

This identity follows by the exact quadratic-state extraction rule: substitute the displayed inverse coordinates, subtract the five retained squares, collect N_c, N_m, and use the three source identities above. It specifies the full scalar, both complete linear normals and the entire hard block. Its coefficients are polynomials in p, v with bounded source coefficients. In particular $C_s, M_s, \beta_c, \beta_m$ are infinitesimal. The hard diagonal residues are $4/(3 + r_0) > 0$ and

$$(f_A)_0 = (f_B)_0 = (f_C)_0 = (f_D)_0 = \lambda(r_0) = \frac{2(1 + r_0)^2}{3 + r_0}, \qquad e_0 = 4\lambda(r_0).$$

For completeness the finite radical division is given here once. A star means $p = \sqrt{q}$, $v = \sqrt{y}$ and $\sigma = \sum f_i - e$. Because every subtracted square vanishes at the actual contact, the original state and angular contact equations give exactly

$$f_A^* + f_B^* = f_C^* + f_D^* = e^*/2, \quad q^{1/4}\sigma^*_{\sqrt{p}} = -4(f_A^* - f_D^*), \quad y^{1/4}\sigma^*_{\sqrt{v}} = -4(f_B^* - f_D^*),$$

$$C_s^* + \beta_c^* = M_s^* + \beta_m^* = 0.$$

If $sQ > 0$, a pure-testing sign choice makes $s, Q > 0$. Put $(A_1, A_2, A_3, A_4) = (Ps, Vs, TQ, Q)$, $G = (A_1A_2A_3A_4)^{1/4}$, $E_i = A_i - G$. Each radical separates into pure-source and pure-testing radicals. The exact circuit is

$$L = \sum A_i^2 - 4G^2 = \sum E_i^2 + 2G\sum E_i.$$

For every pairing $ij|k\ell$,

$$G\sum E_i = G(\sqrt{A_i} - \sqrt{A_j})^2 + G(\sqrt{A_k} - \sqrt{A_\ell})^2 + 2G\{(A_iA_j)^{1/4} - (A_kA_\ell)^{1/4}\}^2.$$

Also $L = (A_1 - A_3)^2 + (A_2 - A_4)^2 + 2(\sqrt{A_1A_3} - \sqrt{A_2A_4})^2$, and for $rr' = G^2$, $(r - r')^2 = (r - G)^2 + (r' - G)^2 + 2G(\sqrt{r} - \sqrt{r'})^2$. A finite average supplies positive pivots simultaneously on the E_i and on $F_p = \sqrt{A_1A_3} - G$, $F_v = \sqrt{A_2A_3} - G$. The exact relations

$$G(\sqrt{p} - q^{1/4}) = q^{1/4}F_p, \qquad G(\sqrt{v} - y^{1/4}) = y^{1/4}F_v$$

turn finite monic Taylor division in $\sqrt{p} - q^{1/4}$ and $\sqrt{v} - y^{1/4}$ into bounded infinitesimal corrections to these positive pivots. More explicitly put $z = \sqrt{p}$, $w = \sqrt{v}$, $z_0 = q^{1/4}$, $w_0 = y^{1/4}$ and $k = (f_C^* + f_D^*)/2$. Each monomial has the finite expansion

$$z^iw^j = \sum_{a=0}^{i}\sum_{b=0}^{j}\binom{i}{a}\binom{j}{b}z_0^{i-a}w_0^{j-b}(z - z_0)^a(w - w_0)^b.$$

All its coefficients are bounded. In $\mathcal{G} - kL$, terms with two differences multiplying G^2 become bounded infinitesimal multiples of F_p^2, F_pF_v, F_v^2, because each replacement contributes a source factor z_0 or w_0. Terms with one difference multiplying GE_i become infinitesimal multiples of F_pE_i, F_vE_i. Terms multiplying E_i^2 have infinitesimal coefficients since $f_i - k$ has zero residue. The only possible first-jet expression is

$$G\{2\sum_i(f_i^* - k)E_i + z_0\sigma_z^*F_p + w_0\sigma_w^*F_v\}.$$

Using the displayed contact jets, it equals exactly

$$2(f_D^* - k)G\sum_i E_i - \frac{z_0\sigma_z^*}{2}\{\sqrt{G}(\sqrt{A_1} - \sqrt{A_3})\}^2$$
$$- \frac{w_0\sigma_w^*}{2}\{\sqrt{G}(\sqrt{A_2} - \sqrt{A_3})\}^2.$$

All three source coefficients are infinitesimal, and the finite circuit average already supplies strictly positive pivots on their roots. For each normal, its two infinitesimal coefficients have zero sum at contact. Write their two raw amplitudes as $G + E_i$; finite monic division of that sum and the same two relations express the whole normal coupling by infinitesimal entries on the already positive roots. Thus the hard block and circuit block have strict residue. For $sQ < 0$ the raw diagonal and $\sqrt{T|sQ|}$ already have positive residue coefficients $\lambda, \lambda, \lambda, \lambda, 4\lambda$; for $sQ = 0$ omit the last root. The strict-Gram lemma proves all three cases. Restore the five positive squares. Section C.9.3 then proves the other measurements, including the literal axes and infinities, for every $n \geq 0$.

C.9.3 Tests outside the HI critical corner

Theorem C.23 (All tests outside the HI critical corner)**.** *Let K_c, K_t be independent real closed extensions of $\mathbb{R}$. Consider the original C2 parent with $d = 1$, $\lambda = \beta = 0$, $n, A \geq 0$, and m, C, D, α positive unbounded source numbers. Then at every projective testing pair except possibly $1/u$ and v both nonzero infinitesimal, and every real or complex testing state, the original parent is a sum of squares in $K_c \otimes_{\mathbb{R}} K_t$. No comparison of source and testing rates is made.*

Proof. For any positive real μ, retain the four nonnegative event positive square terms with coefficients $m - \mu, C - \mu, D - \mu, \alpha - \mu$, and retain the n and A positive square terms in their entirety. It suffices to prove the capped parent obtained by setting $m = C = D = \alpha = \mu$ and $n = A = 0$ is tensor SOS at the given test. The real constant μ may be chosen after the real testing residue is fixed, since each of the original four source coefficients is larger than every real number.

Write $D_u = 1 + u^2$, $D_v = 1 + v^2$. First suppose v is infinitesimal, including zero. Use the roots

$$S = vW, \qquad E = Y - uW, \qquad X, \quad Z.$$

All four capped event amplitudes are bounded linear forms in these roots:

$$M = D_u S - X - uZ + uvE, \quad uW - Y = -E, \quad J = -E + uvX - vZ, \quad vY - Z = uS + vE - Z.$$

The remaining core depends only on the same four roots, so the exact capped parent has a bounded four-root Gram. Let $a = \bar{u}$ and $d_a = 1 + a^2$. Its residue is

$$(8\mu + 2)E^2 + 4\mu(d_a S - X - aZ)^2 + (4\mu + 2)d_a(aS - Z)^2 + 2(X + aZ)^2 - 2d_a Z^2.$$

Put $T = X + aZ$ and $R = aS - Z$; this is an invertible real change of coordinates. The last three-variable form is

$$4\mu(d_a S - T)^2 + 4\mu d_a R^2 + 2T^2 - 2d_a a^2 S^2 + 4d_a aSR.$$

The T and R pivots are $4\mu + 2$ and $4\mu d_a$. The remaining Schur pivot is exactly

$$\frac{d_a[4\mu^2 - a^2(4\mu + 1)]}{\mu(2\mu + 1)}.$$

Taking $\mu = 1 + a^2$ makes its numerator positive, because $4\mu^2 - a^2(4\mu + 1) = 3a^2 + 4$. Thus this bounded Gram has strictly positive real residue and is tensor SOS by the inherited strict-residue Gram lemma.

Next suppose v is a bounded unit. At the real residues $(a, b) = (\bar{u}, \bar{v})$ the four event forms have common kernel

$$(W, X, Y, Z) = W(1, b, a, ab).$$

The core restricted to this line is $2(1 + a^2)(1 + b^2)b^2W^2$, which is strictly positive away from zero. Consequently a sufficiently large positive real μ makes the whole capped residue form positive definite. For completeness, this elementary assertion has a direct quantitative proof. Let $E_0 \succeq 0$ be the event Gram, split the state space into $\ker E_0$ and its orthogonal complement, and let $\sigma > 0$ be its smallest eigenvalue on that complement. Let $\kappa > 0$ be the core's eigenvalue on the one-dimensional kernel and bound the norms of the two remaining core blocks by M. Then $4\mu\sigma > M + M^2/\kappa$ gives a positive Schur complement. Every constant here belongs to $\mathbb{R}$. The exact state Gram is bounded and has this positive residue.

Finally suppose v is unbounded, including the projective endpoint. Set $w = 1/v$, homogenizing before setting $w = 0$. The four event forms become

$$W - wX + uY - uwZ, \quad uW - Y, \quad uwW + uX - wY - Z, \quad Y - wZ,$$

with bounded positive factors D_w or D_u where appropriate. At $w = 0$ their common kernel is $(W, X, Y, Z) = (0, X, 0, aX)$, on which the homogenized core is $2(1 + a^2)X^2$. The same real Schur argument gives a positive residue Gram for a sufficiently large μ. For finite unbounded v, undo homogenization by the pure positive testing square v^2; for the projective endpoint the homogeneous expression is already the required parent. These three cases exhaust every projective Bob test when u is bounded.

It remains to consider $p = 1/u$ infinitesimal, after homogenizing by p^2. If v is a bounded unit, use the four roots

$$S = pY, \qquad T = vY - Z, \qquad W, \quad X.$$

The exact event amplitudes are

$$W - S, \quad J = W + vX - D_v S + pvT, \quad M = pvW - pX + T, \quad T.$$

The signed core also has bounded coefficients in these roots:

$$2J^2 + 2D_pT^2 + 2D_v[p^2X^2 + 2vSX - 2pXT - v^2S^2 + 2pvST - p^2T^2].$$

At $p = 0$, set $b = \bar{v} \neq 0$, $A_0 = W - S$, and $B_0 = X - bS$. The full capped residue is

$$(8\mu + 2)T^2 + 4\mu(1 + b^2)A_0^2 + (4\mu + 2)(A_0 + bB_0)^2 + 4(1 + b^2)bSB_0 + 2(1 + b^2)b^2S^2.$$

As μ grows, the first three positive penalties constrain $T = A_0 = B_0 = 0$, and the residual kernel has positive coefficient $2(1 + b^2)b^2S^2$. The quantitative real Schur argument above therefore gives a finite μ with a strictly positive residue.

If u, v are both projectively unbounded, put $w = 1/v$ and homogenize by p^2w^2. Use $P = Y - wZ$, $S = pZ$, W, X. Now the exact event amplitudes are

$$M = pW - pwX + P, \quad W - pP - wS, \quad J = wW + X - pwP - D_wS, \quad P.$$

The exact core is

$$2J^2 + 2D_pP^2 + 2D_w[p^2X^2 + 2SX - S^2].$$

At $p = w = 0$ the capped residue is exactly

$$4\mu W^2 + (8\mu + 2)P^2 + 4\mu(X - S)^2 + 2X^2,$$

which is strict for every $\mu > 0$. The literal projective endpoints cause no division. The literal critical axes also have direct exact certificates:

$$F|_{p=0} = 4nD_vX^2 + (4m + 4\alpha + 2)(vY - Z)^2 + 4CD_vW^2 + 4AD_vY^2 + (4D + 2)(W + vX)^2,$$
$$F|_{v=0} = 4nD_pX^2 + (4m + 2)(pX + Z)^2 + (4C + 4D + 2)(W - pY)^2 + 4AD_pY^2 + 4\alpha D_pZ^2.$$

Thus only p, v both nonzero infinitesimal remain untreated. Applying the real-state result separately to real and imaginary state components proves the complex-state claim. □

For a strong C2 HI source, the exact source-entry lemma below proves that all four coefficients in the theorem are unbounded. Hence the theorem applies without any additional source-shape assumption. Its precise uncovered complement is the critical testing corner $1/u$ and v both nonzero infinitesimal. Exact sign gauges may make both positive. The original HI source leaf is therefore narrowed but not closed.

C.9.4 Further source reductions

All angles in this section are actual source contacts. They are strictly positive finite field elements; "infinite" below means unbounded in the coefficient field, not a literal projective endpoint. Write $z = c^2 = u_c^2$, $y = a^2z = v_c^2$, and $A_c = 1 + zr$, $S = A_c + a^2zt$. The exact source identities used are

$$n \geq 0 \iff (b + r)A_cS - r\Delta \geq 0, \qquad N = (1 + z)(b + r)h + a^2\{g + j + rt(1 + y)\} > 0.$$

Theorem C.24 (The ordinary–ordinary source orbit is already closed)**.** *If both actual C2 half-angles are positive bounded units, then the source belongs either to the two-weak-complementary-event all-tests exit or to the five-consecutive-strong-event all-tests exit. No upper bound on an original event coefficient is required.*

Proof. Here a, z are units. The positive gaps give $0 < h < 1/z$ and $0 < t < 1 + zr$, so $h, t, b = h + a^2t$ are bounded. From $g = b - 1 + r + zh > 0$ we obtain $1 < b + r + zh \leq (1 + z)b + r$; thus $b + r$ is a positive unit. The positive expansion of Δ makes it a bounded unit too. The original A formula has unit numerator and bounded denominator h, so A is strong.

The exact numerator for n, after substituting $t = (b - h)/a^2$, satisfies the polynomial specialization

$$[(b + r)A_cS - r\Delta]_{r=1} = -z(a^2 + h)(b + 1)(z + 1).$$

If $1 - r$ were infinitesimal, taking residues would contradict $n \geq 0$: the right side is strictly negative. Hence $1 - r$ is a unit. The original m numerator is now a unit and $g\Delta$ is bounded, so m is strong. Apply the exact C2 reflection; it preserves ordinary actual angles and exchanges m, D. Thus D is strong.

If either C or α is weak, its original rational formula implies N infinitesimal: all the multiplying denominator factors are bounded and $z, a, b + r, \Delta$ have bounded inverses. The positive identity for N then forces h, j infinitesimal. Consequently $t = A_c - j$ and $1 - zh$ are units. Both original complementary formulas therefore give C, α infinitesimal. The two-weak theorem applies. Otherwise C, α are both strong; with m, A, D strong, the five-consecutive-event theorem applies. These cases exhaust the source. □

Some directed one-weak angle types are empty. Orient an exactly-one-weak source so that C is weak and α strong. In addition to the ordinary–ordinary exclusion above, its actual angle types cannot be $(\text{unit}, \text{Inf})$, $(\text{Inf}, \text{unit})$ or (Inf, Inf). Here Inf denotes positive infinitesimal.

For $(\text{unit}, \text{Inf})$, z is a unit and a^2 is infinitesimal. Again t, h, b, Δ are bounded, Δ is a unit, and $b + r$ is a unit. Weak C implies N/a^2 infinitesimal. The positive identity forces j and rt infinitesimal. Since $t = A_c - j$ is a unit, r is infinitesimal. Then $g > 0$ makes $b + zh$ a unit, and $b = h + a^2 t$ makes h a unit. The first positive summand of N is consequently a unit, contradicting N/a^2 infinitesimal.

For $(\text{Inf}, \text{unit})$, bounded b is a unit by the inherited source-entry inequality. Explicitly,

$$xC(b - h) = \frac{A_c(xN)}{2\Delta}, \qquad xN = (1 + x)(b + r)(xh + k^2) - x(1 + k^2)(1 - r)(b - h).$$

The last numerator has residue $k_0^2(b_0 + r_0) > 0$, independently of any lower bound on g. Thus $xC(b - h)$ is a unit and C is strong. For unbounded b, (SR1) gives $C \asymp 1/L$ with bounded L, again strong. This exhausts b and excludes weak C in this angle orientation.

For (Inf, Inf), both z, y are infinitesimal and $t < A_c$ is bounded. Put $H = zh$, $L = z(b - h) = yt$ and $B = zb = H + L$. Here $0 < H < 1$, L is infinitesimal, and B is bounded. If H is a unit then b is unbounded; direct division of the original Δ by b gives a positive unit of residue $1 + B_0$. The positive first summand of N consequently gives $C \geq c_0 H/L$ for some positive real c_0, contradicting weak C. Thus weak C forces H, B infinitesimal. The exact ratio is then

$$\frac{\alpha}{C} = \frac{z^2(b + r)(b - h)}{A_c(1 - H)} = \frac{(B + zr)L}{A_c(1 - H)},$$

which is infinitesimal. Therefore α is weak as well, contrary to the selected one-weak orientation.

After these exclusions, an exactly-one-weak source oriented with weak C has only the following candidate actual types:

$$(\text{Inf}, \text{unbounded}), \quad (\text{unit}, \text{unbounded}), \quad (\text{unbounded}, \text{Inf}),$$
$$(\text{unbounded}, \text{unit}), \quad (\text{unbounded}, \text{unbounded}).$$

The second type is the reflection of the single-angle domain and is now fully covered by the homogeneous unbounded-b theorem and the other single-angle cases. The four other listed directed types remain conditional source leaves; no nonemptiness is inferred from their listing.

Other strong-event angle types are not all empty. For $x > 0$ infinitesimal, the exact positive chart

$$c = \sqrt{x}, \quad a = 1, \quad b = 1/x, \quad r = 1/4, \quad t = 1/2$$

has $h = 1/x - 1/2$, $g = 1/x + 1/4 - x/2$, $j = 1/2 + x/4$ and all remaining chart gaps positive. Its original coefficients are

$$n = \frac{8 - 5x^2}{16(1 + x)^2}, \qquad m = \frac{3x^2(3x + 4)}{16(1 + x)(4 + x - 2x^2)},$$
$$C = \alpha = \frac{4 - x}{8x}, \qquad A = \frac{x(x + 4)(3x + 4)}{32(2 - x)(1 + x)^2},$$
$$D = \frac{(2 - x)(3x + 4)}{16x(1 + x)}.$$

Every coefficient is positive. The unchanged exact contact is

$$(u_c, v_c; W, X, Y, Z) = (\sqrt{x}, \sqrt{x}; 1/x, -\sqrt{x}/4, 1/\sqrt{x} - \sqrt{x}/2, 1).$$

Original source stationarity, positive transverse state inertia and Laurent convexity prove physicality, including projective limits. Here C, α, D are unbounded, $m \sim 3x^2/16$, $A \sim x/4$, and $n \to 1/2$. Thus a genuine strong-complementary source can have two infinitesimal actual angles and weak m, A. This is an obstruction to deleting that angle orbit, not a counterexample to tensor SOS.

C.9.5 A strong C event from an unbounded first angle

Theorem C.25 (The directed weak-C types HI and HU are empty)**.** *Use the exact positive rational C2 source, with $d = 1$ and $n \geq 0$. If the first actual half-angle u_c is unbounded and the second actual half-angle v_c is bounded, then $1/C$ is bounded. Consequently no exactly-one-weak source oriented with C weak can have directed actual type HI or HU. Literal $n = 0$ is included.*

This is an original-source impossibility theorem. It makes no claim that a scalar contact identity by itself is a tensor-SOS certificate. All independent testing variables and state coordinates remain unrestricted; the two source leaves are empty before any testing restriction is imposed.

Proof. Write

$$q = u_c^{-2} > 0, \qquad y = v_c^2 > 0, \qquad H = u_c^2 h = v_c Y_c \in (0,1),$$

and retain the original $r \in (0,1)$. Here q is infinitesimal and y is bounded; y may itself be infinitesimal. Put

$$R = r/q = -u_c X_c, \quad B = u_c v_c W_c, \quad L = v_c(u_c W_c - Y_c), \quad T = B - yR.$$

The exact positive gaps give

$$B = H + L = yR + T, \qquad 0 < T < H + y, \qquad L > 0.$$

In particular H, T, y, r are bounded. The original source coordinates become exactly

$$z = 1/q, \quad a^2 = yq, \quad b = yr + qT, \quad h = qH, \quad t = r/q + (T - H)/y,$$

and the two gaps satisfy

$$g = H + (1+y)r - 1 + qT > 0, \qquad j = (H + y - T)/y > 0. \tag{WE1}$$

The displayed expressions are pure coefficient identities; no inverse of a mixed source/testing element is used.

We first show that R is bounded. Set $Q = q^2\Delta > 0$. Direct expansion of the unchanged original formula for n gives

$$n = -\frac{qP}{2rQ}, \tag{WE2}$$

where the following exact decomposition of P is sufficient:

$$\begin{aligned} P &= r^2(1+y)(H+yr) - Tqr(y+2) - q^2(T^2+T) + P_+, \\ P_+ &= HTq^2 + HTqr + Hqr(y+1) + T^2q^2r \\ &\quad + Tq^2r(y+1) + 2Tqr^2(y+1) + qr^2y(y+1) > 0. \end{aligned} \tag{WE3}$$

Suppose for contradiction that R is unbounded, and put $\kappa = q/r = 1/R$, which is infinitesimal. Let $S = H + yr > 0$. The ratio T/S is bounded. Indeed, if r is a unit, then

$$T < H + y \le S/r.$$

If r is infinitesimal, (WE1) and boundedness of y, T imply $H_0 = 1$, so S is a unit and T/S is again bounded. Since T is bounded, $(T^2+T)/S$ is bounded as well. Dividing the lower bound in (WE3) by the positive r^2S gives

$$\frac{P}{r^2S} \ge (1+y) - \kappa\frac{T(y+2)}{S} - \kappa^2\frac{T^2+T}{S}.$$

The last two terms are infinitesimal, while $1 + y \ge 1$. Thus $P > 0$. Equation (WE2) contradicts $n \ge 0$, including the equality case $n = 0$. Therefore R is bounded.

Now $r = qR$ is infinitesimal. Equation (WE1) becomes

$$g = H - 1 + q\{(1+y)R + T\} > 0,$$

so $H_0 = 1$. Both $B = yR + T$ and $L = B - H$ are bounded. The original denominator has the exact form

$$\Delta = (1+q)\{B^2 + 2BR + (1+y)B\} + (1-yq)R^2 + 2(1+y)R + 1 + y. \tag{WE4}$$

It is a positive bounded unit: all quantities are bounded, and $1 - yq$ is a positive unit, with constant contribution $1 + y \ge 1$. The unchanged numerator N_{src} satisfies

$$\frac{N_{\mathrm{src}}}{q} = (1+q)(B+R)(H+y) - (1+y)(1-qR)L. \tag{WE5}$$

Since $H_0 = 1$ and $L = B - H$, its residue is

$$(N_{\mathrm{src}}/q)_0 = (1+y_0)(1+R_0) > 0.$$

Thus N_{src}/q is a positive bounded unit. Finally, $A_c = 1 + R$ and $a^2 t = qL$, so the original coefficient is exactly

$$C = \frac{(1+R)(N_{\mathrm{src}}/q)}{2L\Delta}. \tag{WE6}$$

The numerator has bounded positive inverse, and $L\Delta$ is bounded. Therefore $1/C$ is bounded, as asserted. No lower bound on L was required; if L is infinitesimal the conclusion is stronger. □

The exact C2 reflection preserves directed type HI and exchanges C, α. Applying the theorem in both orientations shows that every physical HI source has both complementary positive square terms strong. For type HU the reflected type is UI; the reflected consequence is that α is automatically strong at every UI actual contact. These observations respect the directed weak-C orientation; they do not identify IH with HI.

C.10 Sources with five lower-bounded events

C.10.1 Five consecutive lower-bounded events

The following argument permits every retained event coefficient to be unbounded. It requires no canonical omission, actual-contact coordinates, or source-shape entry. Five consecutive events in the eight-cycle suffice.

Theorem C.26 (Five consecutive lower bounds)**.** *Let the original canonical source be physical, with $d > 0$ and all eight event coefficients nonnegative. Suppose*

$$d/m,\ d/c,\ d/A,\ d/D,\ d/\alpha \quad \textit{are positive bounded coefficient elements.}$$

Then the entire original parent is tensor SOS at every independent projective measurement pair and every real or complex projective state. There is no upper bound on any event coefficient divided by d. By exact CHSH relabeling the same conclusion holds if any five consecutive event coefficients in the cycle $(\lambda, D, c, A, \alpha, m, \beta, n)$ have these lower bounds.

Proof. Divide by the positive pure coefficient d, so $d = 1$. Let $\mu = \min(1, m, c, A, D, \alpha)$. It is a positive bounded unit. Select the parent F_μ with $m = c = A = D = \alpha = \mu$ and $\lambda = n = \beta = 0$. Keep every nonnegative excess and all three unselected coefficients as their original event squares. This uses only pure coefficient square roots. It makes no assertion that F_μ is globally physical.

At noncommuting residue tests apply the inherited noncommuting-residue theorem to the full physical source. It performs its own positive coefficient normalization if the event weights are unbounded. It remains to certify F_μ near every commuting residue test. Its coefficients are bounded, so a positive-definite real residue Gram is sufficient by the bounded positive-residue Gram lemma.

Write $\mathcal{H}$ for the signed core with $d = 1$ and $D_u = 1 + u^2, D_v = 1 + v^2$. On the four open commuting axes its exact homogeneous restrictions are

$$\begin{aligned} \mathcal{H}|_{u=0} &= 2D_v(X^2 + Y^2), \\ \mathcal{H}|_{v=0} &= 2\{(uW - Y)^2 + (X + uZ)^2\}, \\ \mathcal{H}|_{u=\infty} &= 2\{(W + vX)^2 + (vY - Z)^2\}, \\ \mathcal{H}|_{v=\infty} &= 2D_u(X^2 + Y^2). \end{aligned} \tag{NW1}$$

The remaining finite angle has nonzero residue. At $u = 0$, the retained m, D positive square terms force $W = Z = 0$ on the core kernel. At $v = 0$, the retained A, α positive square terms force $Y = Z = 0$, and then $W = X = 0$. At $u = \infty$, the retained c, A positive square terms force $W = Y = 0$, and then $X = Z = 0$. At $v = \infty$, the retained c, D positive square terms force $W = Z = 0$ after the core has forced $X = Y = 0$. Thus the selected residue state matrix is positive definite on every open axis, for either sign of the other finite angle.

Normalize a nonzero testing state by a coordinate of largest absolute value. Its coordinates are then bounded and one is a unit. At the four double corners, write F_{ab} for the homogeneous restriction of F_μ to $(u, v) = (a, b)$. Directly,

$$\begin{aligned} F_{00} &= (2 + 4\mu)X^2 + (2 + 12\mu)Y^2 + 4\mu Z^2, \\ F_{\infty 0} &= (2 + 8\mu)(W^2 + Z^2) + 4\mu Y^2, \\ F_{0\infty} &= 4\mu(W^2 + Z^2) + 2X^2 + (2 + 12\mu)Y^2, \\ F_{\infty\infty} &= 4\mu W^2 + (2 + 4\mu)X^2 + (2 + 12\mu)Y^2. \end{aligned} \tag{NW2}$$

Here infinity means the polynomial homogeneous chart, obtained by dividing the finite parent by the appropriate pure-testing squared angle. Every form is strict except its explicitly missing state coordinate. Consequently the only nontrivial residue states are the W ray at $(0, 0)$, the X ray at $(\infty, 0)$, and the Z ray at (∞, ∞).

At $(0, 0)$ normalize $W = 1$. The five local coordinates (u, v, X, Y, Z) are infinitesimal. The complete polynomial F_μ has no constant or linear term, and its degree-two part is

$$(2 + 8\mu)(u - Y)^2 + 2X^2 + 4\mu\{(v - X)^2 + Y^2 + Z^2\}. \tag{NW3}$$

Its residue is positive definite: vanishing successively gives $X = Y = Z = v = u = 0$.

At the corner $(\infty, 0)$ normalize $X = 1$ and put $p = 1/u$. The five infinitesimal coordinates are (p, v, W, Y, Z). The complete homogeneous polynomial again has no terms below degree two, and its quadratic part is

$$(2 + 4\mu)(W + v)^2 + (2 + 4\mu)(p + Z)^2 + 4\mu(W^2 + Y^2 + Z^2). \tag{NW4}$$

This is positive definite. At the corner (∞, ∞), normalize $Z = 1$ and put $p = 1/u, q = 1/v$. On (p, q, W, X, Y) the exact quadratic part is

$$(2 + 8\mu)(q - Y)^2 + 2X^2 + 4\mu\{(p - X)^2 + Y^2 + W^2\}. \tag{NW5}$$

It is positive definite as well. One can obtain (NW5) directly from (NW3) by the exact symmetry $(u, v; W, X, Y, Z) \mapsto (1/v, 1/u; Z, X, Y, W)$, which permutes the five equal retained weights.

For completeness, these quadratic calculations control the entire polynomial. In each local chart, allocate every higher monomial to a product of two of the five coordinate roots. The remaining monomial has positive degree, hence is infinitesimal; its coefficient is bounded because it is a real polynomial in the bounded unit μ. Symmetrizing gives an exact finite five-root Gram whose residue is (NW3), (NW4), or (NW5). No inversion, root, or comparison mixing the two fields occurs. The bounded positive-residue Gram lemma therefore certifies the full selected polynomial, including all higher terms.

Restore the excess event squares, the positive pure-testing homogeneous and projective state factors, and the positive coefficient d. The zero state is immediate. Splitting a complex state into real and imaginary parts proves the complex assertion.

Finally the exact full-parent Alice reflection acts on the displayed cycle indices as $i \mapsto 7-i$, and the swap/reflection $(u, v; W, X, Y, Z) \mapsto (1/v, 1/u; Z, X, Y, W)$ acts as $i \mapsto 6 - i$. Their composition gives a one-step cycle rotation. Their full-parent identities have only positive pure-testing homogeneous factors, so they transport the five-positive square term conclusion to every cyclic translate. □

This theorem is a sufficient source domain, not an exhaustive zero-d source cover. It requires five selected events to be positive and noninfinitesimal relative to d. Its all-tests conclusion is independent of actual contact angles and of all original event hierarchies. In particular, it contains both C1 and C2 sources with all six retained events bounded below relative to d.

C.11 Two small actual angles: common identities

C.11.1 Source coefficients and normal arrays

The following are the exact native scalar and state-normal coefficient arrays. Put $r = 1 - e$, $z = aB$, $y = tB$, and

$$c_a = 1 + aBr, \quad S_c = 1 + B(1 - h + ar), \quad g = 1 + a(hB + r - 1), \quad j = c_a - (1 - h)/t,$$

$$\begin{aligned}\Delta &= a + (1 + aB) + (1 + aB)B + 2aB(1 + aB)r + a^3B^2r^2 + 2a^2Br \\ &\quad + tB\{1 + a + aB + a^2Br(2 - r)\}, \\ M &= (1 + aB)(1 + ar)(h + t) - a(1 + tB)e(1 - h).\end{aligned}$$

The physical stationary event coefficients are

$$n = \frac{(1 + ar)c_aS_c - r\Delta}{2r\Delta}, \qquad m = \frac{a^2(1 + tB)ec_aS_c}{2g\Delta},$$
$$C = \frac{c_aM}{2(1 - h)\Delta}, \qquad Ah = \frac{tc_a(1 + ar)S_c}{2\Delta},$$
$$D = \frac{(1 + aB)(1 + ar)(1 - atBr)S_c}{2tj\Delta}, \qquad \alpha = \frac{B^2(1 + ar)M}{2(1 - hB)\Delta}.$$

The following scalar and normal arrays are exact rational functions of the native stationary source. Each case certificate proves the source bounds it needs at its own independent openings; the arrays themselves have no bounded-a or old R9-domain restriction.

Scalar arrays. The notation Ah in these formulas means the product of A and h.

$$\begin{aligned}
q_s &= nr^2 + (m+\tfrac12)e^2, \qquad q = q_s + \alpha,\\
a_w &= (C+D+\tfrac12)(1-h)^2 + Ah\,h,\\
b_2 &= 2mh + C(1-h)^2 + Ah\,h + (\alpha+\tfrac12)h^2,\\
b_4 &= h^2(m+\alpha+\tfrac12) + Ah\,h,\\
d_2 &= (n+\tfrac12)r^2 + 2Dr, \qquad d_4 = (n+D+\tfrac12)r^2,\\
e_0 &= me + D(1-h) + \alpha h + \tfrac12,\\
e_1 &= mhe + r(D+\tfrac12)(1-h) + (\alpha+\tfrac12)h,\\
c_F &= (b_2 - a^2nr^2)/t^2, \qquad \lambda = (m - a^2q)/(ta^2B),\\
f_{A0} &= Ah\,h/t^2 + d_2 - 2e_1/t + 2anr^2/t,\\
f_{C0} &= (\alpha - c_Ft^2B^2)/(taB^2),\\
f'_G &= -\{a_w/t^2 + D - 2e_0/t + 2aq/t + 2c_FtB\}/(aB).
\end{aligned}$$

The potentially singular a, B factors cancel in the combinations used in RC1. One direct proof is to clear the denominators in the displayed source coefficients and the printed monomials, and then expand; all three resulting numerator polynomials are zero. Equivalently, the original value and two first derivatives at contact give precisely RC1. Thus RC1 is an exact identity, including arbitrarily small a and literal $n = 0$.

All first-normal arrays. For the angular weight $q = q_s + \alpha$ define

$$\begin{aligned}
K'_X &= -nr + (m+\tfrac12)e - aq,\\
K'_E &= a_w/t^2 - e_0/t + aq/t,\\
K_Y &= \{-(C+D+\tfrac12)(1-h) + Ah\}/t + D - \alpha,\\
x_3 &= 2\{(D+\tfrac12)[(1-h)/t - 1] + e/2 - nr(1+a/t) - mh/t\},\\
x_5 &= -2r(n+D+\tfrac12),\\
a_E &= 2Ah\,h/t^2 - 2e_1/t, \qquad b_E = 2b_2/t^2, \qquad c_E = 2b_4/t^2,\\
a_Y &= 2\{Ah/t - me + r(D+\tfrac12) - \alpha - \tfrac12\},\\
b_Y &= 2\{-C(1-h) + Ah + m + (\alpha+\tfrac12)h\}/t,\\
c_Y &= 2\{Ah + h(m+\alpha+\tfrac12)\}/t.
\end{aligned}$$

The six exact contact balances. With $x = aB$, $y = tB$, the scalar balances are

$$f_{A0} = \lambda - 2d_4x - b_4y/t^2, \qquad f_{C0} = \lambda + b_4y/t^2, \qquad f'_G = 3\lambda - d_4x + b_4y/t^2. \tag{RC1}$$

The complete first-normal balances are

$$2K'_E + a_Ex + b_Ey + c_Exy + 2\lambda x = 0,$$

$$2K'_X + x_3y + x_5xy - 2t\lambda x = 0, \qquad 2K_Y + a_Yx + b_Yy + c_Yxy = 0. \tag{RC6}$$

Substitution of the native weights and multiplication by the displayed source denominators gives six zero polynomial numerators. Equivalently these are the original contact value and its five first derivatives. They underlie the generic whole-parent identity BA7 (also printed as R10.9). BA6 explicitly gives its entire hard quadratic. Those identities hold before any source size assumption; the bounded-a, R8 and R10 certificates separately establish their coefficient bounds and pivots.

C.11.2 The five-root identity

Set $\kappa = 1/t$, $\ell = 1/a$. The letters e_0, e_1 are the native scalar coefficients above. In this subsection a_g denotes the scalar coefficient called a in the five-root table, and not the source ratio.

Five homogeneous contact roots. Put $U = u^2$, $V = v^2$, and define

$$\xi = X + ruZ, \qquad q = \sqrt{A}\,(Y - huW), \qquad R_1 = uW - \kappa vZ, \quad R_2 = vW - \ell uZ, \quad T = uvW - BZ. \tag{WA3}$$

Every root vanishes at the actual source contact. Inverses and radicals in their coefficients are pure source. The large A positive square term is retained in q; no positive square term is subtracted from the parent and no physicality hypothesis is transferred to a modified remainder.

Use the following source coefficients for the exact scalar graph:

$$\begin{aligned} a_g &= (C+D+\tfrac12)(1-h)^2+Ah^2, & a_4 &= Ah^2,\\ b_2 &= 2mh+C(1-h)^2+(A+\alpha+\tfrac12)h^2, & b_4 &= h^2(m+A+\alpha+\tfrac12),\\ c &= \alpha, & c_2 &= nr^2+(m+\tfrac12)(1-r)^2+\alpha,\\ c_4 &= nr^2, & d_2 &= (n+\tfrac12)r^2+2Dr,\\ d_4 &= (n+D+\tfrac12)r^2. \end{aligned}$$

Thus the graph on $\xi = q = 0$ is

$$\begin{aligned} F_g/4 = {}& [a_gU + a_4U^2 + (m + b_2U + b_4U^2)V]W^2\\ &+ [c + c_2U + c_4U^2 + (D + d_2U + d_4U^2)V]Z^2 - 2uv(e_0 + e_1U)WZ, \end{aligned}$$

where e_0, e_1 are exactly the original coefficients in (WS3). Set

$$\begin{gathered} \beta_s = \frac{Bd_4+\kappa c_4}{\kappa\ell^2}, \qquad \gamma_s = b_4+\frac{d_4}{\kappa\ell},\\ j_0 = \frac{Bd_4-a_4\kappa^2\ell+d_2\ell}{2\kappa\ell^2}, \qquad j_1 = \frac{d_4}{2\kappa\ell}, \qquad J_s(U) = j_0+j_1U,\\ f_s = \frac12\left(b_2-\frac{c_4}{\ell^2}-\frac{c}{B^2}-\frac{Bd_4}{\kappa\ell^2}\right). \end{gathered} \tag{WA4}$$

The four original scalar contact equations (value and the three logarithmic derivatives in $u, v, W/Z$) give the exact identity

$$F_g/4 = (R_1,R_2,T)\,\mathsf{S}\,(R_1,R_2,T)^T, \qquad \mathsf{S} = \begin{pmatrix} a_g+a_4U & uvJ_s & -vJ_s\\ uvJ_s & m+\beta_sU & u(f_s-j_1U)\\ -vJ_s & u(f_s-j_1U) & c/B^2+\gamma_sU \end{pmatrix}. \tag{WA5}$$

This is a different scalar identity from the excluded polynomial two-block splitter (JG7). In particular, the last diagonal retains the full radial mass α/B^2 and the Gram depends on the independent testing variables. The source bounds for this exact identity are established in each coefficient domain below.

Both full first normals and the complete hard block. For the ξ normal define

$$\begin{gathered} p_0 = \frac{D+r(n+1/2)}{\kappa}, \qquad p_1 = \frac{r(n+D+1/2)}{\kappa}, \qquad q_1 = \frac{nr+Bp_1}{\ell},\\ f_x = -mh+(D+\tfrac12)(1-h)-p_0-q_1,\\ \mathsf{L}_x = \big(uv(p_0+p_1U),\ -m+q_1U,\ u(f_x-p_1U)\big). \end{gathered} \tag{WA6}$$

For the unnormalized $\eta = Y - huW$ normal set

$$\begin{gathered} p_y = -(C+D+\tfrac12)(1-h)+Ah, \qquad p_y' = Ah,\\ k_v = -C(1-h)+Ah+m+(\alpha+\tfrac12)h, \qquad k_{uv} = h(A+m+\alpha+\tfrac12),\\ k_z = -m(1-r)+r(D+\tfrac12)-\alpha-\tfrac12,\\ q_y = \frac{-k_z-\kappa p_y'-Bk_{uv}}{\ell}, \qquad f_y = k_v-q_y,\\ \mathsf{L}_y = \big(p_y+p_y'U,\ uvq_y,\ v(f_y+k_{uv}U)\big). \end{gathered} \tag{WA7}$$

The two remaining original source stationarity equations, in X, Y, give these exact representations of the complete first normals. No first-normal term is discarded. Define also

$$\begin{aligned} h_{xx} &= n(1+U)(1+V)+m+(D+\tfrac12)UV+\tfrac12(1+V),\\ h_{xy} &= -uv(m+D+\tfrac12),\\ h_{yy} &= A(1+U)(1+V)+C(1+V)+D+\tfrac12+mUV+(\alpha+\tfrac12)(1+U)V. \end{aligned}$$

Then the *entire original four-state parent* is exactly

$$\begin{gathered} F/4 = \mathcal{R}^T\mathsf{G}(u,v)\mathcal{R}, \qquad \mathcal{R} = (\xi,q,R_1,R_2,T)^T,\\ \mathsf{G} = \begin{pmatrix} h_{xx} & h_{xy}/\sqrt{A} & \mathsf{L}_x\\ h_{xy}/\sqrt{A} & h_{yy}/A & \mathsf{L}_y/\sqrt{A}\\ \mathsf{L}_x^T & \mathsf{L}_y^T/\sqrt{A} & \mathsf{S} \end{pmatrix}. \end{gathered} \tag{WA8}$$

Block notation in (WA8) has sizes $1, 1, 3$. Every coefficient of this finite polynomial matrix is pure source. It then verifies (WA5)–(WA7) from the corresponding four plus two contact equations. Thus (WA8) is not merely a graph or Hessian certificate.

C.11.3 Division at the source-ratio zero

Use the native coefficient dictionary and put $\mu = m/a^2$, $\alpha_* = \alpha/B^2$.

The only new potentially unbounded local coefficient is $(b_2 - \alpha_*)/a$, where

$$b_2 = 2a^2\mu h + C(1-h)^2 + (Ah)h + (B^2\alpha_* + 1/2)h^2.$$

At $a = 0$, with $d_* = 1 + B(1+t)$, the exact values are

$$C = \frac{h+t}{2(1-h)d_*}, \quad Ah = \frac{t[1+B(1-h)]}{2d_*}, \quad \alpha_* = \frac{h+t}{2(1-Bh)d_*}.$$

Substitution gives $b_2 = \alpha_*$ identically for arbitrary B, h, t. Clearing the unit denominators in (WE1), the numerator of $b_2 - \alpha_*$ is therefore divisible by a in the polynomial ring. Its quotient has a unit denominator at $B = a = h = 0$. Hence

$$Q_* = (b_2 - \alpha_*)/a \quad \text{is bounded.} \tag{WE3}$$

This is a joint regularity argument, so it permits arbitrary h/a, including an unbounded ratio.

The complete local five-root Gram survives. Keep $\kappa = B/y$ and every coefficient and normal in (WA3)–(WA8), but set $\ell = 1/a$ and replace R_2 by

$$\widetilde{R}_2 = aR_2 = avW - uZ.$$

The entire full-parent matrix is the exact congruence

$$\widetilde{G} = EGE, \qquad E = \operatorname{diag}(1,1,1,a^{-1},1), \qquad \widetilde{\mathcal{R}} = (\xi, q, R_1, \widetilde{R}_2, T). \tag{WE4}$$

Its newly rescaled coefficients are

$$\begin{gathered} m/a^2 = \mu, \qquad \beta_s/a^2 = (Bd_4 + \kappa c_4)/\kappa, \\ J_s(U)/a = \frac{aBd_4 - a_4\kappa^2 + d_2 + d_4U}{2\kappa}, \qquad f_s/a = \tfrac{1}{2}[Q_* - ac_4 - aBd_4/\kappa], \\ (-m + q_1U)/a = -a\mu + (nr + Bp_1)U, \\ q_y/a = -k_z - \kappa p_y' - Bk_{uv}. \end{gathered} \tag{WE5}$$

C.11.4 The angular circuit

For $\lambda = (m - a^2q)/(ta^2B) > 0$ and $q = nr^2 + (m + 1/2)e^2 + \alpha$, the following identity holds before any source-size assumption. Set $U = u^2$, $V = v^2$, $k = \sqrt{x}$, and $c = \sqrt{a/t}$. Use the state-linear forms

$$\begin{gathered} E = tuW - vZ, \quad \eta = Y - huW = \sqrt{h}\,N_y, \quad \xi = X + ruZ, \quad F_v = (V - y)Z, \quad R_D = v(U - x)Z, \\ J = (uv - x/c)Z, \quad Q = \sqrt{q}(avW - uZ - a\xi), \quad N = \sqrt{n}\{r[(a/t)V - U]Z + u\xi\}, \\ R = \sqrt{\lambda}\,tk(cv - u)W, \qquad T = \sqrt{2\lambda/c}\,k\sqrt{uv}(tcW - Z). \end{gathered} \tag{BA4}$$

The square root in T is pure testing. Initially take $u, v > 0$; literal axes are addressed below.

For a compact coefficient table put

$$\begin{aligned} b_2 &= 2mh + Cl^2 + (A + \alpha + \tfrac{1}{2})h^2, & b_4 &= h^2(m + A + \alpha + \tfrac{1}{2}), \\ d_4 &= (n + D + \tfrac{1}{2})r^2, & e_1 &= mhe + r(D + \tfrac{1}{2})l + (\alpha + \tfrac{1}{2})h, \\ c_F &= (b_2 - a^2nr^2)/t^2, & a_E &= 2Ah^2/t^2 - 2e_1/t, \\ b_E &= 2b_2/t^2, & c_E &= 2b_4/t^2, \\ a_Y &= 2[Ah/t - me + r(D + \tfrac{1}{2}) - \alpha - \tfrac{1}{2}], & & \\ b_Y &= 2[-Cl + Ah + m + (\alpha + \tfrac{1}{2})h]/t, & & \\ c_Y &= 2[Ah + h(m + \alpha + \tfrac{1}{2})]/t, & & \\ x_3 &= 2[(D + \tfrac{1}{2})(l/t - 1) + e/2 - nr(1 + a/t) - mh/t], & x_5 &= -2r(n + D + \tfrac{1}{2}). \end{aligned} \tag{BA5}$$

Let $Y_* = hE/t + \eta$ and define the entire hard quadratic explicitly by

$$\begin{aligned} H_0 = {} & CD_v(lE/t - \eta)^2 + AD_uD_vY_*^2 + (D + \tfrac{1}{2})(lE/t - \eta + uv\xi)^2 \\ & + (\alpha + \tfrac{1}{2})D_uVY_*^2 + [n(1 + V + UV) + D_v/2]\xi^2 \\ & + 2m(VE/t - uv\xi)Y_* + mUVY_*^2. \end{aligned} \tag{BA6}$$

In particular BA6 has no inverse testing angle and no a denominator. Upon replacing η by $\sqrt{h}N_y$, every occurrence of A is through Ah, $\sqrt{h}Ah$, or hAh.

With $\sigma = t\lambda x/\sqrt{q}$, the exact identity is

$$\begin{aligned} P = {} & Q^2 + N^2 + T^2 + R^2 + \lambda J^2 + d_4 R_D^2 + (c_F + b_4 x/t^2)F_v^2 + (b_4/t^2)vR_D F_v \\ & + H_0 - \lambda x E^2 - \mu\xi^2 - 2\sigma Q\xi \\ & + E[(a_E + c_E V)R_D + (b_E + c_E x)vF_v] \\ & + \sqrt{h}N_y[(a_Y + c_Y V)R_D + (b_Y + c_Y x)vF_v] \\ & + \xi[x_5 uvR_D + u(x_3 + x_5 x)F_v]. \end{aligned} \tag{BA7}$$

This is the generic RC/R10 identity, not a scalar restriction. One pen-and-paper verification is to substitute BA4 into BA7 and use

$$R^2 + T^2 + \lambda J^2 = \lambda\{t^2 x[U + (a/t)V]W^2 - 4txuvWZ + (UV + xy)Z^2\}.$$

The hard quadratic is then exactly BA6. The remaining six scalar and first-normal coefficient identities follow by substituting the native source fractions in Section C.11.1; a common denominator leaves zero polynomial numerators. Equivalently these six identities are the original value and all five first-contact conditions. Both the full identity and those six numerator equalities have also been independently expanded during verification.

C.11.5 Homogeneous identities for unit $t > 1$

The following polynomial identities hold before any source-size specialization. Their strict residues in this subsection use $t_0 = \tau > 1$, $C \sim c_0 = \tau/2$, $D \sim d_0 = 1/[2(\tau - 1)]$, $A^{-1/2}$ infinitesimal and n, m, α infinitesimal. Each source calculation establishes its own coefficient bounds; in particular R10.6 below supplies the exact modification when m is merely bounded. Put $\widehat{Y} = \sqrt{A}Y$, so $Y = A^{-1/2}\widehat{Y}$.

First suppose u is a bounded unit or unbounded and v is infinitesimal. Put $p = 1/u$, $D_p = 1 + p^2$ and use

$$N_\infty = \sqrt{n}X, \quad V_X = vX, \quad K = pX + Z, \quad R_\infty = \sqrt{\alpha}pX, \quad W, \quad \widehat{Y}.$$

Let $J_\infty = W + D_p V_X - pY - pvK$. The entire homogeneous parent is

$$\begin{aligned} p^2 P(1/p, v) = {} & D_p D_v N_\infty^2 + m(pvW + vY - K)^2 + CD_v(W - pY)^2 + D_p D_v \widehat{Y}^2 \\ & + DJ_\infty^2 + D_p[R_\infty + \sqrt{\alpha}(vY - K)]^2 \\ & + \tfrac{1}{2}\{J_\infty^2 + (1 - p^2 v^2)K^2 \\ & \quad + D_p[v^2 Y^2 - 2vYK + 2pYV_X + 2pvKV_X - p^2 V_X^2]\}. \end{aligned} \tag{BA11}$$

It follows directly by substituting $Z = K - pX$ in the quarter-parent $P = F/4$ defined by (C.1). Its bounded six-root residue, with $d_p = 1 + p_0^2$, is

$$d_p(N_\infty^2 + R_\infty^2 + \widehat{Y}^2) + K^2/2 + c_0 W^2 + (d_0 + \tfrac{1}{2})(W + d_p V_X)^2 - d_p p_0^2 V_X^2/2.$$

The W, V_X determinant is $d_p c_0(d_0 + 1/2) > 0$, using $c_0(d_0 + 1/2) = (c_0 + d_0 + 1/2)/2$. All other pivots are positive. No inverse of n or α occurs, and either root may vanish. This covers literal $u = \infty$ by $p = 0$ as well.

Next suppose u is infinitesimal and v is a bounded unit. Use the five roots

$$(S, R_u, X, \widehat{Y}, Z) = (\sqrt{m}W, uW, X, \sqrt{A}Y, Z).$$

The exact full polynomial is obtained from the quarter-parent $P = F/4$ defined by (C.1) by the two replacements

$$m(vW - X + uvY - uZ)^2 = (vS - \sqrt{m}X + \sqrt{m}\,uvY - \sqrt{m}\,uZ)^2, \qquad uW = R_u$$

in all other terms. This operation uses no inverse of m or u. Its separate residue pivots on $S, X, \widehat{Y}$ are $V, (1 + V)/2, 1 + V$, where $V = v_0^2 > 0$. The remaining R_u, Z block is

$$\begin{pmatrix} c_0(1 + V) + d_0 + 1/2 & -(d_0 + 1/2)v_0 \\ -(d_0 + 1/2)v_0 & d_0 V \end{pmatrix}, \qquad \det = c_0 d_0 V^2 > 0. \tag{BA12}$$

Here $c_0 d_0 - d_0/2 - 1/4 = 0$. If instead v is unbounded, use $q_v = 1/v$ and the same five roots. Multiplying the same exact polynomial by q_v^2 gives, at $u = q_v = 0$, the diagonal residue

$$(S, R_u, X, \widehat{Y}, Z): \quad (1, c_0, 1/2, 1, d_0) > 0. \tag{BA13}$$

This homogeneous substitution is polynomial and also applies at literal $q_v = 0$.

If both finite testing residues are nonzero, the hard $\widehat{Y}$ pivot is $(1+U)(1+V)$. The complete remaining three-state residue on (W, X, Z) is

$$c_0(1+V)UW^2 + (d_0+\tfrac{1}{2})(u_0W + u_0v_0X - v_0Z)^2 + \tfrac{1}{2}(1+V)(X+u_0Z)^2 - \tfrac{1}{2}(1+U)VZ^2,$$

where $U = u_0^2, V = v_0^2$. Its determinant is

$$(1+U)(1+V)UV\left\{\frac{c_0(2d_0+1)}{4}U + \frac{c_0d_0}{2}V\right\} > 0. \tag{BA14}$$

The $Z = 0$ restriction is strictly positive, so the full matrix is positive definite. Every coefficient is bounded after the hard-state normalization.

The one-infinity cases not already covered by BA11 or BA13 are strict directly. At $u = \infty$ and a bounded unit v, the residue is

$$c_0(1+V)W^2 + (d_0+\tfrac{1}{2})(W+v_0X)^2 + Z^2/2 + (1+V)\widehat{Y}^2 \succ 0.$$

At $v = \infty$ and a bounded unit u, it is

$$c_0UW^2 + (1+U)X^2/2 + d_0(u_0X - Z)^2 + (1+U)\widehat{Y}^2 \succ 0.$$

These are the degree-two homogeneous values of the quarter-parent $P = F/4$ defined by (C.1) in the respective angles; nearby reciprocal coordinates are pure-testing infinitesimals.

For two infinite angles put $p = 1/u$, $q_v = 1/v$ and introduce $P_\infty = pZ$, $Q_\infty = q_vZ$. The entire exact polynomial is

$$\begin{aligned} p^2q_v^2P(1/p, 1/q_v) &= n(1+p^2)(1+q_v^2)X^2 + m(pW - pq_vX + Y - Q_\infty)^2 \\ &\quad + C(1+q_v^2)(W - pY)^2 + (1+p^2)(1+q_v^2)\widehat{Y}^2 \\ &\quad + D(q_vW + X - pq_vY - P_\infty)^2 + \alpha(1+p^2)(Y - Q_\infty)^2 \\ &\quad + \tfrac{1}{2}\{(q_vW + X - pq_vY - P_\infty)^2 + (1+p^2)(Y - Q_\infty)^2 \\ &\qquad + (1+q_v^2)[p^2X^2 + 2XP_\infty - P_\infty^2]\}. \end{aligned} \tag{BA15}$$

Its residue on $(W, X, \widehat{Y}, P_\infty, Q_\infty)$ is

$$c_0W^2 + (d_0+\tfrac{1}{2})X^2 - 2d_0XP_\infty + d_0P_\infty^2 + \widehat{Y}^2 + Q_\infty^2/2 \succ 0.$$

The nontrivial determinant is $d_0/2 > 0$. At $p = q_v = 0$ the two extra roots vanish, respecting the original null state rather than inventing a positive Z pivot.

C.11.6 Bounded source ratio

Let K_c, K_t be independently labelled real closed extensions of $\mathbb{R}$. All source letters below belong to K_c, and all testing letters belong to K_t. No algebraic-disjointness hypothesis is imposed. Write I for positive infinitesimals and call a positive bounded element a unit when its reciprocal is bounded.

We use the actual stationary C2 source, normalized by $d = 1$, with $\lambda_{\text{event}} = \beta = 0$. Put

$$x = aB, \quad y = tB, \quad l = 1 - h, \quad e = 1 - r, \quad g = 1 - ae + haB, \quad j = 1 + aBr - l/t.$$

The source assumptions are

$$\begin{aligned} &B, x, y \in \mathrm{I}, \qquad a > 0 \text{ bounded}, \qquad t \text{ a positive bounded unit}, \\ &0 < h, r < 1, \qquad g, j, 1 - txr > 0, \qquad n \geq 0, \qquad C \geq c_* > 0 \quad (c_* \in \mathbb{R}). \end{aligned} \tag{C.24}$$

The last strict gap is exactly the original $B - xyr > 0$ divided by B. The source formulas, which also fix our normalization, are

$$\begin{aligned} c_a &= 1 + aBr, \quad S = 1 + B(l + ar), \\ \Delta &= a + (1+aB) + (1+aB)B + 2aB(1+aB)r + a^3B^2r^2 + 2a^2Br \\ &\quad + tB\{1 + a + aB + a^2Br(2-r)\}, \\ M &= (1+aB)(1+ar)(h+t) - a(1+tB)el, \\ n &= \frac{(1+ar)c_aS - r\Delta}{2r\Delta}, \qquad m = \frac{a^2(1+tB)ec_aS}{2g\Delta}, \\ C &= \frac{c_aM}{2l\Delta}, \qquad Ah = \frac{tc_a(1+ar)S}{2\Delta}, \\ D &= \frac{(1+aB)(1+ar)(1-taBr)S}{2tj\Delta}, \qquad \alpha = \frac{B^2(1+ar)M}{2(1-hB)\Delta}. \end{aligned} \tag{C.25}$$

The original source is physical; no physicality assertion about a polynomial obtained by a formal coefficient substitution will be used.

Proposition C.27. *Under* (C.24)*, the entire C2 parent is a finite sum of squares in $K_c \otimes_{\mathbb{R}} K_t$ for every independent projective testing pair and every state. The conclusion includes $n = 0$, every relative rate of g, j, h, l, r, e, and a infinitesimal in residue.*

For finite tests set $D_u = 1 + u^2$, $D_v = 1 + v^2$ and define the quarter-parent

$$\begin{aligned} P = {} & m(vW - X + uvY - uZ)^2 + nD_uD_vX^2 + CD_v(uW - Y)^2 + AD_uD_vY^2 \\ & + (D + \tfrac{1}{2})(uW + uvX - Y - vZ)^2 + (\alpha + \tfrac{1}{2})D_u(vY - Z)^2 \\ & + \tfrac{1}{2}D_v(X + uZ)^2 - \tfrac{1}{2}D_uD_vZ^2. \end{aligned} \tag{C.26}$$

All identities below are identities of this four-state polynomial.

Gram completion and real caps. Apply Lemma 3.2 to the bounded remainder after retaining the displayed nonnegative squares. The roots may be unbounded; only its matrix and positive-definite real residue enter this lemma.

For real caps, let a real quadratic form be positive definite on the common kernel of finitely many squared rows. A sufficiently large real multiple of those squares makes the form positive definite: decompose into the kernel and its orthogonal complement and take the Schur complement. Each unbounded source weight admits that cap; a finite weight admits a sufficiently close smaller cap when the finite-residue form is strict. Retain every nonnegative excess.

Source bounds when e is a unit. The denominator has residue $1 + a_0 > 0$. The physical numerator has the exact identity

$$\begin{aligned} 2r\Delta n = eQ_s - BK_s \ge 0, \qquad & Q_s = 1 + B(1 + t + ah + 2ar + at) \\ & \quad + B^2\{a(h + t) + a^2[h(r + 1) + t(1 + r - r^2)]\}, \\ & K_s = (1 + a)(h + t)(1 + aB). \end{aligned} \tag{C.27}$$

Consequently $(nr)_0 = e_0/[2(1 + a_0)] > 0$. In particular n has a positive real lower bound and $n = 0$ is impossible on this branch. The displayed source formulas give positive bounded units

$$nr, \qquad mg/a^2, \qquad Ah, \qquad Cl, \qquad Dj, \qquad \alpha_* := \alpha/B^2. \tag{C.28}$$

For Cl and α_* one needs the actual strong-C assumption: if l is a unit it makes M a unit; if l is infinitesimal then $h_0 = 1$ and $M_0 = (1 + a_0r_0)(1 + t_0) > 0$. This explicitly excludes the weak-C corner where $M_0 = 0$. Thus A, C, D, n have positive real lower bounds. If g is a unit, m/a^2 is a positive unit; if g is infinitesimal, a is a unit and m is unbounded.

One exact five-root matrix for the unit-e local chart. Use the complete WA matrix, including both normal rows, in the roots

$$z = (\xi, \eta, R_1, R_2', T)^T, \quad \xi = X + ruZ, \quad \eta = Y - huW,$$

$$R_1 = uW - vZ/t, \qquad R_2' = avW - uZ, \qquad T = uvW - BZ, \qquad P = z^TGz.$$

The precise generic input here is WA3–WA8: replace its normalized $q = \sqrt{A}\eta$ by η, and its R_2 by R_2'/a. It is an exact full-parent identity derived from the six original contact equations, not an assertion about a Hessian.

The full divergent D channel has the exact identity

$$P = (D - d_0)L_D^2 + z^T\{G|_{D=d_0} + (D - d_0)jH_D\}z, \tag{C.29}$$

where $0 < d_0 < D$ is real,

$$L_D = uv\xi - \eta + lR_1 + ruvR_2' - arvT,$$

and the only nonzero symmetric entries of H_D are

$$(H_D)_{13} = tuv, \quad (H_D)_{15} = -tu, \quad (H_D)_{34} = rtuv, \quad (H_D)_{35} = -artv.$$

For completeness the coefficient identity proving this is

$$\partial_D G = dd^T + \tfrac{1}{2}(sk^T + ks^T) + jH_D,$$

$$d = (uv, -1, l, ruv, -arv)^T,$$
$$s = (0, 0, (aB - u^2)v, -(B - v^2/t)u, u^2 - av^2/t)^T,$$
$$k = (-2rtu, 0, 0, -r^2tu, ar^2t)^T.$$

The root syzygy $s^T z = 0$ and $d^T z = L_D$ prove (C.29) including every normal entry. Since Dj is bounded, the correction is bounded and vanishes in local residue.

If g is infinitesimal, also retain $(m - K)R_{mc}^2$, where $K > 0$ is real and

$$R_{mc} = -\xi + uv\eta + \frac{1 - g}{a} R_2' + huT, \qquad \tau = (m - K)g.$$

The remaining exact matrix is $G|_{m=K} + \tau H_m$, before the independent D operation. In unscaled $R_2 = R_2'/a$ coordinates the only nonzero symmetric entries are

$$(H_m)_{14} = -1, \quad (H_m)_{25} = v, \quad (H_m)_{44} = 2 - g, \quad (H_m)_{45} = hu.$$

This is the full centered-m identity R5.2–R5.5. It commutes with (C.29); neither correction depends on the other coefficient. Here a is a unit and τ is bounded.

Put $N = n + 1/2$, $L = A + C + d_0 + 1/2$ and normalize $\widehat{\xi} = \sqrt{N}\xi$, $\widehat{\eta} = \sqrt{L}\eta$. Every coefficient of the resulting five-root matrix is bounded. Here are the complete bounds needed from the WA table. Its scalar entries involve nr^2, Cl^2, Ah^2, α_*, the capped m, and bounded D; its first normals involve nr, Cl, Ah and these same bounded products; the hard block uses only n/N, A/L, C/L and bounded entries divided by $\sqrt{N}$ or $\sqrt{L}$. All are bounded by (C.28). If a is infinitesimal, then g is a unit and the only apparently singular scaled entry is

$$\frac{b_2 - \alpha_*}{a}, \qquad b_2 = 2mh + Cl^2 + (Ah)h + (B^2\alpha_* + 1/2)h^2. \tag{C.30}$$

The WE3 numerator vanishes exactly at $a = 0$, for arbitrary B, h, r, t. Cancel that factor in the actual source formula. Its remaining denominator uses only $g, \Delta, t, 1 - hB$, all units here; the h, l, r, j denominators have already canceled in the displayed products. The WE5 rewrites cover both normal rows. Thus this is joint regularity, including all simultaneous openings, rather than separate one-variable limits.

At $u = v = 0$ the residue splits into two 2×2 blocks and a scalar. The $(\widehat{\eta}, R_1)$ block has determinant

$$\frac{A(C + d_0 + 1/2)}{A + C + d_0 + 1/2}, \qquad \frac{1}{\det} = \frac{1}{A} + \frac{1}{C + d_0 + 1/2}. \tag{C.31}$$

Its upper-left entry is 1, its other entries are bounded, and the determinant is a positive unit. If g is a unit, the $(\widehat{\xi}, R_2')$ block is

$$\begin{pmatrix} 1 + m/N & -m/(a\sqrt{N}) \\ -m/(a\sqrt{N}) & m/a^2 \end{pmatrix}, \qquad \det = m/a^2 > 0.$$

If g is infinitesimal, its determinant is

$$\frac{K + 2\tau_0 - \tau_0^2\nu}{a_0^2} > 0,$$

where $\nu = \mathrm{st}(N^{-1})$, also when N is unbounded. Choose $K > 2\tau_0^2 + 1$, since $0 \le \nu \le 2$. The remaining T pivot is $\alpha_* > 0$. Therefore the complete bounded matrix has strict residue. This proves every local independent test on the unit-e branch without any contact conductor.

The small-e branch: source limits and physical Schur bound. Now e is infinitesimal and $r_0 = 1$. The actual source formulas imply

$$n, m, \alpha \in \mathrm{I} \cup \{0\}, \quad g_0 = 1, \quad (Ah)_0 = t_0/2, \quad (Cl)_0 = (h_0 + t_0)/2, \quad (Dj)_0 = 1/(2t_0), \quad (\alpha_*)_0 = (1 + a_0)(h_0 + t_0)/2. \tag{C.32}$$

In particular A, C, D have positive real lower bounds, even if the strong-C hypothesis had not been imposed. From $j > 0$ we have $t_0 \ge l_0$. Define

$$q = nr^2 + (m + 1/2)e^2 + \alpha, \quad \mu = m - a^2 q, \quad \lambda = \frac{\mu}{ta^2 B}, \quad \sigma = \frac{t\lambda aB}{\sqrt{q}}.$$

The $a^2 B$ factor in μ cancels exactly in the native rational formula. Its canceled denominator at $B = 0, r = 1$, with h free, is $2t(a + 1)$; hence

$$\lambda_0 = \frac{h_0 + t_0}{2t_0} > 0, \qquad \mu \in \mathrm{I}. \tag{C.33}$$

This includes $a_0 = 0$ and does not assume h infinitesimal.

The exact inequality (C.27) gives $e/B \geq K_s/Q_s$, where $Q_s \sim 1$ and $K_s \sim (1+a)(h+t)$. Since $q \geq e^2/2 + \alpha$, it follows that

$$\mathrm{st}(\sigma^2) \leq \frac{a_0^2(h_0+t_0)}{2(1+a_0)\{1+(1+a_0)(h_0+t_0)\}}. \tag{C.34}$$

If e/B is unbounded the left side is zero. In all cases

$$\mathrm{st}(1/2-\mu-\sigma^2) \geq \frac{1+a_0+(h_0+t_0)(1+2a_0)}{2(1+a_0)\{1+(1+a_0)(h_0+t_0)\}} > 0. \tag{C.35}$$

No division by n occurs, so $n = 0$ is retained.

The exact small-e full-parent decomposition. Use the generic BA4–BA7 identity and its complete displayed H_0. In particular take

$$E = tuW - vZ, \quad \eta = Y - huW, \quad \xi = X + ruZ, \quad F_v = (v^2-y)Z, \quad R_D = v(u^2-x)Z,$$
$$J = (uv - \sqrt{xy})Z, \quad Q = \sqrt{q}(avW - uZ - a\xi), \quad N = \sqrt{n}\{r[(a/t)v^2 - u^2]Z + u\xi\}.$$

With $k = \sqrt{x}$ and $c = \sqrt{a/t}$ retain the whole nonnegative positive square term

$$T^2 = \frac{2\lambda}{c}k^2uv(tcW - Z)^2, \qquad R = \sqrt{\lambda}\,tk(cv-u)W.$$

Initially $u, v > 0$; the literal axes and the other signs are restored below. The generic identity has the exact circuit

$$R^2 + T^2 + \lambda J^2 = \lambda\{t^2x[u^2 + (a/t)v^2]W^2 - 4txuvWZ + (u^2v^2 + xy)Z^2\}.$$

Let $\mathscr{P} = P - N^2 - T^2$. Its entire D channel is

$$\mathscr{P} = DL_D^2 + \mathscr{P}|_{D=0} - 2Dj\,uF_v\xi, \qquad L_D = lE/t - rR_D - \eta + uv\xi. \tag{C.36}$$

This identity holds before any source specialization. Indeed the D terms are

$$D(lE/t - \eta + uv\xi)^2 + Dr^2R_D^2 - 2Dr(lE/t - \eta + uv\xi)R_D + 2D(l/t - 1 - rx)uF_v\xi,$$

whose sum is (C.36) by $j = 1 + xr - l/t$.

Choose positive real caps $A_c < A$, $C_c < C$, $d_0 < D$. Subtract the following exact nonnegative positive square terms:

$$N^2 + T^2 + (D-d_0)L_D^2 + (A-A_c)D_uD_v\eta^2 + (C-C_c)D_v\eta^2. \tag{C.37}$$

The remainder is the full BA quadratic with D formally set to d_0, minus $2(D-d_0)j\,uF_v\xi$, and with precisely the two displayed centered-η squares removed. Every coefficient is bounded. The apparent A, C poles are now only Ah, Cl; for example

$$AD_uD_v(hE/t+\eta)^2 - (A-A_c)D_uD_v\eta^2 = D_uD_v\{A_c\eta^2 + 2(Ah)E\eta/t + (Ah)hE^2/t^2\},$$
$$CD_v(lE/t-\eta)^2 - (C-C_c)D_v\eta^2 = D_v\{C_c\eta^2 - 2(Cl)E\eta/t + (Cl)lE^2/t^2\}.$$

Every other entry of BA5–BA7 uses the bounded products in (C.32), a, t, t^{-1}, or the bounded λ, σ. The correction in (C.36) is bounded because Dj is bounded. There is no a, h, l, or j inverse left in the matrix coefficients. Pure-source inverse factors inside the separately retained positive square term T^2 do not need to be bounded.

At $u = v = 0$ write $d_+ = d_0 + 1/2$ and use residues of the bounded source letters on the right. The entire nontrivial hard block is exactly

$$d_+(R_D - lE/t + \eta)^2 + \frac{t+hl}{2t^2}E^2 - \frac{h}{t}E\eta + (A_c+C_c)\eta^2. \tag{C.38}$$

The remaining blocks are

$$Q^2 - 2\sigma_0Q\xi + \tfrac{1}{2}\xi^2, \quad \frac{h_0+t_0}{2t_0^2}F_v^2, \quad \lambda_0J^2, \quad R^2.$$

The first is strict by (C.35). To make (C.38) strict it suffices to choose the caps so that $2(A_c+C_c)(t_0+h_0l_0) > h_0^2$. If $h_0, l_0 > 0$, the actual finite source residues satisfy

$$A_0 = \frac{t_0}{2h_0}, \quad C_0 = \frac{h_0+t_0}{2l_0}, \quad 2(A_0+C_0)(t_0+h_0l_0) - h_0^2 = \frac{t_0(t_0+h_0)}{h_0l_0} > 0.$$

Thus sufficiently close smaller real caps work. If $h_0 = 0$ or $l_0 = 0$, respectively A or C is unbounded, so a sufficiently large real cap works. This proves strictness also when j is infinitesimal or opens simultaneously with either endpoint. The full bounded-matrix completion principle proves the local small-e claim.

A single critical family supplies every weak-m outer join. When e is infinitesimal, or when e is a unit and a infinitesimal, m is infinitesimal. The finite coefficients A, C, D or their unbounded limits have the common critical source residues

$$A_0 = \frac{t_0}{2h_0}, \qquad C_0 = \frac{h_0 + t_0}{2l_0}, \qquad D_0 = \frac{1}{2(t_0 - l_0)}, \qquad h_0 + l_0 = 1, \quad t_0 \geq l_0. \tag{C.39}$$

Each quotient is used only when its denominator is positive. Its zero denominator means the corresponding coefficient is unbounded. We retain the whole α positive square term. We retain the n positive square term too, or cap it at a positive real lower bound on the unit-e branch. In both cases the remaining X mass from the signed core is at least $D_v/2$.

For u infinitesimal and v a bounded unit put $S_m = \sqrt{m}W$ and $R_u = uW$. Rewrite the entire m positive square term as

$$(vS_m - \sqrt{m}X + \sqrt{m}\,uvY - \sqrt{m}\,uZ)^2$$

and replace every other uW by R_u. These operations introduce no inverse. The S_m pivot is $v_0^2 > 0$ and the X pivot is at least $D_v/2 > 0$. The remaining (R_u, Y, Z) residue is

$$\begin{aligned} F_v^{\mathrm{crit}} &= C_0(1+V)(R_u - Y)^2 + A_0(1+V)Y^2 + D_0(R_u - Y - v_0 Z)^2 \\ &\quad + \tfrac{1}{2}\{(R_u - Y)^2 + VY^2 - 2v_0 R_u Z\}, \qquad V = v_0^2 > 0. \end{aligned} \tag{C.40}$$

In the finite interior its $Z = 0$ restriction is strict and

$$\det F_v^{\mathrm{crit}} = \frac{V^2(h_0 + t_0)^2(1+V)}{8h_0 l_0(t_0 - l_0)} > 0.$$

At $h_0 = 0 < t_0 - 1$, the unbounded A row imposes $Y = 0$ and the remaining determinant is $C_0 D_0 V^2 > 0$. At $l_0 = 0$, the unbounded C row imposes $R_u = Y$ and the determinant is $V^2(t_0 + 1)/(4t_0) > 0$. At $t_0 = l_0$, $h_0 > 0$, the unbounded D row imposes $R_u = Y + v_0 Z$ and the determinant is $V^2(1+V)/(4h_0 l_0) > 0$. At the only simultaneous endpoint $h_0 = 0$, $t_0 = l_0 = 1$, both A, D are unbounded; on their common kernel $Y = 0$, $R_u = v_0 Z$ the remaining form is $V^2 Z^2/2 > 0$. These exhaust the boundary of (C.39) because $t_0 > 0$. The cap principle now supplies a bounded strict full matrix in every case, with all positive original excesses restored.

For u a bounded unit, or unbounded, and v infinitesimal set $p = 1/u$, $K = pX + Z$, $B_v = vX$, and $D_p = 1 + p^2$. The complete identity is

$$\begin{aligned} p^2 P(1/p, v) &= nD_p D_v X^2 + m(pvW + vY - K)^2 + CD_v(W - pY)^2 + AD_p D_v Y^2 \\ &\quad + DJ_p^2 + \alpha D_p(pX + vY - K)^2 \\ &\quad + \tfrac{1}{2}\{J_p^2 + (1 - p^2 v^2)K^2 \\ &\qquad + D_p[v^2 Y^2 - 2vYK + 2pYB_v + 2pvKB_v - p^2 B_v^2]\}, \\ J_p &= W + D_p B_v - pY - pvK. \end{aligned} \tag{C.41}$$

The n, m, α positive square terms are retained exactly. On writing $L = W - pY$, the remaining residue is $K^2/2$ plus

$$C_0 L^2 + D_p A_0 Y^2 + (D_0 + 1/2)(L + D_p B_v)^2 + D_p pY B_v - \tfrac{1}{2} D_p p^2 B_v^2. \tag{C.42}$$

In the finite interior its L, Y diagonal block is strict and its B_v Schur complement is $D_p(h_0 + t_0)/(2t_0) > 0$, since

$$\frac{C_0(D_0 + 1/2)}{C_0 + D_0 + 1/2} = \tfrac{1}{2} + \frac{1}{4A_0} = \frac{h_0 + t_0}{2t_0}.$$

The same four boundary kernels give the following positive transverse Schur coefficients: $D_p/2$ when $h_0 = 0 < t_0 - 1$; $D_p(D_0 + 1/2)$ when $l_0 = 0$; $D_p C_0$ when $t_0 = l_0$, $h_0 > 0$; and $D_p/2$ at $h_0 = 0$, $t_0 = l_0 = 1$. Thus the cap principle proves every such chart, including literal $p = 0$. There is no inverse of v or of a source opening in these roots.

If e is a unit and a is a unit, m has a positive real lower bound instead. For finite tests with only one nonzero residue, cap n, m, A, C, D at positive real lower bounds and restore their exact excess positive square terms. At $v_0 = 0$, $u_0 \neq 0$ a strict selected residue is

$$n_c D_u X^2 + A_c D_u Y^2 + (C_c + D_c + 1/2)(u_0 W - Y)^2 + \tfrac{1}{2}(X + u_0 Z)^2.$$

At $u_0 = 0$, $v_0 \neq 0$ it is

$$(n_c + 1/2)D_v X^2 + m_c(v_0 W - X)^2 + (A_c + C_c + 1/2)D_v Y^2 + D_c(Y + v_0 Z)^2.$$

For both finite testing residues nonzero, apply the common all-noncommuting-residues theorem to the unchanged physical parent. Its arbitrary-coefficient-scale version is the only remote physical theorem imported here; the cap arguments above do not assert that an arbitrary clipped source is physical.

All remaining projective charts. On the unit-e branch, n, A, C, D have positive real lower bounds. After capping them and retaining all excesses and the whole m, α positive square terms, the three exact homogeneous templates R4E14–R4E16 have strict residues

$$n_c D_v X^2 + C_c D_v W^2 + A_c D_v Y^2 + (D_c + 1/2)(W + v_0 X)^2 + \tfrac{1}{2}(v_0 Y - Z)^2 \quad (u = \infty),$$

$$(n_c + 1/2) D_u X^2 + C_c (R_u - Y)^2 + (A_c + 1/2) D_u Y^2 + D_c (u_0 X - Z)^2 \quad (v = \infty),$$

$$(n_c + 1/2) X^2 + C_c W^2 + A_c Y^2 + D_c (X - P_\infty)^2 + \tfrac{1}{2}(Y - Q_\infty)^2 \quad (u = v = \infty),$$

where $R_u = uW$, $P_\infty = Z/u$, $Q_\infty = Z/v$ are legal roots; the displayed matrices are strict on their respective root lists. Their entire homogeneous polynomials have bounded coefficients.

On the small-e branch, m is infinitesimal and n is retained whole. The u-infinite, v-small chart is already (C.41) at $p_0 = 0$. For u infinite and v a bounded unit, the selected residue is

$$C_c D_v W^2 + A_c D_v Y^2 + (D_c + 1/2)(W + v_0 X)^2 + \tfrac{1}{2}(v_0 Y - Z)^2 \succ 0.$$

For v infinite and u bounded use $S_m = \sqrt{m} W$ and $R_u = uW$. The complete homogeneous polynomial obtained from (C.26) has strict selected residue

$$S_m^2 + \tfrac{1}{2} D_u X^2 + C_c (R_u - Y)^2 + (A_c + 1/2) D_u Y^2 + D_c (u_0 X - Z)^2 \succ 0.$$

For two infinite tests put $p = 1/u$, $q_v = 1/v$, $P_\infty = pZ$, $Q_\infty = q_v Z$. The exact half-core is

$$\tfrac{1}{2}\{(q_v W + X - p q_v Y - P_\infty)^2 + (1 + p^2)(Y - Q_\infty)^2 + (1 + q_v^2)(p^2 X^2 + 2 X P_\infty - P_\infty^2)\}.$$

Together with the capped A, C, D positive square terms its residue is

$$C_c W^2 + (D_c + 1/2) X^2 - 2 D_c X P_\infty + D_c P_\infty^2 + A_c Y^2 + \tfrac{1}{2}(Y - Q_\infty)^2 \succ 0;$$

the X, P_∞ determinant is $D_c/2 > 0$. All omitted original positive square terms are exact nonnegative squares. Reciprocal deviations are pure testing infinitesimals, so the same completion principle applies. The literal endpoints are the homogeneous polynomial values, not a division by a zero coordinate.

Finally, the literal finite axes obey the complete identities

$$P|_{u=0} = (n + 1/2) D_v X^2 + m(vW - X)^2 + (C + A + 1/2) D_v Y^2 + D(Y + vZ)^2 + \alpha(vY - Z)^2, \tag{C.43}$$

$$P|_{v=0} = n D_u X^2 + (m + 1/2)(X + uZ)^2 + (C + D + 1/2)(uW - Y)^2 + A D_u Y^2 + \alpha D_u Z^2. \tag{C.44}$$

The sign gauges $(u, Y, Z) \mapsto (-u, -Y, -Z)$ and $(v, X, Z) \mapsto (-v, -X, -Z)$ preserve the parent. Applying the real identity to the real and imaginary parts proves the complex-state statement. The two local branches, the two finite one-unit-angle branches, the noncommuting finite branch, and the displayed homogeneous charts exhaust every independent source and test type. This proves the proposition.

C.11.7 State signs and projective endpoints

We reduce the complete weak-C2 parent to a positive state-ratio chart and evaluate its projective measurement endpoints. The subsequent local certificates treat the remaining positive chart.

Let $K_c, K_t \supset \mathbb{R}$ be real closed fields, occupying separately labelled tensor factors. All quantities u, v, W, X, Y, Z below belong to K_t; all six coefficients n, m, A, C, D, α belong to K_c and are nonnegative. Write $D_u = 1 + u^2$, $D_v = 1 + v^2$, and define the complete parent

$$F = n D_u D_v X^2 + m\{v(W + uY) - X - uZ\}^2 + A D_u D_v Y^2 + C D_v (uW - Y)^2 \tag{C.45}$$

$$+ D(uW + uvX - Y - vZ)^2 + \alpha D_u (vY - Z)^2 + K, \tag{SJ1}$$

$$K = \tfrac{1}{2}(uW + uvX - Y - vZ)^2 + \tfrac{1}{2} D_u (vY - Z)^2 + \tfrac{1}{2} D_v (X + uZ)^2 - \tfrac{1}{2} D_u D_v Z^2. \tag{SJ2}$$

Products involving source and testing quantities mean tensor products. For example, $n D_u D_v X^2$ means $n \otimes (D_u D_v X^2)$; a testing-only term means $1 \otimes K$. The positive square terms in (SJ1) are sums of squares in $K_c \otimes_{\mathbb{R}} K_t$, since each coefficient and each testing multiplier has a square root in its own real closed field.

Sign gauges preserve the entire parent. For $\varepsilon, \eta \in \{1, -1\}$, the substitution

$$(u, v, W, X, Y, Z) \longmapsto (\varepsilon u, \eta v, W, \eta X, \varepsilon Y, \varepsilon\eta Z) \tag{SJ3}$$

preserves every summand of (SJ1), including (SJ2). Indeed the roots of the m, C, D, α positive square terms acquire, respectively, the signs $\eta, \varepsilon, \varepsilon, \varepsilon\eta$, and the other roots also change only by a sign. Thus u, v may both be taken nonnegative without altering the source coefficients or the full expression.

An exact positive hard block. Set

$$t = \frac{X + uZ}{D_u}, \qquad r = \frac{Y - uW}{D_u}, \qquad w = \frac{W + uY}{D_u}, \qquad z = \frac{Z - uX}{D_u}. \tag{SJ4}$$

The inverse transformation is $X = t - uz$, $Z = z + ut$, $Y = r + uw$, $W = w - ur$. Let $H = 1 + u^2 + v^2$ and $d = H^2 - u^2v^2$. Both $H - uv = 1 + (u - v)^2 + uv$ and $H + uv$ are positive for $u, v \geq 0$; in fact $H > |uv|$ for arbitrary u, v. Consequently $d > 0$. Define

$$\mathsf{A} = \frac{1}{2}\begin{pmatrix} H & -uv \\ -uv & H \end{pmatrix}, \qquad \mathsf{B} = \frac{1}{2}\begin{pmatrix} -u^2v & -uv^2 \\ uv^2 & u^2v \end{pmatrix}.$$

The matrix A is positive definite. Direct expansion and square completion give the exact identity

$$\frac{K}{D_u} = \left(\begin{pmatrix} t \\ r \end{pmatrix} + \mathsf{A}^{-1}\mathsf{B}\begin{pmatrix} w \\ z \end{pmatrix}\right)^T \mathsf{A}\left(\begin{pmatrix} t \\ r \end{pmatrix} + \mathsf{A}^{-1}\mathsf{B}\begin{pmatrix} w \\ z \end{pmatrix}\right) \tag{C.46}$$

$$+ \frac{D_uD_v}{2d}\{u^2v^2(w^2 + z^2) - 2uvHwz\}. \tag{SJ5}$$

The rational matrix used in the shift is explicitly

$$\mathsf{A}^{-1}\mathsf{B} = \frac{1}{d}\begin{pmatrix} -u^2v(1 + u^2) & -uv^2(1 + v^2) \\ uv^2(1 + v^2) & u^2v(1 + u^2) \end{pmatrix}. \tag{SJ6}$$

For any $a, b \in K_t$, the hard quadratic form is

$$\begin{pmatrix} a \\ b \end{pmatrix}^T \mathsf{A}\begin{pmatrix} a \\ b \end{pmatrix} = \frac{H - uv}{4}(a + b)^2 + \frac{H + uv}{4}(a - b)^2. \tag{SJ7}$$

This also gives an explicit square decomposition of the first term of (SJ5), with square roots taken only in K_t.

If $u, v \geq 0$ and $wz \leq 0$, the second term of (SJ5) is nonnegative. It too is a square in K_t (or can be split into its three nonnegative terms). Hence $K \geq 0$, and (SJ1) is a tensor sum of squares immediately. In particular the axes $w = 0$ and $z = 0$ are included, without division by either state coordinate. The same identity, without sign gauges, proves the sharper implication $K < 0 \Longrightarrow uvwz > 0$.

Reduction of the remaining states. The only remaining case after (SJ3) is $K < 0$, $u, v > 0$, $wz > 0$. Then $w \neq 0$. Divide all four state coordinates by w. Equation (SJ4) now has $w' = 1$ and $z' = z/w > 0$. The ratio z' is arbitrary: it may be infinitesimal, bounded, or unbounded in K_t. No estimate in this reduction replaces z' by a bounded coefficient.

The parent is homogeneous of degree two in the state coordinates, so

$$F(u, v; W, X, Y, Z) = w^2F(u, v; W/w, X/w, Y/w, Z/w). \tag{SJ8}$$

If the parent on the right has a certificate $\sum_i h_i^2$ in $K_c \otimes_{\mathbb{R}} K_t$, then the parent on the left has certificate $\sum_i((1 \otimes w)h_i)^2$. This proves that certificates for the full parent with $u, v > 0$, $w = 1$, and arbitrary positive z suffice for all finite affine testing parameters and all real states.

Literal projective endpoints. For clarity, the projective endpoints can be checked without a limiting argument. Homogenize $u = U_1/U_0$, $v = V_1/V_0$, and put $D_U = U_0^2 + U_1^2$, $D_V = V_0^2 + V_1^2$, with neither pair identically zero. Multiplication by $U_0^2V_0^2$ turns (SJ2) into

$$K_h = \tfrac{1}{2}(U_1V_0W + U_1V_1X - U_0V_0Y - U_0V_1Z)^2 \tag{C.47}$$

$$+ \tfrac{1}{2}D_U(V_1Y - V_0Z)^2 + \tfrac{1}{2}D_V(U_0X + U_1Z)^2 - \tfrac{1}{2}D_UD_VZ^2. \tag{SJ9}$$

At the four axes this is, exactly,

$$U_0 = 0: \quad K_h = \tfrac{1}{2}U_1^2\{(V_0W + V_1X)^2 + (V_1Y - V_0Z)^2\}, \tag{C.48}$$

$$V_0 = 0: \quad K_h = \tfrac{1}{2}V_1^2D_U(X^2 + Y^2), \tag{C.49}$$

$$U_1 = 0: \quad K_h = \tfrac{1}{2}U_0^2D_V(X^2 + Y^2), \tag{C.50}$$

$$V_1 = 0: \quad K_h = \tfrac{1}{2}V_0^2\{(U_1W - U_0Y)^2 + (U_0X + U_1Z)^2\}. \tag{SJ10}$$

All other homogenized positive square terms in (SJ1) retain nonnegative coefficients and are sums of squares. Thus the endpoints are proved directly, including the intersections of these axes. Equations (SJ3)–(SJ10) establish the stated full-state, sign, and endpoint reduction. They do not establish the still-required positive-ratio chart certificate.

C.11.8 Source coordinates and the two-square decomposition

Actual source and conventions. The native source formulas give the following coordinate identities and boundedness estimates. They provide the source parameters used in the subsequent R2 contact and state-normal certificates.

Use the original weak-C2 native source (a, B, t, r, h), with $0 < r, h < 1$, $a, B, t > 0$, the original strict source gaps, and $n \geq 0$. In R2, t is positive unbounded while

$$B, \qquad z = aB, \qquad y = tB$$

are positive infinitesimals. There is no bound on a and no lower bound on r, h, $1-h$, or $e = 1-r$. Put $l = 1-h$ and $S = 1/t$. Define the actual native arrays

$$c = 1 + zr, \quad s_c = 1 + B(l + ar), \quad R = 1 + hz, \quad g = R - ae > 0, \quad j_D = c - l/t > 0,$$

$$\begin{aligned}\Delta &= a + (1+z) + (1+z)B + 2z(1+z)r + a^3B^2r^2 + 2a^2Br \\ &\quad + tB\{1 + a + z + a^2Br(2-r)\}, \\ M &= (1+z)(1+ar)(h+t) - a(1+y)e\,l.\end{aligned}$$

The six weights used throughout this section are exactly

$$\begin{aligned} n &= \frac{(1+ar)cs_c - r\Delta}{2r\Delta}, & m &= \frac{a^2(1+y)ecs_c}{2g\Delta}, \\ C &= \frac{cM}{2l\Delta}, & A &= \frac{tc(1+ar)s_c}{2h\Delta}, \\ D &= \frac{(1+z)(1+ar)(1-atBr)s_c}{2tj_D\Delta}, & \alpha &= \frac{B^2(1+ar)M}{2(1-hB)\Delta}. \end{aligned} \tag{RER1}$$

All denominators are pure source. The original positivity assumptions are retained without adding a rate comparison.

Physicality controls the apparently unbounded source scale. The exact numerator identity is

$$(1+ar)(1+zr)\{1 + B(l+ar)\} - r\Delta = eQ - BK, \tag{RER2}$$

where

$$\begin{aligned} K &= (a+1)(h+t)(1+z), \\ Q &= 1 + B + y + z(h+t+2r) + z^2\{t(1+er) + h(1+r)\} + Bz(h+t). \end{aligned} \tag{RER3}$$

Every term of Q is nonnegative. Since $r\Delta > 0$, original physicality is exactly $eQ \geq BK$, including equality when $n = 0$. Put $\kappa = ay = tz$. Before assuming that κ is bounded, direct termwise inspection of (RER3), using B, y, z infinitesimal, gives $Q \leq C_0(1+\kappa)$ for a fixed positive real C_0. The strict gap $g > 0$ gives $ae < R \sim 1$, so multiplication of the physicality inequality by a yields

$$\kappa(a+1)(1+hS)(1+z) \leq aeQ < RQ \leq C_1(1+\kappa) \tag{RER4}$$

for a fixed positive real C_1.

If a is bounded, $\kappa = ay$ is infinitesimal. If a is unbounded, move $C_1\kappa$ to the left in (RER4). Since $a + 1 - C_1 \geq (a+1)/2$, one obtains $\kappa \leq 2C_1/(a+1)$, again infinitesimal. In either case, (RER4) also bounds $(a+1)\kappa$. Returning to (RER3) and then to $eQ \geq BK$ proves

$$ay \in \mathrm{I}, \qquad a(a+1)y = O(1), \qquad Q \sim 1, \qquad \frac{(a+1)y}{e} = O(1). \tag{RER5}$$

No division by n occurred; the argument includes the face $n = 0$.

Uniform source normalizations. Put

$$k = \frac{1+ar}{1+a}, \qquad p = \frac{ae}{R} \in (0,1), \qquad \mu = m(1-p) = \frac{mg}{R}, \qquad \delta_n = nr.$$

The quantity k is a positive bounded unit. Indeed $k \leq 1$; if $a \leq 1$, then $k \geq 1/(1+a) \geq 1/2$. If $a > 1$, use $ae < R = 1 + hz$ to get $k > (a - hz)/(1+a)$, which has a positive real lower bound. Direct termwise division of the displayed polynomial Δ by $1 + a$ gives $\Delta/(1+a) \sim 1$. Also

$$\frac{M}{t(1+ar)} \sim 1 :$$

the first term has ratio $(1+z)(1+hS)\sim 1$, while the absolute ratio of the second is at most $R(1+y)S/(1+ar)$ and hence is infinitesimal. Since $c, s_c, j_D, 1-hB\sim 1$, (RER1) now gives

$$\frac{SC\,l}{k}\sim\frac{1}{2},\quad \frac{SA\,h}{k}\sim\frac{1}{2},\quad \frac{D}{Sk}\sim\frac{1}{2},\quad \frac{\alpha}{Sy^2(1+a)k^2}\sim\frac{1}{2}. \tag{RER6}$$

For the D relation use $atB=ay=\kappa\in \mathrm{I}$ from (RER5). Thus all original h and l openings are present.

Two exact bounded products remove the remaining nonuniform source poles:

$$mg=\frac{a^2e(1+y)cs_c}{2\Delta},\qquad \delta_n=\frac{eQ-BK}{2\Delta}. \tag{RER7}$$

The quantities mg, mg/a, mg/a^2, μ, δ_n, and nr^2 are bounded. For example $a^2e/(1+a)\le Ra/(1+a)=O(1)$, while $0\le\delta_n\le eQ/(2\Delta)=O(1)$. These estimates do not imply that m or n themselves are bounded.

Exact extraction from the entire parent. Let testing u, v, W, X, Y, Z be arbitrary, independently of the source; put $D_u=1+u^2$, $D_v=1+v^2$. The complete original quarter-parent is

$$\begin{aligned}\mathcal{P}&=m(vW-X+uvY-uZ)^2+nD_uD_vX^2+CD_v(uW-Y)^2+AD_uD_vY^2\\&\quad+(D+\tfrac12)(uW+uvX-Y-vZ)^2+(\alpha+\tfrac12)D_u(vY-Z)^2\\&\quad+\tfrac12D_v(X+uZ)^2-\tfrac12D_uD_vZ^2.\end{aligned}$$

Define the mixed rows

$$q_a=v(W+uY),\qquad N=X+uZ-pq_a,\qquad \xi=X+ruZ.$$

Then the following identity retains both actual positive square terms, without setting either normal to zero:

$$\mathcal{P}=mpN^2+nD_uD_v\xi^2+G, \tag{RER8}$$

where

$$\begin{aligned}G&=\mu\{N^2-2q_aN+(1-p)q_a^2\}+\delta_nD_uD_v(-2uZX-ru^2Z^2)\\&\quad+CD_v(uW-Y)^2+AD_uD_vY^2\\&\quad+(D+\tfrac12)(uW+uvX-Y-vZ)^2+(\alpha+\tfrac12)D_u(vY-Z)^2\\&\quad+\tfrac12D_v(X+uZ)^2-\tfrac12D_uD_vZ^2.\end{aligned}$$

It follows simply by expanding $m\{(1-p)q_a-N\}^2-mpN^2$ and $n(X^2-\xi^2)=nr(-2uZX-ru^2Z^2)$. Both retained positive square terms are finite separated SOS: their source weights are nonnegative, D_uD_v is pure testing, and N,ξ are allowed mixed rows. When $n=0$ the second positive square term vanishes literally. The exact identity and all estimates above are the exact structural conclusion; no positivity conclusion about G is asserted.

C.11.9 The C1–C2 boundary identity

This section records an exact source-only identity. It does not by itself admit any new independent testing domain. Its purpose is to identify the same physical boundary in the two remaining source charts and to retain the nonnegative slack when that boundary opens.

Use the original weak-C2 variables $B=H+L$, $z,y>0$ and $0<r<1$, with the unchanged source gaps and nonnegative n. Write $e=1-r$, $K=1+z$, $T_c=B+zr$, and let δ be the original positive C2 denominator. Define the positive source polynomial

$$P_n=B+B^2+2Bzr+BHz+yB(1+z)+Hrz^2+yz^2re. \tag{OP1}$$

Then the following identity holds on the entire original source:

$$\boxed{eP_n=(1+z)(H+y)(B+zr)+2r\delta n.} \tag{OP2}$$

Every factor in the first summand is positive, and the second summand is nonnegative, including literal $n=0$. In particular the pure-source weights

$$q=\frac{(1+z)(H+y)(B+zr)}{eP_n}>0,\qquad s_n=\frac{2r\delta n}{eP_n}\ge 0\quad\text{satisfy}\quad q+s_n=1. \tag{OP3}$$

Both square roots are legitimate source-field elements. Neither weight is assumed to have nonzero residue.

Proof. The unchanged formula for n is

$$2r\delta n=(B+zr)(1+zr)(1+zr+B-H)-r\delta.$$

Substitution of the printed polynomial for δ, followed by collecting terms, gives OP2. This is a polynomial identity, so no source boundary is obtained by continuity and no opening has been inverted except the displayed positive pure-source factors. □

Exact correspondence of the source contacts. Use a subscript 1 for native C1 notation in this paragraph. The map

$$x_1 = z, \qquad \eta_1 = \frac{y}{B}, \qquad \zeta_1 = \frac{B}{z}, \qquad r_1 = \frac{H}{B}, \qquad \theta_1 = e, \tag{OP4}$$

with $k_1 = \sqrt{z}$ and the native parameter $\Delta_1 = B/(z\sqrt{y})$, gives

$$(u_c, v_c; W_c, X_c, Y_c, Z_c) = \left(\sqrt{z}, \sqrt{y}; \frac{B}{\sqrt{zy}}, -r\sqrt{z}, \frac{H}{\sqrt{y}}, 1\right)$$

from the original native C1 contact formula. Thus it is the same contact, including all state coordinates. Direct substitution in the native positive polynomial P_d gives $BP_d = P_n$. The native C1 beta coefficient at this formal contact therefore obeys

$$\beta_1 = -\frac{r\delta n}{eP_n} = -\frac{s_n}{2}, \qquad \beta_1 + \tfrac{1}{2} = \frac{q}{2} > 0. \tag{OP5}$$

For $n = 0$, the original C2 parent has $\lambda = n = \beta = 0$. It is literally an original C1 parent, up to an optional positive normalization of C. The native source entry and uniqueness at stationary contact identify its coefficients with OP4. For $n > 0$, OP5 makes the formal C1 beta negative. Hence OP4 does *not* map that source to the nonnegative C1 event cone, and no physical C1 all-tests theorem may be imported there.

C.12 Two small actual angles: the R1 sector

C.12.1 Scalar completion at unbounded source ratio

Scope of the dictionary and budgets. The identities below use the actual native weak-C2 stationary coefficients, with $d = 1$, $\lambda_{\text{event}} = \beta = 0$, all strict native gaps and $n \geq 0$. Source quantities lie in K_c and testing quantities in a separately labelled real closed extension K_t; all inverses and radicals are pure source or pure testing. The rational identities have no source-size restriction. Their R1 estimates assume B, T, x, y infinitesimal; the later RL and EN certificates establish their own replacement estimates.

For the native source write a_s, B_s, t_s, r, h, and put $c_s = 1 + a_s B_s r$, $R_s = 1 + h a_s B_s$, $K_s = 1 + a_s B_s$. The formerly used auxiliary ratio is simply

$$q_{\text{old}} = R_s^2/(a_s t_s K_s^2). \tag{RSR1}$$

No size restriction on q_{old} or q_z is imposed in this refined dictionary. The actual full-R1 estimates, including arbitrary positive $q_z = jT$, are justified by R1N1–R1N11.

The exact compact source and its uniform bounds

Define pure-source quantities

$$\begin{gathered} j = \frac{a_s c_s}{R_s}, \quad B = \frac{B_s R_s}{c_s}, \quad T = \frac{t_s c_s}{R_s}, \\ p = \frac{a_s(1-r)}{R_s}, \quad H = \frac{1-h}{t_s c_s}, \\ x = jB, \quad y = TB, \quad q_z = jT. \end{gathered} \tag{RSR2}$$

The native gaps give $0 < p < \min(1, j)$ and $0 < H < 1$. In the R1 estimates B, T, x, y are positive infinitesimals, while j and $q_z = jT$ are arbitrary positive. The exact dictionary itself also applies at the different source scales used in RL and EN. Direct inversion gives

$$r = \frac{j-p}{j+px}, \quad h = \frac{1-HT}{1+HTx}, \quad a_s = \frac{j+px}{1+HTx}, \quad B_s = \frac{B(1+HTx)}{1+pB}, \quad t_s = \frac{T(1+pB)}{1+HTx}. \tag{RSR3}$$

In particular the compact variables describe the actual source, including its $r \to 0$, $p \to 1$, $H \to 0$, and $H \to 1$ openings. Set

$$K = j + 1 - p, \qquad E = 1 + B(HT + p - 1), \qquad W_0 = (1-H)j + 1 - p. \tag{RSR4}$$

Then $\frac{1}{2}(j+1) < K < j+1$, $E \sim 1$, and $W_0 > 0$. The following polynomials contain the entire stationary dictionary:

$$\begin{aligned}
\mathcal{P} &= j + 1 + B\{(j+1)^2 - p\} \\
&\quad + TB\{Hj + j + 1 + B[Hj(2j+2-p) + p(2j+1-p)]\} \\
&\quad + HjT^2B^2(1 + Bp + BHj), \\
\mathcal{L} &= K + T(W_0 - H) + BT[pK + Hj(1-p)] + BHT^2(W_0 - 1), \\
\mathcal{N} &= p + B(-j^2 + 3jp - j - p^2 + p) \\
&\quad + BT(Hj^2 + Hj - j^2 + jp - j + p) \\
&\quad + B^2T(2Hj^2p - Hj^2 - Hjp^2 + 2Hjp - 2j^2p + 3jp^2 - jp - p^3 + p^2) \\
&\quad + B^2HT^2j(Hj - j + p)(1 + Bp).
\end{aligned} \tag{RSR5}$$

With $\mu = m(1-p)$, the exact six original stationary weights are

$$\begin{aligned}
\mu &= \frac{jp(1+y)(1+x)(1+Hy)}{2\mathcal{P}}, & n &= \frac{\mathcal{N}}{2(j-p)\mathcal{P}}, \\
C &= \frac{\mathcal{L}}{2HT\mathcal{P}}, & A &= \frac{T(1+x)(1+Hy)(K+Hjy)}{2(1-HT)\mathcal{P}}, \\
D &= \frac{(1+Hy)(K+Hjy)\{1 - y[(1-H)j - p]\}}{2T(1-H)\mathcal{P}}, & \alpha &= \frac{B^2(K+Hjy)\mathcal{L}}{2E\mathcal{P}}.
\end{aligned} \tag{RSR6}$$

For example, the original stationary denominator and its source mass satisfy

$$\Delta = \frac{(1+x)\mathcal{P}}{(1+Bp)(1+HTx)}, \qquad M = \frac{(1+x)\mathcal{L}}{(1+HTx)^2}.$$

The two native strict gaps become

$$g = \frac{(1-p)(1+x)}{1+HTx}, \qquad j_{\mathrm{gap}} = \frac{(1-H)(1+x)}{1+Bp}.$$

These equations verify the source origin of RSR6 directly; no new stationary solution has been substituted for the original one.

Every term after $j+1$ in $\mathcal{P}$, divided by $j+1$, is infinitesimal: use $B(j+1) = B + x \in \mathrm{I}$, $T \in \mathrm{I}$, $p < 1$, and $H < 1$. The same termwise estimate, using $K > (j+1)/2$, applies to $\mathcal{L} - K$. Thus

$$\mathcal{P} \sim j + 1, \qquad \mathcal{L} \sim K. \tag{RSR7}$$

These are relative equivalences, valid also when j is infinitesimal or unbounded. They show, in particular, that A/T, THC, and $T(1-H)D$ are positive bounded units, μ/p is bounded, and $h_0 = \mu + 1/2$ is a positive bounded unit. Moreover α/B is infinitesimal. Only the displayed source openings can make m, n, C, D unbounded.

The original physicality condition gives a useful uniform restriction. Regroup the exact polynomial $\mathcal{N}$ as

$$\mathcal{N} = p\{1 + B(3j + 1 - p) + BT(j+1)\} - Bj(j+1)\{1 + (1-H)T\} + \mathcal{R}. \tag{RSR8}$$

Here $\mathcal{R}$ is precisely the last two lines of $\mathcal{N}$ in RSR5. Since $0 < p < \min(1, j)$, each coefficient in its B^2T line is bounded in absolute value by a fixed multiple of $j(j+1)$. The B^2HT^2 line has the same bound, multiplied by the bounded factor $T(1+Bp)$. Consequently

$$|\mathcal{R}| \le C_0 BT\, Bj(j+1)$$

for a fixed real C_0. All corrections multiplying p and $Bj(j+1)$ in RSR8 are infinitesimal. As $n \ge 0$ is exactly $\mathcal{N} \ge 0$, it follows that

$$\eta_* := \frac{Bj(j+1)}{p} \text{ is bounded}, \qquad \eta_* \in \mathrm{I} \implies n \sim \frac{p}{2(j-p)(j+1)} > 0. \tag{RSR9}$$

Indeed $\mathcal{N} = p(1+\varepsilon_1) - Bj(j+1)(1+\varepsilon_2)$ with $\varepsilon_i \in \mathrm{I}$. This proves both assertions without dividing by n and includes literal $n = 0$ in the bounded-η_* alternative. It also gives nr bounded.

The complete graph and its actual contact circuit

The quarter-parent, before any positive square term is removed, is

$$\begin{aligned}\mathscr{P} &= m\{vW + uvY - X - uZ\}^2 + CD_v(uW - Y)^2 + AD_uD_vY^2 \\ &\quad + (D + \tfrac{1}{2})(uW + uvX - Y - vZ)^2 + (\alpha + \tfrac{1}{2})D_u(vY - Z)^2 + nD_uD_vX^2 \\ &\quad + \tfrac{1}{2}D_v(X + uZ)^2 - \tfrac{1}{2}D_uD_vZ^2, \qquad D_u = 1 + u^2,\ D_v = 1 + v^2. \end{aligned} \tag{RSR10}$$

The source-free rotated state coordinate is $w = (W+uY)/D_u$. On its nonzero chart set $w = 1$ and $z = (Z-uX)/D_u$. Define the two complete state normals

$$N = X + uZ - pD_uv, \qquad M = uW - Y - HD_uvz.$$

Their exact inverse is

$$\begin{aligned} X &= N/D_u + pv - uz, \qquad Z = z + uN/D_u + puv, \\ Y &= u - M/D_u - Hvz, \quad W = 1 + uM/D_u + uHvz. \end{aligned} \tag{RSR11}$$

The scalar studied here is exactly $F_{\text{full}} = (\mathscr{P}/D_u)|_{N=M=0}$. No n positive square term, m positive square term, or state-normal term is silently removed. Substitution of RSR11 into RSR10 gives the complete full-parent normal expression RFN7 in Section C.12.2, with $N_0 = N/D_u$, $M_0 = M/D_u$ and the retained positive square term $mpD_u^2N_0^2$ restored.

Use the separated rows

$$U = u/\sqrt{x}, \quad V = v/\sqrt{y}, \quad Z_* = z/\sqrt{q_z}, \qquad Q_* = U - VZ_*, \quad B_* = V - UZ_*, \quad R_* = Z_* - UV,$$

$$\Psi = U^2 + V^2 + Z_*^2 + U^2V^2Z_*^2 - 4UVZ_*. \tag{RSR12}$$

These are rows; U, V, Z_* are not assumed bounded coefficients. The actual contact is $(U, V, Z_*; N, M) = (1, 1, 1; 0, 0)$. The exact native formulas make the complete parent value and all five first derivatives vanish there. On the positive core,

$$\Psi = (U - V)^2 + Z_*^2(1 - UV)^2 + 2UV(1 - Z_*)^2 \tag{RSR13}$$

is a legal separated sum of squares: the last root is $\sqrt{2}\,(uv)^{1/2}(xy)^{-1/4}(1 - Z_*)$, with its radicals in the two separate real closed fields.

Let the ten positive monomial coefficients and the seven circuit masses be exactly RB6–RB8. In particular

$$M_a = Ax, \quad M_b = \{\mu(1 - p) + (n + \tfrac{1}{2})p^2\}y, \quad M_c = \alpha q_z, \quad M_{ac} = nj^2y,$$

$$f_c = M_b - M_{ac}, \quad q_c = M_a - f_c, \quad e_c = (n + \tfrac{1}{2})p^2y^2 - g_c, \quad d_c = M_{b^2c} - k_c.$$

The original contact balances give the exact scalar identity

$$\begin{aligned} F_{\text{full}} &= q_cQ_*^2 + M_{ac}B_*^2 + r_cR_*^2 + d_cV^2Q_*^2 + e_cV^2B_*^2 \\ &\quad + f_c\Psi + g_cV^2\Psi + k_cZ_*^2(V^2 - 1)^2. \end{aligned} \tag{RSR14}$$

The identities RB7 and the two odd coefficient balances are independently checked on the actual source. Thus RSR14 is a finite polynomial identity, not a Taylor approximation to the contact.

Here is an explicit source-only formula for its radial mass. Put $\mathcal{F} = K + T\mathcal{F}_1 + T^2\mathcal{F}_2$, where

$$\begin{aligned} \mathcal{F}_1 = {} & -2BHjp + BHj + 2Bjp - 2Bp^2 + 2Bp \\ & - Hj - Hp - H + j + 1 - p^2/j + p/j, \\ \mathcal{F}_2 = {} & -B^2H^2jp + B^2Hjp - B^2Hp^2 + B^2Hp \\ & - BH^2j - BH^2p + BHj - BHp^2/j + BHp/j. \end{aligned} \tag{RSR15}$$

Every displayed term divided by K is bounded, using $p/j < 1$; terms containing Bj are even infinitesimal. Hence $\mathcal{F} \sim K$. Direct source substitution yields

$$\begin{aligned} f_c &= \frac{Tx^2\mathcal{F}}{2\mathcal{P}} > 0, & q_c &= Ax - f_c \sim \frac{TxK}{2(j+1)} > 0, \\ r_c &= \frac{B^2jT(1-p)(1+x)(1+Hy)\mathcal{L}}{2E\mathcal{P}} > 0, & g_c &= \frac{HT^2j^2B^3\mathcal{L}}{2\mathcal{P}} > 0. \end{aligned} \tag{RSR16}$$

In particular

$$f_c \sim \frac{Tx^2K}{2(j+1)}, \qquad \frac{r_c}{f_c} \sim \frac{1-p}{j}, \qquad \frac{g_c}{f_c} = Hy\frac{\mathcal{L}}{\mathcal{F}}. \tag{RSR17}$$

For the remaining signed mass define

$$\begin{aligned}\mathcal{K} &= pW_0 + B(1-p)\{j[H(p+1)-p] - (1-p)(p-H)\}\\ &\quad + TBH\{(p+1)(1-H)j + (1-p)(BHj + p + 1 - H)\}\\ &\quad + T^2B^2H^2j(1-H).\end{aligned} \tag{RSR18}$$

Then

$$k_c = -\frac{T^2B^2j(1+x)\mathcal{K}}{2E\mathcal{P}}.$$

The identity

$$\begin{aligned}\mathcal{K} - pW_0 + Bp(1-p)(j+1) &= B(1-p)\{Hj(p+1) + H(1-p) + p^2\}\\ &\quad + TBH\{(p+1)(1-H)j + (1-p)(BHj + p + 1 - H)\}\\ &\quad + T^2B^2H^2j(1-H)\end{aligned} \tag{RSR19}$$

has a nonnegative right side. Since $W_0 \geq 1-p$ and $B(j+1) \in \mathrm{I}$, it proves $\mathcal{K} > 0$ and $k_c < 0$. The displayed finite arrays also give

$$\mathcal{K}/K = O(p+B), \qquad |k_c|/(q_c y) = O(p+B). \tag{RSR20}$$

All constants in these estimates are real and uniform over the source openings; neither $j-p$ nor $1-H$ has been bounded below.

Paying the signed masses, with the full current reserved

Put

$$h_0 = \mu + \tfrac{1}{2}, \qquad M_e = \frac{nh_0j^2}{n+h_0}, \qquad J_c = M_e y, \qquad e_0 = \tfrac{1}{2}p^2y^2 - g_c. \tag{RSR21}$$

These are pure-source quantities and $0 \leq J_c \leq M_{ac}$. Writing $e_c = np^2y^2 + e_0$ preserves the entire second correlated current positive square term. The conclusion is the following precise scalar statement: after subtracting the reserved positive square terms

$$(M_{ac} - J_c)B_*^2 + np^2y^2V^2B_*^2, \tag{RSR22}$$

the remainder is a separated SOS and retains, for example, one half of $q_cQ_*^2 + r_cR_*^2 + f_c\Psi + J_cB_*^2$. When $n = 0$ the two reserved positive square terms and J_c vanish literally. This assertion supplies precisely the effective current mass needed when the full N normal is later completed against the original n positive square term. It is not a completion of that normal itself.

A budget has infinitesimal cost when its Gram coefficients are bounded and have zero residue. A finite number of such corrections leaves any fixed fraction smaller than one of each protected square. The remaining Gram matrix has positive-definite real residue, so Lemma 3.2 gives its exact sum of squares. Restoring the budget identities and the retained positive source terms proves the required identity.

First $e_0^- \leq g_c$. The exact budget

$$(1+V^2)\Psi - V^2B_*^2 = Q_*^2 + R_*^2 + V^2(1 - UVZ_*)^2 \tag{RSR23}$$

pays its negative part from the circuit at cost

$$\frac{e_0^-}{f_c} + \frac{e_0^-}{f_c y}v^2 \;\leq\; Hy\frac{\mathcal{L}}{\mathcal{F}} + H\frac{\mathcal{L}}{\mathcal{F}}v^2 \in \mathrm{I}.$$

Its positive part is retained as a square. Also $d_c > 0$ because $M_{b^2c} > 0$ and $k_c < 0$; the d_c and g_c terms cause no loss.

For k_c, use the exact identity

$$Z_*(V^2 - 1) = -VQ_* - R_*. \tag{RSR24}$$

The VQ_* part is paid from $q_cQ_*^2$ with coefficient $2|k_c|v^2/(q_cy)$, which is infinitesimal by RSR20. The remaining R_* part is paid in the following exhaustive three cases. Fixed real multipliers such as the factor two do not change any cost claim.

Case 1: $1-p$ has a positive real lower bound. The exact ratio is

$$\frac{|k_c|}{r_c} = \frac{T\mathcal{K}}{(1-p)(1+Hy)\mathcal{L}} \in \mathrm{I}. \tag{RSR25}$$

This uses only $\mathcal{K}/\mathcal{L} = O(p+B)$ and $T \in \mathrm{I}$. Neither current mass nor a bound on j is required.

Case 2: $1-p\in\mathrm{I}$ and η_* has a positive real lower bound. Here $p\sim 1$, so $Bj(j+1)$ is a bounded positive unit. Since B is infinitesimal, j is unbounded and j^2B is a positive bounded unit. Equations RSR16–RSR20 give

$$\frac{|k_c|}{f_c}\in\mathrm{I},\qquad \frac{|k_c|}{f_cx}=O\!\left(\frac{T(p+B)}{j^2B}\right)\in\mathrm{I}. \tag{RSR26}$$

The exact circuit budget

$$(1+U^2)\Psi=R_*^2+B_*^2+U^2Q_*^2+U^2(1-UVZ_*)^2 \tag{RSR27}$$

therefore pays $|k_c|R_*^2$ at cost $|k_c|/f_c+(|k_c|/(f_cx))u^2\in\mathrm{I}$. This case includes any intersection with literal $n=0$.

Case 3: $1-p,\eta_*\in\mathrm{I}$. Now $p\sim 1$, $j>p$, and RSR9 gives $n>0$ together with its exact relative estimate. The source mass M_e is a positive bounded unit. For the lower bound use

$$\frac{p^2}{M_e}=\frac{p^2}{h_0j^2}+\frac{p^2}{nj^2}=O(1).$$

For the upper bound use $M_e\le h_0j^2$ when j is bounded, and $M_e\le nj^2\sim p/2$ when j is unbounded. In particular,

$$\frac{|k_c|}{f_c}\in\mathrm{I},\qquad \frac{|k_c|}{q_cy}=O(1),\qquad \frac{|k_c|}{J_cx}=O\!\left(\frac{T(p+B)(1+1/n)}{j^2}\right)=O(1). \tag{RSR28}$$

The last bound follows directly from $1/n\sim 2(j-p)(j+1)/p$, with $p\sim 1$ and $j>p$. There is an exact separated SOS budget

$$2\{\Psi+V^2Q_*^2+U^2B_*^2\}-R_*^2\quad\text{is SOS.} \tag{RSR29}$$

For completeness, set $t_*=UV$, $\delta_*=U-V$, $E_*=Z_*(1-t_*)$, $T_*=t_*(1-Z_*)$, and $J_*=T_*-\delta_*^2Z_*$. Its SOS decomposition is

$$(E_*+T_*)^2+2\delta_*^2+4t_*(1-Z_*)^2+2J_*^2+4t_*\delta_*^2Z_*^2.$$

All roots are separated as in RSR13. Applying RSR29 costs a multiple of $|k_c|/f_c$ on the circuit, $|k_c|v^2/(q_cy)$ on Q_*, and $|k_c|u^2/(J_cx)$ on B_*. Each is infinitesimal by RSR28.

The three cases exhaust the source because $0<1-p<1$ and η_* is bounded. A finite number of the displayed allocations therefore proves the scalar assertion following RSR22 with a strict retained fraction. All coefficient bounds depend only on the source and u,v; no bound on the independent state coordinate z was used. The proof allows arbitrary relative rates of all source and testing infinitesimals. Literal $n=0$ is covered by Cases 1 and 2; Case 3 excludes it by the original physicality equation itself.

C.12.2 Completion of both state normals

Use of the complete normal calculation. The parent identities below are exact for the compact stationary dictionary. The source estimates and square budgets give the full local R1 certificate under the full-R1 source hypotheses verified in R1N1–R1N11, with arbitrary positive jT. The later RL and EN bodies reuse only the printed exact identities and supply their own source estimates.

Fields, source scope, and notation. Let $K_c,K_t\supset\mathbb{R}$ be real closed fields, with bounded rings B_c,B_t and infinitesimal ideals $\mathfrak{m}_c,\mathfrak{m}_t$. The ring of certificate expressions is $\mathcal{T}=K_c\otimes_{\mathbb{R}}K_t$, and its bounded coefficient ring is $\mathcal{B}=B_c\otimes_{\mathbb{R}}B_t$. A bounded coefficient with zero real residue belongs to $\mathfrak{n}=\mathfrak{m}_c\otimes B_t+B_c\otimes\mathfrak{m}_t$. Every radical or inverse used below is pure source, pure testing, or an explicitly separated product of such factors. No order on $\mathcal{T}$ and no inverse of a mixed expression is used.

Use the positive compact source coordinates

$$x=jB,\qquad y=TB,\qquad q_z=jT,\qquad 0<p<\min\{1,j\},\qquad 0<H<1, \tag{RFN1}$$

where $B,T,x,y\in\mathfrak{m}_c$ and $q_z>0$ is arbitrary. These are the compact coordinates of the complete original R1 sector; the absence of any q_z restriction is verified in R1N1–R1N11. The original strict source gaps remain in force and $n\ge 0$, including $n=0$. For reference, their native substitution is

$$r=\frac{j-p}{j+px},\quad h=\frac{1-HT}{1+HTx},\quad a_s=\frac{j+px}{1+HTx},\quad B_s=\frac{B(1+HTx)}{1+pB},\quad t_s=\frac{T(1+pB)}{1+HTx}. \tag{RFN2}$$

The six weights n,m,C,A,D,α are always the original source weights at (RFN2), rather than weights of an auxiliary stationary source. Put $\mu=m(1-p)$, $h_0=\mu+\frac{1}{2}$, $k=(j+1-p)/(j+1)$, and $\eta_*=Bj(j+1)/p$. In this

source sector RS proves

$$\tfrac{1}{2} < k < 1, \quad \eta_* = O(1), \quad \mu = O(p), \quad h_0 \asymp 1, \quad A = O(T), \quad \alpha \asymp B^2(j+1), \tag{RFN3a}$$

$$TCH \sim k/2, \qquad T(D+\tfrac{1}{2})(1-H) \sim k/2, \qquad \frac{n(j-p)}{j} = O(1), \tag{RFN3b}$$

$$f_c \asymp Tx^2, \qquad q_c \asymp Tx, \qquad r_c \asymp TjB^2(1-p), \tag{RFN3c}$$

$$\eta_* \in \mathfrak{m}_c \quad \Longrightarrow \quad n \sim \frac{p}{2(j-p)(j+1)} > 0. \tag{RFN3d}$$

Here $\asymp$ means that the ratio is a positive bounded unit, and $\sim$ means that the ratio has residue one. All estimates are uniform at the independent openings H, $1-H$, $j-p$, and $1-p$. In particular $B(j+1) = B + x$ and α are infinitesimal. No comparison between source and testing infinitesimals is allowed.

The complete parent and the correlated source mass

Write $D_u = 1 + u^2$, $D_v = 1 + v^2$. The original quarter-parent is

$$\mathcal{P} = m(vW - X + uvY - uZ)^2 + nD_uD_vX^2 + CD_v(uW - Y)^2 + AD_uD_vY^2 \tag{C.51}$$

$$+ (D+\tfrac{1}{2})(uW + uvX - Y - vZ)^2 + (\alpha+\tfrac{1}{2})D_u(vY - Z)^2 \tag{C.52}$$

$$+ \tfrac{1}{2}D_v(X + uZ)^2 - \tfrac{1}{2}D_uD_vZ^2. \tag{RFN4}$$

Use the two mixed normal rows

$$N_0 = \frac{X + uZ - pD_uv}{D_u}, \qquad M_0 = \frac{uW - Y - HD_uvz}{D_u}. \tag{RFN5}$$

Their exact inverse, with $w = 1$, is

$$\begin{aligned} X &= N_0 + pv - uz, & Z &= z + uN_0 + puv, \\ W &= 1 + u(M_0 + Hvz), & Y &= u - M_0 - Hvz. \end{aligned} \tag{RFN6}$$

Only the pure testing unit D_u is inverted. Retain the nonnegative positive square term $mpD_u^2N_0^2$ and define $G = (\mathcal{P} - mpD_u^2N_0^2)/D_u$. Direct substitution gives the *entire* expression

$$G = \mu D_u\{N_0^2 - 2vN_0 + (1-p)v^2\} + nD_v(N_0 + pv - uz)^2 \tag{C.53}$$

$$+ CD_uD_v(M_0 + Hvz)^2 + AD_v(u - M_0 - Hvz)^2 \tag{C.54}$$

$$+ (D+\tfrac{1}{2})D_u(M_0 - (1-H)vz)^2 \tag{C.55}$$

$$+ (\alpha+\tfrac{1}{2})\{u(1-p)v - (1+Hv^2)z - uN_0 - vM_0\}^2 \tag{C.56}$$

$$+ \tfrac{1}{2}D_uD_v(N_0 + pv)^2 - \tfrac{1}{2}D_v(z + uN_0 + puv)^2. \tag{RFN7}$$

Thus G is exactly quadratic in (N_0, M_0):

$$G = F + 2L_NN_0 + 2L_MM_0 + H_NN_0^2 + H_MM_0^2 + 2(\alpha+\tfrac{1}{2})uvN_0M_0, \tag{RFN8}$$

where $F = G|_{N_0=M_0=0}$ and

$$\begin{aligned} H_N &= h_0D_u + \tfrac{1}{2}v^2 + \alpha u^2 + nD_v, \\ H_M &= D_u(CD_v + D + \tfrac{1}{2}) + AD_v + (\alpha+\tfrac{1}{2})v^2. \end{aligned} \tag{RFN9}$$

There are no omitted higher state-normal terms.

Exact scalar interface, before spending the n-mass. Set

$$U = \frac{u}{\sqrt{x}}, \quad V = \frac{v}{\sqrt{y}}, \quad Z_* = \frac{z}{\sqrt{q_z}}, \qquad Q_* = U - VZ_*, \quad B_* = V - UZ_*, \quad R_* = Z_* - UV,$$

$$\Psi = U^2 + V^2 + Z_*^2 + U^2V^2Z_*^2 - 4UVZ_*. \tag{RFN10}$$

All three normalized testing variables are positive, but none is assumed bounded. A legal separated certificate is

$$\Psi = (U - V)^2 + Z_*^2(1 - UV)^2 + 2UV(1 - Z_*)^2. \tag{RFN11}$$

For example, $\sqrt{UV}$ is the product of the source factor $(xy)^{-1/4}$ and the testing factor $\sqrt{uv}$.

The exact RS/RB scalar identity for the actual source is

$$F = q_c Q_*^2 + nj^2 y B_*^2 + r_c R_*^2 + d_c V^2 Q_*^2 + (np^2y^2 + e_0)V^2 B_*^2 \tag{C.57}$$
$$+ f_c\Psi + g_c V^2\Psi + k_c Z_*^2 (V^2-1)^2, \qquad e_0 = \tfrac{1}{2}p^2y^2 - g_c. \tag{RFN12}$$

For an entirely algebraic definition of the masses, put $K_0 = CH^2 + (D+\frac{1}{2})(1-H)^2$ and

$$\begin{aligned}
&M_a = Ax, && M_b = \{\mu(1-p) + (n+\tfrac{1}{2})p^2\}y,\\
&M_c = \alpha q_z, && M_{ab} = \{A + \alpha(1-p)^2 + \mu(1-p) + \tfrac{1}{2}(1-p)^2\}xy,\\
&M_{ac} = nxq_z = nj^2y, && M_{bc} = \{K_0 + AH^2 + 2H\alpha + H - \tfrac{1}{2}\}yq_z,\\
&M_{b^2} = (n+\tfrac{1}{2})p^2y^2, && M_{abc} = (K_0+n)xyq_z,\\
&M_{b^2c} = H^2(A + C + \alpha + \tfrac{1}{2})y^2q_z, && M_{ab^2c} = CH^2xy^2q_z.
\end{aligned}$$

Then

$$\begin{aligned}
&k_c = (M_a - M_b + M_{ac} + M_{ab^2c} - M_{bc})/2,\\
&r_c = M_c - M_b + M_{ac} - k_c, \quad f_c = M_b - M_{ac}, \quad q_c = M_a - f_c,\\
&d_c = M_{b^2c} - k_c, \qquad g_c = M_{ab^2c}.
\end{aligned}$$

The source contact identities make (RFN12) exact, including both odd monomials of F. It is checked coefficientwise against (RFN7), not only at the contact point.

Define the unnormalized current $b = v - uz/j = \sqrt{y}B_*$ and the pure source quantities

$$c_b = \frac{nj}{n+h_0}, \qquad M_e = \frac{nh_0j^2}{n+h_0}. \tag{RFN13}$$

The whole scalar pieces $nj^2b^2 + np^2v^2b^2$ in (RFN12) must be kept until their correlations with N_0 have been completed. The full N_0-linear term below contains $njb + npv^2b$, so these pieces and the corresponding hard terms give

$$n(N_0 + jb)^2 + nv^2(N_0 + pb)^2.$$

Complete only the first square, using a pure source inverse:

$$h_0N_0^2 + n(N_0+jb)^2 = (n+h_0)(N_0 + c_bb)^2 + M_eb^2. \tag{RFN14}$$

Keep $nv^2(N_0+pb)^2$ whole. Put $\widehat{N} = \sqrt{n+h_0}(N_0 + c_bb)$. The other hard terms in (RFN9) are the nonnegative whole squares

$$h_0(uN_0)^2 + \tfrac{1}{2}(vN_0)^2 + \alpha(uN_0)^2. \tag{RFN15}$$

They are still kept whole after the shift; no term nu^2 or nv^2 is treated as a bounded perturbation.

For $n > 0$ one has the exact bounded-shift identity

$$\frac{c_b^2}{M_e} = \frac{n}{h_0(n+h_0)} = O(1). \tag{RFN16}$$

At $n = 0$, set $c_b = M_e = 0$ and omit the effective current row. Formula (RFN14) remains literal, and no division by n or $\sqrt{M_e}$ is made in that case.

RS pays the signed e_0 and k_c terms in (RFN12) at infinitesimal cost after (RFN14), with the effective current mass M_ey, and leaves fixed positive real fractions of

$$q_cQ_*^2, \qquad M_eb^2, \qquad r_cR_*^2, \qquad f_c\Psi. \tag{RFN17}$$

The second term is omitted when $n = 0$. This is the precise scalar allocation needed here; an allocation that spends the raw nj^2y mass before (RFN14) would not suffice.

Finite exact budgets for the full normal rows

In the rest of the proof, a row is called free if any prescribed fixed real multiple of its square can be purchased at infinitesimal cost from (RFN17). This has an exact meaning: a finite SOS identity writes its square below a finite sum of those retained squares, with bounded separated coefficients in $\mathfrak{n}$. Subtracting any fixed real multiple changes their strict Gram diagonals only infinitesimally.

Three elementary circuit budgets. Direct expansion gives

$$(1+U^2)\Psi - R_*^2 = B_*^2 + U^2Q_*^2 + U^2(1-UVZ_*)^2, \tag{RFN18a}$$
$$(1+V^2)\Psi - V^2B_*^2 = Q_*^2 + R_*^2 + V^2(1-UVZ_*)^2. \tag{RFN18b}$$

There is also a useful opposite-current budget. Put $t_* = UV$, $\delta_* = U - V$, $E_* = Z_*(1-t_*)$, $T_* = t_*(1-Z_*)$, and $J_* = T_* - \delta_*^2 Z_*$. Then

$$2(\Psi + V^2Q_*^2 + U^2B_*^2) - R_*^2 = (E_*+T_*)^2 + 2\delta_*^2 + 4t_*(1-Z_*)^2 \tag{C.58}$$
$$+ 2J_*^2 + 4t_*\delta_*^2 Z_*^2. \tag{RFN18c}$$

Every term on the right is a legal separated square. Thus a nonnegative multiplier a provides aR_*^2 at infinitesimal cost whenever

$$a/f_c \in \mathfrak{n}, \qquad a/(q_c y) = O(1), \qquad a/(M_e yx) = O(1), \tag{RFN18d}$$

since the last two diagonal costs have the additional testing factors v^2 and u^2. This criterion is used only when $M_e > 0$.

An exact radial budget. The row $R_u = v(u^2 - x)$ is free in all source subcases, including the weak opening $1 - p \in \mathfrak{m}_c$. Here is an explicit finite certificate, so that this assertion does not rest on pointwise nonnegativity. Write $D_* = V(U^2-1)$ and

$$t_7 = (Z_* - U^2V,\ V - U^2V,\ U - U^2V,\ UZ_* - U^2V,$$
$$UV - U^2V,\ UVZ_* - U^2V,\ U^2 - U^2V)^T,$$
$$t_3 = (1 - UV,\ V - UV,\ U - UV)^T,$$

$$G_7 = \begin{pmatrix} 4 & -2 & -3/2 & 0 & -1/2 & 0 & 1/2 \\ -2 & 3 & 0 & -3/2 & 0 & -1 & 2 \\ -3/2 & 0 & 4 & 3/2 & -2 & -3 & 0 \\ 0 & -3/2 & 3/2 & 4 & -3/2 & 0 & -3/2 \\ -1/2 & 0 & -2 & -3/2 & 3 & 2 & -1 \\ 0 & -1 & -3 & 0 & 2 & 8 & -4 \\ 1/2 & 2 & 0 & -3/2 & -1 & -4 & 4 \end{pmatrix}, \qquad G_3 = \begin{pmatrix} 3 & -2 & -2 \\ -2 & 2 & 2 \\ -2 & 2 & 3 \end{pmatrix}.$$

The exact identity is

$$4\Psi + 4U^2Q_*^2 + 4U^2R_*^2 - D_*^2 = t_7^T G_7 t_7 + 4VZ_*(1-U)^2 + UZ_* t_3^T G_3 t_3. \tag{RFN19}$$

The leading principal minors of G_7 are $4, 8, 101/4, 1049/16, 907/16, 3279/16, 743/16$; those of G_3 are $3, 2, 2$. Both are positive definite real matrices. Their real Cholesky factors and the separated square roots of VZ_* and UZ_* therefore make the right side a finite legal SOS. Denote it by $\mathcal{W}(U, V, Z_*)$.

Since $R_u = x\sqrt{y}D_*$, the exhaustive source cases are as follows.

Case 1: $1-p$ is a positive unit. Multiply (RFN19) by x^2y. The costs relative to $f_c\Psi, q_cQ_*^2, r_cR_*^2$ are, respectively,

$$O(B), \qquad O(Bu^2), \qquad O(u^2/(1-p)).$$

All belong to $\mathfrak{n}$.

Case 2: $p \sim 1$ and η_* is a positive unit. Then j has a positive real lower bound, and j^2B is a positive unit. Adding $4U^2$ times (RFN18a) to (RFN19) gives

$$D_*^2 \preceq_{\mathrm{sos}} 4(1+U^2+U^4)\Psi + 4U^2Q_*^2.$$

After multiplication by x^2y, the costs are

$$O\big(B + u^2/j + u^4/(j^2B)\big), \qquad O(Bu^2),$$

again infinitesimal. The notation $a \preceq_{\mathrm{sos}} b$ means $b - a \in \Sigma\mathcal{T}^2$ and is only shorthand for the displayed exact identities.

Case 3: $p \sim 1$ and $\eta_* \in \mathfrak{m}_c$. Formula (RFN3d) gives $n > 0$ and $1/M_e = O(1)$; it also gives $p^2/M_e = O(1)$. Put $E^{\mathrm{rad}} = Z_*(U^2-1)$. Permuting V and Z_* in (RFN19) yields

$$\mathcal{W}(U, Z_*, V) = 4\Psi + 4U^2Q_*^2 + 4U^2B_*^2 - (E^{\mathrm{rad}})^2.$$

The identity $D_* = UE^{\mathrm{rad}} + (U^2-1)B_*$ then gives, exactly,

$$8U^2\Psi + 8U^4Q_*^2 + \{8U^4 + 2(U^2-1)^2\}B_*^2 - D_*^2 \tag{C.59}$$
$$= 2U^2\mathcal{W}(U, Z_*, V) + \{UE^{\mathrm{rad}} - (U^2-1)B_*\}^2. \tag{RFN20}$$

After multiplication by x^2y, the three costs are

$$O(u^2/j), \qquad O(u^4/j), \qquad \frac{8u^4 + 2(u^2-x)^2}{M_e}.$$

They are infinitesimal. These three cases prove the asserted free radial row without assuming a rate for any testing variable.

The smaller current helpers. If $\eta_* \in \mathfrak{m}_c$, (RFN3d) implies

$$\frac{p^2}{M_e} = \frac{p^2}{h_0 j^2} + \frac{p^2}{nj^2} = O(1),$$

because $p/j < 1$ and $p(j-p)(j+1)/j^2 \le p + p/j$. Thus $(pv^2b)^2$ costs only $(p^2/M_e)v^4$ of the effective current. If η_* is a positive unit instead, then

$$\frac{p^2y}{f_c} = O\Big(\frac{p + p/j}{\eta_*}\Big) = O(1).$$

Multiply (RFN18b) by $p^2y^2v^2$. This provides $(pv^2b)^2$ with circuit cost

$$\frac{p^2y}{f_c}(yv^2 + v^4) \in \mathfrak{n}.$$

The same identity multiplied by $Tp^2u^2y^2$ shows that $\sqrt{T}\,puvb$ is free: its cost is either $(p^2/M_e)Tu^2v^2$ or $(p^2y/f_c)Tu^2(y+v^2)$.

The entire two-normal conductor

First normal: exact source cancellation and allocation. Put

$$q = vz - Tu, \qquad R = z - uv/B, \qquad R_u = v(u^2 - x),$$
$$K_c = \frac{p}{2} + j\{H(\alpha + \tfrac12) - \tfrac12\} + n(p-j), \qquad K_R = -\mu - (\alpha+\tfrac12)(1-p) + \frac{T}{j}K_c.$$

Both K_c/j and K_R are bounded by (RFN3). The complete first normal is exactly

$$L_N = \{j(n-\alpha) + p(n+\tfrac12)v^2\}b + K_R R_u + \frac{K_c}{j}uvq. \tag{RFN21}$$

In particular the sign of the μ term in K_R is negative. For a direct source check, the difference between this formula and the derivative of (RFN7) is vS_N, where

$$S_N = np - \mu + \tfrac{p}{2} + \{-\mu - (\alpha+\tfrac12)(1-p)\}x + j(\alpha - n) + yK_c = 0.$$

The last equality is an exact rational identity of the actual native source under (RFN2), not an asymptotic contact equation.

After the two correlated n-squares are formed, the remaining linear coefficient is

$$L_N^{\mathrm{rem}} = -j\alpha b + \tfrac12 pv^2 b + K_R R_u + \frac{K_c}{j}uvq.$$

The q row is strict because $q_cQ_*^2 = (q_c/(Tx))(q^2/T)$, with $q_c/(Tx)$ a positive unit. Its coefficient in L_N^{rem} relative to $q/\sqrt{T}$ is $(K_c/j)uv\sqrt{T} \in \mathfrak{n}$. The pv^2b helper is free by the preceding paragraph. For the remaining small row, the exact identity

$$j\alpha b = -\frac{\alpha}{B}R_u - \alpha uR$$

has $\alpha/B = O(B+x) \in \mathfrak{m}_c$. When $1-p$ is a unit, the square of αuR costs $O(B^2(j+1)^2u^2/(1-p)) \in \mathfrak{n}$ against $r_cR_*^2$. When $p \sim 1$, use $B_*^2 \preceq_{\mathrm{sos}} (1+U^2)\Psi$ from (RFN18a). The circuit cost of $(j\alpha b)^2$ is

$$O\big(B^3(j+1)^2 + B^2(j+1)^2u^2/j\big) \in \mathfrak{n},$$

since j is bounded below and $B(j+1) = B + x$ is infinitesimal. Finally, K_RR_u has bounded coefficient; provide an arbitrarily large fixed real R_u^2 pivot using (RFN19)–(RFN20).

The shift $N_0 = \widehat{N}/\sqrt{n+h_0} - c_bb$ introduces no new unbounded Gram coefficient: $1/\sqrt{n+h_0} = O(1)$, and for $n > 0$ $c_b/\sqrt{M_e} = O(1)$ by (RFN16). The only bounded, potentially non-infinitesimal couplings are therefore those of R_u to $\widehat{N}$ and $\sqrt{M_e}b$. A sufficiently large fixed real radial pivot absorbs both while preserving a strict positive real residue. For $n = 0$ the latter coupling is exactly absent.

Second normal: cancellation of both source openings. Define

$$K_d = CH - (D + \tfrac{1}{2})(1 - H), \qquad C_q = K_d + AH + \tfrac{1}{2} + v^2\{CH(1 + u^2) + AH + (\alpha + \tfrac{1}{2})H\}.$$

The complete second normal is

$$L_M = C_q q + \mu uvb + v(\alpha - BTCHu^2)R. \tag{RFN22}$$

The actual source satisfies the three exact identities

$$\begin{aligned} S_0 &= T(K_d + AH + \tfrac{1}{2}) - A = 0, \\ S_1 &= jK_d + \mu + xTCH = 0, \\ S_2 &= jK_d + xTCH + T\{CH + AH + (\alpha + \tfrac{1}{2})H\} - A - (\alpha + \tfrac{1}{2})(1 - p) + \alpha/B = 0. \end{aligned}$$

Before imposing these identities, the complete derivative minus (RFN22) is exactly $uS_0 + u^2vzS_1/j + uv^2(S_2 - S_1)$. This records all state terms and rules out a hidden j-weighted radial error. In particular S_1 gives $K_d = -\mu/j - BTCH = O(1)$ even when C or D is unbounded. Consequently $TC_q = O(T + v^2) \in \mathfrak{n}$.

Normalize

$$M_0 = \sqrt{TH(1 - H)}\,\widehat{M}.$$

The full diagonal (RFN9), not only its zero-angle value, obeys

$$TH(1 - H)H_M \sim k/2 > 0. \tag{RFN23}$$

Indeed the C and D terms contribute $(k/2)D_u\{(1 - H)D_v + H\}$ up to infinitesimal bounded coefficients, and all remaining terms are infinitesimal after this normalization. This is uniform at both independent openings. The normalized $C_q q$ coupling has coefficient $T\sqrt{H(1 - H)}C_q \in \mathfrak{n}$ on the strict row $q/\sqrt{T}$. The normalized μuvb row is free because $\mu = O(p)$ and $\sqrt{T}\,puvb$ has already been purchased.

It remains to allocate the entire opposite row, including its higher testing terms. Put

$$J = B(j + 1) + u^2 \in \mathfrak{n}, \qquad a_R = Tq_z v^2(\alpha - BTCHu^2)^2.$$

The factor $H(1 - H)$ may be discarded from the cost bound, since it is bounded. Since $\alpha - BTCHu^2 = O(BJ)$, the exact squared multiplier a_R satisfies

$$\frac{a_R}{f_c} = O((T/j)v^2J^2), \qquad \frac{a_R}{q_c y} = O(v^2J^2). \tag{RFN24a}$$

If $1 - p$ is a unit, its direct cost against $r_c R_*^2$ is $O(Tv^2J^2/(1 - p)) \in \mathfrak{n}$. If $p \sim 1$ and η_* is a unit, apply (RFN18a); the cost is $(a_R/f_c)(1 + u^2/x) \in \mathfrak{n}$, using that j^2B is a unit. If $p \sim 1$ and $\eta_* \in \mathfrak{m}_c$, (RFN3d) gives

$$\frac{a_R}{M_e yx} = O\left(\frac{T(1 + 1/n)}{j^2}v^2J^2\right) = O(1), \tag{RFN24b}$$

because $(1 + 1/n)/j^2 = O(1)$ in this source case. The other two conditions in (RFN18d) follow from (RFN24a). The exact budget (RFN18c) therefore provides this row at infinitesimal cost as well. These estimates treat the whole square $(\alpha - BTCHu^2)^2$; there is no truncation at a testing degree.

Full Gram, legal roots, and restoration. After the normalizations, the entire remaining mixed hard term is

$$2(\alpha + \tfrac{1}{2})uv\sqrt{TH(1 - H)}\left(\frac{\widehat{N}}{\sqrt{n + h_0}} - c_b b\right)\widehat{M}.$$

Its coefficients relative to $\widehat{N}$, $\widehat{M}$, and, when present, $\sqrt{M_e}b$ are infinitesimal; (RFN16) controls the shift. All other remaining normal couplings were allocated above. Choose a fixed sufficiently large radial pivot, and then purchase all the finitely many free helpers. Their scalar costs lie in $\mathfrak{n}$, so fixed positive fractions of (RFN17) survive. The complete bounded Gram has strict positive real residue: the $\widehat{N}$ pivot is one, the $\widehat{M}$ pivot has residue $\overline{k}/2$, the retained scalar pivots are positive, and the only bounded radial off-diagonals have been absorbed by the chosen real radial pivot.

The separated strict bounded-Gram lemma now gives a finite sum of squares in $\mathcal{T}$. Its row entries need not be bounded. All budget remainders are explicit finite separated sums of squares, all source-only completions use positive pure source denominators, and all state-coordinate inverses are pure testing. Restore the whole squares (RFN15), the retained $nv^2(N_0 + pb)^2$ square, the scalar budget remainders, the $mpD_u^2N_0^2$ positive square term, and the pure testing factor D_u. This certifies the complete original parent (RFN4). Every bounded coefficient used in the argument depends only on the source and on u, v. The arbitrary state ratio z remains inside retained rows throughout. Hence the proof has no bounded-state restriction, and it includes literal $n = 0$ without any limiting argument.

C.12.3 All source ratios and testing angles

Statement and exact source. Use the original common-II source, with $d = 1$, $\lambda = \beta = 0$, the six native coefficients of the actual Section C.11.1, every original strict gap, and $n \geq 0$. In the original R1 sector,

$$t_s, \quad B_s, \quad a_s B_s, \quad t_s B_s > 0 \quad \text{are infinitesimal.} \tag{R1N1}$$

There is no bound on a_s, on any original positive opening, or on the source ratio previously called q_{old}. The conclusion is: the entire original parent is a finite separated tensor sum of squares at every independent real projective measurement pair and every real or complex state. This proof treats R1 only.

Source and testing quantities belong to separate real closed fields $K_c, K_t \supset \mathbb{R}$. All SOS assertions are in $K_c \otimes_{\mathbb{R}} K_t$. We use the actual actual strict bounded-Gram lemma: a bounded finite Gram with positive definite real residue gives tensor SOS, with no boundedness requirement on its row entries. No order or mixed inverse is imposed on the tensor ring.

Put $c_s = 1 + a_s B_s r$, $R_s = 1 + h a_s B_s$, and define

$$\begin{aligned} j &= \frac{a_s c_s}{R_s}, \quad B = \frac{B_s R_s}{c_s}, \quad T = \frac{t_s c_s}{R_s}, \quad p = \frac{a_s(1-r)}{R_s}, \\ H &= \frac{1-h}{t_s c_s}, \qquad x = jB, \quad y = TB, \quad q_z = jT. \end{aligned} \tag{R1N2}$$

The native strict gaps give $0 < p < \min(1, j)$ and $0 < H < 1$. They also give $1 - h < t_s c_s$, so $h \sim 1$. Consequently B, T, x, y are infinitesimal, while q_z *is an arbitrary positive source quantity.* In fact $x = a_s B_s$ and $y = t_s B_s$ exactly. The inverse source substitution is

$$r = \frac{j-p}{j+px}, \quad h = \frac{1-HT}{1+HTx}, \quad a_s = \frac{j+px}{1+HTx}, \quad B_s = \frac{B(1+HTx)}{1+pB}, \quad t_s = \frac{T(1+pB)}{1+HTx}. \tag{R1N3}$$

All arguments below therefore concern the actual original source.

The source bounds are RSR7–RSR9: their proofs use only B, T, Bj infinitesimal and $p, H < 1$, and impose no condition on jT.

The complete small-angle theorem extends to all R1 sources

The exact proof Section C.12.2, RFN4–RFN24, is a full-parent proof, not just a restriction to a scalar graph. We now justify its application under the larger source domain (R1N1), rather than importing its former domain label.

First, all of its exact identities hold for the rational dictionary (R1N3), without any size assumption on q_z. The parent, its two normals, their quadratic block, the scalar circuit, the radial Gram and the three source cancellation identities were all proved as identities of this dictionary. Second, every uniform source bound used in that proof is exactly (R1N4)–(R1N7) or the following scalar bound:

$$f_c \asymp Tx^2, \quad q_c \asymp Tx, \quad r_c \asymp TjB^2(1-p), \quad g_c/f_c = O(Hy), \quad |k_c|/(q_c y) = O(p+B). \tag{R1N9}$$

To check the latter bounds without $jT \in \mathrm{I}$, use the explicit finite polynomials RSR15 and RSR18: each term of $\mathcal{F}_1/K, \mathcal{F}_2/K$ is bounded using $p/j < 1$ and $Bj = x$, and $\mathcal{K}/K = O(p+B)$ by the same inequalities. Thus $\mathcal{F} \sim K$ and all statements (R1N9) follow from RSR16–RSR20. No jT factor occurs in these estimates.

All source and normal estimates in RFN18–RFN24 use precisely these source bounds. In particular the apparently unbounded factor $q_z = jT$ cancels in the exact ratios RFN24a–RFN24b; their proofs use $Bj = x$, B, T, x, y infinitesimal, and no restriction on jT. The effective n square is completed before the signed scalar terms are allocated. Consequently that full scalar-and-two-normal calculation proves every R1 source in the local testing chart, including $n = 0$.

Outer testing angles: two exact current identities

We include the outer mechanism and its precise positivity condition. Let

$$R = uW - Y, \quad C_t = v(uX - Z), \quad B_t = X + uZ, \quad N = R + h_* C_t, \quad D_u = 1 + u^2, \quad D_v = 1 + v^2,$$

and let K be the signed quarter-core

$$K = \tfrac{1}{2}(R + C_t)^2 + \tfrac{1}{2} D_u (vY - Z)^2 + \tfrac{1}{2} D_v B_t^2 - \tfrac{1}{2} D_u D_v Z^2. \tag{R1N12}$$

All unselected original nonnegative events are kept as whole squares.

If $1/v$ is bounded, use

$$K_0 = 2\sqrt{g_*} C_t, \quad B_0 = \sqrt{2D_v/D_u}\, B_t, \quad c_u = D_u^{-1/2}, \quad t_u = u D_u^{-1/2}, \quad \eta_v = \sqrt{D_v}/v.$$

The exact full selected-parent identity is

$$\begin{aligned}4CD_vR^2+4D(R+C_t)^2+4K&=4(C+D)N^2+4Cv^2R^2+2(R+C_t)^2\\&\quad+2D_u(vY-Z)^2+(K_0,B_0)Q(K_0,B_0)^T,\\Q&=\begin{pmatrix}1-c_u^2\eta_v^2/(2g_*) & t_u\eta_v/\sqrt{2g_*}\\ t_u\eta_v/\sqrt{2g_*} & 1\end{pmatrix}.\end{aligned}\tag{R1N13}$$

Since $1/g_*$ is infinitesimal and all the testing factors are bounded, $\overline{Q}=I$. This treats every other angle and every state, including Bob infinity by homogeneous clearing.

For v infinitesimal put $\epsilon=1/g_*$, $q=\sqrt{\epsilon}$, $\gamma=Ag_*$, and

$$K_1=2\sqrt{g_*}C_t+\sqrt{\epsilon}Y,\quad B_1=B_t-uvY,\quad V_0=vY,\quad Y_1=\sqrt{\epsilon D_u}\,Y,$$
$$L_0=\tfrac{1}{2}qc_u(K_1-c_uY_1)-t_uvB_1+c_uvV_0.$$

Then $L_0=\sqrt{D_u}\,v(vY-Z)$ exactly, and

$$\begin{aligned}&4CD_vR^2+4D(R+C_t)^2+4AD_uD_vY^2+4K\\&\quad=4(C+D)N^2+4Cv^2R^2+2(R+C_t)^2+\mathcal{G},\\\mathcal{G}&=K_1^2+2D_v(B_1^2+V_0^2)+(4\gamma D_v-c_u^2)Y_1^2\\&\quad+2qv(K_1-c_uY_1)V_0-2L_0^2.\end{aligned}\tag{R1N14}$$

The complete four-root Gram has residue

$$\operatorname{diag}(1,2,2,\overline{k}^{\,2}-\overline{c_u}^{\,2}).\tag{R1N15}$$

Thus it is strict whenever the last entry is positive. In particular it covers every unbounded u, and covers every noninfinitesimal u whenever $\overline{k}=1$. All coefficients in the identity are bounded functions of the displayed normalized homogeneous Alice coordinates, so it includes the literal Alice-infinite endpoint.

The remaining bounded nonzero Alice chart

The only outer chart left by (R1N13)–(R1N15) has v infinitesimal, u a bounded positive unit, and $\overline{k}<1$. Since $1-k=p/(j+1)$, this implies that p,j are positive units and j is bounded. Therefore $\eta_*\in\mathrm{I}$ and (R1N6) gives the exact relative critical limits

$$m\sim\frac{jp}{2(1-p)(j+1)},\qquad n\sim\frac{p}{2(j-p)(j+1)}.\tag{R1N16}$$

These statements allow either or both coefficients to be unbounded.

Choose bounded positive real caps $m_c\le m$, $n_c\le n$ as specified below and keep their nonnegative excesses. Retain all of α as its original square. Define six rows

$$\widehat{N}=\sqrt{C+D}(R+h_*C_t),\quad P=C_t/\sqrt{T},\quad Y_0=\sqrt{T}\,Y,\quad X,\quad Z,\quad V=vY.\tag{R1N17}$$

They need not be independent or bounded. Put $\delta=(C+D)^{-1/2}=O(\sqrt{T})$, so $R=\delta\widehat{N}-h_*\sqrt{T}\,P$. The selected *entire* parent is exactly

$$\begin{aligned}\mathcal{P}_{\mathrm{sel}}&=\widehat{N}^2+Cv^2R^2+\tfrac{1}{2}(R+C_t)^2+(Tg_*)P^2+(A/T)D_uD_vY_0^2\\&\quad+m_c\{(v/u)R+(D_u/u)V-X-uZ\}^2+n_cD_uD_vX^2\\&\quad+\tfrac{1}{2}D_u(V-Z)^2+\tfrac{1}{2}D_v(X+uZ)^2-\tfrac{1}{2}D_uD_vZ^2.\end{aligned}\tag{R1N18}$$

This is obtained by the exact completion $CD_vR^2+D(R+C_t)^2=(C+D)N^2+g_*C_t^2+Cv^2R^2$. It keeps every state-normal term.

The row relation

$$PY_0=V(uX-Z)\tag{R1N19}$$

allows an arbitrary real zero shift $2b(PY_0-V(uX-Z))$. Keep the two explicitly nonnegative squares Cv^2R^2 and $(R+C_t)^2/2$ whole. The remaining complete Gram on (R1N17) is bounded: $u,1/u$ are bounded, $Tg_*,A/T$ are units, and the R contribution in the m_c square has coefficients $\delta v/u$ and $h_*\sqrt{T}v/u$, both infinitesimal. Its residue is the direct sum of a unit pivot and the two blocks

$$\begin{pmatrix}k_0/2 & b\\ b & k_0(1+u_0^2)/2\end{pmatrix},\qquad Q_3(m_c,n_c;b),\tag{R1N20}$$

where subscripts denote real residues and Q_3 is the quadratic form in (X,Z,V)

$$\begin{aligned}Q_3&=m_c\{(D_u/u)V-X-uZ\}^2+n_cD_uX^2\\&\quad+\tfrac{1}{2}D_uV^2-D_uVZ+\tfrac{1}{2}(X+uZ)^2-2bV(uX-Z).\end{aligned}\tag{R1N21}$$

In (R1N21), $u=u_0>0$ is real. The choices of b,m_c,n_c that make both blocks strict are now explicit.

Interior openings. Suppose $p_0 < 1$ and $j_0 > p_0$. Use initially $m_0 = j_0 p_0/[2(1-p_0)(j_0+1)]$ and $n_0 = p_0/[2(j_0-p_0)(j_0+1)]$, and take $b = k_0/2$. The first leading principal minor of Q_3 is $m_0 + n_0 D_u + 1/2 > 0$; the other two are

$$\frac{p_0 u_0^2 D_u K_0}{4(j_0-p_0)(j_0+1)^2(1-p_0)}, \qquad \frac{p_0 u_0^2 D_u K_0^2}{8(j_0-p_0)(j_0+1)^3}, \tag{R1N22}$$

where $K_0 = j_0 + 1 - p_0$. They are positive. The first block in (R1N20) has determinant $k_0^2 u_0^2/4 > 0$. Positive definiteness is open, so fixed real caps slightly below m_0, n_0 preserve strictness and are smaller than the actual source coefficients.

Only the n opening is infinite. If $j_0 = p_0 < 1$, then n is unbounded and $m_0 = j_0^2/[2(1-j_0)(j_0+1)]$ is finite positive. At $b = k_0/2 = 1/[2(j_0+1)]$, restrict (R1N21) to the kernel $X = 0$ of the n positive square term. Its first diagonal on (Z, V) is $u_0^2(m_0 + 1/2) > 0$, and its determinant is $u_0^2 k_0^2/4 > 0$. Choose $m_c < m_0$ sufficiently close and then a sufficiently large fixed real n_c. The whole block is strict, with both caps below the original weights.

Only the m opening is infinite. If $p_0 = 1 < j_0$, then m is unbounded and $n_0 = 1/[2(j_0^2-1)]$ is finite positive. Take a real b just above $k_0/2 = j_0/[2(j_0+1)]$, with

$$b < \min\{k_0\sqrt{D_u}/2,\ j_0/[2(j_0-1)]\}.$$

This interval is nonempty. The first block in (R1N20) is strict. On the kernel $(D_u/u)V - X - uZ = 0$ of the m positive square term, use (X, V) coordinates. The first diagonal of (R1N21) is $D_u/[2(j_0^2-1)] > 0$, and its determinant is

$$-\frac{D_u^2[\,2b(j_0-1)-j_0\,][\,2b(j_0+1)-j_0\,]}{4u_0^2(j_0^2-1)} > 0. \tag{R1N23}$$

Choose $n_c < n_0$ sufficiently close and then a sufficiently large real m_c. This gives the required full strict block.

Both openings are infinite. If $p_0 = j_0 = 1$, both m, n are unbounded and $k_0 = 1/2$. Choose $1/4 < b < \sqrt{D_u}/4$. The common kernel of the two event positive square terms is $X = 0$, $Z = D_u V/u_0^2$. On it the rest of (R1N21) is exactly

$$\frac{D_u(4b-1)}{2u_0^2} V^2 > 0. \tag{R1N24}$$

A sufficiently large real cap on both positive square terms therefore gives a strict block.

The elementary finite-cap fact used in these cases is as follows: if a real positive semidefinite event Gram is positive definite on the complement of its kernel, and the remaining real quadratic form is positive definite on that kernel, a sufficiently large real multiple of the event Gram makes their sum positive definite. It follows by a block Schur complement, and works for the sum of two event Grams as well. Thus the choices above prove strictness of the *complete* bounded Gram (R1N18), without restricting any normal or state coordinate.

Exhaustive testing and state join

The sign gauges SJ3 preserve every original source positive square term and the whole core, so affine u, v may be taken nonnegative. If v is not infinitesimal, (R1N13) applies. If v is infinitesimal and u is not infinitesimal, either (R1N15) applies or the complete six-root argument (R1N16)–(R1N24) applies. These cases exhaust the outer angles. If $u, v > 0$ are both infinitesimal, the enlarged local FN theorem proved above applies after SJ's full-state reduction to $w = 1, z > 0$; SJ gives the other state signs and the axes directly. Literal homogeneous measurement endpoints are given by SJ10 or the homogeneous current identities, without a limiting argument. The zero state is immediate. For a complex state, the real parent matrix splits its value into its real and imaginary quadratic evaluations, to which the same proof applies.

Restore every capped excess, the original n, m, α positive square terms not used in a particular chart, and all cleared positive pure testing homogeneous factors. All normalizations, radicals and inverses are pure source or pure testing. This proves the complete original R1 source row at all independent tests and states, without a source-ratio split or a comparison of source and testing rates.

C.13 Two small actual angles: the R2 sector

C.13.1 Bounded a or an infinitesimal paired-event parameter

Exact statement. Use the actual original weak-C2 source and all its original strict gaps, with $d = 1$, $\lambda = \beta = 0$, and $n \geq 0$. Suppose t_s is positive unbounded and $B_s, a_s B_s, t_s B_s$ are positive infinitesimals. Put $e = 1 - r$, $R_s = 1 + h a_s B_s$. On the source subset

$$a_s = O(1) \quad \text{or} \quad p = a_s e/R_s \in \mathrm{I}_c, \tag{RL1}$$

the entire parent is separated tensor SOS for every independent projective testing pair and every real or complex state. Literal $n = 0$ and every opening of $r, h, 1 - h, g$ are retained. For the all-testing conclusion, the local proof below is joined explicitly to R2O1–R2O24.

Use the field and bounded-Gram conventions of EN1–EN4.

Actual source entry and bounds

Use the exact compact source dictionary RSR2–RSR8, with

$$j = a_s c_s/R_s, \quad B = B_s R_s/c_s, \quad T = t_s c_s/R_s, \quad p = a_s e/R_s, \quad H = (1-h)/(t_s c_s), \quad c_s = 1 + a_s B_s r.$$

Set

$$S = 1/T, \quad \ell = HT, \quad \bar{h} = 1 - \ell, \quad x = jB, \quad y = TB, \quad B = Sy, \quad q_z = j/S. \tag{RL2}$$

Thus S, x, y are positive infinitesimals, $0 < p < \min(1, j)$, $0 < \ell < 1$, and $j \asymp a_s$. Both ℓ and $\bar{h}$ may be arbitrarily small. The original physicality inequality RER2–RER7 gives $jy \in \mathrm{I}_c$. In terms of the actual finite polynomials $\mathcal{P}, \mathcal{L}, \mathcal{N}, \mathcal{F}, \mathcal{K}$ of RSR, put

$$K = j + 1 - p, \quad k = K/(j+1), \quad F_b = j + 1 + p(1-p)/j, \quad E = 1 + B(\ell + p - 1). \tag{RL3}$$

Here $1/2 < k < 1$ and $1 \le F_b/(j+1) < 2$. Direct substitution $B = Sy, T = 1/S, H = S\ell$ in those finite polynomials gives

$$\mathcal{P}/(j+1) \sim 1, \qquad S\mathcal{L}/K \sim 1, \qquad S\mathcal{F}/F_b \sim 1. \tag{RL4}$$

For clarity, the terms after division contain only bounded $p/j, j/(j+1), K/(j+1)$, multiplied by S, y, or jy. No inverse of ℓ, $\bar{h}$, or $j - p$ is involved.

The same substitution in the exact numerator of n yields

$$\mathcal{N} = p(1+\epsilon_p) - yj(j+1)(1+\epsilon_y), \qquad \epsilon_p, \epsilon_y \in \mathrm{I}_c. \tag{RL5}$$

For an explicit finite division certificate, the zero-p term is

$$\mathcal{N}|_{p=0} = -yj(j+1)\{1 + S(1-\ell)\}\{1 + S\ell yj/(j+1)\}.$$

The quotient $(\mathcal{N} - \mathcal{N}|_{p=0})/p$ is a finite polynomial. This grouping can be read termwise from RSR5: the first coefficient is $1 + y(j+1) + O(Sy(j+1) + Sy^2(j+1))$; the second coefficient is $1 + O(S + y + Syj)$. Using $\mathcal{N} \ge 0$, therefore,

$$\eta_2 = \frac{yj(j+1)}{p} = O(1), \qquad \eta_2 \in \mathrm{I}_c \implies n \sim \frac{p}{2(j-p)(j+1)} > 0. \tag{RL6}$$

The estimate includes the face $\mathcal{N} = 0$, which can occur only in the positive-unit η_2 case.

Let $\mu = m(1-p)$ and $h_0 = \mu + 1/2$. The complete source formulas now imply

$$\begin{gathered} \mu \sim \frac{jp}{2(j+1)}, \quad h_0 \asymp 1, \quad CH \sim k/2, \quad SA\bar{h} \sim k/2, \quad D/S \sim k/2, \\ \alpha \sim \frac{Sy^2K^2}{2(j+1)}, \quad f_c \sim \frac{j^2y^2F_b}{2(j+1)}, \quad r_c \sim \frac{jy^2(1-p)K}{2(j+1)}. \end{gathered} \tag{RL7}$$

These use the exact masses in RFN12 and the closed RSR formulas. All displayed positive masses are strictly positive. Write $\kappa = -k_c > 0$. The exact positive-surplus formula RSR19 proves this sign. Its substituted finite polynomial gives

$$\frac{\kappa}{r_c} = O(p + Sy) \quad (p \in \mathrm{I}_c), \qquad \frac{\kappa}{jy^2} = O(1). \tag{RL8}$$

For the remaining case in (RL1), p and j are positive bounded units, and

$$\frac{\kappa}{f_c} \sim \frac{pK}{jF_b} \le \frac{1}{2}, \qquad jF_b - 2pK = (j-p)K \ge 0. \tag{RL9}$$

The inequality is a bound on real residues when combined with the preceding asymptotic equality. On this case η_2 is infinitesimal, and the effective current defined below is a positive bounded unit.

The entire parent and its harmonic shift

Retain the complete original quarter-parent RFN4. Its exact inverse state change at $w = (W + uY)/(1 + u^2) = 1$ is

$$X = N_0 + pv - uz, \quad Z = z + uN_0 + puv, \quad W = 1 + u(M_0 + Hvz), \quad Y = u - M_0 - Hvz. \tag{RL10}$$

Here $z > 0$ is arbitrary. Retain $mpD_u^2N_0^2$ and divide the remaining whole parent by $D_u = 1 + u^2$. The exact expression is RFN7, with both complete state normals, hard entries RFN9, and scalar RFN12.

Use the additional exact change

$$q = vz - Tu, \qquad L = M_0 + Hq, \qquad M_0 = L - Hq, \qquad L = \sqrt{S\ell\bar{h}}\,\widehat{L}. \tag{RL11}$$

All divisions are pure source or pure testing. In particular this change does not invert the harmonic mixed unit $1 + \rho u^2$. The complete diagonal of $\widehat{L}$ is

$$S\ell\bar{h}\{D_u(CD_v + D + \tfrac{1}{2}) + AD_v + (\alpha + \tfrac{1}{2})v^2\}, \tag{RL12}$$

with positive real residue $\overline{k}/2$ and bounded coefficient array through both openings.

Use the circuit coordinates $U, V, Z_*, Q_*, B_*, R_*, \Psi$ of SH1–SH2, and $b = v - uz/j = \sqrt{y}B_*$. Before spending any scalar n-mass, complete the exact correlated block, as in RFN13–RFN16:

$$h_0N_0^2 + n(N_0 + jb)^2 = (n + h_0)(N_0 + c_bb)^2 + M_eb^2, \quad c_b = \frac{nj}{n + h_0}, \quad M_e = \frac{nh_0j^2}{n + h_0}. \tag{RL13}$$

Keep $nv^2(N_0 + pb)^2$ and all three additional positive hard squares RFN15 whole, including after the change $\widehat{N} = \sqrt{n + h_0}(N_0 + c_bb)$. At $n = 0$, put $c_b = M_e = 0$ and omit the current row. For $n > 0$, $c_b^2/M_e = n/[h_0(n + h_0)]$ is bounded.

The exact shifted scalar. Let F be RFN12. The shifted scalar is $F - 2HqL_M + H^2q^2H_M$. Use the whole $d_cV^2Q_*^2$ term in its q^2 diagonal. Using the exact actual-source identity $S_0 = 0$ in RFN22, the resulting coefficient on $(\sqrt{S}q)^2$ is exactly

$$D_q = S\{A\bar{h}^2 + \ell^2(C + D + \tfrac{1}{2})D_u\} - \frac{f_c}{Tx} - TH^2C\,u^2v^2 - \frac{k_cv^2}{yTx}. \tag{RL14}$$

It has bounded array and positive real residue $\overline{k}/2$. Indeed $f_c/(Tx) = O(jy)$ is infinitesimal, $TH^2C = \ell CH$ is bounded, and $\kappa/(yTx) = \kappa/(jy^2)$ is bounded. Its equivalent Q_*^2 mass has order jy. The only extra scalar cross terms are

$$-2Hq\{\mu uvb + v(\alpha - yCHu^2)R\}, \qquad R = z - uv/(Sy). \tag{RL15}$$

The budgets below allocate their entire expressions.

The remaining signed scalar terms are exactly $e_0V^2B_*^2 + k_cZ_*^2(V^2 - 1)^2$, where $e_0 = p^2y^2/2 - g_c$. The negative part of e_0 is at most g_c. SH2 purchases it at relative circuit cost

$$\frac{g_c}{f_c}(1 + V^2) = O\big(S(y + v^2)\big) \in \mathrm{I}. \tag{RL16}$$

Use $Z_*(V^2 - 1) = -VQ_* - R_*$ for the other signed term. If p is infinitesimal, (RL8) pays its opposite part directly from $r_cR_*^2$, and its VQ_* cost is $O(v^2)$. Otherwise p, j, M_e are positive units. Young's exact square identity with parameter $1/6$, followed by SH1 at $a = 3/2$, gives the legal SOS domination

$$\kappa Z_*^2(V^2 - 1)^2 \preceq_{\mathrm{sos}} \frac{7}{4}\kappa\Psi + \frac{21}{2}\kappa V^2Q_*^2 + \frac{7}{2}\kappa V^2B_*^2. \tag{RL17}$$

The circuit cost is at most $7/8 + o(1)$, by (RL9). The two other relative costs are $O(v^2)$. Thus fixed positive real fractions of $D_q(\sqrt{S}q)^2$, $r_cR_*^2$, $f_c\Psi$, and M_eb^2 remain; the last is absent only at $n = 0$. The opposite mass r_c was not spent in the second case.

All helper rows and both complete normals

As usual, a free row permits any fixed real square pivot at infinitesimal cost from the preceding retained scalar forms. SH2 proves that both $E_u = u(uvz - jy)$ and $E_v = v(uvz - jy)$ are free. The row $R_u = v(u^2 - x)$ is free as well. For p infinitesimal, use the actual RFN19 rational Gram: its three costs are $O(S^2y)$, $O(Syu^2)$, and $O(Su^2/(1 - p))$. When p, j, M_e are units, apply SH1 to the R_*^2 term in RFN19. The resulting costs are

$$O(S^2y + Su^2/j), \quad O(Syu^2 + Su^2v^2), \quad O(jSu^2v^2/M_e), \tag{RL18}$$

all infinitesimal.

The rows pv^2b and $\sqrt{S}\,\mu uvb$ are free. If η_2 is infinitesimal, (RL6) and (RL13) imply

$$\frac{p^2}{M_e} = \frac{p^2}{h_0 j^2} + \frac{p^2}{nj^2} = O(1). \tag{RL19}$$

They then cost $O(v^4)$ and $O(Su^2v^2)$. If η_2 is a positive unit, the actual source gives $p^2y/f_c = O(1)$ and $\mu = O(p)$. The second identity in SH2 provides the same rows with costs

$$O\big(v^2(y+v^2)\big), \qquad O\big(Su^2(y+v^2)\big), \tag{RL20}$$

including $n = 0$.

The full opposite term $\sqrt{S}\,v(\alpha - yCHu^2)R$ is free by SH4–SH8. Also $j\alpha b$ is free:

$$j\alpha b = -(\alpha/B)R_u - \alpha uR, \qquad \alpha/B = O(y(j+1)) \in \mathrm{I}_c. \tag{RL21}$$

For p infinitesimal, the direct r_c cost for αuR is $O(Sy^2(j+1)^2u^2)$. In the unit p, j, M_e case use SH1; the three costs are $O(Sy^2u^2)$, $O(Sy^2u^2v^2)$, and $O(Sjy^2u^2v^2/M_e)$. All are infinitesimal. These statements account for the whole source α, not only its leading coefficient.

The first normal. The exact RFN21 source cancellations give, after (RL11),

$$L'_N = [j(n-\alpha) + p(n+\tfrac{1}{2})v^2]b + K_sR_u + \kappa_N E_v, \tag{RL22}$$

$$K_s = -\mu + (\alpha+\tfrac{1}{2})(\ell+p-1), \qquad \kappa_N = -(n+\tfrac{1}{2})(1-p/j).$$

Both coefficients are bounded. For κ_N, use the exact numerator of n and $n(j-p)/j = O(p/[j(j+1)])$; this cancels the r opening. After the entire correlated completion (RL13), the remaining normal is $-j\alpha b + \frac{1}{2}pv^2b + K_sR_u + \kappa_N E_v$. Every row is free by (RL18)–(RL21). Division by $\sqrt{n+h_0}$ is bounded. The shift generates couplings to $\sqrt{M_e}b$ with bounded coefficients $c_b/\sqrt{M_e}$; they are absent at $n = 0$. Large fixed real helper pivots absorb these couplings by exact Young identities.

The second normal and the entire hard block. The complete RFN22 normal after (RL11) is

$$L_L = (C_q - HH_M)q + \mu uvb + v(\alpha - yCHu^2)R, \qquad C_q - HH_M = -D - H(C+D+\tfrac{1}{2})u^2. \tag{RL23}$$

The coefficient of $\sqrt{S}q$ after the normalization of L is infinitesimal; D is infinitesimal and $H(C+D+1/2)$ is bounded. The other two terms are free by (RL20) and SH6–SH8. Exactly the same rows pay (RL15), whose first factor is $\sqrt{S}q$ with a bounded source multiplier.

The remaining hard off-diagonal is $(\alpha+1/2)uv\sqrt{S\ell\bar{h}}/\sqrt{n+h_0}$, which is infinitesimal. Its c_bb correction has the same bound after division by $\sqrt{M_e}$. The complete finite Gram therefore has a positive definite real residue, with principal normal pivots

$$1, \quad \overline{k}/2, \quad \overline{k}/2 \tag{RL24}$$

on $\widehat{N}, \widehat{L}, \sqrt{S}q$, together with the retained strict scalar and purchased helper pivots. All extra positive hard squares and both original retained current positive square terms are restored whole. The strict bounded-Gram lemma proves tensor SOS for the entire parent.

State and outer-chart exhaustion

The fresh full-state sign lemma SJ1–SJ10 permits positive u, v, and on the remaining negative-core chart gives $wz > 0$. Divide by the nonzero pure-testing w to obtain the normalization used above with arbitrary $z > 0$. No estimate in this proof uses its size. Literal axes and homogeneous endpoints have the direct original-core SOS exit. Real and imaginary state parts separate. The exact outer theorem R2O1–R2O24 covers every other testing pair on all original R2 sources. This proves the stated all-tests conclusion on (RL1).

The complementary source a_s unbounded and p a positive unit is not inferred from this proof. It requires the separate TU source comparison and additional-current conductor.

C.13.2 Circuit budgets for the opposite normal

Use the compact actual-source dictionary of RSR1–RSR8. In the R2 source put $S = 1/T$, $x = jSy$, $B = Sy$, and $H = S\ell$, where $0 < \ell < 1$. Both testing angles u, v are positive infinitesimals. Normalize the rotated testing coordinate $w = 1$; the other rotated coordinate $z > 0$ is arbitrary, including unbounded values. No coefficient estimate below involves z.

Set

$$U = \frac{u}{\sqrt{x}}, \quad V = \frac{v}{\sqrt{y}}, \quad Z_* = \frac{z}{\sqrt{j/S}}, \quad Q_* = U - VZ_*, \quad B_* = V - UZ_*, \quad R_* = Z_* - UV,$$
$$\Psi = U^2 + V^2 + Z_*^2 + U^2V^2Z_*^2 - 4UVZ_*.$$

All monomials and radicals separate between the two fields.

A parameterized opposite-square identity. For every positive real $a > 1$, let $K = a/(a-1)$, $d = U - V$, $t = UV$, $E = Z_*(1-t)$, $T_* = t(Z_* - 1)$. Exact multiplication gives

$$\begin{aligned} a\Psi + KV^2(Q_*^2 + B_*^2) - R_*^2 = ad^2 + (a-1)(E - T_*/(a-1))^2 \\ + (2at + KV^4)(1 - Z_*)^2 + 2KZ_*(t - V^2)^2. \end{aligned} \tag{SH1}$$

The right side is a finite legal separated SOS. For example $a = 3/2$ gives $K = 3$. This is a different budget from the older one with $U^2B_*^2$; neither budget dominates the other uniformly at independent source and testing rates.

The two symmetric current budgets are

$$\begin{aligned} (1+U^2)\Psi &= R_*^2 + B_*^2 + U^2Q_*^2 + U^2(1 - UVZ_*)^2, \\ (1+V^2)\Psi &= Q_*^2 + R_*^2 + V^2B_*^2 + V^2(1 - UVZ_*)^2. \end{aligned} \tag{SH2}$$

In particular the two actual rows

$$E_u = u(uvz - jy), \qquad E_v = v(uvz - jy) \tag{SH3}$$

are free from a fixed positive fraction of $f_c\Psi$ whenever $f_c \asymp j^2y^2$. Their respective relative circuit costs are $O(x + u^2)$ and $O(y + v^2)$, both infinitesimal. Here *free* means that an arbitrarily large fixed real multiple of the row square can be inserted at infinitesimal cost in the strict circuit Gram. It is not a pointwise bound without an SOS witness: multiply (SH2) by j^2y^2x or j^2y^3.

The full opposite row after the harmonic shift. Let

$$q = vz - u/S, \quad R = z - \frac{uv}{Sy}, \quad R_v = u(v^2 - y),$$
$$d_s = \frac{BCH(1-p)(1+x)}{E}, \qquad E = 1 + B(HT + p - 1).$$

The exact actual source identity is

$$\frac{\alpha}{y} - CHx = d_s. \tag{SH4}$$

Indeed it is equivalent to $(K + Hjy) - jE = (1-p)(1+x)$, after canceling the displayed positive source factors in the RSR dictionary. The two exact testing identities

$$u(u^2 - x) = SE_u - Su^2q, \qquad v(u^2 - x) = SE_v - Suvq \tag{SH5}$$

now give the complete identity

$$\begin{aligned} \sqrt{S}\,v(\alpha - yCHu^2)R = [\alpha - yCHu^2 - CHu^2(v^2 - y)]\,(\sqrt{S}q) \\ + \sqrt{S}\,CH(v^2 - y)E_u - \frac{d_s}{\sqrt{S}}R_v. \end{aligned} \tag{SH6}$$

There is no restriction on z, and no normal has been set to zero. The first coefficient in (SH6) is infinitesimal, because $CH = O(1)$, y, u, v are infinitesimal, and $\alpha = O(Sy^2(j+1))$ is infinitesimal by $jy \in \mathrm{I}$. The second coefficient is infinitesimal and (SH3) is free.

For the last term write

$$R_v = -\sqrt{x}\,y(Q_* + VR_*), \qquad \frac{d_s}{\sqrt{S}} = \sqrt{S}\,y\,CH(1-p)(1+x)/E. \tag{SH7}$$

Suppose the scalar has retained strict pivots of respective orders jy and $jy^2(1-p)$ on Q_*^2, R_*^2, as in the actual R2 source. Twice Young's identity and (SH7) provide this row at relative costs

$$O(S^2y^4(1-p)^2), \qquad O(S^2y^2(1-p)v^2), \tag{SH8}$$

respectively. Both are infinitesimal even when $1-p$ has arbitrarily small positive order. No inverse of that opening remains. Multiplication by the additional bounded source factor $\sqrt{\ell(1-\ell)}$, used to normalize the complete second normal, preserves the estimate.

This controls the opposite normal uniformly through all R2 source ratios once the stated scalar pivots are retained. The signed scalar allocation remains a separate step.

C.13.3 Unbounded source ratio and bounded relative current

Statement and domain. Use the actual original weak-C2 source RER1–RER8, normalized by $d = 1$, with $\lambda = \beta = 0$ and every original strict gap. Suppose t, a are positive unbounded, while B, aB, tB are positive infinitesimals, and put $e = 1 - r$. Assume

$$\lambda_n = n/(tB) = O(1), \qquad n \geq 0. \tag{TU1}$$

The entire original parent is tensor SOS for every independent projective test and every state. Literal $n = 0$ is included. No comparison of a testing quantity with a source quantity is made. This theorem treats precisely this source sector; it is not a completion of the other R2 sectors or the global finite-lift proof.

The bounded-Gram and finite toric allocation lemmas used below are the actual retained SC16–SC20 and QI1–QI10 proofs. We use their printed finite algebra and prove its source bounds on the present domain. We do not apply the physical C1 theorem to a formal source whose event coefficient could be negative.

An exact common contact and bounded source comparison

Set

$$S = 1/t, \quad z = 1/a, \quad q = a(tB), \quad w = ae, \qquad b = \sqrt{S}, \quad g = \sqrt{z}, \quad c = \sqrt{q}. \tag{TU2}$$

Thus $B = Szq$, $e = zw$, the actual squared angles are $u_c^2 = Sq$ and $v_c^2 = zq$, and S, z, q are positive infinitesimals. Keep h as the original C2 source parameter. The strict m gap gives

$$R = 1 + Sqh, \quad M = R - w > 0, \qquad 0 < w < R \sim 1. \tag{TU3}$$

Both original openings $h, 1 - h$ remain arbitrary positive quantities.

The exact RER2 identity, divided by $e > 0$, gives

$$Q_{\mathrm{RER}} = \tau\{(1 + z)(1 + Sh)(1 + Sq) + 2r(z\Delta)\lambda_n\}, \qquad \tau = q/(zw), \quad \lambda_n = n/(qz).$$

Here $Q_{\mathrm{RER}} \sim 1$, $z\Delta \sim 1$ and $r \sim 1$. The bracket is a positive bounded unit, since $\lambda_n \geq 0$ is bounded and its first term is a positive unit. Consequently

$$\tau \text{ is a positive bounded unit}, \qquad \overline{\tau} = \frac{1}{1 + 2\overline{\lambda}_n}, \qquad q = zw\tau. \tag{TU4}$$

No division by n has occurred. This sector includes the earlier thin condition $\rho_n = n/e^2 \in \mathrm{I}$: in that case the same RER identity gives $\tau \sim 1$ and $\lambda_n = \rho_n w/\tau \in \mathrm{I}$ directly, without first assuming (TU1).

For the comparison only, substitute into the native C1 rational formulas NC1–NC4

$$k^2 = Sq, \quad \eta = 1/S, \quad \zeta = z, \quad r_{\mathrm{C1}} = h, \quad t_{\mathrm{C1}} = w, \quad a_{\mathrm{C1}} = S(1 - h). \tag{TU5}$$

Their θ is $e = zw$ and their R, M are exactly (TU3). Write a superscript 0 for these formal C1 coefficients. All their source denominators have the requisite positive signs:

$$K = 1 + Sq, \quad J = 1 + Sq(1 - e) - S(1 - h), \quad E = 1 - q(1 - e), \quad Q = 1 - Sqzh,$$

and J, E, Q are positive bounded units. Their common polynomial is

$$P = 1 + qz + Sq(2 + z - 2e) + Sq^2(z + e - e^2) + S^2hq^2(1 + z - e). \tag{TU6}$$

In particular $P \sim 1$. We make no sign claim for β^0.

The two sources have exactly the same state contact. With $Z_c = 1$,

$$u_c = bc, \quad v_c = cg, \quad W_c = bg, \quad Y_c = hu_cW_c, \quad X_c = -(1 - e)u_c. \tag{TU7}$$

This follows either by substitution in the two native dictionaries, or from $Y_c = hu_cW_c$, $X_c = -ru_cZ_c$ and $Z_c/W_c = \sqrt{at}$ in the actual C2 source. In particular the comparison does not move the contact.

Use the actual, not formal, normalized coefficients

$$c_b = SC, \ a_b = SA, \ D_b = D/S, \ \alpha_b = \alpha/(Sz), \quad h_2 = mzM, \quad h_b = zw/2. \tag{TU8}$$

They satisfy $m = h_2/(zM)$ and $\beta + 1/2 = h_b/(zw)$. Use the same normalizations with superscript 0, and let δ mean actual minus formal. The following explicit polynomials make all needed cancellations finite and opening-safe.

Put $G = 1 + z - e$, and define

$$\begin{aligned}
P_C &= 1 + qz + S\{1 + q(2 + 2z - e) + q^2 z\} \\
&\quad + S^2\{q[h(1 + z) + 1 - e] + q^2[hez - e(1 + z) + 2z + 1]\} + S^3 hq^2 G, \\
P_A &= 1 + qz + S\{q(2 + z - 2e) + q^2 z(1 - e)\} + S^2 q^2 G(G - hz), \\
P_D &= 1 + q(z + e) + S\{q[h(1 + z) + 1 - e] + q^2(z + e - e^2)\} + S^2 hq^2 G, \\
P_m &= 1 + qz + S\{q(2 + z - 2e) + q^2 z\} + S^2 q^2\{(1 - e)^2 + z(1 - e) + hez\}, \\
P_b &= (1 + z)(1 + qz) + Sq(1 + z)(2 + z - 2e) \\
&\quad + Sq^2 z(1 + z - e^2) + S^2 q^2 G^2.
\end{aligned}$$

Then the exact rational identities are

$$\begin{aligned}
(1 - h)\delta c_b &= -\frac{n(1 - e)P_C}{zP}, & h\delta a_b &= -\frac{n(1 - e)P_A}{zP}, \\
\delta D_b &= -\frac{n(1 - e)(1 + qz)P_D}{zJP}, & \delta h_2 &= \frac{n(1 - e)(1 + qz)P_m}{zP}, \\
\delta h_b &= \frac{n(1 - e)P_b}{zP}, & \delta\alpha_b &= \frac{n(1 - e)q^2(1 + qz)[(1 + S)e - S(1 - h)(1 + z)]}{zQP}.
\end{aligned} \tag{TU9}$$

These identities are proved simply by substituting the six printed RER1 weights and the six NC4 weights, clearing the indicated pure source denominators, and multiplying. They include $n = 0$; the right sides define no quotient by n.

After substituting $e = zw$, the polynomial

$$P_\Delta = \frac{P_b - (1 + qz)P_m}{z} \tag{TU10}$$

is a genuine finite polynomial in S, z, q, w, h. Indeed the numerator specialized at $z = 0$ is zero identically, so ordinary monic division by z has zero remainder. Consequently

$$\delta(h_b - h_2) = n(1 - e)P_\Delta/P. \tag{TU11}$$

Every polynomial in (TU6), (TU9), (TU10) has bounded coefficients and bounded arguments. From (TU4) and (TU9)–(TU11) we obtain

$$\begin{aligned}
(1 - h)\delta c_b,\ h\delta a_b,\ \delta D_b = q\lambda_n\, O(1), \qquad \delta h_2, \delta h_b = q\lambda_n\, O(1), \\
\delta\alpha_b/q^2 = \lambda_n q\, O(S + e), \qquad \delta(h_b - h_2)/(qz) = \lambda_n O(1).
\end{aligned} \tag{TU12}$$

Here $O(1)$ denotes an explicit bounded rational source array with denominators among the units P, J, Q, R; it does not hide a state or testing factor. These conclusions require no lower bound on h, $1 - h$, or M.

In particular $c_b \asymp 1/(1 - h)$, $a_b \asymp 1/h$, $D_b \sim 1/2$, $\alpha_b/q^2 \sim 1/2$, and $h_2/q, h_b/q$ positive bounded units with common residue $1/2 + \overline{\lambda}_n$. We also have

$$h_b/w = z/2, \qquad h_b - h_2 = O(qz), \qquad H_0 := 4h_2/R + 4h_b/w \geq 2z, \quad H_0 \asymp q + z. \tag{TU13}$$

The last statements use the formal exact SC2 difference together with (TU11), and hold also when M is arbitrarily small.

The complete additional current, including its normal

Let the testing u, v be positive infinitesimals and first take $Z = 1$ on its pure-testing state chart. Use the exact SC coordinates

$$\begin{aligned}
Q_b = (W + uY)/b, \quad Y_b = Y/S, \quad p = w/R, \quad N = ((X + u)/b - pvQ_b)/g, \\
H = uQ_b/b, \quad L = vQ_b/g, \quad \sigma = c_b + a_b, \quad \rho_0 = Kh/R, \quad N_y = \sqrt{\sigma}\{D_u Y_b - \rho_0 H\}.
\end{aligned} \tag{TU14}$$

Both N, N_y vanish at (TU7). The exact SC3 full-parent identity is an algebraic identity for arbitrary $c_b, a_b, D_b, \alpha_b, h_2, h_b$; no C1 coefficient relation is used in deriving it. With (TU8), it represents all the actual C2 parent except its additional term $4nD_uD_vX^2/S$. Retain its original nonnegative m positive square term $4h_2wN^2/(MR)$, incorporate the second paired positive square term into H_0N^2, and retain $4h_b(vN)^2/w$ exactly as in the uniform QI normal lemma. Define $\widehat{N} = \sqrt{H_0}N$.

Put $h_t = H/R$, and introduce the same four positive separated monomials and their geometric mean as in QI:

$$A_* = ch_t, \quad B_* = cL/R, \quad C_* = uv/b, \quad D_* = qg, \qquad \Gamma = (A_*B_*C_*D_*)^{1/4}. \tag{TU15}$$

Their exact relations include $\Gamma^2 = qh_t v$ and $B_*C_* = (c/g)v^2 h_t$. The extra monomial

$$U_* = cgu/b$$

has the two exact identities

$$U_*B_* = \Gamma^2, \qquad cgX/b = czN + eB_* - U_*. \tag{TU16}$$

The entire additional current therefore has the following square completion, with no state normal set to zero:

$$\begin{aligned}\frac{4n}{S} D_u D_v X^2 &= 4\lambda_n D_u D_v (U_* - B_* - czN)^2 \\ &\quad + 4\lambda_n D_u D_v \{2(1-e)\Gamma^2 - (1-e^2)B_*^2 - 2(1-e)czB_*N\}.\end{aligned} \tag{TU17}$$

The first line is a nonnegative separated SOS and vanishes at the actual contact, including its unrestricted state-normal coordinate. Its coefficient is literally zero if $n = 0$. The remaining scalar changes have bounded coefficients on B_*^2 and Γ^2. Their possibly nonzero real residues will cancel exactly with the source comparison below. The full additional first-normal coefficient, on $B_*\widehat{N}$, is infinitesimal because

$$\lambda_n cz/\sqrt{H_0} = O(\lambda_n c\sqrt{z}) \in \mathrm{I}. \tag{TU18}$$

In particular (TU17) does not estimate or discard u/b, which can be arbitrarily large. The new positive square contains that entire row, while the exact product relation in (TU16) places its cross term on the old angular circuit.

Why the full QI allocation is stable here

We verify the actual finite hypotheses of QI1–QI10 rather than an abstract continuity assertion. Its coefficient arrays have only the pure-testing denominators D_u and positive pure-source units after cancellation of the two h openings. The latter cancellation is preserved exactly because

$$1 - \rho_0 = (1-h)/R, \qquad \rho_0 = Kh/R. \tag{TU19}$$

For example all perturbations of $c_b(1-\rho_0)^2$, $a_b\rho_0^2$, $c_b(1-\rho_0)$ and $a_b\rho_0$ are $q\lambda_n$ times bounded rational arrays, by (TU9). Moreover

$$\delta(\sigma\rho_0 - c_b) = \rho_0\delta a_b - (1-\rho_0)\delta c_b = q\lambda_n O(1). \tag{TU20}$$

This is the complete potentially singular harmonic first-normal coefficient after multiplication by $\sqrt{\sigma}$. Both σ and its formal comparison are comparable to $[h(1-h)]^{-1}$, and their ratio has residue one. Their square-root ratios and inverses are consequently bounded pure-source units. Thus this change of hard normal loses no opening bound.

In the exact generic scalar QI2, the changes of a, b_*, c_* are $q\lambda_n$ times bounded arrays in u, v. This follows term by term from (TU12), (TU19): its only C, A occurrences are the first two expressions in (TU19); its D_b occurrences have bounded factors; its α_b occurrences have bounded factors and the additional small factors printed in QI2. Consequently the formal specialization a_0, b_0, c_0 and its exact U, V divisions in QI3–QI4 can be used unchanged, while the remaining q coquotients change by bounded arrays. Write $\lambda_0 = \overline{\lambda}_n$. Their complete leading changes are respectively $0, -8\lambda_0, 4\lambda_0, 0$: substituting $S = z = q = u = v = 0$ in the bounded coquotients (TU9) gives $\delta a/q = -4\lambda_0$, $\delta b_*/q = 0$, $\delta c_*/q = -4\lambda_0$, and $\delta(R^2P_L)/q = 4\lambda_0$. Thus before the extra current the four residues are

$$\overline{a_1 - c_1} = 2, \qquad \overline{b_1 + 2c_1} = -8 - 8\lambda_0, \qquad \overline{R^2P_L/q} = 2 + 4\lambda_0, \qquad \overline{4\alpha_b D_u/q^2} = 2. \tag{TU21}$$

Equation (TU17) adds $-4\lambda_0$ to the B_*^2 residue and $+8\lambda_0$ to the Γ^2 residue. Consequently the *combined* full scalar has exactly the original four residues $(2, -8, 2, 2)$ on $(A_*^2, \Gamma^2, B_*^2, D_*^2)$. All differences from those residues are infinitesimal bounded arrays, because λ_n itself is bounded. This cancellation is why a thinness assumption is unnecessary on (TU1). The one additional bound QI uses from physical $\beta \geq 0$ is also available directly, without any sign for β^0:

$$w/(c/g) = \sqrt{w/\tau} = O(1), \qquad w = O(1). \tag{TU22}$$

Thus the exact dangerous cross term $-4(wv/(c/g))D_uB_*C_*$ still has infinitesimal coefficient.

For the complete first N normal, (TU13) gives exactly the two bounds used in the uniform QI normal lemma: $g/\sqrt{H_0}$ and $c/\sqrt{H_0}$ are bounded, and

$$\frac{h_b D_v - h_2}{\sqrt{qH_0}} = O(z + v^2). \tag{TU23}$$

The H term and all other terms in SC14 therefore have the same infinitesimal coefficients on the QI rows; (TU18) handles the whole extra term. For the harmonic normal, (TU20) adds a $q\lambda_n$ bounded array to the left side of QI10. It can be absorbed into its existing qK_q coquotient. All other terms are the same bounded rational arrays

multiplied by the opening-safe quantities in (TU12), (TU19). The exact D_b root and its two normal terms in QI9 remain unchanged as algebraic identities. The complete hard block therefore retains the real residue $\mathrm{diag}(1,4)$ in $(\widehat{N}, N_y)$.

For clarity, here is the finite allocation being invoked. Use the QI roots

$$E_* = h_t - v, \quad F_* = vh_t - gD_*, \quad R_F = \frac{\sqrt{g\Gamma}}{c}(\Gamma - D_*), \quad R_E = \sqrt{\Gamma}(\sqrt{A_*} - \sqrt{cv}),$$

together with the complete finite toric root list SC16. The exact identities QI7–QI8 give strict real square pivots on F_*, R_F, R_E; QI6 gives the strict E_* pivot and twice the four-monomial circuit. Every remaining coefficient is an infinitesimal bounded rational array in the testing u, v, with the same finite denominators and divisions just verified. All removed positive square terms, including (TU17), vanish at (TU7). The actual source contact value, its three scalar first jets, and both full first-normal contact values consequently vanish exactly for this remaining expression. Applying the two finite monic divisions SC17–SC20 and the displayed QI7 first-jet identities therefore puts the entire scalar correction and both complete first normals on this strict root list with infinitesimal coefficients. The coefficient functions are independent of the state ratio Q_b/Z; no boundedness of that ratio is assumed. This is precisely the constructive QI allocation, with its source inequalities checked in (TU12)–(TU23), not a conclusion from a scalar restriction alone.

The resulting finite Gram is bounded and has positive definite real residue. The strict bounded-Gram lemma supplies tensor SOS. Restore the original positive paired positive square terms and the new full square in (TU17). This proves the local assertion for the entire parent on the positive small-angle, negative-core chart.

States, signs, and all remaining tests

Every inverse and radical used above is pure source, pure testing, or the explicitly separated positive monomial radical in (TU15). For $Z \neq 0$ the normalization $Z = 1$ is pure testing; multiplying back by Z^2 restores homogeneity. At $Z = 0$ the original pure-testing core restriction is nonnegative. The exact core Schur identity and sign gauges used in the retained QI proof force $W + uY > 0$ on the remaining negative-core chart, and hence all four monomials are positive. Otherwise the core and all original event positive square terms already give the direct exit. Literal zero-angle charts have that same original exit. Complex states split into their real and imaginary parts, since the real measurement matrices and full quadratic parent are unchanged.

The new R2 outer theorem R2O1–R2O24 treats every projective testing pair outside the double-infinitesimal original chart, with no condition on a or n/e^2. Its hypotheses are exactly the original RER source conditions already retained here. Combining that result with this local proof gives the claimed all-tests, all-states theorem on (TU1). No other source branch is proved by this conclusion.

C.13.4 Unbounded source ratio and relative current

Precise assertion. Retain the original RER source, all its strict gaps, and the notation TU2–TU16 of the preceding bounded-relative-current proof. Assume the native a, t are positive unbounded and

$$\lambda = \frac{n}{tB} = \frac{n}{qz} \text{ is positive unbounded.} \tag{UL1}$$

Then the whole original weak-C2 parent is tensor SOS for every independent projective test and every state. The argument below complements TU1; it never assigns a sign to the formal C1 coefficient β^0, and never replaces the original physical source by that formal source.

Source bounds without a thinness assumption

The exact RER physicality equation already used in TU4 is

$$Q_{\mathrm{RER}} = \tau\{A + 2r(z\Delta)\lambda\}, \qquad \tau = \frac{q}{zw}, \qquad A = (1+z)(1+Sh)(1+Sq).$$

Here $A, Q_{\mathrm{RER}}, r, z\Delta$ are positive units with residue one. Since λ is unbounded, this proves, with no cancellation of mixed factors,

$$\tau \in \mathrm{I}, \qquad \lambda\tau \sim \frac{1}{2}, \qquad \lambda q \sim \frac{zw}{2}, \qquad \frac{n}{z^2 w} \sim \frac{1}{2}. \tag{UL2}$$

All four divisions are pure source. Recall $0 < w < R \sim 1$. The exact TU9 formulas therefore imply

$$\begin{gathered}
(1-h)\delta c_b,\ h\delta a_b,\ \delta D_b,\ \delta h_2, \delta h_b = \lambda q\, O(1), \\
\delta\alpha_b/q^2 = \lambda q\, O(S + zw), \qquad h_2 \asymp zw, \quad h_b = zw/2, \\
H_0 = 4h_2/R + 2z \asymp z, \qquad H_0 \geq 2z, \\
\frac{g}{\sqrt{H_0}} = O(1), \quad \frac{c}{\sqrt{H_0}} = O(1), \quad \frac{\lambda q}{H_0} = O(1), \quad \frac{n}{z^2} = O(1).
\end{gathered} \tag{UL3}$$

In particular λq is infinitesimal. The displayed coefficient arrays have denominators only among the pure-source units P, J, Q, R after the original $h, 1-h$ openings cancel. The actual and formal $\sigma = c_b + a_b$ differ by a relative infinitesimal. Their square-root ratio is a bounded unit. No lower bound on $w, h, 1-h$ or the strict gap M is assumed.

Use the complete generic SC3 identity with the actual coefficients. Retain its positive m positive square term and use $\widehat{N} = \sqrt{H_0}N$. The formal comparison is used only for finite scalar and normal identities. Its q-coquotients and the QI coefficient arrays are bounded here: their derivation requires S, z, q infinitesimal and w bounded, but not $\beta^0 \geq 0$. The sole QI bound that is no longer available, $w/(c/g) = O(1)$, is replaced explicitly below.

The entire extra current supplies a new angular pivot

Use the exact positive-chart rows TU15–TU16, denoted here by A_*, B_*, C_*, D_*, U_* and Γ. Define

$$J_3 = \sqrt{\lambda}(U_* - B_*), \qquad E_* = h_t - v, \qquad F_* = vh_t - gD_*. \tag{UL4}$$

The first line of the exact whole-current identity TU17 is

$$4D_uD_v\left(J_3 - d_N\widehat{N}\right)^2, \qquad d_N = \frac{\sqrt{\lambda}\,cz}{\sqrt{H_0}} = \sqrt{\frac{nz}{H_0}} \in \mathrm{I}. \tag{UL5}$$

Thus it supplies a strict real pivot on J_3, with only infinitesimal changes to the hard normal Gram. Its scalar remainder is still exactly $8\lambda D_uD_v(1-e)\Gamma^2 - 4\lambda D_uD_v(1-e^2)B_*^2$. Its complete first-normal remainder, retained below, is $-8\lambda D_uD_v(1-e)czB_*N$.

Write $\widetilde{c}, \widetilde{a}, \widetilde{d}, \widetilde{\alpha}, \widetilde{m}, \widetilde{b}$ for the six differences $\delta c_b, \delta a_b, \delta D_b, \delta\alpha_b, \delta h_2, \delta h_b$ divided by λq. They are defined explicitly by the six right sides in TU9, using $n = \lambda qz$; their apparent opening poles cancel in every combination below. Put $U = u^2, V = v^2$, and retain $\rho_0 = Kh/R$, $a_t = 1 - \rho_0 + (w/R)V$. Define the exact rational arrays

$$\begin{aligned}
\mathsf{a} &= 4D_vR^2\{\widetilde{c}(1-\rho_0)^2 + \widetilde{a}\rho_0^2/D_u\} + 4\widetilde{\alpha}zS^2VR^2\rho_0^2/D_u + 4\widetilde{d}S^2R^2a_t^2,\\
\mathsf{b} &= -8\widetilde{\alpha}zSR\rho_0 - 8\widetilde{d}SRa_tD_u,\\
\mathsf{c} &= 4\widetilde{d}D_u^2,\\
\mathsf{l} &= 4R\{\widetilde{m}(1-w/R) + \widetilde{b}(w/R)D_v\}.
\end{aligned} \tag{UL6}$$

For example $\delta a/q = \lambda\mathsf{a}$ in QI2, and $\delta(R^2P_L/q) = \lambda\mathsf{l}$. Direct specialization of TU9 gives, keeping S, z, w, h exact,

$$\begin{aligned}
(\mathsf{a} - \mathsf{c})|_{q=U=V=0} &= 0,\\
(\mathsf{b} + 2\mathsf{c} + 8D_uD_v(1-e))|_{q=U=V=0} &= 0,\\
(\mathsf{l} - 4D_uD_v(1-e^2))|_{q=U=V=0} &= 0.
\end{aligned} \tag{UL7}$$

These are stronger than residue equalities. Clear the unit denominators and divide successively by the monic variables q, U, V. Each of the three expressions is consequently $qF_q + UF_U + VF_V$, with finite bounded rational coefficient arrays. The division does not invert q, U, V in the tensor ring.

The combined full scalar correction to QI6 therefore consists of bounded-array multiples of $\lambda q, \lambda U, \lambda V$ on A_*^2, Γ^2, B_*^2, an infinitesimal coefficient on E_*^2, and the infinitesimal α correction on D_*^2. The following exact identities allocate every apparently unbounded testing term:

$$\begin{aligned}
\lambda U A_*^2 &= \lambda q(uE_* + bC_*)^2,\\
\lambda V A_*^2 &= \lambda q(F_* + gD_*)^2,\\
\lambda U\Gamma^2 &= \lambda q\, bC_*(uE_* + bC_*),\\
\lambda V\Gamma^2 &= \lambda q(F_* + gD_*)(F_* + gD_* - vE_*),\\
\sqrt{\lambda}\,uB_* &= \sqrt{nS/z^2}\,(F_* + gD_*),\\
\sqrt{\lambda}\,vB_* &= \sqrt{n}\,C_* - vJ_3.
\end{aligned} \tag{UL8}$$

Every coefficient introduced on squaring the last two identities is infinitesimal by UL2–UL3 and $S, u, v \in \mathrm{I}$. Finally the formerly dangerous QI6 term has the exact replacement

$$\frac{wg}{c}\,vB_*C_* = wz\,C_*^2 - \frac{wg}{c\sqrt{\lambda}}\,vC_*J_3, \qquad \frac{wg}{c\sqrt{\lambda}} = w\sqrt{\frac{z^2}{n}} = O(1). \tag{UL9}$$

Both resulting coefficients are infinitesimal. This retains the entire term for arbitrary independent source and testing rates.

Both unrestricted state-normal corrections

The new complete first N coefficient, including the remainder of TU17, differs from the formal generic SC coefficient by

$$\begin{aligned}\delta\mathcal{L}_N &= 8\lambda c(1-e)\mathsf{F}_N B_* + 8\lambda q\widetilde{d}\,ubvg(Sa_tH - D_u v),\\ \mathsf{F}_N &= \frac{P_b D_v - (1+qz)P_m}{P} - zD_uD_v = V - zUD_v + qF_{Nq}.\end{aligned} \tag{UL10}$$

The last equality is a finite monic q division; its coquotient has bounded arrays because the numerator at $q=0$ is exactly $V - zUD_v$. After division by $\sqrt{H_0}$, its q part is infinitesimal on B_* by UL3. The other parts use

$$\begin{aligned}\lambda cVB_* &= \lambda qg\,vC_* - \sqrt{\lambda q}\,v^2 J_3,\\ \lambda czUB_* &= \lambda qbg\,v(uE_* + bC_*).\end{aligned} \tag{UL11}$$

Their coefficients on $\widehat{N}$ are infinitesimal since $g/\sqrt{H_0}$ and $\sqrt{\lambda q/H_0}$ are bounded. The $\widetilde{d}$ term in UL10 uses $uH = R(uE_* + bC_*)$ and $uv = bC_*$; it has the same infinitesimal coefficient bounds. Thus this is a lift of the complete first normal, with no state component put to zero.

For the second normal it is convenient to compare the coefficient after multiplication by $\sqrt{\sigma}$. Differentiating the full generic SC3 identity, including the restored D positive square term, gives

$$\sqrt{\sigma}\,\delta\mathcal{L}_y = 8\lambda q\{\mathsf{F}_H H + \mathsf{F}_v v\}, \tag{UL12}$$

where

$$\begin{aligned}\mathsf{F}_H &= D_v\{\rho_0\widetilde{a} - (1-\rho_0)\widetilde{c}\} - D_v\widetilde{a}U\rho_0/D_u + \widetilde{\alpha}zS^2V\rho_0/D_u - \widetilde{d}S^2a_t,\\ \mathsf{F}_v &= \widetilde{d}SD_u - \widetilde{\alpha}zS.\end{aligned}$$

These arrays are bounded through both source openings. Exactly,

$$\mathsf{F}_H|_{q=U=V=0} = \frac{(1-e)S}{1-S(1-h)}, \qquad \mathsf{F}_v|_{q=U=V=0} = -\frac{(1-e)S}{1-S(1-h)}. \tag{UL13}$$

As $R=1$ there, finite monic division gives $R\mathsf{F}_H + \mathsf{F}_v = qF_q + UF_U + VF_V$ with bounded arrays. Since $H = R(E_* + v)$, the whole UL12 coefficient is therefore an infinitesimal combination of E_*, F_*, A_*, C_*, D_*: use only

$$qv = cA_* - qE_*, \qquad Uv = buC_*, \qquad Vv = v(F_* + gD_* - vE_*). \tag{UL14}$$

The factor $1/\sqrt{\sigma}$ is bounded. Changing the formal harmonic-normal normalization to the actual one multiplies its previously bounded arrays by a pure-source unit with residue one.

For completeness, the unchanged formal first N coefficient is bounded after the actual $\sqrt{H_0}$ normalization: $h_2^0, h_b^0 = O(q)$ and $h_b^0 - h_2^0 = O(qz)$, while $g/\sqrt{H_0}$ and $c/\sqrt{H_0}$ are bounded. These are exactly the two estimates used in the QI normal allocation. All remaining hard coefficients from the original C, A, D, α lines have the same residues as before; the relative C, A changes and the absolute D, α changes are infinitesimal. Together with UL5 the complete hard block on $\widehat{N}, N_y, J_3$ has a positive definite real residue.

A finite contact lift for the additional row

It remains to allocate the full scalar correction on roots that vanish at contact. Merely bounding a coefficient on A_*, B_*, C_*, D_* would not suffice. Keep the complete strict toric root list SC16, the QI rows E_*, F_*, R_F, R_E, and add

$$V_3 = \sqrt{\lambda\Gamma}(\sqrt{U_*} - \sqrt{B_*}).$$

Since $U_*B_* = \Gamma^2$, the exact identities are

$$\begin{aligned}J_3^2 &= 4V_3^2 + \lambda(\sqrt{U_*} - \sqrt{B_*})^4,\\ \Gamma J_3 &= \frac{V_3^2}{\sqrt{\lambda}} - 2\sqrt{\lambda}\,\Gamma(B_* - \Gamma).\end{aligned} \tag{UL15}$$

Reserve half the J_3^2 coefficient in UL5 for the first identity. It supplies strict real pivots on both J_3, V_3, with the quartic square retained nonnegative. The normal coupling in UL5 remains an infinitesimal hard Gram correction.

Here are the additional first-jet coefficient checks needed for the second identity, so that its factor $\sqrt{\lambda}$ causes no hidden loss. The only new scalar terms linear in J_3 in UL8–UL9 have coefficients, up to bounded arrays, of the forms

$$\epsilon_B = -2\sqrt{n}\,v, \qquad \epsilon_T = \frac{wg}{c\sqrt{\lambda}}\,v. \tag{UL16}$$

Both have bounded finite testing difference arrays and are infinitesimal. At the actual pure-source contact $v_c = cg$,

$$\sqrt{\lambda}\,\epsilon_B(u_c, v_c) = -2n \in \mathrm{I}, \qquad \sqrt{\lambda}\,\epsilon_T(u_c, v_c) = wz = e \in \mathrm{I}. \tag{UL17}$$

Multiplication by any of the bounded arrays produced in UL7 preserves these conclusions. No other scalar term introduces J_3 linearly: the other five identities in UL8, and the λq scalar corrections, use only the old QI rows.

Write a term $\epsilon J_3 A_i$ as $\epsilon J_3(A_i - \Gamma) + \epsilon\Gamma J_3$. The first summand is already an infinitesimal quadratic on the strict list. Divide $\epsilon - \epsilon(u_c, v_c)$ finitely by $u - u_c, v - v_c$ and apply the exact SC18 conductors to $\Gamma(u - u_c)$ and $\Gamma(v - v_c)$. This allocates the nonconstant part as infinitesimal quadratic corrections involving J_3. For the constant part use UL15. Its V_3^2 coefficient is infinitesimal, and its remaining coefficient on $\Gamma(B_* - \Gamma)$ is infinitesimal by UL17.

There are also new bare-Γ E_*, F_* terms when the first five identities in UL8 are expanded and their C_*, D_* factors are written as $(A_i - \Gamma) + \Gamma$. Put $L = \lambda q$ and $M_1 = nS/z^2$, both infinitesimal. The additional E_* coefficients are bounded-array multiples of Lbu and Lgv; the additional F_* coefficients are bounded-array multiples of Lg and $M_1 g$. At $u_c = bc, v_c = cg$ their required contact ratios are

$$\frac{Lbu_c}{c} = LS, \qquad \frac{Lgv_c}{c} = Lz, \qquad \frac{Lg}{g} = L, \qquad \frac{M_1 g}{g} = M_1. \tag{UL18}$$

All are infinitesimal, with bounded finite difference arrays before these contact substitutions. The exact QI7 identities therefore allocate these additional terms by the same finite division as the old E_*, F_* terms. This check is necessary: the two new normal rows cannot be ignored when applying the contact-jet argument.

After these explicit additional steps, the remainder has exactly the QI form: a bounded infinitesimal quadratic on the listed roots, the now-verified bare-Γ E_*, F_* couplings, and a toric expression $\Gamma \sum_i \mu_i(A_i - \Gamma) + \Gamma^2\sigma$. The old E_*, F_* contact coefficient ratios remain the bounded ones in QI7–QI8: their unchanged $q = 0$ arrays depend on bounded w and do not require $w/(c/g)$ bounded. Every retained positive square term, including UL5 and UL15, vanishes at the actual contact. The complete actual scalar consequently has zero contact value and all three zero first jets. Apply the two explicit finite divisions SC17–SC20. Their three-jet balance makes the constant toric first-jet coefficients equal; SC19 and the variance roots then allocate this common term. This puts the entire scalar on a bounded Gram with strictly positive real residue.

The full normal lifts UL10–UL14 have exact contact zero as well. Their J_3, E_*, F_* terms already vanish there. The sum of their remaining four monomial coefficients is zero at contact, so one finite SC18 division allocates them on the same strict list. Every coefficient function in these steps depends only on the testing u, v. No bound on the state ratio Q_b/Z is used, and the third contact jet is used only for its exact toric balance.

The strict bounded-Gram lemma now proves tensor SOS for the whole original parent on the small-angle positive negative-core chart. All its radicals are radicals of separated positive monomials; every inverse is pure source or pure testing.

Projective completion and scope

The state, sign, zero-axis and real/imaginary-state arguments are the same exact original-core arguments at the end of TU. In particular $Z = 0$ is an original positive restriction, and the negative-core $Z = 1$ chart forces $W + uY > 0$, permitting the separated radicals above. The theorem R2O covers every testing angle chart outside u, v both infinitesimal without any condition on λ or the native a. This proves UL1 for all tests and states.

Together with TU this covers native a unbounded: the nonnegative $\lambda = n/(tB)$ is bounded, including zero, or unbounded. The bounded-a theorem supplies the complementary source domain.

C.13.5 Complementary testing angles

This proof treats every independent testing pair outside the positive double-infinitesimal angle chart of the original R2 source. It uses an explicit full-parent Bell reflection and the two-current and six-root identities that were freshly proved for R1. Only those universal algebraic identities and their real matrix arguments are reused. No assertion that the reflected source is an original R1 source is made.

Actual source and exact reflected parent

The source is the original weak-C2 tuple, with $d = 1$, $\lambda = \beta = 0$, the six coefficients in (RER1), all original strict gaps, $n \geq 0$, $a, B, t_s > 0$, and $0 < r, h < 1$. R2 means

$$t_s \text{ unbounded}, \qquad B, z = aB, y = t_s B \text{ infinitesimal}. \tag{R2O1}$$

Put $S = 1/t_s$, $l = 1 - h$, $e = 1 - r$, $R_s = 1 + hz$, $g_s = R_s - ae > 0$, and $k = (1 + ar)/(1 + a)$. The rederived exact source lemma (RER1)–(RER7) gives

$$k \text{ a positive bounded unit}, \qquad k < 1, \qquad SCl \sim SAh \sim D/S \sim k/2, \qquad \Delta/(1 + a) \sim 1. \tag{R2O2}$$

It also gives ay infinitesimal, $Q \sim 1$, and the exact expressions

$$n = \frac{eQ - BK_s}{2r\Delta}, \qquad m = \frac{a^2 e(1+y)cs_c}{2g_s\Delta}, \qquad BK_s = (1+a)y(1+hS)(1+z), \tag{R2O3}$$

where $c = 1 + zr \sim 1$ and $s_c = 1 + B(l + ar) \sim 1$. There is no positive lower bound assumed on r, h, l, or g_s.

For arbitrary finite testing angles define the unnormalized testing state transformation

$$\begin{aligned} G\psi = (&uvW - uX - vY + Z,\ -vW + X - uvY + uZ,\\ &-uW - uvX + Y + vZ,\ W + vX + uY + uvZ). \end{aligned} \tag{R2O4}$$

It has $G^T G = D_u D_v I$. The Bell transformation on observables is

$$(A'_0, A'_1, B'_0, B'_1) = (-B_1, -B_0, -A_1, -A_0).$$

After the displayed real local basis gauges it sends the angles to $(u', v') = (v, u)$ and the source coefficients to

$$(\lambda', n', m', C', A', D', \alpha', \beta') = (\alpha, m, n, C, D, A, \lambda, \beta). \tag{R2O5}$$

Direct substitution in the whole parent gives

$$F_{c'}(v, u; G\psi) = D_u D_v F_c(u, v; \psi). \tag{R2O6}$$

The transformed $\lambda' = \alpha$ event is kept as its whole positive square. Because $D_u D_v > 0$ is pure testing, a certificate on the left restores the right by dividing each root by $\sqrt{D_u D_v}$ within K_t. No source or testing contact is identified in this operation.

In the next two subsections write the transformed testing angles again as u, v, and denote the reflected coefficients by

$$\mathsf{C} = C, \quad \mathsf{D} = A, \quad \mathsf{A} = D, \quad \mathsf{m} = n, \quad \mathsf{n} = m.$$

Thus the symbols on the right refer to the original R2 source. Define the pure-source quantities

$$g_* = \frac{\mathsf{CD}}{\mathsf{C} + \mathsf{D}}, \qquad h_* = \frac{\mathsf{D}}{\mathsf{C} + \mathsf{D}}, \qquad \gamma = \mathsf{A} g_*. \tag{R2O7}$$

If $U_C = SCl$ and $U_A = SAh$, then exactly

$$Sg_* = \frac{U_C U_A}{hU_C + lU_A} \sim k/2, \qquad \gamma \sim k^2/4. \tag{R2O8}$$

The denominator is a positive bounded unit even when h or l is infinitesimal. In particular $1/\mathsf{C}$, $1/\mathsf{D}$, $1/g_*$ are infinitesimal, $(\mathsf{C} + \mathsf{D})^{-1/2} = O(\sqrt{S})$, and $0 < h_* < 1$. These are the complete source bounds needed for the first two outer current identities.

Universal outer current identities

Set

$$R_t = uW - Y, \quad C_t = v(uX - Z), \quad B_t = X + uZ, \quad J = R_t + C_t, \quad N = R_t + h_* C_t,$$

and let K be the exact signed quarter-core

$$K = \tfrac{1}{2}J^2 + \tfrac{1}{2}D_u(vY - Z)^2 + \tfrac{1}{2}D_v B_t^2 - \tfrac{1}{2}D_u D_v Z^2. \tag{R2O9}$$

All unselected original events, including λ', are retained as whole nonnegative positive square terms.

If $1/v$ is bounded, put

$$K_0 = 2\sqrt{g_*}C_t, \quad B_0 = \sqrt{2D_v/D_u}\, B_t, \quad c_u = D_u^{-1/2}, \quad t_u = uD_u^{-1/2}, \quad \eta_v = \sqrt{D_v}/v.$$

The entire selected-parent identity is

$$\begin{aligned} 4\mathsf{C}D_v R_t^2 + 4\mathsf{D}J^2 + 4K = {}& 4(\mathsf{C} + \mathsf{D})N^2 + 4\mathsf{C}v^2 R_t^2 + 2J^2 + 2D_u(vY - Z)^2 \\ &+ (K_0, B_0) \begin{pmatrix} 1 - c_u^2 \eta_v^2/(2g_*) & t_u \eta_v/\sqrt{2g_*} \\ t_u \eta_v/\sqrt{2g_*} & 1 \end{pmatrix} \begin{pmatrix} K_0 \\ B_0 \end{pmatrix}. \end{aligned} \tag{R2O10}$$

The finite Gram is bounded and has real residue I, because c_u, t_u and η_v are bounded while $1/g_*$ is infinitesimal. The actual separated bounded-Gram lemma makes it tensor SOS. The roots themselves need not be bounded.

This gives every state and every other finite angle when transformed Bob is noninfinitesimal, equivalently when the original Alice angle is noninfinitesimal.

For positive infinitesimal transformed v, put $q = g_*^{-1/2}$ and define

$$K_1 = 2\sqrt{g_*}C_t + qY, \quad B_1 = B_t - uvY, \quad V_0 = vY, \quad Y_1 = q\sqrt{D_u}Y,$$

$$L_0 = \tfrac{1}{2}qc_u(K_1 - c_uY_1) - t_uvB_1 + c_uvV_0.$$

Then $L_0 = \sqrt{D_u}\,v(vY - Z)$ exactly, and

$$\begin{aligned} &4\mathsf{C}D_vR_t^2 + 4\mathsf{D}J^2 + 4\mathsf{A}D_uD_vY^2 + 4K \\ &\quad = 4(\mathsf{C}+\mathsf{D})N^2 + 4\mathsf{C}v^2R_t^2 + 2J^2 + \mathcal{G}, \\ \mathcal{G} &= K_1^2 + 2D_v(B_1^2 + V_0^2) + (4\gamma D_v - c_u^2)Y_1^2 \\ &\quad + 2qv(K_1 - c_uY_1)V_0 - 2L_0^2. \end{aligned} \tag{R2O11}$$

This is the complete four-root Gram, including all normal terms. Its residue in the ordered frame (K_1, B_1, V_0, Y_1) is

$$\operatorname{diag}(1, 2, 2, k_0^2 - c_{u,0}^2). \tag{R2O12}$$

Subscript 0 denotes real residue. Hence it is strict whenever its last entry is positive. This includes every unbounded transformed u and every bounded positive-unit transformed u when $k_0 = 1$. All inverse and square-root factors in these identities belong to one field at a time.

The remaining unit angle and both coefficient openings

It remains to treat transformed v infinitesimal, transformed u a bounded positive unit, and $k_0 < 1$. The source identity

$$1 - k = \frac{ae}{1+a}, \qquad 0 < ae < R_s \sim 1 \tag{R2O13}$$

shows that a, e are positive bounded units: if a were infinitesimal or unbounded, or if e were infinitesimal while a is bounded, then $1 - k$ would be infinitesimal. Thus in the current chart a is a positive unit, e is a positive unit, and both $1 + a$ and Δ are units.

Use the opening-safe source coordinates

$$j = R_s/a, \qquad p = e, \qquad j - p = g_s/a > 0. \tag{R2O14}$$

They satisfy $0 < p < \min(1, j)$, j, p are positive bounded units, and the residue of $(j + 1 - p)/(j + 1)$ equals k_0. The exact relative critical coefficient identities are

$$\frac{\mathsf{m}}{jp/[2(1-p)(j+1)]} = \left(Q - \frac{BK_s}{e}\right)\frac{R_s + a}{R_s\Delta}, \qquad \frac{\mathsf{n}}{p/[2(j-p)(j+1)]} = (1+y)cs_c\frac{R_s+a}{\Delta}. \tag{R2O15}$$

Their right sides have residue one. Indeed BK_s/e is infinitesimal here by (R2O3), and all other normalized factors have the limits stated in (RER). No division by r, g_s, n, or an infinitesimal gap was used to assert a bounded error. The displayed quotients cancel those openings exactly before the limit is taken. In particular the critical chart itself forces original $n > 0$, even though the theorem as a whole retains $n = 0$.

Choose positive real caps $m_c \le \mathsf{m}$ and $n_c \le \mathsf{n}$, with the choices specified below. Keep both nonnegative excesses as the original whole positive square terms. Define six rows

$$\widehat{N} = \sqrt{\mathsf{C}+\mathsf{D}}\,N, \quad P = C_t/\sqrt{S}, \quad Y_0 = \sqrt{S}\,Y, \quad X, \quad Z, \quad V = vY. \tag{R2O16}$$

They can be unbounded or linearly dependent. If $\delta = (\mathsf{C}+\mathsf{D})^{-1/2}$, then $R_t = \delta\widehat{N} - h_*\sqrt{S}P$. The selected entire quarter-parent is exactly

$$\begin{aligned} \mathcal{P}_{\mathrm{sel}} &= \widehat{N}^2 + \mathsf{C}v^2R_t^2 + \tfrac{1}{2}J^2 + (Sg_*)P^2 + (\mathsf{A}/S)D_uD_vY_0^2 \\ &\quad + m_c\{(v/u)R_t + (D_u/u)V - X - uZ\}^2 + n_cD_uD_vX^2 \\ &\quad + \tfrac{1}{2}D_u(V-Z)^2 + \tfrac{1}{2}D_v(X+uZ)^2 - \tfrac{1}{2}D_uD_vZ^2. \end{aligned} \tag{R2O17}$$

The full-state row relation is

$$PY_0 = V(uX - Z). \tag{R2O18}$$

Add $2b(PY_0 - V(uX - Z)) = 0$ for a real b. Keep the two displayed positive squares $\mathsf{C}v^2R_t^2$ and $J^2/2$ whole. The remaining Gram is bounded: $u, 1/u$ are bounded, Sg_* and A/S are units, and the R_t contributions in the m_c

square have coefficients $\delta v/u$ and $h_*\sqrt{S}v/u$, both infinitesimal. Its residue is the direct sum of a unit pivot, the matrix

$$\begin{pmatrix} k_0/2 & b \\ b & k_0(1+u_0^2)/2 \end{pmatrix}, \tag{R2O19}$$

and the following real quadratic form on (X, Z, V):

$$Q_3 = m_c\{(D_u/u)V - X - uZ\}^2 + n_c D_u X^2 + \tfrac{1}{2}D_u V^2 - D_u VZ + \tfrac{1}{2}(X+uZ)^2 - 2bV(uX - Z). \tag{R2O20}$$

In this formula $u = u_0 > 0$ is real. We now give all finite-cap choices; no opening of either source coefficient is suppressed.

If $p_0 < 1$ and $j_0 > p_0$, use initially

$$m_0 = \frac{j_0 p_0}{2(1-p_0)(j_0+1)}, \quad n_0 = \frac{p_0}{2(j_0-p_0)(j_0+1)}, \quad b = k_0/2.$$

The three leading principal minors of Q_3 are

$$m_0 + n_0 D_u + \tfrac{1}{2}, \quad \frac{p_0 u_0^2 D_u(j_0+1-p_0)}{4(j_0-p_0)(j_0+1)^2(1-p_0)}, \quad \frac{p_0 u_0^2 D_u(j_0+1-p_0)^2}{8(j_0-p_0)(j_0+1)^3}, \tag{R2O21}$$

all strictly positive. Matrix (R2O19) has determinant $k_0^2 u_0^2/4 > 0$. Choose real caps slightly below m_0, n_0; openness of real positive definiteness and (R2O15) ensure they remain strict and are below the actual coefficients.

If $j_0 = p_0 < 1$, the coefficient n is unbounded and $m_0 = j_0^2/[2(1-j_0)(j_0+1)]$ is finite positive. Take $b = 1/[2(j_0+1)]$. On the kernel $X = 0$ of the n positive square term, the first diagonal and determinant of Q_3 on (Z, V) are

$$u_0^2(m_0 + 1/2) > 0, \qquad u_0^2/[4(j_0+1)^2] > 0. \tag{R2O22}$$

Choose $m_c < m_0$ sufficiently close and then a sufficiently large real n_c. A real block Schur complement makes the full matrix strict.

If $p_0 = 1 < j_0$, the coefficient m is unbounded and $n_0 = 1/[2(j_0^2-1)]$ is finite positive. Choose a real b with

$$\frac{j_0}{2(j_0+1)} < b < \min\left\{\frac{k_0\sqrt{D_u}}{2}, \frac{j_0}{2(j_0-1)}\right\}.$$

The interval is nonempty because $u_0 > 0$. On the kernel $(D_u/u)V - X - uZ = 0$ of the m positive square term, the first diagonal and determinant in (X, V) are

$$\frac{D_u}{2(j_0^2-1)} > 0, \qquad -\frac{D_u^2[2b(j_0-1)-j_0][2b(j_0+1)-j_0]}{4u_0^2(j_0^2-1)} > 0. \tag{R2O23}$$

Choose $n_c < n_0$ sufficiently close, then a sufficiently large real m_c. Matrix (R2O19) and the full Q_3 are strict.

If $p_0 = j_0 = 1$, both coefficients are unbounded and $k_0 = 1/2$. Choose $1/4 < b < \sqrt{D_u}/4$. On the common event kernel $X = 0$, $Z = D_u V/u_0^2$, the remaining form is

$$\frac{D_u(4b-1)}{2u_0^2}V^2 > 0. \tag{R2O24}$$

Sufficiently large real caps on both events make the full form strict. The finite-cap step in each case is the elementary real fact that a quadratic form positive definite on the kernel of a nonnegative event Gram becomes positive definite after a sufficiently large multiple of that event Gram is added; a block Schur complement proves it.

These four cases exhaust $0 < p_0 \le \min(1, j_0)$ with positive j_0, p_0. The separated bounded-Gram lemma now proves the entire selected parent (R2O17), with every state normal retained. Restore the original excesses and all unused positive square terms. The real caps are used only on this testing chart: no globally physical capped source is asserted, so this argument does not conflict with the finite-contact cap exclusion (CAP15).

Exact outer and projective join

Return to the original testing angles. The full-parent sign gauges (SJ3) permit $u, v \ge 0$ without changing source coefficients. Literal zeros and projective infinities are the exact signed-core squares (SJ10), plus the untouched event squares. For finite positive angles:

- if the original u is noninfinitesimal, use (R2O10);
- if the original u is infinitesimal and v is unbounded, use (R2O11)–(R2O12);

- if the original u is infinitesimal and v is a positive bounded unit, use (R2O11) when $k_0 = 1$, and the complete six-root argument (R2O13)–(R2O24) when $k_0 < 1$.

These are exactly all finite positive angle pairs outside the case in which both are infinitesimal. Every argument is a whole-parent state Gram, so no additional state chart or state-ratio restriction is present. Equation (R2O6) restores the original frame and positive pure-testing factor. The zero state and the real/imaginary split for complex states are immediate. No source-to-testing rate comparison was made.

C.14 Two small actual angles: the R3 and R4 sectors

C.14.1 All ratios at infinitesimal e

Statement and exact domain. For the present source partition it suffices to impose additionally a unbounded; the identities below remain valid on this smaller domain.

Use the original common-II stationary source, its six native weights n, m, C, A, D, α, all its strict source gaps, $d = 1$, $\lambda = \beta = 0$, and $n \geq 0$. Suppose B_s, a_sB_s, t_sB_s are positive infinitesimals, t_s, h are positive bounded units, and $e = 1 - r$ is positive infinitesimal. In addition impose either

$$\begin{aligned} &\text{R3}: \quad 1 - h \text{ is infinitesimal,} \\ &\text{R4}: \quad 1 - h \text{ is a positive unit and } 1 + a_sB_sr - (1-h)/t_s \text{ is infinitesimal.} \end{aligned} \tag{EN1}$$

There is no condition on the size of a_s, $a_s^2B_s$, a_se, or their ratios. The conclusion is finite separated tensor SOS for the entire original parent at every independent projective testing pair and every state, including signs, axes, and literal $n = 0$. The bounded-a parts of R3 and R4 are covered by Proposition C.27; the present calculation supplies their unbounded-a parts.

The source and testing fields are independent real closed extensions $K_c, K_t \supset \mathbb{R}$. A bounded coefficient means an element of $B_c \otimes B_t$; its residue is the homomorphism to $\mathbb{R}$. The strict bounded-Gram lemma is used only for matrices with positive definite real residue. Row entries may be arbitrarily large. Every inverse and radical below is pure source, pure testing, or a product of such factors.

One exact compact source for both ends

Use the exact compact variables of the exact actual-source dictionary RSR2–RSR6:

$$\begin{aligned} &c_s = 1 + a_sB_sr, \quad R_s = 1 + ha_sB_s, \quad j = a_sc_s/R_s, \quad B = B_sR_s/c_s, \quad T = t_sc_s/R_s, \\ &p = a_se/R_s, \quad H = (1-h)/(t_sc_s), \qquad x = jB = a_sB_s, \quad y = TB = t_sB_s, \quad q_z = jT. \end{aligned} \tag{EN2}$$

Thus $0 < p < \min(1, j)$, $0 < H < 1$, and

$$B, x, y \in \mathrm{I}_c, \quad T \in \mathrm{U}_c, \quad 1 - HT \in \mathrm{U}_c, \quad p/j \in \mathrm{I}_c. \tag{EN3}$$

The last assertion is exact source entry: $e = p(1+x)/(j+px) \sim p/j$. In R3, H is infinitesimal. In R4, $1 - H$ is infinitesimal and $0 < \overline{T} < 1$. In particular $H(1-H) \in \mathrm{I}_c$ in both rows. The native inverse is

$$r = \frac{j-p}{j+px}, \quad h = \frac{1-HT}{1+HTx}, \quad a_s = \frac{j+px}{1+HTx}, \quad B_s = \frac{B(1+HTx)}{1+pB}, \quad t_s = \frac{T(1+pB)}{1+HTx}. \tag{EN4}$$

The exact six weights are RSR6 at this substitution. These rational identities, including the contact first jets, hold at the displayed substitution.

Put $\sigma = 1 + (1-H)T$, $K = j + 1 - p$, $E = 1 + B(HT + p - 1)$, and use the explicit polynomials $\mathcal{P}, \mathcal{L}, \mathcal{N}, \mathcal{F}, \mathcal{K}$ in RSR5, RSR15, RSR18. Then

$$\frac{\mathcal{P}}{j+1} \sim 1, \qquad \frac{\mathcal{L}}{j+1} \sim \sigma, \qquad \frac{\mathcal{F}}{j+1} \sim \sigma. \tag{EN5}$$

Here $T, \sigma, (1-HT)^{-1}$ are bounded positive units. To prove these uniform statements, divide the printed finite polynomials by $j + 1$. The corrections in $\mathcal{P}$ contain $B + x$ or TB; those in $\mathcal{L}$ after $(j+1)\sigma$ contain $p/(j+1)$ or $B + x$. In $\mathcal{F}$ the only additional quotient is $p(1-p)/(j(j+1))$, infinitesimal by (EN3). Every other term contains B or $Bj/(j+1)$. Thus no restriction on j or jT is needed.

The exact physicality numerator has the grouping

$$\mathcal{N} = p\{1 + B(3j + 1 - p) + BT(j+1)\} - Bj(j+1)\{1 + (1-H)T\} + \mathcal{R}, \quad |\mathcal{R}| \leq C_0BT\,Bj(j+1). \tag{EN6}$$

The bound follows termwise from RSR5 using $p < \min(1, j)$ and bounded T. Since $n = \mathcal{N}/[2(j-p)\mathcal{P}] \geq 0$,

$$\begin{aligned} &\eta_* = \frac{Bj(j+1)}{p} = O(1), \qquad 0 \leq n = O\Big(\frac{p}{j(j+1)}\Big) \in \mathrm{I}_c, \\ &\eta_* \in \mathrm{I}_c \Longrightarrow n \sim \frac{p}{2(j-p)(j+1)} > 0. \end{aligned} \tag{EN7}$$

This includes $\mathcal{N} = 0$ literally.

Write $\mu = m(1-p)$, $h_0 = \mu + 1/2$. The full source formulas give

$$\begin{aligned} &\mu = O(p), \quad h_0 \asymp 1, \quad \alpha/[B^2(j+1)] \sim \sigma/2, \quad \alpha/B \in \mathrm{I}_c, \\ &A \sim \frac{T}{2(1-HT)}, \qquad CH \sim \frac{\sigma}{2T}, \qquad D(1-H) \sim \frac{1}{2T}. \end{aligned} \tag{EN8}$$

Consequently A and the finite one of C, D have positive real lower bounds. The other coefficient is unbounded. The exact normalized source limits are

	C	A	D
$R3$	$+\infty$	$T/2$	$1/(2T)$
$R4$	$1/(2T)$	$T/[2(1-T)]$	$+\infty$.

(EN9)

Only real residues of the bounded entries are meant in this table. The coefficient m may be infinitesimal, a unit, or unbounded.

The scalar allocation in the complete two-normal frame

Use the entire parent RFN4 and the exact full-state change RFN5–RFN9, with $w = (W + uY)/(1+u^2) = 1$ and arbitrary $z = (Z - uX)/(1+u^2) > 0$. The identities are rational identities of (EN4), so they retain both complete normals and every state term in the present source. The tests in this subsection satisfy $u, v > 0$ infinitesimal. For clarity, the inverse is

$$X = N_0 + pv - uz, \quad Z = z + uN_0 + puv, \quad W = 1 + u(M_0 + Hvz), \quad Y = u - M_0 - Hvz. \tag{EN10}$$

Retain the actual positive square term $mpD_u^2N_0^2$, and divide the remaining entire parent by the pure testing factor $D_u = 1 + u^2$. Its exact expansion is RFN7–RFN9. In particular the hard entries are

$$\begin{aligned} &H_N = h_0 D_u + \tfrac{1}{2}v^2 + \alpha u^2 + nD_v, \\ &H_M = D_u(CD_v + D + \tfrac{1}{2}) + AD_v + (\alpha + \tfrac{1}{2})v^2, \qquad H_{NM} = (\alpha + \tfrac{1}{2})uv. \end{aligned} \tag{EN11}$$

Keep the separated rows $U = u/\sqrt{x}$, $V = v/\sqrt{y}$, $Z_* = z/\sqrt{q_z}$, $Q_* = U - VZ_*$, $B_* = V - UZ_*$, $R_* = Z_* - UV$, and

$$\Psi = U^2 + V^2 + Z_*^2 + U^2V^2Z_*^2 - 4UVZ_* = (U-V)^2 + Z_*^2(1-UV)^2 + 2UV(1-Z_*)^2. \tag{EN12}$$

The last radical is separated, for example $\sqrt{UV} = (xy)^{-1/4}\sqrt{uv}$. The exact scalar identity is RFN12 with the same actual circuit masses. From (EN5), the closed formulas RSR16–RSR20 now give

$$\begin{aligned} &f_c \sim Tx^2\sigma/2, \quad q_c \sim \frac{Tx}{2(1-HT)}, \quad r_c \sim TjB^2(1-p)\sigma/2, \\ &g_c/f_c = Hy\,\mathcal{L}/\mathcal{F} \in \mathrm{I}_c, \quad k_c < 0, \quad |k_c|/(q_c y) = O(p+B). \end{aligned} \tag{EN13}$$

In particular all three displayed positive masses are positive. The sign of k_c follows from the same exact positive-surplus identity RSR19: its error bound uses $B(j+1) \in \mathrm{I}_c$, not T infinitesimal.

Before spending any current mass, perform the exact source-only completion

$$\begin{aligned} h_0N_0^2 + n(N_0 + jb)^2 &= (n + h_0)(N_0 + c_b b)^2 + M_e b^2, \\ b = v - uz/j, \quad c_b &= \frac{nj}{n+h_0}, \quad M_e = \frac{nh_0j^2}{n+h_0}. \end{aligned} \tag{EN14}$$

Keep $nv^2(N_0 + pb)^2$ and the positive extra hard terms RFN15 as whole squares. This reserves exactly the correlated n positive square terms of the complete parent. If $n = 0$, set $c_b = M_e = 0$ and omit that row. For $n > 0$, $c_b^2/M_e = n/[h_0(n+h_0)]$ is bounded.

The signed scalar allocation differs from R1 in one source split. The e_0 cost is unchanged: its negative part is at most g_c, and RFN18b provides it at cost $H(\mathcal{L}/\mathcal{F})(y+v^2) \in \mathrm{I}$. Also $d_c > 0$. Use $Z_*(V^2-1) = -VQ_* - R_*$ for k_c. The VQ_* cost is $O((p+B)v^2)$, infinitesimal. For its opposite-row part the exhaustive cases are:

$p \in \mathrm{I}_c$	$\lvert k_c\rvert/r_c = T\mathcal{K}/[(1-p)(1+Hy)\mathcal{L}] = O(p+B) \in \mathrm{I}_c$
$p \in \mathrm{U}_c,\ \eta_* \in \mathrm{U}_c$	j unbounded, $j^2B \in \mathrm{U}_c$, $\lvert k_c\rvert/f_c \in \mathrm{I}_c$, $\lvert k_c\rvert/(f_c x) = O(1)$
$p \in \mathrm{U}_c,\ \eta_* \in \mathrm{I}_c$	j unbounded, $M_e \in \mathrm{U}_c$, $\lvert k_c\rvert/f_c \in \mathrm{I}_c$, $\lvert k_c\rvert/(q_c y) = O(1)$, $\lvert k_c\rvert/(M_e yx) = O(1)$.

(EN15)

In the second case j is unbounded because p/j is infinitesimal; $j^2B \asymp p\eta_*$. In the third, $nj^2 \sim p/2$ and n is infinitesimal, so $M_e \sim p/2$. The closed formulas show $|k_c|/f_c = O((p+B)/j)$ in the last two cases and $|k_c|/(M_e yx) = O((p+B)(1+1/n)/j^2)$ in the third. These bounds are uniform through $p \to 1$.

Case two uses RFN18a: $R_*^2 \preceq_{\mathrm{sos}} (1+U^2)\Psi$. Case three uses RFN18c: $R_*^2 \preceq_{\mathrm{sos}} 2(\Psi + V^2Q_*^2 + U^2B_*^2)$. The non-infinitesimal source ratios in (EN15) acquire the testing factors u^2 or v^2. Thus all costs are infinitesimal. Literal $n = 0$ lies in the first or second case; it is not discarded. After this finite allocation, fixed positive fractions of

$$q_cQ_*^2, \quad r_cR_*^2, \quad f_c\Psi, \quad M_eb^2 \tag{EN16}$$

remain, with the last row absent exactly at $n = 0$.

Both complete normal arrays, with the new endpoint weights

The exact derivatives RFN21–RFN22, and their exact source cancellations $S_N = S_0 = S_1 = S_2 = 0$, hold unchanged. Let $q = vz - Tu$, $R = z - uv/B$, $R_u = v(u^2 - x)$. In particular

$$\begin{aligned}
L_N^{\mathrm{rem}} &= -j\alpha b + \tfrac{1}{2}pv^2b + K_RR_u + (K_c/j)uvq,\\
L_M &= C_qq + \mu uvb + v(\alpha - BTCHu^2)R,\\
K_c &= p/2 + j\{H(\alpha+1/2) - 1/2\} + n(p-j),\\
K_R &= -\mu - (\alpha+1/2)(1-p) + (T/j)K_c,\\
K_d &= CH - (D+1/2)(1-H) = -\mu/j - BTCH,\\
C_q &= K_d + AH + 1/2 + v^2\{CH(1+u^2) + AH + (\alpha+1/2)H\}.
\end{aligned} \tag{EN17}$$

Every displayed coefficient other than a retained row is bounded: p/j and n are infinitesimal, $\mu/j = O(p/(j+1))$ is bounded, and $A, CH, D(1-H), T$ are bounded by (EN8). In particular K_R, K_c/j, and C_q are bounded.

The complete bounds for these costs are as follows; they use the actual $T \in \mathrm{I}_c$ hypothesis of RFN. The exact radial Gram RFN19–RFN20 purchases R_u^2 at infinitesimal cost from (EN16). Its three cases are still $1-p$ a unit; $p \sim 1, \eta_*$ a unit; and $p \sim 1, \eta_*$ infinitesimal. Their costs are respectively

$$\begin{array}{r|l}
1-p \in \mathrm{U}_c & O(B),\ O(Bu^2),\ O(u^2/(1-p))\\
p \sim 1, \eta_* \in \mathrm{U}_c & O(B + u^2/j + u^4/(j^2B)),\ O(Bu^2)\\
p \sim 1, \eta_* \in \mathrm{I}_c & O(u^2/j),\ O(u^4/j),\ [8u^4 + 2(u^2-x)^2]/M_e.
\end{array} \tag{EN18}$$

They require only T bounded above and below, because T cancels between $y = TB$ and the masses (EN13). The small-current budgets NSR3–NSR5 likewise apply unchanged: both pv^2b and $\sqrt{T}\,\mu uvb$ are free. For η_* infinitesimal their coefficients are $O(v^4)$ and $O(Tu^2v^2)$, using $p^2/M_e = O(1)$. For η_* a unit the coefficients are $O(v^2(y+v^2))$ and $O(Tu^2(y+v^2))$, using $p^2y/f_c = O(1)$. Each is infinitesimal.

The row $q/\sqrt{T}$ is strict by (EN13). Its N coupling has coefficient $(K_c/j)uv\sqrt{T}$ infinitesimal. The identity $j\alpha b = -(\alpha/B)R_u - \alpha uR$ handles the remaining N term. For $1-p$ a unit the latter square costs $O(B^2(j+1)^2u^2/(1-p))$ against $r_cR_*^2$. For $p \sim 1$, RFN18a bounds the entire $j\alpha b$ square by $O(B^3(j+1)^2 + B^2(j+1)^2u^2/j)$ of the circuit. These are infinitesimal. A sufficiently large fixed real radial pivot pays the bounded K_R coupling, including the bounded shift $c_b/\sqrt{M_e}$ from (EN14).

Normalize the complete second normal by $M_0 = \sqrt{TH(1-H)}\,\widehat{M}$. The full hard diagonal (EN11) then has residue

$$\overline{TH(1-H)H_M} = \begin{cases} (1+\overline{T})/2, & R3,\\ 1/2, & R4. \end{cases} \tag{EN19}$$

Every term of the complete diagonal is bounded: the only divergent factors occur in CH and $D(1-H)$. The normalized q coupling is $T\sqrt{H(1-H)}C_q$, now infinitesimal because $H(1-H)$ is infinitesimal. This is the replacement for the T-infinitesimal argument in R1. The normalized current coupling is a bounded multiple of the already purchased $\sqrt{T}\,\mu uvb$.

For the entire remaining opposite row put $J = B(j+1) + u^2$ and $a_R = Tq_zv^2(\alpha - BTCHu^2)^2$. Since $\alpha - BTCHu^2 = O(BJ)$, its costs are

$$\begin{aligned}
a_R/f_c &= O((T/j)v^2J^2), & a_R/(q_cy) &= O(v^2J^2),\\
a_R/r_c &= O(Tv^2J^2/(1-p)), & a_R/(M_eyx) &= O(T(1+1/n)v^2J^2/j^2).
\end{aligned} \tag{EN20}$$

The direct r_c cost applies for $1-p$ a unit. For $p \sim 1, \eta_*$ a unit, RFN18a applies using j^2B a unit. For $p \sim 1, \eta_*$ infinitesimal, RFN18c applies using $(1+1/n)/j^2 = O(1)$. Thus the entire square, including every higher testing degree, is free. No state ratio occurs in a cost.

Finally the complete mixed hard term is $2(\alpha+1/2)uv\sqrt{TH(1-H)}(\widehat{N}/\sqrt{n+h_0} - c_b\widehat{b})\widehat{M}$. Its bounded coefficients have zero residue. The strict bounded-Gram argument therefore applies exactly as in the final paragraph of RFN:

choose a large fixed radial pivot, provide the finitely many free rows, and retain positive fractions of (EN16). The two hard pivots are 1 and the positive number (EN19). Restore all whole positive square terms in (EN14), RFN15, and $mpD_u^2N_0^2$, and the pure testing factor D_u. This proves the full local parent for all infinitesimal $u, v > 0$ and arbitrary $z > 0$, with both unrestricted state normals. The exact SJ state/sign reduction supplies every state and the literal axes in this local chart.

Every other testing angle, with weak m retained

This outer join uses only (EN9), positivity of the six native weights, and exact full-parent identities. Set $\widetilde{m} = m/(1+m)$, a bounded positive source coefficient below m, and retain the difference as the whole original m square. Keep the entire n and α events throughout. For the single unbounded coefficient C or D, retain its excess over a sufficiently large positive real cap. Other selected coefficients have the actual bounded positive residues (EN9).

Finite Bob with nonzero residue and bounded Alice. Let v have nonzero finite residue, u be bounded, and use the five full-state rows

$$P = uW, \quad Y, \quad X, \quad Z, \quad S = \sqrt{\widetilde{m}}\, W. \tag{EN21}$$

The selected m event is the entire square $[vS - \sqrt{\widetilde{m}}(X - uvY + uZ)]^2$. All other terms use P, Y, X, Z with bounded coefficients. In R3 the strong row is $P - Y$; on its kernel the selected remaining form on (Y, X, Z) is

$$\frac{T}{2} D_u D_v Y^2 + \frac{1+T}{2T} v^2 (uX - Z)^2 + K_0.$$

In R4 the strong row is $P + uvX - Y - vZ$; on its kernel it is

$$\frac{D_v v^2}{2T}(Z - uX)^2 + \frac{T}{2(1-T)} D_u D_v Y^2 + K_0, \quad K_0 = \frac{D_u(vY - Z)^2 + D_v(X + uZ)^2 - D_u D_v Z^2}{2}.$$

The leading principal minors in the order (Y, X, Z) are

	$R3$	$R4$
1	$\dfrac{D_u(TD_v + v^2)}{2}$	$\dfrac{D_u(T + v^2)}{2(1-T)}$
2	$\dfrac{D_u(TD_v + v^2)(Tu^2v^2 + Tv^2 + T + u^2v^2)}{4T}$	$\dfrac{D_u D_v(T + v^2)(T + u^2v^2)}{4T(1-T)}$
3	$\dfrac{v^2(1+T)D_u^2 D_v(Tu^2 + v^2)}{8T}$	$\dfrac{v^2 D_u^2 D_v^2(Tu^2 + v^2)}{8T(1-T)}$.

(EN22)

All are strictly positive at the real residues, even $u = 0$. The S direction has the positive coefficient v^2 in the selected m event. A sufficiently large real cap on the strong row makes the entire five-row residue positive definite. This is the real finite-cap Schur lemma: if a bounded real quadratic form is positive on the kernel of a linear map, adding a sufficiently large real multiple of the squared map makes it positive definite. The unbounded actual coefficient permits every such cap. This argument includes m infinitesimal.

Infinitesimal Bob and Alice a finite positive unit. Use $R = uW - Y$, $B_t = X + uZ$, $V_Z = vZ$, and Y; the testing inverse $1/u$ is bounded. The full selected parent is a bounded quadratic on these four rows because $v(W + uY) = (v/u)(R + D_u Y)$ and $J = R + uvB_t - D_u V_Z$. Its residue splits off $(\widetilde{m} + 1/2)B_t^2$ and the form

$$CR^2 + AD_u Y^2 + (D + 1/2)(R - D_u V_Z)^2 - D_u Y V_Z - \tfrac{1}{2} D_u V_Z^2. \tag{EN23}$$

In R3 restrict to the strong-row kernel $R = 0$. After removing a positive factor D_u, the matrix on (Y, V_Z) is $\left(\begin{smallmatrix} T/2 & -1/2 \\ -1/2 & (1+T)D_u/(2T) - 1/2 \end{smallmatrix}\right)$; its determinant is $(1+T)u^2/4 > 0$. In R4 the strong kernel is $R = D_u V_Z$. On (R, Y) its first diagonal is $1/(2T) - 1/(2D_u) > 0$, and its determinant is $u^2/[4(1-T)] > 0$. A large real cap again makes (EN23) strict.

Infinite Bob and bounded Alice. Put $w = 1/v$, divide the homogeneous parent by v^2, and use the five rows (EN21). Select any fixed positive real lower caps c_0, a_0, d_0 below C, A, D, and the actual $\widetilde{m}$ event. The full quadratic is bounded; its residue is

$$[S + \sqrt{\widetilde{m}}\, uY]^2 + c_0(P - Y)^2 + (a_0 + \tfrac{1}{2}) D_u Y^2 + d_0(uX - Z)^2 + \tfrac{1}{2} D_u X^2. \tag{EN24}$$

This is positive definite on the five rows, for every bounded residue of u. It includes $u = 0$ and m infinitesimal.

Infinite Alice. Let $s = 1/u$ and keep the entire m, n, α events. For finite v of nonzero residue, after division by u^2 the selected residue on (W, Y, X, Z) is

$$c_0 D_v W^2 + a_0 D_v Y^2 + (d_0 + \tfrac{1}{2})(W + vX)^2 + \tfrac{1}{2}(vY - Z)^2,$$

which is positive definite. For v infinitesimal use instead $(N, W, Y, V) = (sX + Z, W, Y, vX)$. The complete signed core after division by u^2 is

$$\frac{1}{2}\{W + (1+s^2)V - sY - svN\}^2 + \frac{1}{2}(1+s^2)v^2Y^2 - (1+s^2)Y(vN - sV) - \frac{1}{2}(1+s^2)(vN - sV)^2 + \frac{1}{2}D_v N^2.$$

Adding the selected C, A, D positive square terms gives strict residue $c_0W^2 + a_0Y^2 + (d_0 + 1/2)(W + V)^2 + N^2/2$. Every full coefficient is bounded; no s/v is used. If also Bob is infinite, divide by u^2v^2 and use $(W, X, Y, S_Z, T_Z) = (W, X, Y, sZ, wZ)$. The complete selected residue is

$$c_0 W^2 + a_0 Y^2 + d_0(X - S_Z)^2 + \tfrac{1}{2}X^2 + \tfrac{1}{2}(Y - T_Z)^2, \tag{EN25}$$

again positive definite. The full quadratic follows directly from $J/(uv) = wW + X - swY - S_Z$ and the exact signed core; its coefficients are bounded polynomials in s, w.

These charts exhaust all nonnegative projective angles: both infinitesimal are the complete local theorem, and every other pair is one of (EN21)–(EN25) or (EN23). All strict-matrix coefficients use only bounded pure source or pure testing factors. Restore every selected excess, the untouched m, n, α positive square terms, and the homogeneous testing squares. The exact SJ sign gauges, literal projective endpoints and real/imaginary state split finish the asserted all-tests theorem.

C.15 Two small actual angles: the R6 sector

C.15.1 Positive residue of a^2B

Scope. All notation and all physical source constraints are the original weak-C2 ones. In particular $d = 1$, $\lambda = \beta = 0$, $n \geq 0$, and the actual finite stationary contact, every strict gap, the total-scale condition, and the complement of earlier cases are retained. On R6, $t, h, l = 1-h, j, r$ are positive bounded units, $B, x = aB, y = tB, g$ are positive infinitesimals, and here a is unbounded. Define

$$s = a^{-1}, \quad w_s = ae, \quad b = a^2B, \quad e = 1 - r, \qquad B = s^2 b, \quad x = sb, \quad y = ts^2 b. \tag{RB1}$$

The existing exact source identity gives

$$2r\Delta_* n/s^2 = w_s Q_* - bK_*, \qquad Q_* \sim \Delta_* \sim 1, \quad K_* \sim h + t. \tag{RB2}$$

The gap $g = 1 - w_s + hsb$ gives $w_s \sim 1$. Hence b is bounded and

$$\rho = n/e^2 \sim \frac{1 - b_0(h_0 + t_0)}{2}, \qquad 0 \leq b_0(h_0 + t_0) \leq 1. \tag{RB3}$$

Here and below a subscript zero means the real residue; a displayed residue formula with bounded source letters has the same meaning. No testing variable occurs in these source estimates.

Theorem C.28. *On the entire original R6 source intersection, assume in addition that $b = a^2B$ is a positive bounded unit. Then the complete original parent is a finite sum of squares in the separated source–testing tensor ring at every independent projective testing pair and every real or complex state, with either angle sign and every literal projective endpoint. All positive s, g opening rates are independent. Literal $n = 0$ is included. The theorem makes no assertion about b infinitesimal.*

Source bounds needed by the proof. Put $R = 1 + hx$, $K = 1 + x$ and $T = 1 + x(1 - e)$. The original rational source formulas, after the pure-source substitutions (RB1), have only unit denominators after mg, n/s^2 and $\alpha/(s^3b^2)$ are formed. Their joint residues are

$$mg \sim \tfrac{1}{2}, \quad C \sim \frac{h + t}{2l}, \quad A \sim \frac{t}{2h}, \quad D \sim \frac{1}{2(t - l)}, \quad \frac{\alpha}{s^3b^2} \sim \frac{h + t}{2}. \tag{RB4}$$

Thus m is unbounded, C, A, D are positive units, and all four have positive real lower bounds. There is no s/g or g/s assumption. These are rational cancellations in the actual physical source; no formal negative-beta source is imported.

Two contact-zero positive square terms remove the two troublesome terms. Write the complete parent in the normalization $P = F/4$:

$$\begin{aligned} P &= nD_uD_vX^2 + mM^2 + CD_v(uW - Y)^2 + AD_uD_vY^2 \\ &\quad + (D + \tfrac{1}{2})J^2 + (\alpha + \tfrac{1}{2})D_u(vY - Z)^2 + \tfrac{1}{2}D_v(X + uZ)^2 - \tfrac{1}{2}D_uD_vZ^2, \\ M &= vW - X + uvY - uZ, \qquad J = uW + uvX - Y - vZ, \quad D_u = 1 + u^2, \quad D_v = 1 + v^2. \end{aligned} \tag{RB5}$$

In the finite state chart $Z = 1$, use the pure-testing coordinate $\omega = (W + uY)/D_u$ and the exact normals

$$\begin{gathered} p = \frac{w_s}{R} = 1 - \frac{g}{R}, \quad c = \frac{l}{tT}, \quad q_A = D_uv\omega, \quad q_B = v(1 - uX), \\ N = X + u - pq_A, \quad M_B = uW - Y - cq_B. \end{gathered} \tag{RB6}$$

Their full inverse is

$$\begin{gathered} X = N + pD_uv\omega - u, \quad q_B = v(D_u - uN - puD_uv\omega), \\ W = \omega + \frac{u(M_B + cq_B)}{D_u}, \quad Y = u\omega - \frac{M_B + cq_B}{D_u}. \end{gathered} \tag{RB7}$$

The only testing denominator is the positive pure-testing unit D_u. Set

$$\delta = \frac{R}{K\sqrt{bt}}, \quad E = \frac{tK}{R}, \quad k = \sqrt{x}, \quad K_c = \delta^2Ek^2 = \frac{sR}{K}, \quad \gamma = \frac{r}{K_c} = \frac{rK}{sR}. \tag{RB8}$$

All of δ, E, b are positive bounded units in this theorem. The actual contact in these coordinates is

$$(u, \omega, v; N, M_B) = (k, \delta k, \delta k^2E; 0, 0). \tag{RB9}$$

For direct verification, its original state coordinates with $Z = 1$ are

$$u_c = \sqrt{sb}, \quad v_c = s\sqrt{bt}, \quad W_c = \sqrt{s/t}, \quad X_c = -r\sqrt{sb}, \quad Y_c = hs\sqrt{b/t}, \quad \omega_c = \sqrt{s/t}\,R/K. \tag{RB9a}$$

They give $u_cW_c - Y_c = ls\sqrt{b/t} = c\,v_c(1 - u_cX_c)$ and $X_c + u_c = e\sqrt{sb} = pD_{u_c}v_c\omega_c$. Also $\gamma v_c\omega_c = r\sqrt{sb} = -X_c$. Thus both normals and the retained n row vanish exactly, before any residue calculation. The same contact substitution also gives $\omega_c = \delta k$, $v_c = \delta k^2E$, and $K_c = sR/K$. For any fixed positive real K_m, retain the two whole squares

$$(m - K_m)N^2, \qquad nD_uD_v(X + \gamma v\omega)^2. \tag{RB10}$$

The first coefficient is positive because m is unbounded. The second vanishes identically when $n = 0$. Both rows vanish at (RB9), with every first jet of their squares. No inverse of n is used.

The complete remaining polynomial is exactly

$$\begin{aligned} G &= K_mN^2 - 2\frac{mg}{R}q_AN + \frac{mg^2}{R^2}q_A^2 \\ &\quad - 2n\gamma D_uD_vX(v\omega) - n\gamma^2D_uD_v(v\omega)^2 \\ &\quad + CD_v(uW - Y)^2 + AD_uD_vY^2 + (D + \tfrac{1}{2})J^2 \\ &\quad + (\alpha + \tfrac{1}{2})D_u(vY - 1)^2 + \tfrac{1}{2}D_v(X + u)^2 - \tfrac{1}{2}D_uD_v. \end{aligned} \tag{RB11}$$

This is the entire parent minus precisely (RB10), including every first normal and every hard quadratic term. The exact cancellation $M = (g/R)q_A - N$ removes the whole g^{-1} pole before an estimate. Moreover $n\gamma = O(s)$ and $n\gamma^2$ is bounded. All coefficients in (RB11), after (RB7), are therefore bounded. The only nonconstant state-normal degree is one or two.

Raw support before the conductor. Define

$$A_0 = \omega v, \quad B_0 = \delta v, \quad C_0 = \delta Eu\omega, \quad D_0 = \delta^2k^3E, \quad X_0 = B_0 - C_0. \tag{RB12}$$

At $N = M_B = 0$ let $V_0 = 1 - puv\omega$. Then

$$q_B = D_uvV_0, \quad uW = u\omega + u^2cvV_0, \quad Y = u\omega - cvV_0, \quad J = -(1 - c)D_uvV_0, \quad X + u = pD_uA_0. \tag{RB13}$$

Thus every term of the scalar of (RB11) lies, with bounded coefficients, in $(A_0, B_0, C_0)^2 + (D_0^2)$. The only constant is from α, and α/D_0^2 is bounded with residue $b(h + t)/2$. For the new negative n contribution use exactly

$$\begin{aligned} -2n\gamma D_uD_vXA_0 - n\gamma^2D_uD_vA_0^2 &= 2n\gamma D_uD_vuA_0 \\ &\quad - (2n\gamma pD_u + n\gamma^2)D_uD_vA_0^2 - 2n\gamma D_uD_vA_0N, \end{aligned} \tag{RB14}$$

where $uA_0 = B_0C_0/(\delta^2E)$. Both complete first normals lie in (A_0, B_0, C_0) with bounded coefficients. Every remaining hard entry is bounded. These are full rational support statements, not a scalar contact-value test. Clearing D_u^2 makes them polynomial; its inverse is a pure-testing square factor when restored.

The contact Gram and its exact leading calculation. Let

$$Y_0 = A_0 - kC_0, \quad Z_0 = uB_0 - kC_0, \quad H_0 = kC_0 - D_0.$$

Use the separated nonnegative monomials $a_i^2 = (A_0, uB_0, kC_0, D_0)$ and the two pairings $Q_1 = a_1a_2 - a_3a_4$, $Q_2 = a_1a_3 - a_2a_4$. Their exact circuit is

$$\begin{aligned} T_4 &= A_0^2 + (uB_0)^2 + (kC_0)^2 + D_0^2 - 4k^2B_0C_0 \\ &= \tfrac{1}{2}\{(Y_0 - Z_0)^2 + H_0^2 + Y_0^2 + (Z_0 + H_0)^2\} + Q_1^2 + Q_2^2. \end{aligned} \tag{RB15}$$

Its matrix on (Y_0, Z_0, H_0) is $\begin{pmatrix} 1 & -1/2 & 0 \\ -1/2 & 1 & 1/2 \\ 0 & 1/2 & 1 \end{pmatrix}$, which is positive definite with determinant $1/2$. All radicals separate: a_1 is pure testing, a_2, a_3 are products of pure-source and pure-testing radicals, and a_4 is pure source.

Here is enough explicit algebra to reproduce the leading collection. At $g = 0$, the canceled source expansions through the needed orders are

$$\begin{aligned} n/s^2 &= \tfrac{1}{2}\{1 - b(h+t)\} + O(s), & mg &= \tfrac{1}{2} + \tfrac{1}{2}(bh-1)s + O(s^2), \\ C &= \frac{h+t}{2l} - \frac{t+1}{2l}s + O(s^2), & A &= \frac{t}{2h} - \frac{t}{2h}s + O(s^2), \\ D &= \frac{1}{2(t-l)} - \frac{bt(h+t) + t - l}{2(t-l)^2}s + O(s^2), & \alpha &= \tfrac{1}{2}b^2(h+t)s^3 + O(s^4). \end{aligned} \tag{RB16}$$

Here $O(s^j)$ means exact divisibility by s^j in a rational ring with unit denominator, not an asymptotic substitution in the proof identity. Give k, u, ω weight one and v weight two. Put $q = b(h+t)$. Using $s = k^2/b$ and $\delta = (1 - lk^2 + O(k^4))/\sqrt{bt}$, $\delta E = \sqrt{t/b}$, the scalar $G_0 = G|_{N=M_B=0}$ satisfies

$$G_0 = \frac{b}{2h}(1+u^2)X_0^2 + \frac{q}{2}T_4 - \frac{k^2[1 + bh(h+t)]}{2h}X_0^2 + O_{\mathrm{wt}}(8) \qquad (\mathrm{mod}\ g). \tag{RB17}$$

The complete first-normal coefficients obey

$$L_N = O_{\mathrm{wt}}(5), \qquad L_{M_B} = \frac{X_0}{h\delta} + O_{\mathrm{wt}}(4) \qquad (\mathrm{mod}\ g), \tag{RB18}$$

and the hard block has residue

$$\mathrm{diag}\left(K_m + \tfrac{1}{2},\ A_* + C_* + D_* + \tfrac{1}{2}\right), \quad A_* = \frac{t}{2h},\ C_* = \frac{h+t}{2l},\ D_* = \frac{1}{2(t-l)}. \tag{RB19}$$

Apply the inherited monic contact–conductor identities CC1–CC6 to the exact difference in (RB17), and to the exact first-normal differences in (RB18). Their hypotheses are exactly the raw support in (RB13)–(RB14), together with zero contact value and full first jet. Both follow from the original stationary parent and (RB10); the model X_0^2, T_4 also has these zeros. No contact difference is inverted. The conductor maps preserve this grading: X_0 has weight two, the five circuit roots have weight three. The scalar difference of weight at least eight and normal differences of the displayed weights therefore have strictly positive-weight Gram coefficients. Every such coefficient is infinitesimal. Terms divisible by g also lift with bounded coefficients and remain infinitesimal: the contact (RB9), δ and E are independent of g, and the canceled coefficient ring has unit denominators at $g = 0$. Clearing and restoring its finite unit denominators changes none of these conclusions.

Consequently the residue on the eight roots $(X_0, Y_0, Z_0, H_0, Q_1, Q_2, N, \delta M_B)$ is the strict circuit with coefficient $q/2 > 0$, the N pivot $K_m + 1/2 > 0$, and the coupled block

$$\begin{pmatrix} b/(2h) & bt/(2h) \\ bt/(2h) & bt(A_* + C_* + D_* + 1/2) \end{pmatrix}. \tag{RB20}$$

Its determinant is

$$\frac{b^2t}{2h}(C_* + D_* + 1/2) > 0. \tag{RB21}$$

All pivots are positive bounded units. The separated strict-residue Gram lemma therefore gives finite tensor SOS for the exact remainder. Restore (RB10) and the pure-source/pure-testing clearing factors. This proves the whole local contact parent, for every independent rate and every normal direction.

All remaining finite state and angle charts. The following details make the join an all-tests result. For bounded u with nonzero residue and infinitesimal v, retain $nD_uD_vX^2$ and $\alpha D_u(vY-Z)^2$, and the centered m excess $(m-K_m)N^2$ from (RB6), now homogeneous in Z. Put $V_Z = vZ$ and $R_t = uW - Y$. After $X = N - uZ + pv(W+uY)$, the whole remainder is a bounded quadratic on (N, R_t, Y, V_Z). Its residue splits off $K_m + 1/2$ on N, with the other form

$$C_*R_t^2 + A_*(1+u^2)Y^2 + (D_*+1/2)\{R_t - (1+u^2)V_Z\}^2 - (1+u^2)YV_Z - \tfrac{1}{2}(1+u^2)V_Z^2. \tag{RB22}$$

Its first two leading principal minors are positive, and its determinant is exactly

$$(1+u^2)^2A_*C_*(D_*+1/2)u^2 > 0. \tag{RB23}$$

This direct four-root proof covers every state, including $Z=0$, with no negative-beta or small-ρ argument.

It remains to consider both angles infinitesimal. The exact testing-core sign reduction from SD5 handles nonnegative core and literal axes by original-event squares. On its negative-core branch one may use the standard exact gauges $u, v > 0$. Normalizing the original state by a maximal coordinate, the case Z infinitesimal has W a unit and is covered by BE4, with positive real caps of m, A supplied by (RB4). Retain the actual nonnegative n, α positive square terms. That exact selected full-parent frame is (X, Y, uW, vW) and has strict residue; no coefficient bound involving b enters its proof.

If Z is a unit normalize $Z = 1$. The same core identity gives $\omega > 0$, and ω is bounded. If ω is infinitesimal, use (RB17)–(RB21). If ω is a positive unit, use the exact relations $B_0 = (\delta/\omega)A_0$ and $C_0 = B_0 - X_0$ in the raw support (RB13)–(RB14). The inverse is pure testing. Its complete residue on $(A_0, X_0, D_0, N, \delta M_B)$ consists of the positive $q/2$ pivots on A_0, D_0, the $K_m + 1/2$ normal pivot, and precisely (RB20). Direct substitution in (RB11) gives these residues; all other coefficients contain k, g, u, v after the cancellations already recorded. Thus this whole state chart is strict as well.

Every other projective test. Use the exact centered-event identity (R5.5) for the finite remote pairs $v_0 \neq 0$. All source residues are (RB4) and $n_0 = \alpha_0 = 0$. On the kernel of the limiting centered m row, the determinant is

$$\frac{(1+U)^2(1+V)^2(h+t)}{8hl(t-l)}\{V^2 + t(h+t)U^2V + hUV^2\} > 0. \tag{RB24}$$

The restriction to $Z=0$ is positive by the original A, C positive square terms and the nonnegative signed-core restriction, exactly as in RT10. A sufficiently large positive real cap K_m makes the full bounded residue strict. Its choice depends on the real testing residues only; unbounded m permits every such real cap.

For unbounded u and bounded unit v, R6.5 at $n_0 = 0$ is strict because C_*W^2, A_*Y^2 and $(D_*+1/2)(W+vX)^2$ have common kernel zero. For unbounded u and infinitesimal v, the exact BE $(\infty, 0)$ frame $(pX + Z, W, Y, vX)$ retains every event except positive caps of C, A, and has strict residue. For unbounded v and bounded u, R6.6 at $n_0 = 0$ is strict; the centered correction vanishes on its m-row kernel. For both angles unbounded, R6.7 at $n_0 = 0$ has block determinants $D_*/2$ and $A_*(K_m + 1/2)$, both positive. These are exact homogeneous full-parent identities. Their literal reciprocal zeros include the actual projective endpoints; undoing a nonzero homogeneous factor uses a pure-testing square only.

The charts exhaust the stated b-unit source domain. Restore all retained squares and normalizations and apply the real identities to the real and imaginary parts of a complex state.

Imported exact identities. The BE4 four-root state identity and the unnumbered $(\infty, 0)$ identity in Section B.4.13 supply the two selected full-parent frames used above. The latter is the paragraph beginning "At $(\infty, 0)$ retain all events except c, A". The remote kernel determinant and its $Z = 0$ restriction are printed in (RB24) and the paragraph following it. The centered-event identity is (R5.5) in Section C.17.5. That subsection derives the full-state kernel determinant and real-cap argument under the present R6 hypotheses. The homogeneous kernel and root matrices R6.5–R6.7, and the separate $(\infty, 0)$ identity (C.70), are proved in Section C.17.6; they are used at $n_0 = 0$ with the source residues (RB4). Finally SD5 and its following sign Schur identity in Section B.2.1 give the pure-testing inverse and the exact sign reduction. Only these algebraic identities are used with the present source estimates; no C1 physical-source or negative-beta perturbation theorem is imported.

C.15.2 Infinitesimal a^2B

Source scope and relation to the preceding theorem. Retain the complete original weak-C2 source: $d = 1$, $\lambda = \beta = 0$, $n \geq 0$, the actual stationary contact, every strict gap, total scale, and the complement of prior cases. On R6, $t, h, l = 1 - h, r$ and the original j gap are positive bounded units, while $g, B, x = aB, y = tB$ are positive infinitesimals. This section treats a unbounded and $b = a^2B$ positive infinitesimal. The notation $s = 1/a$, $w_s = ae$, $B = s^2b$, $x = sb$, $e = s(1 + hsb - g)$ is unchanged from (RB1). The exact physical identity (RB2), equivalently RT2, remains in force. It bounds b on all unbounded-a R6 and gives $\rho_0 = 1/2$ here, where $\rho = n/e^2$. In particular this subdomain has $n > 0$; the literal $n = 0$ source face is covered by the preceding b-unit theorem.

Theorem C.29. *On this entire original R6 source intersection with $b = a^2B$ infinitesimal, the complete original parent is a finite separated source–testing tensor sum of squares at every independent projective testing pair and every real or complex state. Both angle signs, all literal projective endpoints, and all relative opening rates of s, b, g are included. No mixed inverse or mixed square root is used.*

Together with the b-unit theorem and the inherited bounded-a R6 theorem, this proves the entire R6 row. It does not by itself admit another weak-C2 row or the global finite-lift theorem.

A regular source chart. Keep $R = 1 + hx$, $K = 1 + x$, $T = 1 + xr$, $k = \sqrt{x}$ and the contact parameters $\delta = R/(K\sqrt{bt})$, $E = tK/R$ from (RB8). Define

$$\varepsilon = \delta^{-1}, \qquad \jmath = \delta k, \qquad k = \varepsilon\jmath. \tag{RI1}$$

Both ε and $\jmath$ are positive infinitesimals. In this section $\jmath$ denotes the contact scale, not the original source gap called j. The exact source substitution is

$$\begin{gathered} x = \varepsilon^2\jmath^2, \quad K = 1 + x, \quad R = 1 + hx, \quad E = tK/R, \\ b = \frac{\varepsilon^2 R^2}{K^2 t}, \qquad s = \frac{t\jmath^2 K^2}{R^2}. \end{gathered} \tag{RI2}$$

All denominators in (RI2) are positive source units. There is no comparison of a testing variable with either source infinitesimal.

Use precisely the complete contact-centered remainder G in (RB11), with the full coordinate inverse (RB7). Thus the two retained positive square terms on the contact chart are $(m - K_m)N^2$ and $nD_uD_v(X + \gamma v\omega)^2$, where

$$p = (1 + hx - g)/R, \quad c = l/(tT), \quad \gamma = rK/(sR), \quad D_u = 1 + u^2, \quad D_v = 1 + v^2.$$

Its exact contact is now

$$(u, \omega, v; N, M_B) = (\varepsilon\jmath, \jmath, \varepsilon\jmath^2 E; 0, 0). \tag{RI3}$$

Substitution of (RI3) in the complete contact formulas (RB7)–(RB9a) makes both normals and the retained n row vanish.

Two exact source cancellations. The quantities mg, $n\gamma$, $n\gamma^2$ and $\alpha/(s^3b^2)$ have canceled denominators that are units at $s = b = g = 0$. Introduce

$$\mu_1 = mg/R, \quad \mu_2 = (mg)g/R^2, \quad F_0 = p^2/2 - n\gamma^2 - 2n\gamma p + \mu_2, \qquad A_N = p - 2\mu_1 - 2n\gamma. \tag{RI4}$$

Both F_0 and A_N are divisible by b in this rational unit ring. Here is a short exact verification, independent of a series expansion. At $b = 0$, put $q = 1 - g$. Direct cancellation in the original source gives

$$p = q, \quad mg = \frac{q}{2(1 + s)}, \quad n\gamma = \frac{sq}{2(1 + s)}, \quad n\gamma^2 = \frac{q(1 - sq)}{2(1 + s)}. \tag{RI5}$$

Substitution makes both expressions in (RI4) identically zero. The denominators are units before this specialization; hence the polynomial numerators are divisible by b, without introducing a pole. Their needed residues are

$$\frac{F_0}{b} \sim \frac{h + t}{2}, \qquad \left.\frac{A_N}{b}\right|_{s=g=0} = 0, \qquad \left.\frac{A_N}{bs}\right|_{g=0} \sim h + t. \tag{RI6}$$

The last quotient also has a unit canceled denominator at $s = b = 0$. Every statement in (RI5)–(RI6) is separately checked from the original source formulas. In particular the source estimate is uniform in g/s, s/b and all other opening ratios.

Complete bounded raw support, including both first normals. Use the separated raw monomials

$$\mathsf{A} = \varepsilon\omega v, \quad \mathsf{B} = v, \quad \mathsf{C} = Eu\omega, \quad \mathsf{D} = \varepsilon^2\jmath^3 E, \qquad \mathsf{x} = \mathsf{B} - \mathsf{C}. \tag{RI7}$$

Set $H = \alpha + 1/2$, $L = AD_v + Hv^2$, and $V_0 = 1 - puv\omega$. At zero normals, (RB7) gives

$$q_B = D_u v V_0, \quad uW - Y = cD_u v V_0, \quad Y = u\omega - cvV_0, \quad J = -(1 - c)D_u v V_0,$$

and the entire scalar has the following exact allocation:

$$G_0 = f_A\mathsf{A}^2 + f_{BB}\mathsf{B}^2 + f_{BC}\mathsf{BC} + f_{CC}\mathsf{C}^2 + \varrho\mathsf{D}^2, \tag{RI8}$$

where

$$\begin{aligned} f_A &= D_u^2 D_v F_0/\varepsilon^2,\\ f_{BB} &= [Cc^2 D_u^2 D_v + (D+1/2)(1-c)^2 D_u^2 + D_u L c^2]V_0^2\\ &\quad + 2HD_u c V_0 - D_u/2 - D_u^2\mu_2 v^2\omega^2,\\ f_{BC} &= [-2D_u L c V_0 - 2HD_u + 2n\gamma D_u D_v]/E,\\ f_{CC} &= [D_u L + D_u D_v n\gamma^2 v^2]/E^2,\\ \varrho &= \alpha D_u/\mathsf{D}^2. \end{aligned} \tag{RI9}$$

These coefficients are bounded. In particular

$$\frac{F_0}{\varepsilon^2} = \frac{R^2}{K^2 t}\frac{F_0}{b}, \qquad \frac{\alpha}{\mathsf{D}^2} = \frac{1}{t}\frac{\alpha}{s^3 b^2}. \tag{RI10}$$

The term that would give u^2/ε^2 in the first raw coefficient has instead been placed in f_{CC}, using the exact identity $u^2(v\omega)^2 = (v\mathsf{C}/E)^2$. The final term of f_{BB} likewise allocates the whole $-D_u^2\mu_2 v^4\omega^2$ contribution without dividing by ε. These are identities for the full scalar.

Write the exact normal expansion as $G = G_0 + L_N N + L_M M_B + G_{\mathrm{hard}}$, with G_{hard} quadratic. The two complete linear coefficients are

$$L_N = R_A\mathsf{A} + R_B\mathsf{B} + R_C\mathsf{C}, \qquad L_M = S_B\mathsf{B} + S_C\mathsf{C}, \tag{RI11}$$

where

$$\begin{aligned} R_A &= D_u A_N/\varepsilon, \qquad R_C = 2cuvL/E, \qquad S_C = -2L/E,\\ R_B &= \Delta_{v,0}(L_N - R_A\mathsf{A} - R_C\mathsf{C}), \qquad S_B = \Delta_{v,0}(L_M - S_C\mathsf{C}). \end{aligned} \tag{RI12}$$

Here L_N, L_M are the complete coefficient extractions from the bounded full parent in the stated state coordinates. The displayed source cancellations make both quotient numerators vanish identically at $v = 0$. Thus these are finite monic polynomial divisions, with the pure testing denominator D_u^2 cleared first. They give all normal coefficients, including the higher testing terms. Their coefficients are bounded; division shifts testing degree by one and preserves the weighted coefficient estimates used below.

Here $A_N/\varepsilon = \varepsilon R^2(A_N/b)/(K^2 t)$ is bounded. Thus no first-normal pole has been discarded. Every hard coefficient is bounded as well: the only testing inverse in the full change of state is D_u^{-1}, and all original source coefficients in (RB11) are bounded after the displayed cancellation. At $u = v = 0$ the complete hard matrix is exactly

$$\mathrm{diag}(K_m + 1/2, A + C + D + 1/2). \tag{RI13}$$

A bounded conductor for these complete coefficients. Define the contact differences

$$\mathsf{y} = \mathsf{A} - k\mathsf{C}, \quad \mathsf{z} = u\mathsf{B} - k\mathsf{C}, \quad \mathsf{h} = k\mathsf{C} - \mathsf{D}.$$

The scaled conductor identities are

$$\begin{aligned} \mathsf{C}(u-k) &= \mathsf{z} - u\mathsf{x},\\ \varepsilon\mathsf{C}(\omega - \jmath) &= \mathsf{y} - \varepsilon\omega\mathsf{x},\\ \mathsf{C}(v - \varepsilon\jmath^2 E) &= E(\omega\mathsf{z} + \jmath\mathsf{y} - k\omega\mathsf{x}),\\ \mathsf{C}v(\omega - \jmath) &= E\omega(\omega-\jmath)\mathsf{z} + E\jmath\omega\mathsf{y} - Ek\omega^2\mathsf{x}. \end{aligned} \tag{RI14}$$

These are SE5–SE6 with $w = \omega$, also checked directly here. For a monic contact division

$$S = (u-k)T_u + (\omega - \jmath)T_\omega + (v - \varepsilon\jmath^2 E)T_v, \qquad T_\omega = \varepsilon U + vV,$$

the exact bounded lift is

$$\begin{aligned} \mathsf{C}S &= \mathsf{x}[-uT_u - \varepsilon\omega U - Ek\omega^2 V - Ek\omega T_v]\\ &\quad + \mathsf{y}[U + E\jmath\omega V + E\jmath T_v]\\ &\quad + \mathsf{z}[T_u + E\omega(\omega-\jmath)V + E\omega T_v]. \end{aligned} \tag{RI15}$$

No inverse of a small source parameter or a contact difference occurs.

We verify the hypothesis of (RI15) for the full coefficients, rather than import the different small-e source assumptions of SE8. Clear the finite source unit denominators and D_u^2. They can be chosen independent of ω and even in u, v. In (RI9), all ω dependence is through $uv\omega$ and $v^2\omega^2$; the other testing dependence is even in u, v. The same property holds after subtracting any model of the form used below.

For the scalar square-lift algorithm CC4, substitute $\mathsf{A} = \mathsf{y} + k\mathsf{C}$, $\mathsf{B} = \mathsf{x} + \mathsf{C}$, $\mathsf{D} = k\mathsf{C} - \mathsf{h}$ into (RI8). The coefficient requiring its first conductor is

$$S = k^2 f_A + f_{BB} + f_{BC} + f_{CC} + k^2\varrho.$$

Its zero contact value follows by domain cancellation, as in CC4. Divide monically in the order u, then ω, then v. After $u = k = \varepsilon\jmath$, the ω divided difference belongs to $(\varepsilon v, v^2)$. Accordingly choose $T_\omega = \varepsilon U + vV$ with $U, V \in (v)$. Modulo ε, the remaining v polynomial is even, so $T_v \in (\varepsilon, v)$; also $T_u|_{\varepsilon=u=v=0} = 0$.

After this first conductor the coefficient of h remains $-2k\varrho$. The sole gradient relation is $(-2k, 1, 1, -1)$ on $(\mathsf{x}, \mathsf{y}, \mathsf{z}, \mathsf{h})$; the determinant of the first three contact gradients is $-\varepsilon^3\jmath^4E^2 \neq 0$. The adjugate cancellation of CC5 therefore removes the contact values without putting an inverse of this determinant into any coefficient. Subtract the contact multiple $2k\varrho_c(-2k, 1, 1, -1)$. Each of the four resulting coefficient remainders J_i vanishes at contact. Its restriction at $\varepsilon = u = v = 0$ is independent of ω: for the x coefficient this follows from $2f_{BB} + f_{BC}$; the other three restrictions are zero by the U, V, T_u, T_v properties just established and the factor k. Consequently the next monic ω divided difference of every J_i belongs to (ε, v). Formula (RI15) applies again.

The last scalar repair uses the exact circuit relation

$$k\mathsf{C}(\mathsf{y} + \mathsf{z} - \mathsf{h} - 2k\mathsf{x}) = Q_1^2 - \mathsf{yz},$$

where Q_1 is the first of the separated radical pairings below. This completes the same restricted scalar square lift as CC6. For (RI11), one conductor suffices: after replacing A, B as above, the coefficient multiplying C, restricted at $\varepsilon = u = v = 0$, is independent of ω directly from (RI12). Its divided difference therefore lies in (ε, v) and (RI15) applies. The full scalar and both full normals have thus been lifted over the bounded coefficient ring. The restricted raw support $\mathsf{A}^2, \mathsf{B}^2, \mathsf{BC}, \mathsf{C}^2, \mathsf{D}^2$ is essential; an unrestricted ideal claim is neither needed nor asserted.

The original stationary parent and the two retained contact-zero positive square terms give the exact scalar value and full first-jet identities used here. The contact (RI3) is independent of g. Hence taking the exact coefficient difference divisible by g preserves these identities and every raw-support condition.

Strict residues of the complete contact lift. Suppose now that u, v, ω are all positive infinitesimals, as in the remaining contact state chart. Give

$$(\varepsilon, \jmath, u, \omega, v) \text{ weights } (1, 1, 2, 1, 3).$$

Then x has weight three and the five circuit roots have weight five. Every contact substitution and (RI14)–(RI15) preserves the weights. Put $\kappa = (h + t)/(2t)$ and

$$\Lambda = \frac{1 - s + s^2 + u^2 - [2l + h(h + t)]k^2}{2ht}, \qquad \mu = \frac{1}{h(1 + s)}.$$

Direct exact collection of (RI9) through weight ten gives

$$G_0 = \Lambda\mathsf{x}^2 + \kappa T_4 + O_{\mathrm{wt}}(12) \qquad (\mathrm{mod}\ g), \tag{RI16}$$

where

$$T_4 = \mathsf{A}^2 + (u\mathsf{B})^2 + (k\mathsf{C})^2 + \mathsf{D}^2 - 4k^2\mathsf{BC}.$$

The two complete normal collections are

$$L_N = O_{\mathrm{wt}}(8), \qquad L_M = \mu\mathsf{x} + O_{\mathrm{wt}}(7) \qquad (\mathrm{mod}\ g). \tag{RI17}$$

For transparency, the only source terms needed for (RI16)–(RI17), besides (RI6), are the following exact Taylor coefficients at $g = 0$:

$$\begin{aligned} A &= \frac{t}{2h}(1 - s + s^2) + O_{\deg(s,b)}(3), \\ C &= \frac{h + t}{2l} - \frac{t + 1}{2l}s + \frac{t + 1}{2l}s^2 + O_{\deg(s,b)}(3), \\ D &= \frac{1 - s + s^2}{2(t - l)} - \frac{sbt(h + t)}{2(t - l)^2} + O_{\deg(s,b)}(3), \\ n\gamma &= \frac{s}{2} - \frac{s^2}{2} - \frac{sb(h + t)}{2} + O_{\deg(s,b)}(3). \end{aligned} \tag{RI18}$$

Here the error denotes exact degree filtration in a rational ring with unit denominators. Also $s = t\jmath^2 + O_{\mathrm{wt}}(6)$, $c = (l/t)(1 - k^2) + O_{\mathrm{wt}}(6)$ and $E = t(1 + lk^2) + O_{\mathrm{wt}}(8)$. The weight eight bound for L_N follows term by term from (RI6), (RI12). At $g = 0$, after the displayed unit denominators are cleared, all scalar weighted monomials

have even weight, and all M_B linear monomials have odd weight. Hence the next possible degrees are twelve and seven, respectively.

The stronger coefficient-wise statement also holds for the displayed raw representation: $f_A - \kappa$ and $\varrho - \kappa$ have weight at least two, while

$$f_{BB} - \Lambda - \kappa u^2, \qquad f_{BC} + 2\Lambda + 4\kappa k^2, \qquad f_{CC} - \Lambda - \kappa k^2$$

each has weight at least six. Similarly $S_B - \mu, S_C + \mu$ have weight at least four separately, and (R_A, R_B, R_C) have weights at least $(3, 5, 5)$. Thus the chosen raw allocation of each exact difference already has the claimed filtration; no low-weight Gram coefficients are hidden by cancellations between raw products.

Apply the bounded conductor just proved to the exact differences in (RI16)–(RI17). The scalar correction Gram coefficients have weight at least two; the M_B normal corrections have weight at least two; the N normal corrections have weight at least three. They are all infinitesimal. The coefficients divisible by g lift boundedly by the same argument and are infinitesimal independently of every other opening. Restoring the finite unit denominators preserves this fact.

For strictness use the two pairings on the separated monomials

$$a_i^2 = (\mathsf{A}, u\mathsf{B}, k\mathsf{C}, \mathsf{D}), \qquad Q_1 = a_1a_2 - a_3a_4, \quad Q_2 = a_1a_3 - a_2a_4.$$

Their product relation gives the exact two-pairing identity (RB15), with the scaled roots. Its three-dimensional difference block has determinant $1/2$, and both pairing pivots are positive. The complete residue on $(\mathsf{x}, \mathsf{y}, \mathsf{z}, \mathsf{h}, Q_1, Q_2, N, M_B)$ therefore consists of this strict circuit with coefficient $\kappa > 0$, the N pivot $K_m + 1/2 > 0$, and

$$\begin{pmatrix} 1/(2ht) & 1/(2h) \\ 1/(2h) & A_* + C_* + D_* + 1/2 \end{pmatrix}, \quad A_* = t/(2h),\ C_* = (h+t)/(2l),\ D_* = 1/[2(t-l)]. \tag{RI19}$$

Its determinant is exactly

$$\frac{C_* + D_* + 1/2}{2ht} > 0. \tag{RI20}$$

All pivots are positive bounded units. The strict separated Gram lemma WA10 yields a finite tensor-SOS identity for the complete remainder. Restore the two actual nonnegative retained positive square terms and all separated clearing factors. This proves the entire local contact chart, including every normal direction and arbitrary relative rates.

The ω-unit state chart uses the original n positive square term. When u, v are infinitesimal, $Z = 1$, and ω is a positive unit, retain $nD_uD_vX^2$ itself, together with $(m - K_m)N^2$. The complete remainder is (RB11) with its two negative $n\gamma$ and $n\gamma^2$ contributions absent. This choice is legitimate directly from $n \geq 0$ and does not require a contact lift away from the contact state scale.

Use the raw roots

$$(A_o, \mathsf{x}, \mathsf{D}, N, M_B), \qquad A_o = \omega v.$$

The only new inverse is $1/\omega$, which is pure testing and bounded. The relations $\mathsf{B} = A_o/\omega$ and $\mathsf{C} = A_o/\omega - \mathsf{x}$ put the entire scalar and both first normals in this frame, by the same full formulas (RI9), (RI12) with the absent n terms deleted. All coefficients are bounded; the constant term is still $\alpha D_u = \varrho\mathsf{D}^2$. Explicitly $\mathsf{A} = \varepsilon A_o$, so the first scalar term becomes $D_u^2D_v(p^2/2 + \mu_2)A_o^2$ and the first N-normal term becomes $D_u(p - 2\mu_1)A_o$. The printed ε denominators therefore cancel before any coefficient estimate. Its exact residue is the A_o pivot $1/2$, the D pivot κ, the N pivot $K_m + 1/2$, and the coupled block (RI19). The first N normal has zero residue and the first M_B normal has residue x/h. Thus this entire state chart also has a strict finite separated Gram.

All other independent tests and states. Every source bound used by the noncontact joins of the preceding b-unit theorem is (RB4), which is valid here without a lower bound on b. For bounded unit u and infinitesimal v, retain the original n, α events and use the exact full-state form (RB22), whose determinant (RB23) is positive. For both angles infinitesimal, the SD5 exact sign reduction disposes of the nonnegative-core branch and literal zero axes by original-event squares. On the negative branch choose the exact gauges $u, v > 0$. If the normalized state has Z infinitesimal, BE4 gives the strict full-state frame (X, Y, uW, vW) using positive real caps of m, A. Otherwise normalize $Z = 1$; $\omega > 0$ is bounded, and its two possibilities are exactly the contact chart and the unit chart just proved. No source/testing ratio split is used.

For finite remote $v_0 \neq 0$, use the actual centered-event identity R5.5, the strict kernel determinant RT10 (printed as RB24), and a sufficiently large positive real cap below unbounded m. For unbounded u and bounded unit v, use R6.5 specialized to $n_0 = 0$ and the source residues (RB4). For unbounded u and infinitesimal v, use the exact BE $(\infty, 0)$ full frame $(p_uX + Z, W, Y, vX)$, $p_u = 1/u$, with positive caps of C, A. For unbounded v and bounded u, R6.6 at $n_0 = 0$ is strict. For both angles unbounded, R6.7 at $n_0 = 0$ has determinants $D_*/2$ and $A_*(K_m + 1/2)$

on its two nontrivial blocks. These uses import the displayed homogeneous identities with the source bounds (RB4), without importing a source theorem from another stratum. Literal reciprocal zeros are polynomial values of those charts. All signs and complex states follow by the same exact gauges and the real/imaginary split. This exhausts all independent tests.

The contact identities. Equations (RI8)–(RI15) establish the rational-support hypotheses of SE5–SE7 in Section B.4.18 for the present coefficients, with the bounds proved here. We then apply the monic square-lift and gradient repair CC4–CC6 in Section 6.2.

C.16 Two small actual angles: the R7 sector

C.16.1 Unbounded a, including the physical zero face

Quantified source and conclusion. Keep the entire original weak-C2 source: $d = 1$, $\lambda = \beta = 0$, the actual rational stationary coefficients, $n \geq 0$, every strict gap, the original finite stationary contact, total scale, and all prior exclusions. In R7, B, z, y are positive infinitesimals, $a = z/B$, $t = y/B$ is a positive bounded unit, and h, g are positive infinitesimals. The original exclusion of simultaneously infinitesimal r and the source gap j_{gap} is retained.

Theorem C.30. *On the whole original R7 source intersection with a unbounded, the complete original parent is a finite separated source–testing tensor sum of squares for every independent projective testing pair and every real or complex state. Both angle signs, all literal zero and infinite angles, literal $n = 0$, and every relative opening rate of the source infinitesimals are included. No inverse or radical of a mixed source–testing expression is used.*

Together with the bounded-a theorem this covers R7.

Actual source scale and the physical inequality. Use

$$\begin{gathered} s = 1/a, \quad b = a^2 B, \quad x = sb = z, \quad B = s^2 b, \\ w_s = ae = 1 + hx - g, \quad e = sw_s, \quad r = 1 - e, \quad l = 1 - h. \end{gathered} \tag{RU1}$$

Set $K = 1 + x$, $R = 1 + hx$, $T = 1 + xr$ and $j_{\mathrm{gap}} = T - l/t$. The exact original source identity is

$$\begin{gathered} 2r\Delta n = eQ - BK_{\mathrm{phys}}, \qquad 2r\Delta_* n/s^2 = w_s Q - bK_*, \qquad \Delta_* = s\Delta, \\ Q = 1 + B(1 + t + ah + 2ar + at) + B^2\{a(h + t) + a^2[h(r + 1) + t(1 + r - r^2)]\}, \\ K_{\mathrm{phys}} = (a + 1)(h + t)(1 + x), \qquad K_* = (1 + s)(h + t)(1 + x). \end{gathered} \tag{RU2}$$

Before bounding b, $Q \sim \Delta_* \sim 1$ since all their nonconstant terms contain s or x. Here $w_s \sim 1$ and $K_* \sim t$ is a positive unit. The original inequality $eQ \geq BK_{\mathrm{phys}}$ therefore gives b bounded and $n = O(s^2)$, including its literal equality case. In particular $r \sim 1$, so the original R7 exclusion is respected. The source gap gives $t_0 \geq 1$; j_{gap} is a unit exactly when $t_0 > 1$, and otherwise $t_0 = 1$.

The original rational formulas, after cancellation, give jointly

$$mg \sim \tfrac{1}{2}, \quad C \sim \tfrac{t}{2}, \quad A_h := Ah \sim \tfrac{t}{2}, \quad \beta_D := D(1 - c) = Dj_{\mathrm{gap}}/T \sim \frac{1}{2t}, \quad \frac{\alpha}{s^3 b^2} \sim \tfrac{t}{2}, \qquad c = \frac{l}{tT}. \tag{RU3}$$

Thus m, A are unbounded; C is a positive unit. If $t_0 > 1$, $D \sim 1/[2(t - 1)]$ is a positive unit. If $t_0 = 1$, D is unbounded. The canceled coefficients $mg, C, A_h, \beta_D, n\gamma, n\gamma^2$ below have unit denominators through both small-gap ends. No ratio h/g, j_{gap}/h, or comparison with a testing rate is used.

Four whole positive square terms and a bounded full-parent remainder. Use the complete parent $P = F/4$:

$$\begin{aligned} P &= nD_u D_v X^2 + mM^2 + CD_v(uW - Y)^2 + AD_u D_v Y^2 \\ &\quad + (D + 1/2)J^2 + (\alpha + 1/2)D_u(vY - Z)^2 + D_v(X + uZ)^2/2 - D_u D_v Z^2/2, \\ M &= vW - X + uvY - uZ, \qquad J = uW + uvX - Y - vZ. \end{aligned}$$

The exact coordinate inverse (RB6)–(RB7) is

$$\begin{gathered} D_u = 1 + u^2, \quad D_v = 1 + v^2, \quad Z = 1, \quad p = w_s/R, \quad q_A = D_u v\omega, \\ N = X + u - pq_A, \qquad q_B = v(1 - uX), \qquad M_B = uW - Y - cq_B, \\ X = N + pD_u v\omega - u, \quad q_B = v(D_u - uN - puD_u v\omega), \\ W = \omega + u(M_B + cq_B)/D_u, \qquad Y = u\omega - (M_B + cq_B)/D_u. \end{gathered} \tag{RU4}$$

The sole testing inverse is D_u^{-1}. Put

$$\gamma = \frac{rK}{sR}, \quad \eta = Y - huW, \quad q = 1 - c = j_{\mathrm{gap}}/T,$$
$$\mu_1 = mg/R, \quad \mu_2 = (mg)g/R^2, \quad n_1 = n\gamma, \quad n_2 = n\gamma^2.$$

Retain the four entire nonnegative squares

$$mN^2, \qquad nD_uD_v(X + \gamma v\omega)^2, \qquad AD_uD_v\eta^2, \qquad DM_B^2. \tag{RU5}$$

All four rows vanish at the actual contact; (RB9a) verifies the first two, and directly $Y_c = hu_cW_c$, $M_{B,c} = 0$. The n positive square term vanishes identically on the literal $n = 0$ face.

The full A, D cancellations are the exact identities

$$\begin{aligned} AD_uD_v(Y^2 - \eta^2) &= 2A_hD_uD_vuWY - A_hhD_uD_v(uW)^2, \\ D(J^2 - M_B^2) &= -2\beta_DM_Bq_B + \beta_Dqq_B^2, \qquad J = M_B - qq_B. \end{aligned} \tag{RU6}$$

Since $M = (g/R)q_A - N$, subtraction of precisely (RU5) leaves

$$\begin{aligned} G = \ & -2\mu_1q_AN + \mu_2q_A^2 - 2n_1D_uD_vX(v\omega) - n_2D_uD_v(v\omega)^2 \\ & + CD_v(uW - Y)^2 + 2A_hD_uD_vuWY - A_hhD_uD_v(uW)^2 \\ & - 2\beta_DM_Bq_B + \beta_Dqq_B^2 + J^2/2 \\ & + (\alpha + 1/2)D_u(vY - 1)^2 + D_v(X + u)^2/2 - D_uD_v/2. \end{aligned} \tag{RU7}$$

This is a whole-parent identity, including every scalar, normal and hard coefficient. Equations (RU3), (RU6) bound all of them before any contact restriction. In particular neither h^{-1} nor j_{gap}^{-1} survives.

One regular contact frame for the two b alternatives. Use the pure-source parameters

$$\varepsilon = \frac{K\sqrt{bt}}{R}, \quad \jmath = \sqrt{s/t}\,\frac{R}{K}, \quad k = \varepsilon\jmath = \sqrt{x}, \quad E = tK/R. \tag{RU8}$$

The symbol $\jmath$ is a contact scale, not j_{gap}. Here $\jmath$ is infinitesimal; ε is a unit or infinitesimal exactly as b is. The exact inverse source chart is

$$x = \varepsilon^2\jmath^2, \qquad b = \frac{\varepsilon^2R^2}{K^2t}, \qquad s = \frac{t\jmath^2K^2}{R^2}. \tag{RU9}$$

Only source units are inverted. The contact is $(u, \omega, v; N, M_B) = (k, \jmath, \varepsilon\jmath^2E; 0, 0)$.

Define

$$\begin{aligned} \mathsf{A} = \varepsilon\omega v, \quad \mathsf{B} = v, \quad \mathsf{C} = Eu\omega, \quad \mathsf{D} = \varepsilon^2\jmath^3E, \\ \mathsf{x} = \mathsf{B} - \mathsf{C}, \quad \mathsf{y} = \mathsf{A} - k\mathsf{C}, \quad \mathsf{z} = u\mathsf{B} - k\mathsf{C}, \quad \mathsf{h} = k\mathsf{C} - \mathsf{D}. \end{aligned} \tag{RU10}$$

The two potentially singular source combinations are

$$F_0 = p^2/2 - n_2 - 2n_1p + \mu_2, \qquad A_N = p - 2\mu_1 - 2n_1.$$

Their exact canceled quotients satisfy

$$F_0/b \sim t/2, \qquad A_N/(bs) \sim t. \tag{RU11}$$

Both denominators are units through $s = b = h = g = 0$. These assertions follow from the original rational source, not from the inequality $n \geq 0$. In particular

$$F_0/\varepsilon^2 = \frac{R^2}{K^2t}\frac{F_0}{b}, \qquad \alpha/\mathsf{D}^2 = \frac{1}{t}\frac{\alpha}{s^3b^2}, \qquad A_N/\varepsilon = \frac{\varepsilon R^2s}{K^2t}\frac{A_N}{bs} \tag{RU12}$$

are bounded without losing the zero-n face.

Exact full scalar and both first normals. Put $H = \alpha + 1/2$, $V_0 = 1 - puv\omega$. At $N = M_B = 0$, $uW = \mathsf{C}/E + u^2c\mathsf{B}V_0$ and $Y = \mathsf{C}/E - c\mathsf{B}V_0$. The complete scalar is

$$G_0 = f_A\mathsf{A}^2 + f_{BB}\mathsf{B}^2 + f_{BC}\mathsf{BC} + f_{CC}\mathsf{C}^2 + \varrho\mathsf{D}^2, \tag{RU13}$$

with the bounded coefficients

$$\begin{aligned} f_A &= D_u^2 D_v F_0/\varepsilon^2, \qquad \varrho = \alpha D_u/\mathsf{D}^2, \\ f_{BB} &= [Cc^2 D_u^2 D_v + (\beta_D q + q^2/2)D_u^2 + HD_u v^2 c^2 - A_h D_u D_v(2u^2 + hu^4)c^2]V_0^2 \\ &\quad + 2HD_u c V_0 - D_u/2 - D_u^2 \mu_2 v^2 \omega^2, \\ f_{BC} &= [2A_h D_u D_v(lu^2 - 1)cV_0 - 2HD_u v^2 c V_0 - 2HD_u + 2n_1 D_u D_v]/E, \\ f_{CC} &= [A_h(2-h)D_u D_v + HD_u v^2 + D_u D_v n_2 v^2]/E^2. \end{aligned} \tag{RU14}$$

For the full normal expansion $G = G_0 + L_N N + L_M M_B + G_{\mathrm{hard}}$, the exact coefficients are

$$L_N = R_A\mathsf{A} + R_B\mathsf{B} + R_C\mathsf{C}, \qquad L_M = S_B\mathsf{B} + S_C\mathsf{C}, \tag{RU15}$$

where

$$R_A = D_u A_N/\varepsilon, \qquad R_C = 2cuv\{A_h D_v(1-lu^2) + Hv^2\}/E, \qquad S_C = 2\{A_h D_v(lu^2-1) - Hv^2\}/E,$$

$$R_B = \Delta_{v,0}(L_N - R_A\mathsf{A} - R_C\mathsf{C}), \qquad S_B = \Delta_{v,0}(L_M - S_C\mathsf{C}). \tag{RU16}$$

Here L_N, L_M are the complete coefficient extractions from the bounded full parent in the stated state coordinates. The displayed source cancellations make both quotient numerators vanish identically at $v = 0$. Thus these are finite monic polynomial divisions, with the pure testing denominator D_u^2 cleared first. They give all normal coefficients, including the higher testing terms. Their coefficients are bounded; division shifts testing degree by one and preserves the weighted coefficient estimates used below.

At $u = v = 0$ the complete hard matrix on (N, M_B) is exactly $\mathrm{diag}(1/2, C + 1/2)$.

Bounded monic conductor, including the small-ε end. The identities (RI14)–(RI15) apply to (RU10) without a source-shape assumption. Their crucial inputs are checked here on the new complete coefficients. After finite source unit denominators and D_u^2 are cleared, every ω occurrence of (RU14) is through $uv\omega$ or $v^2\omega^2$, and all remaining testing powers are even in u, v. The scalar raw support is exactly the five terms in (RU13).

For completeness, first replace $\mathsf{A} = \mathsf{y} + k\mathsf{C}$, $\mathsf{B} = \mathsf{x} + \mathsf{C}$, and $\mathsf{D} = k\mathsf{C} - \mathsf{h}$. The coefficient requiring a first conductor is $S = k^2 f_A + f_{BB} + f_{BC} + f_{CC} + k^2\varrho$. Divide its contact-zero difference monically in order u, ω, v. At $u = k = \varepsilon\jmath$, the ω divided difference is in $(\varepsilon v, v^2)$; write it as $\varepsilon U + vV$ with $U, V \in (v)$. The last v divided difference is in (ε, v) because, modulo ε, the remaining v dependence is even. The first u divided difference restricts to zero at $\varepsilon = u = v = 0$.

These facts permit the exact bounded lift (RI15), which uses the identity for $\mathsf{C}v(\omega - \jmath)$ and never divides by ε. Subtract the gradient multiple $2k\varrho_c(-2k, 1, 1, -1)$. The first three contact gradients have determinant $-\varepsilon^3\jmath^4E^2 \neq 0$; the CC5 adjugate argument cancels a zero identity, and introduces no inverse of that determinant. Each second-stage coefficient at $\varepsilon = u = v = 0$ is independent of ω: the x one restricts to $2f_{BB} + f_{BC}$, and the other three to zero. Hence every second ω divided difference is in (ε, v) and has the same bounded lift. The final repair is

$$k\mathsf{C}(\mathsf{y} + \mathsf{z} - \mathsf{h} - 2k\mathsf{x}) = Q_1^2 - \mathsf{yz}. \tag{RU17}$$

The pairings Q_i are defined below. For each complete normal in (RU15), one conductor suffices: after replacing A, B, its C coefficient restricted at $\varepsilon = u = v = 0$ is independent of ω, directly from (RU16). This is precisely the normal criterion in the same proof. When b is a unit, ε^{-1} is bounded and the ordinary CC1–CC6 conductor is also available.

All contact value and first-jet hypotheses follow from the stationary original parent and the four exact contact-zero positive square terms (RU5). The algorithm is a bounded, source-linear map on the stated raw coefficient class and preserves the filtrations used next. In particular it preserves the source ideal (h, g); no separate contact-zero assertion for its individual coefficient summands is needed. The source substitution in each monic step uses the actual E, which is a bounded unit through $h = 0$.

The complete filtered leading model. Set

$$\Lambda = \frac{1}{t(1+s)}, \qquad \mu = \frac{1}{1+s}, \qquad T_4 = \mathsf{A}^2 + (u\mathsf{B})^2 + (k\mathsf{C})^2 + \mathsf{D}^2 - 4k^2\mathsf{BC}.$$

The model scalar is

$$\begin{aligned} G_{\mathrm{mod}} &= \Lambda\mathsf{x}^2 + T_4/2 - (1/t + 1/2)k^2\mathsf{x}^2 + (k^2 - u^2)\mathsf{Cx}/t \\ &= \left[\Lambda + \frac{u(u+k)}{t} - (1/t + 1/2)k^2\right]\mathsf{x}^2 + T_4/2 - \frac{u+k}{t}\mathsf{xz}. \end{aligned} \tag{RU18}$$

The second equality is the exact conductor identity $\mathsf{C}(u-k)=\mathsf{z}-u\mathsf{x}$. Use the first expression for the raw coefficient filtration, since it retains the even-u class. Its angular raw coefficients are

$$\begin{aligned} F_{BB}&=\Lambda+u^2/2-(1/t+1/2)k^2,\\ F_{BC}&=-2\Lambda-u^2/t+(3/t-1)k^2,\\ F_{CC}&=\Lambda+u^2/t-2k^2/t. \end{aligned} \tag{RU19}$$

At $h=g=0$, exact source collection gives the following two coefficientwise filtrations:

	b a unit	b infinitesimal
variable weights	$(k,u,\omega,v)=(1,1,1,2)$	$(\varepsilon,\jmath,u,\omega,v)=(1,1,2,1,3)$
$\mathrm{wt}(\mathsf{x})$	2	3
$\mathrm{wt}(\mathsf{y},\mathsf{z},\mathsf{h},Q_1,Q_2)$	3	5
$f_A-1/2,\ \varrho-1/2$	≥ 2	≥ 2
$f_{BB}-F_{BB},\ f_{BC}-F_{BC},\ f_{CC}-F_{CC}$	≥ 4	≥ 6
$S_B-\mu,\ S_C+\mu$	≥ 2	≥ 4
(R_A,R_B,R_C)	$\geq(2,3,3)$	$\geq(3,5,5)$.

(RU20)

These statements concern the chosen raw coefficients separately; polynomial cancellation between raw products is not used to infer them. Consequently

	b a unit	b infinitesimal
G_0-G_{mod}	$O_{\mathrm{wt}}(8)$	$O_{\mathrm{wt}}(12)$
$L_M-\mu\mathsf{x}$	$O_{\mathrm{wt}}(4)$	$O_{\mathrm{wt}}(7)$
L_N	$O_{\mathrm{wt}}(5)$	$O_{\mathrm{wt}}(8)$.

(RU21)

Here each equality is modulo the source ideal (h,g).

To reproduce the collection, the necessary b-infinitesimal source jets at $h=g=0$ are

$$\begin{aligned} C&=t/2-(t+1)s/2+(t+1)s^2/2+O_{\deg(s,b)}(3),\\ A_h&=t(1-s+s^2)/2+O_{\deg(s,b)}(3),\\ \beta_D&=[1-s+s^2-(t+1)sb]/(2t)+O_{\deg(s,b)}(3),\\ n_1&=(s-s^2-tsb)/2+O_{\deg(s,b)}(3). \end{aligned} \tag{RU22}$$

Also $c=(1-k^2)/t+O_{\mathrm{wt}}(6)$, $E=t(1+k^2)$ at $h=0$, and $s=t\jmath^2+O_{\mathrm{wt}}(6)$. For the unit-$b$ filtration use $s=k^2/b$ and the respective coefficients of s: $-(t+1)/2,-t/2,-[1+(t+1)b]/(2t),(1-bt)/2$. The source cancellations (RU11)–(RU12) supply the first and last rows of (RU20). Scalar weights are even in both charts. The M_B weights are even in the first and odd in the second, giving exactly the next weights in (RU21).

Apply the bounded conductor to the exact differences in (RU21). Their coefficientwise raw filtration makes every correction Gram coefficient have strictly positive weight. Terms in (h,g) lift boundedly and remain infinitesimal at arbitrary relative rates. The whole hard block differs infinitesimally from $\mathrm{diag}(1/2,(t+1)/2)$, since all its coefficients are bounded and u,v,h,g,s are infinitesimal.

On the negative-core contact chart take $u,v,\omega>0$. The separated nonnegative monomials

$$a_i^2=(\mathsf{A},u\mathsf{B},k\mathsf{C},\mathsf{D}),\qquad Q_1=a_1a_2-a_3a_4,\quad Q_2=a_1a_3-a_2a_4$$

have the exact product relation yielding

$$T_4=\tfrac{1}{2}\{(\mathsf{y}-\mathsf{z})^2+\mathsf{h}^2+\mathsf{y}^2+(\mathsf{z}+\mathsf{h})^2\}+Q_1^2+Q_2^2. \tag{RU23}$$

Its three-dimensional difference matrix has determinant $1/2>0$. Each radical separates into source and testing factors. The full eight-root residue on $(\mathsf{x},\mathsf{y},\mathsf{z},\mathsf{h},Q_1,Q_2,N,M_B)$ therefore consists of half this strict circuit, the N pivot $1/2$, and the complete coupled block

$$\begin{pmatrix}1/t & 1/2\\ 1/2 & (t+1)/2\end{pmatrix},\qquad \det=\frac{t+2}{4t}>0. \tag{RU24}$$

The additional cross term in the second line of (RU18) has infinitesimal coefficient. WA10 now gives a finite separated SOS for the exact remainder. Restore all four actual squares (RU5). This proves the whole local contact chart in both b alternatives.

The ω-unit state chart. If u, v are infinitesimal, $Z = 1$, and ω is a positive unit, replace the retained centered n positive square term by $nD_uD_vX^2$ itself. The complete remainder is (RU7) with the two n_1, n_2 terms removed. Use the frame $(A_o, \mathsf{x}, \mathsf{D}, N, M_B)$, $A_o = \omega v$. The exact identities $\mathsf{A} = \varepsilon A_o$, $\mathsf{B} = A_o/\omega$, $\mathsf{C} = A_o/\omega - \mathsf{x}$ use only a pure-testing unit inverse. In particular the first scalar coefficient becomes $D_u^2 D_v(p^2/2 + \mu_2)A_o^2$, and the first N coefficient becomes $D_u(p - 2\mu_1)A_o$; the printed ε denominators cancel completely. The full residue has the A_o, D, N pivots $1/2, 1/2, 1/2$ and the same block (RU24). The complete normal residues are $0, \mathsf{x}$. Hence this chart also has a strict finite separated Gram.

A remote cap lemma and complete bounded parents. Away from the contact chart there is no need for centered excesses. Retain the original nonnegative n, α positive square terms and the positive excesses of m, A above chosen positive real caps m_c, A_c. If D is unbounded, retain its original excess above a real cap D_c too; otherwise leave D at its actual bounded value. The remainder is the full original parent with these three coefficients capped, and all its coefficients are bounded.

We use the elementary finite-dimensional fact that a real quadratic form positive definite on the common kernel of finitely many real linear rows becomes positive definite after sufficiently large positive real multiples of the squared rows are added. Choose coordinates on the common kernel and its orthogonal complement; the assertion follows by a Schur complement. Unbounded actual m, A, D, when used, exceed each chosen real cap. Thus every cap is an exact nonnegative positive square term subtraction in the original parent, and a strict residue gives WA10 for the entire bounded remainder. The real caps depend only on real testing residues, never on a source/testing rate comparison.

Finite remote v with nonzero residue. Write $U = u^2, V = v^2$ for real testing residues, with $V > 0$. On the common kernel $Y = M = 0$, $X = vW - uZ$, and the remaining two-state form is exactly

$$CD_vUW^2 + (D + 1/2)(uD_vW - vD_uZ)^2 + VD_vW^2/2 - D_uVZ^2/2. \tag{RU25}$$

Its determinant is

$$VD_uD_v\{DV/2 + U[CD - (D + 1/2)/2] + C(D + 1/2)U^2\}. \tag{RU26}$$

For $t_0 > 1$, insert $C = t/2, D = 1/[2(t-1)]$; it equals $D_uD_v[V^2 + t^2U^2V]/[4(t-1)] > 0$. Its W^2 coefficient is positive since $V > 0$. For $t_0 = 1$, $C = 1/2$ and one may choose a real D cap: the braces are $D(V + U^2)/2 + (U^2 - U)/4$, positive for sufficiently large D. The cap lemma then chooses m_c, A_c for a strict full state matrix.

Finite unit u with infinitesimal v, including zero. Use $N = -M = X + uZ - v(W + uY)$, $R_t = uW - Y$, and $V_Z = vZ$. After the original n, α positive square terms and allowed positive excesses are retained, the entire remainder lies in the square of the four roots (N, R_t, Y, V_Z). Only the pure-testing unit u is inverted to write $W = (R_t + Y)/u$; no inverse of v is needed. Its exact residue is

$$(m_c + 1/2)N^2 + CR_t^2 + A_cD_uY^2 + (D + 1/2)(R_t - D_uV_Z)^2 - D_uYV_Z - D_uV_Z^2/2. \tag{RU27}$$

For $t_0 > 1$, its restriction to $Y = 0$ has determinant $t^2D_u(D_u - 1)/[4(t-1)] > 0$ on (R_t, V_Z). Choose A_c sufficiently large. For $t_0 = 1$, the common kernel $Y = 0, R_t - D_uV_Z = 0$ has value $D_uu^2V_Z^2/2 > 0$; large real D_c, A_c give strictness. The normal pivot is positive. The full four-root support and the residue are checked directly; literal $v = 0$ is their polynomial value.

Both finite angles infinitesimal, and all state charts. The exact SD5 pure-core sign identity deals with the nonnegative-core branch and literal zero axes by original-event squares. On the negative branch use its standard gauges $u, v > 0$ and normalize the state by a maximal coordinate. If Z is infinitesimal, W is a unit and the exact BE4 selected four-root identity applies to (X, Y, uW, vW), with positive caps of m, A provided by (RU3). All other original events remain as nonnegative positive square terms. If Z is a unit normalize $Z = 1$; the same sign identity gives bounded positive ω. Its infinitesimal and unit alternatives are exactly (RU18)–(RU24) and the subsequent ω-unit chart.

Every reciprocal chart and literal projective infinity. For unbounded u, put $p_u = 1/u$ and use the entire homogeneous polynomial p_u^2P. If bounded v has nonzero residue, the full capped four-state residue is

$$CD_vW^2 + A_cD_vY^2 + (D_c + 1/2)(W + vX)^2 + (m_c + 1/2)(vY - Z)^2 \succ 0. \tag{RU28}$$

Here D_c denotes either a positive real cap or the actual positive unit residue. The four square rows have common kernel zero because v is a unit. If instead v is infinitesimal, including zero, use the exact BE $(\infty, 0)$ selected full frame $(p_uX + Z, W, Y, vX)$ with positive caps of C, A. Its strict residue is the sum of positive multiples of

$(p_uX+Z)^2,(W+vX)^2,W^2,Y^2$; the other original positive square terms are retained. This handles the zero-n degeneration at that corner.

For unbounded v and bounded u, put $q_v=1/v$. The full capped residue of q_v^2P is

$$m_c(W+uY)^2+C(uW-Y)^2+(A_c+1/2)D_uY^2+D_c(uX-Z)^2+D_uX^2/2\succ 0. \tag{RU29}$$

Its common square kernel is zero for every bounded real u, including zero. Finally, if both angles are unbounded, use $p_u^2q_v^2P$ and the full frame $(X,W,Y,P_Z=p_uZ,Q_Z=q_vZ)$. The exact residue is

$$CW^2+A_cY^2+D_c(X-P_Z)^2+X^2/2+(m_c+1/2)(Y-Q_Z)^2\succ 0. \tag{RU30}$$

The two nontrivial block determinants are $D_c/2$ and $A_c(m_c+1/2)$, both positive. The entire polynomial has this five-root support, with no inverse of p_u or q_v after homogenization. All three formulas are checked from the original parent. Their literal reciprocal zeros are polynomial values; restoring a nonzero homogeneous factor uses a pure-testing square.

The charts exhaust independent testing pairs and states. Apply each real identity to real and imaginary parts of a complex state, restore every retained actual positive square term and separated clearing factor, and use the exact sign gauges for both angle signs. This completes the stated all-tests theorem without a restriction on any opening ratio.

C.17 Two small actual angles: the R8 and R10 sectors

C.17.1 The opening D gap

Assume t,g units, h,j infinitesimal, and a unbounded. Then e is infinitesimal, $r\sim 1$, and $t\sim 1$. Use the generic BA identity above and the exact D identity in (C.36). The following normalization and residue formula hold before the thick/thin source alternatives are used. Put $L_D=lE/t-rR_D-\sqrt{h}N_y+uv\xi$ and $M_D=\sqrt{D}L_D$. Use the polynomial inverse

$$E=\frac{t}{l}\{D^{-1/2}M_D+rR_D+\sqrt{h}N_y-uv\xi\}. \tag{JD5}$$

On R8, t/l is a unit, $D^{-1/2}$ is infinitesimal, and Dj is bounded. Every coefficient of the new remainder is bounded whenever the remaining source arrays of R10.9 are bounded. In particular no ratio h/j, x/j, or testing angle divided by a source opening occurs. The full divergent rank-one block has been normalized jointly.

The exact residue after the joint normalization. Put $\sigma=t\lambda x/\sqrt{q}$ and $\mu=m-a^2q$ as in R10.9. At the R8 source/testing corner, (C.36) and (JD5) have the complete residue

$$M_D^2+r_0^2(C_0+n_0)R_D^2+(Ah)_0N_y^2+(n_0+\tfrac12-\mu_0)\xi^2+(c_F)_0F_v^2+\lambda_0J^2+R^2+Q^2-2\sigma_0Q\xi. \tag{JD6}$$

This formula applies when r,n are bounded and r a unit. Its derivation can retain $n,C,Ah,\mu,c_F,\lambda,\sigma$ as free source letters: specialize $u=v=\sqrt{h}=D^{-1/2}=x=\alpha=0$ and $t=1$ in (C.36) and (JD5). In particular the E,R_D,N_y terms become $r^2(C+n)R_D^2+AhN_y^2$, with no missing D contribution. The separate N^2,T^2 remain intact.

Unbounded a: the thick layer. Put $s=1/a$, $w=ae$, $b=a^2B$. The exact R10 source identity still reads

$$n=s^2\frac{wQ_*-bK_*}{2r\Delta_*},\qquad Q_*\sim 1,\ K_*\sim t,\ \Delta_*\sim 1. \tag{JD11}$$

Thus b/w is bounded, b is bounded, $g_0=1-w_0>0$, and

$$\begin{gathered}(m/w)_0=\frac{1}{2(1-w_0)},\quad C_0=(Ah)_0=1/2,\quad (Dj)_0=1/2,\\ \lambda_0=1/2,\quad (c_F)_0=(1-w_0+b_0)/2>0.\end{gathered} \tag{JD12}$$

The source formulas for these quantities have no j denominator; only D itself is unbounded, and (C.36) removes that pole. For $\rho=n/e^2$ positive noninfinitesimal, retain the exact R10.10 identities and R10.11 Schur pivot:

$$\begin{gathered}\bar q=a^2q=w^2\{\rho r^2+H\}+a^2\alpha,\quad H=m+\tfrac12,\quad \chi=m/w,\quad \chi-H\in\mathrm{Inf},\\ S=\tfrac12-\mu-\frac{(t\lambda x)^2}{q}=H-\frac{m^2}{\bar q}\ge\frac{H\rho r^2}{\rho r^2+H}+\frac{H^2-\chi^2}{\rho r^2+H}.\end{gathered} \tag{JD13}$$

The first term has positive residue and the second is infinitesimal, also for unbounded ρ. Moreover $x/\sqrt{q}=b/\sqrt{\bar q}$ is bounded by JD11 and $\bar q\ge Hw^2$. JD6 consequently has strict residue on $(M_D,R_D,N_y,\xi,F_v,J,R,Q)$, with the only possible nonzero cross residue in its strictly positive Q,ξ block. Restore N^2,T^2. This proves the full local parent, not just its contact-state restriction.

Unbounded a: the thin layer and the exact paired-D repair. Assume $0 < \rho = n/e^2$ infinitesimal. JD11 gives $b/w \sim 1/t$, so $\psi = w/b$ is a unit. The formal C1 contact is exactly OP4, with native parameters

$$x_1 = x, \quad \eta_1 = t, \quad \zeta_1 = s, \quad r_1 = h, \quad \theta_1 = e, \quad t_1 = w.$$

TL2 still gives $\beta_1 = -s_n/2$ infinitesimal and $\kappa = (1+2n)/(1-s_n) \sim 1$. The native gaps satisfy $M_1 = g$ a unit and $J_1 = j > 0$. The canceled CA/CS/SE remainders have no inverse J_1; their divergent paired-D mass is retained in SD3. The formal source has $C_1, A_1 h$ positive units, $D_1 j$ a positive unit, positive m_1, α_1, and positive $\beta_1 + 1/2$. These facts follow directly from TL3 and JD12. Its bounded opening ratio $t_1/e_{1,\mathrm{scale}}^2 = w/(tb)$ is a unit. Thus the explicit CA/CS/SE strict base and their finite conductors apply exactly as in the R16 thin proof; the reservation $(2+4\beta_1)\nu - \nu$ is positive because β_1 is infinitesimal. This is an algebraic use of those formulas, not an application of a physical C1 theorem with negative beta.

Here is the one change needed to the R16 perturbation. In native notation put

$$T_1 = c_a = 1 + xr, \quad a_1 = l/t, \quad b_D = \frac{J_1}{T_1} = \frac{j}{c_a}, \quad h_1 = \frac{a_1}{T_1} = 1 - b_D,$$

$$q_B = v(Z - uX), \quad M_B = (uW - Y) - h_1 q_B, \quad R_{D,\mathrm{event}} = M_B - b_D q_B. \tag{JD14}$$

The old retained paired-D positive square term in $\kappa F_1/4$ is $\kappa(D_1 + 1/2)h_1 M_B^2$. In the actual parent retain instead the positive contact-zero positive square term $(D_2 + 1/2)h_1 M_B^2$. For $\delta_D = (D_2 + 1/2) - \kappa(D_1 + 1/2)$ the entire resulting difference is

$$\boxed{\delta_D\{R_{D,\mathrm{event}}^2 - h_1 M_B^2\} = \eta_D\{M_B^2 - 2M_B q_B + b_D q_B^2\}, \quad \eta_D = \delta_D b_D.} \tag{JD15}$$

By the exact TL3 coefficients, with $\Theta = n/B$,

$$\eta_D = \frac{\Theta\{S_c P_D/t - jP_\kappa\}}{K(h+t)c_a}, \qquad \eta_D/\rho = O(w). \tag{JD16}$$

Every denominator is a unit. In particular the previously unbounded $\rho w/j$ correction is absent; no relative w/j bound is needed.

Keep the R16 positive contact-zero n row TL4, the actual centered A positive square term, and every other unchanged retained positive square term. The exact full stress is TL6 with its $\delta_D R_{D,\mathrm{event}}^2$ line replaced by the right side of JD15. Both parents and all retained positive square terms have zero contact value and full first jet, so this new entire stress has the same exact vanishing.

For completeness its new term satisfies every finite-support premise of the inherited conductor. The exact inverse state formulas give

$$q_B = v\{D_u - uN - puD_u v\mathfrak{w}\}, \qquad p = t_1/R_1. \tag{JD17}$$

At zero normals $q_B = vD_u V_0$. Hence the new scalar term is in $(v)^2$, both complete first normals are in (v), and the remaining hard entries are bounded. There are no terms above normal degree two. Every dependence on $\mathfrak{w}$ has a factor p. In the CA chart JD16 divided by ρ has bounded coefficients on the same raw roots as TL8. In the SE chart

$$\Theta/\rho = \varepsilon^2 \tau^2 QE/K$$

gives the same extra ε^2 factor to every new coefficient. Modulo ε, scalar coefficients are even in u, v and the normal coefficients are independent of $\mathfrak{w}$. Thus the exact support, full-jet, and finite divisibility hypotheses of CC3–CC6 and SE5–SE8 are unchanged. The complete stress divided by ρ has a bounded Gram on the same eight roots. Multiplication by ρ makes it an infinitesimal perturbation of the strict formal C1 base. Restoring the positive square terms proves the actual C2 parent on the entire thin contact chart, including both full state normals.

At literal $n = 0$, OP4 is a physical C1 identification with k_1, ζ_1, r_1 infinitesimal, $\eta_1 = t$ a unit and $e_{1,\mathrm{scale}}^2 = tb$ bounded. The physical C1.2/C1.3 all-tests theorems allow arbitrary positive J_1 and literal beta zero, so they cover this boundary exactly.

The unit-W entry on the unbounded-a thin layer. The actual physical C2 critical coefficient is

$$C_* = \frac{(h + 2Ah)(1 + 2D)}{2(4AhD - h)}.$$

Use the polynomial UW3 numerator/denominator with $b = wz_*$, $z_* = b/w$ bounded. Its numerator is divisible by sw because it vanishes identically at $s = 0$ and $w = 0$. The denominator after canceling the m numerator has residue $16t_0 = 16$, also when $j_0 = 0$. Consequently

$$(C_* - C)/m = s \times O(1). \tag{JD18}$$

There is no j inverse in this estimate. SEO12 permits both A, D unbounded: its entire hard block remains strict and its two radial masses tend to $2, 2$. With $W = 1$, the exact testing identity $vZ = Z(R_m + X + uZ)/(1 + uY)$ absorbs the full negative critical correction by JD18 and the retained original m square. All denominators are pure testing units. This proves every local normalized unit-W state, including $Z = 0$; negative-core state normalization leaves exactly the $Z = 1$, $u, v, \mathfrak{w} > 0$ infinitesimal chart proved above. This is the same actual-C2 state entry as UW, with its source bound checked at the new $t_0 = 1$ boundary.

Finite remote tests when e is infinitesimal. Here $C_* - C$ is infinitesimal by JD18 and bounded m. The full critical-wall identities SEO17, SEO18, SEO20, in the normalized SEO12 frame, are strict whenever at least one finite angle has nonzero residue. They explicitly allow A, D unbounded. The entire correction $-4(C_* - C)_+ D_v(uW - Y)^2$ has infinitesimal bounded entries in that frame. Retain original n, m, α and any positive excess. This proves every finite remote test, with all states, for the entire e-infinitesimal R8 domain.

Projective endpoints, signs, and the complete join. On all of R8, C is a positive unit and A, D are unbounded. For infinity charts retain the whole original m, n, α positive square terms, clip C, A, D at suitable positive real lower bounds, and retain their positive excesses. The exact homogeneous infinity identities used in SEO19 and its accompanying BE2 routing require only these lower bounds and $d = 1$; they do not require D bounded or m bounded below. In particular the delicate $(u, 1/v) = (0, 0)$ corner has the strict selected residue

$$2X^2 + 4c(uW - Y)^2 + (4A_* + 2)Y^2 + 4D_*Z^2$$

on (X, Y, uW, Z) after retaining m. All leftover W occurrences carry u. The remaining single and double infinities are exactly the same homogeneous event identities. Literal endpoint values are taken in those original homogeneous formulas; no higher-degree contact graph is substituted at infinity.

Here a is unbounded, so e is infinitesimal and $r \sim 1$. The thick, positive-thin, and literal-zero alternatives for n/e^2 are exhaustive. The finite outer calculation above uses the exact coefficient repair JD18 and the strict forms SEO17, SEO18, SEO20 in Section B.4.19. The displayed homogeneous forms prove every infinite test. The sign gauges, exact axis identities, state homogenization and real/imaginary splitting therefore give all independent tests and states on this unbounded-a R8 domain. The bounded-a part of R8 is Proposition C.27.

C.17.2 Source scale, remote tests, and weighted local completion

Use the native coefficient dictionary, the exact full-parent identity BA4–BA7, the homogeneous identities BA11–BA15, and the physical source identity (C.27). The following bounds are derived on the present unbounded-a coefficient domain.

Scope and result. Retain the original weak-C2 R10 physical stationary source, every strict source gap, $n \geq 0$, the positive-infinitesimal total-scale intersection, and every inherited exclusion. Thus t, g, j are positive bounded units, B, h, e are infinitesimal, a is unbounded, and $x = aB$ is infinitesimal. The new all-tests case proved here is the source subdomain

$$n/e^2 \notin \mathrm{Inf}\,. \tag{R10.1}$$

Every nonlocal independent testing chart is also proved on all of R10. The positive-n, infinitesimal-n/e^2 local complement is handled by the following thin-layer and state-entry sections; it is not covered by this weighted pivot. The literal $n = 0$ boundary is already an inherited C1 case, by the exact map specified below. These statements retain all independent rates.

The decisive exact source scale. Use the pure-source variables

$$s = 1/a, \qquad w = ae, \qquad b = a^2B = x/s, \quad B = s^2b, \quad x = sb, \quad e = sw, \quad r = 1 - sw.$$

Initially b is not assumed bounded. The gap $g = 1 - w + hx$ makes w bounded and $1 - w$ a positive unit in residue. In the exact positive polynomials Q_s, K_s in (C.27) put

$$Q_* = 1 + B(1 + t + ah + 2ar + at) + B^2\{a(h + t) + a^2[h(r + 1) + t(1 + r - r^2)]\},$$

$$K_* = (1 + s)(h + t)(1 + x), \qquad \Delta_* = s\Delta.$$

Before imposing a bound on b, $Q_* \sim 1$ and $\Delta_* \sim 1$: write them using bounded w, infinitesimal s, x, h; their only nonconstant monomials contain s or x. Equation (C.27) gives exactly

$$\boxed{2r\Delta_*\, n/s^2 = wQ_* - bK_*.} \tag{R10.2}$$

Consequently

$$0 < b/w \le Q_*/K_*, \qquad 0 \le n/s^2 \le wQ_*/(2r\Delta_*). \tag{R10.3}$$

In particular $b = O(w)$ is bounded, $n = O(s^2 w)$, an is infinitesimal, and $ck = \sqrt{b/t}$ is bounded for $c = \sqrt{a/t}$, $k = \sqrt{x}$. This is an original-source bound, including equality $n = 0$, not a comparison with either testing angle.

After this bound every displayed source denominator is a unit in (s, b, w, h, t) except the explicit positive monomials already canceled. Direct finite rational cancellation in the source formulas gives

$$\begin{gathered} m/w \sim \frac{1}{2(1-w)}, \quad C \sim Ah \sim t/2, \quad D \sim \frac{1}{2(t-1)}, \quad \alpha/(s^3 b^2) \sim t/2, \\ \lambda = \frac{m - a^2 q}{t a^2 B} \sim \frac{1}{2}, \qquad c_F \sim \frac{t - w + tb}{2t^2} > 0, \qquad q = nr^2 + (m + \tfrac{1}{2})e^2 + \alpha. \end{gathered} \tag{R10.4}$$

The last strictness follows from $t_0 > 1$ and $w_0 < 1$. These are joint residue identities in bounded letters; they do not set w or b to a positive unit. In particular they hold when either is infinitesimal. The cancellation defining λ leaves no b or s denominator; the new checker verifies this explicitly. We also have m bounded, $n, \alpha, A^{-1/2}$ infinitesimal, and

$$d_4 \sim \frac{t}{2(t-1)}, \quad b_4 \sim 0, \quad a_E \sim -\frac{1}{t-1}, \quad x_3 \sim -1. \tag{R10.5}$$

All first-normal arrays printed in Section C.11.1 are bounded: the only newly unbounded letter a occurs there through an or aq, both infinitesimal. The factors $a^2 n$ in c_F are bounded.

All remote tests on all of R10. The exact full-parent identities printed with BA11–BA15 and the exact identities (C.43)–(C.44) apply algebraically. The bounded-a proof used m infinitesimal; under R10.4 its entire nonnegative event positive square term is retained. In BA11 this adds a nonnegative rank-one form to the strict six-root residue displayed immediately after that identity; all its coefficients are bounded. For the chart u infinitesimal, v a bounded unit, keep $S = \sqrt{m}W$. The complete (S, X) residue is now

$$(v_0 S - \sqrt{m_0}X)^2 + \frac{1 + v_0^2}{2}X^2, \qquad \det = \frac{v_0^2(1 + v_0^2)}{2} > 0. \tag{R10.6}$$

The (uW, Z) block remains BA12 and the normalized Y pivot remains strict; the extra uvY, uZ terms in the m positive square term are infinitesimal. For unbounded v its reciprocal identity has the same diagonal residue BA13 because the potential $\sqrt{m}X$ term carries $1/v$. On the remaining remote charts retain mR_m^2 as a full positive square and apply the unchanged strict residue to the remainder. The original homogeneous identities include either literal infinity, and the exact literal-axis identities (C.43)–(C.44) retain $n = 0$ without a division. The original sign gauges and real/imaginary split are unchanged. Hence every test outside both angles infinitesimal is covered for the entire R10 source.

The exact local full-parent identity. Use the generic contact balances RC1, RC6 in Section C.11.1 and the complete identity BA4–BA7. These are rational identities of the original source and never require bounded a. Set $U = u^2, V = v^2$, $y = tB$,

$$\begin{gathered} E = tuW - vZ, \quad N_y = (Y - huW)/\sqrt{h}, \quad \xi = X + ruZ, \\ F_v = (V - y)Z, \quad R_D = v(U - x)Z, \quad J = (uv - x/c)Z, \\ Q = \sqrt{q}(avW - uZ - a\xi), \qquad N = \sqrt{n}\{r[(a/t)V - U]Z + u\xi\}, \\ R = \sqrt{\lambda}\, tk(cv - u)W, \qquad T = \sqrt{2\lambda/c}\, k\sqrt{uv}(tcW - Z), \quad \mu = m - a^2 q = ta\lambda x. \end{gathered}$$

At positive nonzero angles these are exactly the state-linear roots in BA4, with no testing-ratio inverse. The entire hard quadratic $\mathcal{H}_0$ is the explicit expression BA6. It has no inverse testing angle or a denominator; after $\eta = \sqrt{h}N_y$, every occurrence of A is through Ah, $\sqrt{h}Ah$, or hAh. Its R10 local residue is

$$\operatorname{diag}\left(\frac{1}{2(t_0 - 1)}, \frac{t_0}{2}, \frac{1}{2}\right) \quad \text{on } (E, N_y, \xi). \tag{R10.7}$$

This permits bounded m; every occurrence of m in the remaining hard entries carries a positive testing degree or h.

The principal angular–ξ coupling is resolved as a whole. Writing $R_c = kv(tcW - Z)$, direct multiplication gives

$$-2t\lambda ckR_c\xi + 2t\lambda c(R_D - uJ)\xi = -2t\lambda xQ\xi/\sqrt{q} - 2\mu\xi^2. \tag{R10.8}$$

Thus neither cu nor cR_D is left as a Gram coefficient. The complete generic identity BA7, with the source arrays just specified, gives the entire original $F/4$ exactly as

$$\begin{aligned} &Q^2+N^2+T^2+R^2+\lambda J^2+d_4R_D^2+(c_F+b_4x/t^2)F_v^2+(b_4/t^2)vR_DF_v\\ &\quad+\mathcal{H}_0-\lambda xE^2-\mu\xi^2-2(t\lambda x/\sqrt{q})Q\xi\\ &\quad+E\{(a_E+c_EV)R_D+(b_E+c_Ex)vF_v\}\\ &\quad+\sqrt{h}N_y\{(a_Y+c_YV)R_D+(b_Y+c_Yx)vF_v\}\\ &\quad+\xi\{x_5uvR_D+u(x_3+x_5x)F_v\}. \end{aligned} \tag{R10.9}$$

All coefficients are the exact source arrays in Section C.11.1 and BA5. This identity retains every scalar, first-normal and hard-normal term.

A strict weighted pivot when n/e^2 is noninfinitesimal. Put $\rho=n/e^2$, $H=m+1/2$, $\chi=m/w$, and $\overline{q}=a^2q$. Then

$$\overline{q}=w^2(\rho r^2+H)+a^2\alpha,\qquad \chi-H\in\mathrm{Inf}\,. \tag{R10.10}$$

The Q,ξ source block in R10.9 has Schur pivot

$$S=\frac{1}{2}-\mu-\frac{(t\lambda x)^2}{q}=H-\frac{m^2}{\overline{q}}. \tag{R10.11}$$

The equality uses $a^2q+\mu=m$ exactly. Since $\alpha\ge 0$,

$$S\ge H-\frac{\chi^2}{\rho r^2+H}=\frac{H\rho r^2}{\rho r^2+H}+\frac{H^2-\chi^2}{\rho r^2+H}. \tag{R10.12}$$

Here H is a positive bounded unit. Under R10.1 the first summand has positive residue and the second is infinitesimal, also when ρ is unbounded. Hence S is a positive bounded unit. Moreover b/w is bounded and $\overline{q}\ge Hw^2$, so $x/\sqrt{q}=b/\sqrt{\overline{q}}$ is bounded. All entries of this strict source block are therefore bounded.

In the rest of R10.9 the only nonzero off-diagonal residue is a_E between E and R_D, giving

$$\frac{1}{2(t_0-1)}\begin{pmatrix}1&-1\\-1&t_0\end{pmatrix}\succ 0.$$

The N_y,F_v,J,R pivots have positive residues $t_0/2,(t_0-w_0+t_0b_0)/(2t_0^2),1/2,1$. Every other entry is infinitesimal by R10.4–R10.5 and the explicit positive testing degrees in R10.9. Retain N^2,T^2 as exact positive squares. The bounded-Gram lemma supplies finite tensor SOS for the entire parent on R10.1. No source/testing rate comparison occurs. All roots homogenize without Z or testing-ratio inverses, so literal axes and $Z=0$ are included directly. The original signs, complex-state split and the remote theorem complete the all-tests case.

Literal $n=0$ and the exact remaining complement. At $n=0$, OP4 is a physical C1 identification with

$$x_1=x,\quad \eta_1=t,\quad \zeta_1=s,\quad r_1=h,\quad \theta_1=e,\quad e_{1,\mathrm{scale}}=\sqrt{tb}.$$

Thus k_1,ζ_1,r_1 are infinitesimal, η_1 a unit, and $e_{1,\mathrm{scale}}$ bounded. Its infinitesimal type is covered by the local calculation in Section B.4.18 together with the outer calculation in Section B.4.19; its positive-unit type is covered by the neighboring Section B.4.15. Both retain $\beta=0$. This is C1.2/C1.3 routing, not the unbounded-η C1.1 theorem. For $n>0$ OP5 gives formal $\beta_1<0$, so this physical C1 theorem cannot be extended by continuity or imported into the neighborhood. The complement left for the companion thin-layer proof is exactly

$$0<n/e^2\in\mathrm{Inf},\qquad u,v\in\mathrm{Inf},$$

with every original physical and total-scale intersection retained. In this complement R10.11 can have zero residue: at a unit w and $b_0=w_0/t_0$, its residue is zero. Thus R10.9 is a useful full-parent identity there, but its stated strict-Gram argument does not prove that complement. The companion thin-layer proof and direct state join complete this complement; the final join states the whole-R10 theorem.

C.17.3 The positive-n thin layer

Domain. All polynomial identities below use the convention $F/4$, as in R10.9; in particular the original n positive square term is $nD_uD_vX^2$. All parent and positive square term identities in this section use the common normalization $F/4$. Keep every original R10 condition and assume $\rho=n/e^2>0$ infinitesimal. Use s,w,b,x,B from R10.2. Dividing that exact identity by w gives

$$Q_*-(b/w)K_*=2r\Delta_*\rho w,\qquad b/w\sim 1/t. \tag{TL1}$$

Thus $\psi = w/b$ is a positive bounded unit. No testing rate occurs. The formal native C1 map OP4 is

$$x_1 = x, \quad \eta_1 = t, \quad \zeta_1 = s, \quad r_1 = h, \quad \theta_1 = e, \quad t_1 = \theta_1/\zeta_1 = w.$$

Write $P_d = P_n/B$. OP3–OP5 give the exact slack

$$s_n = 2r\Delta_*\rho w/P_d, \quad \beta_1 = -s_n/2, \quad \pi = 1 - s_n = 1 + 2\beta_1 > 0, \quad \kappa = (1+2n)/\pi. \tag{TL2}$$

Hence $s_n/w = O(\rho)$, β_1 is infinitesimal, and $\kappa \sim 1$. We substitute these coefficients into the C1 polynomial identities to obtain the comparison below.

Exact coefficient comparison. Let F_1 be the native C1 parent at this same contact, normalized by $d_1 = 1$, and F_2 the actual C2 parent, with $d_2 = 1$. For a coefficient L put $\delta L = L_2 - \kappa L_1$. Set $K = 1 + x$, $c_a = 1 + xr$, $S_c = 1 + B(1-h) + xr$, $Q_c = 1 - hB$, and $\Theta = n/B = \rho w\psi$. The following short polynomials are bounded with the displayed unit denominators:

$$P_\kappa = c_a^2 + sx(1+t) + sx^2(r + he + t),$$

$$P_D = tx^2r^2 + x\{t - 1 + e(1 - h - 2t)\} - 1, \quad P_\alpha = 1 + s - 2e + x\{r^2 + s(r + he)\}.$$

Exact cancellation of the original C2 and native C1 formulas gives

$$\begin{array}{c|l} \delta m & 0 \\ \delta C & -\Theta c_a/(1-h) \\ \delta(Ah) & -\Theta t c_a S_c/[K(h+t)] \\ \delta D & \Theta S_c P_D/[K(h+t)tj] \\ \delta\alpha & -nxP_\alpha/(KQ_c) \\ \delta d = 1-\kappa & -2\Theta P_\kappa/[K(h+t)] \\ \delta\beta + \delta d/2 & -n. \end{array} \tag{TL3}$$

In particular $\delta C, \delta D, \delta d, \delta(Ah) = O(\rho w)$, while $\delta\alpha = O(nx)$. There is no surviving $1/h$ after the large A event is centered.

The formal C1 local identities still supply a strict base. The native η_1 is a unit, k_1, ζ_1, r_1 are infinitesimal, and $e_{1,\mathrm{scale}}^2 = tb$ is bounded. Separate its positive-unit and infinitesimal types. Use respectively the exact centered-A CA5/CA8 construction (or its CS1 scaling) and the SE4/SE9–SE11 construction. This use concerns the printed formulas and conductor lemmas; the physical-source theorem is not being invoked at $\beta_1 < 0$. Here are all changes to their source premises:

The formulas and native stationary identities are rational identities, valid at the formal OP4 contact. All native gaps remain positive; $M = g$ and $J = j$ are units. The other native event coefficients are positive, with $C_1, D_1, A_1 h$ positive units and A_1 unbounded. The opening ratio $t_1/e_{1,\mathrm{scale}}^2 = w/(tb)$ is a positive bounded unit by TL1, which supplies the only scale bound previously obtained from $\beta_1 \geq 0$. All four retained positive square terms SD3 involve the positive paired coefficient $\beta_1 + 1/2 = \pi/2$, never a separate β_1 square. In SE4, the coefficient of the reserved normal positive square term is $(2 + 4\beta_1)\nu$, so its subtraction of ν is still positive, since β_1 is infinitesimal. The remaining source bounds, finite coquotients and strict residue matrices are unchanged. A positive-unit normalization to $C_1 = 2$, when used by those formulas, changes none of these bounds. Consequently their exact local construction yields the same strict bounded eight-root Gram and positive retained positive square terms for κF_1 on its contact chart.

A new positive contact-zero n row. Use the inherited pure-testing coordinate and normals

$$\mathfrak{w} = (W + uY)/D_u, \quad A_t = v\mathfrak{w}, \quad X = N + pD_uA_t - uZ, \quad p = t_1/R_1,$$

with $D_u = 1 + u^2, D_v = 1 + v^2$. In the exact CA chart the contact is

$$(u, \mathfrak{w}, v) = (k, \delta k, \delta k^2 E), \quad K_c = \delta^2 E k^2 = \zeta_1 K/Q.$$

Here Q, E denote the inherited CA source letters, not the angular weight of R10. The SE chart has $k = \varepsilon j$, $\delta = 1/\varepsilon$, so $K_c = j^2 E$. Put $\gamma = r/K_c$ and retain the entire state-linear square

$$\boxed{\mathcal{N}^2 = nD_uD_v(X + \gamma A_t)^2.} \tag{TL4}$$

At the actual contact $A_t = K_c k$ and $X = -rk$, so its value and all first derivatives vanish exactly. The square root of $D_u D_v$ is pure testing; every other radical or inverse is pure source or pure testing. The following entire-parent identity includes its full normal:

$$\begin{aligned} nD_uD_vX^2 - \mathcal{N}^2 = {} & 2n\gamma D_uD_vuA_tZ - (2n\gamma pD_u + n\gamma^2)D_uD_vA_t^2 \\ & - 2n\gamma D_uD_vA_tN. \end{aligned} \tag{TL5}$$

There is no remaining hard quadratic from this subtraction. This row is necessary: without it the scalar nu^2 lies outside the old raw square ideal, even if n/k^4 is infinitesimal.

Exact bounded stress after the positive square terms. Let $A_{*,1}$ be the inherited bounded centered-A cap. In the actual parent retain $(A_2 - \kappa A_{*,1})D_uD_v(Y - huW)^2$, which is positive because A_2 is unbounded and $\kappa A_{*,1}$ bounded. Comparing with the corresponding retained positive square term of κF_1 replaces the whole A difference by

$$\delta(Ah)D_uD_v(2uWY - hu^2W^2).$$

After retaining TL4, the full difference of the bounded remainders is exactly

$$\begin{aligned} \mathscr{E} = {} & \left[nD_uD_vX^2 - \mathcal{N}^2\right] - nD_v(X + uZ)^2 + \delta CD_v(uW - Y)^2 \\ & + (\delta D + \delta d/2)R_{D,\text{event}}^2 + \delta\alpha D_u(vY - Z)^2 \\ & + (\delta d/2)D_u\{(vY - Z)^2 - D_vZ^2\} \\ & + \delta(Ah)D_uD_v(2uWY - hu^2W^2), \end{aligned} \tag{TL6}$$

where $R_{D,\text{event}} = uW + uvX - Y - vZ$ is the original event row. This follows from TL3; in particular the paired beta difference is exactly $-n$. Every scalar, first-normal and hard-normal term is present. The full expression and its first derivatives vanish at the unchanged source contact, because both full parents and every retained square do.

Finite support and bounds for the unchanged conductor. Work on the local chart $Z = 1$, with $u, v, \mathfrak{w}$ positive infinitesimal. At zero normals set

$$V_0 = 1 - puv\mathfrak{w}, \quad q_B = D_uvV_0, \quad uW = u\mathfrak{w} + u^2h_1vV_0, \quad Y = u\mathfrak{w} - h_1vV_0, \quad h_1 = a_1/T_1. \tag{TL7}$$

These are exact inherited state inverses. The rows $uW - Y$ and $R_{D,\text{event}}$ are in (v) at zero normals; $X + u = pD_uA_t$. The centered A expression in TL6 is a quadratic in $(v, u\mathfrak{w})$ with bounded coefficients. The signed-core line has no constant: $(vY - 1)^2 - D_v$ is explicitly $(vY)^2 - 2vY - v^2$. The sole state-constant term is $\delta\alpha D_u$. Finally TL5 is a combination of $uA_t = (v)(u\mathfrak{w})$ and A_t^2, with first normal in (A_t). Thus this is an explicit raw allocation, not an ideal-membership guess.

For the unit-$e_{1,\text{scale}}$ chart use

$$(A, B_c, C_c, D_0) = (\mathfrak{w}v, \delta v, \delta Eu\mathfrak{w}, \delta^2k^3E).$$

Every scalar coefficient of $\mathscr{E}/\rho$ in $(A, B_c, C_c)^2 + (D_0^2)$ is bounded with unit denominators. Both complete first normals belong to (A, B_c, C_c) with bounded coefficients, and the hard-normal entries are bounded. The decisive source coquotients are

$$\Theta/\rho = \delta^2QEw^2/K, \quad n\gamma^2/\rho = w^2r^2Q^2/K^2, \quad nx/(\rho D_0^2) = Q^2w^2/K^2. \tag{TL8}$$

The remaining $n\gamma$ coquotient has an additional δ^2k^2E. All apparent negative k powers have therefore canceled before the conductor is used.

For the infinitesimal-$e_{1,\text{scale}}$ chart use the exact SE parameters $w = t_1 = \varepsilon^2\tau$, $\zeta_1 = j^2QE/K$, $k = \varepsilon j$, and

$$(A, B_c, C_c, D_0) = (\varepsilon\mathfrak{w}v, v, Eu\mathfrak{w}, \varepsilon^2j^3E).$$

The full raw scalar allocation of $\mathscr{E}/\rho$ is

$$f_AA^2 + f_{BB}B_c^2 + f_{BC}B_cC_c + f_{CC}C_c^2 + f_0D_0^2. \tag{TL9}$$

All five coefficients are bounded; after division by ρ each has a factor ε^2. The A^2 coefficient has this factor by TL10; all its other terms carry $p = \varepsilon^2\tau/R_1$, because $\delta m = 0$. The coefficients of B_c^2, B_cC_c, C_c^2 have it from Θ/ρ, and the constant coefficient from the final identity in TL10. In particular

$$\Theta/\rho = \varepsilon^2\tau^2QE/K,$$

$$\frac{n\gamma^2}{\rho\varepsilon^2}=\frac{\varepsilon^2\tau^2r^2Q^2}{K^2},\quad \frac{n\gamma}{\rho\varepsilon}=\frac{\varepsilon^3j^2E\tau^2rQ^2}{K^2},\quad \frac{nx}{\rho D_0^2}=\frac{\varepsilon^2\tau^2Q^2}{K^2}. \tag{TL10}$$

The first-normal coefficients are in (A,B_c,C_c) with bounded arrays, also divisible by ε^2 after division by ρ; the new A coefficient has the stronger ε^3j^2 factor printed in TL10. The extra term $-nD_v(X+u)^2$ has scalar coefficient $-np^2D_u^2D_v$ on A_t^2, first normal $-2npD_uD_vA_tN$, and hard term $-nD_vN^2$; TL10 and $p=\varepsilon^2\tau/R_1$ bound each normalized coefficient.

Exactly as in SE8, all dependence of the raw coefficients on $\mathfrak{w}$ comes through V_0 and has a factor $p=O(\varepsilon^2)$. The additional TL5 coefficients are independent of $\mathfrak{w}$. Modulo ε, scalar raw coefficients are even in u,v; the normal raw coefficients are independent of $\mathfrak{w}$. Clearing powers of D_u and the source unit denominators preserves these properties. These are precisely the finite divisibility premises of SE7: the first scalar conductor has $T_{\mathfrak{w}}\in(\varepsilon)$, the second-stage remainders have $T_{\mathfrak{w}}\in(\varepsilon,v)$, and each full normal has $T_{\mathfrak{w}}\in(\varepsilon)$.

For either chart, TL6 has exact contact value and first-jet zero. Apply the inherited monic contact lift to $\mathscr{E}/\rho$: CC3–CC6 in the first chart and the division-free SE7 lift in the second. The raw support restriction retains only D_0^2, not arbitrary mixed D_0 products. The proofs use only finite monic division and bounded unit denominators; extending their coefficient ring by the bounded source letters in TL8–TL10 changes nothing. They give a bounded Gram matrix on the same eight inherited roots for $\mathscr{E}/\rho$. Hence the exact stress $\mathscr{E}$ has an infinitesimal Gram matrix on that list. No inverse of ρ occurs in its final coefficients: division by ρ here records the common positive source factor before a linear finite lift.

Adding this infinitesimal matrix to the strict formal-C1 base Gram preserves its positive residue. Restore all its retained positive square terms, the modified positive centered-A positive square term, and TL4. This proves the full actual C2 parent on the thin-layer contact chart, with both complete normals, all hard couplings and every independent rate. The direct actual-C2 outer/state argument and the all-R10 remote argument are required for the all-tests join; no negative formal-beta positive square term is used in either join.

C.17.4 The critical wall for unit W

Statement. On the entire original R10 source, every independent test with u,v infinitesimal and normalized W a testing unit is tensor SOS. This includes bounded Z of every sign, infinitesimal Z, and $Z=0$. The theorem uses the actual physical C2 coefficients. No continuation of a negative-beta C1 source is involved.

Use the R10 source scale $s=1/a$, $w=ae$, $b=a^2B$, and put $z=b/w$. The source constraint proves z bounded. Write $A_h=Ah$. The source estimates give A_h,C,D positive units, h infinitesimal, m/w a positive unit, and $t_0>1$, $w_0<1$. Define the positive pure-source critical coefficient

$$C_*=\frac{(h+2A_h)(1+2D)}{2(4A_hD-h)}=\frac{(1+2A)(1+2D)}{2(4AD-1)}. \tag{UW1}$$

Its denominator after the first normalization is a positive unit. We prove the opening-uniform estimate

$$\frac{C_*-C}{m}=s\,R,\qquad R\text{ bounded}. \tag{UW2}$$

A short polynomial proof of the source estimate. All the following letters are polynomials in bounded source variables (s,w,z,h,t) after $b=wz$, with $x=sb$, $B=s^2b$, $r=1-sw$:

$$c_a=1+xr,\quad S_c=1+B(1-h)+sbr,\quad g=1-w+hx,\quad J=tc_a-(1-h),\quad \Delta_*=s\Delta,$$

$$M_*=(1+x)(s+r)(h+t)-sw(1+tB)(1-h).$$

Use the exact source numerators and denominators

$$\begin{aligned}
&N_C=c_aM_*, && D_C=2(1-h)\Delta_*,\\
&N_A=tc_a(s+r)S_c, && D_A=2\Delta_*,\\
&N_D=(1+x)(s+r)(1-txr)S_c, && D_D=2J\Delta_*,\\
&N_m=w(1+tB)c_aS_c, && D_m=2g\Delta_*.
\end{aligned}$$

Thus $C=N_C/D_C$, $A_h=N_A/D_A$, $D=N_D/D_D$, and $m=N_m/D_m$. Clear the denominator in UW1:

$$\begin{aligned}
P&=(hD_A+2N_A)(D_D+2N_D)D_C\\
&\quad-2N_C(4N_AN_D-hD_AD_D),\\
Q&=2(4N_AN_D-hD_AD_D)D_C.
\end{aligned} \tag{UW3}$$

Then $C_*-C=P/Q$. At $s=0$ one has

$$C=\frac{t+h}{2(1-h)},\quad A_h=t/2,\quad D=\frac{1}{2(t-1+h)},$$

and substitution gives $C_* = C$ identically. At $w = 0$ (hence $b = 0$), the same three formulas hold for every s, because all factors $1 + s$ cancel. Therefore the polynomial P is divisible by both s and w, hence $P = swP_1$. Since $N_m = wN_{m,1}$,

$$\frac{C_* - C}{m} = s\,\frac{P_1 D_m}{QN_{m,1}}.$$

The denominator $QN_{m,1}$ at $s = h = 0$ is $16t > 0$. This proves UW2 by finite polynomial cancellation, without comparing w to s or h. Its displayed residue also gives $R \sim t(1 - w)(tz + 2)$, though the proof only needs boundedness.

Complete parent and state allocation. Set $\epsilon = (C_* - C)_+$. By UW2, ϵ/m is infinitesimal. If $C > C_*$, retain the nonnegative excess $C - C_*$. Retain the entire original n and α positive square terms; beta is zero. The remaining signed core and C_*, A, D positive square terms are exactly the critical parent SEO11, with $c = C_*$ and $d = 1$, because

$$2C_*(4AD - 1) = (1 + 2A)(1 + 2D).$$

The exact critical-wall identities SEO12–SEO15 give a bounded strict Gram on

$$B_t = X + uZ, \quad H_A = \sqrt{A}(Y - yV), \quad H_D = \sqrt{4D + 2}\{uW - Y + uvB_t - (b_* + u^2)V\}, \quad uV,\ vV,$$

where $V = vZ$, $y = 1/(2A)$ and $b_* = (4AD - 1)/(A(4D + 2))$. Here y, b_* are bounded; C_*, D are units and A may be unbounded. The hard residue is strict and the two radial masses have positive residues 2 and $2/t_0$. This is an application of those full polynomial identities, not of their original C1 source theorem.

Normalize $W = 1$ by a pure-testing factor. The entire m event obeys

$$R_m = v(1 + uY) - B_t, \qquad V = \frac{Z}{1 + uY}(R_m + B_t). \tag{UW4}$$

The denominator is a positive testing unit. From the displayed H_D root,

$$uW - Y = (b_* + u^2)V - uvB_t + H_D/\sqrt{4D + 2}.$$

Thus $\sqrt{\epsilon}(uW - Y)$ is a linear combination of the strict critical roots and $\sqrt{m}R_m$ with infinitesimal coefficients. The only potentially small denominator is accounted for by $\sqrt{\epsilon/m}$, infinitesimal by UW2. Consequently the complete negative correction $-4\epsilon(1 + v^2)(uW - Y)^2$ has an infinitesimal bounded Gram. The m positive square term supplies a strict additional pivot. Restore all retained original positive square terms to obtain the entire C2 parent, including every hard state direction.

State entry for the remaining local work. Normalize a real testing state by a maximal coordinate. On the negative pure-testing-core branch at u, v infinitesimal, X, Y are infinitesimal. If W is a unit the theorem applies. Otherwise Z is a unit; set $Z = 1$, and then $w_{\text{test}} = (W + uY)/(1 + u^2)$ is infinitesimal. The inherited exact Schur sign identity gives the positive contact chart. Literal axes have the original event-square exits; complex states split into real and imaginary parts. Together with the R10 remote theorem, the only remaining task is the contact chart $u, v, w_{\text{test}} > 0$ infinitesimal, $Z = 1$, on any source not already proved locally.

C.17.5 Exact extraction of the centered m event

The identities in this subsection concern the entire four-state parent (C.26). They do not transfer a source admission from one stratum to another. All source quantities belong to K_c, all testing quantities belong to K_t, and the two labelled factors remain separate. Put $H = hB$, $a = x/B$ and $g = 1 - a(1 - r - H)$, as in the original stationary source. Write

$$M = vW - X + uvY - uZ, \qquad \xi = X + ruZ, \quad \eta = Y - huW, \quad R_2 = vW - a^{-1}uZ, \quad T = uvW - BZ.$$

Direct expansion gives the contact-centered row

$$R_{mc} = M - gvW = -\xi + uv\eta + (1 - g)R_2 + huT. \tag{R5.2}$$

Indeed $1 - g = a(1 - r - H)$ and $H = hB$. At the original contact the four roots on the right vanish. Thus retaining a multiple of R_{mc}^2 preserves the contact value and its full first jet.

For any positive real $K < m$, subtract this whole square from the original parent. With P denoting (C.26), the exact remainder is

$$P - (m - K)R_{mc}^2 = P\big|_{m=K}^{\text{event}} + 2\tau vWM - \tau g v^2 W^2,$$
$$\tau = (m - K)g.$$

Here $P|^{\mathrm{event}}_{m=K}$ means changing only the coefficient of M^2 in the displayed event parent; all other actual source coefficients remain fixed. Equivalently, its complete m contribution is the elementary identity

$$mM^2 - (m-K)R_{mc}^2 = KM^2 + 2(m-K)g\,vWM - (m-K)g^2v^2W^2. \tag{R5.5}$$

This is an identity for arbitrary W, X, Y, Z, before restricting a state or taking a residue. In particular, when g is infinitesimal, mg is bounded and the other event coefficients are bounded, its remainder has bounded coefficients at every bounded testing pair. It does not require a to be bounded. The retained term is a square in the tensor ring because $\sqrt{m-K}$ belongs to the source field.

The same extraction has an exact five-root matrix form. In the unnormalized WA coordinates $\mathcal{R} = (\xi, \eta, R_1, R_2, T)^T$, where $R_1 = uW - t^{-1}vZ$, let G be the full WA matrix, including both complete normal rows and its hard block. Put $L_m = (-1, uv, 0, 1-g, hu)^T$. Then

$$G = G\big|^{\mathrm{table}}_{m=K} + (m-K)L_mL_m^T + \tau H_m, \tag{R5.3}$$

where the only nonzero entries of the symmetric matrix H_m are

$$(H_m)_{14} = -1, \qquad (H_m)_{25} = v, \qquad (H_m)_{44} = 2-g, \qquad (H_m)_{45} = hu. \tag{R5.4}$$

The full WA table is affine in m. Its entrywise derivative, with every other actual source coefficient fixed, is $\partial_m G = L_mL_m^T + gH_m$, which proves (R5.3). This formal table substitution differs from the event substitution in (R5.5): the WA table has already used the original contact balances. Neither substitution asserts that a modified source is physical or stationary. Both expressions describe the same original parent after removing $(m-K)R_{mc}^2$.

A real cap on a squared row. Let B_0 be a real symmetric form and let M_0 be a nonzero real linear row. If B_0 is positive definite on $\ker M_0$, then $B_0 + KM_0^2$ is positive definite for all sufficiently large positive real K. To see this, choose real coordinates (z, q) with $q = M_0$ and write the form as

$$z^TAz + 2q\,b^Tz + (c+K)q^2, \qquad A \succ 0. \tag{C.60}$$

Its real Schur complement is $c + K - b^TA^{-1}b$, which is positive once K is sufficiently large. This uses only real matrices. The argument also applies after adding a fixed real multiple of LM_0 for any real row L, because that addition vanishes on $\ker M_0$.

The actual unbounded-a R6 finite remote source. Retain the original physical R6 source, its stationary contact, all strict gaps, coefficient signs, total scale and prior exclusions. Thus $t, h, l = 1-h, j, r$ are positive bounded units, $B, x = aB, y = tB, g$ are positive infinitesimals, a is unbounded, and $n \geq 0$. No lower bound on $b = a^2B$ is imposed. Put $s = a^{-1}$ and $w = a(1-r) = 1 + hx - g$, so $w \sim 1$, $x = sb$ and $B = s^2b$. The physical identity (C.27) gives

$$2r\Delta_* n/s^2 = wQ_* - bK_*, \qquad \Delta_* = s\Delta \sim 1, \quad Q_* \sim 1, \quad K_* = (1+s)(h+t)(1+x) \sim h+t.$$

Here Q_* is exactly Q_s in that identity. Since $n \geq 0$, this forces b to be bounded. Substitution in the actual rational source formulas (C.25), followed by cancellation before taking residues, yields jointly

$$\begin{gathered} (mg)_0 = \tfrac{1}{2}, \qquad n_0 = \alpha_0 = 0, \\ A_0 = \frac{t_0}{2h_0}, \qquad C_0 = \frac{h_0+t_0}{2l_0}, \qquad D_0 = \frac{1}{2(t_0-l_0)}, \\ (n/s^2)_0 = \frac{1 - b_0(h_0+t_0)}{2}, \qquad \Big(\frac{\alpha}{s^3b^2}\Big)_0 = \frac{h_0+t_0}{2}. \end{gathered} \tag{C.61}$$

The canceled denominators are units jointly at $s = g = 0$ for bounded b. In particular these formulas cover b a unit or infinitesimal, and literal $n = 0$. The positive unit j gives $t_0 > l_0$, while $h_0, l_0 > 0$. Thus A_0, C_0, D_0 are strictly positive and m is unbounded. No estimate compares s, g or b with a testing variable.

Consider a bounded testing pair with $v_0 \neq 0$, and write u, v, h, t, l for its real testing residues and the real source residues in this paragraph. Set $U = u^2$, $V = v^2$, $D_u = 1+U$, $D_v = 1+V$. By (R5.5), the complete bounded remainder has residue

$$B_0 + KM_0^2 + vWM_0, \qquad M_0 = vW - X + uvY - uZ,$$

where the full four-state form B_0 is

$$\begin{aligned} B_0 = {} & C_0D_v(uW-Y)^2 + A_0D_uD_vY^2 + (D_0+\tfrac{1}{2})(uW+uvX-Y-vZ)^2 \\ & + \tfrac{1}{2}D_u(vY-Z)^2 + \tfrac{1}{2}D_v(X+uZ)^2 - \tfrac{1}{2}D_uD_vZ^2. \end{aligned}$$

On $M_0 = 0$, substitute $X = vW + uvY - uZ$. The determinant of this three-state form in (W, Y, Z) is exactly

$$\frac{D_u^2 D_v^2 (h+t)}{8hl(t-l)}\{V^2 + t(h+t)U^2V + hUV^2\} > 0. \tag{C.62}$$

At $Z = 0$ its signed core is $[J^2 + D_u V Y^2 + D_v X^2]/2$, with $J = uW + uvX - Y$. The positive A_0 term forces $Y = 0$, the X^2 term then forces $X = 0$, and $M_0 = 0$ with $v \neq 0$ forces $W = 0$. Hence the restriction to $Z = 0$ is positive definite. Together with the determinant this proves $B_0|_{M_0=0} \succ 0$. The centered correction vWM_0 vanishes on that same kernel. The real-cap argument above therefore gives a positive real K for which the full four-state residue is strictly positive definite.

Unbounded actual m permits this real choice with $K < m$. All coefficients of the exact remainder belong to $B_c \otimes_{\mathbb{R}} B_t$: the source coefficients $n, A, C, D, \alpha, g, \tau$ are bounded, and u, v are bounded in their own field. Lemma 3.2 now gives its exact finite tensor sum of squares for arbitrary state coordinates. Restoring the retained square proves precisely this finite remote join. The same real identity applies separately to real and imaginary state parts. The projective endpoint identities follow in the next subsection.

C.17.6 Exact projective full-state identities for R6 and R7

The following polynomial identities concern the original parent (C.1), rather than a source theorem on another stratum. They also give an endpoint proof that applies when the actual coefficient n is zero or the actual coefficients A, D are unbounded. Write $D_z = 1 + z^2$ for each angular variable z, and continue to use $P = F/4$. Thus

$$\begin{aligned} P &= nD_uD_vX^2 + mM^2 + CD_v(uW - Y)^2 + AD_uD_vY^2 + DJ^2 + \alpha D_u(vY - Z)^2 \\ &\quad + \tfrac{1}{2}\{J^2 + D_u(vY - Z)^2 + D_v(X + uZ)^2 - D_uD_vZ^2\}, \\ M &= vW - X + uvY - uZ, \qquad J = uW + uvX - Y - vZ. \end{aligned}$$

Three whole homogeneous polynomials. For $p = 1/u$ define

$$M_u = pvW - pX + vY - Z, \qquad J_u = W + vX - pY - pvZ.$$

Multiplication by p^2 gives the exact identity

$$\begin{aligned} P_u := p^2P &= nD_pD_vX^2 + mM_u^2 + CD_v(W - pY)^2 + AD_pD_vY^2 \\ &\quad + DJ_u^2 + \alpha D_p(vY - Z)^2 + \tfrac{1}{2}\mathcal{C}_u, \\ \mathcal{C}_u &= J_u^2 + D_p(vY - Z)^2 + D_v(pX + Z)^2 - D_pD_vZ^2. \end{aligned} \tag{C.63}$$

For $q = 1/v$ put

$$M_v = W - qX + uY - quZ, \qquad J_v = quW + uX - qY - Z.$$

The other single homogeneous polynomial is

$$\begin{aligned} P_v := q^2P &= nD_uD_qX^2 + mM_v^2 + CD_q(uW - Y)^2 + AD_uD_qY^2 \\ &\quad + DJ_v^2 + \alpha D_u(Y - qZ)^2 + \tfrac{1}{2}\mathcal{C}_v, \\ \mathcal{C}_v &= J_v^2 + D_u(Y - qZ)^2 + D_q(X + uZ)^2 - D_uD_qZ^2. \end{aligned} \tag{C.64}$$

For both reciprocal coordinates use the five roots (X, W, Y, P_Z, Q_Z), where $P_Z = pZ$ and $Q_Z = qZ$, and set

$$M_b = pW - pqX + Y - Q_Z, \qquad J_b = qW + X - pqY - P_Z.$$

Then the complete polynomial is

$$\begin{aligned} P_b := p^2q^2P &= nD_pD_qX^2 + mM_b^2 + CD_q(W - pY)^2 + AD_pD_qY^2 \\ &\quad + DJ_b^2 + \alpha D_p(Y - Q_Z)^2 + \tfrac{1}{2}\mathcal{C}_b, \\ \mathcal{C}_b &= J_b^2 + D_p(Y - Q_Z)^2 + D_q(p^2X^2 + 2XP_Z - P_Z^2). \end{aligned} \tag{C.65}$$

In particular the apparently remaining Z^2 term has canceled in (C.65). This is an identity on the five displayed roots; their independence is unnecessary. All three right-hand sides are polynomials at $p = 0$ or $q = 0$. They therefore define the literal projective endpoints without division by a reciprocal coordinate.

The R6 kernel identities, including zero residue of n. Suppose g is infinitesimal, m is unbounded, mg is bounded, n, C, A, D, α are bounded, $n \geq 0$, and $C_0, A_0, D_0 > 0$, $\alpha_0 = 0$. These are precisely the coefficient properties needed here. Both the original bounded-a R6 source and the unbounded-a R6 sources supply them; in the latter, $n_0 = 0$, including when n is literally zero.

For any positive real $K < m$, apply the exact identity

$$mH^2 - (m-K)(H-gL)^2 = KH^2 + 2\mu LH - \mu g L^2, \qquad \mu = (m-K)g, \tag{C.66}$$

with $(H, L) = (M_u, pvW), (M_v, W), (M_b, pW)$ in the three homogeneous polynomials. The remainder has bounded coefficients. Write $\kappa = \mu_0 = (mg)_0$.

At $p = 0$ the centered correction has zero residue. On the kernel $vY - Z = 0$ of the limiting m row, the complete residue is

$$n_0 D_v X^2 + C_0 D_v W^2 + A_0 D_v Y^2 + (D_0 + \tfrac{1}{2})(W + vX)^2. \tag{R6.5}$$

Here and in the next two displays the remaining testing letters denote their real residues. The full four-state residue is this form plus $(K + \frac{1}{2})(vY - Z)^2$. If $n_0 > 0$, it is strict for every finite v. If $n_0 = 0$, the determinant of the kernel form is

$$A_0 C_0 (D_0 + \tfrac{1}{2}) v^2 D_v^2.$$

It is strict for $v \neq 0$. At $v = 0$ its X direction is absent; the separate four-root calculation below treats that corner.

At $q = 0$ the centered correction is $2\kappa W(W + uY)$, which vanishes on the limiting m-row kernel $W + uY = 0$. On this kernel the complete residue is

$$(n_0 + \tfrac{1}{2}) D_u X^2 + C_0 D_u^2 Y^2 + (A_0 + \tfrac{1}{2}) D_u Y^2 + D_0 (uX - Z)^2. \tag{R6.6}$$

Its determinant on (X, Y, Z) is

$$D_0 (n_0 + \tfrac{1}{2}) D_u^2 (C_0 D_u + A_0 + \tfrac{1}{2}) > 0.$$

Its Y coefficient is positive and its (X, Z) block is strict, so the kernel form is positive definite for every finite u, including $u = 0$ and $n_0 = 0$. The real-cap argument in Section C.17.5 chooses a sufficiently large real K for the full four-state residue; its inverses occur only in a real matrix, never in the tensor ring.

At $p = q = 0$ the centered correction again has zero residue and the complete five-root form is

$$(n_0 + \tfrac{1}{2}) X^2 + C_0 W^2 + A_0 Y^2 + D_0 (X - P_Z)^2 + (K + \tfrac{1}{2})(Y - Q_Z)^2. \tag{R6.7}$$

The two nontrivial block determinants are $D_0(n_0 + \frac{1}{2})$ and $A_0(K + \frac{1}{2})$. Both remain positive at $n_0 = 0$; the whole five-root determinant is $A_0 C_0 D_0 (n_0 + \frac{1}{2})(K + \frac{1}{2}) > 0$.

A common capped proof for the R6 and R7 endpoints. There is also an endpoint proof using only positive real lower bounds on m, C, A, D. It avoids imposing boundedness on A or D. Choose positive real numbers

$$m_c < m, \qquad C_c < C, \qquad A_c < A, \qquad D_c < D.$$

Retain the whole n, α event squares and the four nonnegative excess event squares with coefficients $m - m_c, C - C_c, A - A_c, D - D_c$. In each homogeneous chart the remaining polynomial is exactly (C.63), (C.64), or (C.65) with

$$(n, \alpha, m, C, A, D) = (0, 0, m_c, C_c, A_c, D_c).$$

Thus all its coefficients are bounded pure-testing elements. Its complete residues in the three charts are, respectively,

$$C_c D_v W^2 + A_c D_v Y^2 + (D_c + \tfrac{1}{2})(W + vX)^2 + (m_c + \tfrac{1}{2})(vY - Z)^2, \tag{C.67}$$

$$m_c (W + uY)^2 + C_c (uW - Y)^2 + (A_c + \tfrac{1}{2}) D_u Y^2 + D_c (uX - Z)^2 + \tfrac{1}{2} D_u X^2, \tag{C.68}$$

$$C_c W^2 + A_c Y^2 + D_c (X - P_Z)^2 + \tfrac{1}{2} X^2 + (m_c + \tfrac{1}{2})(Y - Q_Z)^2. \tag{C.69}$$

The square rows in (C.67) have common kernel zero when $v \neq 0$. The square rows in (C.68) have common kernel zero for every finite u: first $X = Y = 0$, then the m_c and D_c rows give $W = Z = 0$. In (C.69), the separate W, X, Y squares and the last two rows force all five roots to vanish. These are strict Gram matrices for the entire homogeneous state polynomial.

The R6 bounds give the required positive real lower bounds. In unbounded-a R7, m, A are unbounded and $C_0 = t_0/2 > 0$. When $t_0 > 1$, $D_0 = 1/[2(t_0 - 1)] > 0$; when $t_0 = 1$, D is unbounded. Hence these same real caps exist on both R7 gap boundaries. The actual n, α packets were retained, so neither of them was replaced by its zero residue in the full polynomial.

The complete $(\infty, 0)$ allocation. Assume p, v are infinitesimal, allowing either to be literally zero. Retain all the original event squares except positive real caps C_c, A_c of the C, A events. Put

$$H = pX + Z, \qquad T = vX, \qquad \widehat{J} = W + D_p T - pY - pvH.$$

The entire selected polynomial has the following four-root identity:

$$\begin{aligned} Q_{\infty 0} &= C_c D_v (W - pY)^2 + A_c D_p D_v Y^2 + \tfrac{1}{2}\widehat{\mathcal{C}}, \\ \widehat{\mathcal{C}} &= \widehat{J}^2 + D_p v^2 Y^2 - 2D_p vYH + (1 - p^2 v^2)H^2 \\ &\quad + 2pD_p TY + 2pvD_p TH - p^2 D_p T^2. \end{aligned} \tag{C.70}$$

Indeed substitute $Z = H - pX$ into $\mathcal{C}_u$ and collect every remaining vX as T. This gives (C.70) by multiplication and addition only. In particular it gives every state cross term, with no inverse of p or v. On (H, W, Y, T) its residue is

$$\tfrac{1}{2}H^2 + \tfrac{1}{2}(W + T)^2 + C_c W^2 + A_c Y^2 \succ 0, \qquad \det = \tfrac{1}{4} A_c C_c > 0.$$

This replaces the nonstrict ordinary state residue at $n_0 = v_0 = 0$. It also treats the literal endpoint, where some of the four roots are dependent or zero.

Passage from the residues to tensor sums of squares. Each polynomial above has an explicit finite root frame with bounded coefficients. For the centered R6 formulas the coefficients belong to $B_c \otimes_{\mathbb{R}} B_t$ because $\mu = (m - K)g$ is bounded; for the capped proof they even belong to B_t. The strict real matrices therefore meet every hypothesis of Lemma 3.2. That lemma gives a finite tensor sum of squares for the complete remainder, rather than merely pointwise real positivity. Restore the retained positive event squares, or the positive centered excess square in (C.66). For nonzero reciprocal coordinates, restoring homogeneity multiplies squares only by pure-testing factors. At literal endpoints the displayed polynomial value is used directly. All identities hold for either angle sign and every real state; a complex state is treated by adding the identities for its real and imaginary parts. These calculations cover every projective test with at least one angle at infinity or unbounded, without using a finite-contact theorem on another source stratum.

Reference

Bibliography

[1] C. Scheiderer, *Second-order cone representation for convex sets in the plane*, SIAM J. Appl. Algebra Geom. **5**, 114–139 (2021), arXiv:2004.04196v2. The general tensor-evaluation criterion used here is Corollary 3.19, p. 15; the quantitative tensor-rank criterion is Theorem 3.10. `https://arxiv.org/abs/2004.04196`.